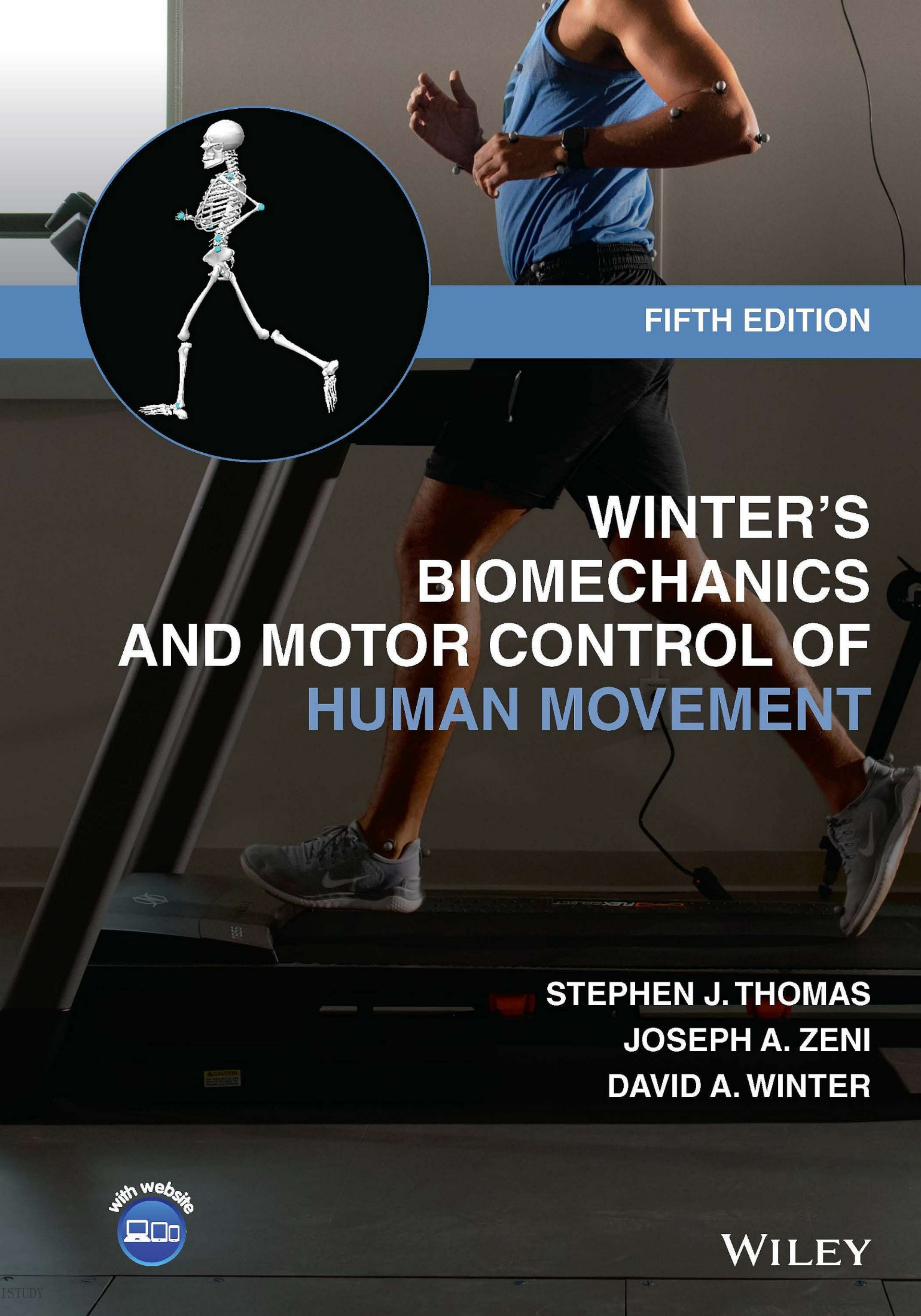

FIFTH EDITION

WINTER'S BIOMECHANICS AND MOTOR CONTROL OF HUMAN MOVEMENT

STEPHEN J. THOMAS
JOSEPH A. ZENI
DAVID A. WINTER

with website

WILEY

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Fifth Edition

STEPHEN J. THOMAS
Thomas Jefferson University, Philadelphia, PA, USA

JOSEPH A. ZENI
Rutgers University, Newark, NJ, USA

DAVID A. WINTER (DECEASED)
University of Waterloo, Waterloo, Ontario, Canada

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DEDICATION

We would like to dedicate this edition to David A. Winter and his family. To those inside the field of biomechanics, David Winter needs no introduction. He is often described as the founding father of modern biomechanics and rightfully so. His education was in the field of electrical engineering and he went on to become a Fellow of the Institute of Electrical and Electronics Engineers. However, his education, training, and service were not constrained to electrical engineering. He served in professorship roles in biomedical engineering, surgery, and kinesiology. He was a founding member of the Canadian Society for Biomechanics and later, became the first recipient of the Career Investigators Award by the same society. He received the Lifetime Achievement Award by the Gait and Clinical Movement Analysis Society, as well as the Muybridge Medal by the International Society of Biomechanics, which is the society's most prestigious honor. He solely authored four previous editions of this textbook, which is considered the seminal technical biomechanics textbook and is still readily used by professionals in biomechanics, engineering, kinesiology, rehabilitation sciences, and many other related fields. He also authored three additional textbooks and over 100 peer-reviewed research publications. He finished his career as a Distinguished Professor Emeritus in Kinesiology at the University of Waterloo.

His accomplishments in the field of biomechanics demonstrate his drive and passion for his profession. However, his passion and love for his family never took a backseat to his career. We were very fortunate to meet and speak with his three children Merriam, Andrew, and Bruce. The stories they shared about their father made clear that he was a constant and loving presence in their lives. The Winter family always ate dinner together, which served as a roundtable for insightful and thought-provoking conversations about the things the children were currently involved in at school and life. His children revealed that their father never missed an event or activity in their busy lives. His daughter Merriam shared a memory of her father from when she was six years old. When she happened upon her father working in the basement, she asked "What are you doing, dad?" He responded "Making wood!" What he really was doing was building a life-size playhouse for the backyard that included a porch, flower boxes in the windows, benches, a table, and all painted green and pink. She also shared her memories about how David Winter built the family a cottage on the outskirts of Halifax from scratch. His son Bruce shared stories of visiting his father in his gait lab when he was attending college and being able to see David Winter's passion for teaching students and working with patients. David Winter poured his devotion to teaching his students, and Bruce shared that these students became part of the family and would often join the family dinner at their house. These heartwarming stories demonstrate that it is possible to achieve so much in your career, while still putting your family in the forefront. This is an important lesson to anyone in academia or industry, vocations which continually compete for our precious time.

In speaking with David Winter's family, a key lesson became clear; the foundation for a successful life and career is to do what you love, while supporting those around you; whether that be your patients, students, colleagues, or most importantly, your family and friends. We would like to thank his children and grandchildren for sharing their dad and grandfather with all of us, in the field of biomechanics, as we are all better because of him.

Stephen J. Thomas, PhD, ATC
Joseph A. Zeni, PhD, PT

CONTENTS

APPENDICES

LIST OF CONTRIBUTORS

Garrett S. Bullock, PT, DPT, DPhil, Instructor, Department of Orthopaedic Surgery, Wake Forest School of Medicine, Winston-Salem, NC, USA

Christopher J. Burcal, PhD, ATC, Assistant Professor, School of Health and Kinesiology, University of Nebraska at Omaha, Omaha, NE, USA

Thomas Hulcher, M.S., Biomechanist, Department of Physical Therapy, Jefferson College of Rehabilitation Science, Thomas Jefferson University, Philadelphia, PA, USA

Namwoong Kim, PhD, Research Associate, Division of Health & Kinesiology, Incheon National University, Incheon, Republic of Korea

Brian A. Knarr, PhD, Associate Professor, Department of Biomechanics, College of Education, Health, and Human Sciences, University of Nebraska at Omaha, Omaha, NE, USA

Amy L. Lenz, PhD, Research Instructor, Department of Orthopaedics, School of Medicine, University of Utah, Salt Lake City, UT, USA

Todd J. Leutzinger, PhD, Instructor, Department of Biomechanics, College of Education, Health, and Human Sciences, University of Nebraska at Omaha, Omaha, NE, USA

Alan R. Needle, PhD, ATC, LAT, CSCS, Associate Professor, Department of Public Health & Exercise Science and Department of Rehabilitation Science, Appalachian State University, Boone, NC, USA.

Kristen F. Nicholson, PhD, Assistant Professor, Department of Orthopaedic Surgery, Wake Forest School of Medicine, Winston-Salem, NC, USA

Ankur Padhye PT, **PhD Student**, Rehabilitation Sciences/Movement Science and Disorders Concentration, Department of Physical Therapy, East Carolina University, Greenville, NC, USA

Stephen J. Thomas, PhD, ATC, Associate Professor and Department Chair, Department of Exercise Science, Jefferson College of Rehabilitation Science, Thomas Jefferson University, Philadelphia, PA, USA

John D. Willson, PhD, PT, Associate Professor, Department of Physical Therapy, East Carolina University, Greenville, NC, USA

David A. Winters (Deceased), University of Waterloo, Waterloo, Ontario, Canada

Joseph A. Zeni, PhD, PT, Associate Professor, Department of Rehabilitation and Movement Sciences, School of Health Professions, Rutgers University, Newark, NJ, USA

PREFACE

The field of biomechanics has experienced a wealth of growth since the printing of the fourth edition. With sweeping technological advancements, biomechanics has become an integral component of robotics, healthcare and medicine, athletic performance, movies and animation, and numerous other fields. With these changes, the discipline of biomechanics has become more tangible, recognizable, and accessible to individuals across the world. This new fifth edition reflects the expanded breadth and depth of biomechanics and includes updated references and research.

The original text, *Biomechanics of Human Movement*, published in 1979, had its title changed to *Biomechanics and Motor Control of Human Movement* when the second edition was published in 1990. This change is to acknowledge the new directions and research of the 1980s. In that second edition, five of eight chapters addressed various aspects of muscles and motor systems. The third edition, published in 2004, added information about three-dimensional (3D) kinematics and kinetics, reflecting the continued emphasis on motor control and the close links between biomechanics and movement analysis. The fourth edition, published in 2009, added updated discussions of anatomy, muscle physiology, and electromyography, with a strong emphasis on dynamic movements and in vivo data and analyses.

In keeping with the past updates, this updated edition integrates the new directions of biomechanics. These new directions include a strong emphasis on the role of the central nervous system in motor control and biomechanical analyses, as well as a focus on practical applications of biomechanics. The fifth edition has two new chapters, Chapter 12, "Central Nervous System's Role in Biomechanics," and Chapter 13, "A Case-Based Approach to Interpreting Biomechanical Data." Chapter 3, "Kinematics," has undergone significant updates to incorporate new technology and approaches to measuring human movement, including, but not limited to markerless technology. Chapter 7, "Understanding 3-D Kinematic and Kinetic Variables," includes new sections on markersets design, event detection, the incorporation of ISB standards, and induced acceleration analysis. Chapter 10, "Modeling of Human Movement" has been completely rewritten to include the current modeling approaches and technologies used in biomechanical research. Chapter 11 has been renamed to "Static and Dynamic Balance" to better reflect the content in this chapter. All chapters have also undergone edits to include updated information in each of the topic areas. The appendices, which underwent major additions in the second edition, remain intact. The extensive numerical tables contained in Appendix A: "Kinematic, Kinetic, and Energy Data" can also be found online in excel format to facilitate learning experiences in the modern classroom.

As was stated in the original editions, it is expected that the student has had basic courses in anatomy, mechanics, calculus, and electrical science. The major disciplines to which the book is directed are kinesiology, biomechanics, engineering (bioengineering, mechanical, and rehabilitation engineering), physical education, ergonomics, physical, and occupational therapy, athletic training, neuroscientists, sport scientists, and exercise scientists. The text should also prove valuable to researchers in orthopedics, muscle physiology, and rehabilitation medicine. Lastly, this text can provide valuable information to athletic coaches to help guide them in the appropriate use of biomechanical data to make training and coaching decisions.

Stephen J. Thomas, PhD, ATC
Thomas Jefferson University
Philadelphia, PA, USA
March 2022

Joseph A. Zeni, PhD, PT
Rutgers University
Newark, NJ, USA
March 2022

ACKNOWLEDGMENTS

My first acknowledgment goes to my family. To my parents, Howard and Carol for instilling in me the Scranton, PA grit, and work ethic that has guided me through my career. To my wife Regina, the love of my life, for always supporting our family, and making us laugh. To my son Jaxson and daughter Penelope for always making me proud and putting a smile on my face.

My second acknowledgment goes to my mentors, colleagues, and students who have guided me, inspired me, and helped me tremendously along the way. Although I cannot include everyone I would like to acknowledge a few. Buz and Kathy Swanik who were my mentors from undergrad through my PhD and played a significant role in my career. John Kelly has been a lifelong mentor and big brother not only in my career by also my personal life. Katie Reuther has been a great friend and colleague who taught me many engineering principles starting during my postdoc at the University of Pennsylvania. Tommy Nowakowski, one of my best friends, who is responsible for me becoming an Athletic Trainer. My first cohort of students (Justin Cobb, Ohsana Valle, Morgan Wambold, and Ryan Paul) who I am proud to say I have learned more from then they learned from me. Lastly, all of the professors in the Biomechanics PhD program at the University of Delaware.

Without all of you, this would not be possible!

Stephen J. Thomas, PhD, ATC

This work would not be possible without those who have supported, educated, and nurtured me throughout my life and career. I would like to acknowledge the consistent support of my parents, Kathleen and Joe, who have always worked tirelessly to provide me with opportunities to succeed. My wife, Courtney, has been an unwavering rock in our household, balancing our family, her work, and my endeavors for the past 15 years. I must also acknowledge those who have not only taught me foundational biomechanics, but also passed along a passion for research, education, and the academic world. Foremost, my PhD advisor, Jill Higginson, a biomechanical engineer who took a chance and welcomed me, a clinician with little to no understanding of mechanical systems, into her lab. I learned so much during that time and fell in love with the application of engineering principles to the human body. To Anil Bhave, who introduced me to the clinical world of motion analysis, I am forever grateful for your enthusiasm and patience as my mentor and boss. Numerous professors from the University of Delaware Biomechanics and Movement Science program played a critical role and pushed me to be a better student and enjoy the process of learning: Jim Richards, Bill Rose, Lynn Snyder-Mackler, Thomas Buchanan, Stuart Binder-Macleod, and many others; I thank you for everything. These professors are educators, mentors, and colleagues without equal.

JOSEPH A. ZENI, PHD, PT

ABOUT THE COMPANION WEBSITE

This book is accompanied by a companion website

www.wiley.com/go/Thomas/WintersBiomechanics5e

The website includes:

- Powerpoints which contains the key point for all the individual chapters.
- Appendix contains table regarding the kinematic, kinetic, and energy.

1

BIOMECHANICS AS AN INTERDISCIPLINE

Stephen J. Thomas[1], Joseph A. Zeni[2], and David A. Winters[3,†]
[1]*Thomas Jefferson University, Philadelphia, PA, USA*
[2]*Rutgers University, Newark, NJ, USA*
[3]*University of Waterloo, Waterloo, Ontario, Canada*

1.0 INTRODUCTION

Biomechanics of human movement is a field that has grown and advanced significantly in the past decade and includes observing, measuring, analyzing, assessing, and interpreting human movement. A wide variety of physical movements are involved – everything from the gait of the physically handicapped to the lifting of a load by a factory worker to the performance of a superior athlete. The physical and biological principles that apply are the same in all cases. What changes from case to case are the specific movement tasks and the level of detail that is being asked about the assessment and interpretation of each movement.

The list of professionals interested in biomechanics is quite long: orthopedic surgeons, athletic trainers, biomedical and mechanical engineers, occupational and physical therapists, kinesiologists, sport scientists, prosthetists, psychiatrists, orthotists, athletic coaches, sports equipment designers, and so on. At the basic level, the name given to the science dedicated to the area of human movement is kinesiology. It is a broad discipline blending aspects of psychology, sports nutrition, motor development and learning, and exercise physiology as well as biomechanics. Biomechanics, as an outgrowth of both life and physical sciences, is built on the basic body of knowledge of physics, chemistry, mathematics, physiology, and anatomy. It is amazing to note that the first real "biomechanicians" date back to Leonardo da Vinci, Galileo, Lagrange, Bernoulli, Euler, and Young. All these scientists had primary interests in the application of mechanics to biological problems.

1.0.1 Importance of Human Movement Analysis

The first question that one may ask is "What is the benefit of assessing and interpreting human movement." The answer to this can vary depending on the specific movement being studied and the expected outcomes for that specific individual. At the most basic level, it is important to

[†]Deceased.

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

understand the underlying mechanisms responsible for the development of movement and compensation due to injury or pain. These mechanisms will act as a roadmap or equation that clinicians can utilize during rehabilitation to optimize movement, which will result in long-term recovery and injury prevention. When asking this question in reference to athletics, the goal is typically to increase performance and also to prevent overuse injuries. The current conundrum in many sports is that as performance increases the risk of injury also increases. This can result in an exponentially challenging situation for biomechanists to resolve when working with athletes.

1.0.2 The Interprofessional Team

As the field of biomechanics continues to evolve it is becoming much more evident that a team approach needs to be used when working with patients/clients. As mentioned previously, the professions interested in human movement are very broad and span many disciplines that have unique skill sets to help patients/clients. This includes surgery, rehabilitation, exercise, mental health, nutrition, equipment and prosthetic designs, etc. It is unfortunate that often times these professions work in silos, which often result in conflicting advice and direction. This not only confuses the patients/clients but also causes set backs in their progression. The first step in developing an interprofessional team is identifying and respecting the similarities and differences in expertise across the continuum of care for the patient/client. The focus always needs to be directed toward the patient/client. The next step is effective communication between team members and the patient/client. Each team member needs to communicate collectively following assessments so that all of the collected information can be discussed and interpreted. This interpretation will then be used by the team to create an optimal plan to achieve the patient's/client's goals. It is also very important to communicate this plan to the patient/client in a digestible form. Educational programs in biomechanics often focus on the science and technology used to measure and assess human movement very accurately. However, it often falls short on providing training for interpreting and communicating the results of human movement assessments to the patient/client. This is an incredibly important and valuable skill that needs to be taught in educational programs, which will result in the expansion of the field outside the traditional settings of research labs and provide value to patients/clients from all walks of life.

1.1 MEASUREMENT, DESCRIPTION, ANALYSIS, AND ASSESSMENT

The scientific approach as applied to biomechanics has been characterized by a fair amount of confusion. Some descriptions of human movement have been passed off as assessments, some studies involving only measurements have been falsely advertised as analyses, and so on. It is, therefore, important to clarify these terms. Any quantitative assessment of human movement must be preceded by a measurement and description phase, and if more meaningful diagnostics are needed, a biomechanical analysis is usually necessary. Most of the material in this text is aimed at the technology of measurement and description and the modeling process required for analysis. The final interpretation, assessment, or diagnosis is movement specific and is limited to the examples given.

Figure 1.1, which has been prepared for the assessment of the physically handicapped, depicts the relationships between these various phases of assessment. All levels of assessment involve a human being and are based on his or her visual observation of a patient or subject, recorded data, or some resulting biomechanical analysis. The primary assessment level uses direct observation, which places tremendous "overload" even on the most experienced observer. All measures are subjective and are almost impossible to compare with those obtained previously. Observers are then faced with the tasks of documenting (describing) what they see, monitoring changes, analyzing the information, and diagnosing the causes. If measurements can be made during the patient's movement, then data can be presented in a convenient manner to describe the movement quantitatively. Here the assessor's task is considerably

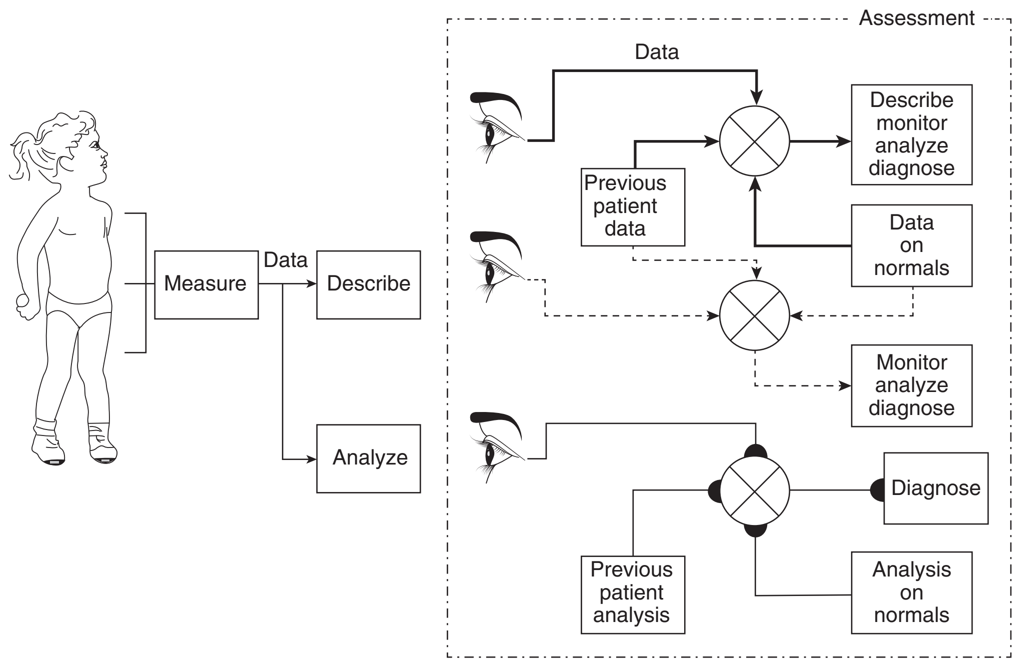

Figure 1.1 Schematic diagram showing the three levels of assessment of human movement.

changes, carry out simple analyses, and try to reach a more objective diagnosis. At the highest level of assessment, the observer can view biomechanical analyses that are extremely powerful in diagnosing the exact cause of the problem, compare these analyses with the normal population, and monitor their detailed changes with time.

The measurement and analysis techniques used in an athletic event could be identical to the techniques used to evaluate an amputee's gait. However, the assessment of the optimization of the energetics of the athlete is quite different from the assessment of the stability of the amputee. Athletes are looking for very detailed but minor changes that will improve their performance by a few percentage points, sufficient to move them from fourth to first place. Their training and exercise programs and reassessment normally continue over an extended period of time. The amputee, on the other hand, is looking for major improvements, probably related to safe walking, but not fine and detailed differences. This person is quite happy to be able to walk at less than maximum capability, although techniques are available to permit training and have the prosthesis readjusted until the amputee reaches some perceived maximum. When working with Para-athletes these approaches and goals are often are blended together. In ergonomic studies, assessors are likely looking for maximum stresses in specific tissues during a given task, to thereby ascertain whether the tissue is working within safe limits. If not, they will analyze possible changes in the workplace or task in order to reduce the stress or fatigue.

1.1.1 Measurement, Description, and Monitoring

It is difficult to separate the two functions of measurement and description. However, for clarity, the student should be aware that a given measurement device can have its data presented in a number of different ways. Conversely, a given description could have come from several different measurement devices.

Earlier biomechanical studies had the sole purpose of describing a given movement, and any assessments that were made resulted from visual inspection of the data. The description of the data can take many forms: plots of body coordinates, stick such as gait velocity, load lifted, or height of a jump. A video camera, by itself, is a measurement device, and the resulting plots form the description of the event in time and space. The coordinates of key anatomical landmarks can be extracted and plotted at regular intervals in time. Time history plots of one or more coordinates are useful in describing detailed changes in a particular landmark. They also can reveal changes in velocity and acceleration. A total description in the plane of the movement is provided by the stick diagram, in which each body segment is represented by a straight line or stick. Joining the sticks together gives the spatial orientation of all segments at any point in time. Repetition of this plot at equal intervals of time gives a pictorial and anatomical description of the dynamics of the movement. Here, trajectories, velocities, and accelerations can be visualized. To get some idea of the volume of the data present in a stick diagram, the student should note that one full page of coordinate data is required to make the complete plot for the description of the event. The coordinate data can be used directly for any desired analysis: reaction forces, muscle moments, energy changes, efficiency, and so on. Conversely, an assessment can occasionally be made directly from the description. A trained observer, for example, can scan a stick diagram and extract useful information that will give some directions for training or therapy, or give the researcher some insight into basic mechanisms of movement.

The term *monitor* needs to be introduced in conjunction with the term *describe*. To monitor means to note changes over time. Thus, a physical therapist will monitor the progress (or the lack of it) for each physically disabled person undergoing therapy. Only through accurate and reliable measurements will the therapist be able to monitor any improvement and thereby make inferences to the validity of the current therapy. What monitoring does not tell us is why an improvement is or is not taking place; it merely documents the change. All too many coaches or therapists document the changes with the inferred assumption that their intervention has been the cause. However, the scientific rationale behind such inferences is missing. Unless a detailed analysis is done, we cannot document the detailed motor-level changes that will reflect the results of therapy or training.

1.1.2 Analysis

The measurement system yields data that are suitable for analysis. This means that data have been calibrated and are as free as possible from noise and artifacts. *Analysis* can be defined as any mathematical operation that is performed on a set of data to present them in another form or to combine the data from several sources to produce a variable that is not directly measurable. From the analyzed data, information may be extracted to assist in the assessment stage. In some cases, the mathematical operation can be very simple, such as the processing of an electromyographic (EMG) signal to yield an *envelope* signal. The mathematical operation performed here can be described in two stages. The first is a full-wave *rectifier* (the electronic term for a circuit that gives the absolute value). The second stage is a low-pass filter (which mathematically has the same transfer function as that between a neural pulse and its resultant muscle twitch). A more complex biomechanical analysis could involve a link-segment model, and with appropriate kinematic, anthropometric, and kinetic output data, we can carry out analyses that could yield a multitude of significant time-course curves. Figure 1.2 depicts the relationships between some of these variables. The output of the movement is what we see. It can be described by a large number of kinematic variables: displacements, joint angles, velocities, and accelerations. If we have an accurate model of the human body in terms of anthropometric variables, we can develop a reliable link-segment model. With this model and accurate kinematic data, we can predict the net forces and muscle moments that caused the movement we just observed. Such an analysis technique is called an *inverse solution*. It is extremely valuable, as it allows us to estimate variables such as joint reaction forces and moments. Such variables are not measurable directly. In a similar manner, individual muscle forces might be predicted through the development of a mathematical model of a muscle, which could have neural drive, length, velocity, and cross-sectional area as inputs.

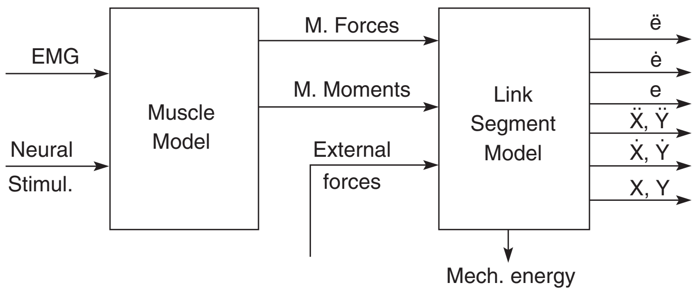

Figure 1.2 Schematic diagram to show the relationship between the neural, kinetic, and kinematic variables required to describe and analyze human movement.

1.1.3 Assessment and Interpretation

The entire purpose of any assessment is to make a positive decision about a physical movement. An athletic coach might ask, "Is the mechanical energy of the movement better or worse than before the new training program was instigated, and why?" Or the orthopedic surgeon may wish to see the improvement in the knee muscle moments of a patient a month after surgery. Or a basic researcher may wish to interpret the motor changes resulting from certain perturbations and thereby verify or negate different theories of neural control. In all cases, if the questions asked yield no answers, it can be said that there was no information present in the analysis. The decision may be positive in that it may confirm that the coaching, surgery, or therapy has been correct and should continue exactly as before. Or, if this is an initial assessment, the decision may be to proceed with a definite plan based on new information from the analysis. The information can also cause a negative decision, for example, to cancel a planned surgical procedure and to prescribe therapy instead.

Some biomechanical assessments involve a look at the description itself rather than some analyzed version of it. Commonly, ground reaction force curves from a force plate are examined. This electromechanical device gives an electrical signal that is proportional to the weight (force) of the body acting on it. Such patterns appear in Figure 1.3. A trained observer can detect pattern changes as a result of pathological gait and may come to some conclusions as to whether the

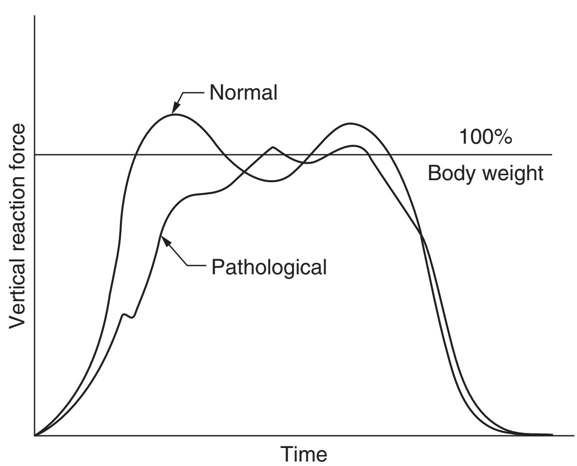

Figure 1.3 Example of a ground reaction force curve that has sometimes been used in the diagnostic assessment of pathological gait.

patient is improving, but he or she will not be able to assess why. At best, this approach is speculative and yields little information regarding the underlying cause of the observed patterns.

1.2 BIOMECHANICS AND ITS RELATIONSHIP WITH PHYSIOLOGY AND ANATOMY

Because biomechanics is still an expanding discipline, it is important to identify its interaction with other areas of movement science: neurophysiology, exercise physiology, and anatomy. The neuromuscular system acts to control the release of metabolic energy for the purpose of generating controlled patterns of tension at the tendon. That tension waveform is a function of the physiological characteristics of the muscle (i.e. fiber type) and of its metabolic state (rested vs. fatigued). The tendon tension is generated in the presence of passive anatomical structures (ligaments, articulating surfaces, and skeletal structures). Figure 1.4 depicts the relationship between the sensory system, the neurological pathways, the muscles, the skeletal system, and the link-segment model that we analyze. The essential characteristic of this total system is that it is converging in nature. The structure of the neural system has many excitatory and inhibitory synaptic junctions, all summing their control on a final synaptic junction in the spinal cord to control individual motor units. The α motoneuron (1), which is often described as the final common pathway, has its synapse on the motor end point of the muscle motor unit. A second level of convergence is the summation of all twitches from all active motor units at the level of the tendon (2). This summation results from the neural recruitment of motor units based on the size principle (DeLuca et al., 1982; Henneman and Olson, 1965). The resultant tension is a temporal superposition of twitches of all active motor units, modulated by the length and velocity characteristics of the muscle. A third level of musculoskeletal integration at each joint center where the moment-of-force (3) is the algebraic summation of the force/moment products of all muscles crossing that joint plus the moments generated by the passive anatomical structures at the joint. The moments

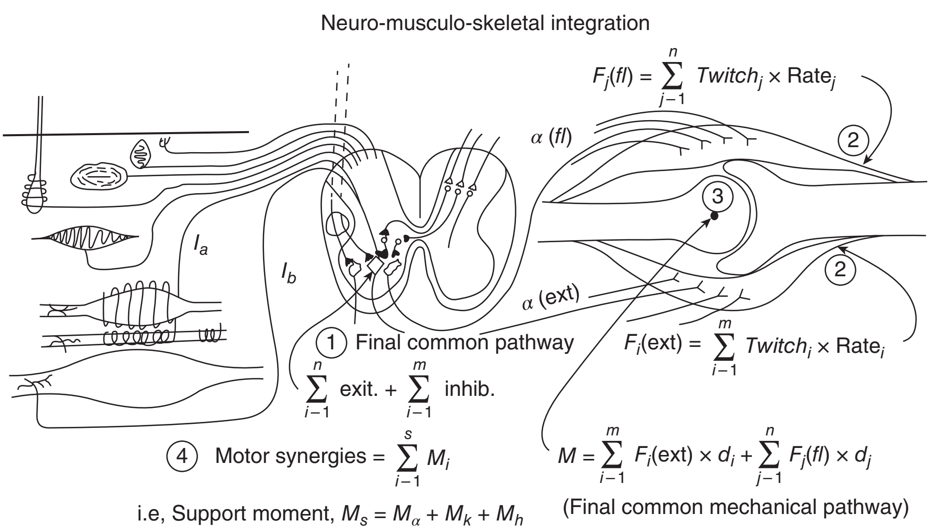

Figure 1.4 Four levels of integration in the neuromusculoskeletal system provide control of human movement. The first is the neural summation of all excitatory/inhibitory inputs to the α motoneuron (1). The second is the summation of all motor twitches from the recruitment of all active motor units within the muscle and is seen as a tendon force (2). The third is the algebraic summation of all agonist and antagonist muscle moments at the joint axis (3). Finally, integrations are evident in combined moments acting synergistically toward a common goal (4).

we routinely calculate include the net summation of all agonist and antagonist muscles crossing that joint, whether they are single- or double-joint muscles. In spite of the fact that this moment signal has mechanical units ($N \cdot m$), we must consider the moment signal as a neurological signal because it represents the final desired central nervous system (CNS) control. Finally, an intersegment integration may be present when the moments at two or more joints collaborate toward a common goal. This collaboration is called a synergy. One such synergy (4), referred to as the support moment, quantifies the integrated activity of all muscles of the lower limb in their defense against a gravity-induced collapse during walking (Winter and Overall, 1980; Winter, 1984).

Bernstein (1967) predicted that the CNS exerts control at the level of the joints or at the synergy level when he postulated the "principle of equal simplicity" because "it would be incredibly complex to control each and every muscle." One of the by-products of these many levels of integration and convergence is that there is considerably more variability at the neural (EMG) level than at the motor level and more variability at the motor level than at the kinematic level. The resultant variability can frustrate researchers at the neural (EMG) level, but the positive aspect of this redundancy is that the neuromuscular system is, therefore, very adaptable (Winter, 1984). This adaptability is very meaningful in pathological gait as a compensation for motor or skeletal deficits. For example, a major adaptation took place in a patient who underwent a knee replacement because of osteoarthritic degeneration (Winter, 1989). For years prior to the surgery, this patient had refrained from using her quadriceps to support her during walking; the resultant increase in bone-on-bone forces induced pain in her arthritic knee joint. She compensated by using her hip extensors instead of her knee extensors and maintained a near-normal walking pattern; these altered patterns were retained by her CNS long after the painful arthritic knee was replaced. Therefore, this moment-of-force must be considered the final desired pattern of CNS control, or in the case of pathological movement, it must be interpreted either as a disturbed pattern or as a CNS adaptation to the disturbed patterns. This adaptability is discussed further in Chapters 5, 8, and 12.

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2

SIGNAL PROCESSING

JOSEPH A. ZENI[1], STEPHEN J. THOMAS[2], AND DAVID A. WINTERS[3,†]

[1]*Rutgers University, Newark, NJ, USA*
[2]*Thomas Jefferson University, Philadelphia, PA, USA*
[3]*University of Waterloo, Waterloo, Ontario, Canada*

2.0 INTRODUCTION

All of the biomechanical variables are time-varying, and it does not matter whether the measure is kinematic, kinetic, or electromyographic (EMG); it must be processed like any other signal. Some of these variables are directly measured: acceleration and force signals from transducers or EMG from bioamplifiers. Others are a product of our analyses: moments-of-force, joint reaction forces, mechanical energy, and power. All can benefit from further signal processing to extract cleaner or averaged waveforms, correlated to find similarities or differences, or even transformed into the frequency domain.

This chapter will summarize the analysis techniques associated with auto- and cross-correlations, frequency (Fourier) analysis, and their applications to correct data record length and sampling frequency. The theory of digital filtering is presented here; however, the specific applications of digital filtering of kinematics appear in Chapter 3 and analog filtering of EMG in Chapter 9. The applications of ensemble averaging of variables associated with repetitive movements are also presented.

2.1 AUTO- AND CROSS-CORRELATION ANALYSES

Autocorrelation analyzes how well a signal is correlated with itself, between the present point in time and past and future points in time. Cross-correlation analyses evaluate how well a given signal is correlated with another signal over past, present, and future points in time. We are familiar in statistics with the Pearson product-moment correlation. It is a measure of relationship between two variables and allows us to determine whether a variable x increases or decreases as the variable y increases. The strength and polarity of this relationship are given by the correlation coefficient: the higher the value the stronger the relationship, while the sign indicates if

[†]Deceased.

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

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variables x and y are increasing and decreasing together (positive correlation) or if one is increasing while the other is decreasing (negative correlation). The correlation coefficient is a normalized dimensionless number varying from −1 to +1.

2.1.1 Similarity to the Pearson Correlation

Consider the formula for the Pearson product-moment correlation coefficient relating two variables, x and y:

$$r = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \overline{x})(y_i - \overline{y})}{s_x s_y} \tag{2.1}$$

where x_i and y_i are the ith samples of x and y, x and y are the means of x and y, and s_x and s_y are the standard deviations of x and y.

The numerator of the formula is the sum of the product of the two variables after the mean value of each variable has been subtracted. It is easy to appreciate that if x and y are random and unrelated then $(x_i - x)$ and $(y_i - y)$ will be scattered in the $x - y$ plane about zero (see Figure 2.1). These products will be +ve in quadrants 1 and 3 and –ve in quadrants 2 and 4, and provided there are enough points their sum, r, will tend toward zero, indicating no relationship between the two variables.

Now if the variables are related and tend to increase and decrease together $(x_i - x)$ and $(y_i - y)$ will fall along a line with a positive slope in the $x - y$ plane (see Figure 2.2). When we sum the products in Equation (2.1), we will get a finite +ve sum, and when this sum is divided by N, we remove the influence of the number of data points. This product will have the units of the product of the two variables, and its magnitude will also be scaled by those units. To remove those two factors, we divide by $s_x s_y$, which normalizes the correlation coefficient so that it is dimensionless and lies between −1 and +1.

There is an estimation error in the correlation coefficient if we have a finite number of data points, which is true of most all physiological samples. Therefore, the level of significance will increase or decrease with the number of data points. The calculation for the t-statistic of the correlation coefficient includes the number of observations (n) in the numerator and strength of relationship in the denominator (r). As the number of observed samples increases (n) for a

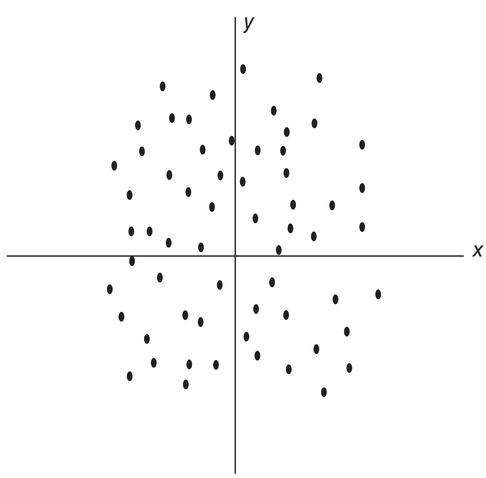

Figure 2.1 Scatter diagram of variable x against variable y

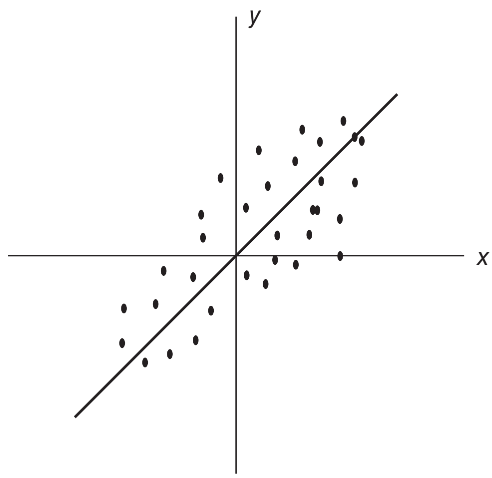

Figure 2.2 Scatter diagram showing a positive correlation between variable x and variable y.

given r-value, there is a commensurate increase in the t-statistic and an increased likelihood of achieving statistical significance for a given relationship. Any standard statistics textbook includes a table of significance for the coefficient r for any given number of data points.

2.1.2 Formulae for Auto- and Cross-Correlation Coefficients

The auto- and cross-correlation coefficient is simply the Pearson product-moment correlation calculated on two time series of data rather than on individual measures of data. Autocorrelation, as the name suggests, involves correlating a time series with itself. Cross-correlation, on the other hand, correlates two independent time series. The major difference is that a correlation of time series data does not yield a single correlation coefficient, but rather a whole series of correlation values. This series of values is achieved by shifting one of the series forward and backward in time, the value of this shifting will be evident later. The magnitude (+ve or –ve) of this shifting is decided by the user and the time series of correlations is a function of the phase shift, τ. The formula for the autocorrelation of $x(t)$ is $R_{xx}(\tau)$:

$$R_{xx}(\tau) = \frac{\frac{1}{T}\int_0^T x(t)x(t+\tau)dt}{R_{xx}(0)} \tag{2.2}$$

where $x(t)$ has zero mean.

The formula for the cross-correlation of $x(t)$ and $y(t)$ is $R_{xy}(\tau)$:

$$R_{xy}(\tau) = \frac{\frac{1}{T}\int_0^T x(t)y(t+\tau)dt}{\sqrt{R_{xx}(0)R_{yy}(0)}} \tag{2.3}$$

where $x(t)$ and $y(t)$ have zero means.

It is easy to see the similarities between these formulae and the formula for the Pearson product-moment coefficient. The summation sign is replaced by we now divide by T rather than N. The denominator in these two equations, as in the Pearson equation, normalizes the correlation to be dimensionless from −1 to +1. Also, the two time series must have a zero mean, as was the case in the Pearson formula, when the means of x and y were subtracted. Note that the Pearson correlation is a *single* coefficient, while these auto- and cross-correlations are a series of correlation scores overtime at each value of τ.

2.1.3 Four Properties of the Autocorrelation Function

Property #1. The maximum value of $R_{xx}(\tau)$ is $R_{xx}(0)$ which, in effect, is the mean square of $x(t)$. For all values of the phase shift, τ, either +ve or −ve $R_{xx}(\tau)$ is less than $R_{xx}(0)$, which can be seen from the following proof.

From basic mathematics we know:

$$\int_0^T (x(t) - x(t-\tau))^2 dt \geq 0$$

Expanding, we get:

$$\int_0^T \left(x(t)^2 + x(t-\tau)^2 - 2x(t)x(t-\tau)\right) dt \geq 0$$

$$\int_0^T x(t)^2 dt + \int_0^T x(t-\tau)^2 dt - 2\int_0^T x(t)x(t-\tau) dt \geq 0$$

For these integrations, τ is constant; thus, the second term is equal to the first term, and the denominator for the autocorrelation is the same for all terms and is not shown. Thus:

$$\begin{aligned} &R_{xx}(0) + R_{xx}(0) - 2R_{xx}(\tau) \geq 0 \\ &R_{xx}(0) - R_{xx}(\tau) \geq 0 \end{aligned} \tag{2.4}$$

Property #2. An autocorrelation function is an even function, which means that the function for a −ve phase shift is a mirror image of the function for a +ve phase shift. This can be easily derived as follows; for simplicity, we will only derive the numerator of the equation:

$$R_{xx}(\tau) = \frac{1}{T}\int_0^T x(t)x(t+\tau)dt$$

Substituting $t = (t' - \tau)$ and taking the derivative, we have $dt = dt'$:

$$R_{xx}(\tau) = \frac{1}{T}\int_0^T x(t'-\tau)x(t')dt' = R_{xx}(-\tau) \tag{2.5}$$

Therefore, we have to calculate only the function for +ve phase shifts because the function is a mirror image for −ve phase shifts.

Property #3. The autocorrelation function for a periodic signal is also periodic, but the phase of the function is lost. Consider the autocorrelation of a sine wave; again we derive only the numerator of the equation.

$$x(t) = E\sin(\omega t)$$

$$R_{xx}(\tau) = \frac{1}{T}\int_0^T E\sin(\omega t)E\sin(\omega(t-\tau))dt$$

Using the common trig identity: $\sin(a)\sin(b) = 1/2(\cos(a-b) - \cos(a+b))$, we get:

$$R_{xx}(\tau) = \frac{E^2}{2T}\left[t\cos(\omega t) - \frac{1}{2\omega}\sin(2\omega t + \omega\tau)\right]_0^T$$

$$R_{xx}(\tau) = \frac{E^2}{2T}\left[(T\cos(\omega t) - 0) - \frac{1}{2\omega}(\sin(2\omega T + \omega\tau) - \sin(\omega t))\right]$$

Since T is one period of $\sin(\omega t)$, $\therefore \sin(2\omega T + \omega\tau) - \sin(\omega\tau) = 0$ for all τ.

$$\therefore R_{xx}(\tau) = \frac{E^2}{2}\cos(\omega\tau) \tag{2.6}$$

Similarly if $x(t) = E\cos(\omega t)$ also $R_{xx}(\tau) = \frac{E^2}{2}\cos(\omega\tau)$.

Note that Equation (2.6) is an even function as predicted by Property #2; a plot of this $R_{xx}(\tau)$ after normalization is presented in Figure 2.3.

This property is useful in detecting the presence of periodic signals buried in white noise. White noise is defined as a signal made up of a series of random points, where there is zero

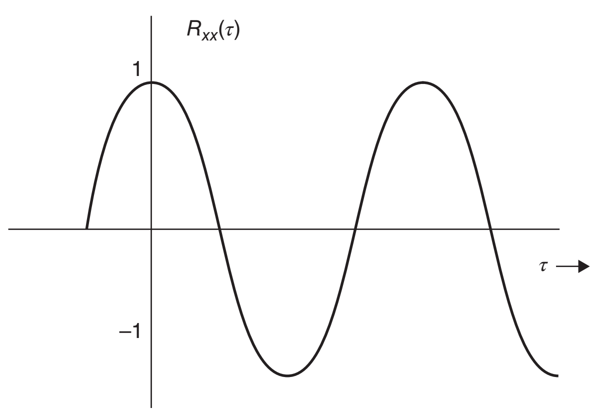

Figure 2.3 Autocorrelation of a sine or cosine signal. Note that this is an even function and the repetitive nature of $R_{xx}(\tau)$ at the frequency of the sine and cosine wave.

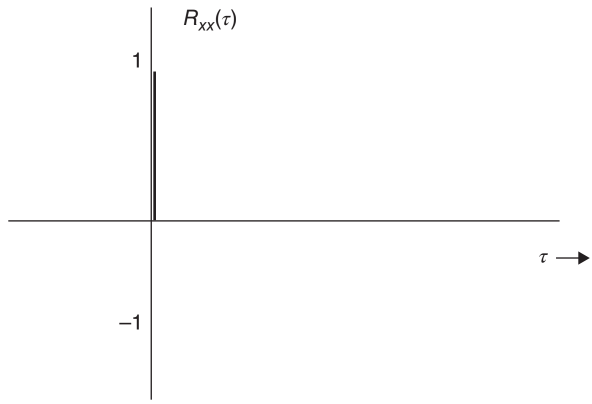

Figure 2.4 Autocorrelation of white noise. Note that $R_{xx}(\tau) = 0$ at all $\tau \neq 0$, indicating that each data point has 0 correlation with all other data points ahead and behind it in time.

correlation between the signal at any point with the signal at any point ahead of or behind it in time. Therefore, at any $\tau \neq 0$ $R_{xx}(\tau) = 0$ and at $\tau = 0$ $R_{xx}(\tau) = 1$. Thus, the autocorrelation of white noise is an impulse, as shown in Figure 2.4.

If we have a signal, $s(t)$, with added noise, $n(t)$, we can express $x(t) = s(t) + n(t)$, and substituting in the numerator of Equation (2.2) we get:

$$R_{xx}(\tau) = \int_0^T (s(t) + n(t))(s(t+\tau) + n(t+\tau))dt$$

$$= \int_0^T s(t)s(t+\tau)dt + \int_0^T n(t)s(t+\tau)dt$$

$$+ \int_0^T s(t)n(t+\tau)dt + \int_0^T n(t)n(t+\tau)dt$$

Since the signal and noise are uncorrelated, the second and third terms will = 0.

$$\therefore R_{xx}(\tau) = R_{ss}(\tau) + R_{nn}(\tau) \tag{2.7}$$

Property #4. As seen in Property #3 the frequency content of $x(t)$ is present in $R_{xx}(\tau)$. The power spectral density function is the Fourier transform of $R_{xx}(\tau)$; more will be said about this in the next section on frequency analysis. However, it is sometimes valuable to use the autocorrelation function to identify any periodicity present in $x(t)$ or to identify the presence of an interfering signal (e.g. hum) in our biological signal. Even if there were no periodicity in $x(t)$, the duration of $R_{xx}(\tau)$ would give an indication of the frequency spectra of $x(t)$; lower frequencies result in $R_{xx}(\tau)$ remaining above zero for longer phase shifts, while high frequencies tend to zero for small phase shifts.

2.1.4 Three Properties of the Cross-Correlation Function

Property #1. The cross-correlation of $x(t)$ and $y(t)$ is not an even function. Because the two signals are completely different, the phase shifting in the +ve direction will not result in the same "cross products" as shifting in the –ve direction. Thus, $R_{xy}(\tau) \neq R_{xy}(-\tau)$.

Property #2. The maximum value of $R_{xy}(\tau)$ is not necessarily at $\tau = 0$. The maximum +ve or negative peak of $R_{xy}(\tau)$ will occur when the two signals are most in phase or most out of phase. For example, if $x(t)$ is a sine wave and $y(t)$ is a cosine wave of the same frequency at $\tau = 0$, the signals are 90° out of phase with each other, and the cross products over one cycle will sum to zero. However, shifting the cosine wave forward 90° will bring the two signals into phase such that all the cross products are +ve and $R_{xy}(\tau) = 1$. Shifting the cosine wave backward 90° will bring the two signals 180° out of phase so that all the "cross products" are −ve and $R_{xy}(\tau) = -1$. A physiological example is the measurement of transmission delays (neural or muscular) to determine the conduction velocity of the signal. Consider Figure 2.5, where the signal is stimulated and is recorded at sites S1 and S2; the distance between the sites is d. The time delay between the S1 and S2 is t as determined from $R_{S1S2}(\tau)$, the cross-correlation of S1 and S2. Figure 2.5 shows a peak at $\tau_1 = t$ when S2 is shifted so that it is in phase with S1.

Property #3. The Fourier transform of the cross-correlation function is the cross-spectral density function, which is used to calculate the coherence function, which is a measure of the common frequencies present in the two signals. This is a valuable tool in determining the transfer function of a system in which you cannot control the frequency content of the input signal. For example, in determining the transfer function of a muscle with EMG as an input and force an output, we cannot control the input frequencies (Bobet and Norman, 1990).

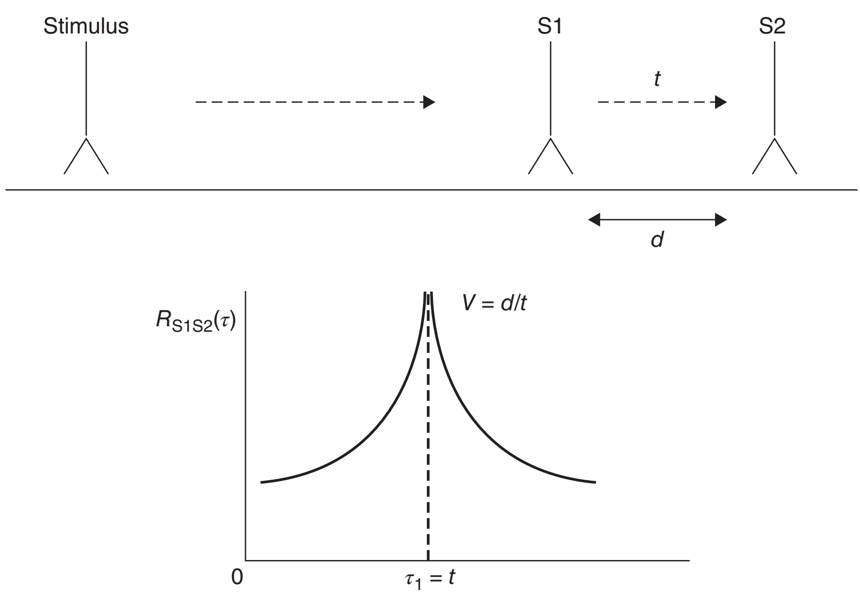

Figure 2.5 Cross-correlation of a neural or muscular signal recorded at two sites, S1 and S2, separated by a distance, d. $R_{xy}(\tau)$ reaches a peak when S2 record is shifted $\tau_1 = t$ second. Thus, the velocity of the transmission $V = d/t$.

2.1.5 Importance in Removing the Mean Bias from the Signal

A caution that must be heeded when cross-correlating two signals is that the mean amplitude of the waveform (DC bias) in both signals must be removed prior to calculating $R_{xy}(\tau)$. Most standard programs do this without your knowledge, but if you are writing your own program, you must do so or a major error will result. Consider $x(t) = s_1(t) + m_1$ and $y(t) = s_2(t) + m_2$, where m_1 and m_2 are the means of s_1 and s_2, respectively.

$$
\begin{aligned}
R_{xy}(\tau) &= \int_0^T (s_1(t) + m_1)(s_2(t+\tau) + m_2)dt \\
&= \int_0^T s_1(t)s_2(t+\tau)dt + \int_0^T m_1 s_2(t+\tau)dt \\
&\quad + \int_0^T m_2 s_1(t)dt + \int_0^T m_1 m_2 dt
\end{aligned}
$$

Since the signals and m_1 and m_2 are uncorrelated, the second and third terms will = 0.

$$
\therefore R_{xy}(\tau) = \int_0^T s_1(t)s_2(t+\tau)dt + \int_0^T m_1 m_2 dt
$$

The first term is the desired cross-correlation, but a major bias will be added by the second term, and the peak of $R_{xy}(\tau)$ may be grossly exaggerated.

2.1.6 Digital Implementation of Auto- and Cross-Correlation Functions

Since data are now routinely collected and stored in a computer, the implementation of the auto- and cross-correlation is the digital equivalent of Equations (2.2) and (2.3), shown below in Equations (2.8) and (2.9)

$$
R_{xx}(\tau) = \frac{\frac{1}{N}\sum_{n=1}^{N}[(x(n)-\bar{x})(x(n+\tau)-\bar{x})]}{\frac{1}{N}\sum_{n=1}^{N}(x(n)-\bar{x})^2} \tag{2.8}
$$

$$
R_{xy}(\tau) = \frac{\frac{1}{N}\sum_{n=1}^{N}[(x(n)-\bar{x})(y(n+\tau)-\bar{y})]}{\frac{1}{N}\sum_{n=1}^{N}(x(n)-\bar{x})(y(n)-\bar{y})} \tag{2.9}
$$

Both auto- and cross-correlations are calculated for various phase shifts that a priori must be specified by the user, and this will have an impact on the number of data points used in the formulae. If, for example, $x(n)$ and $y(n)$ are 1000 data points, and it is desired that $\tau = \pm 100$, then we can only get 800 cross products and, therefore, N will be set to 800. Sometimes the signals of interest are periodic (such as gait); then, we can wrap the signal on itself and calculate the correlations using all the data points. Such an analysis is known as a circular correlation.

2.1.7 Application of Autocorrelations

As indicated in Property #3, an autocorrelation indicates the frequency content of $x(t)$. Figure 2.6 presents an EMG record and its autocorrelation. The upper trace (a) is the raw EMG signal, which

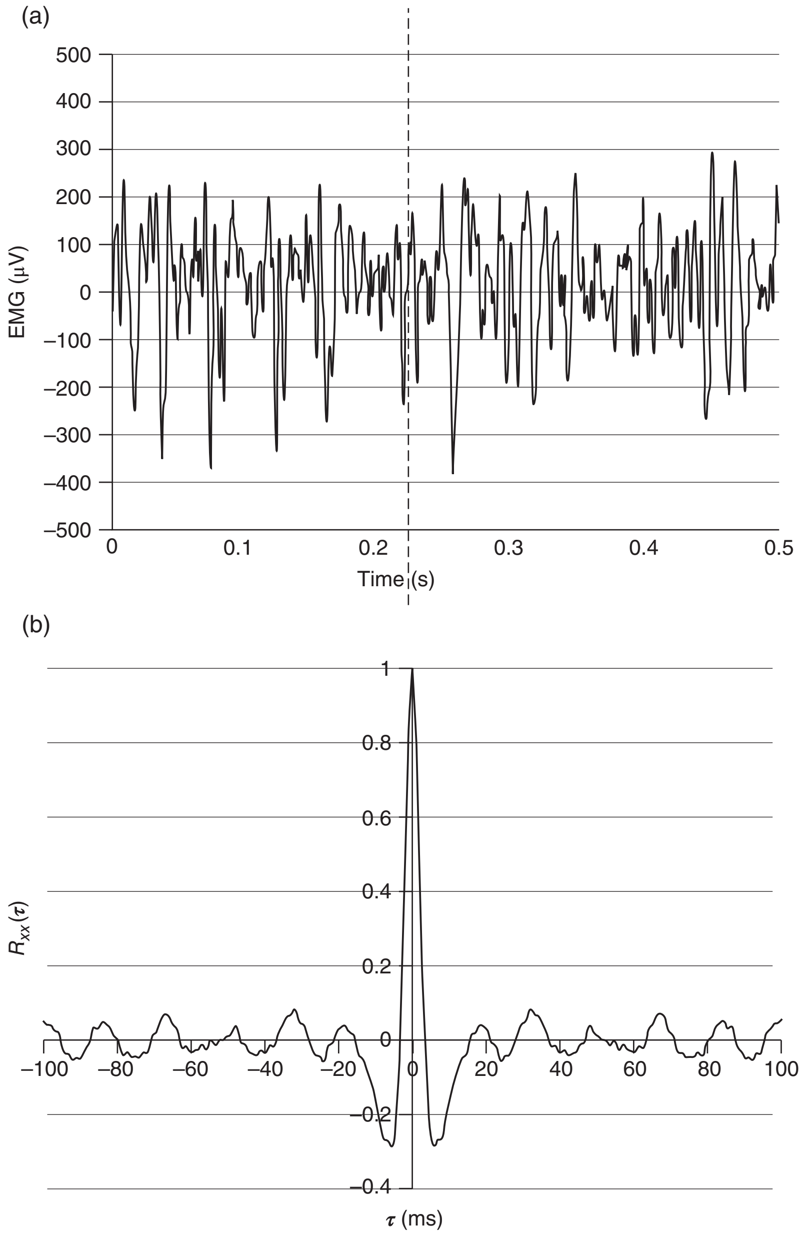

Figure 2.6 (a) is a surface EMG signal recorded for 0.5 second that does not show the presence of any 60 Hz hum pickup. (b) is the autocorrelation of this EMG over a $\tau = \pm 100$ ms. Again, note that this is an even function and observe the presence of a periodic component closely resembling a sinusoidal wave with peaks equal to the period of a 60 Hz (approximately 17 ms).

does not show any visible evidence of hum, but the autocorrelation seen in the lower trace (b) is an even function as predicted by Property #2 and shows the presence of 60 Hz hum. This consistent electrical hum is a common source of interference in electrophysiological experiments, including those that use EMG and electroencephalograms. Note from Figure 2.3 that $R_{xx}(\tau)$ for a sinusoidal wave has its first zero crossing at ¼ of a cycle of the sinusoidal frequency; thus, we can use that first zero crossing of $R_{xx}(\tau)$ to estimate the average frequency in the EMG. The first zero crossing for this $R_{xx}(\tau)$ occurred at about 3 ms, representing an average period of 12 ms, or an average frequency of about 83 Hz.

2.1.8 Applications of Cross-Correlations

2.1.8.1 Quantification of Cross-Talk in Surface Electromyography. Cross-correlations quantify what is in common in the profiles of $x(t)$ and $y(t)$ but also any common signal present in both $x(t)$ and $y(t)$. This may be true in the recordings from surface electrodes that are close enough to be subject to cross-talk. In brief, cross-talk occurs when the EMG signal of an active muscle is inadvertently detected in the recording electrode of an inactive muscle. Because a knowledge of surface recording techniques and the biophysical basis of the EMG signal is necessary to understand cross-talk, the student is referred to Section 9.2.5 in Chapter 9 for a detailed description of how $R_{xy}(\tau)$ has been used to quantify cross-talk.

2.1.8.2 Measurement of Delay Between Physiological Signals. Experimental research conducted to find the phase advance of one EMG signal ahead of another has been used to find balance strategies in walking (Prince et al., 1994). Balance of the head and trunk during gait against large inertial forces is achieved by the paraspinal muscles. It was noted that the head anterior/posterior (A/P) accelerations were severely attenuated (0.48 m/s^2) compared with hip accelerations (1.91 m/s^2), and it was important to determine how the activity of the paraspinal muscles contributed to this reduced head acceleration. The EMG profiles at nine vertebral levels from C7 down to L4 were analyzed to find the time delays between those balance muscles. Figure 2.7 presents the ensemble average (see Section 2.3 later) of L4 and C7 muscle profiles over the stride

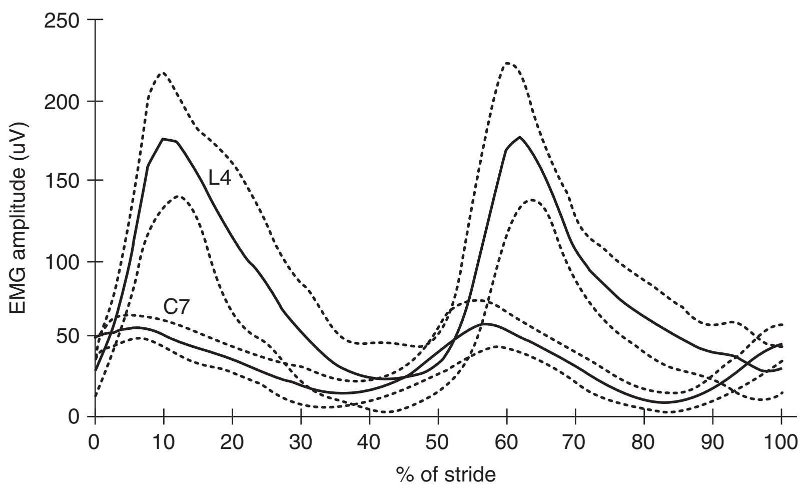

Figure 2.7 Ensemble-averaged profiles over the stride period of EMG signals of paraspinal muscles at C_7 and L_4 levels for one of the subjects. Note the second harmonic peaks occurring during the weight acceptance periods of the left and right feet to balance the trunk and head. The C_7 amplitude is lower than the L_4 amplitude because the inertial load above C_7 is considerably lower than that above L_4. More important is the timing of C_7 so that it is ahead of L_4, indicating that the head is balanced first, ahead of the trunk. (With

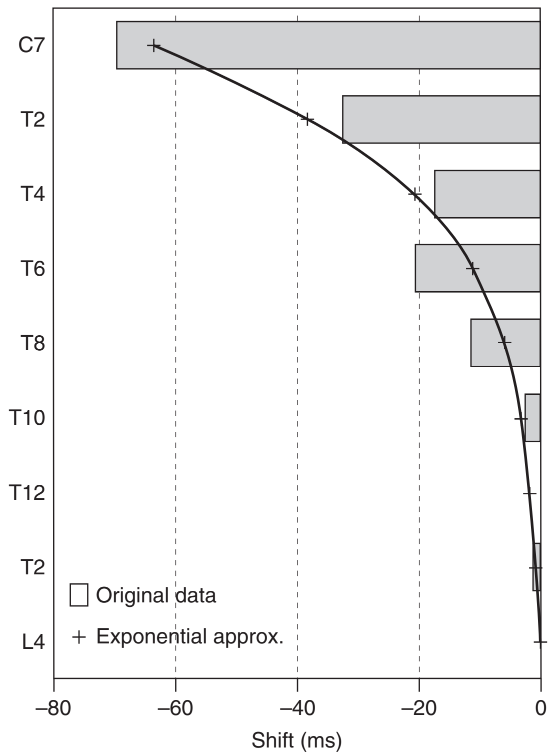

Figure 2.8 Phase shift (ms) of the activation profiles of the paraspinal muscles relative to the profile at the L_4 level. The negative shift indicates the activation was in advance of L_4. The curve fit was exponential. (With Permission of Elsevier.)

period for one subject). A cross-correlation of these two signals showed that C7 was in advance of L4 by about 70 ms. For all 10 young adults in this study, all signals at C7, T2, T4, T6, T8, T10, T12, and L2 were separately cross-correlated with the L4 profile. The phase shift of these signals is presented in Figure 2.8. The earlier turn-on of the more superior paraspinal muscles indicates a "top-down" anticipatory strategy to stabilize the head first, then the cervical level, the thoracic level, and finally the lumbar level. This strategy resulted in a dramatic decrease in the A/P head acceleration over the stride period compared to the A/P acceleration of the pelvis. In a subsequent study on fit and healthy elderly (Wieman, 1991) the head/hip acceleration (%) in the elderly (41.9%) was significantly higher ($p < 0.02$) compared with that of young adults (22.7%); this indicated that the elderly had lost this "top-down" anticipatory strategy, and the paraspinal EMG profiles bore this out.

2.1.8.3 Measurement of Synergistic and Coactivation EMG Profiles. There is considerable information in EMG profiles regarding the action of agonist/antagonist muscle groups during any given activity. Recently, cross-correlation techniques have been used to quantify coactivation patterns (agonist/antagonist active at the same time) and non-coactivation patterns (agonist/antagonist having synergistic out of phase patterns): Nelson-Wong et al. (2008) reported a study of left and right gluteus medius patterns during a long duration standing manual task. Because these patterns are an excellent example of motor synergies and are also related to another medial/lateral postural strategy their details are presented in Chapter 11.

2.2 FREQUENCY ANALYSIS

2.2.1 Introduction – Time Domain vs. Frequency Domain

All the signals that we measure and analyze have a characteristic frequency content, which we refer to as the signal spectrum; this is a plot of all the harmonics in the signal from the lowest to the highest. The purpose of this section is to provide a conceptual background with sufficient mathematical derivations to help the student collect and process data and be an intelligent collector and consumer of commercial software. Frequency domain analysis uses a powerful transform called the Fourier transform, named after Baron Jean-Baptiste-Joseph Fourier, a French mathematician who developed the technique in 1807.

The knowledge of the frequency spectrum of any given signal is mandatory in making decisions about collection and processing of any given signal. The spectrum is a prime determinant of the sampling rate and number of observations that are appropriate before any analog-to-digital conversion. The spectrum also influences decisions pertaining to the frequency of data filtering to remove undesirable noise and movement artifacts. All these factors will be discussed in the sections to come.

2.2.2 Discrete Fourier (Harmonic) Analysis

1. **Alternating Signals.** An alternating signal (often called AC, for alternating current) is one that continuously changes over time. It may be periodic or completely random, or a combination of both. Also, any signal may have a DC (direct current) component, which may be defined as the bias value about which the AC component fluctuates. Figure 2.9 shows example signals.
2. **Frequency Content.** Any of these signals can also be discussed in terms of their frequency content. A sine (or cosine) waveform is a single frequency; any other waveform can be the sum of a number of sine and cosine waves.

Note that the Fourier transformation (see Figure 2.10) of periodic signals has discrete frequencies, while nonperiodic signals have a continuous spectrum defined by its lowest frequency, f_1, and its highest frequency, f_2. To analyze a periodic signal, we must express the frequency content in multiples of the fundamental frequency f_0. These higher frequencies are called *harmonics*. The third harmonic is $3f_0$, and the tenth harmonic is $10f_0$. Any perfectly periodic signal can be broken

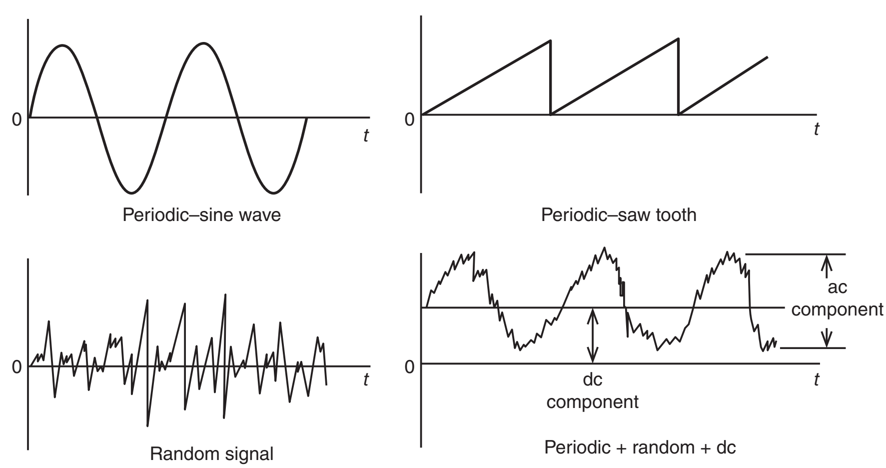

Figure 2.9 Time-related waveforms demonstrate the different types of signals that may be processed.

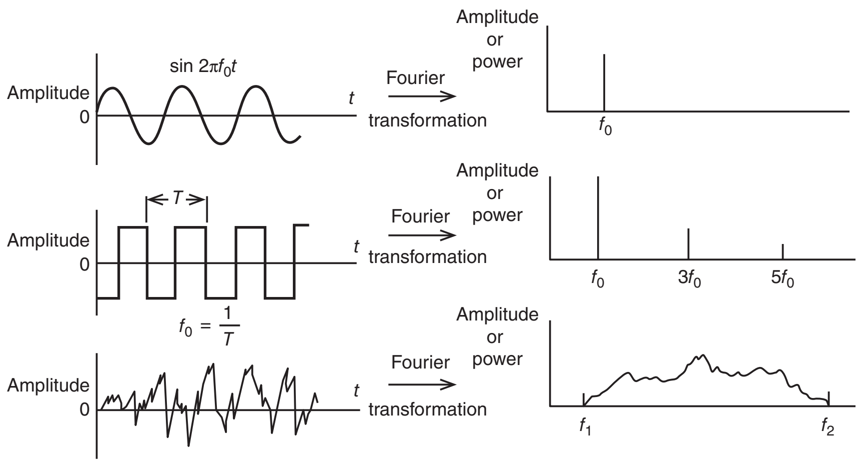

Figure 2.10 Relationship between a signal as seen in the time domain and its equivalent in the frequency domain.

down into its harmonic components. The sum of the proper amplitudes of these harmonics is called a *Fourier series*.

Thus, a given signal $V(t)$ can be expressed as:

$$V(t) = V_{dc} + V_1 \sin(\omega_0 t + \theta_1) + V_2 \sin(2\omega_0 t + \theta_2) + \cdots + V_n \sin(n\omega_0 t + \theta_n) \tag{2.10}$$

where $\omega_0 = 2\pi f_0$, and θ_n is the phase angle of the nth harmonic.

For example, a square wave of amplitude V can be described by the Fourier series of odd harmonics:

$$V(t) = \frac{4V}{\pi}\left(\sin\omega_0 t + \frac{1}{3}\sin 3\,\omega_0 t + \frac{1}{5}\sin 5\,\omega_0 t + \cdots\right) \tag{2.11}$$

A triangular wave of duration $2t$ and repeating itself every T seconds is:

$$V(t) = \frac{2Vt}{T}\left[\frac{1}{2} + \left(\frac{2}{\pi}\right)^2 \cos\omega_0 t + \left(\frac{2}{3\pi}\right)^2 \cos 3\omega_0 t + \cdots\right] \tag{2.12}$$

Several names are given to the graph showing these frequency components: *spectral plots, harmonic plots*, and *spectral density functions*. Each shows the amplitude or power of each frequency component plotted against frequency; the mathematical process to accomplish this is called a *Fourier transformation* or *harmonic analysis*. Figure 2.10 shows plots of time-domain signals and their equivalents in the frequency domain.

Care must be used when analyzing or interpreting the results of any harmonic analysis. Such analyses assume that each harmonic component is present with a constant amplitude and phase over the total analysis period. Such consistency is evident in Equation (2.10), where amplitude V_n and phase θ_n are assumed constant. However, in real life, each harmonic is not constant in either amplitude or phase. A look at the calculation of the Fourier coefficients is needed for any signal $x(t)$. Over the period of time T, using the *discrete Fourier transform*, we calculate n harmonic coefficients.

$$a_n = \frac{2}{T}\int_0^T x(t)\cos n\omega_0 t\, dt \tag{2.13}$$

$$b_n = \frac{2}{T}\int_0^T x(t)\sin n\omega_0 t\, dt \tag{2.14}$$

$$c_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1}\left(\frac{a_n}{b_n}\right) \tag{2.15}$$

It should be noted that a_n and b_n are calculated *average* values over the period of time T. Thus, the amplitude c_n and the phase θ_n of the nth harmonic are average values as well. A certain harmonic may be present only for part of the time T, but the computer analysis will return an average value, assuming that it is present over the entire time. The fact that a_n and b_n are average values is important when we attempt to reconstitute the original signal as is demonstrated in Section 2.2.4.5.

The digital equivalent of the Fourier transform is important to review because it gives us some insight into the number of calculations that are necessary. In digital form, Equations (2.13) and (2.14) for N samples during the period T:

$$a_n = \frac{2}{N}\sum_{i=0}^{N} x_i \cos\left(n\omega_0 i/N\right) \tag{2.16}$$

$$b_n = \frac{2}{N}\sum_{i=0}^{N} x_i \sin\left(n\omega_0 i/N\right) \tag{2.17}$$

For each of the n harmonics, N calculations are necessary. The number of harmonics that can be analyzed is from the fundamental ($n = 1$) up to the Nyquist frequency, which is when there are two samples per sine or cosine wave or when $n = N/2$. Therefore, for $N/2$ harmonics, there are $N^2/2$ calculations necessary for each of the sine or cosine coefficients. The total number of calculations is N^2. It should be noted that the major expense in computer time is looking up the sine and cosine values for each of the N angles.

2.2.3 Fast Fourier Transform (FFT)

The fast Fourier transform (FFT) became necessary because of the extremely large number of calculations necessary in the Discrete Fourier Transform. As early as 1942, Danielson and Lanczos introduced the *Danielson-Lanczos Lemma*, which showed that the Discrete Fourier Transform of length N can be broken into two separate odd and even-numbered components of length $N/2$ each. In a similar manner, each $N/2$ component can be broken into two more odd- and even-numbered components of length $N/4$ each, and each of these can be broken into two more odd and even components of length $N/8$ each. Thus, the basis of the FFT is a data record that must be a power of two (2^n). Therefore, if you collect data files that are not a power of two in length, the FFT can only accept the largest 2^n length file within your data file. For example, if you collected 1000 data points, the largest file length that could be used would be 512 points; thus, 488 points would be wasted. Therefore, it is advisable to prearrange data collection files to be binary in length; in the case of the previous example, a data file of 1024 points would be appropriate. With the advent of computers, many FFT algorithms appeared (Brigham, 1974), and in the mid-1960s, J. W. Cooley and J. W. Tukey at IBM developed the most commonly used FFT algorithm (Cooley and Tukey, 1965).

One of the major savings in the FFT is to avoid repetitive and time-consuming calculations especially sines and cosines. If we look at Equations (2.16) and (2.17), we see that for the fundamental frequency ($n = 1$), we must calculate N sine and N cosine values. For the second harmonic, we recalculate every second sine and cosine value, and for the third harmonic we recalculate every third sine and cosine value, and so on up to the highest harmonic. The FFT calculates all sine and cosine values for the fundamental and this forms a "look-up" table for the fundamental plus all higher harmonics. Further computational savings are achieved by clustering all the products of x_i and the same sine value, then summing all the x_i values, and then carrying out one product with the sine value. The number of calculations for the FFT = $N \log_2 N$, which is considerably less than N^2 for the Discrete Fourier Transform. For example, for $N = 65{,}536$ and a CPU cycle time of 1 ns (1 GHz), the DFT would take $(65{,}536^2)^{-9} = 4.29$ s, while the FFT would take $(65536 \log_2 65536)^{-9} = 0.001$ s.

2.2.4 Applications of Spectrum Analyses

2.2.4.1 Analog-to-Digital Converters. To students not familiar with electronics, the process that takes place during conversion of a physiological signal into a digital computer can be somewhat mystifying. A short schematic description of that process is now given. An electrical signal representing a force, an acceleration, an EMG potential, or similar is fed into the input terminals of the analog-to-digital converter. The computer controls the rate at which the signal is sampled; the optimal rate is governed by the sampling theorem (see Section 2.2.4.2).

Figure 2.11 depicts the various stages in the conversion process. The first is a sample/hold circuit in which the analog input signal is changed into a series of short-duration pulses, each one equal in amplitude to the original analog signal at the time of sampling. The final stage of conversion is to translate the amplitude and polarity of the sampled pulse into digital format. This is usually a binary code in which the signal is represented by a number of bits. For example, a 16-bit code represents $2^{16} = 65{,}536$ levels. This means that the original sampled analog signal can be broken into 65,536 discrete amplitude levels with a unique code representing each of these levels. Each coded sample (consisting of zero and one second) forms a 16-bit "word," which is rapidly stored in computer memory for recall at a later time. If five seconds of signal were converted at a sampling rate of 100 Hz, there would be 500 data words stored in memory to represent the original five seconds signal.

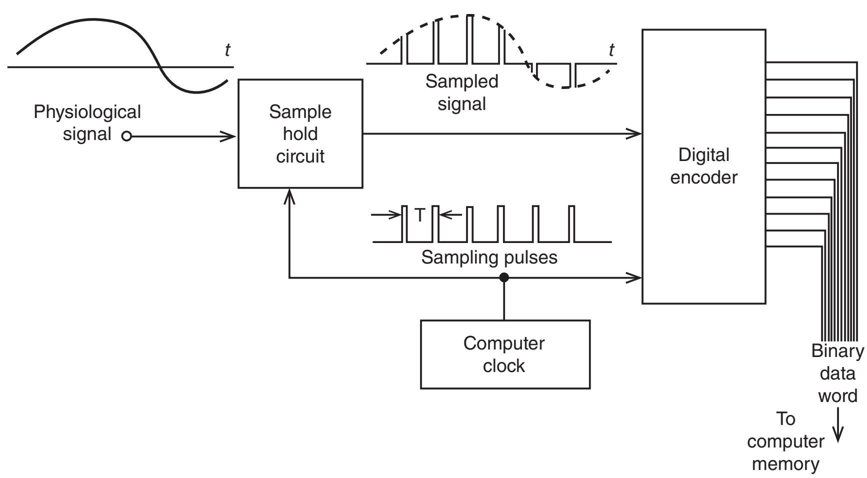

Figure 2.11 Schematic diagram showing the steps involved in an analog-to-digital conversion of a physiological signal.

2.2.4.2 Deciding the Sampling Rate – The Sampling Theorem. In the processing of any time-varying data, no matter what their source, the sampling theorem must not be violated. Without going into the mathematics of the sampling process, the theorem states that "the process signal must be sampled at a frequency at least twice as high as the highest frequency present in the signal itself." If we sample a signal at too low a frequency, we get aliasing errors. This results in false frequencies, frequencies that were not present in the original signal, being generated in the sample data. Figure 2.12 illustrates this effect. Both signals are being sampled at the same interval T. Signal 1 is being sampled about ten times per cycle, while signal 2 is being sampled less than twice per cycle. Note that the amplitudes of the samples taken from signal 2 are identical to those sampled from signal 1. A false set of sampled data has been generated from signal 2 because the sample rate is too low – the sampling theorem has been violated.

When performing motion analysis, researchers and clinicians often use digital video to assess joint positions or angles. There is a trade-off between using a high sampling rate (or more frames per second) and quality of the image. At higher frames per second, there is a drop in resolution of the images and a greater computational expense to capture the frames. This was more problematic before the advent of digital video, but higher frame rates require greater computational processing during analysis and greater storage space for larger files. For normal and pathological gait studies, it has been shown that kinetic and energy analyses can be done with negligible error using a standard 24-frame per second movie camera, in part because of the frequency of joint movements and physiological signals (accelerations, reaction forces) during walking is low (Winter, 1982).

Figure 2.13 compares the results of kinematic analysis of the foot during normal walking, where a 50-Hz capture rate was compared with 25 Hz. The data were collected at 50 Hz, and the acceleration of the foot was calculated using every frame of data, then reanalyzed again, using every second frame of converted data. It can be seen that the difference between the curves is minimal; only at the peak negative acceleration was there a noticeable difference. The final decision as to whether this error is acceptable should not rest in this curve, but in your final goal. If, for example, the final analysis was a hip and knee torque analysis, the acceleration of the foot segment may not be too important, as is evident from another walking trial, shown in Figure 2.14. The minor differences in no way interfere with the general pattern of joint torques over the stride period, and the assessment of the motor patterns would be identical. Thus, for movements such as walking or for slow movements, an inexpensive camera at 24 frames per second appears to be quite adequate. Faster movement, such as sprinting or throwing may require higher frame rates to adequately capture the movement without aliasing errors.

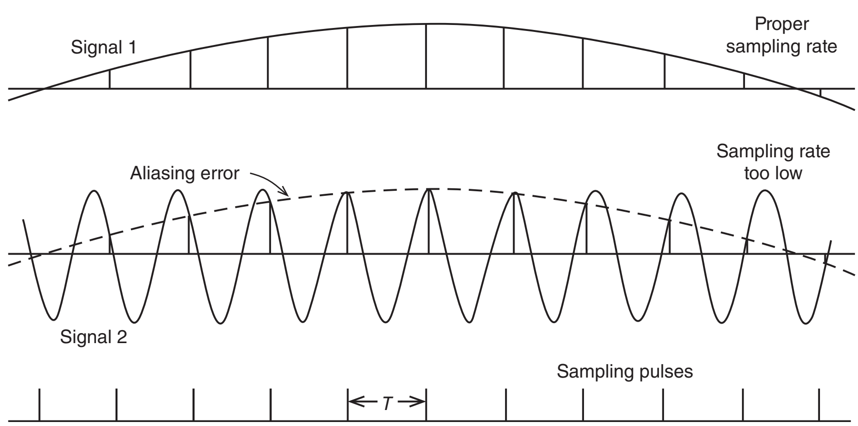

Figure 2.12 Sampling of two signals, one at a proper rate, the other at too low a rate. Signal 2 is sampled at a rate less than twice its frequency, such that its sampled amplitudes are the same as for signal 1. This represents a violation of the sampling theorem and results in an error called *aliasing*.

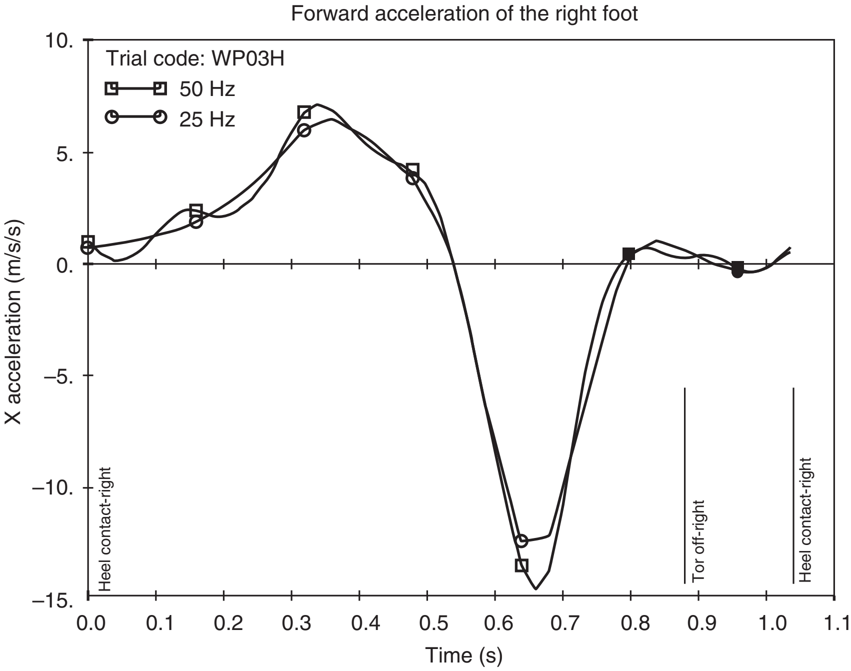

Figure 2.13 Comparison of the forward acceleration of the right foot during walking using the same data sampled at 50 Hz and at 25 Hz (using data from every second frame). The major pattern is maintained with minor errors at the peaks.

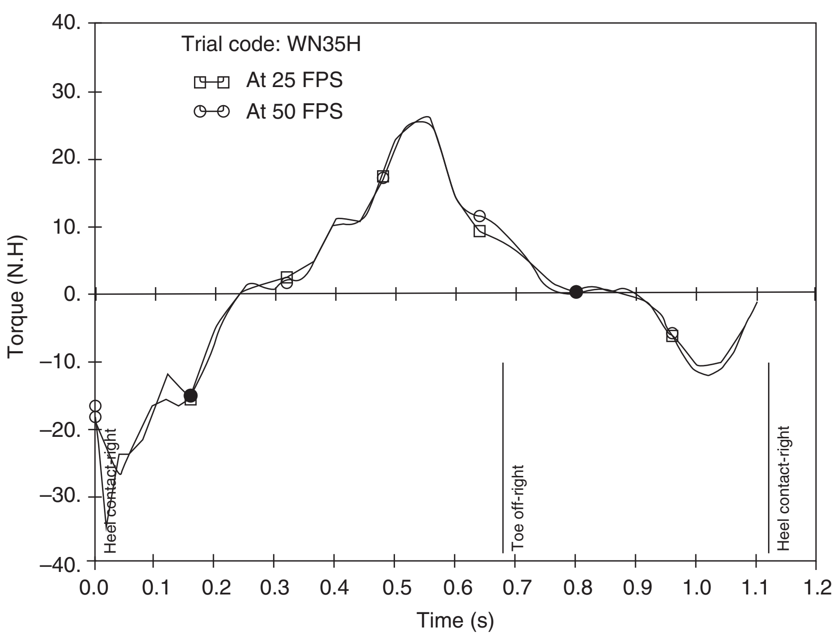

Figure 2.14 Comparison of the hip moment of force during level walking using the same data sampled at 50 Hz and at 25 Hz. The residual error is quite small because the joint reaction forces dominate the inertial contributions to the net moment of force.

2.2.4.3 Deciding the Record Length. The duration of record length is decided by the lowest frequency present in the signal. In cyclical events such as walking, cycling, or swimming the lowest frequency is easy to determine; it is the stride frequency or how often each segment of the body repeats itself. For example, if a patient is walking at 105 steps/minute the step frequency is 105/60 = 1.75 steps/second = 0.875 strides/second. Thus, the fundamental frequency is 0.875 Hz. However, there are a number of noncyclical movements, which do not have a defined lowest frequency. One such "movement" is standing either quietly or in a work-related task. In quiet standing, we model the total body as an inverted pendulum (Gage et al., 2004), which simplifies the total body into a single weighted-average center of mass (COM) and which can be compared with the center of pressure (COP) measured from the force plate. Figure 2.15 presents a typical FFT of the COP and COM in the A/P direction for a subject standing quietly for 137 seconds (8192 samples @ 60 Hz). Note that the FFT plots the amplitude of each harmonic from 0.0073 to 1 Hz. Also note the dominant low-frequency components of both COM and COP below 0.2 Hz. The length of record must be at least a minute or longer. This long record may be compromised when studying patients with balance disorders because they may not be able to stand quietly for that length of time. However, for studies on normal subjects, Carpenter et al. (2001) found that records of at least one minute were required for acceptable reliability.

2.2.4.4 Analog and Digital Filtering of Signals – Noise and Movement Artifacts. The basic approach can be described by analyzing the frequency spectrum of both signal and noise. Figure 2.16a shows a schematic plot of a signal and noise spectrum. As can be seen, the signal is assumed to occupy the lower end of the frequency spectrum and overlaps with the noise, which is usually higher frequency. Filtering of any signal is aimed at the selective rejection, or attenuation, of certain frequencies. In the preceding case, the obvious filter is one that passes, unattenuated, the lower-frequency signals, while at the same time attenuating the higher-frequency noise. Such a filter, called a *low-pass filter*, has a frequency response as shown in Figure 2.16b. The frequency response of the filter is the ratio of the output $X_o(f)$ of the filter to its input $X_i(f)$ at each frequency present. As can be seen, the response at lower frequencies is 1.0. This means that the input signal passes through the filter unattenuated. However, there is a sharp transition at the cutoff frequency f_c so that the signals above f_c are severely attenuated.

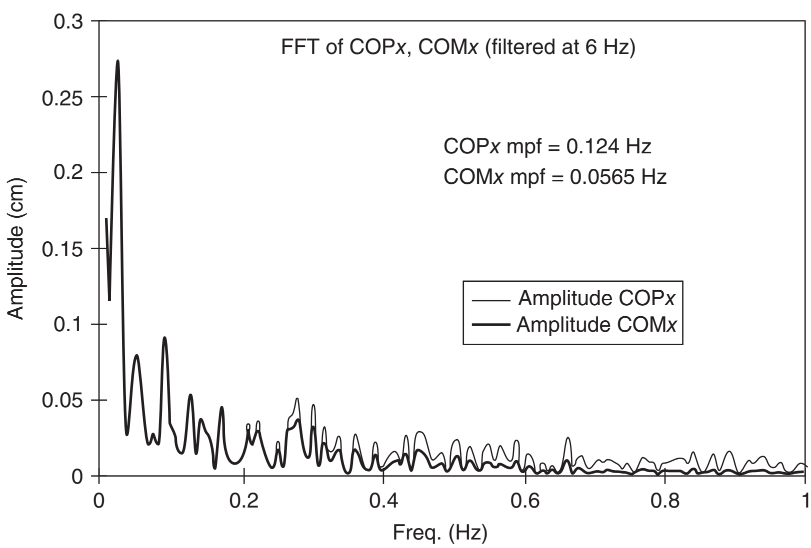

Figure 2.15 FFT of the COM and COP in the anterior/posterior direction of a subject standing quietly for 137 seconds. (Winter (1995).)

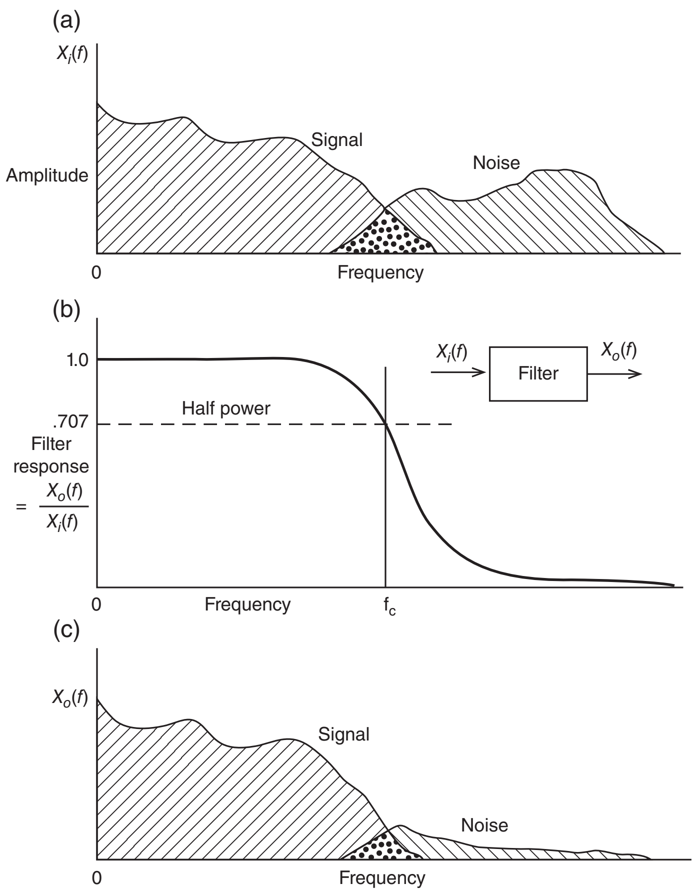

Figure 2.16 (a) Hypothetical frequency spectrum of a waveform consisting of a desired signal and unwanted higher frequency noise. (b) Response of low-pass filter $X_0(f)/X_i(f)$, introduced to attenuate the noise. (c) Spectrum of the output waveform, obtained by multiplying the amplitude of the input by the filter response at each frequency. Higher-frequency noise is severely attenuated, while the signal is passed with only minor distortion in transition region around f_c.

The net result of the filtering process can be seen by plotting the spectrum of the output signal $X_o(f)$ as seen in Figure 2.16c. Two things should be noted. First, the higher-frequency noise has been severely reduced but not completely rejected. Second, the signal, especially in the region where the signal and noise overlap (usually around f_c) is also slightly attenuated. This results in a slight distortion of the signal. Thus, a compromise has to be made in the selection of the cutoff frequency. If f_c is set too high, less signal distortion occurs, but too much noise is allowed to pass. Conversely, if f_c is too low, the noise is reduced drastically, but at the expense of increased signal distortion. A sharper cutoff filter will improve matters, but at an additional expense. In digital filtering, this means a more complex digital filter and, thus, more computer time.

(a) Hypothetical frequency spectrum of a waveform consisting of a desired signal and unwanted higher-frequency noise.
(b) Response of low-pass filter $X_o(f)/X_i(f)$, introduced to attenuate the noise.
(c) Spectrum of the output waveform, obtained by multiplying the amplitude of the input by the filter response at each frequency. Higher-frequency noise is severely attenuated, while the signal is passed with only minor distortion in

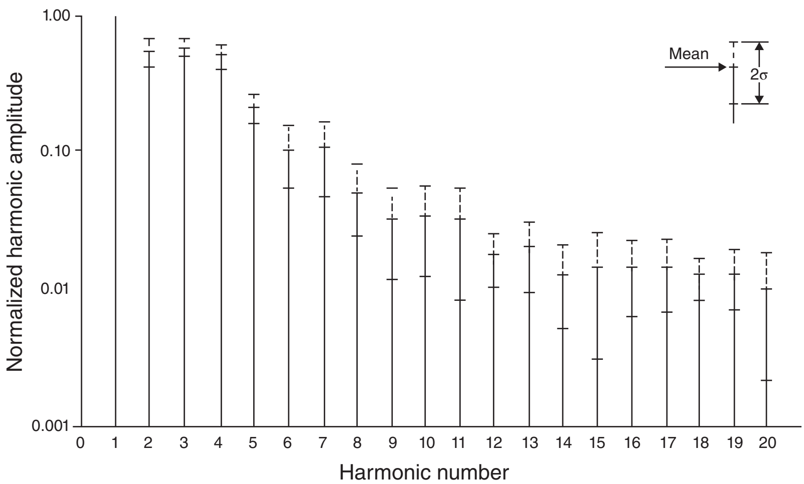

Figure 2.17 Harmonic content of the vertical displacement of a toe marker from 20 subjects during normal walking. Fundamental frequency (harmonic number = 1) is normalized at 1.00. Over 99% of power is contained below the seventh harmonic. (With Permission of Elsevier.)

The first aspect that must be assessed is what the signal spectrum is as opposed to the noise spectrum. This can readily be done, as is seen in the harmonic analysis for 20 subjects presented in Figure 2.17. Here is the harmonic content of the vertical displacement of the toe marker in natural walking (Winter et al., 1974). The highest harmonics were found to be in the toe and heel trajectories, and it was found that 99.7% of the signal power was contained in the lower seven harmonics (below 6 Hz). Above the seventh harmonic, there was still some signal power, but it had the characteristics of "noise." Noise is the term used to describe components of the final signal that are not the result of the process itself (in this case, walking). Noise comes from many sources: electronic noise in optoelectric devices, spatial precision of the scan or digitizing system, and error in marker reconstruction during digital motion analysis. If the total effect of all these errors is random, then the true signal will have an added random component. Usually, the random component is high frequency, as is borne out in Figure 2.17. Here, we see evidence of higher-frequency components extending up to the twentieth harmonic, which was the highest frequency analyzed. The presence of the higher-frequency noise is of considerable importance when we consider the problem of trying to calculate velocities and accelerations from the displacement data, as will be evident in later chapters.

The theory behind digital filtering (Radar and Gold, 1967) will not be covered, but the application of low-pass digital filtering will be described in detail. As a result of the previous discussion for these data on walking, the cutoff frequency of a digital filter should be set at about 6 Hz. The format of a recursive digital filter that processes the raw data in time domain is as follows:

$$\begin{aligned} X^1(nT) &= a_0X(nT) + a_1X(nT-T) + a_2X(nT-2T) \\ &\quad + b_1X^1(nT-T) + b_2X^1(nT-2T) \end{aligned} \tag{2.18}$$

where

X^1 = filtered output coordinates
X = unfiltered coordinate data
nT = nth sample
$(nT - T)$ = $(n - 1)$th sample
$(nT - 2T)$ = $(n - 2)$th sample
$a_0, \ldots, b_0, \ldots$ = filter coefficients.

These filter coefficients a_0, a_1, a_2, b_1, and b_2 are constants that depend on the type and order of the filter, the sampling frequency, and the cutoff frequency. As can be seen, the filter output $X^1(nT)$ is a weighted version of the immediate and past raw data plus a weighted contribution of past filtered output. The exact equations to calculate the coefficients for a Butterworth or a critically damped filter are as follows:

$$\omega_c = \frac{(\tan(\pi f_c/f_s))}{C} \tag{2.19}$$

where C is the correction factor for number of passes required, to be explained shortly. For a single pass filter $C = 1$.

$K = \sqrt{2}\omega_c$ for a Butterworth filter,
or, $2\omega_c$ for a critically damped filter

$$K_2 = \omega_c^2, \quad a_0 = \frac{K_2}{(1 + K_1 + K_2)}, \quad a_1 = 2a_0, \quad a_2 = a_0$$

$$K_3 = \frac{2a_0}{K_2}, \quad b_1 = -2a_0 + K_3$$

$$b_2 = 1 - 2a_0 - K_3, \qquad \text{or} \qquad b_2 = 1 - a_0 - a_1 - a_2 - b_1$$

For example, a Butterworth-type low-pass filter of second order is to be designed to cutoff at 6 Hz using camera data taken at 60 Hz (60 frames per second). As seen in Equation (2.1), the only thing that is required to determine these coefficients is the ratio of sampling frequency to cutoff frequency. In this case, it is 10. The design of such a filter would yield the following coefficients:

$$a_0 = 0.067455, \quad a_1 = 0.13491, \quad a_2 = 0.067455,$$
$$b_1 = 1.14298, \quad b_2 = -0.41280$$

Note that the algebraic sum of all the coefficients equals 1.0000. This gives a response of unity over the passband. Note that the same filter coefficients could be used in many different applications, as long as the ratio f_s/f_c is the same. For example, an EMG signal sampled at 2000 Hz with cutoff desired at 400 Hz would have the same coefficients as one employed for video coordinates where the video rate was 30 Hz and cutoff was 6 Hz. The number of passes, C, in Equation (2.1) is important when filtering kinematic data in order to eliminate the phase shift of the filtered data. This aspect of digital filtering of kinematic data will be detailed later in Chapter 3.

2.2.4.5 Fourier Reconstitution of Original Signal. Figure 2.18 is presented to illustrate a Fourier reconstitution of the vertical trajectory of the heel of an adult walking his or her natural cadence. A total of nine harmonics is represented here because the addition of higher harmonics did not improve the curve of the original data. As can be seen, the harmonic reconstitution is visibly different from the original, sufficiently so as to cause reasonable errors in subsequent biomechanical analyses. This is because each harmonic amplitude and phase values are average values, as we cautioned about in Section 2.2.2, and this is especially true for a foot marker during gait, which has high frequencies during swing and low frequencies during stance.

2.2.4.6 Fourier Analysis of White Noise. White noise was introduced in Figure 2.4, where an autocorrelation showed that each point has zero correlation with any points ahead or behind it in time. In a computer, white noise can be simulated by a random number generator. The other characteristic of white noise is in the frequency domain where

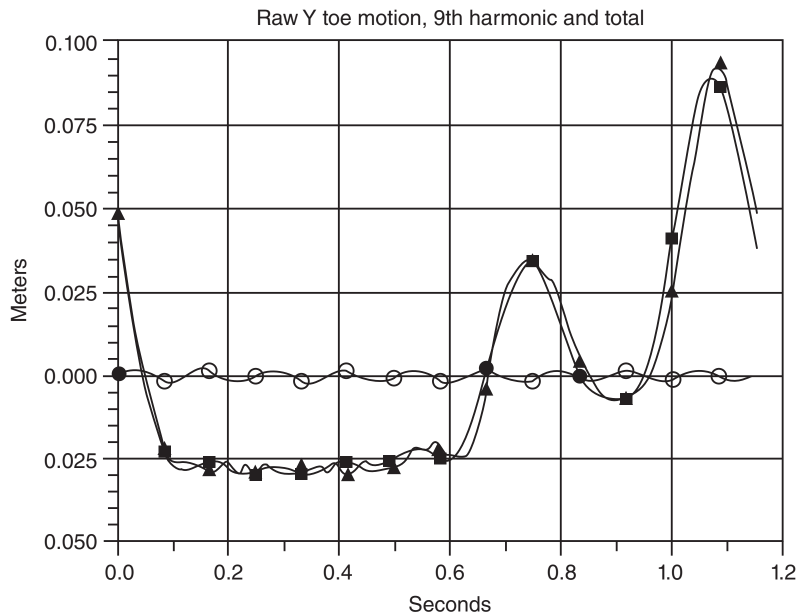

Figure 2.18 Fourier reconstruction of the vertical trajectory of a toe marker during one walking stride. The actual trajectory is shown by the open square, the reconstruction from the first nine harmonics is plotted with open triangles, and the contribution of the ninth harmonic is plotted with open circles. The difference between the actual and the reconstructed waveforms is the result of the lack of stationarity in the original signal.

the whole range of the signal, and this is similar to some of the noise apparent in kinematic data optoelectric systems. To demonstrate this frequency spectrum characteristic we present an analysis of the FFT of white noise. Figure 2.19a is a white noise signal simulated on Excel from a random number generator with amplitude ±1 sampled at a rate of 2048 samples/second. Thus, the highest frequency present in this signal is the Nyquist frequency of 1024 Hz, so our FFT will cover frequencies from 0 to 1024 Hz. An FFT of the signal in Figure 2.19a is presented in Figure 2.19b. Note that the spectrum over that range appears "noisy," but that is because of the fact that the FFT is plotting the amplitude at every single Hz. A more realistic plot of this spectrum is to carry out a moving average across the full spectrum, and this was done with a 40-point moving average, which is the "white" curve in Figure 2.19b. This moving average approaches the theoretical constant amplitude across the full spectrum; its average amplitude is 0.330 and ranges from 0.233 to 0.446. A longer time-domain record than the 2048 points will result in a more constant frequency plot. Also, it should be noted that some of the individual harmonic amplitudes are greater than one, and this is not expected from the white noise signal, which had an amplitude of ±1. What is not shown here are the phase angles of each of the harmonics; each has a different phase angle, so there will be many cancellations as each harmonic is added.

2.3 ENSEMBLE AVERAGING OF REPETITIVE WAVEFORMS

A large number of movements that we study are cyclical in nature and, therefore, can benefit from a cyclical average of its many variables. Gait (walking and running) is the most common repetitive movement, but cycling, rowing, and lifting also benefit from such averaged profiles. Both intra- and inter-subject averaged profiles have been reported on a wide variety of kinematic, kinetic, and EMG variables. The major benefit of such a technique is that the averaged waveform is more reliable and the variation about the mean gives

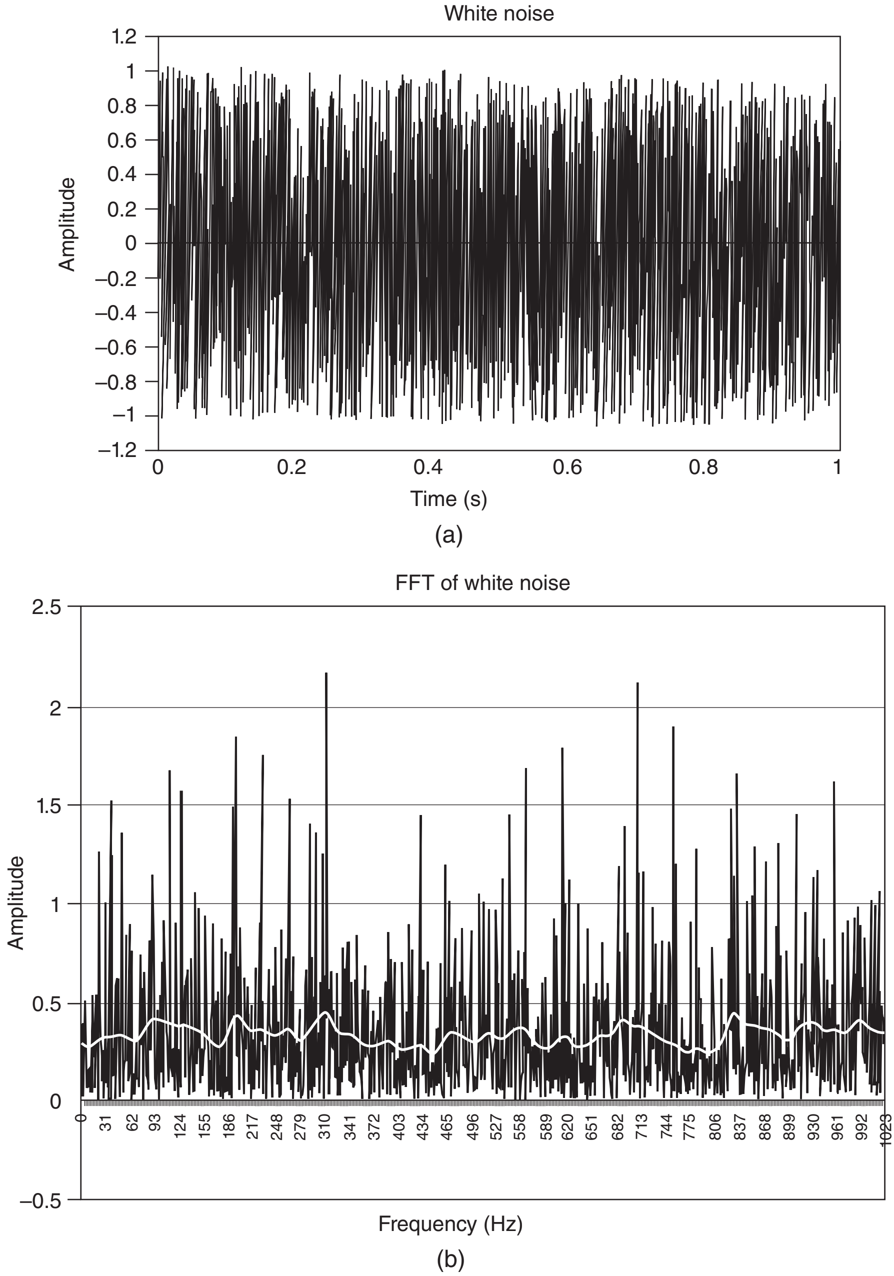

Figure 2.19 (a) A simulated white noise signal from a random number generator with amplitude ± 1 samples at 2048 samples/second. Panel (b) is the FFT of the white noise signal at 1 Hz intervals from 0 to 1024 Hz. The “white” line passing through this FFT plot is a 40-point moving average showing that the signal has approximately equal power across the full spectrum of the signal.

randomness of the variable. For example, in gait, the intra-subject lower limb joint angles have minimal variability, while the moment profiles at these same joints are quite variable. This phenomenon has resulted in a covariance analysis, which can readily be done; we can calculate the mean variance at each of the joints from these ensemble averages, from which the covariances can be determined. As a result of those analyses, a total lower

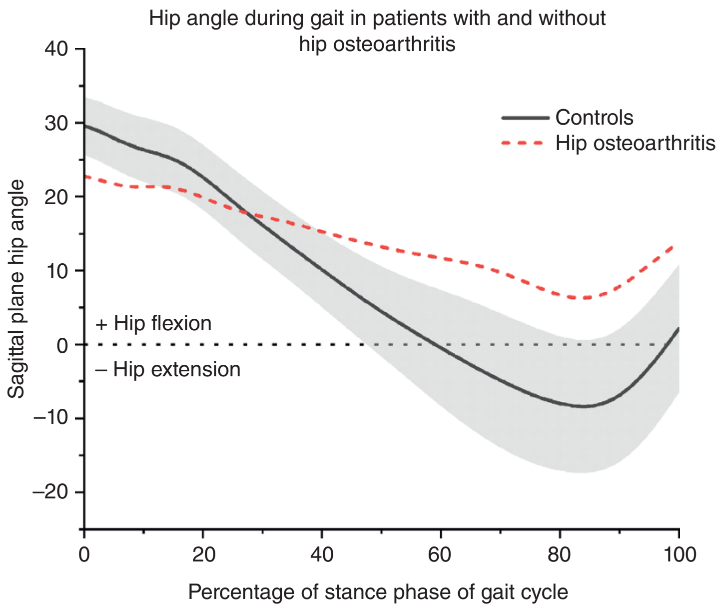

Figure 2.20 Sagittal plane joint angles from an individual with hip osteoarthritis (dashed line) and the ensemble average from 20 age-matched individuals without hip pain (solid line). The gray area represents ±1 standard deviation of the normative sample. Positive values represent hip flexion and negative values represent hip extension.

and this is reported in detail in Chapter 11. Such ensemble-averaged waveforms also form the basis of clinical assessments, where the patient's profile superimposed on the average for a comparable healthy group provides a very powerful tool in diagnosing specific movement abnormalities. An example of such a clinical analysis is presented in the next section, Section 2.3.1.

2.3.1 Examples of Ensemble-Averaged Profiles

Figure 2.20 shows a typical waveform from a clinical gait study of a patient with hip osteoarthritis. The sagittal plane hip joint angles during the stance phase of gait are overlaid on the averaged profiles of 20 age-matched adults. The dark, solid line represents the ensemble average from 20 age-matched individuals who did not have lower extremity pathology, while the shaded gray area represents ±1 standard deviation of that sample. The lighter, dashed curve represents the hip joint angles from the individual with hip osteoarthritis. In this example, it is clear that the individual with hip osteoarthritis has a movement profile that is different than those with pathology. Specifically, this individual has less hip flexion at the start of the stance phase and has a lack of hip extension during the later phases of stance. Collectively, this patient has a lower overall joint excursion and walks with what can be considered "stiff-leg gait." The patient's joint angles, alongside a normative data subset with estimates of variability, allow clinicians and researchers to identify specific movement abnormalities and apply appropriate therapies.

In this case, the stance phase of gait has been normalized to 100% to account for temporospatial differences in stance time between individuals. The time base of each subject's profile must be altered from their individual stance time (in seconds) to a 100% stance baseline. This may condense or expand the individual's curve but allows for consistent averaging when the actual timing may be variable between subjects.

2.3.2 Normalization of Time Bases to 100%

Assume that for one subject there are n samples of a given variable x over the stride period and we wish to normalize this time-domain record to N samples this normalized curve y. Therefore, each interval of the normalized curve would be n/N samples in duration. Assume that $n = 107$ samples and $N = 100$. At $N = 1$, we need the value of the variable at the 1.07th sample; thus, by linear interpolation, $y_1 = 0.93x_1 + 0.07x_2$; for $N = 2$ we need the value at the 2.14th sample, or $y_2 = 0.86x_2 + 0.14x$; for $N = 3$ we need the value at the 3.21th sample, or $y_3 = 0.79x_3 + 0.21x_4$, and so on; for $N = 99$ we need the value at the 105.93th sample, or $y_{99} = 0.07x_{105} + 0.93x_{106}$; and finally $y_{100} = x_{107}$.

It is important to note in clinical studies, normalization to 100% of a gait cycle can have unintended consequences. Removal of the factor of time (in seconds) from physiological data precludes evaluations of measures in which time is a relevant factor. For example, in weight-bearing joints of the lower extremity, the rate and total duration of load are thought to impact the integrity of the biological tissues. Calculations of loading rate or total load should not be performed on time-normalized curves.

2.3.3 Measure of Average Variability about the Mean Waveform

The most common descriptive statistical measure of data where a single measure is taken is the coefficient of variation, CV = $\boldsymbol{\sigma}/\overline{X}$ where $\boldsymbol{\sigma}$ is the standard deviation and $\overline{X}$ is the sample mean. The CV is a variability to mean ratio, and with the ensemble average, we wish to calculate a similar variability score; this score is the average $\boldsymbol{\sigma}$ over the stride period of N points divided by the average signal (absolute value). Some researchers suggest that the rms value of the signal replaces the absolute value in the denominator, but this changes the CV scores very little.

$$\mathrm{CV} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N} \sigma_i^2}}{\frac{1}{N}\sum_{i=1}^{N} |X_i|} \tag{2.20}$$

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3

KINEMATICS

Amy L. Lenz
Department of Orthopaedics, School of Medicine, University of Utah, Salt Lake City, UT, USA

3.0 HISTORICAL DEVELOPMENT AND COMPLEXITY OF PROBLEM

Interest in the actual patterns of movement of humans and animals goes back to prehistoric times and was depicted in cave drawings, statues, and paintings. Such replications were subjective impressions of the artist. It was not until a little over a century ago that the first motion picture cameras recorded locomotion patterns of both humans and animals. Marey, the French physiologist, used a photographic "gun" in 1885 to record displacements in human gait and chronophotographic equipment to get a stick diagram of a runner. About the same time, Muybridge in the United States triggered 24 cameras sequentially to record the patterns of a running man. Since then the technology has rapidly progressed, and we now can record and analyze everything from the gait of a child with cerebral palsy to the performance of an elite athlete.

The term used for these descriptions of human movement is *kinematics*. Kinematics is not concerned with the forces, either internal or external, that cause the movement but rather with the details of the movement itself. A complete and accurate quantitative description of the simplest movement requires a huge volume of data and a large number of calculations, resulting in an enormous number of graphic plots. For example, to describe the movement of the lower limb in the sagittal plane during one stride can require up to 50 variables. These include linear and angular displacements, velocities, and accelerations. It should be understood that any given analysis may use only a small fraction of the available kinematic variables. An assessment of a running broad jump, for example, may require only the velocity and height of the body's center of mass. On the other hand, a mechanical power analysis of an amputee's gait may require almost all the kinematic variables that are available. Therefore, the importance of each kinematic variable and interpretation of the data for each specific patient is commonly thought of as the intersection of science and art.

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

3.1 KINEMATIC CONVENTIONS

In order to keep track of all the kinematic variables (Tables B.1 and B.2), it is important to establish a convention system. In the anatomical literature, a definite convention has been established, and we can completely describe a movement using terms such as *proximal*, *flexion*, and *anterior*. It should be noted that these terms are all relative to the anatomic position and describe the position of one limb relative to another. They do not give us any idea as to where we are in space. Thus, if we wish to analyze movement relative to the ground or the direction of gravity, we must establish an absolute spatial reference system. Such conventions are mandatory when imaging devices are used to record the movement.

3.1.1 Absolute Spatial Reference System

Several spatial reference systems have been proposed. The one utilized throughout the text is the one often used for human gait. The vertical direction (superior-inferior) is Z, the direction of progression (anterior–posterior) is X, and the sideways direction (medial–lateral) is Y. Figure 3.1 depicts this convention. The positive direction is as shown. Angles must also have a zero reference and a positive direction. Angles in the YZ plane are measured from 0° in the Y direction, with positive angles being counterclockwise. Similarly, in the ZX plane, angles start at 0° in the Z direction and increase positively counterclockwise. The convention for velocities and accelerations follows correctly if we maintain the spatial coordinate convention:

$\dot{x}$ = velocity in the X direction, positive when X is increasing
$\dot{y}$ = velocity in the Y direction, positive when Y is increasing

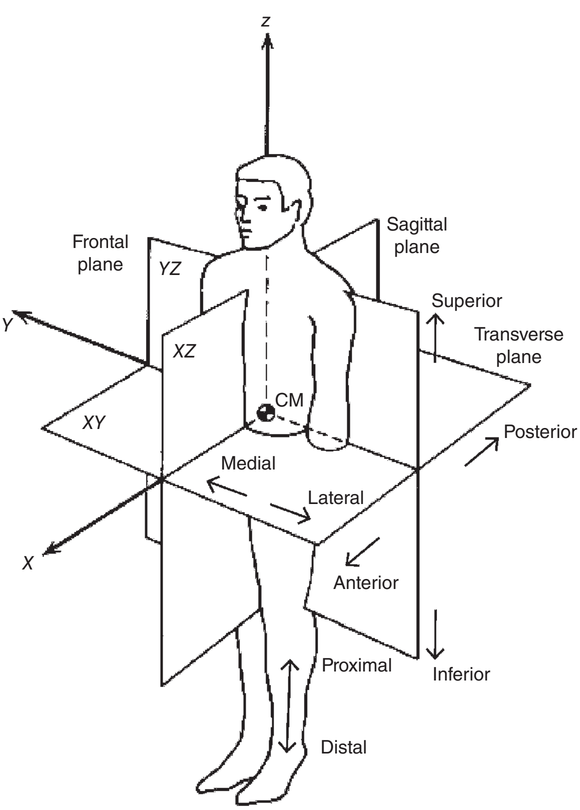

Figure 3.1 Spatial coordinate system for all data and analyses

$\dot{z}$ = velocity in the Z direction, positive when Z is increasing
$\ddot{x}$ = acceleration in the X direction, positive when $\dot{x}$ is increasing
$\ddot{y}$ = acceleration in the Y direction, positive when $\dot{y}$ is increasing
$\ddot{z}$ = acceleration in the Z direction, positive when $\dot{z}$ is increasing

The same applies to angular velocities and angular accelerations. A counterclockwise angular increase is a positive angular velocity, ω. When ω is increasing, we calculate a positive angular acceleration, α.

3.1.2 Total Description of a Body Segment in Space

The complete kinematics of any body segment requires 15 data variables, all of which are changing with time:

1. Position (x, y, z) of segment center of mass
2. Linear velocity ($\dot{x}, \dot{y}, \dot{z}$) of segment center of mass
3. Linear acceleration ($\ddot{x}, \ddot{y}, \ddot{z}$) of segment center of mass
4. Angle of segment in two planes, θ_{xy}, θ_{yz}
5. Angular velocity of segment in two planes, ω_{xy}, ω_{yz}
6. Angular acceleration of segment in two planes, α_{xy}, α_{yz}

Note that the third angle data are redundant; any segment's direction can be completely described in two planes. For a complete description of the total body (feet + legs + thighs + trunk + head + upper arms + forearms and hands = 14 segments), movement in three-dimensional (3D) space required $15 \times 14 = 210$ data variables. It is no small wonder that we have challenges describing and analyzing more complex movements. Certain simplifications can certainly reduce the number of variables to a manageable number. In gait, for example, the head, arms, and trunk (HAT) are often considered to be a single segment. Therefore, the data variables in this case (seven segments) can be reduced to 105.

3.2 DIRECT MEASUREMENT TECHNIQUES

3.2.1 Goniometers

A goniometer is a special name given to the electrical potentiometer that can be attached to measure a joint angle. One arm of the goniometer is attached to one limb segment, the other to the adjacent limb segment, and the axis of the goniometer is aligned to the joint axis. In Figure 3.2, you can see the fitting of the goniometer to a knee joint along with the equivalent electrical circuit. A constant voltage E is applied across the outside terminals, and the wiper arm moves to pick off a fraction of the total voltage. The fraction of the voltage depends on the joint angle θ.

Thus, the voltage on the wiper arm is $\nu = kE\theta = k_1\theta$ volts.

Note that a voltage proportional to θ requires a potentiometer whose resistance varies linearly with θ.

A goniometer designed for clinical studies is shown fitted on a patient in Figure 3.3.

Advantages

1. A goniometer is generally inexpensive.
2. The output signal is available immediately for recording or conversion into a computer.
3. Planar rotation is recorded independently of the plane of movement of the joint.

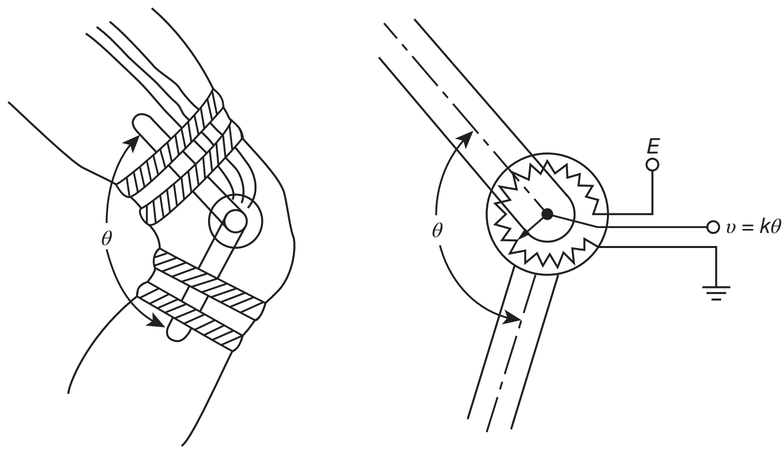

Figure 3.2 Mechanical and electrical arrangement of a goniometer located at the knee joint. Voltage output is proportional to the joint angle.

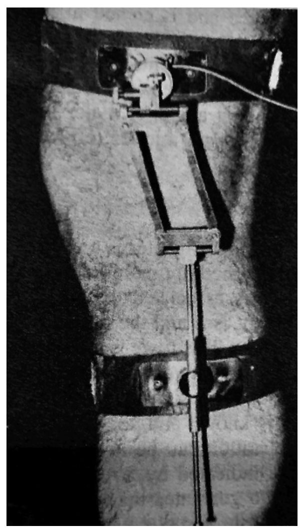

Figure 3.3 Electrogoniometer designed to accommodate changes in the axis of rotation of the knee joint, shown here fitted on a patient. (Reproduced by permission of Chedoke-McMaster Medical Center, Hamilton, Ont. Canada.)

Disadvantages

1. Relative angular data are given, not absolute angles, thus severely limiting the data's assessment value.
2. It may require an excessive length of time to fit and align, and the alignment over fat and muscle tissue can vary over the time of the movement.
3. If a large number are fitted, movement can be encumbered by the straps and cables.
4. More complex goniometers are required for joints that do not move as hinge joints (Finley and Karpovich, 1964).

3.2.2 Accelerometers

As indicated by its name, an accelerometer is a device that measures acceleration. Most accelerometers are nothing more than force transducers designed to measure the reaction forces associated with a given acceleration. If the acceleration of a limb segment is a and the mass inside is m, then the force exerted by the mass is $F = ma$. This force is measured by a force transducer, usually a strain gauge or piezoresistive type. The mass is accelerated against a force transducer that produces a signal voltage V, which is proportional to the force, and since m is known and constant, V is also proportional to the acceleration. Today almost all accelerometers are 3D transducers. A 3D transducer is nothing more than three individual accelerometers mounted at right angles to each other, each one then reacting to the orthogonal component acting along its axis. Even with a 3-axis accelerometer mounted on a limb, there can be problems because of limb rotation, as indicated in Figure 3.4. In both cases, the leg is accelerating in the same *absolute* direction, as indicated by vector a. The measured acceleration component a_n is quite different in each case. Thus, the accelerometer is limited to those movements whose direction in space does not change drastically or to special contrived movements, such as horizontal flexion of the forearm about a fixed elbow joint.

A typical electric circuit of a piezoresistive accelerometer is shown in Figure 3.5. It comprises a half-bridge consisting of two equal resistors R_1. Within the transducer, resistors R_a and R_b

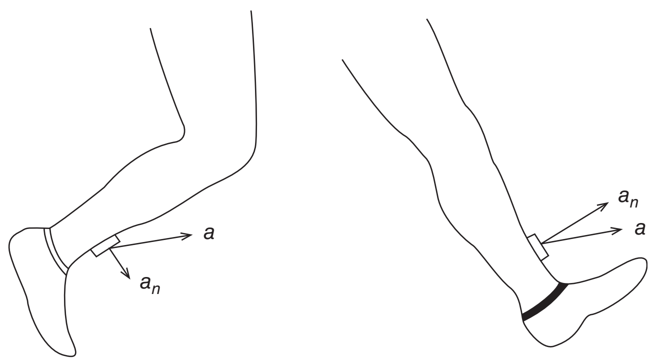

Figure 3.4 Two movement situations where the acceleration in space is identical but the normal components are quite different.

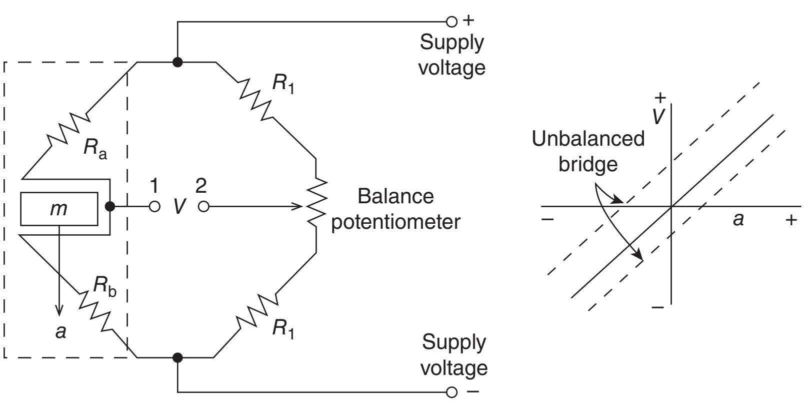

Figure 3.5 Electrical bridge circuit used in most force transducers and accelerometers. See text for detailed operation.

change their resistances proportionally to the acceleration acting against them. With no acceleration, $R_a = R_b = R_1$, and with the balance potentiometer properly adjusted, the voltage at terminal 1 is the same as that at terminal 2. Thus, the output voltage is $V = 0$. With the acceleration in the direction shown, R_b increases and R_a decreases; thus, the voltage at terminal 1 increases. The resultant imbalance in the bridge circuit results in voltage V, proportional to the acceleration. Conversely, if the acceleration is upward, R_b decreases and R_a increases; the bridge unbalances in the reverse direction, giving a signal of the opposite polarity. Thus, over the dynamic range of the accelerometer, the signal is proportional to both the magnitude and the direction of acceleration acting along the axis of the accelerometer. However, if the balance potentiometer is not properly set, we have an unbalanced bridge and we could get a voltage-acceleration relationship like that indicated by the dashed lines.

Advantages

1. Output signal is available immediately for recording or conversion into a computer.

Disadvantages

1. Acceleration is relative to its position on the limb segment.
2. Attaching several to the body may restrict normal movement.
3. The mass of the accelerometer may result in a movement artifact, especially in rapid movements or movements involving impacts (Morris and Accelerometry-A, 1973).

3.2.3 Inertial Sensors

Inertial measurement units (IMUs) are electrical devices that commonly combine the use of 3D accelerometers and 3D gyroscopes to measure segment orientations. Gyroscopes are sensors that measure orientation and angular velocity relative to the earth's gravitational pull. IMUs sometimes are built with magnetometers to measure orientation in the transverse plane but are often subjected to ferromagnetic disturbances, which result in larger errors. Absolute orientation can be determined with the use of one sensor if global segment kinematics are of interest. However, if joint kinematics are desired during everyday life (in which a motion capture system would not be appropriate) then single 3D IMUs can be attached to each body segment of interest. Since gyroscopes suffer from drift over time, corrections often need to be performed. There are several methods to correct drift: (1) a reset at a known static kinematic state (using accelerometer data), (2) with the use of fusion algorithms that integrate the gyroscope and accelerometer data and sometimes magnetometers (the same ferromagnetic disturbances can occur) to make drift corrections, or (3) the use of a high pass filter.

There are several methodological approaches to measuring 3D joint kinematics from IMUs (McGrath et al., 2018): (1) Precise manual alignment technique: IMUs are manually positioned on the segment to align one of the IMU axes of rotations to the axis of rotation of the joint. A static calibration trial is then recorded and used in a weighted least squares optimization routine to determine the IMU to segment relationship. This technique is best used in joints that function as hinge joints and have large ranges of motion like knee flexion and extension. (2) Functional alignment techniques: these techniques use functional calibration trials that involve prescribed movements to determine the IMU to segment relationship. This technique can be limited if the subject is not able to perform the functional calibration movements. This is likely to occur when performing motion analysis on injured patients or those with disabilities. (3) Model-based techniques: these techniques decompose the relative angular velocity between two IMU sensors into their separate axis and position components with a formulated optimization routine. These approaches avoid having precise alignment of sensors on segments. Principle component analysis (PCA) has been used successfully to decompose the sensors to estimate the joint axis and allow for the creation of two segment coordinate systems. Euler angle rotation is then applied to the two segment coordinate systems to calculate anatomic joint angles.

Advantages

1. Continuous collection of kinematic joint data during natural everyday life and in locations where motion capture systems are accessible.

Disadvantages

1. Positioning and alignment of sensors.
2. Need to correct for drift.
3. Ferromagnetic disturbances of magnetometers.
4. Measurement errors with all techniques to translate IMU coordinate systems to segment coordinate systems.

3.2.4 Special Joint Angle Measuring Systems

In the area of ergonomics, special glove systems have been developed to measure the kinematics of the fingers and the thumb. Figure 3.6 shows the construction of a glove transducer, which comprises a lightweight elastic glove with sensors on the proximal two joints of each finger and thumb plus a thumb abductor sensor. Another hand-specific design is constructed of 11 inertial sensors positioned along each of the hand and finger segments and secured with

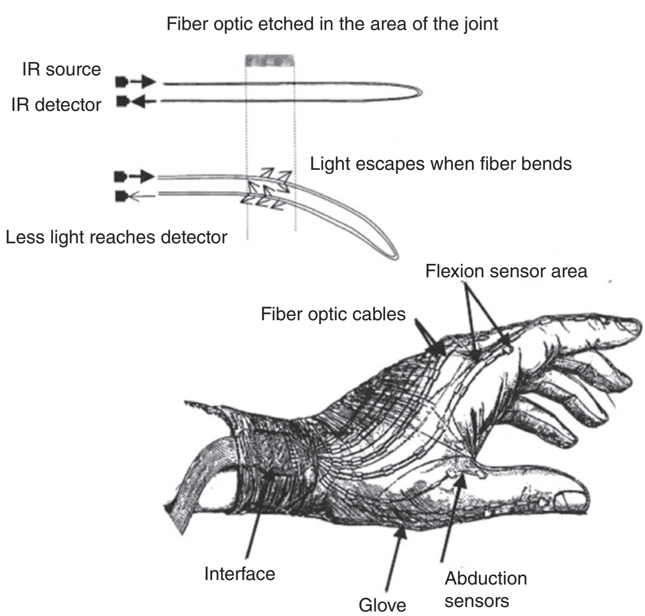

Figure 3.6 Construction and operation of a glove transducer to measure angular displacements of the fingers. Transducer is a loop of fiber-optic cable; the amount of light returning to the detector decreases with increased finger flexion. Each cable is calibrated for angular displacement versus detected light intensity. (Courtesy of the Ergonomics Laboratory, Department of Kinesiology, University of Waterloo, Waterloo, Ont. Canada.)

Velcro (van den Noort et al., 2016). Each sensor contains a 3D gyroscope and 3D accelerometer. Anatomic calibration steps are required to determine sensor to segment coordinate systems, which include both static and dynamic trials for both the fingers and thumb separately. A major use for such a system has been in the study of repetitive strain injuries (Moore et al., 1991).

3.2.5 Electromagnetic Systems

Electromagnetic (EM) tracking systems utilize magnetic fields of known locations and magnitudes with sensors on segments of the human body to compute 6 degrees of freedom position and orientation throughout the EM field. Systems are comprised of EM sensors (i.e. receivers) and EM field generators (i.e. transmitters), not requiring direct line-of-sight between transmitters and receivers (Figure 3.7). Most EM systems utilize digitization of anatomical landmarks that are defined relative to EM sensors' local coordinate systems to build anatomical frames of body segments (Urbanczyk et al., 2021). In practice, EM systems can be limited by tethered cables and allow for a few sensors to be tracked resulting in data collection rates ranging from 40 to 250 Hz. Older systems reduced their data collection rates when more sensors were added, but most newer systems are able to read sensor data in parallel, thus not affecting the data collection rate of a single sensor. Different systems utilize varying technology for field generators which can produce either changing (AC) or quasi-static (pulsed DC) EM fields. The sensors measure magnetic flux, defined as a component of the magnetic field passing through a specific point in space. Notably, a sensor only measures the gradient of the EM field that represents differences in the EM field intensity between different positions. Therefore, disturbances in the EM field due to ferromagnetic objects and other electronic devices within the proximity of the EM tracking system can cause functional limitations of using this technology. On average EM tracking can achieve accuracy of 1.0 mm in good environments, but distortion in the EM field can dramatically drop accuracy (Franz et al., 2014).

Figure 3.7 Electromagnetic system setup on a participant for upper extremity testing. One sensor is applied to each segment to track 3D position data. In this participant, a sensor was applied to the spine, scapula, and the humerus (not pictured), which will allow 3D kinematics for the scapulothoracic articulation and the glenohumeral joint.

Advantages

1. Localized small sensors without the requirement for line-of-sight.

Disadvantages

1. Magnetic field distortion due to ferromagnetic objects and other electronic devices causing interference can compromise tracking accuracy.
2. Small tracking volume and limited range relative to EM field generator.
3. Possible restricted movement from tethered cable connections for participants performing complex motions.
4. Limited number of sensors can reduce total number of tracked body segments.

3.3 IMAGING MEASUREMENT TECHNIQUES

"A picture is worth a thousand words" holds an important message for any human observer, including the biomechanics researcher interested in human movement. Because of the complexity of most movements, the only system that can possibly capture all the data is an imaging system. Given the additional task of describing a dynamic activity, we are further challenged by having to capture data over an extended period of time. This necessitates taking many images at regular intervals during the event.

There are many types of imaging systems that could be used. The discussion will describe different types: camera, optical motion capture, optoelectric, biplane fluoroscopy, and markerless systems. Whichever system is chosen, a lens is involved; therefore, a short review of basic optics is given here.

3.3.1 Review of Basic Lens Optics

A simple converging lens is one that creates an inverted image in focus at a distance v from the lens. As seen in Figure 3.8, if the lens – object distance is u, then the focal length f of the lens is:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \tag{3.1}$$

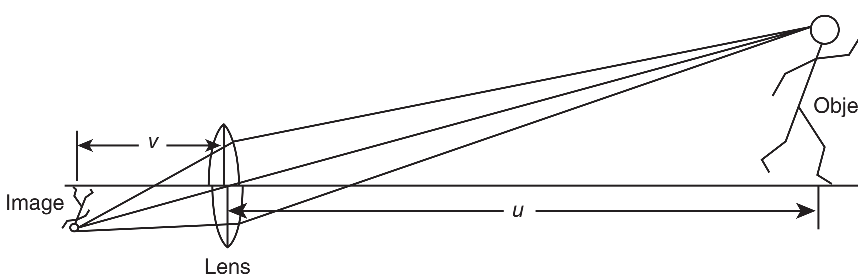

Figure 3.8 Simple focusing lens system showing relationship between the object and image.

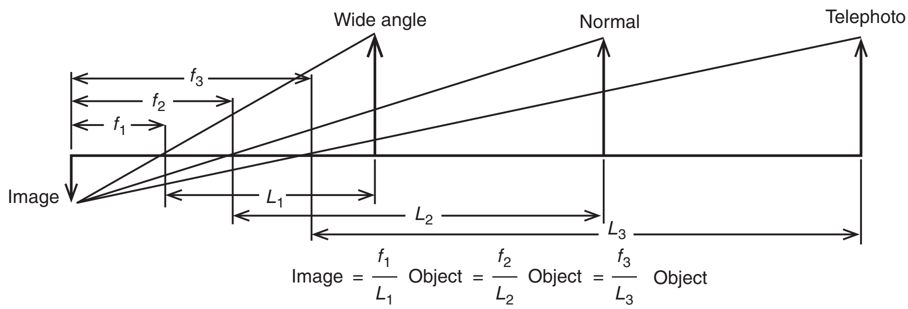

Figure 3.9 Differences in the focal length of wide angle, normal, and telephoto lenses result in an image of the same size.

The imaging systems used for movement studies are such that the object – lens distance is quite large compared with the lens – image distance. Therefore,

$$\frac{1}{u} \approx 0, \quad \frac{1}{f} = \frac{1}{v}, \qquad \text{or} \quad f = v \tag{3.2}$$

Thus, if we know the focal length of the lens system, we can see that the image size is related to the object size by a simple triangulation. A typical focal length is 25 mm, a wide-angle lens is 13 mm, and a telephoto lens is 150 mm. A zoom lens is just one in which the focal length is infinitely variable over a given range. Thus, as L increases, the focal length must increase proportionately to produce the same image size. Figure 3.9 illustrates this principle. For maximum accuracy, it is highly desirable that the image be as large as possible. Thus, it is advantageous to have a zoom lens rather than a series of fixed lenses; individual adjustments can be readily made for each movement to be studied, or even during the course of the event.

3.3.2 *f*-Stop Setting and Field of Focus

The amount of light entering the lens is controlled by the lens opening, which is measured by its *f*-stop (*f* means fraction of lens aperture opening). The larger the opening, the lower the *f*-stop setting. Each *f*-stop setting corresponds to a proportional change in the amount of light allowed in. A lens may have the following settings: 22, 16, 11, 8, 5.6, 4, 2.8, and 2. *f*/22 is 1/22 of the lens diameter, and *f*/11 is 1/11 of the lens diameter. Thus *f*/11 lets in four times the light that *f*/22 does. The fractions are arranged so that each one lets in twice the light of the adjacent higher setting (e.g. *f*/2.8 provides twice the light of *f*/4).

To keep the lighting requirements to a minimum, it is obvious that the lens should be opened as wide as possible with a low *f* setting. However, problems occur with the field of focus. This is defined as the maximum and minimum range of the object that will produce a focused image. The lower the *f* setting, the narrower the range over which an object will be in focus. For example, if we wish to photograph a movement that is to move over a range from 10 to 30 ft, we cannot reduce the *f*-stop below 5.6. The range set on the lens would be about 15 ft, and everything between 10 and 30 ft would remain in focus. The final decision regarding *f*-stop depends on the shutter speed of the movie camera and the film speed.

3.3.3 Television Imaging Camera Historical Development

Historically, the major difference between television and cinematography was the fact that television had a fixed frame rate (60 Hz in North America and 50 Hz in Europe). However, as we have emerged into a modern digital era, television and video frame rates have increased dramatically to the point where our human eye cannot detect discontinuous data for human movements. We are now more limited by resolution, pixels, and capture frame rates. In yet, appreciating the progression and technological advancements are important to contextually evaluate modern-day equipment options for optically capturing human kinematics.

Almost all movement analysis television systems (i.e. motion capture) were developed in university research laboratories (Eberhart and Inman, 1951). In the late 1960s, the first reports of television-based systems started to appear: at Delft University of Technology in The Netherlands (Furnée, 1967, according to Woltring, 1987) and at the Twenty-First Conference engineering in medicine and biology (EMB) in Houston, Texas (Winter et al., 1968). The first published paper on an operational system was by Dinn et al. (1970) from the Technical University of Nova Scotia in Halifax, Nova Scotia, Canada. It was called Computer INterface for TELivision (CINTEL) and was developed for digitizing angiographic images at four bits (16 grey levels) to determine the time course of left ventricular volume (Trenholm et al., 1972). It was also used for gait studies at the University of Manitoba in Winnipeg, Manitoba, Canada, where, with higher spatial resolution and a one-bit (black/white) conversion, the circular image of a reflective hemispheric ping pong ball attached on anatomical landmarks was digitized (Winter et al., 1972). With about 10 pixels within each marker image, it was possible by averaging their coordinates to improve the spatial precision of each marker from 1 cm (distance between scan lines of each field) to about 1 mm. The 3 M Scotch® material that was used as reflective material has been used by most subsequent experimental and commercial systems.

Jarett et al. (1976) reported a system that detected the left edge of the image of a small reflective marker that occupied one or two scan lines. Unfortunately, the spatial precision was equal to the scan line distance, which is about 1 cm. This system was adopted and improved by VIdeo CONvertor (VICON) in their commercial system. Both left and right edges of the marker image were detected, and subsequently, the detected points were curve fitted via software advanced material analysis and simulation software (AMASS) to a circle (Macleod et al., 1990). Based on the circle fit, the centroid was calculated. Other commercial systems, such as the one developed by the Motion Analysis Corporation, use patented edge-detection techniques (Expert Vision). Shape recognition of the entire marker image, rather than edge detection, was used by the Elaboratore di Immagini Televisive (ELITE) system developed in Milan, Italy. A dedicated computer algorithm operating in real-time used a cross-correlation pattern recognition technique based on size and shape (Ferrigno and Pedotti, 1985). These systems used all grey levels in the shape detection, thus at the time improving the spatial resolution to 1/2800 of the viewing field. Considering the field height to be about 2.5 m, this represented a precision of about 0.9 mm.

These systems have continued to advance, now with many major competitor manufacturers, resulting in even higher accuracy and precision systems being available. The next section will describe current standards for optical motion capture technology.

3.3.4 Optical Motion Capture

Modern-day optical motion capture systems (i.e. passive motion capture) are quite common across the university, clinical, and industrial settings for measuring human movement by placing reflective spherical markers on limb segments across the body to approximate the skeletal movement during activities. The applications are widely varied from using motion capture to reproduce human movement for making animations and video games, to detecting abnormal gait when planning surgical interventions to improve functional deficits (Figure 3.10). Independent of application, the technology is the same.

Motion capture systems consist of motion capture cameras (Figure 3.11) placed around a room to establish a calibrated capture volume. From the ring around the camera's lens, the light-emitting diodes (LEDs) typically emit infrared light that is

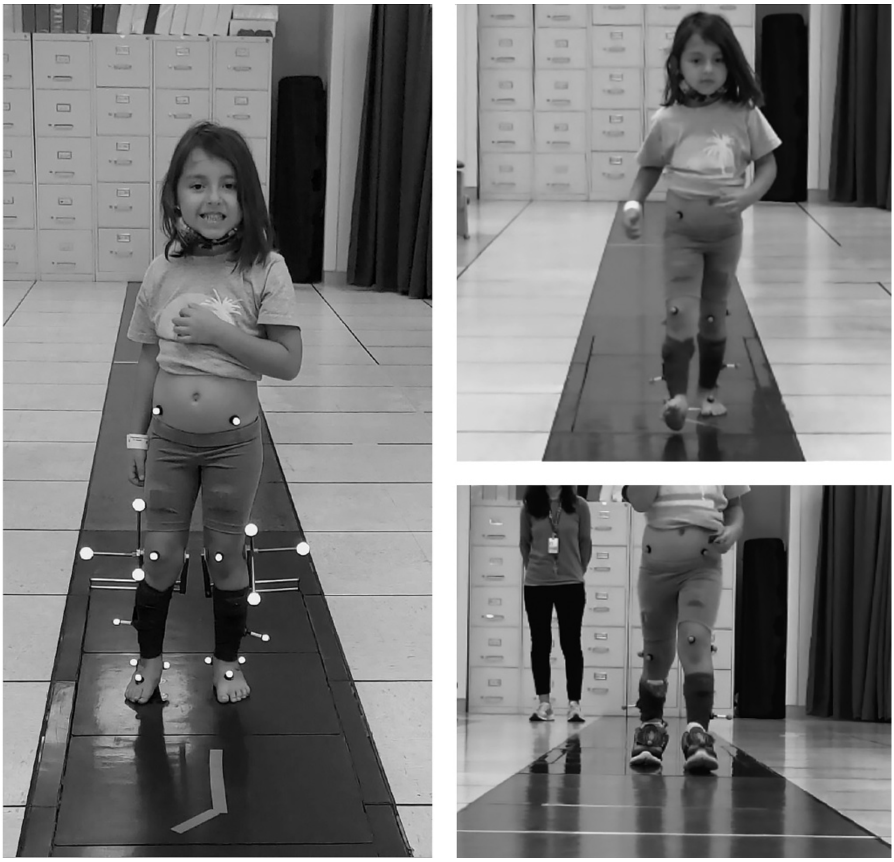

Figure 3.10 Gait assessment of a pediatric patient with a history of club foot in a clinical gait laboratory. Infrared cameras on the ceiling and walls capture the reflected light from the spherical reflective markers mounted on both sides of the body. Studies are performed barefoot and in shod conditions. (Courtesy of the Motion Analysis Center, Shriners Children's Chicago, Chicago, IL.)

Figure 3.11 Typical Vicon infrared motion capture camera rigidly mounted in a motion analysis laboratory. The infrared lights form a ring shape around the lens and can capture data up to 2000 Hz. (Courtesy of the Motion Capture Core Facility, University of Utah, Salt Lake City, UT.)

worn on the individual performing movements in the capture volume. Using wavelengths in the infrared spectrum has benefits such as reducing the influence of ambient light and using pulsed waves to reduce motion-blur artifact. Each camera records a two-dimensional (2D) series of images, capturing each reflective marker location over time. Each camera in the system is assigned a local 2D plane that is used, based on the calibrated relative position and orientation with respect to other cameras, to calculate a 3D location of each marker within the global laboratory reference frame. In a well-calibrated motion capture setup, the individual marker errors can be less than a couple of millimeters. The accuracy also increases by having additional cameras in the system to provide redundant marker tracking and minimize marker dropout due to occlusions of markers throughout more complicated activities, or while tracking multiple people in the capture volume.

There are many different types of cameras available for motion capture, and the specific needs will depend on the experiments being performed within the laboratory. Similar to traditional optical video cameras, those used in motion analysis have varying available frame rates, fields of view, and resolutions. There is often an inverse relationship between the frame rate and the field of view and resolution of the cameras, such that as the frame rate increases to capture very fast movements, the resolution and field of view will be lowered. Rapid movements, therefore, may require additional cameras placed closer to the object of interest.

Most cameras allow orientation in a landscape or portrait direction. To ensure the greatest overlap of the capture volume between cameras, it is best to leave them oriented in the landscape direction. If it is necessary to capture portion of the volume in a higher vertical direction, it would be more beneficial to move the cameras away from the object and place at a higher elevation, rather than change the orientation of one or several of the cameras. Placing the cameras at an elevated position with a downward orientation is beneficial for multiple reasons; it reduces the loss of field of view into the floor and reduces the likelihood of masking a camera view area due to the visualization of another camera across the capture volume. Elevation of the cameras can be done by three primary means: the use of tripods, mounting to a speed rail on the wall, or mounting to a truss system that is separate from the physical wall of the laboratory. Speed rail and truss mountings preclude routine movement of cameras between experiments, but these systems prevent accidental touching or movement of the cameras during data collections.

In most camera-based motion capture systems, the markers are identified through either the reflection of infrared light back to the source camera, or from emission of infrared light from the markers itself, as is the case in an active marker system. In either system, the position of the marker is identified by the light emitted or reflected from it. It is important that the cameras recognize only the markers as light sources and ignore any other reflection in the laboratory that is coming from a source other than the markers. If extraneous reflections are seen during calibration, it can increase the error in the calibration process. If reflections are identified during data collection trials, the system may identify these reflections as markers, and require additional postprocessing to remove unwanted "ghost" markers in the trials. Ideally, the source of these reflections should be identified and corrected prior to the start of a data collection. Reflections can occur for various reasons, most commonly from excess ambient light reflected off shiny objects in the lab or floor, or from reflective clothing material that is common in athletic shoes and apparel. If the reflections cannot be removed, the cameras must be "masked" and a region that contains the reflection will be ignored by each camera. While this may reduce the presence of ghost markers, it also reduces the visible area of the camera, and true markers that occupy the masked camera space will also not be detected.

Over the years, the diameter of spherical markers that these systems could accurately track has been dramatically reduced with systems now being able to track facial movements with 3 mm hemispherical markers. More commonly, 9–14 mm spherical markers are used for gait analysis and full-body evaluation of movements (Figure 3.10). A minimum of three markers needs to be placed on each segment to create a local coordinate system to later calculate joint angles (see Sections 3.6.1 and 3.6.2). More than three markers are preferred to provide redundant tracking of a single segment and ensure more markers are visible for postprocessing the motion data. Most software packages include algorithms to fill gaps in the data based on surrounding marker trajectories and other mathematical approximations. Also, it is important that all the markers on a single segment are not colinear because this can lead to errors in automatic marker tracking, as well as the lack of distinct anatomical directions to create a meaningful local coordinate system for that limb's segment.

Many marker-set definitions currently exist in common practice. Before using any marker-set, new users should be trained on proper placement of markers and a reliability test should be conducted to ensure errors are minimized that could be a result of multiple people placing markers in different manners on study participants. Also, there are some limitations to marker-set definitions because they are assuming a segment is a rigid body, not accounting for underlying complexity (i.e. foot mechanics) or skin, muscle, and fat motion artifact. Therefore, researching a marker-set before using it to fully understand the strengths and weaknesses is critical for a meaningful experimental study design. Marker set design is described in Chapter 7.

Advantages

1. All data are presented in an absolute spatial reference system, in a plane normal to the optical axis of the camera.
2. Systems are not limited to the number of markers used.
3. Multiple individuals can be tracked at the same time.
4. Comparisons can be made between conditions such as barefoot versus shod activities.
5. Marker placement is noninvasive.

Disadvantages

1. Inconsistent marker placement can cause errors in motion tracking results.
2. Motion artifact from skin, muscle, and fat between the surface of the skin and underlying anatomical landmarks causes errors when approximating skeletal movement, particularly in patients with a higher body mass index (BMI).
3. Depending on the marker-set used, placement of markers and postprocessing time can be intensive.
4. Some motion capture systems cannot be used outside in daylight due to infrared light interferences.

3.3.5 Optoelectric Techniques

In contrast to passive motion capture cameras, optoelectric techniques (i.e. active motion capture systems) require the participant to wear tiny infrared lights on each desired anatomical landmark or segment. Additional digitized points with an infrared light-equipped wand can be used to mark anatomical landmarks relative to active markers placed on body segments. The first commercial system was developed by Northern Digital in Waterloo, Ontario, Canada. Active motion capture systems have advanced over the years but are largely the same construct. A system includes a single fixed unit where three cameras are rigidly mounted, as shown in Figure 3.12. The left and right cameras are mounted to face slightly inward to establish a capture volume where the infrared lights emitted from active markers can be seen by all three cameras. Each camera has planar definitions, where the intersection of those three camera planes can identify a unique 3D point in space. Thus, each active marker (i.e. infrared diode) pulses and the x, y, z coordinates in the global reference frame are recorded. The 3D accuracy for the OPTOTRAK camera

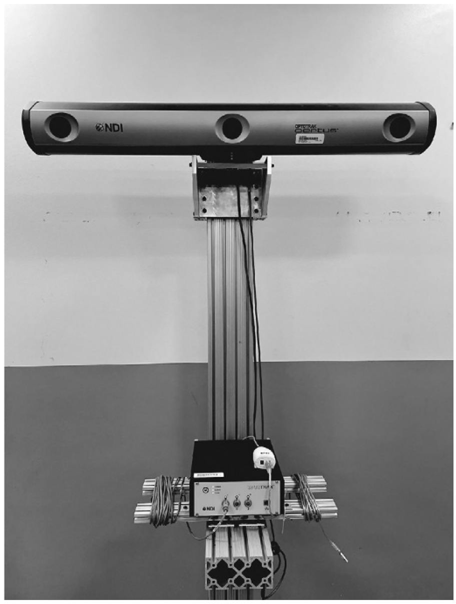

Figure 3.12 An OPTOTRAK system with three lenses. The two outside lenses face slightly inward and define different planes while the middle lens is in more of a horizontal plane. Marker frequency can be pulsed up to 4600 Hz. (Courtesy of the Orthopaedic Research Laboratory, University of Utah, Salt Lake City, UT.)

mounted as shown in Figure 3.12 is 0.1 mm with a 0.01 mm resolution. These mobile systems can also be preferable when conducting research with cadavers where active markers can be placed on a bone pin-based structure to capture individual bone motion with higher precision than is afforded by passive motion capture systems.

Advantages

1. Active markers can pulse at higher frequencies than passive motion capture systems.
2. A single unit is easily mobile for closer more detailed movement assessment.

Disadvantages

1. Encumbrance and time to fit wired light sources (e.g. infrared diodes) can be prohibitive in certain movements.
2. The number of active marker sources is limited.
3. Rotational movements can cause data drop out due to the direct line of sight being interrupted between the active markers and the camera mount.

3.3.6 Biplane Fluoroscopy

Biplane fluoroscopy has emerged as a relatively new and viable tool to capture in vivo bone motion during dynamic activities. The technology consists of a modified system of two fluoroscopes that are detached, allowing for mobility of each X-ray emitter and image intensifier to be set up in a variety of positions depending on the region of the body that is being evaluated (Figure 3.13). Due to the newer nature of this technology, consistent terminology across the field is lacking and this imaging technique goes by many names (i.e. dual fluoroscopy, biplane/biplanar videoradiography, and dual fluoroscopic imaging system).

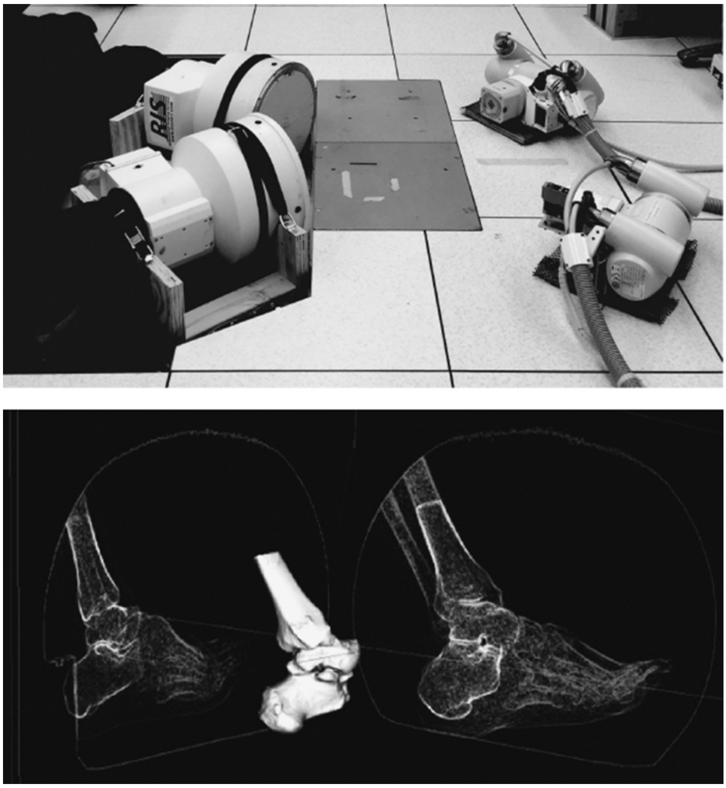

Figure 3.13 On the top, a biplane fluoroscopy setup is showing two X-ray emitters (right) and two image intensifiers (left) positioned at floor level surrounding two force platforms to simultaneously collect ground reaction forces with in vivo ankle kinematics. On the bottom, DSX software shows tracked solutions for the tibia, talus, and calcaneus in 3D with the two biplane fluoroscopy images calibrated with respect to each other, a 3D model of the bones generated from computed tomography, and digitally reconstructed radiographs aligned on the fluoroscopic images. From this in vivo kinematics of tibiotalar and subtalar joint motion can be calculated. (Courtesy of the Orthopaedic Research Laboratory, University of Utah, Salt Lake City, UT.)

With a system positioned, the image intensifiers are approximately 90° with respect to one another but can also be at a wider obtuse angle depending on the desired field of view. To calibrate the system and establish a known location of the two image intensifiers, a calibration cube with tantalum beads is placed in the center of the capture volume. The known locations of all beads are used to establish a system x, y, z position and orientation for a global reference frame. Two additional calibration trials are acquired: a grid is placed on the face of the image intensifiers to account for distortion around the perimeter of the field of view, and a white trial establishes contrast balance when nothing is in the field of view.

Participant data collection begins with a static trial which establishes the pose of bones relative to one another. Then dynamic trials occur either by positioning the participant within the capture volume to perform an activity or proceeding through the capture volume to evaluate various activities that are more dynamic (Figure 3.14). Data frequencies are typically 100–200 Hz for pulsed systems, but continuous X-ray can be used at higher capture frequencies. But an important consideration is the radiation exposure a participant will experience during a study. It is critical to work with a medical physicist to optimize energy settings (kVp and mA) for imaging to reduce potential radiation. Therefore, extensive institutional review board protocols need to be approved

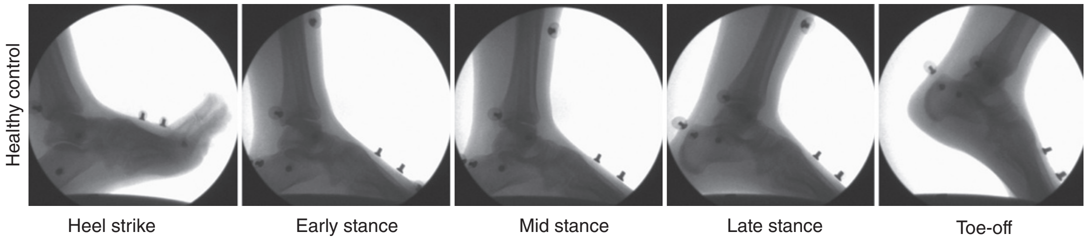

Figure 3.14 Raw fluoroscopy images are essentially X-ray video files which can be evaluated at quasi-static points throughout the gait cycle or evaluated as time domain series data. Here single frames are visualized from a healthy individual performing a barefoot overground walking trial. (Courtesy of the Orthopaedic Research Laboratory, University of Utah, Salt Lake City, UT.)

with calculated dosimetry reports for the body-specific region being studied. The room also needs to be shielded to contain radiation scatter during experiments. An environmental health and safety department will typically monitor radiation in the room as well as for staff conducting the studies regularly.

In order to evaluate the in vivo kinematics, a volumetric imaging dataset (i.e. computed tomography or magnetic resonance imaging) also needs to be acquired for each participant. Participant-specific bone models are segmented on the scan to generate a 3D bone model. Model-based tracking is a method of creating a digitally reconstructed radiograph (DRR) of the 3D bone model and that DRR is projected into both calibrated views of the biplane fluoroscopy data (Figure 3.13). The 3D position is semi-automatically manipulated until the DRR aligns over the bone in both fluoroscopic images for every frame of the collected activity. Because this technique eliminates all error due to soft tissue artifact, typical validated measurements have reported errors of <1 mm and <1° (Kapron et al., 2014; Wang et al., 2015; Miranda et al., 2011).

While biplane fluoroscopy has many advantages, it is also limited by a small capture volume. To compliment the localized data collection of in vivo bone motion, systems can be synced with a passive motion capture system to also collect full-body kinematics that are outside of the fluoroscopic field of view.

Advantages

1. Eliminates skin, muscle, and fat motion artifact errors by measuring underlying bone movement directly.
2. Enables measurement of small articular joints not possible with skin-based motion capture (i.e. foot mechanics).
3. Sub-millimeter and sub-degree accuracy.
4. Can be synced with passive motion capture systems to collect additional data outside of the fluoroscopy field of view.

Disadvantages

1. Exposure to ionizing radiation due to X-ray technology and varies depending on the region of the body being imaged.
2. Limited field of view.
3. Expensive system requiring a room with radiation shielding and protection.
4. Potential for operator-tracking errors.

3.3.7 Markerless Systems

Markerless motion capture using deep learning and neural network approaches has the potential to revolutionize the field of biomechanics. Multiple cameras are setup in any location and calibrated with a grid system seen by multiple cameras. Participants are encouraged to wear tight-fitting clothes to minimize errors due to lack of movement visualization. The software leverages computer vision techniques to address many limitations commonly seen in marker-based motion capture systems (Section 3.3.4). Instead of tracking external markers affixed to participants, it can track any feature of interest, approximating skeletal movement of full-body motion and multiple individuals at once. Comparisons have been made to marker-based motion capture demonstrating joint center root mean squared offsets to be around 2.5 cm for all joints except for the hip which demonstrated higher errors (Kanko et al., 2021a). Additionally, inter-session repeatability for markerless systems has been shown to be within 3° (Kanko et al., 2021b). While markerless tracking systems are still not as accurate as marker-based systems, they provide added ease and ability to measure human movement in a nonrestricted setting.

Advantages

1. No markers are attached to the participant being evaluated.
2. Calibration is quick by using a grid pattern that is seen by multiple cameras.
3. Motion capture can be easily performed outside and in non-laboratory settings, bringing motion capture to the activity of interest, hopefully maintaining the natural execution of human movements being studied.
4. Data can be imported to Visual3D for postprocessing analyses.

Disadvantages

1. Rotational kinematics have higher errors than typical passive motion capture systems.
2. Resolution of the camera will greatly impact the tracking accuracy.
3. Atypical poses may not be recognized by the deep learning approach, therefore requiring additional training of the tracking algorithm.
4. Participants need to wear tight fit clothing for optimal tracking, baggy clothing will result in higher errors.

3.3.8 Summary of Various Kinematic Systems

Since the early years of motion analysis performed by Muybridge, technological advancements have diversified the types of systems that exist to measure human kinematics. Each laboratory must define its special requirements before choosing a particular system to weigh the advantages and disadvantages afforded by each approach. Even a combination of multiple systems may achieve the best experimental goals. A clinical gait lab may settle on passive motion capture because of the encumbrance of optoelectric and EM systems with wires and the reduced precision of markerless systems to evaluate rotational abnormalities for treatment planning. Ergonomic and athletic environments may require instant or near-instant feedback to the participant or athlete, thus dictating the need for an automated system that can be deployed on the field more easily by using IMUs or a markerless system. Basic researchers evaluating targeted joint level kinematics with direct high precision and accuracy measurement of bone motion may require a biplane fluoroscopy system. And, finally, the cost of hardware and software may be the single limiting factor that may force a compromise as to the final decision.

3.4 CLINICAL MEASURES OF KINEMATICS

The ability to ambulate and perform activities of daily living is essential for maintaining independence and a high quality of life. When orthopedic or neurological disorders impair physical function, patients will often seek clinical treatment where a physician, physical therapist, occupational therapist, or athletic trainer may evaluate their range of motion, walking, and mobility. In the clinical setting, motion analysis systems may not be easily readable as described in Section 3.3, and the ability to monitor the patient at home for further evaluation would be highly beneficial. Therefore, the emergence of simple 2D kinematic applications for smartphones and sensors that can be sent home with the patient enables additional data and evaluation of function and mobility before more aggressive treatment options are pursued. However, these types of approaches can generate massive quantities of data that are prone to noise and may oversimplify the complexity of a patient's disorder.

3.4.1 2-D Kinematic Apps/Sensors

With the advancements of smartphones and smart devices, data can be collected and evaluated from our activities of daily living with greater ease than coming to a motion analysis laboratory. However, these approaches are typically limited to 2D estimations of movement. While not as precise and accurate as systems in Section 3.3, they can provide insightful data in real-time. Video annotation tools (i.e. Kinovea, Dartfish™, and Mustard) can provide slow-motion video and overlay of segment long axes to quickly evaluate misalignment that can be corrected with modified motion routines. Additionally, small sensors (i.e. Garmin™ and Blast Motion™) that are synced to a smartphone can provide cadence, number of steps, speed, and acceleration to monitor when someone has possibly fallen or has a sporting technique that can cause problems for overuse injuries. With any of these apps and sensor systems, care should be given to ensure repeatability and reliability of measurement by also providing patients instructions if being sent home to collect more data during daily life.

3.4.2 Sensor-Based Systems

Particularly popular in athletics, sensor-based systems can be worn during practice or gameplay to monitor athletes' performance and evaluate injury risks. For example, a baseball player may wear a compression sleeve with a lightweight sensor (i.e. PULSEthrow) to estimate the torque placed on the ulnar collateral ligament during a bullpen throwing session, long toss program, or as a player is recovering from an injury. Football players may have a smart helmet (i.e. NoMoDiagnostics) that can use head acceleration data to spot a concussion in real-time. Simple sensor applications like this can soon become more widespread in athletics to better understand injury mechanisms and detect problems in real-time.

3.5 PROCESSING OF RAW KINEMATIC DATA

3.5.1 Nature of Unprocessed Image Data

Motion capture technologies are sampling processes. They capture the movement event for a short period of time, after which no further changes are recorded until the next field or frame. Playing a video clip back slowly demonstrates this phenomenon: the image jumps from one position to the next in a distinct step rather than a continuous process. The only reason that video does not appear to jump at normal projection speeds (60 or 120 frames/second for video) is because the eye can retain an image for a period of about 1/15 seconds. The eye's short-term "memory" enables the human observer to average or smooth out the

The converted coordinate data from motion capture recordings are called *raw* data. This means that they contain additive noise from many sources: electronic noise in nonoptical systems and spatial precision from optical systems. All of these will result in random errors in the converted data. It is, therefore, essential that the raw data be smoothed, and in order to understand the techniques used to smooth the data, an appreciation of harmonic (or frequency) analysis is necessary. The theory of harmonic analyses has been covered in Section 2.1; however, there are some additional special problems with the processing of kinematic data that are now discussed.

3.5.2 Signal Versus Noise in Kinematic Data

In the study of movement, the signal may be an anatomical coordinate that changes with time. For example, in running, the Z (vertical) coordinate of the heel will have certain frequencies that will be higher than those associated with the vertical coordinate of the knee or trunk. Similarly, the frequency content of all trajectories will decrease in walking compared with running. In repetitive movements, the frequencies present will be multiples (harmonics) of the fundamental frequency (stride frequency). When walking at 120 steps per minute (2 Hz), the stride frequency is 1 Hz. Therefore, we can expect to find harmonics at 2, 3, 4 Hz, and so on. Normal walking has been analyzed by digital computers, and the harmonic content of the trajectories of seven leg and foot markers was determined (Winter et al., 1974). The highest harmonics were found to be in the toe and heel trajectories, and it was found that 99.7% of the signal power was contained in the lower seven harmonics (below 6 Hz). The harmonic analysis for the toe marker for 20 subjects is shown in Figure 3.15, which is the same as Figure 2.17 and is repeated to show the noise content. Above the seventh harmonic, there was still some signal power, but it had the characteristics of “noise.” Noise is the term used to describe components of the final signal that are not due to the process itself (in this case, walking). Sources of noise were noted in Section 3.4.1, and if the total effect of all these errors is random, then the true signal will have an added random component. Usually, the random component is high frequency, as is borne out in Figure 3.15. Here you can see evidence of higher-frequency components extending up to the twentieth harmonic, which was the highest frequency analyzed.

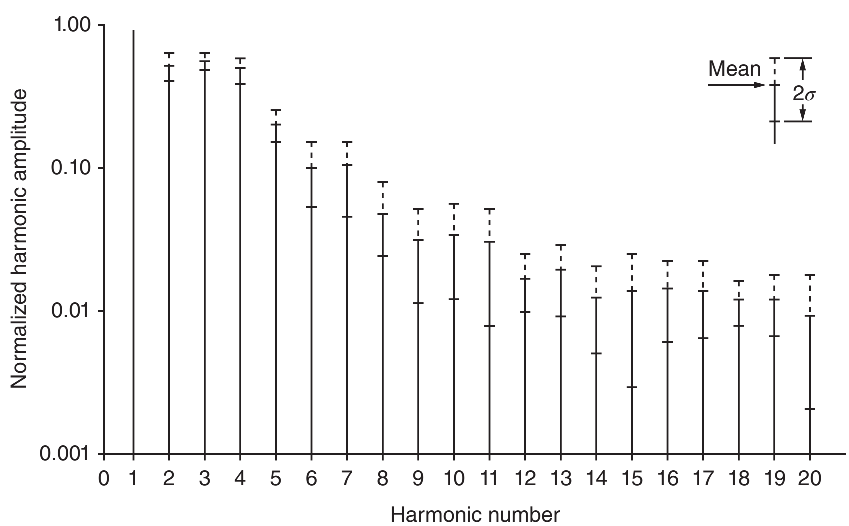

Figure 3.15 Harmonic content of the vertical displacement of a toe marker from 20 subjects during normal walking. Fundamental frequency (harmonic number = 1) is normalized at 1.00. Over 99% of the power is contained below the seventh harmonic. (Reproduced by permission

3.5.3 Problems of Calculating Velocities and Accelerations

The presence of this higher-frequency noise is of considerable importance when we consider the problem of trying to calculate velocities and accelerations. Consider the process of time differentiation of a signal containing additive higher-frequency noise. Suppose that the signal can be represented by a summation of N harmonics:

$$x = \sum_{n=1}^{N} X_n \sin\left(n\omega_0 t + \theta_n\right) \tag{3.3}$$

where

ω_0 = fundamental frequency
n = harmonic number
X_n = amplitude of nth harmonic
θ_n = phase of nth harmonic.

To get the velocity in the x-direction V_x, we differentiate with respect to time:

$$V_x = \frac{dx}{dt} = \sum_{n=1}^{N} n\omega_0 X_n \cos\left(n\omega_0 t + \theta_n\right) \tag{3.4}$$

Similarly, the acceleration A_x is:

$$A_x \frac{dV_x}{dt} = -\sum_{n=1}^{N} \left(n\omega_0\right)^2 X_n \sin\left(n\omega_0 t + \theta_n\right) \tag{3.5}$$

Thus, the amplitude of each of the harmonics increases with its harmonic number; for velocities, they increase linearly, and for accelerations, the increase is proportional to the square of the harmonic number. This phenomenon is demonstrated in Figure 3.15, where the fundamental, second, and third harmonics are shown, along with their first and second-time derivatives. Assuming that the amplitude x of all three components is the same, we can see that the first derivative (velocity) of harmonics increases linearly with increasing frequency. The first derivative of the third harmonic is now three times that of the fundamental. For the second time derivative, the increase repeats itself, and the third harmonic acceleration is now nine times that of the fundamental (Figure 3.16).

In the trajectory data for gait, $x1$ might be 5 cm and $x20$ = 0.5 mm. The twentieth harmonic noise is hardly perceptible in the displacement plot. In the velocity calculation, the twentieth harmonic increases 20-fold so that it is now one-fifth that of the fundamental. In the acceleration calculation, the twentieth harmonic increases by another factor of 20 and now is four times the magnitude of the fundamental. This effect is shown if you look ahead to Figure 3.18, which plots the acceleration of the toe during walking. The random-looking signal is the raw data differentiated twice. The smooth signal is the acceleration calculated after most of the higher-frequency noise has been removed. Techniques to remove this higher-frequency noise are now discussed.

3.5.4 Smoothing and Curve Fitting of Data

The removal of noise can be accomplished in several ways. The aims of each technique are basically the same. However, the results differ somewhat.

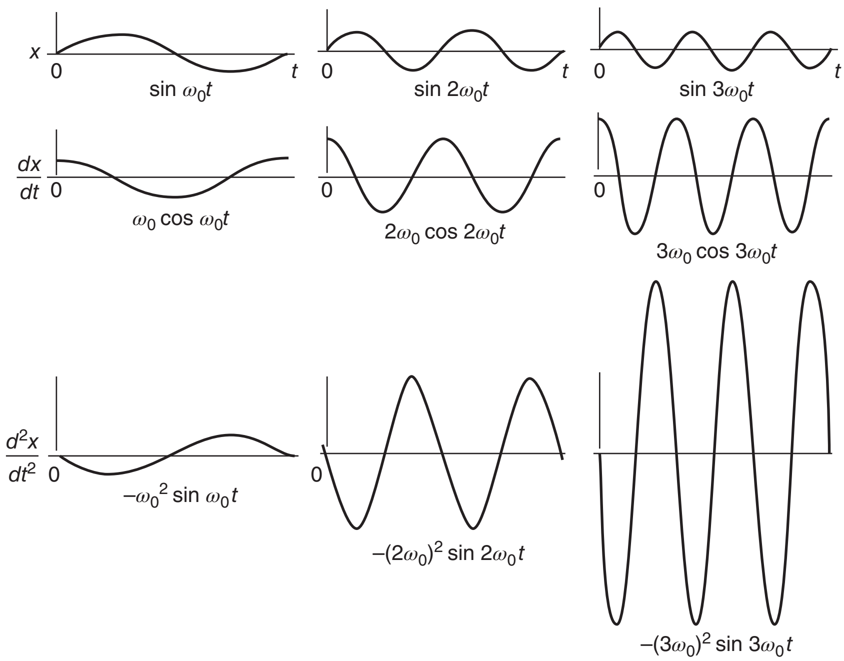

Figure 3.16 Relative amplitude changes as a result of the time differentiation of signals of increasing frequency. The first derivative increases the amplitude proportional to frequency; the second derivative increases the amplitude proportionally to frequency squared. Such a rapid increase has severe implications in calculating accelerations when the original displacement signal has high-frequency noise present.

3.5.4.1 Curve-Fitting Techniques. The basic assumption here is that the trajectory signal has a predetermined shape and that by fitting the assumed shape to a "best fit" with the raw noisy data, a smooth signal will result. For example, it may be assumed that the data are a certain order polynomial:

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \cdots + a_n t^n \tag{3.6}$$

By using computer techniques, the coefficients $a_0, \ldots, a_n$ can be selected to give a best fit, using such criteria as minimum mean square error.

A second type of curve fit can be made assuming that a certain number of harmonics are present in the signal. Reconstituting the final signal as a sum of N lowest harmonics,

$$x(t) = a_0 + \sum_{n=1}^{N} a_n \sin(n\omega_0 t + \theta_n) \tag{3.7}$$

This model has a better basis, especially in repetitive movement, while the polynomial may be better in certain nonrepetitive movement such as broad jumping. However, there are severe assumptions regarding the consistency (stationarity) of a_n and θ_n, as was discussed previously in Chapter 2.

A third technique, spline curve fitting, is a modification of the polynomial technique. The curve to be fitted is broken into sections, each section starting and ending with an inflection point, with special fitting being done between adjacent sections. The major problem with this technique is the error introduced by the improper selection of the must be determined from the noisy data and, thus, are strongly influenced by the very noise that we are trying to eliminate.

3.5.4.2 Digital Filtering – Refiltering to Remove Phase Lag of Low-Pass Filter. The fourth and most common technique used to attenuate the noise is digital filtering, which was introduced in Chapter 2. Digital filtering is not a curve-fitting technique like the three discussed above but is a noise attenuation technique based on differences in the frequency content of the signal versus the noise. However, there are some additional problems related to the low-pass filtering of the raw kinematic coordinates, and these are now discussed. For the sake of convenience, the formulae necessary to calculate the five coefficients of a second-order filter are repeated here:

$$\omega_c = \frac{(\tan(\pi f_c/f_s))}{C} \tag{3.8}$$

where C is the correction factor for number of passes required, to be explained shortly. For a single-pass, filter $C = 1$.

$$K = \sqrt{2}\omega_c \text{ for a Butterworth filter}$$
$$\text{or, } 2\omega_c \text{ for a critically damped filter}$$
$$K_2 = \omega_c^2, \quad a_0 = \frac{K_2}{(1 + K_1 + K_2)}, \quad a_1 = 2a_0, \quad a_2 = a_0$$
$$K_3 = \frac{2a_0}{K_2}, \quad b_1 = -2a_0 + K_3$$
$$b_2 = 1 - 2a_0 - K_3, \quad \text{or} \quad b_2 = 1 - a_0 - a_1 - a_2 - b_1$$

As well as attenuating the signal, there is a phase shift of the output signal relative to the input. For this second-order filter, there is a 90° phase lag at the cutoff frequency. This will cause a second form distortion, called *phase distortion*, to the higher harmonics within the bandpass region. Even more phase distortion will occur to those harmonics above f_c, but these components are mainly noise, and they are being severely attenuated. This phase distortion may be more serious than the amplitude distortion that occurs to the signal in the transition region. To cancel out this phase lag, the once-filtered data was filtered again, but this time in the reverse direction of time (Winter et al., 1974). This introduces an equal and opposite phase lead so that the net phase shift is zero. Also, the cutoff of the filter will be twice as sharp as that for single filtering. In effect, by this second filtering in the reverse direction, we have created a fourth-order zero-phase-shift filter, which yields a filtered signal that is back in phase with the raw data but with most of the noise removed.

In Figure 3.17, we see the frequency response of a second-order Butter-worth filter normalized with respect to the cutoff frequency. Superimposed on this curve is the response of the fourth-order zero-phase-shift filter. Thus, the new cutoff frequency is lower than that of the original single-pass filter; in this case, it is about 80% of the original. The correction factor for each additional pass of a Butterworth filter is $C = (2^{1/n} - 1)^{0.25}$, where n is the number of passes. Thus, for a dual pass, $C = 0.802$. For a critically damped filter, $C = (2^{1/2n} - 1)^{0.5}$; thus, for a dual pass, $C = 0.435$. This correction factor is applied to Equation (3.8) and results in the cutoff frequency for the original single-pass filter being set higher, so that after the second pass the desired cutoff frequency is achieved. The major difference between these two filters is a compromise in the response in the time domain. Butterworth filters have a slight overshoot in response to step- or impulse-type inputs, but they have a much shorter rise time. Critically damped filters have no overshoot but suffer from a slower rise time. Because impulsive-type inputs are rarely seen in human movement data, the Butterworth filter is preferred.

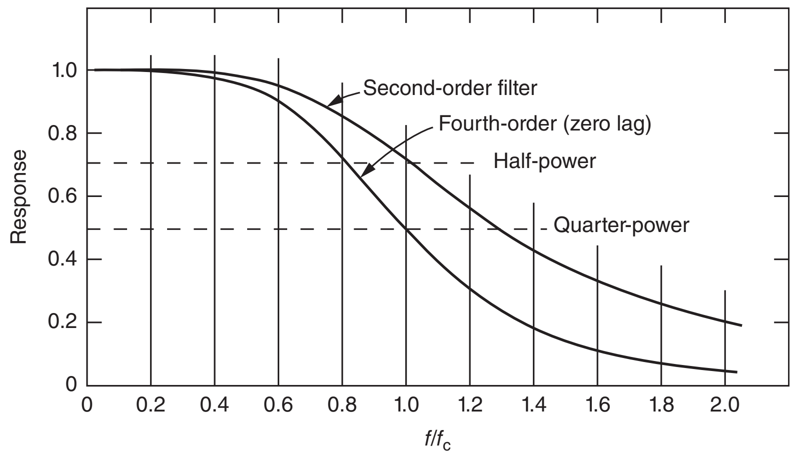

Figure 3.17 Response of a second-order low-pass digital filter. Curve is normalized at 1.0 at the cutoff frequency, f_c. Because of the phase lag characteristics of the filter, a second refiltering is done in the reverse direction in time, which results in a fourth-order zero-lag filter.

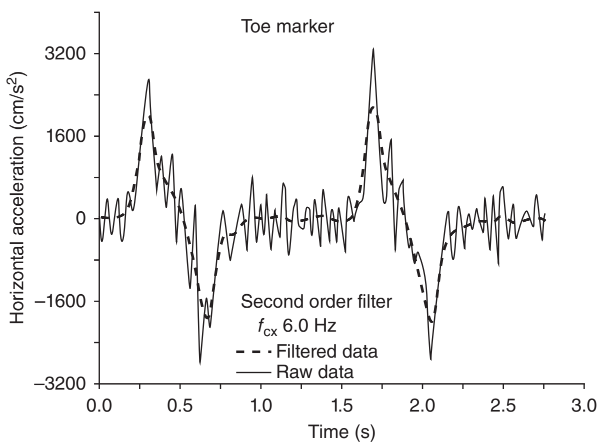

Figure 3.18 Horizontal acceleration of the toe marker during normal walking as calculated from displacement data from television. The solid line is the acceleration based on the unprocessed "raw" data; the dotted line is that calculated after the data has been filtered with a fourth-order zero-lag low-pass digital filter. (Reproduced by permission from the *Journal of Biomechanics*.)

The application of one of these filters in smoothing raw coordinate data can now be seen by examining the data that yielded the harmonic plot in Figure 3.15. The horizontal acceleration of this toe marker, as calculated by finite differences from the filtered data, is plotted in Figure 3.18. Note how repetitive the filtered acceleration is and how it passes through the "middle" of the noisy curve, as calculated using the unfiltered data. Also, note that there is no phase lag in these filtered data because of the dual forward and reverse filtering processes.

3.5.4.3 Choice of Cutoff Frequency – Residual Analysis. There are several ways to choose the best cutoff frequency. The first is to carry out a harmonic analysis as depicted in Figure 3.15. By analyzing the power in each of the components, a decision can be made as to how much power to accept and how much to reject. However, such a decision assumes that the filter is ideal and has an infinitely sharp cutoff. A better

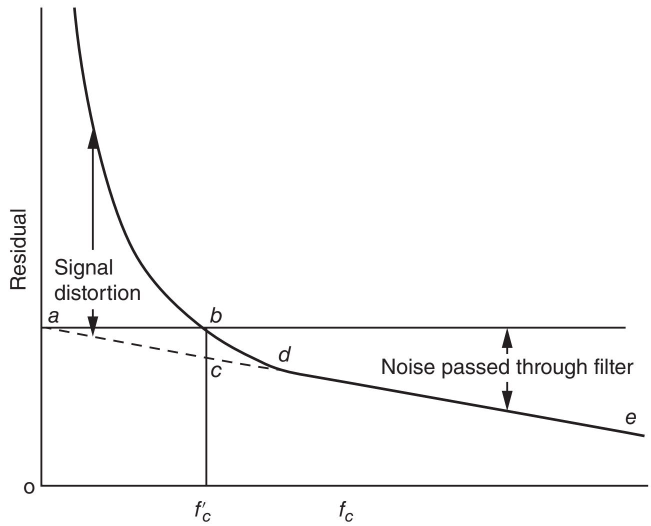

Figure 3.19 Plot of the residual between a filtered and an unfiltered signal as a function of the filter cutoff frequency. See text for the interpretation as to where to set the cutoff frequency of the filter.

of the difference between filtered and unfiltered signals over a wide range of cutoff frequencies (Wells and Winter, 1980). In this way, the characteristics of the filter in the transition region are reflected in the decision process. Figure 3.19 shows a theoretical plot of residual versus frequency. The residual at any cutoff frequency is calculated as follows [see Equation (3.9)] for a signal of N sample points in time:

$$R(f_c) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \left(X_i - \hat{X}_i\right)^2} \tag{3.9}$$

where

f_c = is the cutoff frequency of the fourth-order dual-pass filter
X_i = is raw data at ith sample
$\hat{X}_i$ = is filtered data at the ith sample using a fourth-order zero-lag filter.

If our data contained no signal, just random noise, the residual plot would be a straight line decreasing from an intercept at 0 Hz to an intercept on the abscissa at the Nyquist frequency (0.5 fs). The line de represents our best estimate of that noise residual. The intercept a on the ordinate (at 0 Hz) is nothing more than the rms value of the noise, because $\hat{X}_i$ for a 0-Hz filter is nothing more than the mean of the noise over the N samples. When the data consist of true signal plus noise, the residual will be seen to rise above the straight (dashed) line as the cutoff frequency is reduced. This rise above the dashed line represents the signal distortion that is taking place as the cutoff is reduced more and more.

The final decision is where f_c should be chosen. The compromise is always a balance between the signal distortion and the amount of noise allowed through. If we decide that both should be equal, then we simply project a line horizontally from a to intersect the residual line at b. The frequency chosen is f_c^1, and at this frequency, the signal distortion is represented by bc. This is also an estimate of the noise that is passed through the filter. Figure 3.20 is a plot of the residual of four markers from one stride of gait data, and both vertical and horizontal coordinates were analyzed

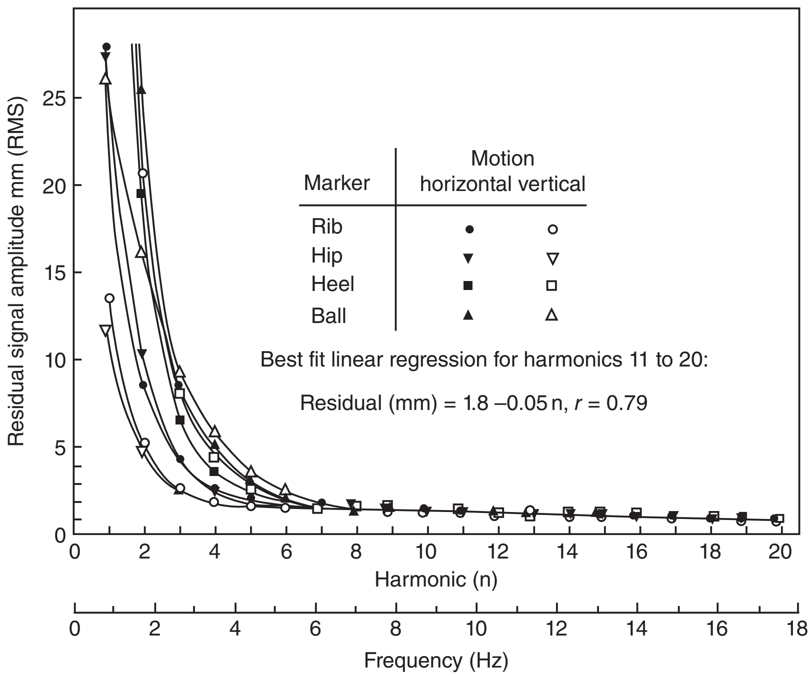

Figure 3.20 Plot of the residual of four markers from a walking trial; both vertical and horizontal displacement data. Motion data were digitized with the camera 5 m from the subject.

seen, the straight regression line that represents the noise is essentially the same for both coordinates on all markers. This tells us that the noise content, mainly introduced by the human digitizing process, is the same for all markers. This regression line has an intercept of 1.8 mm, which indicates that the rms of the noise is 1.8 mm. In this case, the motion capture camera was 5 m from the subject and the image was 2 m high by 3 m wide. Thus, the rms noise is less than one part in 1000.

Also, we see distinct differences in the frequency content of different markers. The residual shows the more rapidly moving markers on the heel and ball to have power up to about 6 Hz, while the vertical displacements of the rib and hip markers were limited to about 3 Hz. Thus, through this selection technique, we could have different cutoff frequencies specified for each marker displacement.

3.5.4.4 Optimal Cutoff Frequency. The residual analysis technique described in the previous section suggested the choice of a frequency where the signal distortion was equal to the residual noise. This optimal cutoff frequency applies to displacement data only. However, this may not be the optimum frequency for all amplitudes of signal and noise, all sampling frequencies, and all levels of differentiation: velocities versus accelerations. Giakas and Baltzopoulos (1997) showed that the optimal cutoff frequencies depended on noise level and whether displacements, velocities, or accelerations were being considered. Unfortunately, their reference displacement signal was reconstituted from a harmonic analysis, and in Chapter 2 this technique was shown to have major problems because of the lack of stationarity of each harmonics amplitude and phase. Yu et al. (1999) carried out a detailed analysis to estimate the optimum cutoff frequency for higher-order derivatives, especially accelerations. They found the optimum cutoff frequencies to be somewhat higher than those estimated for the displacement residual analysis. This is not surprising when we consider that the acceleration increases the higher-frequency noise in the acceleration waveform will increase far more rapidly than the signal itself. Also, when the sampling frequency, f_s, increases, the sampling period, $\Delta t = 1/f_s$, decreases, and thus the noise as calculated by finite differences increases [see Equations (3.17) and (3.18c)]. Thus, Yu et al. (1999) estimated that the optimum cutoff frequency was not only a function of the residual between the filtered and unfiltered data but also a function of f_s. Their estimated optimal cutoff frequency, $f_{c,2}$, was:

$$f_{c,2} = 0.06 f_s - 0.000022 f_s^2 + 5.95/\varepsilon \tag{3.10}$$

where f_s is the sampling frequency and ε is the relative mean residual between X_i and $\hat{X}_i$ [terms defined in Equation (3.9)]. These authors present example acceleration curves (see Figure 3.4 in Yu et al., 1999) that show a reasonable match between the accelerometer data and the filtered motion capture data, except that the lag of the filtered data suggests that a second-order low-pass filter was used rather than the desired fourth-order zero-lag filter.

3.5.5 Comparison of Some Smoothing Techniques

It is valuable to see the effect of several different curve-fitting techniques on the same set of noisy data. The following summary of a validation experiment, which was conducted to compare (Pezzack et al., 1977) three commonly used techniques, illustrates the wide differences in the calculated accelerations.

Data obtained from the horizontal movement of a lever arm about a vertical axis were recorded in three different ways. A goniometer on the axis recorded angular position, an accelerometer mounted at the end of the arm gave tangential acceleration and thus angular acceleration, and motion capture data gave image information that could be compared with the angular and acceleration records. The comparisons are given in Figure 3.21. Figure 3.21a compares the angular position of the lever arm as it was manually moved from rest through about 130° and back to the original position. The goniometer signal and the lever angle as analyzed from the motion capture data are plotted and compared closely. The only difference is that the goniometer record is somewhat noisy compared with the motion capture data.

Figure 3.21b compares the directly recorded angular acceleration, which can be calculated by dividing the tangential acceleration by the radius of the accelerometer from the center of rotation, with the angular acceleration as calculated via the second derivative of the digitally filtered coordinate data (Winter et al., 1974). The two curves match extremely well, and the finite-difference acceleration exhibits less noise than the directly recorded acceleration. Figure 3.21c compares the directly recorded acceleration with the calculated angular acceleration, using a polynomial fit on the raw angular data. A ninth-order polynomial was fitted to the angular displacement curve to yield the following fit:

$$\begin{aligned} \theta(t) = 0.064 + 2.0t - 35t^2 + 210t^3 - 430t^4 + 400t^5 \\ - 170t^6 + 25t^7 + 2.2t^8 - 0.41t^9 \ \text{rad} \end{aligned} \tag{3.11}$$

Note that θ is in radians and t in seconds. To get the curve for angular acceleration, all we need to do is take the second time derivative to yield:

$$\begin{aligned} \alpha(t) = -70 + 1260t - 5160t^2 + 8000t^3 - 5100t^4 \\ + 1050t^5 + 123t^6 - 29.5t^7 \ \text{rad/s}^2 \end{aligned} \tag{3.12}$$

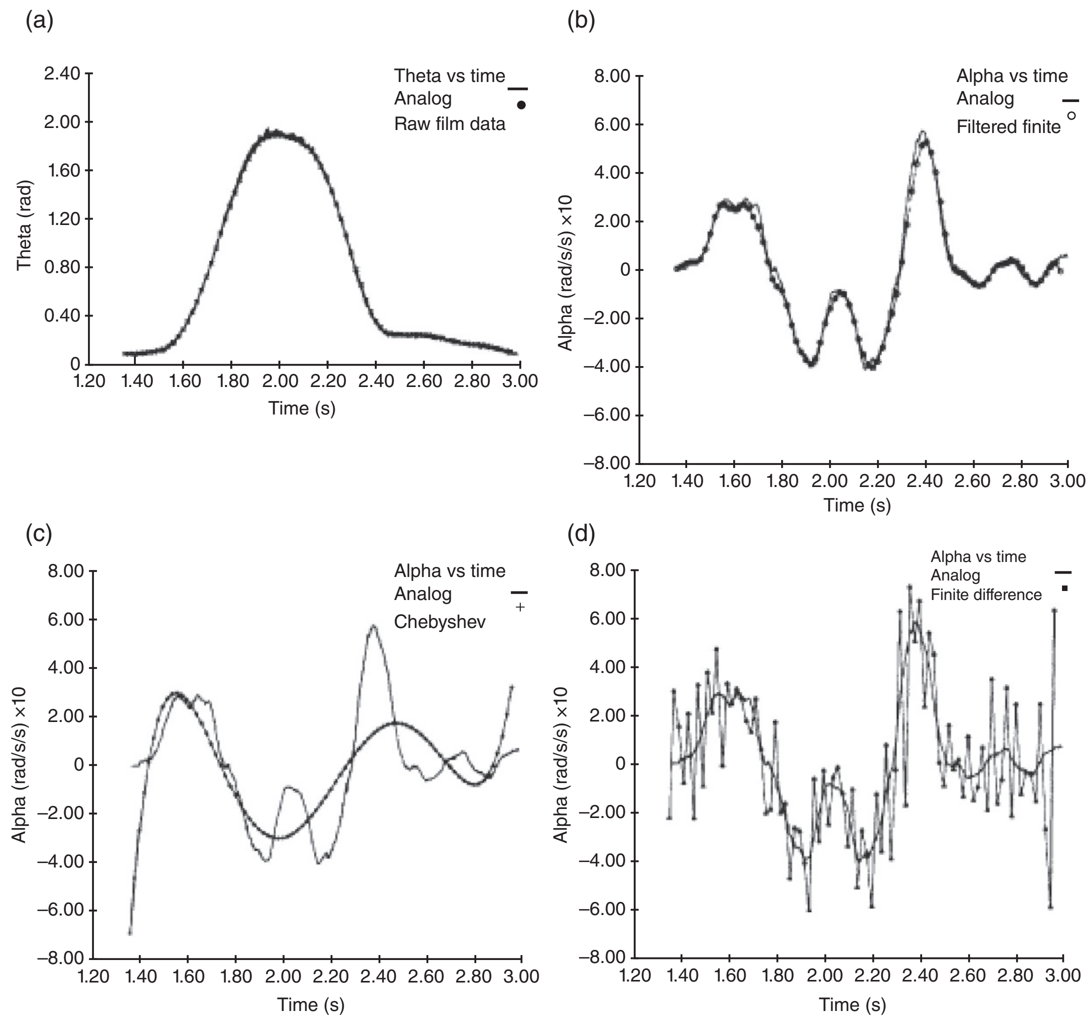

Figure 3.21 Comparison of several techniques used to determine the acceleration of a movement based on motion capture displacement data. (a) Displacement angle of a simple extension/flexion as plotted from motion capture and goniometer data. (b) Acceleration of the movement in (a) as measured by an accelerometer and as calculated from motion capture coordinates after digital filtering. (c) Acceleration as determined from a ninth-order polynomial fit of the displacement data compared with the directly recorded acceleration. (d) Acceleration as determined by finite-difference technique of the raw coordinate data compared with the accelerometer curve. (Reproduced by permission from the *Journal of Biomechanics*.)

This acceleration curve, compared with the accelerometer signal, shows considerable discrepancy, enough to cast doubt on the value of the polynomial fit technique. The polynomial is fitted to the displacement data in order to get an analytic curve, which can be differentiated to yield another smooth curve. Unfortunately, it appears that a considerably higher-order polynomial would be required to achieve even a crude fit, and the computer time might become too prohibitive.

Finally, in Figure 3.21d, you can see the accelerometer signal plotted against angular acceleration as calculated by second-order finite-difference techniques using raw coordinate data. The plot speaks for itself – the accelerations are too noisy to mean anything.

3.6 CALCULATION OF OTHER KINEMATIC VARIABLES

3.6.1 Limb-Segment Angles

Given the coordinate data from anatomical markers at either end of a limb segment, it is an easy step to calculate the absolute angle of that segment in space. It is not necessary that the two markers be at the extreme ends of the limb segment, as long as they are in line with the long-bone axis. Figure 3.22 shows the outline of a leg with seven anatomical markers in a four-segment three-joint system. Markers 1 and 2 define the thigh in the sagittal plane. Note that, by convention, all angles are measured in a counterclockwise direction, starting with the horizontal equal to 0°. Thus, θ_{43} is the angle of the leg in space and can be calculated from:

$$\theta_{43} = \arctan \frac{y_3 - y_4}{x_3 - x_4} \tag{3.13}$$

or, in more general notation,

$$\theta_{ij} = \arctan \frac{y_j - y_i}{x_j - x_i} \tag{3.14}$$

As has already been noted, these segment angles are absolute in the defined spatial reference system. It is, therefore, quite easy to calculate the joint angles from the angles of the two adjacent segments.

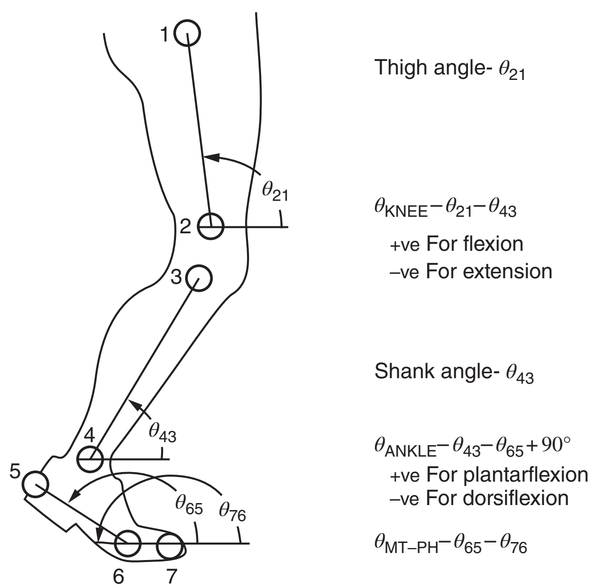

Figure 3.22 Marker location and limb and joint angles using an established convention. Limb angles in the spatial reference system are determined using counterclockwise from the horizontal as positive. Thus, angular velocities and accelerations are also positive in the counterclockwise direction in the plane of movement; this is essential for consistent convention use in subsequent kinetic analyses. Conventions for joint angles (which are relative) are subject to wide variation among researchers; thus, the convention used must be clarified.

3.6.2 Joint Angles

Each joint has a convention for describing its magnitude and polarity. For example, when the knee is fully extended, it is described as 0° flexion, and when the leg moves in a posterior direction relative to the thigh, the knee is said to be in flexion. In terms of the absolute angles described previously,

$$\text{knee angle} = \theta_k = \theta_{21} - \theta_{43}$$

If $\theta_{21} > \theta_{43}$, the knee is flexed; if $\theta_{21} < \theta_{43}$, the knee is extended.

The convention for the ankle is slightly different in that 90° between the leg and the foot is boundary between plantarflexion and dorsiflexion. Therefore,

$$\text{ankle angle} = \theta_a = \theta_{43} - \theta_{65} + 90^\circ$$

If θ_a is positive, the foot is plantarflexed; if θ_a is negative, the foot is dorsiflexed.

3.6.3 Velocities – Linear and Angular

There can be severe problems associated with the determination of velocity and acceleration information. For the reasons outlined, we will assume that the raw displacement data have been suitably smoothed by digital filtering and we have a set of smoothed coordinates and angles to operate upon. To calculate the velocity from displacement data, all that is needed is to take the finite difference. For example, to determine the velocity in the x-direction, we calculate $\Delta x/\Delta t$, where $\Delta x = x_{i+1} - x_i$, and Δt is the time between adjacent samples x_{i+1} and x_i.

The velocity calculated this way does not represent the velocity at either of the sample times. Rather, it represents the velocity of a point in time halfway between the two samples. This can result in errors later on when we try to relate the velocity-derived information to displacement data, and both results do not occur at the same point in time. A way around this problem is to calculate the velocity and accelerations on the basis of $2\Delta t$ rather than Δt. Thus, the velocity at the ith sample is:

$$Vx_i = \frac{x_{i+1} - x_{i-1}}{2\Delta t}\ \text{m/s} \tag{3.15}$$

Note that the velocity is at a point halfway between the two samples, as depicted in Figure 3.23. The assumption is that the line joining x_{i-1} to x_{i+1} has the same slope as the line drawn tangent to the curve at x_i.

For angular velocities, the formula is the same except that we use angular data rather than displacement data in Equation (3.14); the angular acceleration at the ith sample is:

$$\omega_i = \frac{\theta_{i+1} - \theta_{i-1}}{2\Delta t}\ \text{rad/s} \tag{3.16}$$

3.6.4 Accelerations – Linear and Angular

Similarly, the acceleration is:

$$Ax_i = \frac{Vx_{i+1} - Vx_{i-1}}{2\Delta t}\ \text{m/s}^2 \tag{3.17}$$

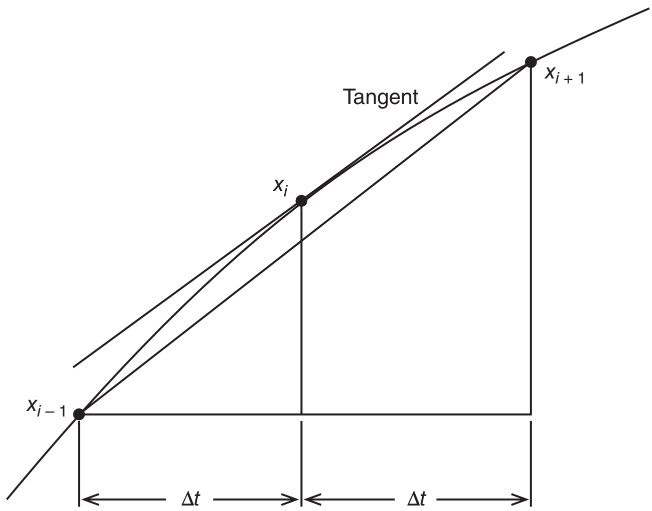

Figure 3.23 Finite-difference technique for calculating the slope of a curve at the *i*th sample point.

Note that Equation (3.16) requires displacement data from samples $i + 2$ and $i - 2$; thus, a total of five successive data points go into the acceleration. An alternative and slightly better calculation of acceleration uses only three successive data coordinates and utilizes the calculated velocities halfway between sample times:

$$Vx_{i+1/2} = \frac{x_{i+1} - x_i}{\Delta t} \text{ m/s} \tag{3.18a}$$

$$Vx_{i-1/2} = \frac{x_i - x_{i-1}}{\Delta t} \text{ m/s} \tag{3.18b}$$

Substituting these "halfway" velocities into Equation (3.17) we get:

$$Ax_i = \frac{x_{i+1} - 2x_i + x_{i-1}}{\Delta t^2} \text{ m/s}^2 \tag{3.18c}$$

For angular accelerations merely replace displacement data with angular data in Equations (3.17) or (3.18a–c).

3.7 PROBLEMS BASED ON KINEMATIC DATA

1. Tables A.1 and A.2a in Appendix A, plot the vertical (Y) displacement of the raw and filtered data (in centimeters) for the greater trochanter (hip) marker for frames 1 to 30. Use a vertical scale as large as possible so as to identify the noise content of the raw data. In a few lines, describe the results of the smoothing by the digital filter.
2. Using filtered coordinate data (see Table A.2c), plot the vertical displacement of the heel marker from toe off right (TOR) (frame 1) to the next TOR (frame 70).
   - **(a)** Estimate the instant of heel-off during midstance. (*Hint:* Consider the elastic compression and release of the shoe material when arriving at your answer.)
   - **(b)** Determine the maximum height of the heel above ground level during swing. When does this occur during the swing phase? (*Hint:* Consider the lowest displacement of the heel marker during stance as an indication of

- **(c)** Describe the vertical heel trajectory during the latter half of swing (frames 14–27), especially the four frames immediately prior to heel right contact (HRC).
- **(d)** Calculate the vertical heel velocity at HRC.
- **(e)** Calculate from the horizontal displacement data the horizontal heel velocity at HCR.
- **(f)** From the horizontal coordinate data of the heel during the first foot flat period (frames 35–40) and the second foot flat period (frames 102–106), estimate the stride length.
- **(g)** If one stride period is 69 frames, estimate the forward velocity of this subject.

3. Plot the trajectory of the trunk marker (rib cage) over one stride (frames 28–97).

- **(a)** Is the shape of this trajectory what you would expect in walking?
- **(b)** Is there any evidence of conservation of mechanical energy over the stride period? (That is, is potential energy being converted to kinetic energy and vice versa?)

4. Determine the vertical displacement of the toe marker (Table A.2d) when it reaches its lowest point in late stance and compare that with the lowest point during swing, and thereby determine how much toe clearance took place. Answer: ytoe(fr.13) = 0.0485 m, ytoe(fr.66) = 0.0333 m, clearance = 0.0152 m = 1.52 cm.

5. From the filtered coordinate data (see Table A.2b), calculate the following and check your answer with that listed in the appropriate listings (see Tables A.3a, A.3b, A.3c, A.4).

- **(a)** The velocity of the knee in the *X* direction for frame 10.
- **(b)** The acceleration of the knee in the *X* direction for frame 10.
- **(c)** The angle of the thigh and leg in the spatial reference system for frame 30.
- **(d)** From (c) calculate the knee angle for frame 30.
- **(e)** The absolute angular velocity of the leg for frame 30 (use angular data, Table A.3b).
- **(f)** Using the tabulated vertical velocities of the toe, calculate its vertical acceleration for frames 25 and 33.

6. From the filtered coordinate data in Table A.2b and A.2c, calculate the following and check your answer from the results tabulated in Table A.3a and A.3b.

- **(a)** The center of mass of the foot segment for frame 80.
- **(b)** The velocity of the center of mass of the leg for frame 70. Give the answer in both coordinate and polar form.

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4

ANTHROPOMETRY

JOSEPH A. ZENI[1], STEPHEN J. THOMAS[2], AND DAVID A. WINTERS[3,†]
[1]*Rutgers University, Newark, NJ, USA*
[2]*Thomas Jefferson University, Philadelphia, PA, USA*
[3]*University of Waterloo, Waterloo, Ontario, Canada*

4.0 SCOPE OF ANTHROPOMETRY IN MOVEMENT BIOMECHANICS

Anthropometry is the major branch of anthropology that studies the physical measurements of the human body to determine differences in individuals and groups. A wide variety of physical measurements are required to describe and differentiate the characteristics of race, sex, age, and body type. In the past, the major emphasis of these studies has been evolutionary and historical. However, more recently, a major impetus has come from the needs of technological developments, especially man-machine interfaces: workspace design, cockpits, pressure suits, armor, and so on. Most of these needs are satisfied by basic linear, area, and volume measures. However, human movement analysis requires kinetic measures as well: masses, moments of inertia, and their locations. There exists also a moderate body of knowledge regarding the joint centers of rotation, the origin and insertion of muscles, the angles of pull of tendons, and the length and cross-sectional area of muscles.

4.0.1 Segment Dimensions

The most basic body dimension is the length of the segments between each joint. These vary with body build, sex, and racial origin. Dempster (1955) and Dempster et al. (1959) have summarized estimates of segment lengths and joint center locations relative to anatomical landmarks. An average set of segment lengths expressed as a percentage of body height was prepared by Drillis and Contini (1966) and is shown in Figure 4.1. These segment proportions serve as a good approximation in the absence of better data, preferably measured directly from the individual.

[†]Deceased

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

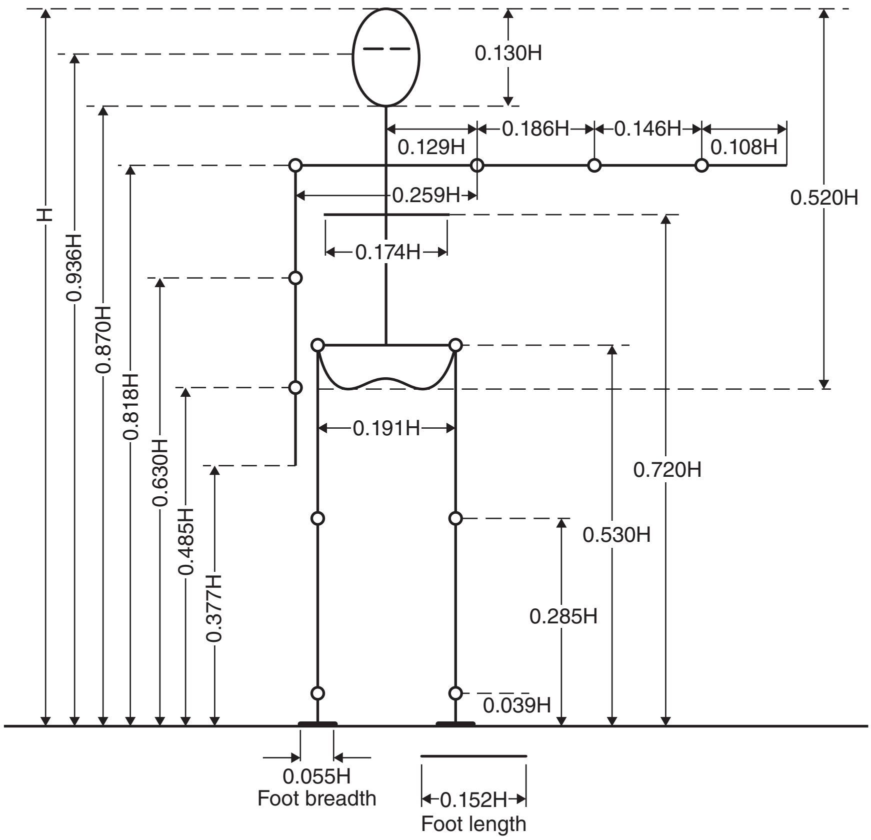


Figure 4.1 Body segment lengths expressed as a fraction of body height *H*.

4.1 DENSITY, MASS, AND INERTIAL PROPERTIES

Kinematic and kinetic analyses require data regarding mass distributions, mass centers, moments of inertia, and the like. Some of these measures have been determined directly from cadavers; others have utilized measured segment volumes in conjunction with density tables, and more modern techniques use scanning systems that produce the cross-sectional image at many intervals across the segment.

4.1.1 Whole-Body Density

The human body consists of many types of tissue, each with a different density. Cortical bone has a specific gravity greater than 1.8, muscle tissue is just over 1.0, fat is less than 1.0, and the lungs contain light respiratory gases. The average density of the whole body is a function of body build, called *somatotype*. Drillis and Contini (1966) developed an expression for body density d as a function of ponderal index $c = h/w^{1/3}$, where w is body weight (pounds) and h is body height (inches):

$$d = 0.69 + 0.0297c \text{ kg/L} \tag{4.1}$$

The equivalent expression in metric units, where body mass is expressed in kilograms and height in meters, is:

$$d = 0.69 + 0.9c \text{ kg/L} \tag{4.2}$$

It can be seen that a shorter individual with higher body weight has a lower ponderal index than a taller individual with a lower relative body weight and, therefore, the shorter individual has a lower body density.

Example 4.1. Using Equations (4.1) and (4.2), calculate the whole-body density of an adult whose height is 5′10″ and who weighs 170 lb.

$$c = h/w^{1/3} = 70/170^{1/3} = 12.64$$

Using Equation (4.1),

$$d = 0.69 + 0.0297c = 0.69 + 0.0297 \times 12.64 = 1.065 \text{ kg/L}$$

In metric units,

$$h = 70/39.4 = 1.78 \text{ m}, \quad w = 170/2.2 = 77.3 \text{ kg}, \quad \text{and}$$
$$c = 1.78/77.3^{1/3} = 0.418$$

Using Equation (4.2),

$$d = 0.69 + 0.9c = 0.69 + 0.9 \times 0.418 = 1.066 \text{ kg/L}$$

4.1.2 Segment Densities

Each body segment has a unique combination of bone, muscle, fat, and other tissue, and the density within a given segment is not uniform. Generally, because of the higher proportion of bone, the density of distal segments is greater than that of proximal segments, and individual segments increase their densities as the average body density increases. Figure 4.2 shows these trends for six limb segments as a function of whole-body density, as calculated by Equations (4.1) or (4.2) or as measured directly (Drillis and Contini, 1966; Contini, 1972).

4.1.3 Segment Mass and Center of Mass

The terms *center of mass* and *center of gravity* are often used interchangeably. The more general term is center of mass (COM), while the center of gravity refers to the COM in one axis only, that defined by the direction of gravity. In the two horizontal axes, the term COM must be used.

As the total body mass increases, so does the mass of each individual segment. Therefore, it is possible to express the mass of each segment as a percentage of the total body mass. Table 4.1 summarizes the compiled results of several investigators. These values are utilized throughout this text in subsequent kinetic and energy calculations. It should be noted that other tables of inertial properties and segment masses exist (de Leva, 1996; Dumas et al., 2007) and comparisons of these parameters have been reported (Dumas and Wojtusch, 2018). The location of the COM is also given as a percentage of the segment length from either the distal or the proximal end. In cadaver studies, it is quite simple to locate the COM by simply determining the center of balance of each segment. To calculate the COM in vivo, we need the profile of cross-sectional area and

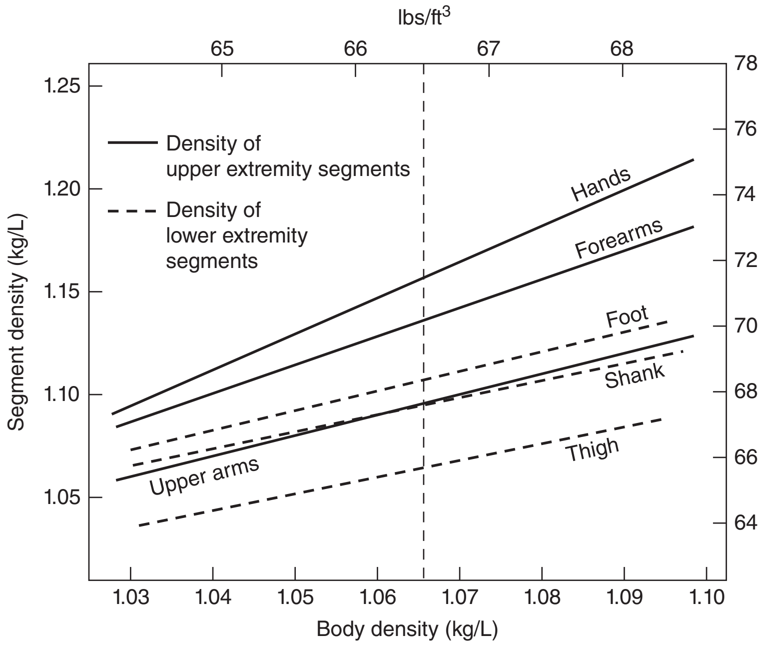


Figure 4.2 Density of limb segments as a function of average body density.

length. Figure 4.3 gives a hypothetical profile where the segment is broken into n sections, each with its mass indicated. The total mass M of the segment is:

$$M = \sum_{i=1}^{n} m_i \tag{4.3}$$

where m_i, is the mass of the ith section.

$$m_i = d_i V_i$$

where d_i density of ith section

V_i volume of ith section.

If the density d is assumed to be uniform over the segment, then $m_i = dV_i$ and:

$$M = d \sum_{i=1}^{n} V_i \tag{4.4}$$

The COM is such that it must create the same net gravitational moment of force about any point along the segment axis as did the original distributed mass. Consider the COM to be located a distance x from the left edge of the segment,

$$\begin{aligned} Mx &= \sum_{i=1}^{n} m_i x_i \\ x &= \frac{1}{M} \sum_{i=1}^{n} m_i x_i \end{aligned} \tag{4.5}$$

We can now represent the complex distributed mass by a single mass M located at a distance x from one end of the segment.

TABLE 4.1 Anthropometric Data

Segment	Definition	Segment Weight/Total Body Weight	Center of Mass/ Segment Length		Radius of Gyration/ Segment Length			Density
			Proximal	Distal	C of G	Proximal	Distal	
Hand	Wrist axis/knuckle II middle finger	0.006 M	0.506	0.494 P	0.297	0.587	0.577 M	1.16
Forearm	Elbow axis/ulnar styloid	0.016 M	0.430	0.570 P	0.303	0.526	0.647 M	1.13
Upper arm	Glenohumeral axis/elbow axis	0.028 M	0.436	0.564 P	0.322	0.542	0.645 M	1.07
Forearm and hand	Elbow axis/ulnar styloid	0.022 M	0.682	0.318 P	0.468	0.827	0.565 P	1.14
Total arm	Glenohumeral joint/ulnar styloid	0.050 M	0.530	0.470 P	0.368	0.645	0.596 P	1.11
Foot	Lateral malleolus/head metatarsal II	0.0145 M	0.50	0.50 P	0.475	0.690	0.690 P	1.10
Leg	Femoral condyles/medial malleolus	0.0465 M	0.433	0.567 P	0.302	0.528	0.643 M	1.09
Thigh	Greater trochanter/femoral condyles	0.100 M	0.433	0.567 P	0.323	0.540	0.653 M	1.05
Foot and leg	Femoral condyles/medial malleolus	0.061 M	0.606	0.394 P	0.416	0.735	0.572 P	1.09
Total leg	Greater trochanter/medial malleolus	0.161 M	0.447	0.553 P	0.326	0.560	0.650 P	1.06
Head and neck	C7–T1 and 1st rib/ear canal	0.081 M	1.000	— PC	0.495	0.116	— PC	1.11
Shoulder mass	Sternoclavicular joint/glenohumeral axis	—	0.712	0.288	—	—	—	1.04
Thorax	C7–T1/T12–L1 and diaphragm*	0.216 PC	0.82	0.18	—	—	—	0.92
Abdomen	T12–L1/L4–L5*	0.139 LC	0.44	0.56	—	—	—	—
Pelvis	L4–L5/greater trochanter*	0.142 LC	0.105	0.895	—	—	—	—
Thorax and abdomen	C7–T1/L4–L5*	0.355 LC	0.63	0.37	—	—	—	—
Abdomen and pelvis	T12–L1/greater trochanter*	0.281 PC	0.27	0.73	—	—	—	1.01
Trunk	Greater trochanter/glenohumeral joint*	0.497 M	0.50	0.50	—	—	—	1.03
Trunk head neck	Greater trochanter/glenohumeral joint*	0.578 MC	0.66	0.34 P	0.503	0.830	0.607 M	—
Head, arms, and trunk (HAT)	Greater trochanter/glenohumeral joint*	0.678 MC	0.626	0.374 PC	0.496	0.798	0.621 PC	—
HAT	Greater trochanter/mid rib	0.678	1.142	—	0.903	1.456	—	—

NOTE: These segments are presented relative to the length between the greater trochanter and the glenohumeral joint.
Source Codes: M, Dempster via Miller and Nelson; *Biomechanics of Sport*, Lea and Febiger, Philadelphia, 1973. P, Dempster via Plagenhoef; *Patterns of Human Motion*, Prentice-Hall, Inc. Englewood Cliffs, NJ, 1971. L, Dempster via Plagenhoef from living subjects; *Patterns of Human Motion*, Prentice-Hall, Inc., Englewood Cliffs, NJ, 1971. C, Calculated.

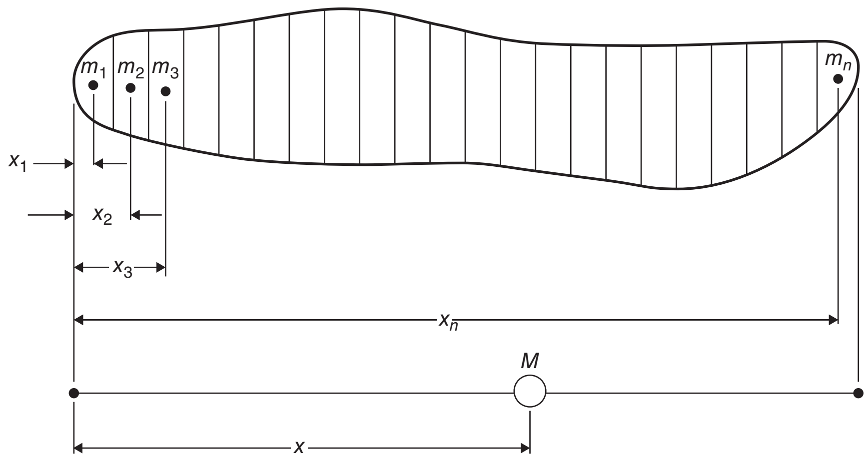


Figure 4.3 Location of the center of mass of a body segment relative to the distributed mass.

Example 4.2. From the anthropometric data in Table 4.1 calculate the coordinates of the COM of the foot and the thigh given the following coordinates: ankle (84.9, 11.0), metatarsal (101.1, 1.3), greater trochanter (72.1, 92.8), and lateral femoral condyle (86.4, 54.9). From Table 4.1, the foot COM is 0.5 of the distance from the lateral malleolus (ankle) to the metatarsal marker. Thus, the COM of the foot is:

$$x = (84.9 + 101.1) \div 2 = 93.0\,\text{cm}$$

$$y = (11.0 + 1.3) \div 2 = 6.15\,\text{cm}$$

The thigh COM is 0.433 from the proximal end of the segment. Thus, the COM of the thigh is:

$$x = 72.1 + 0.433(86.4 - 72.1) = 78.3\,\text{cm}$$

$$y = 92.8 - 0.433(92.8 - 54.9) = 76.4\,\text{cm}$$

4.1.4 Center of Mass of a Multisegment System

With each body segment in motion, the COM of the total body is continuously changing with time. It is, therefore, necessary to recalculate it after each interval of time, and this requires knowledge of the trajectories of the COM of each body segment. Consider at a particular point in time a three-segment system with the centers of mass as indicated in Figure 4.4. The COM of the total system is located at (x_0, y_0), and each of these coordinates can be calculated separately; $M = m_1 + m_2 + m_3$, and:

$$x_0 = \frac{m_1x_1 + m_2x_2 + m_3x_3}{M} \tag{4.6}$$

$$y_0 = \frac{m_1y_1 + m_2y_2 + m_3y_3}{M} \tag{4.7}$$

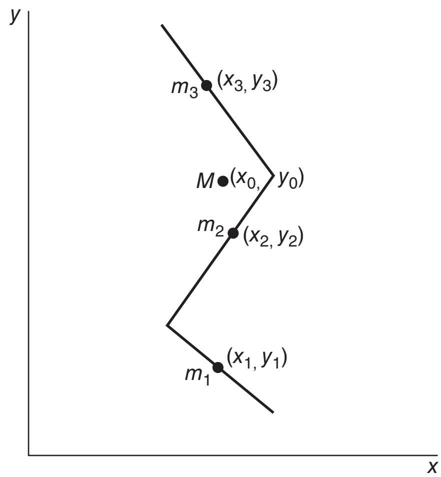


Figure 4.4 Center of mass of a three-segment system relative to the centers of mass of the individual segments.

The COM of the total body is a frequently calculated variable. Its usefulness in the assessment of human movement, however, is quite limited. Some researchers have used the time history COM to calculate the energy changes of the total body. Such a calculation is erroneous, because the COM does not account for energy changes related to reciprocal movements of the limb segments. Thus, the energy changes associated with the forward movement of one leg and the backward movement of another will not be detected in the COM, which may remain relatively unchanged. More about this will be said in Chapter 6. The major use of the body COM is in the analysis of sporting events, especially jumping events, where the path of the COM is critical to the success of the event because its trajectory is decided immediately at takeoff. Also, in studies of body posture and balance, the COM is an essential calculation.

4.1.5 Mass Moment of Inertia and Radius of Gyration

The location of the COM of each segment is needed for an analysis of translational movement through space. If accelerations are involved, we need to know the inertial resistance to such movements. In the linear sense, $F = ma$ describes the relationship between a linear force F and the resultant linear acceleration a. In the rotational sense, $M = I\alpha$. M is the moment of force causing the angular acceleration α. Thus, I is the constant of proportionality that measures the ability of the segment to resist changes in angular velocity. M has units of $\mathrm{N \cdot m}$, α is in $\mathrm{rad/s^2}$, and I is in $\mathrm{kg \cdot m^2}$. The value of I depends on the point about which the rotation is taking place and is a minimum when the rotation takes place about its COM. Consider a distributed mass segment as in Figure 4.3. The moment of inertia about the left end is:

$$
\begin{aligned}
I &= m_1 x_1^2 + m_2 x_2^2 + \cdots + m_n x_n^2 \\
&= \sum_{i=1}^{n} m_i x_i^2
\end{aligned}
\tag{4.8}
$$

It can be seen that the mass close to the center of rotation has very little influence on I, while the furthest mass has a considerable effect. This principle is used in industry to regulate the speed of rotating machines: the mass of a flywheel is concentrated at the perimeter of the wheel with as large a radius as possible. Its large moment of inertia resists changes in velocity and, therefore, tends to keep the machine speed constant.

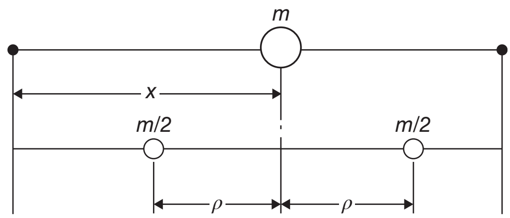


Figure 4.5 Radius of gyration of a limb segment relative to the location of the center of mass of the original system.

Consider the moment of inertia I_0 about the COM. In Figure 4.5 the mass has been broken into two equal point masses. The location of these two equal components is at a distance ρ_0 from the center such that:

$$I_0 = m\rho_0^2 \tag{4.9}$$

ρ_0 is the radius of gyration and is such that the two equal masses shown in Figure 4.5 have the same moment of inertia in the plane of rotation about the COM as the original distributed segment did. Note that the COM of these two equal point masses is still the same as the original single mass.

4.1.6 Parallel Axis Theorem

Most body segments do not rotate about their mass center but rather about the joint at either end. In vivo measures of the moment of inertia can only be taken about a joint center. The relationship between this moment of inertia and that about the COM is given by the parallel axis theorem. A short proof is now given.

$$\begin{aligned} I &= \frac{m}{2}(x-\rho_0)^2 + \frac{m}{2}(x+\rho_0)^2 \\ &= m\rho_0^2 + mx^2 \\ &= I_0 + mx^2 \end{aligned} \tag{4.10}$$

where

I_0 = moment of inertia about COM
x = distance between COM and center of rotation
m = mass of segment.

Actually, x can be any distance in either direction from the COM as long as it lies along the same axis as I_0 was calculated on.

Example 4.3

(a) A prosthetic leg has a mass of 3 kg and a COM of 20 cm from the knee joint. The radius of gyration is 14.1 cm. Calculate I about the knee joint.

$$\begin{aligned} I_0 &= m\rho_0^2 = 3(0.141)^2 = 0.06\ \text{kg}\cdot\text{m}^2 \\ I &= I_0 + mx^2 \\ &= 0.06 + 3(0.2)^2 = 0.18 \end{aligned}$$

(b) If the distance between the knee and hip joints is 42 cm, calculate I_h for this prothesis about the hip joint as the amputee swings through with a locked knee.

$$x = \text{distance from mass center to hip} = 20 + 42 = 62\ \text{cm}$$
$$I = I_0 + mx^2$$
$$= 0.06 + 3(0.62)^2 = 1.21\ \text{kg} \cdot \text{m}^2$$

Note that I_h is about 20 times that calculated about the COM.

4.1.7 Use of Anthropometric Tables and Kinematic Data

Using Table 4.1 in conjunction with kinematic data, we can calculate many variables needed for kinetic energy analyses (see Chapters 5 and 6). This table gives the segment mass as a fraction of body mass and centers of mass as a fraction of their lengths from either the proximal or the distal end. The radius of gyration is also expressed as a fraction of the segment length about the COM, the proximal end, and the distal end.

4.1.7.1 Calculation of Segment Masses and Centers of Mass.

Example 4.4. Calculate the mass of the foot, shank, thigh, and head/arms/trunk (HAT) and its location from the proximal or distal end, assuming that the body mass of the subject is 80 kg. Using the mass fractions for each segment,

$$\begin{aligned}
\text{Mass of foot} &= 0.0145 \times 80 = 1.16\ \text{kg}\\
\text{Mass of leg} &= 0.0465 \times 80 = 3.72\ \text{kg}\\
\text{Mass of thigh} &= 0.10 \times 80 = 8.0\ \text{kg}\\
\text{Mass of HAT} &= 0.678 \times 80 = 54.24\ \text{kg}
\end{aligned}$$

Direct measures yielded the following segment lengths: foot = 0.195 m, leg = 0.435 m, thigh = 0.410 m, HAT = 0.295 m.

$$\begin{aligned}
\text{COM of foot} &= 0.50 \times 0.195 = 0.098\ \text{m between ankle and metatarsal markers}\\
\text{COM of leg} &= 0.433 \times 0.435 = 0.188\ \text{m below femoral condyle marker}\\
\text{COM of thigh} &= 0.433 \times 0.410 = 0.178\ \text{m below greater trochanter marker}\\
\text{COM of HAT} &= 1.142 \times 0.295 = 0.337\ \text{m above greater trochanter marker}
\end{aligned}$$

where COM stands for COM.

*4.1.7.2 **Calculation of Total-Body Center of Mass.*** The calculation of the COM of the total body is a special case of Equations (4.6) and (4.7). For an n-segment body system, the COM in the X direction is:

$$x = \frac{m_1x_1 + m_2x_2 + \cdots + m_nx_n}{m_1 + m_2 + \cdots + m_n} \tag{4.11}$$

where $m_1 + m_2 + \cdots + m_n = M$, the total body mass.

It is quite normal to know the values of $m_1 = f_1 M$, $m_2 = f_2 M$, and so on. Therefore,

$$x = \frac{f_1 M x_1 + f_2 M x_2 + \cdots + f_n M x_n}{M} = f_1 x_1 + f_2 x_2 + \cdots + f_n x_n \tag{4.12}$$

This equation is easier to use because all we require is knowledge of the fraction of total body mass and the coordinates of each segment's COM. These fractions are given in Table 4.1.

It is not always possible to measure the COM of every segment, especially if it is not in full view of the camera. In the sample data that follows, we have the kinematics of the right side of HAT and the right limb during walking. It may be possible to simulate data for the left side of HAT and the left limb. If we assume symmetry of gait, we can say that the trajectory of the left limb is the same as that of the right limb, but out of phase by half a stride. Thus, if we use data for the right limb one-half stride later in time and shift them back in space one-half a stride length, we can simulate data for the left limb and left side of HAT.

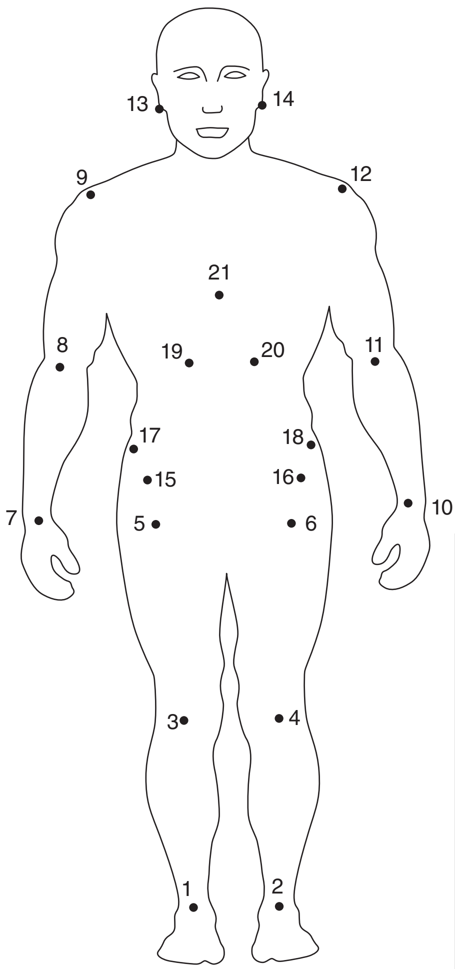
Example 4.5. Calculate the total-body COM at a given frame 15. The time for one stride was 68 frames. Thus, the data from frame 15 become the data for the right lower limb and the right half of HAT, and the data one-half stride (34 frames) later become those for the left side of the body. All coordinates from frame 49 must now be shifted back in the x-direction by a step length. An examination of the x coordinates of the heel during two successive periods of stance showed the stride length to be 264.2 − 122.8 = 141.4 cm. Therefore, the step length is 70.7 cm = 0.707 m. Table 4.2 shows the coordinates of the body segments for both left and right halves of the body for frame 15. The mass fractions for each segment are as follows: foot = 0.0145, leg = 0.0465, thigh = 0.10,1/2 HAT = 0.339. The mass of HAT dominates the body COM, but the energy changes in the lower limbs will be seen to be dominant as far as walking is concerned (see Chapter 6).

COM analyses in three dimensions are not an easy measure to make because every segment of the body must be identified with markers and tracked with a three-dimensional (3D) imaging system. In some studies of standing, the horizontal anterior/posterior displacement of a rod attached to the pelvis has been taken as an estimate of COM movement (Horak et al., 1992). However, in situations when a patient flexes the total body at the hip (called a "hip strategy") to defend against a forward fall, the pelvis moves posteriorly considerably more than the COM (Horak and Nashner, 1986). In 3D assessments of COM displacements, the only technique is optical tracking of markers on all segments

TABLE 4.2 Coordinates for Body Segments, Example 4.5

	X (m)		*Y* (m)	
Segment	Right	Left	Right	Left
Foot	0.791	1.353 − 0.707 = 0.646	0.101	0.067
Leg	0.814	1.355 − 0.707 = 0.648	0.374	0.334
Thigh	0.787	1.402 − 0.707 = 0.695	0.708	0.691
1/2 HAT	0.721	1.424 − 0.707 = 0.717	1.124	1.122

$x = 0.0145(0.791 + 0.645) + 0.0465(0.814 + 0.648) + 0.1(0.787 + 0.695) + 0.339(0.721 + 0.717) = 0.724$ m
$y = 0.0145(0.101 + 0.067) + 0.0465(0.374 + 0.334) + 0.1(0.708 + 0.691) + 0.339(1.124 + 1.122) = 0.937$ m



Segment	Mass fraction	Definition of segment COM
Head	0.081	(13 + 14)/2
Trunk 4	0.136	(9 + 12 + 21)/3
Trunk 3	0.078	((19 + 20)/2 + 21)/2
Trunk 2	0.065	(17 + 18 + 19 + 20)/4
Trunk 1	0.078	(17 + 18 + 15 + 16)/4
Pelvis	0.142	(15 + 16)/2
Thighs	0.100 (2)	0.433 × 3 + 0.567 × 5 and 0.433 × 4 + 0.567 × 6
Legs and feet	0.060 (2)	0.606 × 1 + 0.394 × 3 and 0.606 × 2 + 0.394 × 4
Upper arms	0.028 (2)	0.436 × 8 + 0.564 × 9 and 0.436 × 11 + 0.564 × 12
Lower arms	0.022 (2)	0.682 × 7 + 0.318 × 8 and 0.682 × 10 + 0.318 × 11
Total	1.00	

Figure 4.6 A 21-marker, 14-segment model to estimate the 3D center of mass of the total body in balance control experiments. Four trunk segments were necessary to track the internal mass shifts of the thoracic/lumbar volumes.

(or as many segments as possible). MacKinnon and Winter (1993) used a seven-segment total body estimate of the lower limbs and of the HAT to identify balance mechanisms in the frontal plane during level walking. Jian et al. (1993) reported a 3D analysis of a similar seven-segment estimate of the total body COM in conjunction with the center of pressure during initiation and termination of gait and identified the motor mechanisms responsible for that common movement.

One of the most complete measures of COM to date has been a 21-marker, 14-segment model that has been used to determine the mechanisms of balance during quiet standing (Winter et al., 1998). Figure 4.6 shows the location of the markers, and the accompanying table gives the definition of each of the 14 segments, along with mass fraction of each segment. It is worth noting that most of the segments are fairly rigid segments (head, pelvis, upper and lower limbs). However, the trunk is not that rigid, and it required four separate segments to achieve a reliable estimate, mainly because these trunk segments undergo internal mass shifts due to respiratory and cardiac functions. The validity of any COM estimate can be checked with the equation for the inverted pendulum model during the movement: $\mathrm{COP} - \mathrm{COM} = -K \cdot \mathrm{C\ddot{O}M}$ (Winter et al., 1998).

COP is the center of pressure recorded from force plate data, $\ddot{C}OM$ is the horizontal acceleration of COM in either the anterior/posterior or medial/lateral direction, and $K = I/Wh$, where I is the moment of inertia of the total body about the ankles, W is body weight, and h is the height of COM above the ankles.

4.1.7.3 Calculation of Moment of Inertia.

Example 4.6. Calculate the moment of inertia of the leg about its COM, its distal end, and its proximal end. From Table 4.1, the mass of the leg is $0.0465 \times 80 = 3.72$ kg. The leg length is given as 0.435 m. The radius of gyration/segment length is 0.302 for the COM, 0.528 for the proximal end, and 0.643 for the distal end.

$$I_0 = 3.72(0.435 \times 0.302)^2 = 0.064\,\text{kg} \cdot \text{m}^2$$

About the proximal end,

$$I_p = 3.72(0.435 \times 0.528)^2 = 0.196\,\text{kg} \cdot \text{m}^2$$

About the distal end,

$$I_d = 3.72(0.435 \times 0.643)^2 = 0.291\,\text{kg} \cdot \text{m}^2$$

Note that the moment of inertia about either end could also have been calculated using the parallel axis theorem. For example, the distance of the COM of the leg from the proximal end is $0.433 \times 0.435 = 0.188$ m, and:

$$I_p = I_0 + mx^2 = 0.064 + 3.72(0.188)^2 = 0.196\,\text{kg} \cdot \text{m}^2 0.1pc$$

Example 4.7. Calculate the moment of inertia of HAT about its proximal end and about its COM. From Table 4.1, the mass of HAT is $0.678 \times 80 = 54.24$ kg. The HAT length is given as 0.295 m. The radius of gyration about the proximal end/segment length is 1.456.

$$I_p = 54.24(0.295 \times 1.456)^2 = 10.01\,\text{kg} \cdot \text{m}^2 0.1pc$$

From Table 4.1, the COM/segment length = 1.142 from the proximal end.

$$I_0 = I_p - mx^2 = 10.01 - 54.24(0.295 \times 1.142)^2 = 3.85\,\text{kg} \cdot \text{m}^2$$

We could also use the radius of gyration/segment length about the COM = 0.903.

$$I_0 = mp^2 = 54.24(0.295 \times 0.903)^2 = 3.85\,\text{kg} \cdot \text{m}^2 0.1pc$$

4.2 DIRECT EXPERIMENTAL MEASURES

For more exact kinematic and kinetic calculations, it is preferable to have directly measured anthropometric values. The equipment and techniques that have been developed have limited capability and sometimes are not much of an improvement over the values obtained from tables.

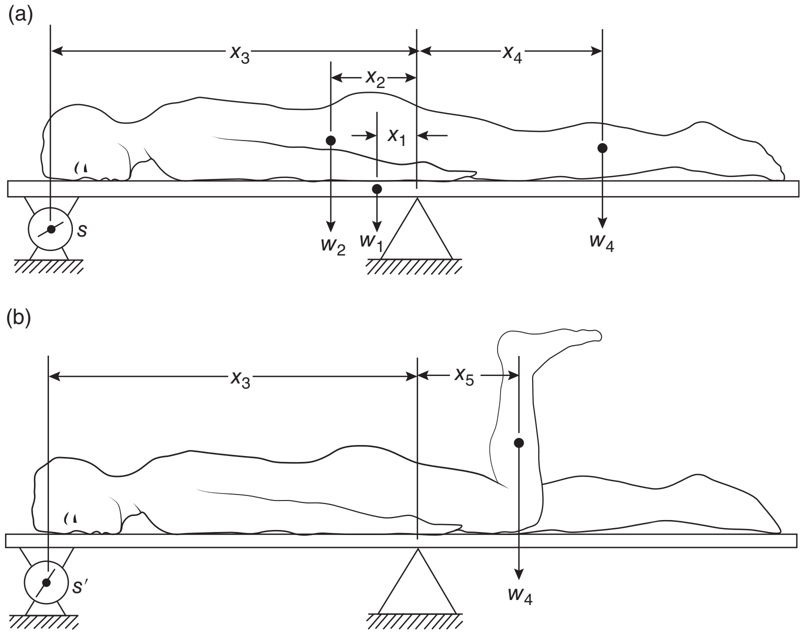


Figure 4.7 Balance board technique. (a) In vivo determination of mass of the location of the anatomical center of mass of the body. (b) Mass of a distal segment. See text for details.

4.2.1 Location of the Anatomical Center of Mass of the Body

The COM of the total body, called the *anatomical center of mass*, is readily measured using a balance board, as shown in Figure 4.7a. It consists of a rigid board mounted on a scale at one end and a pivot point at the other end, or at some convenient point on the other side of the body's COM. There is an advantage in locating the pivot as close as possible to the COM. A more sensitive scale (0–5 kg) rather than a 50- or 100-kg scale is possible, which will result in greater accuracy. It is presumed that the weight of the balance board, w_1, and its location, x_1, from the pivot are both known along with the bodyweight, w_2. With the body lying prone the scale reading is S (an upward force acting at a distance x_3 from the pivot). Taking moments about the pivot:

$$w_1 x_1 + w_2 x_2 = S x_3$$
$$x_2 = \frac{S x_3 - w_1 x_1}{w_2} \tag{4.13}$$

4.2.2 Calculation of the Mass of a Distal Segment

The mass or weight of a distal segment can be determined by the technique demonstrated in Figure 4.7b. The desired segment, here the leg and foot, is lifted to a vertical position so that its COM lies over the joint center. Prior to lifting, the COM was x_4 from the pivot point, with the scale reading S. After lifting, the leg COM is x_5 from the pivot, and the scale reading has increased to S^1. The decrease in the clockwise moment due to the leg movement is equal to the increase in the scale reaction force moment about the pivot point,

$$W_4(x_4 - x_5) = (S^1 - S)x_3$$
$$w_4 = \frac{(S^1 - S)x_3}{(x_4 - x_5)} \tag{4.14}$$

The major error in this calculation is the result of errors in x_4, usually obtained from anthropometric tables. To get the mass of the total limb, this experiment can be repeated with the subject lying on his back and the limb flexed at an angle of 90°. From the mass of the total limb, we can now subtract that of the leg and foot to get the thigh mass.

4.2.3 Moment of Inertia of a Distal Segment

The equation for the moment of inertia, described in Section 4.1.5, can be used to calculate I at a given joint center of rotation. I is the constant of proportionality that relates the joint moment to the segment's angular acceleration, assuming that the proximal segment is fixed. A method called the *quick release experiment* can be used to calculate I directly and requires the arrangement pictured in Figure 4.8. We know that $I = M/\alpha$, so if we can measure the moment M that causes an angular acceleration α, we can calculate I directly. A horizontal force F pulls on a convenient rope or cable at a distance y_1 from the joint center and is restrained by an equal and opposite force acting on a release mechanism. An accelerometer is attached to the leg at a distance y_2 from the joint center. The tangential acceleration a is related to the angular acceleration of the leg α by $a = y_2\alpha$.

With the forces in balance as shown, the leg is held in a neutral position and no acceleration occurs. If the release mechanism is actuated, the restraining force suddenly drops to zero and the net moment acting on the leg is Fy_1, which causes an instantaneous acceleration α. F and a can be recorded on a dual-beam storage oscilloscope; most pen recorders have too low a frequency response to capture the acceleration impulse. The moment of inertia can now be calculated,

$$I = \frac{M}{\alpha} = \frac{Fy_1y_2}{a} \tag{4.15}$$

Figure 4.8 shows the sudden burst of acceleration accompanied by a rapid decrease in the applied force F. This force drops after the peak of acceleration and does so because the forward displacement of the limb causes the tension to drop in the pulling cable. A convenient release mechanism can be achieved by suddenly cutting the cable or rope that holds back the leg.

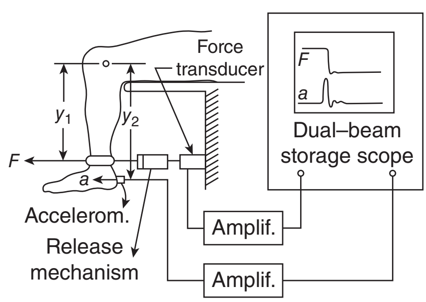


Figure 4.8 Quick-release technique for the determination of the mass moment of inertia of a distal segment. Force F applied horizontally results, after release of the segment, in an initial acceleration a. Moment of inertia can then be calculated from F, y_1, y_2.

The sudden accelerometer burst can also be used to trigger the oscilloscope sweep so that the rapidly changing force and acceleration can be captured.

More sophisticated experiments have been devised to measure more than one parameter simultaneously. Such techniques were developed by Hatze (1975) and are capable of determining the moment of inertia, the location of the COM, and the damping coefficient simultaneously.

4.2.4 Joint Axes of Rotation

Markers attached to the body are usually placed to represent our best estimate of a joint center. However, because of anatomical constraints, our location can be somewhat in error. The lateral malleolus, for example, is a common location for ankle joint markers. However, the articulation of the tibial/talus surfaces is such that the distal end of the tibia (and the fibula) move in a small arc over the talus. The true axis of rotation is actually a few centimeters distal of the lateral malleolus. Even more drastic differences are evident at some other joints. The hip joint is often identified in the sagittal plane by a marker on the upper border of the greater trochanter. However, it is quite evident that the marker is somewhat more lateral than the center of the hip joint such that internal and external rotations of the thigh relative to the pelvis may cause considerable errors, as will abduction/adduction at that joint.

Thus, it is important that the true axes of rotation be identified relative to anatomical markers that we have placed on the skin. There are two primary methods to estimate joint centers, predictive and functional models. In a predictive model, a regression equation derived from medical imaging is used to estimate the location of the internal joint center based on the location of one or more surface markers on the segment. For example, Harrington et al. (2007) used created regression equations based on the subject characteristics of pelvic width, pelvic depth, and leg length to estimate the 3D locations of the hip joint center using markers placed on anatomical landmarks. In the functional model, joint center is estimated based on the movement of markers on adjacent segments. There are a variety of functional models, and the specifics of these techniques are in part dependent on the degrees of freedom of the joint of interest. Geometric sphere fitting is one such functional method that is commonly applied to determine the center of the hip joint (Leardini et al., 1999). In this method, the subject has markers attached to the pelvis and along the thigh. The subject moves the limb through sequential flexion and extension, abduction and adduction, and circumduction to create a spherical path of the thigh markers. The path of the markers is tracked and the center of this motion is calculated and defined as the hip joint center. Geometric sphere fitting algorithms may be best applied to the ball and socket motion of the hip joint, whereas a more constrained approach may better fit the knee or ankle joint.

One example of this functional technique is described here. Figure 4.9 shows two segments in a planar movement. First, they must be translated and rotated in space so that one segment is fixed in space and the second rotates as shown. At any given instant in time, the true axis of rotation is at (x_c, y_c) within the fixed segment, and we are interested in the location of (x_c, y_c) relative to anatomical coordinates (x_3, y_3) and (x_4, y_4) of that segment. Markers (x_1, y_1) and (x_2, y_2) are located as shown; (x_1, y_1) has an instantaneous tangential velocity V and is located at a radius R from the axis of rotation. From the line joining (x_1, y_1) to (x_2, y_2), we calculate the angular velocity of the rotating segment $\overline{\omega}_z$. With one segment fixed in space, $\overline{\omega}_z$ is nothing more than the joint angular velocity,

$$\overline{V} = \overline{\omega}_z \times \overline{R} \tag{4.16a}$$

or, in Cartesian coordinates,

$$V_x\hat{i} + V_y\hat{j} = \left(R_y\omega_z\right)\hat{i} - \left(R_x\omega_z\right)\hat{j}$$

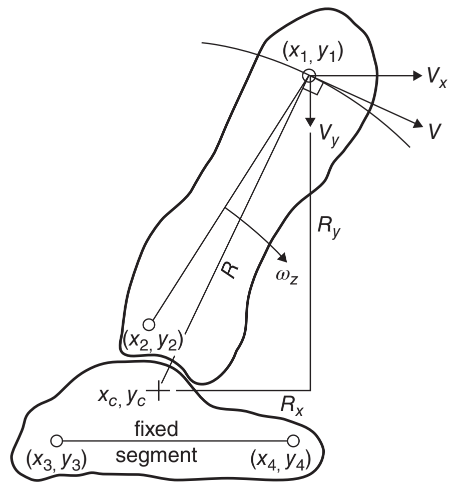


Figure 4.9 Technique used to calculate the axis of rotation between two adjacent segments x_c, y_c. Each segment must have two markers in the plane of movement. After data collection, the segments are rotated and translated so that one segment is fixed in space. Thus, the moving-segment kinematics reflects the relative movement between the two segments, and the axis of rotation can be located relative to the anatomical location of markers on that fixed segment. See text for complete details.

Therefore,

$$V_x = R_y\omega_z \quad \text{and} \quad V_y = -R_x\omega_z \tag{4.16b}$$

Since V_x, V_y, and ω_z can be calculated from the marker trajectory data, R_y and R_x can be determined. Since x_1, y_1 is known, the axis of rotation x_c, y_c can be calculated. Care must be taken when ω_z approaches 0 or reverses its polarity, because R, as calculated by Equation (4.16a), becomes indeterminate or falsely approaches very large values. In practice, we have found errors become significant when ω_z falls below 0.5 r/s.

Currently, the International Society of Biomechanics recommends the use of functional algorithms for individuals with sufficient range of motion to perform the necessary calibration movements. For individuals without adequate motion, predictive models are recommended (Wu et al., 2002).

4.3 MUSCLE ANTHROPOMETRY

Before we can calculate the forces produced by individual muscles during normal movement, we usually need some dimensions from the muscles themselves. If muscles of the same group share the load, they probably do so proportionally to their relative cross-sectional areas. Also, the mechanical advantage of each muscle can be different, depending on the moment arm length at its origin and insertion, and on other structures beneath the muscle or tendon that alter the angle of pull of the tendon.

4.3.1 Cross-Sectional Area of Muscles

The functional or physiologic cross-sectional area (PCA) of a muscle is a measure of the number of sarcomeres in parallel with the angle of pull of the muscles. In pennate muscles, the fibers act at an angle from the long axis and, therefore, are not as muscle. The angle between the long axis of the muscle and the fiber angle is called *pennation angle*. In parallel-fibered muscle, the PCA is:

$$\text{PCA} = \frac{m}{dl}\text{cm}^2 \tag{4.17}$$

where

m = mass of muscle fibers, g
d = density of muscle, g/cm^3, =1.056 g/cm^3
l = length of muscle fibers, cm.

In pennate muscles, the physiological cross-sectional area becomes:

$$\text{PCA} = \frac{m\cos\theta}{dl}\text{cm}^2 \tag{4.18}$$

where θ is the pennation angle, which increases as the muscle shortens.

Wickiewcz et al. (1983), using data from three cadavers, measured muscle mass, fiber lengths, and pennation angle for 27 muscles of the lower extremity. Representative values are given in Table 4.3. The PCA as a percentage of the total cross-sectional area of all muscles crossing a given joint is presented in Table 4.4. In this way, the relative potential contribution of a group of agonist muscles can be determined, assuming that each is generating the same stress. Note that a double-joint muscle, such as the gastrocnemius, may represent different percentages at different joints because of the different total PCA of all muscles crossing each joint.

4.3.2 Change in Muscle Length During Movement

A few studies have investigated the changes in the length of muscles as a function of the angles of the joints they cross. Grieve et al. (1978), in a study on eight cadavers, reported percentage length changes of the gastrocnemius muscle as a function of the knee and ankle angle. The resting length of the gastrocs was assumed to be when the knee was flexed 90° and the ankle was in an intermediate position, neither plantarflexed nor dorsiflexed. With 40° plantarflexion, the muscle shortened 8.5% and linearly changed its length to a 4% increase at 20° dorsiflexion. An almost linear curve described the changes at the knee: 6.5% at full extension to a 3% decrease at 150° flexion.

TABLE 4.3 Mass, Length, and PCA of Some Muscles

Muscle	Mass (g)	Fiber Length (cm)	PCA (cm^2)	Pennation Angle (°)
Sartorius	75	38	1.9	0
Biceps femoris (long)	150	9	15.8	0
Semitendinosus	75	16	4.4	0
Soleus	215	3.0	58	30
Gastrocnemius	158	4.8	30	15
Tibialis posterior	55	2.4	21	15
Tibialis anterior	70	7.3	9.1	5
Rectus femoris	90	6.8	12.5	5
Vastus lateralis	210	6.7	30	5
Vastus medialis	200	7.2	26	5
Vastus intermedius	180	6.8	25	5

TABLE 4.4 Four Percent PCA of Muscles Crossing Ankle, Knee, and Hip Joints

Ankle		Knee		Hip	
Muscle	%PCA	Muscle	%PCA	Muscle	%PCA
Soleus	41	Gastrocnemius	19	Iliopsoas	9
Gastrocnemius	22	Biceps femoris (small)	3	Sartorius	1
Flexor hallucis longus	6	Biceps femoris (long)	7	Pectineus	1
Flexor digitorum longus	3	Semitendinosus	3	Rectus femoris	7
Tibialis posterior	10	Semimembranosus	10	Gluteus maximus	16
Peroneus brevis	9	Vastus lateralis	20	Gluteus medius	12
Tibialis anterior	5	Vastus medialis	15	Gluteus minimus	6
Extensor digitorum longus	3	Vastus intermedius	13	Adductor magnus	11
Extensor hallucis longus	1	Rectus femoris	8	Adductor longus	3
		Sartorius	1	Adductor brevis	3
		Gracilis	1	Tensor fasciae latae	1
				Biceps femoris (long)	6
				Semitendinosus	3
				Semimembranosus	8
				Piriformis	2
				Lateral rotators	13

4.3.3 Force per Unit Cross-Sectional Area (Stress)

A wide range of stress values for skeletal muscles has been reported (Haxton, 1944; Alexander and Vernon, 1975; Maughan et al., 1983). Most of these stress values were measured during isometric conditions and range from 20 to 100 N/cm^2. These higher values were recorded in pennate muscles, which are those whose fibers lie at an angle from the main axis of the muscle. Such an orientation effectively increases the cross-sectional area above that measured and used in the stress calculation. Haxton (1944) related force to stress in two pennate muscles (gastrocs and soleus) and found stresses as high as 38 N/cm^2. Dynamic stresses have been calculated in the quadriceps during running and jumping to be about 70 N/cm^2 (based on a peak knee extensor moment of 210 Nm in adult males) and about 100 N/cm^2 in isometric maximum voluntary contractions (MVCs) (Maughan et al., 1983).

4.3.4 Mechanical Advantage of Muscle

The origin and insertion of each muscle define the angle of pull of the tendon on the bone and, therefore, the mechanical leverage it has at the joint center. Each muscle has its unique moment arm length, which is the length of a line normal to the muscle passing through the joint center. This moment arm length changes with the joint angle. One of the first studies done in this area (Smidt, 1973) reported the average moment arm length (26 subjects) for the knee extensors and for the hamstrings acting at the knee. Both these muscle groups showed an increase in the moment length as the knee was flexed, reaching a peak at 45°, then decreasing again as flexion increased to 90°. As musculoskeletal imaging has improved, researchers are able to measure in vivo moment arms during dynamic activities. This is an important consideration as muscle force during activity may affect the distance between its line of action and center of joint rotation. Accurate estimation of moment arm length throughout the course of range of motion and during active muscle contraction is a key component for simulation studies, as discussed in detail in Chapter 10.

4.3.5 Multijoint Muscles

A large number of the muscles in the human body pass over more than one joint. In the lower limbs, the hamstrings are extensors of the hip and flexors of the knee, the rectus femoris is a combined hip flexor and knee extensor, and the gastrocnemius is knee flexors and ankle plantarflexors. The fiber length of many of these muscles may be insufficient to allow a complete range of movement of both joints involved. Elftman (1966) has suggested that many normal movements require lengthening at one joint simultaneously with shortening at the other. Consider the action of the rectus femoris, for example, during early swing in running. This muscle shortens as a result of hip flexion and lengthens at the knee as the leg swings backward in preparation for swing. The tension in the rectus femoris simultaneously creates a flexor hip moment (positive work) and an extensor knee moment to decelerate the swinging leg (negative work) and start accelerating it forward. In this way, the net change in muscle length is reduced compared with 2 equivalent single-joint muscles, and excessive positive and negative work within the muscle can be reduced. A double-joint muscle could even be totally isometric in such situations and would effectively be transferring energy from the leg to the pelvis in the example just described. In running during the critical push-off phase, when the plantarflexors are generating energy at a high rate, the knee is continuing to extend. Thus, the gastrocnemii may be essentially isometric (they may appear to be shortening at the distal end and lengthening at the proximal end). Similarly, toward the end of swing in running, the knee is rapidly extending while the hip has reached full flexion and is beginning to reverse (i.e. it has an extensor velocity). Thus, the hamstrings appear to be rapidly lengthening at the distal end and shortening at the proximal end, with the net result that they may be lengthening at a slower rate than a single joint would.

It is also critical to understand the role of the major biarticulate muscles of the lower limb during stance phase of walking or running. Figure 4.10 shows the gastrocnemii, hamstrings, and rectus femoris and their moment-arm lengths at their respective proximal and distal ends. The hamstrings have a 5-cm moment-arm at the ankle and 3.5-cm moment-arm at the knee. Thus,

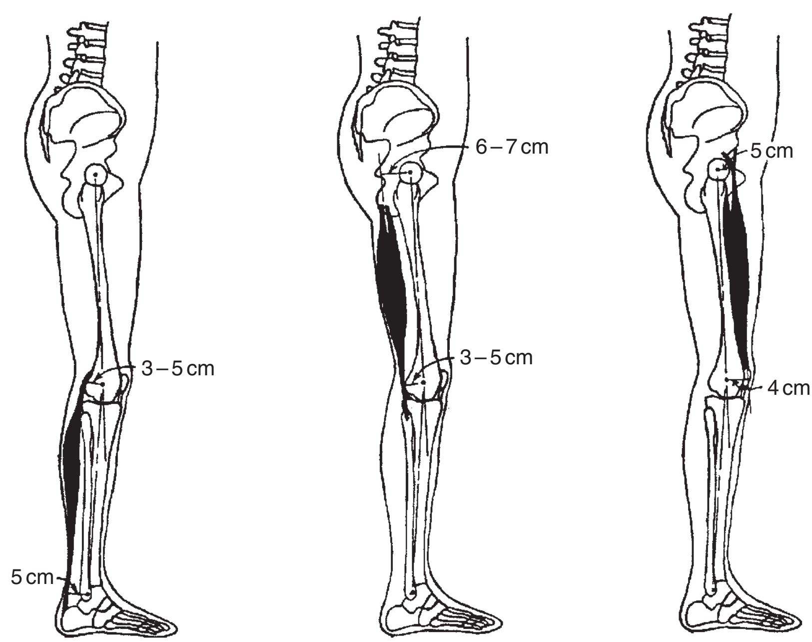


Figure 4.10 Three major biarticulate muscles of the lower limb. Shown are the gastrocnemii, hamstrings, and rectus femoris, and their moment-arm lengths at their proximal and distal ends. These moment-arms are critical to the functional role of these muscle groups during weight bearing;

when they are active during stance, their contribution to the ankle extensor moment is about 50% greater than their contribution to the knee flexor moment. The net effect of these two contributions is to cause the leg to rotate posteriorly and prevent the knee from collapsing. The hamstrings, with the exception of the short head of the biceps femoris, have moment-arms of 6–7 cm at the hip but only 3.5 cm at the knee. Thus, when these muscles are active during stance, their contribution to hip extension is about twice their contribution to knee flexion. The net effect of these two actions is to cause the thigh to rotate posteriorly and prevent the knee from collapsing. Finally, the rectus femoris is the only biarticulate muscle of the large quadriceps group, and its moment-arm at the hip is slightly larger than at the knee. However, the quadriceps activate as a group, and because the uniarticulate quadriceps comprise 84% of the PCA of the quadriceps (see Table 4.3), the dominant action is knee extension. Thus, the net effect of the major biarticulate muscles of the lower limb is extension at all three joints, and therefore they contribute, along with all the uniarticulate extensors, to defending against a gravity-induced collapse. The algebraic summation of all three moments during stance phase of gait has been calculated and has been found to be dominantly extensor (Winter, 1984). This summation has been labeled the support moment and is discussed further in Section 11.1.

4.4 PROBLEMS BASED ON ANTHROPOMETRIC DATA

1. (a) Calculate the average body density of a young adult whose height is 1.68 m and whose mass is 68.5 kg. *Answer:* 1.059 kg/L.

(b) For the adult in (a), determine the density of the forearm and use it to estimate the mass of the forearm that measures 24.0 cm from the ulnar styloid to the elbow axis. Circumference measures (in cm) taken at 1-cm intervals starting at the wrist are 20.1, 20.3, 20.5, 20.7, 20.9, 21.2, 21.5, 21.9, 22.5, 23.2, 23.9, 24.6, 25.1, 25.7, 26.4, 27.0, 27.5, 27.9, 28.2, 28.4, 28.4, 28.3, 28.2, and 28.0. Assuming the forearm to have a circular cross-sectional area over its entire length, calculate the volume of the forearm and its mass. Compare the mass as calculated with that estimated using averaged anthropometric data (Table 4.1). *Answer:* Forearm density = 1.13 kg/L; volume = 1.174 L; mass = 1.33 kg. Mass calculated from Table 4.1 = 1.10 kg.

(c) Calculate the location of the COM of the forearm along its long axis and give its distance from the elbow axis. Compare that with the COM as determined from Table 4.1. *Answer:* COM = 10.34 cm from the elbow; from Table 4.1, COM = 10.32 cm from the elbow.

(d) Calculate the moment of inertia of the forearm about the elbow axis. Then calculate its radius of gyration about the elbow and compare it with the value calculated from Table 4.1. *Answer:* $I_p = 0.0201\ \text{kg}\cdot\text{m}^2$; radius of gyration = 12.27 cm; radius of gyration from Table 4.1 = 12.62 cm.

2. (a) From the data listed in Table A.3b in Appendix A, calculate the COM of the lower limb for frame 70. *Answer:* $x = 1.755$ m; $y = 0.522$ m.

(b) Using your data of stride length (Problem 2(f) in Section 3.8) and a stride time of 68 frames, create an estimate (assuming symmetrical gait) of the coordinates of the left half of the body for frame 30. From the segment centers of mass (Table A.4), calculate the COM of the right half of the body (foot + leg + high + 1/2 HAT), and of the left half of the body (using segment data suitably shifted in time and space). Average the two centers of mass to get the COM of the total body for frame 30. *Answer:* $x = 1.025$ m, $y = 0.904$ m.

3. (a) Calculate the moment of inertia of the HAT about its COM for the subject described in Appendix A. *Answer:* $I_0 = 1.96\ \text{kg}\cdot\text{m}^2$.

(b) Assuming that the subject is standing erect with the two feet together, calculate the moment of inertia of HAT about the hip joint, the knee joint, and the ankle joint. What does the relative size of these moments of inertia tell us about the relative magnitude of the joint moments required to control the inertial load of HAT. *Answer:* $I_h = 5.09\ \text{kg} \cdot \text{m}^2$, $I_k = 15.78\ \text{kg} \cdot \text{m}^2$, $I_a = 42.31\ \text{kg} \cdot \text{m}^2$.

(c) Assuming that the COM of the head is 1.65 m from the ankle, what percentage does it contribute to the moment of inertia of HAT about the ankle? *Answer:* I_{head} about ankle $=12.50\ \text{kg} \cdot \text{m}^2$, which is 29.6% of I_{hat} about the ankle.

4. (a) Calculate the moment of inertia of the lower limb of the subject in Appendix A about the hip joint. Assume that the knee is not flexed and the foot is a point mass located 6 cm distal to the ankle. *Answer:* $I_{\text{lower limb}}$ about hip $=1.39\ \text{kg} \cdot \text{m}^2$.

(b) Calculate the increase in the moment of inertia as calculated in (a) when a ski boot is worn. The mass of the ski boot is 1.8 kg, and assume it to be a point mass located 1 cm distal to the ankle. *Answer*: I_{boot} about hip $= 1.01\ \text{kg} \cdot \text{m}^2$.

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5

KINETICS: FORCES AND MOMENTS OF FORCE

Stephen J. Thomas[1], Joseph A. Zeni[2], and David A. Winters[3,†]

[1]*Thomas Jefferson University, Philadelphia, PA, USA*
[2]*Rutgers University, Newark, NJ, USA*
[3]*University of Waterloo, Waterloo, Ontario, Canada*

5.0 BIOMECHANICAL MODELS

Chapter 3 dealt at length with the movement itself, without regard to the forces that cause the movement. The study of these forces and the resultant energetics is called *kinetics*. Knowledge of the patterns of the forces is necessary for an understanding of the cause of any movement. Transducers have been developed that can be implanted surgically to measure the force exerted by a muscle at the tendon. However, such techniques have applications only in animal experiments and, even then, only to a limited extent. It, therefore, remains that we attempt to calculate these forces indirectly, using readily available kinematic and anthropometric data. The process by which the reaction forces and muscle moments are calculated is called *link-segment modeling*. Such a process is depicted in Figure 5.1. If we have a full kinematic description, accurate anthropometric measures, and the external forces, we can calculate the joint reaction forces and muscle moments. This prediction is called an *inverse solution* and is a very powerful tool in gaining insight into the net summation of all muscle activity at each joint. Such information is very useful to the surgeon, therapist, athletic trainer, kinesiologist, exercise scientist, and coach in their diagnostic assessments. The effect of training, therapy, or surgery is extremely evident at this level of assessment, although it is often obscured in the original kinematics.

5.0.1 Link-Segment Model Development

Since we are not able to measure the joint forces and moments of force directly, a model is required to calculate them indirectly. The validity of any assessment is only as good as the model itself. Accurate measures of segment masses, centers of mass (COMs), joint centers, and moments of inertia are required. Such data can be obtained from statistical tables based on the person's height, weight, and, sometimes, sex, as was detailed in Chapter 4. A limited number of these variables can be measured directly, but some of the techniques are time-consuming and

[†]Deceased.

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

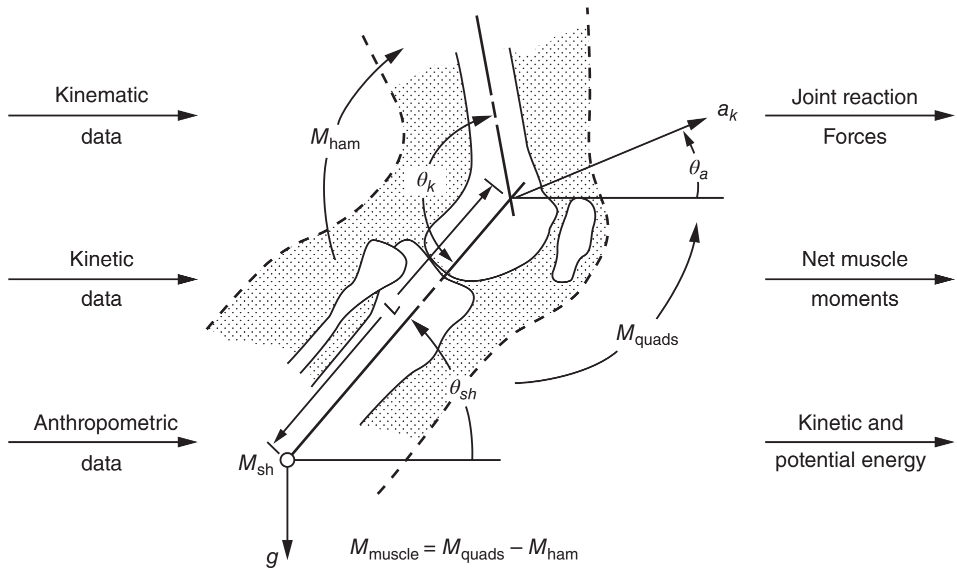

Figure 5.1 Schematic diagram of the relationship between kinematic, kinetic, and anthropometric data and the calculated forces, moments, energies, and powers using an inverse solution of a link-segment model.

have limited accuracy. Regardless of the source of the anthropometric data, the following assumptions are made with respect to the model:

1. Each segment has a fixed mass located as a point mass at its COM (which will be the center of gravity in the vertical direction).
2. The location of each segment's COM remains fixed during the movement.
3. The joints are considered to be hinge (or ball-and-socket) joints.
4. The mass moment of inertia of each segment about its mass center (or about either proximal or distal joints) is constant during the movement.
5. The length of each segment remains constant during the movement (e.g. the distance between hinge or ball-and-socket joints remains constant).

Figure 5.2 shows the equivalence between the anatomical and the link-segment models for the lower limb. The segment masses m_1, m_2, and m_3 are considered to be concentrated at points. The distance from the proximal joint to the mass centers is considered to be fixed, as are the length of the segments and each segment's moment of inertia I_1, I_2, and I_3 about each COM.

5.0.2 Forces Acting on the Link-Segment Model

1. *Gravitational Forces*. The forces of gravity act downward through the COMs of each segment and are equal to the magnitude of the mass times acceleration due to gravity (normally 9.8 m/s^2).
2. *Ground Reaction or External Forces*. Any external forces must be measured by a force transducer. Such forces are distributed over an area of the body (such as the ground reaction forces under the area of the foot). In order to represent such forces as vectors, they must be considered to act at a point that is usually called the center of pressure (COP). A suitably constructed force plate, for example, yields signals from which the COP can be calculated.

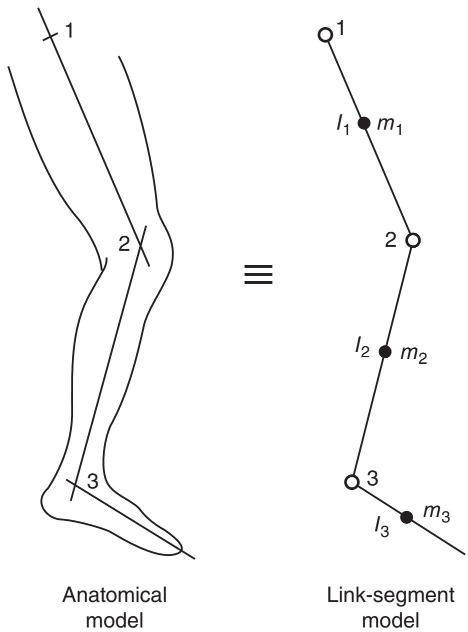

Figure 5.2 Relationship between anatomical and link-segment models. Joints are replaced by hinge (pin) joints and segments are replaced by masses and moments of inertia located at each segment's center of mass.

3. *Muscle and Ligament Forces*. The net effect of muscle activity at a joint can be calculated in terms of net muscle moments. If a co-contraction is taking place at a given joint, the analysis yields only the net effect of both agonist and antagonistic muscles. Also, any friction effects at the joints or within the muscle cannot be separated from this net value. Increased friction merely reduces the effective "muscle" moment; the muscle contractile elements themselves are actually creating moments higher than that analyzed at the tendon. However, the error at low and moderate speeds of movement is usually only a few percent. At the extreme range of movement of any joint, passive structures such as joint capsule and ligaments come into play to contain the range. The moments generated by these tissues will add to or subtract from those generated by the muscles. Thus, unless the muscle is silent, it is impossible to determine the contribution of these passive structures.

5.0.3 Joint Reaction Forces and Bone-on-Bone Forces

The three forces described in the preceding sections constitute all the forces acting on the total body system itself. However, our analysis examines the segments one at a time and, therefore, must calculate the reaction between segments that occur within the joints (Paul, 1966). A free-body diagram of each segment is required, as shown in Figure 5.3. Here the original link-segment model is broken into its segmental parts. For convenience, we make the break at the joints and the forces that act across each joint must be shown on the resultant free-body diagram. This procedure now permits us to look at each segment and calculate all unknown joint reaction forces. In accordance with Newton's 3rd law, there is an equal and opposite force acting at each hinge joint in our model. For example, when a leg is held off the ground in a static condition, the foot is exerting a downward force on the joint capsule, tendons, and ligaments crossing the ankle joint. This is seen as a downward force acting on the leg equal to the weight of the foot. Likewise, those passive joint tissues are exerting an equal upward force on

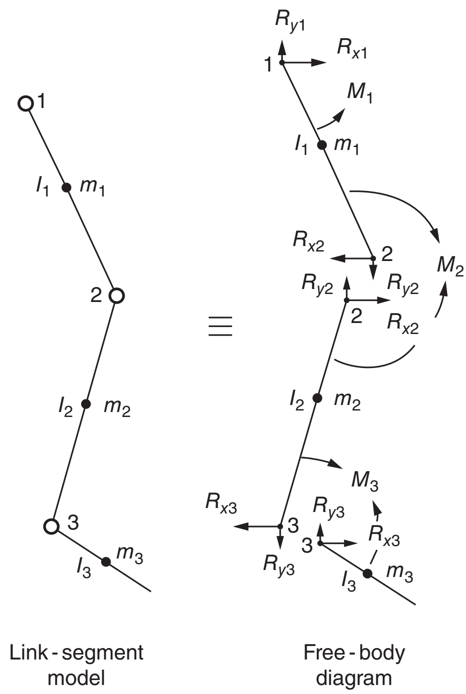

Figure 5.3 Relationship between the free-body diagram and the link-segment model. Each segment is "broken" at the joints, and the reaction forces and moments of force acting at each joint are indicated.

Considerable confusion exists regarding the relationship between joint reaction forces and joint bone-on-bone forces. The latter forces are the actual forces seen across the articulating surfaces and include the effect of muscle activity and the action of joint capsules and ligaments. Actively contracting muscles pull the articulating surfaces together, creating compressive forces and, sometimes, shear forces. In addition, when passive joint structures reach their maximal length they can also create compressive and shear forces. In the simplest situation, the bone-on-bone force equals the active force due to the muscles and passive structures plus the joint reaction forces. Figure 5.4 illustrates these differences. Case 1 has the lower segment with a weight of 100 N hanging passively from the muscles originating in the upper segment. The two muscles are not contracting but are, assisted by joint capsule and ligamentous tissue, pulling upward with an equal and opposite 100 N. The link-segment model shows these equal and opposite reaction forces. The bone-on-bone force is zero, indicating that the joint articulating surfaces are under neither tension nor compression. In case 2, there is an active contraction of the muscles, so that the total upward force is now 170 N. The bone-on-bone force is 70 N compression. This means that a force of 70 N exists across the articulating surfaces. As far as the lower segment is concerned, there is still a net reaction force of 100 N upward (170 N upward through muscles and 70 N downward across the articulating surfaces). The lower segment is still acting downward with a force of 100 N; thus, the free-body diagram remains the same. Generally, the anatomy is not as simple as depicted. More than one muscle is usually active on each side of the joint, so it is difficult to apportion the forces among the muscles. Also, the exact angle of pull of each tendon, joint capsule, and ligament in addition to the geometry of the articulating surfaces is not always readily available and change throughout the joints range of motion. Thus, a much more complex free-body diagram must be used, and the techniques and examples are described in Section 5.3.

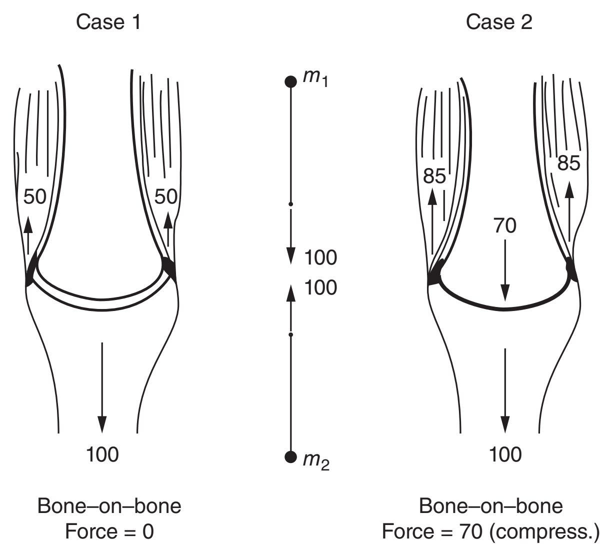

Figure 5.4 Diagrams to illustrate the difference between joint reaction forces and bone-on-bone forces. In both cases, the reaction force is 100 N acting upward on M2 and downward on M1. With no muscle activity (case 1), the bone-on-bone force is zero; with muscle activity (case 2) it is 70 N.

5.1 BASIC LINK-SEGMENT EQUATIONS – THE FREE-BODY DIAGRAM

Each body segment acts independently under the influence of reaction forces and muscle moments, which act at either end, plus the forces due to gravity (Bresler and Frankel, 1950). Consider the planar movement of a segment in which the kinematics, anthropometrics, and reaction forces at the distal end are known (see Figure 5.5).

Known

a_x, a_y = acceleration of segment COM
θ = angle of segment in plane of movement
α = angular acceleration of segment in plane of movement

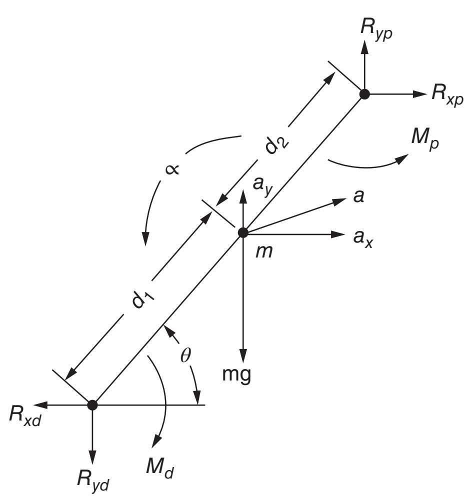

Figure 5.5 Complete free-body diagram of a single segment, showing reaction and gravitational forces, net moments of force, and all linear and angular accelerations.

R_{xd}, R_{yd} = reaction forces acting at distal end of segment, usually determined from a prior analysis of the proximal forces acting on distal segment
M_d = net muscle moment acting at distal joint, usually determined from an analysis of the proximal muscle acting on distal segment

Unknown

R_{xp}, R_{yp} = reaction forces acting at proximal joint
M_p = net muscle moment acting on segment at proximal joint

Equations

1. $\Sigma F_x = ma_x$

$$R_{xp} - R_{xd} = ma_x \tag{5.1}$$

2. $\Sigma F_y = ma_y$

$$R_{yp} - R_{yd} - mg = ma_y \tag{5.2}$$

3. About the segment COM, $\Sigma M = I_0\alpha$ (5.3)

Note that the muscle moment at the proximal end cannot be calculated until the proximal reaction forces R_{xp} and R_{yp} have first been calculated.

Example 5.1 (see Figure 5.6). In a static situation, a person is standing on one foot on a force plate. The ground reaction force is found to act 4 cm anterior to the ankle joint. Note that convention has the ground reaction force R_{y1} always acting upward. We also show the horizontal reaction force R_{x1} to be acting in the positive direction (to the right). If this force actually acts to the left, it will be recorded as a negative number. The subject's mass is 60 kg, and the mass of the foot is 0.9 kg. Calculate the joint reaction forces and net muscle moment at the ankle. R_{y1} = body weight = $60 \times 9.8 = 588\ N$.

1. $\Sigma F_x = ma_x$,

$$R_{x2} + R_{x1} = ma_x = 0$$

Note that this is a redundant calculation in static conditions.

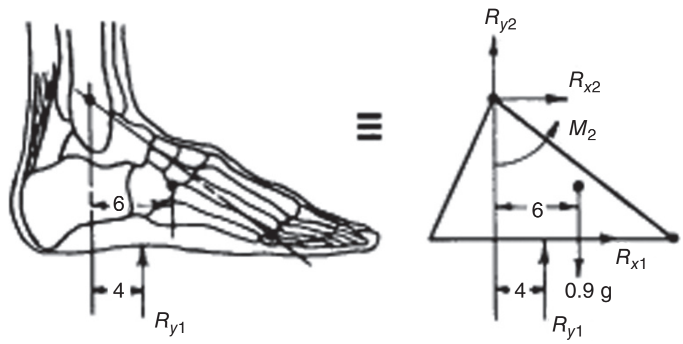

Figure 5.6 Anatomical and free-body diagram of foot during weight bearing.

2. $\Sigma F_y = ma_y$,

$$R_{y2} + R_{y1} - mg = ma_y$$

$$R_{y2} + 588 - 0.9 \times 9.8 = 0$$

$$R_{y2} = -579.2\,N$$

The negative sign means that the force acting on the foot at the ankle joint acts *downward*. This is not surprising because the entire body weight, less that of the foot, must be acting downward on the ankle joint.

3. About the COM, $\Sigma M = I_0\alpha$,

$$M_2 - R_{y1} \times 0.02 - R_{y2} \times 0.06 = 0$$

$$M_2 = 588 \times 0.02 + (-579.2 \times 0.06) = -22.99\,\mathrm{N \cdot m}$$

The negative sign means that the real direction of the muscle moment acting on the foot at the ankle joint is clockwise, which means that the plantarflexors are active at the ankle joint to maintain the static position. These muscles have created an action force that resulted in the ground reaction force that was measured, and whose COP was 4 cm anterior to the ankle joint.

Example 5.2 (see Figure 5.7). From the data collected during the swing of the foot, calculate the muscle moment and reaction forces at the ankle. The subject's mass was 80 kg and the ankle-metatarsal length was 20.0 cm. From Table 4.1, the inertial characteristics of the foot are calculated:

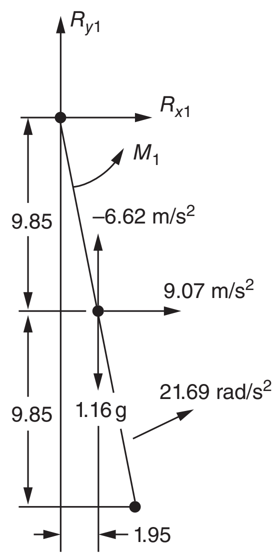

Figure 5.7 Free-body diagram of foot during swing showing the linear accelerations of the center of mass and the angular acceleration of the segment. Distances are in centimeters. Three unknowns, R_{x1}, R_{y1}, and M1, are to be calculated assuming a positive direction as shown.

$$m = 0.0145 \times 80 = 1.16\,\text{kg}$$
$$\rho_0 = 0.475 \times 0.20 = 0.095\,\text{m}$$
$$I_0 = 1.16\,(0.095)^2 = 0.0105\,\text{kg·m}^2$$
$$\alpha = 21.69\,\text{rad/s}^2$$

1. $\Sigma F_x = ma_x$,

   $$R_{x1} = 1.16 \times 9.07 = 10.52\,N$$

2. $\Sigma F_y = ma_y$,

   $$R_{y1} - 1.16\text{g} = m(-6.62)$$
   $$R_{y1} = 1.16 \times 9.8 - 1.16 \times 6.62 = 3.69\,N$$

3. At the COM of the foot, $\Sigma M = I_0\alpha$,

   $$M_1 - R_{x1} \times 0.0985 - R_{y1} \times 0.0195 = 0.0105 \times 21.69$$
   $$\begin{aligned} M1 &= 0.0105 \times 21.69 + 10.52 \times 0.0985 + 3.69 \times 0.0195 \\ &= 0.23 + 1.04 + 0.07 = 1.34\,\text{N} \cdot \text{m} \end{aligned}$$

Discussion

1. The horizontal reaction force of 10.52 N at the ankle is the cause of the horizontal acceleration that we calculated for the foot.
2. The foot is decelerating its upward rise at the end of lift-off. Thus, the vertical reaction force at the ankle is somewhat less than the static gravitational force.
3. The ankle muscle moment is positive, indicating net dorsiflexor activity (tibialis anterior), and most of this moment (1.04 out of 1.34 N · m) is required to cause the horizontal acceleration of the foot's center of gravity, with very little needed (0.23 N · m) to angularly accelerate the low moment of inertia of the foot.

Example 5.3 (see Figure 5.8). For the same instant in time, calculate the muscle moments and reaction forces at the knee joint. The leg segment was 43.5 cm long.

$$m = 0.0465 \times 80 = 3.72\,\text{kg}$$
$$\rho_0 = 0.302 \times 0.435 = 0.131\,\text{m}$$
$$I_0 = 3.72\,(0.131)^2 = 0.0638\,\text{kg·m}^2$$
$$\alpha = 36.9\,\text{rad/s}^2$$

From Example 5.2, $R_{x1} = 10.52\ N$, $R_{y1} = 3.69\ N$, and $M_1 = 1.34 N \cdot m$

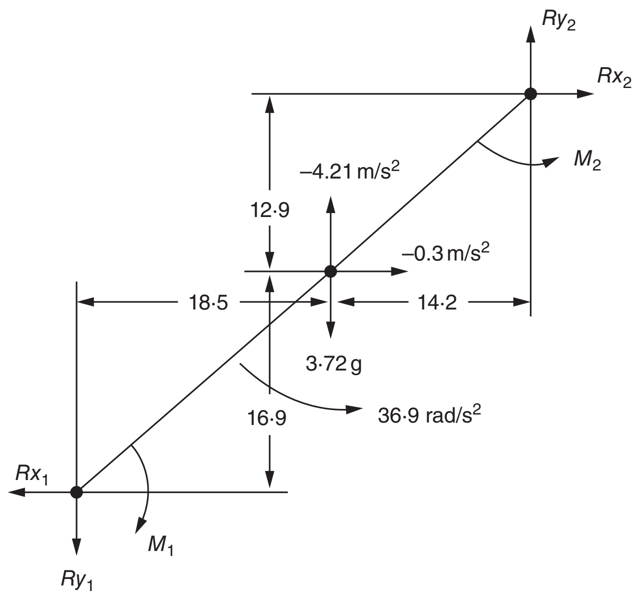

Figure 5.8 Free-body diagram of leg at the same instant in time as the foot in Figure 5.7. Linear and angular accelerations are as shown. Distances are in centimeters. The distal end and reaction forces and moments have been reversed, recognizing Newton's third law. Again, the three unknowns R_{x2}, R_{y2}, and M2 are assumed to be positive with direction as indicated.

1. $\Sigma F_x = ma_x$,

   $R_{x2} - R_{x1} = ma_x$

   $R_{x2} = 10.52 + 3.72(-0.03) = 10.41N$

2. $\Sigma F_y = ma_y$,

   $R_{y2} - R_{y1} - mg = ma_y$

   $R_{y2} = 3.69 + 3.72 \times 9.8 + 3.72(-4.21) = 24.48N$

3. About the COM of the leg, $\Sigma M = I\alpha$,

   $$M_2 - M_1 - 0.169R_{x1} + 0.185\,R_{y1} - 0.129\,R_{x2} + 0.142\,R_{y2} = I\alpha$$

   $$M_2 = 1.34 + 0.169 \times 10.52 - 0.185 \times 3.69 + 0.129 \times 10.41$$
   $$-0.142 \times 24.48 + 0.0638 \times 36.9$$
   $$= 1.34 + 1.78 - 0.68 + 1.34 - 3.48 + 2.35 = 2.65\ \mathrm{N{\cdot}m}$$

Discussion

1. M_2 is positive. This represents a counterclockwise (extensor) moment acting at the knee. The quadricep muscles at this time are rapidly extending the swinging leg.
2. The angular acceleration of the leg is the net result of two reaction forces and one muscle moment acting at each end of the segment. Thus, there may not be a single primary force causing the movement we observe. In this case, each force and moment had a significant influence on the final acceleration.

5.2 FORCE TRANSDUCERS AND FORCE PLATES

In order to measure the force exerted by the body on an external body or load, we need a suitable force-measuring device. Such a device, called a *force transducer*, gives an electrical signal proportional to the applied force. There are many kinds available: strain gauge, piezoelectric, piezoresistive, capacitive, and others. All these work on the principle that the applied force causes a certain amount of strain within the transducer. For the strain gauge type, a calibrated metal plate or beam within the transducer undergoes a very small change (strain) in one of its dimensions. This mechanical deflection, usually a fraction of 1%, causes a change in resistances connected as a bridge circuit (see Chapter 3), resulting in an unbalance of voltages proportional to the force. Piezoelectric and piezoresistive types require minute deformations of the atomic structure within a block of special crystalline material. Quartz, for example, is a naturally found piezoelectrical material, and deformation of its crystalline structure changes the electrical characteristics such that the electrical charge across appropriate surfaces of the block is altered and can be translated via suitable electronics to a signal proportional to the applied force. Piezoresistive types exhibit a change in resistance which, like the strain gauge, upset the balance of a bridge circuit.

5.2.1 Multidirectional Force Transducers

In order to measure forces in two or more directions, it is necessary to use a bi- or tri-directional force transducer. Such a device is nothing more than two or more force transducers mounted at right angles to each other. The major problem is to ensure that the applied force acts through the central axis of each of the individual transducers. As discussed previously there are several technologies used to measure force. The two most commonly used for commercial force plates are strain gauges and piezoelectric crystals. Both technologies provide very accurate force measures important for inverse solutions in biomechanics; however, it is important to note the advantages and disadvantages of both technologies. This information will allow the user to identify the force measuring technology most appropriate for the activities being studied.

Strain gauge force transducers

Advantages

- No drift over long data collections (i.e. postural stability)
- High sampling rate

Disadvantages

- Mid amplitude range
- Ballistic activities

Piezoelectric crystals force transducers

Advantages

- High amplitude range
- High sampling rate
- Ballistic activities

Disadvantages

- Drift over long data collections (i.e. postural stability)

5.2.2 Force Plates

The most common force acting on the body is the ground reaction force, which acts on the foot during standing, walking, or running. This force vector is three-dimensional (3D) and consists of a vertical component plus two shear components acting along the force plate surface. These shear forces are usually resolved into anterior–posterior and medial–lateral directions (Elftman, 1939).

The fourth variable needed is the location of the COP of this ground reaction vector. The foot is supported over a varying surface area with different pressures at each part. Even if we knew the individual pressures under every part of the foot, we would be faced with the expensive problem of calculating the net effect of all these pressures as they change with time. It is, therefore, necessary to use a force plate to give us all the forces necessary for a complete inverse solution.

Two common types of force plates are now explained. The first and most commonly used is a flat plate supported by four triaxial transducers, depicted in Figure 5.9. Consider the coordinates of each transducer to be (0, 0), (0, Z), (X, 0), and (X, Z). The location of the COP is determined by the relative vertical forces seen at each of these corner transducers. If we designate the vertical forces as F_{00}, F_{XO}, F_{OZ}, and F_{XZ}, the total vertical force is $F_Y = F_{00} + F_{XO} + F_{OZ} + F_{XZ}$. If all four forces are equal, the COP is at the exact center of the force plate, at (X/2, Z/2). In general,

$$x = \frac{X}{2}\left[1 + \frac{(F_{X0} + F_{XZ}) - (F_{00} + F_{0Z})}{F_Y}\right] \tag{5.4}$$

$$z = \frac{Z}{2}\left[1 + \frac{(F_{0Z} + F_{XZ}) - (F_{00} + F_{X0})}{F_Y}\right] \tag{5.5}$$

A second type of force plate and currently very rare has one centrally instrumented pillar that supports an upper flat plate. Figure 5.10 shows the forces that act on this instrumented support. The action force of the foot F_y acts downward, and the anterior-posterior shear force can act either forward or backward. Consider a reverse shear force F_x, as shown. If we sum the moments acting about the central axis of the support, we get:

$$\begin{aligned} M_z - F_y \cdot x + F_x \cdot y_0 &= 0 \\ x &= \frac{F_x \cdot y_0 + M_z}{F_y} \end{aligned} \tag{5.6}$$

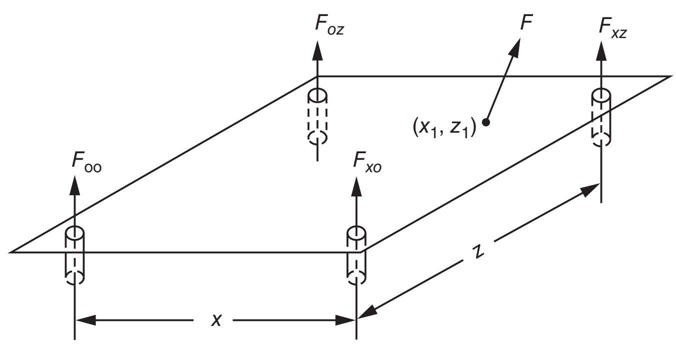

Figure 5.9 Force plate with force transducers in the four corners. Magnitude and location of ground reaction force F can be determined from the signals from the load cells in each of the support bases.

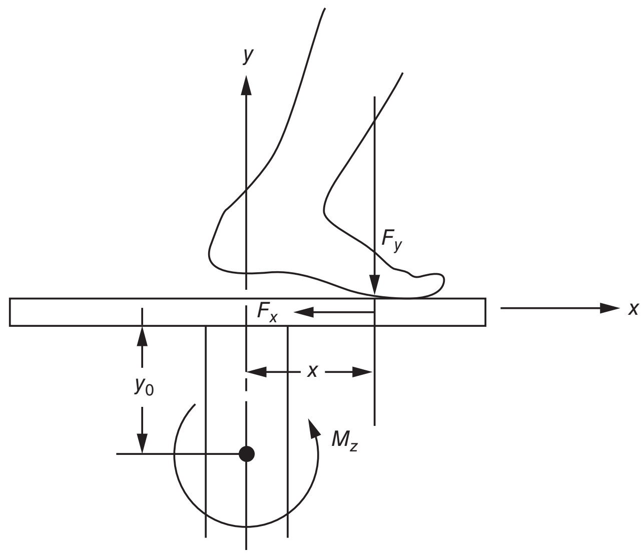

Figure 5.10 Central support type force plate, showing the location of the center of pressure of the foot and the forces and moments involved.

where

M_z = bending moment about axis of rotation of support

y_0 = distance from support axis to force plate surface.

Since F_x, F_y, and M_z continuously change with time, x can be calculated to show how the COP moves across the force plate. Caution is required for both types of force platforms when F_y is very low (<2% of body weight). During this time, a small error in F_y represents a large percentage error in x and z, as calculated by Equations (5.4), (5.5), and (5.6).

Typical force plate data are shown in Figure 5.11 plotted against time for a subject walking at a normal speed. The vertical reaction force F_y is very characteristic in that it shows a rapid rise at heel contact to a value in excess of body weight as full weight bearing takes place (i.e. the body mass is being accelerated upward). Then as the knee flexes during midstance, the plate is partially unloaded and F_y drops below body weight. At push off, the plantarflexors are active in causing ankle plantarflexion. This causes a second peak greater than body weight. Finally, the weight drops to zero as the opposite limb takes up the body weight. The anterior-posterior reaction force F_x is the horizontal force exerted by the force plate on the foot. Immediately after heel contact, it is negative, indicating a backward horizontal friction force between the floor and the shoe. If this force were not present, the foot would slide forward, as sometimes happens when walking on icy or slippery surfaces. Near midstance, F_x goes positive, indicating that the force plate reaction is acting forward as the muscle forces (mainly plantarflexor) cause the foot to push back against the plate.

The COP typically starts at the lateral heel, assuming that initial contact is made by the heel, and then progresses forward toward the metatarsal heads and the first toe. The position of the COP relative to the foot cannot be obtained from the force plate data themselves. We must first know where the foot is, relative to the midline of the plate. The type of force plate used to collect these data is the centrally instrumented pillar type depicted in Figure 5.10. It should be noted that M_z is positive at heel contact and then becomes negative as the bodyweight moves forward. The centers of pressure X_{cp} and Y_{cp} are calculated in absolute coordinates to match those given in the kinematics listing. Y_{cp} was set at 0 to indicate ground level.

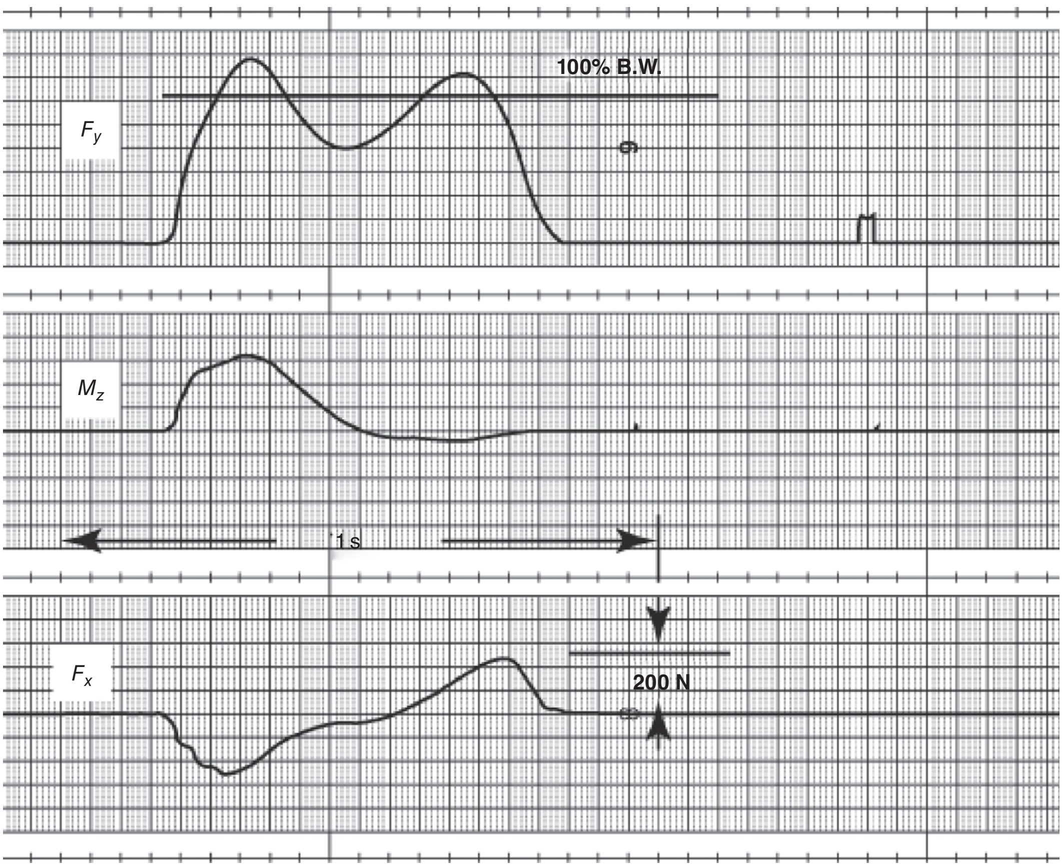

Figure 5.11 Force plate record obtained during gait, using a central support type as shown in Figure 5.10.

It is very important to realize that these reaction forces are merely the algebraic summation of all mass-acceleration products of all body segments. In other words, for an N-segment system the reaction force in the x-direction is:

$$F_x = \sum_{i=1}^{N} m_i a_{xi} \tag{5.7}$$

where
m_i = mass of ith segment
a_{xi} = acceleration of ith-segment COM in the x-direction.

Similarly, in the vertical direction,

$$F_y = \sum_{i=1}^{N} m_i \left(a_{yi} + g\right) \tag{5.8}$$

where
a_{yi} = acceleration of ith-segment COM in the y-direction
g = acceleration due to gravity.

With a person standing perfectly still on the platform, F_y will be nothing more than the sum of all segment masses times g, which is the body weight. The interpretation of the ground reaction forces as far as what individual segments are doing is virtually impossible. In the algebraic summation of mass-acceleration products, there can be many cancellations. For example, F_Y could remain constant as one arm is accelerating upward, while the other has an equal and opposite downward acceleration. Thus, to interpret the complete meaning of any reaction force, we would be forced to determine the accelerations of the COMs of all segments and carry out the summations seen in Equations (5.7) and (5.8).

5.2.2.1 Alternative Force Collection Devices

5.2.2.1.1 Instrumented Treadmills The implementation of instrumented treadmills has helped to alleviate many of the limitations present when collecting walking and running biomechanical data. First, many biomechanical labs are often small, which will not allow for an adequate amount of strides to produce a natural gait pattern especially with running. Next, even when enough lab space is available for walking or running unnatural strides often occur when attempting to target the force plate. Lastly, treadmills allow researchers to control walking and running speed, which can influence kinematics and kinetics.

There are two main styles of instrumented treadmills: (1) side-by-side treadmills (Figure 5.12) and (2) tandem treadmills. Side-by-side treadmills are designed to provide consistent sagittal plane kinematics but slight alterations in frontal plane kinematics may occur attempting to position each foot on the belt. Side-by-side treadmills also provide researchers with the ability to independently change speeds to create random perturbations or when testing pathological gait as seen in stroke patients. Tandem treadmills are designed to maintain consistent kinematics and prevent the frontal plane alteration seen in side-by-side treadmills. However, having the belts split front to back can slightly alter sagittal plane kinematics due to maintaining the body's COM at the treadmill split. This design also avoids force recording problems that occur during double limb support during walking. Tandem treadmills are able to alter belt speed but are not typically done independently.

The technical challenge of instrumented treadmills is the numerous external loads placed on the treadmill during walking and running that have the potential to cause inaccurate ground reaction force data. To avoid these errors instrumented treadmills have been designed to have the

Figure 5.12 Example of a Bertec instrumented side-by-side treadmill. (Photo provided with permission from Fernando Santos of Bertec LLC.)

entire treadmill mounted on the force transducers so that these external loads essentially cancel each other out. The external load that is of most concern is the frictional force between the belt and the surface of the force plate caused by the motors pulling the belt. Willems and Gosseye (2013) used a model to demonstrate all forces that are applied to the treadmill during motion and how these external forces are nil or cancel each other out including the friction between the moving belt and the treadmill surface. As they reported, mechanical vibration occurs when a participant is walking or running on the treadmill as the connections between components and the material itself is not completely ridged. In addition, the deflections of the material can also store and release energy altering force readings and accelerations of the participant. The vibrations of the treadmills COM will introduce noise into the force readings. In addition, it will also alter the signal-to-noise ratio, which ultimately can decrease accuracy in the COP calculation. One solution for overcoming the noise is to average or integrate ground reaction forces over several strides; however, this removes the ability to examine peak forces. Another solution is to use a low pass filter to remove the vibration noise. Commercially available instrumented treadmills are designed to have high natural frequencies so that the frequencies of the participant walking or running don't converge. This design feature minimizes the vibration noise and also allows the use of a low pass filter to eliminate the high frequency vibration noise. In light of some of the technical challenges, several studies have demonstrated accurate and reliable data collected on instrumented treadmills as compared to overground walking and running.

5.2.2.1.2 Instrumented Handrails Instrumented handrails have been integrated into instrumented treadmills as a way to calculate both lower and upper extremity inverse dynamics in populations that require gait assistance either unilaterally or bilaterally. Instrumented handrails have also been used to examine inverse dynamics during stair ascent and descent. The handrails utilize standard three-component force transducers. Therefore, the mathematical approach is the same with the exception of correcting for the angle of the handrail when mounted on instrumented stairs or when treadmills are in inclined positions. However, most commercially available treadmills that incline will account for this within their collection software.

5.2.2.1.3 Insole Pressure Sensors The COP measured by the force plate is a weighted average of the distributed COPs under the two feet if both are in contact with the ground or under the one foot that is in contact. However, the COP signal tells us nothing about the pressure at any of the contact points under the foot. For example, during midstance in walking or running, there are two main pressure areas: the ball of the foot and the heel, but the force plate records the COP as being under the arch of the foot, where there may in fact be negligible pressure. To get some insight into the distributed pressures around all contact points, a number of special pressure measurement systems have been developed. A typical pressure measurement system was introduced a number of years ago by Tekscan and is schematically shown in Figure 5.13. It is a flexible sheet of tactile force sensors that is cut in the shape of a shoe insole and that can be trimmed down to any shoe size. Each sensel shown in the exploded view is about 0.2 in. × 0.2 in. (5 mm × 5 mm), giving 25 sensels/sq. in. Two thin flexible polyester sheets have electrically conductive electrodes deposited in rows and columns. A thin semiconductive coating is applied between the conductive rows and columns, and its electrical resistance changes with the pressure applied. The matrix of sensels results in a matrix of voltage changes that are computer displayed as different colors for different pressure levels. Colors range from blue through green, yellow, orange, and red (1–125 psi). Thus, the high- and low-pressure areas under the foot are visually evident as the pressure moves from the heel to the toes over the stance period (Hsiao et al., 2002). Such devices have been valuable in identifying high-pressure points in various foot deformities and in diabetic feet (Pitie et al., 1999). The results of surgery and shoe orthotics to relieve the pressure points are also immediately available. Lastly, these sensors have greatly enhanced the development of footwear (including various forms of athletic footwear) and

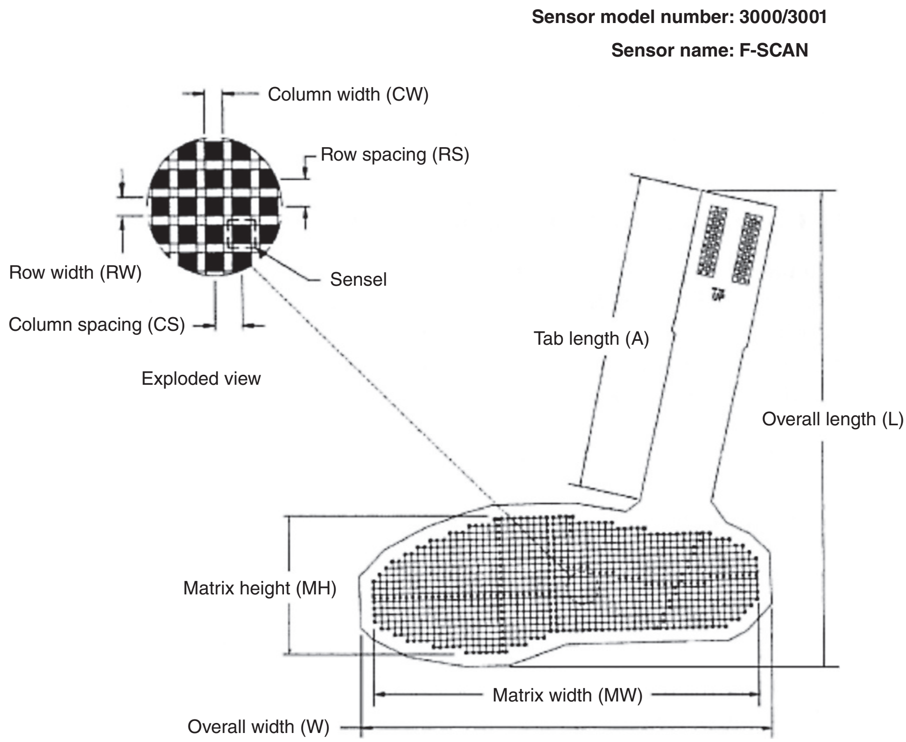

Figure 5.13 Foot pressure measurement system introduced by Tekscan, Inc. It is a shoe insole insert with an array of pressure sensors that vary their resistance with applied pressure. See the text for detailed operation of sensor and computer display system. (Reproduced with permission of Tekscan, Inc.)

5.2.2.1.4 Instrumented Walking Boots An instrumented walking boot was designed (Hullfish et al., 2020) and used to estimate Achilles tendon load. The boot was outfitted with an insole pressure sensor and a load cell mounted to the posterior strut of the walking boot to calculate the overall resistance moment created from the boot. In this study, the instrumented walking boot was used in conjunction with a motion capture system including a force plate. The motion capture system was used to both validate the walking boot and was also required to calibrate the insole sensor. Therefore, in its current form, the walking boot is not able to be used outside of a motion capture lab. Nonetheless, this walking boot can provide useful information specifically to Achilles tendon loading. It is important to note that this experimental approach has several assumptions: (1) the plantar loads measured with the insole sensor are all created from the plantar flexor muscles of the ankle and passed directly through the Achilles tendon and (2) the posterior strut of the boot resists all of the reaction loads during walking. It is currently unknown if these assumptions are true as the toe flexor muscles can create plantar forces and there are also altered neuromuscular strategies in patients with pain or injury. However, the current approach was found to detect decreases in Achilles tendon loads that occur when the ankle is positioned in greater amounts of plantarflexion within the boot.

5.2.3 Combined Force Plate and Kinematic Data

It is valuable to see how the reaction force data from the force plate are combined with the segment kinematics to calculate the muscle moments and reaction forces at the ankle joint during dynamic stance. This is best illustrated in a sample calculation. For a subject in late stance (see Figure 5.14), during toe-off the following foot accelerations were recorded: $a_x = 3.25$ m/s^2,

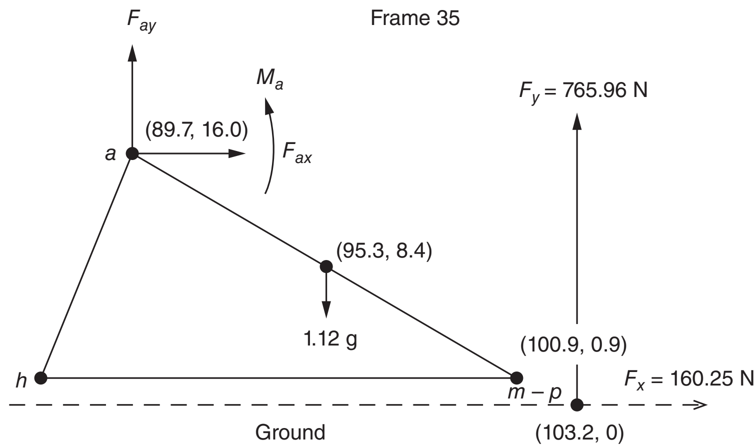

Figure 5.14 Free-body diagram of foot during weight bearing with the ground reaction forces, F_x and F_y, shown as being located at the COP.

$a_y = 1.78$ m/s^2, and $\alpha = -45.35$ rad/s^2. The mass of the foot is 1.12 kg, and the moment of inertia is 0.01 kg $\cdot$ m^2.

Example 5.4 (see Figure 5.14). From Equation (5.1),

$$F_{ax} + F_x = ma_x$$
$$F_{ax} = 1.12 \times 3.25 - 160.25 = -156.6\,\text{N}$$

From Equation (5.2),

$$F_{ay} + F_y - mg = ma_y$$
$$F_{ay} = 1.12 \times 1.78 - 765.96 + 1.12 \times 9.81 = -753.0\,\text{N}$$

From Equation (5.3), about the COM of the foot, $\Sigma M = I\alpha$,

$$\begin{aligned} M_a + F_x \times 0.084 + F_y \times 0.079 - F_{ay} \times 0.056 - F_{ax} \times 0.076 \\ = 0.01\,(-45.35) \end{aligned}$$
$$\begin{aligned} M_a = -0.01 \times 45.35 - 0.084 \times 160.25 - 0.079 \times 765.96 - 0.056 \\ \times 753.0 - 0.076 \times 156.6 = -128.5\,\text{N}\cdot\text{m} \end{aligned}$$

The polarity and the magnitude of this ankle moment indicate strong plantarflexor activity acting to push off the foot and cause it to rotate clockwise about the metatarsophalangeal joint.

5.2.4 Interpretation of Moment-of-Force Curves

A complete link-segment analysis yields the net muscle moment at every joint during the time course of movement (Pedotti, 1977). As an example, we discuss in detail the ankle moments-of-force of a hip joint replacement patient, as shown in Figure 5.15. Three repeat trials are plotted.

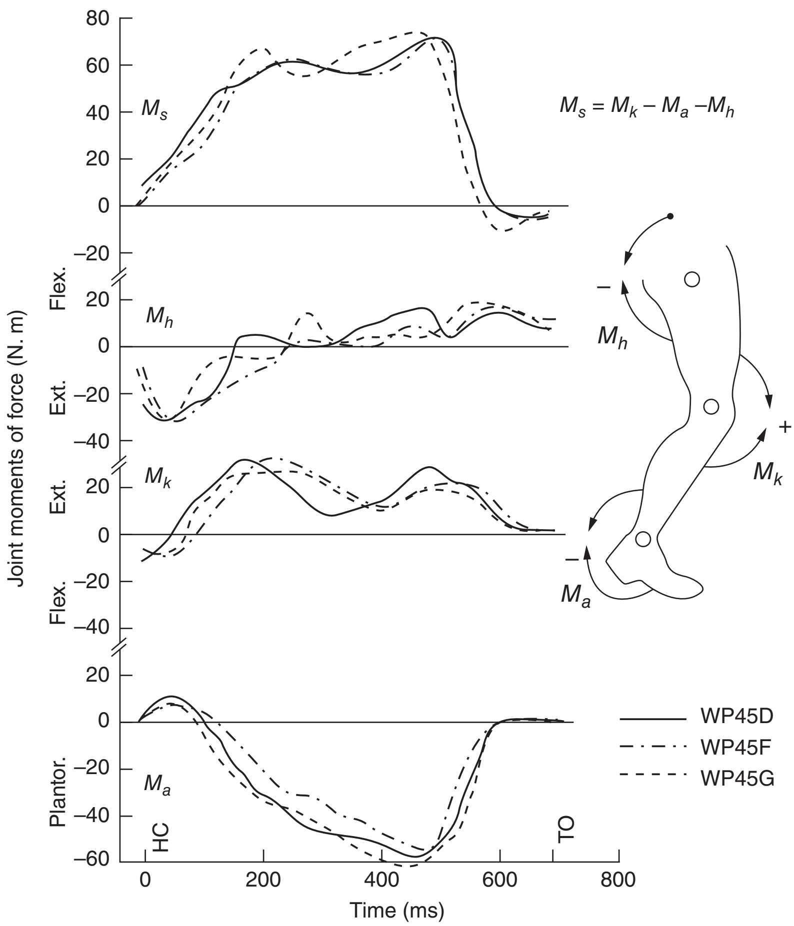

Figure 5.15 Joint moment-of-force profiles from three repeat trials of a patient fitted with a total hip replacement. See the text for a detailed discussion.

The convention for the moments is shown to the right of the plot. Counterclockwise moments acting on a segment distal to the joint are positive; clockwise moments are negative. Thus, a plantarflexor moment (acting on its distal segment) is negative, a knee extensor moment is shown to be positive, and a hip extensor moment is negative.

The moments are plotted during stance, with heel contact at 0 and toe-off at 680 ms. The ankle muscles generate a positive (dorsiflexor) moment for the first 80 ms of stance as the anterior tibial muscles act eccentrically to lower the foot to the ground. Then the plantarflexors increase their activity during mid- and late stance. During midstance, they act to control the amount of forward rotation of the leg over the foot, which is flat on the ground. As the plantarflexors generate their peak of about 60 N·m, they cause the foot to plantarflex and create toe-off. This concentric action results in a major generation of energy in healthy patients (Winter, 1983), but in this patient, it was somewhat reduced because of the pathology related to the patient's hip joint replacement. Prior to toe-off, the plantarflexor moment drops to 0 because that limb is now unloaded due to the fact that the patient is now weight bearing on their good limb and, for the last 90 ms prior to toe-off, the toe is just touching the ground with a light force. At this same point in time, the hip flexors are active to pull this limb upward and forward as a first phase of lower limb swing.

The knee muscles effectively show one pattern during all of stance. The quadriceps are active to generate an extensor moment, which acts to control the stance and also extends the knee during midstance. Even during toe-off, when the knee starts flexing in preparation for swing, the quadriceps act eccentrically to control the amount of knee flexion. At heel contact, the hip moment is negative (extensor) and remains so until midstance. Such activity has two functions. First, the hip muscles act on the thigh to assist the quadriceps in controlling the amount of knee flexion. Second, the hip extensors act to control the forward rotation of the upper body as it attempts to rotate forward over the hip joint (the reaction force at the hip has a backward component during the first half of stance). Then, during the latter half of stance, the hip moment becomes positive (flexor), initially to reverse the backward rotating thigh and then, as described earlier, to pull the thigh forward and upward.

The fourth curve, M_s, bears some explanation. It is the net summation of the moments at all three joints, such that extensor moments are positive. It is called the *support moment* (Winter, 1980) because it represents a total limb pattern to push away from the ground. In scores of walking and running trials on healthy participants and patients, this synergy has been shown to be consistently positive during single support, in spite of considerable variability at individual joints (Winter, 1984). This latter curve is presented as an example to demonstrate that moment-of-force curves should not be looked at in isolation but rather as part of a total integrated synergy in a given movement task. A complete discussion and analysis of this synergy are presented in Chapter 11.

5.2.5 Differences Between Center of Mass and Center of Pressure

For many in the applied and clinical areas, the terms COM and COP are often misinterpreted or even interchanged. The COM of the body is the net location of the COM in 3D space and is the weighted average of the COM of each segment as calculated in Chapter 4. The location of the COM in the vertical direction is sometimes called the *center of gravity* (COG). The trajectory of this vertical line from the COM to the ground allows us to compare the trajectories of the COM and COP. The trajectory of the COP is totally independent of the COM, and it is the location of the vertical ground reaction force vector from a single force platform, assuming that all body contact points are on that platform. The vertical ground reaction force is a weighted average of the location of all downward (action) forces acting on the force plate. These forces depend on the foot placement and the motor control of the ankle muscles. Thus, the COP is the neuromuscular response to the imbalances of the body's COM. The major misuse of the COP comes from researchers who refer to COP as "sway," thereby inferring it to be the kinematic measure COM.

The difference between COM and COP is demonstrated in Figure 5.16. Here, we see a subject swaying back and forth while standing erect on a force plate. Each figure shows the changing situation at one of five different points in time. Time 1 has the body's COM (shown by the vertical bodyweight vector, W) to be ahead of the COP (shown by the vertical ground reaction vector, R). This "parallelogram of forces" acts at distances g and p, respectively, from the ankle joint. The magnitudes of W and R are equal and constant during quiet standing. Assuming that the body to be pivoting about the ankles and neglecting the small mass of the feet, a counterclockwise moment equal to Rp and a clockwise moment equal to Wg will be acting. At Time 1, $Wg > Rp$ and the body will experience a clockwise angular acceleration, α. It will also have a clockwise angular velocity, ω. In order to correct this forward imbalance, the subject will increase their plantarflexor activity, which will increase the COP such that at Time 2 the COP will be anterior of the COM. Now $Rp > Wg$. Thus, α will reverse and will start to decrease ω until, at Time 3, the time integral of α will result in a reversal of ω. Now both ω and α are counterclockwise, and the body will be experiencing a backward sway. The subject's response to the backward sway at Time 4 is to decrease their COP by reduced plantarflexor activation. Now $Wg > Rp$ and α will reverse, and, after a period of time, ω will decrease and reverse, and the body will return to the original conditions, as seen at Time 5. From this sequence of events relating COP to COM, it is evident that the plantarflexors/dorsiflexors vary the net ankle moment to control the COP and thereby regulate the body's COM. However, it is apparent that the COP must

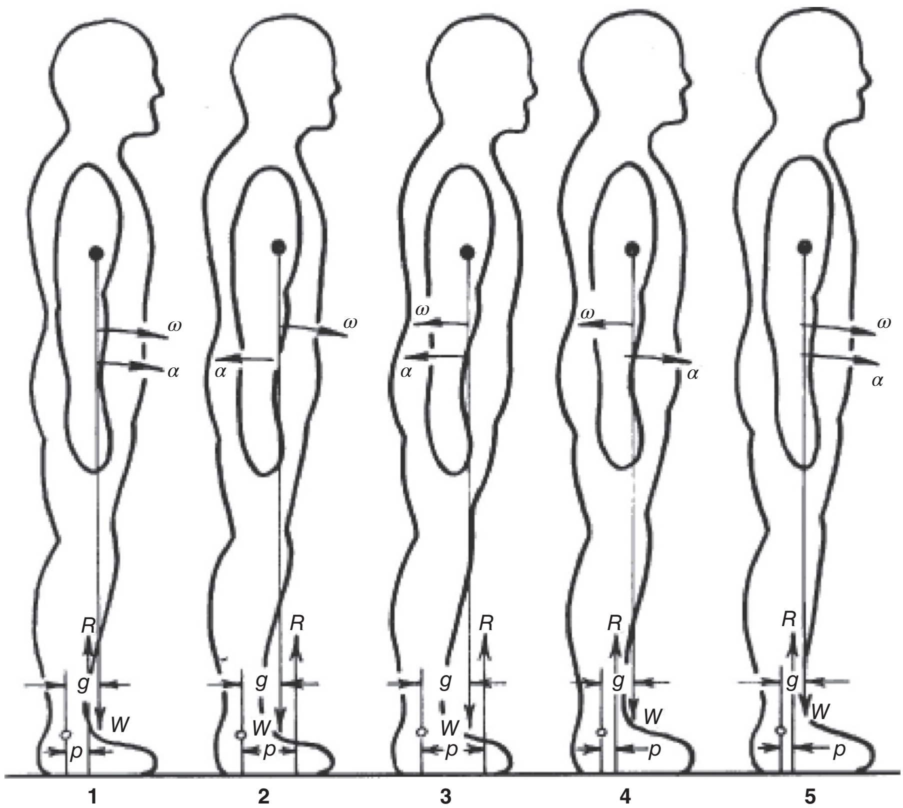

Figure 5.16 A subject swaying back and forth while standing on a force platform. Five different points in time are described, showing the COM and the COP locations along with the associated angular accelerations and velocities of the body. See the text for a detailed description.

posterior of the COM; thus, the dynamic range of the COP must be somewhat greater than that of the COM. If the COM were allowed to move within a few centimeters of the toes, it is possible that a corrective movement of the COP to the extremes of the toes might not be adequate to reverse ω. Here, the subject would be forced to take a step forward to arrest a forward fall.

Figure 5.17 is a typical 40-second record of the center of pressure (COP_x) and center of mass (COM_x) in the anterior/posterior direction of an adult subject standing quietly. Note that both signals are virtually in phase and that COP is slightly greater than COM. As was seen in the discussion regarding Figure 5.16, the COP must move ahead of and behind the COM in order to decelerate it and reverse its direction. Note that all reversals of direction of COM coincide with an overshoot of the COP signal.

5.2.6 Kinematics and Kinetics of the Inverted Pendulum Model

All human movement (except in space) is done in a gravitational environment, and, therefore, posture and balance are continuous tasks that must be accomplished. In normal daily activity at home, at work, and in our sports and recreation, we must maintain a safe posture and balance. The base of support can vary from one foot (running) to a four-point support (football), and it is essential that the COM remain within that base of support or move safely between the two feet if it lies temporarily outside the base of support (as it does in running and during the single-support phase of walking). There is a common model that allows us to analyze the dynamics of balance: the inverted pendulum model, which relates the trajectories of the COP and COM. As was seen in

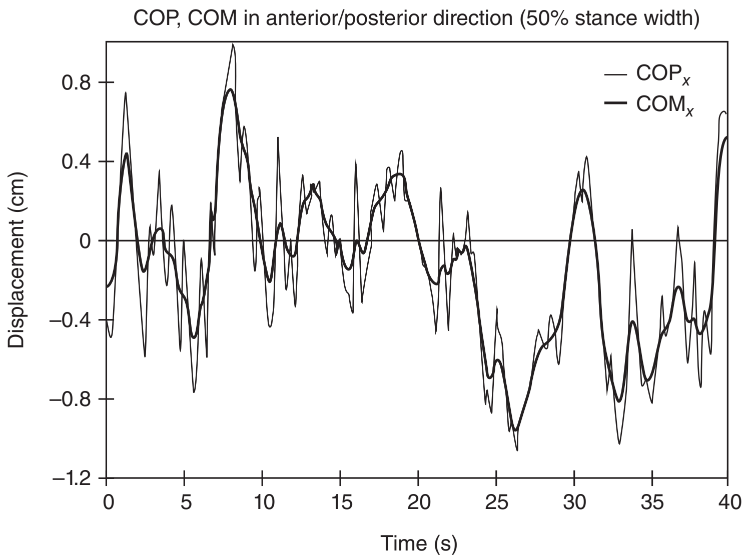

Figure 5.17 Typical 40-second record of the total body center of mass (COM_x) and center of pressure (COP_x) in the anterior/posterior direction during quiet standing. The COP amplitude exceeds that of the COM, and the reversals of direction of the COM (i.e. high + ve or − ve accelerations) are caused by overshoots of COP, as predicted by the discussion related to Figure 5.16.

Section 5.2.5, the position of the COP relative to the COM decides the direction of the angular acceleration of the inverted pendulum. A full biomechanical analysis of the inverted pendulum model in both sagittal and frontal planes has been presented by Winter et al. (1998). In the sagittal plane, assuming that the body swayed about the ankles, it was shown that:

$$\mathrm{COP}_x - \mathrm{COM}_x = -\frac{I_S \mathrm{COM}_x}{Wh} \tag{5.9}$$

where

I_s = moment of inertia of body about ankles in sagittal plane
$\mathrm{C\ddot{O}M}_x$ = forward acceleration of COM
W = body weight above ankles
h = height of COM above ankles.

In the frontal plane, the balance equation is virtually the same:

$$\mathrm{COP}_z - \mathrm{COM}_z = -\frac{I_f \mathrm{COM}_z}{Wh} \tag{5.10}$$

where

I_f = moment of inertia of body about ankles in frontal plane
$\mathrm{C\ddot{O}M}_z$ = medial/lateral acceleration of COM.

The major difference between Equations (5.9) and (5.10) is in the muscle groups that control the COP. In Equation (5.9), $\mathrm{COP}_x = M_a/R$, where M_a is the sum of the right and left plantarflexor moments and R is the total vertical reaction force at the ankles. In Equation (5.10), $\mathrm{COP}_z = M_t/R$, where $M_t = M_{al} + M_{ar} + M_{hl} + M_{hr}$. M_{al} with M_{ar} are the left and right frontal ankle moments, while M_{hl} and M_{hr} are the left and right frontal plane hip moments. Thus, frontal plane balance is controlled by four torque motors, one at each corner of the closed-link parallelogram consisting of the two lower limbs and the pelvis. The hip abductor/adductor moments have been shown to be totally dominant in side-by-side standing (Winter et al., 1996), while the ankle invertor/evertor moments play a negligible role in balance control.

The validity of the inverted pendulum model is evident in the validity of Equations (5.9) and (5.10). As COP and COM are totally independent measures, then the correlation of (COP–COM) with $\ddot{\text{COM}}$ will be a measure of the validity of this simplified model. Validations of the model during quiet standing (Gage et al., 2004) showed a correlation of $r = -0.954$ in the anterior/posterior (A/P) direction and $r = -0.85$ in the medial/lateral (M/L) direction. The lower correlation in the M/L direction was due to the fact that the M/L COM displacement was about 45% of that in the A/P direction. Similar validations have been made to justify the inverted pendulum model during initiation and termination of gait (Jian et al., 1993); correlations averaged –0.94 in both A/P and M/L directions.

5.3 BONE-ON-BONE FORCES DURING DYNAMIC CONDITIONS

Link-segment models assume that each joint is a hinge joint and that the moment of force is generated by a torque motor. In such a model, the reaction force calculated at each joint would be the same as the force across the surface of the hinge joint (i.e. the bone-on-bone forces). However, our muscles are not torque motors; rather, they are linear motors that produce additional compressive and shear forces across the joint surfaces. Thus, we must overlay on the free-body diagram these additional muscle-induced forces. In the extreme range of joint movement, we would also have to consider the forces from the ligaments and anatomical constraints. However, for the purposes of this text, we will limit the analyses to estimated muscle forces.

5.3.1 Indeterminacy in Muscle Force Estimates

Estimating muscle force is a major problem, even if we have good estimates of the moment of force at each joint. The solution is indeterminate. Figure 5.18 demonstrates the number of major muscles responsible for the sagittal plane joint moments of force in the lower limb. At the knee, for example, there are nine muscles whose forces create the net moment from our inverse solution. The line of action of each of these muscles is different and continuously changes with time. Thus, the moment arms are also dynamic variables. Therefore, the extensor moment is a net algebraic sum of the cross product of all force vectors and moment arm vectors,

$$M_j(t) = \sum_{i=1}^{N_e} F_{ei}(t) \times d_{ei}(t) - \sum_{i=1}^{Nf} F_{fi}(t) \times d_{fi}(t) \tag{5.11}$$

where

N_e = number of extensor muscles
N_f = number of flexor muscles
$F_{ei}(t)$ = force in ith extensor muscle at time t
$d_{ei}(t)$ = moment arm of ith extensor muscle at time t.

Thus, a first major step is to make valid estimates of individual muscle forces and to combine them with a detailed kinematic/anatomical model of lines of pull of each muscle relative to the joint's center (or our best estimate of it). Thus, a separate model must be developed for each joint, and a number of simplifying assumptions are necessary in order to resolve the indeterminacy problem. An example is now presented of a runner during the rapid toe-off phase when the plantarflexors are dominant.

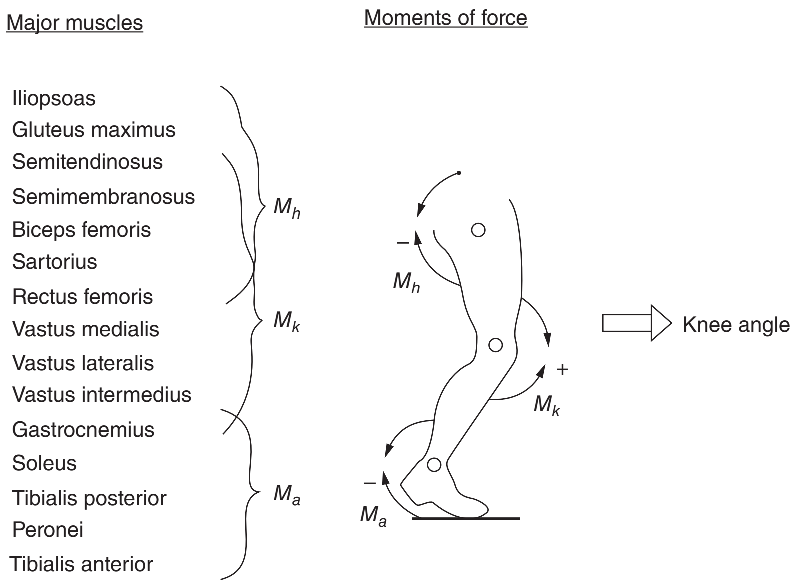

Figure 5.18 Fifteen major muscles are responsible for the sagittal plane moments of force at the ankle, knee, and hip joints. During weight bearing, all three moments control the knee angle. Thus, there is considerable indeterminacy when relating knee angle changes to any single moment pattern or to any unique combination of muscle activity.

5.3.2 Example Problem

During late stance, a runner's foot and ankle are shown (see Figure 5.19) along with the direction of pull of each of the plantarflexors. The indeterminacy problem is solved by assuming that there is no co-contraction and that each active muscle's stress is equal [i.e. its force is proportional to its physiological cross-sectional area (PCA)]. Thus, Equation (5.11) can be modified as follows for the five major muscles of the extensor group acting at the ankle (Scott and Winter, 1990):

$$M_a(t) = \sum_{i=1}^{5} PCA_i \times S_{ei}(t) \times d_{ei}(t) \tag{5.12}$$

Since the PCA for each muscle is known and d_{ei} can be calculated for each point in time, the stress, $S_{ei}(t)$ can be estimated.

For this model, the ankle joint center is assumed to remain fixed and is located as shown in Figure 5.20. The location of each muscle origin and insertion is defined from the anatomical markers on each segment, using polar coordinates. The attachment point of the ith muscle is defined by a distance R_i from the joint center and an angle θ_{mi} between the segment's neutral axis and the line joining the joint center to the attachment point. Thus, the coordinates of any origin or insertion at any instant in time are given by:

$$X_{mi}(t) = X_i(t) + R_i \cos[\theta_{mi}(t) + \theta_s(t)] \tag{5.13a}$$

$$Y_{mi}(t) = Y_i(t) + R_i \sin[\theta_{mi}(t) + \theta_s(t)] \tag{5.13b}$$

where $\theta_s(t)$ is the angle of the foot segment in the spatial coordinate system. In Figure 5.20, the free-body diagram of the foot is presented, showing the internal anatomy to demonstrate the problems of defining the effective insertion point of the muscles and the effective line of pull of each muscle. Four muscle forces are shown here: soleus F_s, gastrocnemius F_g, flexor hallucis longus F_h, and peronei F_p. The ankle joint's center is defined by

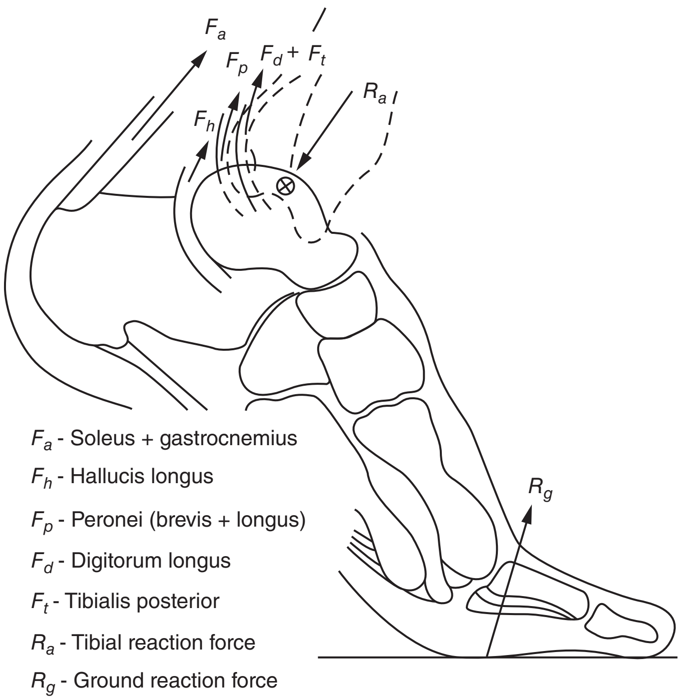

Figure 5.19 Anatomical drawing of foot and ankle during the toe-off phase of a runner. The tendons for the major plantarflexors as they cross the ankle joint are shown, along with ankle and ground reaction forces.

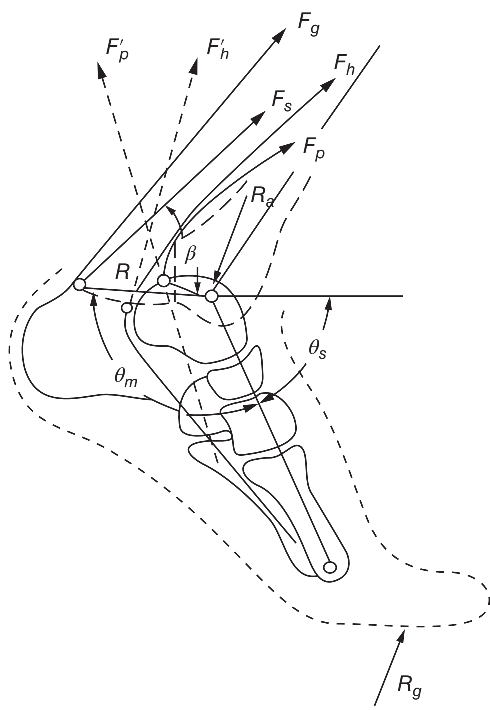

Figure 5.20 Free-body diagram of foot segment showing the actual and effective lines of pull of four of the plantarflexor muscles. Also shown are the reaction forces at the ankle

The insertion of the Achilles tendon is at a distance R with an angle θ_m from the foot segment (defined by a line joining the ankle to the fifth-metatarsal–phalangeal joint). The foot angular position is θ_s in the plane of movement. The angle of pull for the soleus and gastrocnemius muscles from this insertion is rather straightforward. However, for F_h and F_p the situation is quite different. The effective angle of pull is the direction of the muscle force as it leaves the foot segment. The flexor hallucis longus tendon curves under the talus and inserts on the distal phalanx of the big toe. As this tendon leaves the foot, it is rounding the pulleylike groove in the talus. Thus, its effective direction of pull is F'_h. Similarly, the peronei tendon curves around the distal end of the lateral malleolus. However, its effective direction of pull is F'_p.

The moment arm length d_{ei} for any muscle required by Equation (5.12) can now be calculated:

$$d_{ei} = R_i \sin \beta_i \tag{5.14}$$

where β_i is the angle between the effective direction of pull and the line joining the insertion point to the joint center.

In Figure 5.20, β for the soleus only is shown. Thus, it is now possible to calculate $S_{ei}(t)$ over the time that the plantarflexors act during the stance phase of a running cycle. Thus, each muscle force $F_{ei}(t)$ can be estimated by multiplying $S_{ei}(t)$ by each PCA$_i$. With all five muscle forces known, along with R_g and R_a, it is possible to estimate the total compressive and shear forces acting at the ankle joint. Figure 5.21 plots these forces for a middle-distance runner over the stance period of 0.22 s. The compressive forces reach a peak of more than 5500 N, which is

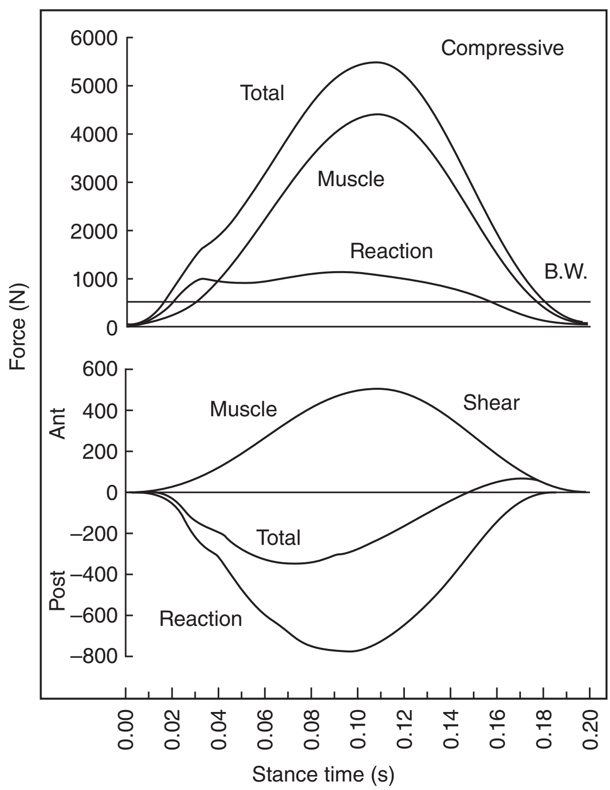

Figure 5.21 Compressive and shear forces at the ankle calculated during the stance phase of a middle-distance runner. The reaction force accounts for less than 20% of the total compressive force. The direction of pull of the major plantarflexors is such as to cause an anterior shear force (the talus is shearing anteriorly with respect to the tibia), which is opposite to that caused by the reaction forces.

an anti-shear mechanism.

in excess of 11 times this runner's body weight. It is interesting to note that the ground reaction force accounts for only 1000 N of the total force, but the muscle forces themselves account for over 4500 N. The shear forces (at right angles to the long axis of the tibia) are actually reduced by the direction of the muscle forces. The reaction shear force is such as to cause the talus to move posteriorly under the tibia. However, the line of pull of the major plantarflexors (soleus and gastrocnemius) is such as to pull the foot in an anterior direction. Thus, the total shear force is drastically reduced from 800 N to about 300 N, and the muscle action has been classified as an anti-shear mechanism.

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6

MECHANICAL WORK, ENERGY, AND POWER

Joseph A. Zeni[1], Stephen J. Thomas[2], and David A. Winters[3,†]

[1]Rutgers University, Newark, NJ, USA
[2]Thomas Jefferson University, Philadelphia, PA, USA
[3]University of Waterloo, Waterloo, Ontario, Canada

6.0 INTRODUCTION

If we were to choose which biomechanical variable contains the most information, we would be forced to look at a variable that relates to the energetics. Without that knowledge, we would know nothing about the energy flows that *cause* the movement we are observing; and no movement would take place without those flows. Diagnostically, we have found joint mechanical powers to be the most discriminating in all our assessment of pathological gait. Without them, we could have made erroneous or incomplete assessments that would not have been detected by electromyographic (EMG) or moment-of-force analyzes alone. Also, valid mechanical work calculations are essential to any efficiency assessments that are made in sports and work-related tasks.

Before proceeding, the student should have in mind certain terms and laws relating to mechanical energy, work, and power, and these will now be reviewed.

6.0.1 Mechanical Energy and Work

Mechanical energy and work have the same units (joules) but have different meanings. Mechanical energy is a measure of the state of a body at an instant in time as to its ability to do work. For example, a body which has 200 J of kinetic energy (KE) and 150 J of potential energy (PE) is capable of doing 350 J of work (on another body). Work, on the other hand, is the measure of energy flow from one body to another, and time must elapse for that work to be done. If energy flows from body *A* to body *B*, we say that body *A* does work on body *B*; or muscle *A* can do work on segment *B* if energy flows from the muscle to the segment.

[†]Deceased.

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

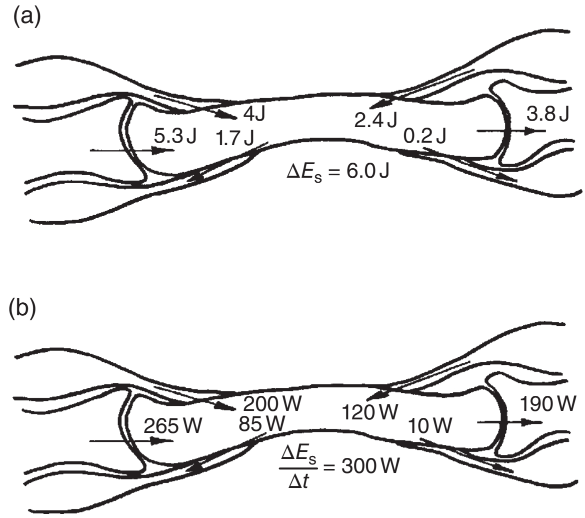

Figure 6.1 (a) Flow of energy into and out of a segment from the adjacent connective tissue and joint contacts over a period of time Δt. (b) Rate of flow of energy (power) for the same segment and same point in time as (a). A "power balance" can be calculated; see the text for discussion.

6.0.2 Law of Conservation of Energy

At all points in the body at all instants in time, the law of conservation of energy applies. For example, any body segment will change its energy only if there is a flow of energy or out of any adjacent structure (tendons, ligaments, or joint contact surfaces). Figure 6.1 depicts a segment that is in contact at the proximal and distal ends and has four muscle attachments. In this situation, there are six possible routes for energy flow. Figure 6.1a shows the work (in joules) done at each of these points over a short period of time Δt. The law of conservation of energy states that the algebraic sum of all the energy flows must equal the energy change of that segment. In the case shown, $\Delta E_s = 4 + 2.4 + 5.3 - 1.7 - 0.2 - 3.8 = 6.0\,\text{J}$. Thus, if we are able to calculate each of the individual energy flows at the six attachment points, we should be able to confirm ΔE_s through an independent analysis of mechanical energy of that segment. The balance will not be perfect because of measurement errors and because our link-segment model does not perfectly satisfy the assumptions inherent in link-segment analyzes. A second way to look at the energy balance is through a *power balance*, which is shown in Figure 6.1b and effectively looks at the-rate of flow of energy into and out of the segment and equates that to the rate of change of energy of the segment. Thus if, Δt is 20 ms, $\Delta E_s/\Delta t = 200 + 120 + 265 - 85 - 10 - 190 = 300\,\text{W}$.

Another aspect of energy conservation takes place within each segment. Energy storage within each segment takes the form of potential and KE (translational and rotational). Thus, the segment energy E_s at any given point in time could be made up of any combination of potential and kinetic energies, quite independent of the energy flows into or out of the segment. In Section 6.3.1, the analysis of these components and the determination of the amount of conservation that occurs within a segment over any given period of time are demonstrated.

6.0.3 Internal Versus External Work

The only source of mechanical energy generation in the human body is the muscles, and the major site of energy absorption is also the muscles. A very small fraction of energy is dissipated into heat as a result of joint friction and viscosity in the connective tissue. Thus, mechanical energy is continuously flowing into and out of muscles and from segment to segment. To reach an external load, there may be many energy changes in the intervening segments between the source and the external load. In a lifting task (see Figure 6.2), the work rate

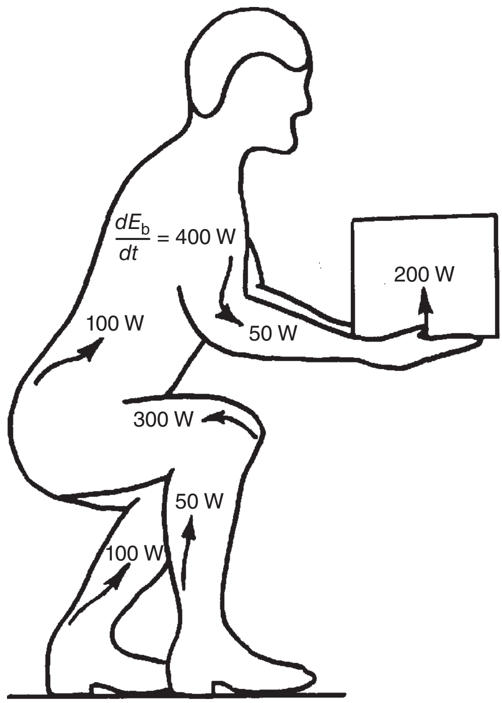

Figure 6.2 Lifting task showing the power generation from a number of muscles and the combined rate of change of energy of the body (internal work) and rate of energy flow to the load (external work).

but the work rate to increase the energy of the total body by the source muscles of the lower limb might be 400 W. Thus, the sum of the internal and external work rates would be 600 W, and this generation of energy might result from many source muscles, as shown. Or, during many movement tasks such as walking and running, there is no external load, and all the energy generation and absorption are required simply to move the body segments themselves. A distinction is made between the work done on the body segments (called *internal* work) and the work done on the load (called *external* work). Thus, lifting weights, pushing a car, or cycling an ergometer have well-defined external loads. One exception to external work definition includes lifting one's own body weight to a new height. Thus, running up a hill involves both internal and external work. External work can be negative if an external force is exerted on the body and the body gives way. Thus, in contact sports, external work is regularly done on players being pushed or tackled. A baseball does work on the catcher as his hand and arms give way.

In bicycle ergometry, the cyclist does internal work just to move his limbs through the cycle (freewheeling). Figure 6.3 shows a situation where the cyclist did both internal and external work. This complex experiment has one bicycle ergometer connected via the chain to a second bicycle. Thus, one cyclist can bicycle in the forward direction (positive work), while the other cycles backward (negative work). The assumption made by the researchers who introduced this novel idea was that each cyclist was doing equal amounts of work (Abbot et al., 1952). This is not true because the positive-work cyclist must do his or her own internal work plus the internal work on the negative-work cyclist plus any additional negative work of that cyclist. Thus, if the internal work of each cyclist were 75 W, the positive-work cyclist would have to do mechanical work at 150 W rate just to "freewheel" both cyclists. Then as the negative-work cyclist contracted his or her muscles, an additional load would be added. Thus, if the negative-work cyclist worked at 150 W, the positive-work cyclists would be loaded to 300 W. It is no wonder that the negative-work cyclist can cycle with ease while the positive-work cyclist rapidly fatigues. They are simply not working at the same mechanical work rate,
work far exceeds that of negative work.

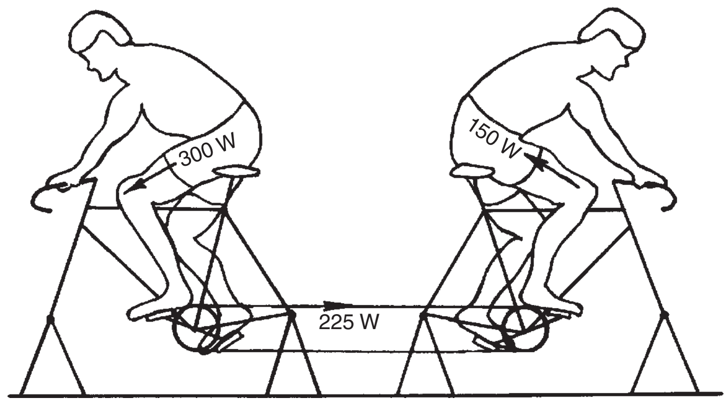

Figure 6.3 Bicycle ergometry situation in which one subject (left) cycles in a forward direction and does work on a second subject who cycles in the reverse direction. The positive-work cyclist not only does external work on the negative-work cyclist but also does the internal work to move the limbs of both cyclists. Contrary to common interpretation, both cyclists are not performing equal magnitudes of mechanical work.

6.0.4 Positive Work of Muscles

Positive work is work done during a concentric contraction, when the muscle moment acts in the same direction as the angular velocity of the joint. If a flexor muscle is causing a shortening, we can consider the flexor moment to be positive and the angular velocity to be positive. The product of muscle moment and angular velocity is positive; thus, power is positive, as depicted in Figure 6.4a. Conversely, if an extensor muscle moment is negative and an extensor angular velocity is negative, the product is still positive, as shown in Figure 6.4b. The integral of the power over the time of the contraction is the net work done by the muscle and represents generated energy transferred from the muscles to the limbs.

6.0.5 Negative Work of Muscles

Negative work is work done during an eccentric contraction when the muscle moment acts in the opposite direction to the movement of the joint. This usually happens when an external force, F_{ext}, acts on the segment and is such that it creates a joint moment greater than the muscle moment. The external force could include gravitational or ground reaction forces. Using the polarity convention as described, we can see in Figure 6.5a that we have a flexor moment

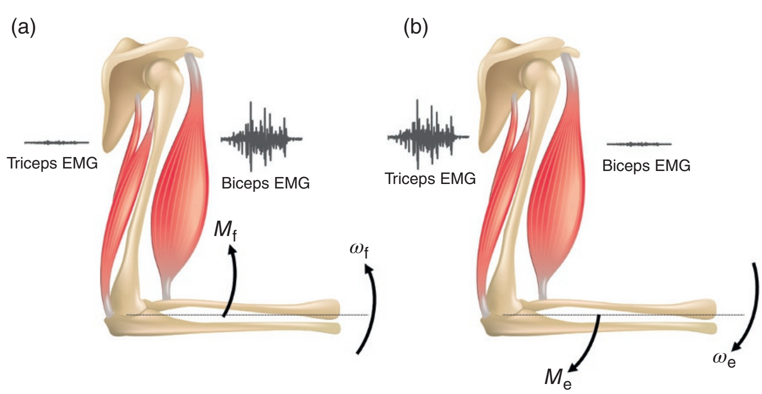

Figure 6.4 Positive power as defined by the net muscle moment and angular velocity. (a) A flexion moment acts while the forearm is flexing. (b) An extension moment acts during an

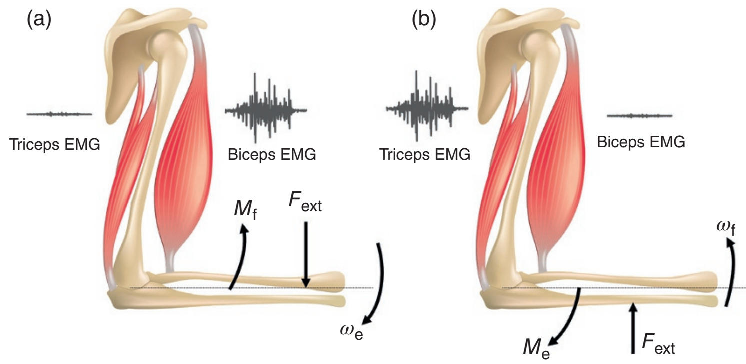

Figure 6.5 Negative power as defined by net muscle moment and angular velocity. (a) An external force causes extension when the flexors are active. (b) An external force causes flexion in the presence of an extensor muscle moment.

(positive) with an extensor angular velocity (negative). The product yields a negative power, so that the work done during this angular change is negative. Similarly, when there is an extensor moment (negative) during a flexor angular change (positive), the product is negative (Figure 6.5b). Here, the net work is being done by the external force on the muscles and represents a flow of energy from the limbs into the muscles (absorption).

6.0.6 Muscle Mechanical Power

The rate of work done by most muscles is rarely constant with time. Because of rapid time-course changes, it has been necessary to calculate muscle power as a function of time (Elftman, 1939; Quanbury et al., 1975; Cappozzo et al., 1976; Winter and Robertson, 1978). At a given joint, muscle power is the product of the net muscle moment and angular velocity,

$$P_m = M_j \omega_j \ \text{W} \tag{6.1}$$

where

P_m = muscle power, W
M_j = net muscle moment, N · m
ω_j = joint angular velocity, rad/s.

As has been described in the previous sections, P_m can be either positive or negative. During even the simplest movements, the power will reverse sign several times. Figure 6.6 depicts the muscle moment, angular velocities, and muscle power as a function of time during a simple extension and flexion of the forearm. As can be seen, the time courses of M_j and ω_j are roughly out of phase by 90°. During the initial extension, there is an extensor moment and an extensor angular velocity as the triceps do positive work on the forearm. During the latter extension phase, the forearm is decelerated by the biceps (flexor moment). Here the biceps are doing negative work (absorbing mechanical energy). Once the forearm is stopped, it starts accelerating in a flexor direction still under the moment created by the biceps, which are now doing positive work. Finally, at the end of the movement, the triceps decelerate the forearm as the extensor muscles lengthen; here, P_m is negative.

6.0.7 Mechanical Work of Muscles

Until now, we have used the terms *power* and *work* almost interchangeably. Power is the rate of doing work. Thus, to calculate work done, we must

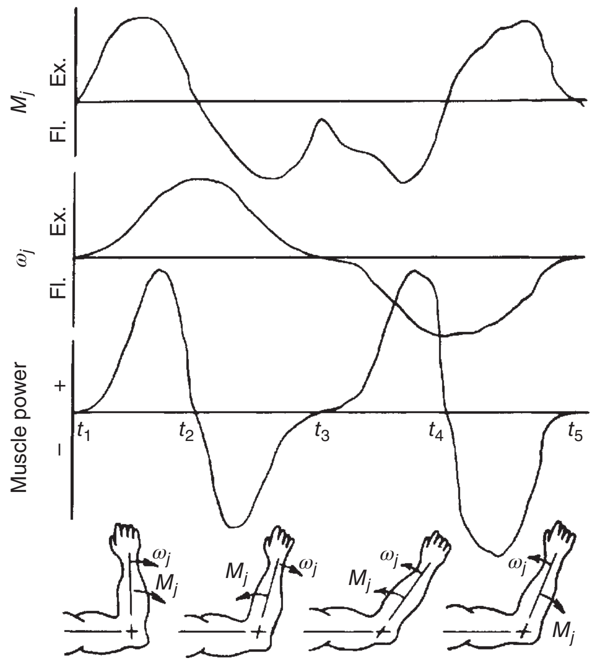

Figure 6.6 Sequence of events during simple extension and flexion of the forearm. Muscle power shows two positive bursts alternating with two negative bursts.

product of power and time is work, and it is measured in joules (1 J = 1 W · s). If a muscle generates 100 W for 0.1 s, the mechanical work done is 10 J. This means that 10 J of mechanical energy has been transferred from the muscle to the limb segments. As the example in Figure 6.6 shows, power is continuously changing with time. Thus, the mechanical work done must be calculated from the time integral of the power curve. The work done by a muscle during a period t_1 to t_2 is:

$$W_m = \int_{t_1}^{t_2} P_m \, dt \ \ \text{J} \tag{6.2}$$

In the example described, the work done from t_1 to t_2 is positive, from t_2 to t_3 it is negative, from t_3 to t_4 it is positive again, and during t_4 to t_5 it is negative. If the forearm returns to the starting position, the net mechanical work done by the muscles is zero, meaning that the time integral of P_m from t_1 to t_5 is zero. It is therefore critical to know the exact times when P_m is reversing polarities in order to calculate the total negative and the total positive work done during the event.

6.0.8 Mechanical Work Done on an External Load

When any part of the body exerts a force on an adjacent segment or on an external body, it can only do work if there is movement. In this case, work is defined as the product of the force acting on a body and the displacement of the body in the direction of the applied force. The work, dW, done when a force causes an infinitesimal displacement, ds, is:

$$dW = F \, ds \tag{6.3}$$

Or the work done when F acts over a distance S_1 is:

$$W = \int_0^{S_1} F\,ds = FS_1 \tag{6.4}$$

If the force is not constant (which is most often the case), then we have two variables that change with time. Therefore, it is necessary to calculate the power as a function of time and integrate the power curve with respect to time to yield the work done. Power is the rate of doing work, or dW/dt.

$$\begin{aligned} P &= \frac{dW}{dt} = F\frac{ds}{dt} \\ &= \overline{F} \cdot \overline{V} \end{aligned} \tag{6.5}$$

where

P = instantaneous power, W
$\overline{F}$ = force, N
$\overline{V}$ = velocity, m/s.

Since both force and velocity are vectors, we must take the dot product, or the product of the force and the component of the velocity that is in the same direction as the force. This will yield:

$$P = FV\cos\theta = F_xV_x + F_yV_y \tag{6.6}$$

where

θ = angle between the force and velocity vectors in the plane defined by those vectors
F_x and F_y = forces in x and y directions
V_x and V_y = velocities in x and y directions.

For the purpose of this initial discussion, let us assume that the force and the velocity are always in the same direction. Therefore, $\cos\theta = 1$ and:

$$\begin{aligned} P &= FV \ \ \text{W} \\ W &= \int_0^t P\,dt = \int_0^t FV\,dt \ \ \text{J} \end{aligned} \tag{6.7}$$

Example 6.1. A baseball is thrown with a constant accelerating force of 100 N for a period of 180 ms. The mass of the baseball is 1.0 kg, and it starts from rest. Calculate the work done on the baseball during the time of force application.

Solution

$$\begin{aligned} S_1 &= ut + \frac{1}{2}at^2 \\ u &= 0 \\ a &= F/m = 100/1.0 = 100 \ \ \text{m/s}^2 \\ S_1 &= \frac{1}{2} \times 100(0.18)^2 = 1.62 \ \ \text{m} \\ W &= \int_0^{S_1} F\,ds = FS_1 = 100 \times 1.62 = 162\ \text{J} \end{aligned}$$

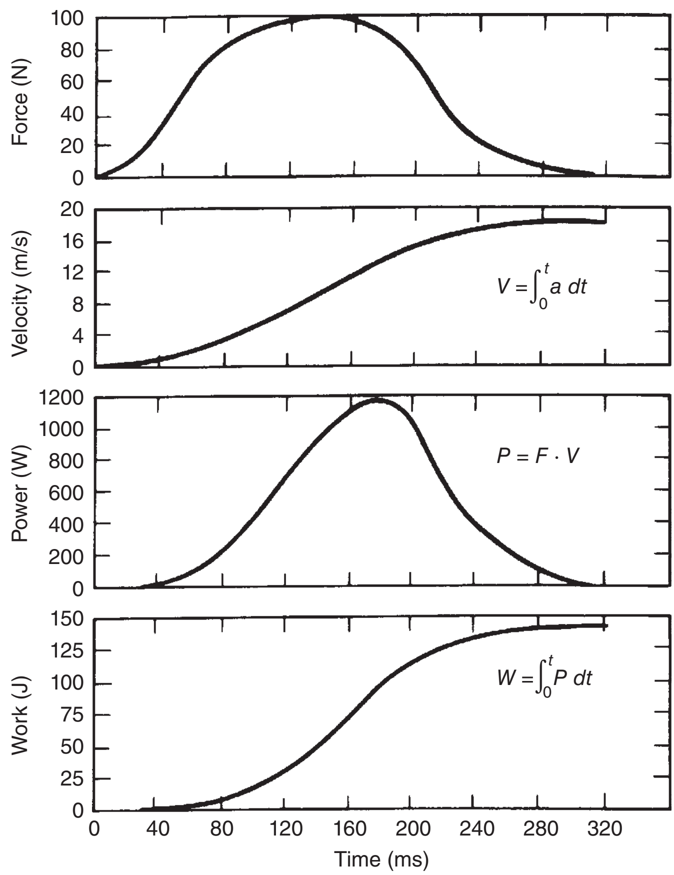

Figure 6.7 Forces, velocity, mechanical power, and work done on a baseball while being thrown. See the text for details.

Example 6.2 A baseball of mass 1 kg is thrown with a force that varies with time, as indicated in Figure 6.7. The velocity of the baseball in the direction of the force is also plotted on the same time base and was calculated from the time integral of the acceleration curve (which has the same numerical value as the force curve because the mass of the baseball is 1 kg). Calculate the instantaneous power to the baseball and the total work done on the baseball during the throwing period.

The peak power calculated here may be considered quite high, but it should be noted that this peak has a short duration. The average power for the throwing period is less than 500 W. In real-life situations, it is highly unlikely that the force will ever be constant; thus, instantaneous power must always be calculated. When the baseball is caught, the force of the hand still acts against the baseball, but the velocity is reversed. The force and velocity vectors are now in opposite directions. Thus, the power is negative and the work done is also negative, indicating that the baseball is doing work on the body.

6.0.9 Mechanical Energy Transfer Between Segments

Each body segment exerts forces on its neighboring segments, and if there is a translational movement of the joints, there is a mechanical energy transfer between segments. In other words, one segment can do work on an adjacent segment by a force

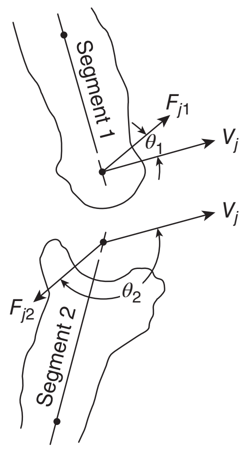

Figure 6.8 Reaction forces and velocities at a joint center during dynamic activity. The dot product of the force and velocity vectors is the mechanical power (rate of mechanical energy transfer) across the joint.

(Quanbury et al., 1975). This work is in addition to the muscular work described in Sections 6.0.4–6.0.7. Equations can be used to calculate the rate of energy transfer (i.e. power) across the joint center. Consider the situation in Figure 6.8 at the joint between two adjacent segments. F_{j1}, the reaction force of segment 2 on segment 1, acts at an angle θ_1 from the velocity vector V_j. The product of $F_{j1}V_j\cos\theta_1$ is positive, indicating that energy is being transferred into segment 1. Conversely, $F_{j2}V_j\cos\theta_2$ is negative, denoting a rate of energy outflow from segment 2. Since $P_{j1} = -P_{j2}$, the outflow from segment 2 equals the inflow to segment 1. In an n-joint system, there will be n power flows, but the algebraic sum of all those power flows will be zero, reinforcing the fact that these flows are passive and, therefore, do not add to or subtract from the total body energy.

This mechanism of energy transfer between adjacent segments is quite important in the conservation of energy of any movement because it is a passive process and does not require muscle activity. In walking, this has been analyzed in detail (Winter and Robertson, 1978). At the end of swing, for example, the swinging foot and leg lose much of their energy by transfer upward through the thigh to the trunk, where it is conserved and converted to KE to accelerate the upper body in the forward direction.

6.1 EFFICIENCY

The term *efficiency* is probably the most abused and misunderstood term in human movement energetics. Confusion and error result from an improper definition of both the numerator and the denominator of the efficiency equation (Gaesser and Brooks, 1975; Whipp and Wasserman, 1969). In the next section, four causes of inefficiency are discussed in detail, and these mechanisms must be recognized in whatever formula evolves. Overlaid on these four mechanisms are two fundamental reasons for inefficiency: inefficiency in the conversion of metabolic energy to mechanical energy, and neurological inefficiency in the control of that energy. Metabolic energy is converted to mechanical energy at the tendon, and the metabolic efficiency depends on the conditioning of each muscle, the metabolic (fatigue) state of muscle, the subject's diet, and any possible metabolic disorder. This conversion of energy would be called *metabolic* or *muscle efficiency* and would be defined as follows:

$$\text{metabolic (muscle) efficiency} = \frac{\Sigma\ \text{mechanical work done by all muscles}}{} \qquad (6.8)$$

Such an efficiency is impossible to calculate at this time because it is currently impossible to calculate the work of each muscle (which would require force and velocity time histories of every muscle involved in the movement) and to isolate the metabolic energy of those muscles. Thus, we are forced to compromise and calculate an efficiency based on segmental work and to correct the metabolic cost by subtracting estimates of overhead costs not associated with the actual mechanical work involved. Thus, an efficiency would be defined as:

$$\text{mechanical efficiency} = \frac{\text{mechanical work (internal + external)}}{\text{metabolic cost} - \text{resting metabolic cost}} \tag{6.9}$$

The resting metabolic cost in bicycling, for example, could be the cost associated with sitting still on the bicycle.

A further modification is *work efficiency*, which is defined as:

$$\text{work efficiency} = \frac{\text{external mechanical work}}{\text{metabolic cost} - \text{zero-work metabolic cost}} \tag{6.10}$$

The zero-work cost would be the cost measured with the cyclist freewheeling.

In all of the efficiency calculations described, there are varying amounts of positive and negative work. The metabolic cost of positive work exceeds that of equal levels of negative work. However, negative work is not negligible in most activities. Level gait has equal amounts of positive and negative work. Running uphill has more positive work than negative work, and vice versa for downhill locomotion. Thus, all of the efficiency calculations yield numbers that are strongly influenced by the relative percentages of positive and negative work. An equation that gets around this problem is:

$$\text{metabolic cost (positive work)} + \text{metabolic cost (negative work)} = \text{metabolic cost}$$

or

$$\frac{\text{positive work}}{\eta_+} + \frac{\text{negative work}}{\eta_-} = \text{metabolic cost} \tag{6.11}$$

where η_+ and η_- are the positive and negative work efficiencies, respectively.

The interpretation of efficiency is faulty if it is assumed to be simply a measure of how well the metabolic system converts biochemical energy into mechanical energy, rather than a measure of how well the neural system is performing to control the conversion of that energy. An example will demonstrate the anomaly that results. A normal healthy adult walks with 100 J mechanical work per stride (half positive, half negative). The metabolic cost is 300 J per stride, and this yields an efficiency of 33%. A neurologically disabled adult would do considerably more mechanical work because of his or her jerky gait pattern, say 200 J per stride. Metabolically, the cost might be 500 J per stride, which would give an efficiency of 40%. Obviously, the healthy adult is a more efficient walker, but our efficiency calculation does not reflect that fact. Neurologically, the disabled person is quite inefficient because he or she is not generating an effective and smooth neural pattern. However, the disabled person is quite efficient in the actual conversion of metabolic energy to mechanical energy (at the tendon), and that is all that is reflected in the higher efficiency score.

6.1.1 Causes of Inefficient Movement

It is often difficult for a therapist or coach to concentrate directly on efficiency. Rather, it is more reasonable to focus on the individual causes of inefficiency and thereby automatically improve the efficiency of the movement. The four major causes of mechanical inefficiency (Winter, 1978) will now be described.

6.1.1.1 Cocontractions. Obviously, it is inefficient to have muscles cocontract because they fight against each other without producing a net movement. Suppose that a certain movement can be accomplished with a flexor moment of 30 N m. The most efficient way to do this is with flexor activity only. However, the same movement can be achieved with 40 N·m flexion and 10 N m extension, or with 50 N m flexion and 20 N m extension. In the latter case, there is an unnecessary 20 N m moment in both the extensors and the flexors. Another way to look at this situation is that the flexors are doing unnecessary positive work to overcome the negative work of the extensors.

Cocontractions occur in many pathologies, notably those involving the central nervous system, as is the case in individuals with hemiplegic stroke and spastic cerebral palsy. Cocontractions also occur to a limited extent during normal movement when it is necessary to stabilize a joint, especially if heavyweights are being lifted or at the ankle joint during walking or running. At present, the measurement of unnecessary cocontractions is only possible by monitoring the EMG activity of the antagonistic muscles. Without an exact EMG calibration versus tension for each muscle, it is impossible to arrive at a quantitative measure of cocontraction. Falconer and Winter (1985) presented a formula by which cocontraction can be quantified,

$$\%\text{COCON} = 2 \times \frac{M_{\text{antag}}}{M_{\text{agon}} + M_{\text{antag}}} \times 100\% \tag{6.12}$$

where M_{antag} and M_{agon} are the moments of force of antagonists and agonists, respectively.

In the example reported, the antagonist activity results in an equal increase in agonist activity; thus, the unnecessary activity must be twice that of the antagonist alone. If the antagonists created a 20 N m extensor moment and the agonists generated a 50 N m flexor moment, %COCON would be 40/70 × 100% = 57%. However, most movements involve continuously changing muscle forces; thus, an agonist muscle at the beginning of the movement will likely reverse its role and become an antagonist later on in the movement. Joint moments of force, as seen in many graphs in Chapter 4, reverse their polarities many times; thus, a modification of Equation (6.12) is needed to cope with these time-varying changes. Figure 6.9 demonstrates the profile of activity of two antagonistic muscles during a given movement. The cross-hatching of muscles *A* and *B* shows a common area of activity that indicates the cocontraction area. Thus, the percent cocontraction is defined as:

$$\%\text{COCON} = 2 \times \frac{\text{common area } A \,\&\, B}{\text{area } A + \text{area } B} \times 100\% \tag{6.13}$$

If EMG is the primary measure of relative tension in the muscle, we can suitably process the raw EMG to yield a tension-related activation profile (Milner-Brown et al., 1973; Winter, 1976).

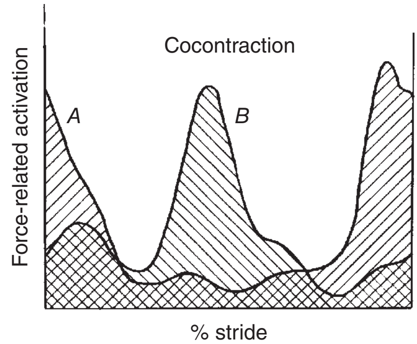

Figure 6.9 Profiles of activity of two antagonist muscles, with cross-hatched area representing the cocontraction. See the text for detailed discussion and analysis.

The activity profiles of many common muscles look very much like those portrayed in Figure 6.9. In this case, muscle A is the tibialis anterior, and muscle B is the soleus as seen over one walking stride. Using Equation (6.13) for the profiles in Figure 6.5, %COCON was calculated to be 24%.

It needs to be noted that additional methods for calculating cocontraction exist. Equation (6.14) has been used to quantify muscle cocontraction in clinical studies.

$$\%\text{COCON} = \left(M_{\text{lower}}/M_{\text{higher}}\right) \times \left(M_{\text{lower}} + M_{\text{higher}}\right) \tag{6.14}$$

M_{lower} and M_{higher} represent the EMG values from the lower and higher amplitude muscle, respectively, at each point in time (Rudolph et al., 2000). In this equation, the magnitude of each muscle is considered, as well as the magnitude of their relative cocontraction. From a clinical perspective, this may be an important consideration because the overall compression force within a joint is dependent not only on the relative cocontration between opposing muscles (ratio) but also the magnitude of the individual muscle force. Interpreting the results from studies in which cocontraction is calculated must be done so cautiously. The equation used to estimate cocontraction can dramatically alter the magnitude of the cocontraction value (Kellis et al., 2003).

6.1.1.2 Isometric Contractions Against Gravity. In normal dynamic movement, there is minimal muscle activity that can be attributed to holding limb segments against the forces of gravity. This is because the momentum of the body and limb segments allows for a smooth interchange of energy. However, in many pathologies, the movement is so slow that there are extended periods of time when limb segments or the trunk are being held in near-isometric contractions. Spastic cerebral palsy patients often crouch with their knee flexed, requiring excessive quadriceps activity to keep them from falling down. Other times, individuals with central nervous system disorders may have poor coordination and hold their limb off the ground for a period of time prior to swing-through. This would require muscle activity to maintain the limb in an unsupported position against gravity.

Most movement assessments, be it clinical or research studies, do not quantify isometric work against gravity because there is no movement involved. While quantification of EMG signals, or the use of EMG-driven models, can be used to provide some measure of the efficiency of movement, each muscle's EMG would have to be calibrated against the extra metabolism required to contract that muscle. At present, no there is no standard technique to separate the metabolic cost of this inefficiency.

6.1.1.3 Generation of Energy at One Joint and Absorption at Another. The least known and understood cause of inefficiency occurs when one muscle group at one joint does positive work at the same time as negative work is being done at others. Such an occurrence is really an extension of what occurs during a cocontraction (e.g. positive work being canceled out by negative work of the antagonistic muscles). It is quite difficult to visualize when this happens. During normal walking, it occurs during double support when the energy increase of the push off leg takes place at the same time as the weight-accepting leg absorbs energy. Figure 6.10 shows this point in gait: the

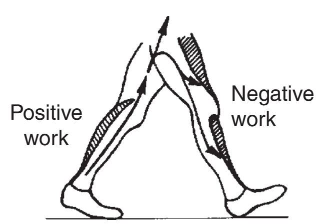

Figure 6.10 Example of a point in time during gait that positive work by the push off muscles can be canceled by negative work of the weight-accepting muscles of the contralateral leg.

left leg's push off (positive work) is due primarily to plantarflexors; the right leg's energy absorption (negative work) takes place in the quadriceps and tibialis anterior. There is no doubt that the instability of pathological gait is a major cause of this type of inefficient muscle activity. The only way to analyze such inefficiencies is to calculate the muscle power at each joint separately and to quantify the overlap of simultaneous phases of positive and negative work.

In spite of the inefficiencies inferred by such events, it must be remembered that many complex movements such as walking or running require that several functional tasks be performed at the same time. The example in Figure 6.10 illustrates such a situation: the plantarflexors are completing push off while the contralateral muscles are involved in weight acceptance. Both these events are essential to a safe walking pattern.

6.1.1.4 Jerky Movements. Efficient energy exchanges are characterized by smooth-looking movements. A ballet dancer and a high jumper execute smooth movements for different reasons, one for artistic purposes, the other for efficient performance. Energy added to the body by positive work at one point in time is conserved, and very little of this energy is lost by muscles doing negative work. The jerky gait of a child with cerebral palsy is quite the opposite. Energy added at one time is removed a fraction of a second later. The movement has a steady succession of stops and starts, and each of these bursts of positive and negative work has a metabolic cost. The energy cost from jerky movements can be assessed in two ways: by work analysis based on a segment-by-segment energy analysis or by a joint-by-joint power analysis. Both of these techniques are described later.

6.1.2 Summary of Energy Flows

It is valuable to summarize the flows of energy from the metabolic level through to an external load. Figure 6.11 depicts this process schematically. Metabolic energy cannot be measured directly but can be calculated indirectly from the amount of O_2 required or by the CO_2 expired. The details of these calculations and their interpretation are the subject of many textbooks and are beyond the scope of this book.

At the basal level (resting, lying down), the muscles are relaxed, but they still require metabolic energy to keep them alive. The measure of this energy level is called *maintenance heat*. Then as a muscle contracts, it requires energy, which shows up as additional heat called *activation heat*. It has been shown to be associated with the rate of build-up of tension within the muscle and is accompanied by an internal shortening of the muscle contractile elements. Stable heat is the heat that measures the energy required to maintain tension within accounted for the muscle. *Labile heat* is a third type of heat seen in isometric contractions and is the heat not generated by either tension or rate of tension generation. The final type of heat loss is *shortening heat*, which is associated with the actual shortening of the muscle under load. Students are referred to the excellent review by Hill (1960).

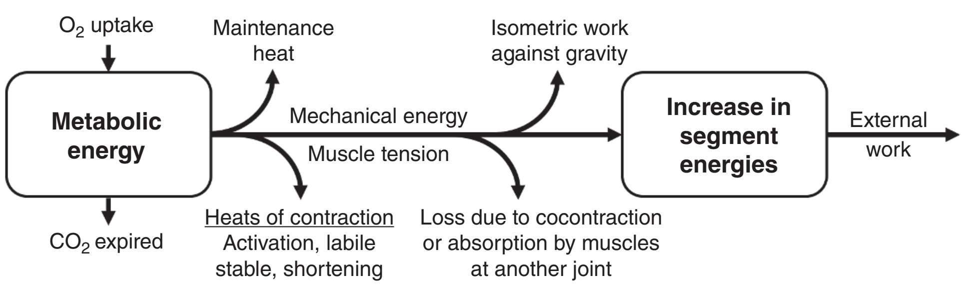

Figure 6.11 Flow of energy from metabolic level of external mechanical work. Energy is lost as heat associated with the contractile process or by inefficiencies after energy has been

Finally, at the tendon, we see the energy in mechanical form. The muscle tension can contribute to four types of mechanical loads. It can be involved with a cocontraction, isometric "work" against gravity, or a simultaneous absorption by another muscle, or it can cause a net change in the body energy. In the latter case, if positive work is done, the net body energy will increase; if negative work is done, there will be a decrease in total body energy. Finally, if the body exerts forces on an external body, some of the energy may be transferred as the body performs external work.

6.2 FORMS OF ENERGY STORAGE

1. **Potential Energy.** PE is the energy due to gravity and, therefore, increases with the height of the body above ground or above some other suitable reference datum,

$$\text{PE} = mgh \text{ J} \tag{6.15}$$

where

m = mass, kg
g = gravitational acceleration, equal to 9.8 m/s^2
h = height of center of mass, m.

With $h = 0$, the PE decreases to zero. However, the ground reference datum should be carefully chosen to fit the problem in question. Normally, it is considered to be the lowest point that the body takes during the given movement. For a diver, it could be the water level; for a person walking, it would be the lowest point in the pathway.

2. **Kinetic Energy.** There are two forms of KE, that due to translational velocity and that due to rotational velocity,

$$\text{translational KE} = 1/2mv^2 \tag{6.16}$$

where v = velocity of center of mass, m/s.

$$\text{rotational KE} = 1/2I\omega^2 \tag{6.17}$$

where

I = rotational moment of inertia, kg·m^2
ω = rotational velocity of segment, rad/s.

Note that these two energies increase as the velocity is squared. The polarity of direction of the velocity is unimportant because velocity squared is always positive. The lowest level of KE is therefore zero when a body is at rest.

3. **Total Energy and Exchange within a Segment.** As mentioned previously, the energy of a body exists in three forms so that the total energy of a body is:

$$\begin{aligned} E_s &= \text{PE} + \text{translational KE} + \text{rotational KE} \\ &= mgh + \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \text{ J} \end{aligned} \tag{6.18}$$

It is possible for a body to exchange energy within itself and still maintain a constant total energy.

Example 6.3. Suppose that the baseball in the Example is thrown vertically. Calculate the potential and kinetic energies at the time of release, at maximum height, and when it reaches the ground. Assume that it is released at a height of 2 m above the ground and that the vertical accelerating force of 100 N is in excess of gravitational force. At release,

$$
\begin{aligned}
a &= 100\ \mathrm{m/s^2} \quad \text{(as calculated previously)}\\
v &= \int_0^{t_1} a\,dt = at_1 = 100t_1\\
t_1 &= 180\ \mathrm{ms}\\
v &= 18\ \mathrm{m/s}\\
\text{translational KE} &= \frac{1}{2}mv^2 = \frac{1}{2} \times 1 \times 18^2 = 162\ \mathrm{J}
\end{aligned}
$$

Note that this 162 J is equal to the work done on the baseball prior to release.

$$
\begin{aligned}
\text{total energy} &= \mathrm{PE}(t_1) + \text{translational KE}(t_1)\\
&= 19.6 + 162 = 181.6\ \mathrm{J}
\end{aligned}
$$

If we ignore air resistance, the total energy remains constant during the flight of the baseball, such that at t_2, when the maximum height is reached, all the energy is PE and KE = 0. Therefore, PE(t_2) = 181.6 J. This means that the baseball reaches a height such that $mgh_2 = 181.6$ J, and:

$$h_2 = \frac{181.6}{1.0 \times 9.8} = 18.5\ \mathrm{m}$$

At t_3, the baseball strikes the ground, and $h = 0$. Thus, PE(t_3) = 0 and KE(t_3) = 181.6 J. This means that the velocity of the baseball is such that $\frac{1}{2}mv^2 = 181.6$ J, or:

$$v = 19.1\ \mathrm{m/s}$$

This velocity is slightly higher than the release velocity of 18 m/s because the ball was released from 2 m above ground level.

6.2.1 Energy of a Body Segment and Exchanges of Energy Within the Segment

Most body segments contain all three energies in various combinations at any point in time during a given movement. A diver at the top of a dive has considerable PE, and during the dive, converts it to KE. Similarly, a boomerang when released has rotational and translational KE, and at peak height, some of the translational KE has been converted to PE. At the end of its travel, the boomerang will have regained most of its translational KE.

In a multisegment system, such as the human body, the exchange of energy can be considerably more complex. There can be exchanges within a segment or between adjacent segments. A good example of energy exchange within a segment is during normal gait. The upper part of the body (head, arms, and trunk [HAT]) has two peaks of PE each stride – during the midstance of each leg. At this time, HAT has slowed its forward velocity to a minimum. Then, as the body falls forward to the double-support position, HAT picks up velocity at the expense of a loss in

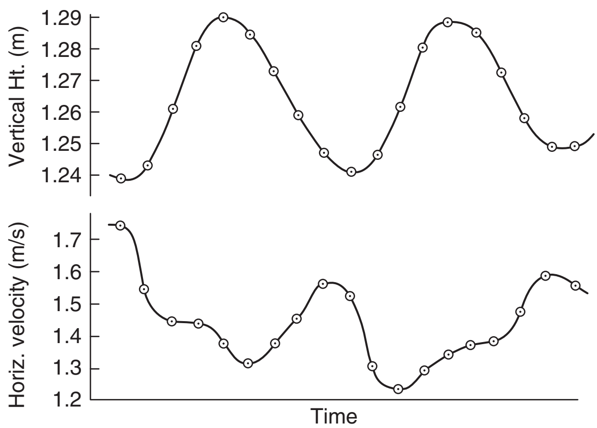

Figure 6.12 Plot of vertical displacement and horizontal velocity of HAT shows evidence of energy exchange within the upper part of the body during gait.

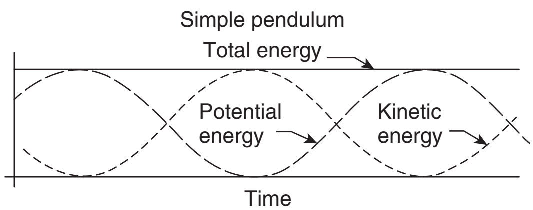

Figure 6.13 Exchange of kinetic and potential energies in a swinging frictionless pendulum. Total energy of the system is constant, indicating that no energy is being added or lost. (With permission of University of Toronto Press.)

height. Evidence of energy exchange should be seen from a plot of the horizontal velocity and the vertical displacement of the center of gravity of HAT (see Figure 6.12). The PE, which varies with height, changes roughly as a sinusoidal wave, with a minimum during double support and reaching a maximum during midstance. The forward velocity is almost completely out of phase, with peaks approximately during double support and minima during midstance.

Exchanges of energy within a segment are characterized by opposite changes of the potential and KE components. Figure 6.13 shows what would happen if a perfect exchange took place, as in a swinging frictionless pendulum. The total energy would remain constant over time in the presence of large changes of potential and KE.

Consider the other extreme, in which no energy exchange takes place. Such a situation would be characterized by totally in-phase energy components, not necessarily of equal magnitude, as depicted in Figure 6.14.

6.2.1.1 Approximate Formula for Energy Exchanges Within a Segment. An approximation of energy exchange can be calculated if we know the peak-to-peak change in each of the energy components over a period of time:

$$E_{\text{ex}} = \Delta E_p + \Delta E_{kt} + \Delta E_{kr} - \Delta E_s \tag{6.19}$$

If there is no exchange, $\Delta E_p + \Delta E_{kt} + \Delta E_{kr} = \Delta E_s$. If there is 100% exchange, $\Delta E = 0$.

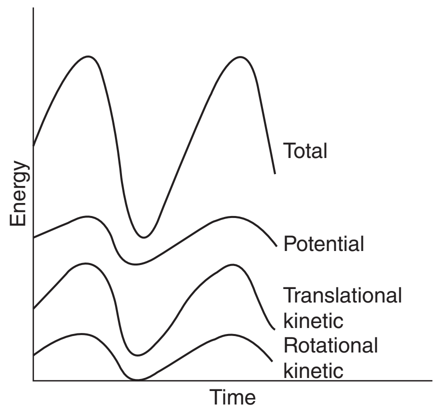

Figure 6.14 Energy patterns for a segment in which no exchanges are taking place. All energy components are perfectly in phase.

Example 6.4. A visual scan of the energies of the leg segment during walking yields the following maximum and minimum energies on the stride period:

$$\begin{aligned}
&E_s(\max) = 29.30\text{ J}, \quad E_s(\min) = 13.14\text{ J},\\
&E_p(\max) = 15.18\text{ J}, \quad E_p(\min) = 13.02\text{ J},\\
&E_{kt}(\max) = 13.63\text{ J}, \quad E_{kt}(\min) = 0.09\text{ J},\\
&E_{kr}(\text{oxmax}) = 0.95\text{ J}, \quad E_{kr}(\min) = 0\text{ J}
\end{aligned}$$

Thus,

$$\begin{aligned}
\Delta E_s &= 29.30 - 13.14 = 16.16\text{ J},\\
\Delta E_p &= 15.18 - 13.02 = 2.16\text{ J},\\
\Delta E_{kt} &= 13.63 - 0.09 = 13.54\text{ J},\\
\Delta E_{kr} &= 0.95 - 0 = 0.95\text{ J}
\end{aligned}$$

Since $\Delta E_p + \Delta E_{kt} + \Delta E_{kr} = 16.65$ J, it can be said that $16.65 - 16.16 = 0.49$ J were exchanged during the stride. Thus, the leg is a highly nonconservative system.

6.2.1.2 Exact Formula for Energy Exchange Within Segments. The example just discussed illustrates a simple situation in which only one minimum and maximum occur over the period of interest. If individual energy components have several maxima and minima, we must calculate the sum of the absolute energy changes over the time period. The work, W_s, done on and by a segment during N sample periods is:

$$W_s = \sum_{i=1}^{N} |\Delta E_s| \text{ J} \tag{6.20}$$

Assuming that there are no energy exchanges between any of the three components (Norman et al., 1976), the work done by the segment during the N sample periods is:

$$W'_s = \sum_{i=1}^{N} \left(|\Delta E_p| + |\Delta E_{kt}| + |\Delta E_{kr}|\right) \text{ J} \tag{6.21}$$

Therefore, the energy W_c conserved within the segment during the time is:

$$W_c = W'_s - W_s \text{ J} \tag{6.22}$$

The percentage energy conservation, C_s, during the time of this event is:

$$C_s = \frac{W_c}{W'_s} \times 100\% \tag{6.23}$$

If $W'_s - W_s$, all three energy components are in phase (they have exactly the same shape and have their minima and maxima at the same time), and there is no energy conservation. Conversely, as demonstrated by an ideal pendulum, if $W_s = 0$, then 100% of the energy is being conserved.

6.2.2 Total Energy of a Multisegment System

As we proceed with the calculation of the total energy of the body, we merely sum the energies of each of the body segments at each point in time (Bresler and Berry, 1951; Ralston and Lukin, 1969; Winter et al., 1976). Thus, the total body energy E_b at a given time is:

$$E_b = \sum_{i=1}^{B} E_{si} \text{ J} \tag{6.24}$$

where

E_{si} = total energy of ith segment at that point in time
B = number of body segments.

The individual segment energies continuously change with time, so it is not surprising that the sum of these energies will also change with time. However, the interpretation of changes in E_b must be done with caution when one considers the potential for transfer of energy between segments and the number of possible generators and absorbers of energy at each joint. For example, transfers of energy between segments (see Section 6.0.9) will not result in either an increase or a loss in total body energy. However, in other than the simplest movements, there may be several simultaneous concentric and eccentric contractions. Thus, over a period of time, two muscle groups may generate 30 J, while one other muscle group may absorb 20 J. The net change in body energy over that time would be an increase of 10 J. Only through a detailed analysis of mechanical power at the joints (see Section 6.3.1.4) are we able to assess the extent of such a cancelation. Such events are obviously inefficient but are necessary to accomplish many desired movement patterns.

Consider now a simple muscular system that can be represented by a pendulum mass with a pair of antagonistic muscle groups, m_1 and m_2, crossing a simple hinge joint. Figure 6.15 shows such an arrangement along with a time history of the total energy of the system. At t_1, the segment is rotating counterclockwise at ω_1 rad/s. No muscle activity occurs until t_2 when m_2 contracts. Between t_1, and t_2, the normal pendulum energy exchange takes place and the total energy remains constant. However, between t_2 and t_3, muscle m_2 causes an increase in both kinetic and potential energies of the segment. The muscle moment has been in the same direction as the direction of rotation, so positive work has been done by the muscle on the limb segment, and the total energy of the limb has increased. Between t_3 and t_4, both muscles are inactive, and the total energy remains at the higher but constant

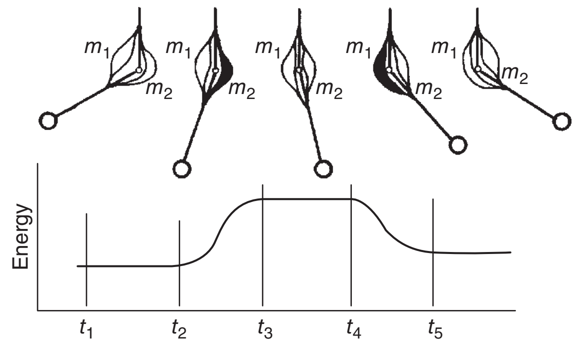

Figure 6.15 Pendulum system with muscles. When positive work is done, the total energy increases; when negative work is done, the total energy decreases.

down the segment. Energy is lost by the segment and is absorbed by m_1. This is negative work being done by the muscle because it is lengthening during its contraction. Thus, at t_5 the segment has a lower total energy than at t_4.

The following major conclusions can be drawn from this example.

1. When muscles do positive work, they increase the total body energy.
2. When muscles do negative work, they decrease the total body energy.
3. During cyclical activity, such as constant velocity level running, the net energy change per stride equals zero (neglecting the small air friction and shoe friction losses). Thus, the internal positive work done per stride should equal the internal negative work done per stride.

6.3 CALCULATION OF INTERNAL AND EXTERNAL WORK

Internal and external work has been calculated in a multitude of ways by different researchers. Some assume that the energy changes in the body center of mass yield the total internal work done by all the muscles. Others look only at the "vertical work" resulting from PE increases in the body center of gravity, while others (especially in the exercise physiology area) even ignore internal work completely. It is, therefore, important to look at all possible sources of muscle activity that have a metabolic cost and keep those readily available on a "checklist" to see how complete any analysis really is. This same list serves a useful purpose in focusing our attention on possible causes of inefficient movement (see Section 6.1.1).

6.3.1 Internal Work Calculation

The various techniques for calculating the internal work have undergone a general improvement over the years. The vast majority of the research has been done in the area of human gait, and because gait is a complex movement, it will serve well as an example of the dos and don'ts.

6.3.1.1 Energy Increases in Segments. A number of early researchers attempted a calculation of work based on increases in potential or kinetic energies of the body or of individual segments. Fenn (1929), in his accounting of the flow of energy from metabolic to mechanical, calculated the kinetic and potential energies of each major segment of a sprinter. He then summed the increases in each of these segment energies over the stride period to yield the net mechanical work. Unfortunately, Fenn's calculations ignored two energy exchanges within segments and passive transfers between segments. Thus, his mechanical work calculations were predictably high: the average power of his sprinters was computed to be three horsepower. Conversely, Saunders et al. (1953), Cotes and Meade (1960), and Liberson (1965) calculated the "vertical work" of the trunk as representing the total work done by the body. These calculations ignored the major energy exchange that takes place within the HAT and also the major work done by the lower limbs.

6.3.1.2 Center-of-Mass Approach. Cavagna and Margaria (1996) proposed a technique to calculate work based on potential and kinetic energies of the body's center of mass. Their data were based on force platform records during walking and running, from which the translational kinetic and potential energies were calculated. Such a model makes the erroneous assumption that the body's center of mass reflects the energy changes in all segments. The body's center of mass is a *vector* sum of all segment mass-acceleration products, and, as such, opposite-polarity accelerations will cancel out. But energies are scalars, not vectors, and, therefore, the reciprocal movements that dominate walking and running will be largely canceled. Thus, simultaneous increases and decreases in oppositely moving segments will go unnoticed. Also, Cavagna's technique is tied to force platform data, and nothing is known about the body's center of gravity during non-weight-bearing phases of running. Thus, this technique has underestimation errors and limitations that have been documented (Winter, 1979). Also, the center-of-mass approach does not account for the energy losses from the simultaneous generation and absorption of energy at different joints.

6.3.1.3 Sum of Segment Energies. A major improvement on the previous techniques was made by Ralston and Lukin (1969) and Winter et al. (1976). Using displacement transducers and television (TV) imaging techniques, the kinetic and potential energies of the major segments were calculated. A sum of the energy components within each segment recognized the conservation of energy within each segment (see Section 6.2.1) and a second summation across all segments recognized energy transfers between adjacent segments (see Section 6.0.9). The total bodywork is calculated (Winter, 1979) to be:

$$W_b = \sum_{i=1}^{N} |\Delta E_b| \text{ J} \tag{6.25}$$

However, this calculation underestimates the simultaneous energy generation and absorption at different joints. Thus, W_b will reflect a low estimate of the positive and negative work done by the human motor system. Williams and Cavanagh (1983) made empirical estimates to correct for these underestimates in running.

6.3.1.4 Joint Power and Work. In Sections 6.0.6 and 6.0.7, techniques for the calculation of the positive and negative work at each joint were presented. Using the time integral of the power curve (Equation 6.6), we are able to get at the "sources" and "sinks" of all the mechanical energy. Figure 6.16 is an example to show the work phases at the knee during slow running. The power bursts are labeled $K_1, \cdots, K_5$, and the energy generation/absorption resulting from the time integral of each phase is shown (Winter, 1983). In this runner, it is evident that the energies absorbed early in stance (53 J) by the knee extensors and by the knee flexors in later swing (24 J) dominate the profile; only 31 J are generated by the knee extensors in middle and late stance.

It should be noted that this technique automatically calculates any external work that is done. The external power will be reflected in increased joint moments, which, when multiplied by the joint angular velocity, will show an increased power equal to that done externally.

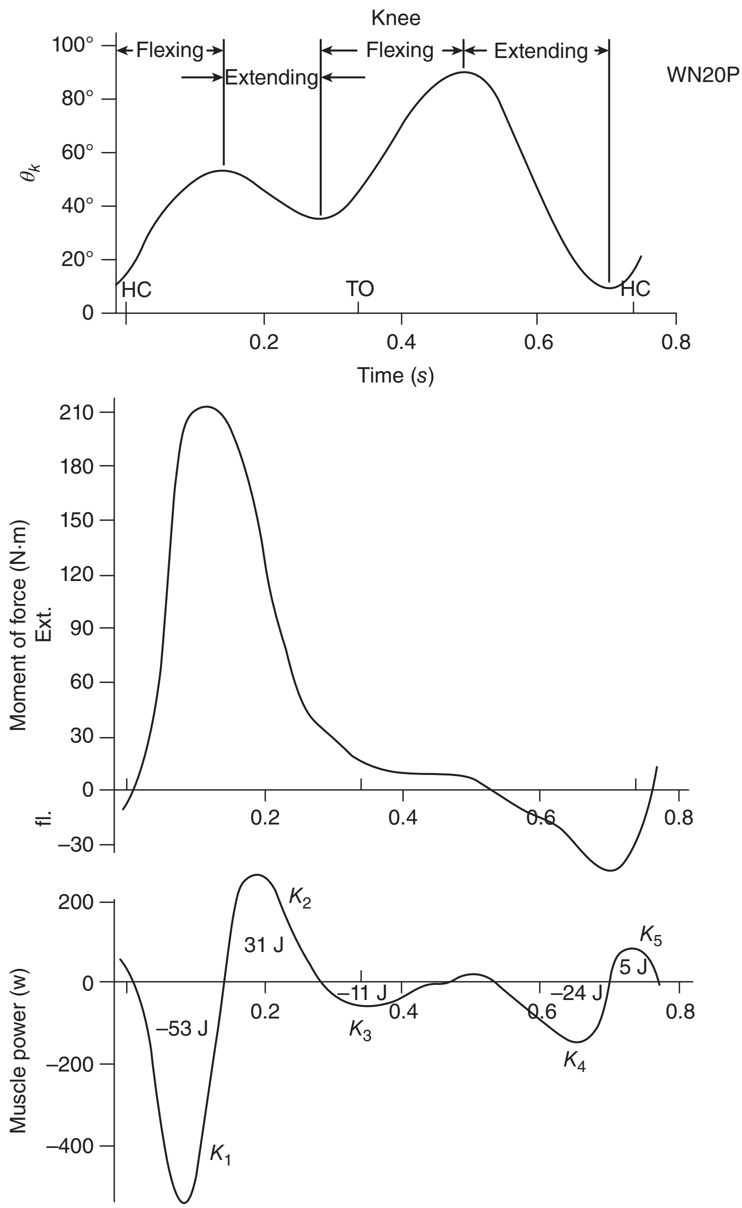

Figure 6.16 Plots of knee angle, moment of force, and power in a slow runner. Five power phases are evident: K_1, energy absorption by knee extensors; K_2, positive work as extensors shorten under tension; K_3, deceleration of backward rotation of leg and foot as thigh drives forward during late stance and early swing; K_4, deceleration of swinging leg and foot by knee flexors prior to heel contact; and K_5, small positive burst to flex the leg slightly and slow down its forward motion to near-zero velocity prior to heel contact. (From Winter (1983)/With permission of Elsevier.)

6.3.1.5 Muscle Power and Work. Even with the detailed analysis described in the previous section, we have underestimated the work done by cocontracting muscles. Joint power, as calculated, is the product of the joint moment of force M_j and the angular velocity ω_j. M_j is the net moment resulting from all agonist and antagonist activity, and therefore, cannot account for simultaneous generation by one muscle group and absorption by the antagonist group, or vice versa. For example, if $M_j = 40\,\mathrm{N \cdot m}$ and $\omega_j = 3$ rad/s, the joint power would be calculated to be 120 W. However, if there were a cocontraction, the antagonists might be producing a resisting moment of 10 N m. Thus, in this case, the agonists would be generating energy at the rate of $50 \times 3 = 150$ W while the antagonists would be absorbing energy at a rate of $10 \times 3 = 30$ W.

Thus, the net power and work calculations as described in Section 6.3.1.4 will underestimate both the positive and the negative work done by the muscle groups at each joint. To date, there has been very limited progress to calculate the power and work associated with each muscle's action. The major problem is to partition the contribution of each muscle to the net moment, and this issue has been addressed in Section 5.3.1. However, if the muscle force F_m and the muscle velocity V_m were known, the muscle work W_m would be calculated as:

$$W_m = \int_{t_1}^{t_2} F_m \cdot V_m \, dt \tag{6.26}$$

Morrison (1970) analyzed the power and work in four muscles in normal walking, and some later work (Yack, 1986) analyzed the muscles forces and powers in the three major biarticulate muscle groups during walking.

6.3.1.6 Summary of Work Calculation Techniques. Table 6.1 summarizes the various approaches described over the past few decades and the different energy components that are not accounted for by each technique.

6.3.2 External Work Calculation

It was noted in Sections 6.3.1.4 and 6.3.1.5 that the work calculations done using Equations (6.6) and (6.26) automatically take into account all work done by the muscles independent of whether that work was internal or external. There is no way to partition the external work except by taking measurements at the interface between human and external load. A cyclist, for example, would require a force transducer on both pedals plus a measure of the velocity of the pedal. Similarly, to analyze a person lifting or lowering a load would need a force transducer between the hands and the load, or an imaging record of the load and the body (from which an inverse solution would calculate the reaction forces and velocity). The external work W_e is calculated as:

$$W_e = \int_{t_1}^{t_2} \overline{F}_r \cdot \overline{V}_c \, dt \tag{6.27}$$

where

$\overline{F}_r$ = reaction force vector, N
$\overline{V}_c$ = velocity of contact point, m/s
t_1, t_2 = times of beginning and end of each power phase.

6.4 POWER BALANCES AT JOINTS AND WITHIN SEGMENTS

In Section 6.1.2, examples were presented to demonstrate the law of conservation of mechanical energy within a segment. Also, in Section 6.0.6, muscle mechanical power was introduced, and in Section 6.0.9, the concept of passive energy transfers across joints was noted. We can now look at one other aspect of muscle energetics that is necessary before we can achieve a complete power

TABLE 6.1 Techniques to Calculate Internal Work in Movement

Technique	Work Components Not Accounted for by Technique				
Increase PE or KE (Fenn, 1929; Saunders et al., 1953; Liberson, 1965)	Energy exchange within segments and transfers between segments	Simultaneous increases or decreases in reciprocally moving segments	Simultaneous generation and absorption at different joints	Cocontractions	Work against g
Center of mass (Cavagna and Margaria, 1996)		Simultaneous increases or decreases in reciprocally moving segments	Simultaneous generation and absorption at different joints	Cocontractions	Work against g
Σ Segment energies (Winter, 1979)			Simultaneous generation and absorption at different joints	Cocontractions	Work against g
Joint power* $\int M_j\omega_j\,dt$ (Winter, 1983)				Cocontractions	Work against g
Muscle power* $\int F_m V_m\,dt$ (Yack, 1986)					Work against g

* Also accounts for external work if it is present.

balance segment by segment: the fact that active muscles can transfer energy from segment to segment in addition to their normal role of generation and absorption of energy.

6.4.1 Energy Transfer via Muscles

Muscles can function to transfer energy from one segment to the other if the two segments are rotating in the same direction. In Figure 6.17, we have two segments rotating in the same direction but with different angular velocities. The product of $M\omega_2$ is positive (both M and ω_2 have the same polarity), and this means that energy is flowing into segment 2 from the muscles responsible for moment M. The reverse is true as far as segment 1 is concerned, $M\omega_1$ is negative, showing that energy is leaving that segment and entering the muscle. If $\omega_1 = \omega_2$ (i.e. an isometric contraction), the same energy rate occurs and a transfer of energy from segment 1 to segment 2 via the isometrically acting muscles. If $\omega_1 > \omega_2$, the muscles are lengthening, and thus, absorption plus a transfer takes place, while if $\omega_1 < \omega_2$, the muscles are shortening and a generation as well as a transfer occur. Figure 6.18, from Robertson and Winter (1980), summarizes all possible power functions that can occur at a given joint, and if we do not account for these energy transfers through the muscles, we will not be able to account for the total power balance within each segment. Thus, we must modify Equation (6.5) to include the angular velocities of the adjacent segments in order to partition the transfer component. Therefore, ω_j is replaced by $(\omega_1 - \omega_2)$,

$$P_m = M_j(\omega_1 - \omega_2)\ \text{W} \tag{6.28}$$

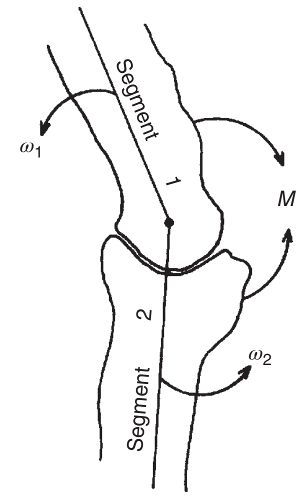

Figure 6.17 Energy transfer between segments occurs when both segments are rotating in the same direction and when there is a net moment of force acting across the joint. See the text for a detailed discussion.

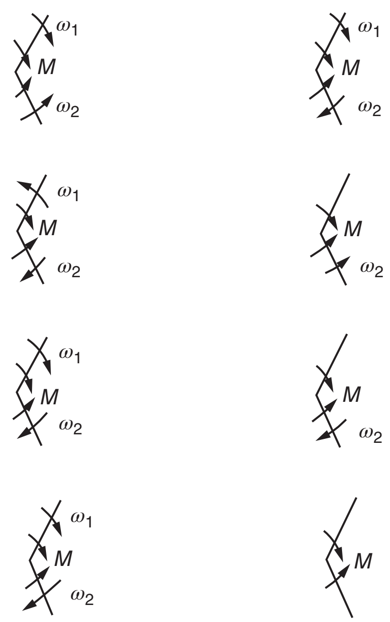

Figure 6.18 Power generation, transfer, and absorption functions. (From Roberston and Winter (1980)/With permission of Elsevier.)

Thus, if ω_1 and ω_2 have the same polarity, the rate of transfer will be the lesser of the two power components. Examples are presented in Section 6.4.2 to demonstrate the calculation and to reinforce the sign convention used.

6.4.2 Power Balance Within Segments

Energy can enter or leave a segment at muscles and across joints at the proximal and distal ends. Passive transfer across the joint (Equation 6.9) and active

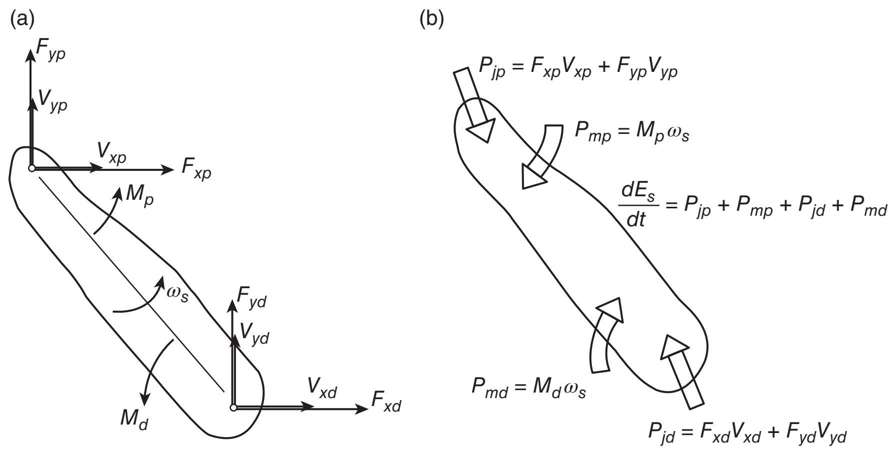

Figure 6.19 (a) Biomechanical variables describing the instantaneous state of a given segment in which passive energy transfers may occur at the proximal and distal joint centers and active transfers through the muscles at the proximal and distal ends. (b) Power balance as calculated using the variables shown in (a). The passive power flow at the proximal end P_{jp}, and the distal end P_{jd}, combined with the active (muscle) power at the proximal end P_{mp} and the distal end P_{md}, must equal the rate of change of energy in the segment dE_s/dt.

(Equation 6.28) must be calculated. Consider Figure 6.19a as the state of a given segment at any given point in time. The reaction forces and the velocities at the joint centers at the proximal and distal ends are shown plus the moments of force acting at the proximal and distal ends along with the segment angular velocity. The total energy of the segment E_s as calculated by Equation (6.18) must also be known. Figure 6.19b is the power balance for that segment, the arrows showing the directions where the powers are positive (energy entering the segment across the joint or through the tendons of the dominant muscles). If the force–velocity or moment–ω product turns out to be negative, this means that energy flow is leaving the segment. According to the law of conservation of energy, the rate of change of energy of the segment should equal the four power terms,

$$\frac{dE_s}{dt} = P_{jp} + P_{mp} + P_{jd} + P_{md} \tag{6.29}$$

(a) Biomechanical variables describing the instantaneous state of a given segment in which passive energy transfers may occur at the proximal and distal joint centers and active transfers through the muscles at the proximal and distal ends.
(b) Power balance as calculated using the variables shown in (*a*). The passive power flow at the proximal end P_{jp}, and the distal end P_{jd}, combined with the active (muscle) power at the proximal end P_{mp} and the distal end P_{md}, must equal the rate of change of energy of the segment dE_s/dt.

A sample calculation for two adjacent segments is necessary to demonstrate the use of such power balances and also to demonstrate the importance of passive transfers across joints and across muscles as major mechanisms in the energetics of human movement.

Example 6.5. Carry out a power balance for the leg and thigh segments for frame 5, that is, deduce the dynamics of energy flow for each segment separately and determine the power dynamics of the knee muscles (generation, absorption,

Table A.2a, hip velocities,

$$V_{xh} = 1.36\,\text{m/s} \quad V_{yh} = 0.27\,\text{m/s}$$

Table A.2b, knee velocities,

$$V_{xk} = 2.61\,\text{m/s} \quad V_{yk} = 0.37\,\text{m/s}$$

Table A.2c, ankle velocities,

$$V_{xa} = 3.02\,\text{m/s} \quad V_{ya} = 0.07\,\text{m/s}$$

Table A.3b, leg angular velocity,

$$\omega_{lg} = 1.24\,\text{rad/s}$$

Table A.3c, thigh angular velocity,

$$\omega_{th} = 3.98\,\text{rad/s}$$

Table A.5a, leg segment reaction forces and moments,

$$F_{xk} = 15.1\,\text{N}, \quad F_{yk} = 14.6\,\text{N}, \quad F_{xa} = -12.3\,\text{N}, \quad F_{ya} = 5.5\,\text{N},$$
$$M_a = -1.1\,\text{N}\cdot\text{m} \quad M_k = 5.8\,\text{N}\cdot\text{m}$$

Table A.5b, thigh segment reaction forces and moments,

$$F_{xk} = -15.1\,\text{N}, \quad F_{yk} = -14.6\,\text{N}, \quad F_{xh} = -9.4\,\text{N}, \quad F_{yk} = 102.8\,\text{N},$$
$$M_k = -5.8\,\text{N}\cdot\text{m}, \quad M_h = 8.5\,\text{N}\cdot\text{m}$$

Table A.6, leg energy,

$$E_{lg}(\text{frame6}) = 20.5\,\text{J}, \quad E_{lg}(\text{frame4}) = 20.0\,\text{J}$$

Table A.6, thigh energy,

$$E_{th}(\text{frame6}) = 47.4\,\text{J}, \quad E_{th}(\text{frame4}) = 47.9\,\text{J}$$

1. Leg Power Balance.

$$\begin{aligned}
\Sigma\ \text{powers} &= F_{xk}V_{xk} + F_{yk}V_{yk} + M_k\omega_{lg} + F_{xa}V_{xa} + F_{ya}V_{ya} + M_a\omega_{lg} \\
&= 15.1 \times 2.61 + 14.6 \times 0.37 + 5.8 \times 1.24 - 12.3 \\
&\quad \times 3.02 + 5.5 \times 0.7 - 1.1 \times 1.24 \\
&= 44.81 + 7.19 - 33.3 - 1.36 \\
&= 17.34\,\text{W} \\
\frac{\Delta E_{lg}}{\Delta t} &= \frac{20.5 - 20.0}{0.0286} = 17.5\,\text{W} \\
\text{balance} &= 17.5 - 17.34 = 0.16\,\text{W}
\end{aligned}$$

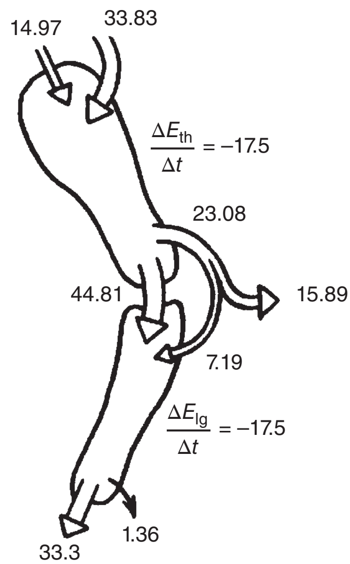

Figure 6.20 Summary of thigh and leg power flows as calculated in Example 6.5. There is power transfer from the thigh to the leg at a rate of 7.19 W through the quadriceps muscles plus passive flow across the knee of 44.81 W.

2. Thigh Power Balance.

$$\begin{aligned}
\Sigma \text{ powers} &= F_{xh}V_{xh} + F_{yh}V_{yh} + M_h\omega_{th} + F_{xk}V_{xk} + F_{yk}V_{yk} + M_k\omega_{th} \\
&= -9.4 \times 1.36 + 102.8 \times 0.27 + 8.5 \times 3.98 \\
&\quad -15.1 \times 2.61 - 14.6 \times 0.37 - 5.8 \times 3.98 \\
&= 14.97 + 33.83 - 44.81 - 23.08 \\
&= -19.09\text{ W} \\
\frac{\Delta E_{th}}{\Delta t} &= \frac{47.4 - 47.9}{0.0286} = -17.5\text{ W} \\
\text{balance} &= -17.5 - (-19.09) = 1.59\text{ W}
\end{aligned}$$

3. *Summary of Power Flows*. The power flows are summarized in Figure 6.20 as follows: 23.08 W leaves the thigh into the knee extensors, and 7.19 W enters the leg from the same extensors. Thus, the knee extensors are actively transferring 7.19 W from the thigh to the leg and are simultaneously absorbing 15.89 W.

6.5 PROBLEMS BASED ON KINETIC AND KINEMATIC DATA

1. **(a)** Calculate the potential, translational, and rotational kinetic energies of the leg segment for frame 20 using appropriate kinematic data, and check your answer with Table A.6 in Appendix A.

   (b) Repeat Problem 1*a* for the thigh segment for frame 70.

2. **(a)** Plot (every second frame) the three components of energy plus the total energy of the leg over the stride period and discuss whether this segment conserves or does not conserve energy over the stride period (frames 28–97).
   (b) Repeat Problem 2*a* for the thigh segment.
   (c) Repeat Problem 2*a* for the HAT segment. Using Equation (6.19), calculate the approximate percentage of energy conservation in the HAT segment over the stride period. Compare this percentage with that calculated using the exact Equations (6.20)–(6.23).
3. **(a)** Assuming symmetrical gait, calculate the total energy of the body for frame 28. [*Hint:* For a stride period of 68 frames, data for the left side of the body can be estimated using right-side data half a stride (34 frames) later.]
   (b) Scan the total energy of all segments, and note the energy changes over the stride period in the lower limb compared with that of the HAT. From your observations, deduce whether the movement of the lower limbs or that of HAT makes the major demands on the metabolic system.
4. **(a)** Using segment angular velocity data in Table A.7 in the appendix plus appropriate data from other tables, calculate the power generation or absorption of the muscles at the following joints. Identify in each case the muscle groups involved that are responsible for the generation or absorption. Check your numerical answers with Table A.7.
   (i) Ankle joint for frame 30.
   (ii) Ankle joint for frame 50.
   (iii) Ankle joint for frame 65.
   (iv) Knee joint for frame 35.
   (v) Knee joint for frame 40.
   (vi) Knee joint for frame 65.
   (vii) Knee joint for frame 20.
   (viii) Hip joint for frame 50.
   (ix) Hip joint for frame 70.
   (x) Hip joint for frame 4.
   (b) **(i)** Scan the listings for muscle power in Table A.7 and identify where the major energy generation occurs during walking. When in the gait cycle does this occur and by what muscles?
   (ii) Do the knee extensors generate any significant energy during walking? If so, when during the walking cycle?
   (iii) What hip muscle group generates energy to assist the swinging of the lower limb? When is this energy generated?
5. Using Equation (6.10) (see Figure 6.8), calculate the passive rate of energy transfer across the following joints, and check your answers with Table A.7. From what segment to what segment is the energy flowing?
   (i) Ankle for frame 20.
   (ii) Ankle for frame 33.
   (iii) Ankle for frame 65.
   (iv) Knee for frame 2.
   (v) Knee for frame 20.
   (vi) Knee for frame 65.
   (vii) Hip for frame 2.
   (viii) Hip for frame 20.
   (ix) Hip for frame 67.

6. **(a)** Using equations in Figure 6.19b, carry out a power balance for the foot segment for frame 20.
   (b) Repeat Problem 6(*a*) for the leg segment for frame 20.
   (c) Repeat Problem 6(*a*) for the leg segment for frame 65.
   (d) Repeat Problem 6(*a*) for the thigh segment for frame 63.
7. Muscles can transfer energy between adjacent segments when they are rotating in the same direction in space. Calculate the power transfer between the following segments, and indicate the direction of energy flow. Compare your answers with those listed in Table A.7.
   (a) Leg/foot for frame 60.
   (b) Thigh/leg for frame 7.
   (c) Thigh/leg for frame 35.

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7

UNDERSTANDING 3D KINEMATIC AND KINETIC VARIABLES

THOMAS HULCHER
Department of Physical Therapy, Jefferson College of Rehabilitation Science, Thomas Jefferson University, Philadelphia, PA, USA

7.0 INTRODUCTION

Over the past 10 years, there have been major commercial advances in three-dimensional (3D) hardware and software, with the most recent being in markerless technology. Chapter 3 included descriptions of some of the 3D imaging and software systems that have been introduced. The majority of the systems are near infrared, with multiple-camera arrangements requiring passive reflective markers, while other systems use video-based cameras and markerless technology. Regardless of the system used, the output of the data collection stage is a file of x, y, z coordinates of each of the markers or each segment in markerless technology at each sampled point in time. These coordinates are in the global reference system (GRS) that is fixed in the laboratory or data collection space. The purpose of this chapter is to go through the steps where these coordinate data are transformed into the anatomical axes of the body segments so that a kinetic analysis can be done in a similar manner, as has been detailed for two-dimensional (2D) analyses in previous chapters.

7.1 AXES SYSTEMS

There are several axes reference systems that must be introduced in addition to the GRS. The markers that are placed on each segment provide a marker axis system that is a local reference system (LRS) for each individual segment. A second LRS is the axis system that defines the principal axis of each segment. Because skeletal landmarks are used to define these axes, this system is referred to as the anatomical axis system.

7.1.1 Global Reference System

For purposes of convenience, we will be consistent in our axis directions for the GRS: X is the forward/backward direction, Z is the vertical (gravitational) axis, and Y is the left/right

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

(medial/lateral) axis. Thus, the *XY* plane is the horizontal plane and, by definition, is orthogonal to the vertical axis. The directions of these GRS axes are the same as those of the axes in the force plate. To ensure that this is so, a spatial calibration system (a rigid L-frame of precisely known dimensions or a rigid 3D mechanical axis) is instrumented with markers and is placed on one of the force plates and aligned along the *X* and *Y* axes of the force platform. The position of each of the markers relative to the origin of the force plate is known and fed into the computer. If more than one force platform is used, the origin of each additional platform is recorded by an *X* and *Y* offset from the primary platform. An additional offset in the *Z* direction would be necessary if the additional platform were at a different height from the first (as would be necessary for a biomechanical analysis of stairway or ramp walking). In many research situations, the cameras are rearranged to best capture the movement of a particular study and, therefore, require a new calibration of the GRS each time they are moved. Once the calibration is complete, the cameras cannot be moved, and care must be taken to ensure they are not accidentally displaced. Even in laboratories where the equipment is not moved, it is important to periodically calibrate the cameras to prevent drift and error propagation.

7.1.2 Local Reference Systems and Rotation of Axes

Within each segment, the anatomical axis system is set with its origin at the center of mass (COM) of the segment, and its principal z axis is usually along the long axis of the segment or, in the case of segments like the pelvis, along a line defined by skeletal landmarks such as posterior superior iliac spine and anterior superior iliac spine (PSIS and ASIS). The other local axis system is constructed on the segment by using a set of surface markers. A total of two transformations are necessary to get from the GRS to the marker axis system and from the marker to the anatomical axis system. Figure 7.1 shows how one of those rotations is done. The axis system x, y, z needs to be rotated into the system denoted by x''', y''', z'''. Many sequences of rotation are possible; here, we use the common Cardan sequence x–y–z. This means that we rotate about the x axis first, about the new y axis second, and about the new z axis last. The first rotation is θ_1 about the x axis to get x', y', z'. Because we have rotated about the x axis, x will not be changed and $x' = x$, while the y axis changes to y' and the z axis to z'. The second rotation is θ_2 about the new y' axis to get x'', y'', z''. Because this rotation has been about the y' axis, $y'' = y'$. The final rotation is θ_3 about the new z''

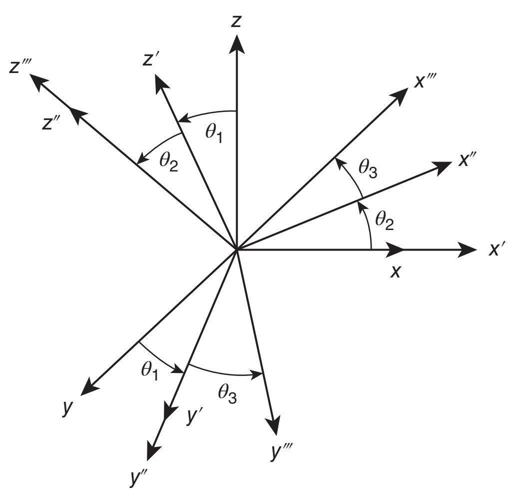

Figure 7.1 Cardan sequence of three rotations about the x, y, z axes. The first rotation, θ_1, about the x axis to get x', y', z'; the second rotation, θ_2, about the new y' axis to get x'', y'', z'', and the final rotation, θ_3, is about the new z'' axis to get the desired x''', y''', z'''.

axis to get the desired x''', y''', z'''. Assuming that we have a point with coordinates x_0, y_0, z_0 in the original x, y, z axis system, that same point will have coordinates x_1, y_1, z_1 in the x', y', z' axis system. Based on the rotation θ_1:

$$\begin{aligned} x_1 &= x_0 \\ y_1 &= y_0 \cos\theta_1 + z_0 \sin\theta_1 \\ z_1 &= -y_0 \sin\theta_1 + z_0 \cos\theta_1 \end{aligned}$$

Using the shorthand notation $c_1 = \cos\theta_1$ and $s_1 = \sin\theta_1$, in matrix notation, this may now be written as:

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_1 & s_1 \\ 0 & -s_1 & c_1 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} = [\Phi_1] \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \tag{7.1}$$

After the second rotation θ_2 about y', this point will have coordinates x_2, y_2, z_2 in the x'', y'', z'' axis system.

$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} c_2 & 0 & -s_2 \\ 0 & 1 & 0 \\ s_2 & 0 & c_2 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = [\Phi_2] \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \tag{7.2}$$

Finally, the third rotation θ_3 about z'' yields the coordinates x_3, y_3, z_3 in the x''', y''', z''' axis system.

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = [\Phi_3] \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \tag{7.3}$$

Combining Equations (7.1)–(7.3), we get:

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = [\Phi_3][\Phi_2][\Phi_1] \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \tag{7.4}$$

Note that the matrix multiplication as shown in Equation (7.4) is not commutative, which means that the order of the transformations must be such that $[\Phi_1]$ is done first, $[\Phi_2]$ second, and $[\Phi_3]$ last. In other words, $[\Phi_1][\Phi_2] \neq [\Phi_2][\Phi_1]$ An expansion of Equation (7.4) yields:

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} c_2c_3 & s_3c_1 + s_1s_2c_3 & s_1s_3 - c_1s_2c_3 \\ -c_2s_3 & c_1c_3 - s_1s_2s_3 & s_1c_3 + c_1s_2s_3 \\ s_2 & -s_1c_2 & c_1c_2 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} \tag{7.5}$$

7.1.3 Other Possible Rotation Sequences

In theory, there are 12 possible correct rotation sequences; all were introduced by the Swiss mathematician, Leonhard Euler (1707–1783). The list that sequences. The example explained previously is generally referred to as the Cardan system, which is commonly used in biomechanics, while the z–x–z rotation sequence, generally referred to as the Euler system, is commonly used in mechanical engineering.

$$\begin{array}{llll}
x-y'-x'' & x-y'-z''\,(\text{Cardan}) & x-z'-x'' & x-z'-y'' \\
y-x'-y'' & y-x'-z'' & y-z'-x'' & y-z'-y'' \\
z-x'-y'' & z-x'-z''\,(\text{Euler}) & z-y'-x'' & z-y'-z''
\end{array}$$

7.1.4 Dot and Cross Products

In three-dimensional motion analysis, we are dealing almost exclusively with vectors. When vectors are multiplied, we must compute the mathematical function called the *dot* or *cross product*. The dot product is also called the scalar product because the result is a scalar while the cross product is also called the vector product because the result is a vector. Dot product was first introduced in Chapter 6 in the calculation of the mechanical power associated with a force and velocity vector. Only the component of the force, F, and the velocity, V, that are parallel result in the power, $P = F \cdot V = |F||V| \cos\theta$ where θ is the angle between F and V in the FV plane. In 3D, the power, $P = F_xV_x + F_yV_y + F_zV_z$.

Cross products are used to find the product of two vectors in one plane where the product is a vector normal to that plane. Suppose we have vectors $\mathrm{A} = (A_x,\ A_y,\ A_z)$ and $\mathrm{B} = (B_x,\ B_y,\ B_z)$ and the cross product is a vector C defined as $\mathrm{C} = (\mathrm{A} \times \mathrm{B})$. C is perpendicular to A and B in the direction defined by the right-hand rule and has a magnitude $\mathrm{C} = |\mathrm{A}||\mathrm{B}| \sin\theta$ where θ is the angle between A and B. The vector C is calculated by the expansion of the determinant where i, j, and k are unit vectors in the x, y, z axes:

$$\mathrm{C} = \mathrm{A} \times \mathrm{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\begin{aligned}
\mathrm{C} &= \left(A_yB_z - A_zB_y\right)i - (A_xB_z - A_zB_x)j + \left(A_xB_y - A_yB_x\right)k \\
&= iC_x + jC_y + kC_z
\end{aligned} \tag{7.6}$$

7.2 MARKER AND ANATOMICAL AXES SYSTEMS

The following description outlines the steps that are necessary to transform the x, y, z marker coordinates from the GRS to the anatomical axes of the segments of the person whose movement is being analyzed. Figure 7.2 presents the axis systems involved for a given segment whose COM is at c and whose axes x–y–z are as shown. The GRS has axes X–Y–Z, and they are fixed for any given camera arrangement. The second axis system, x_m–y_m–z_m, is the marker axis system for each segment, and this can vary from laboratory to laboratory. Even within a given laboratory, each experiment could have a different arrangement of markers. For a correct 3D analysis, there must be at least three independent markers per body segment. The markers on each segment must not be collinear, which means they must not be in a straight line. They must form a plane in 3D space; as shown in Figure 7.2, the three tracking markers m_{T1}, m_{T2}, and m_{T3} define the tracking marker plane. This plane is assumed to contain the x_m and z_m axes such that all three markers are in the +ve x_m and the +ve z_m quadrant. One point on this marker plane is arbitrarily chosen as the origin of the marker axes system; here m_{T1} is chosen and is labeled

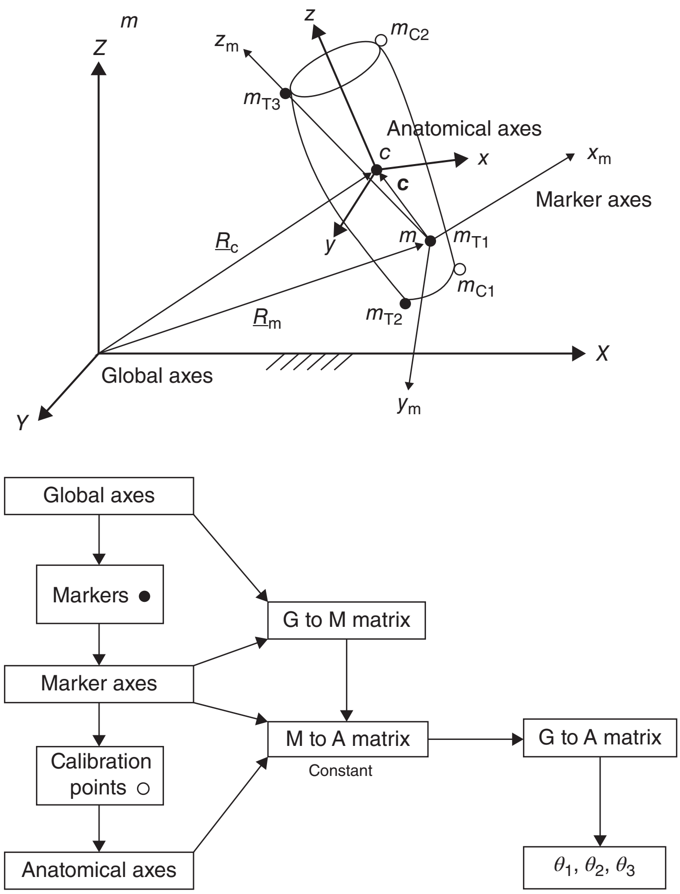

Figure 7.2 An anatomical segment showing the GRS, the marker axes, and anatomical axes. Three tracking markers, m_{T1}, m_{T2}, and m_{T3}, in conjunction with two calibration markers, m_{C1} and m_{C2}, define the constant [M to A] matrix. The three tracking markers in the GRS define the [G to M] matrix. The product of the variable [G to M] and the fixed [M to A] matrices gives the variable [G to A] matrix from which a new θ_1, θ_2, and θ_3 are defined for each frame.

the +ve z_m axis; y_m is normal to the tracking plane, and x_m is orthogonal to the plane defined by y_m–z_m to form a right-hand system.

The purpose of the anatomical calibration process is to find the relation between the marker axes, x_m–y_m–z_m, and the anatomical axes, x–y–z. This process requires the subject to assume a well-defined position; usually, the anatomical position is used or a "T" position. At this time, extra calibration markers may be placed temporarily on the segment to define well-known anatomical points from which the segment's anatomical axes can be defined. For example, for the leg segment, the three tracking markers could be placed on the head of the fibula (m_{T3}), on the lateral malleolus (m_{T2}), and at the midpoint on the anterior surface of the tibia (m_{T1}). During calibration, temporary markers, m_{C1} and m_{C2}, could be placed on the medial malleolus and the medial epicondyle of the tibia, respectively. With the subject standing still for about a second, the coordinates of the three tracking and the two calibration markers are recorded and averaged over the calibration time. The long axis of the leg segment (z axis) would then be defined as the line joining the midpoint between the lateral and medial malleoli (m_{T2} and m_{C1}) to the midpoint between the head of the fibula and the medial epicondyle of the tibia (m_{T3} and m_{C2}). These midpoints are the ankle and knee joint centers, respectively. The leg z axis plane normal to which is the leg x axis. The direction of the leg y axis would be defined as a line normal to the leg x–z plane such that the leg x–y–z is a dextral system. The anatomical axes of the leg are now defined relative to the three tracking markers. The location of the COM of the leg would be a known distance along the z axis of the leg from the ankle joint; thus, the vector, **c**, from m, the origin of the tracking marker axis system, is also known. The two calibration markers are now removed and are no longer needed because the orientation of the axis system of the three tracking markers is now known and assumed to be fixed relative to the newly defined anatomical axes.

In clinical gait laboratories, it is impossible for many patients, such as patients with cerebral palsy or previous stroke, to assume the anatomical position for even a short period of time. Thus, the clinical gait teams have developed a consistent marker arrangement combined with a number of specific anthropometric measures. These include such measures as ankle and knee diameters which, when combined with generic X-ray anthropometric measures, allow the team to input an algorithm so that offset displacements from the tracking markers to the joint centers are known. Patients are then asked to assume a static standing position with a number of temporary calibration markers similar to what has been described previously. In effect, the major single difference in the clinical laboratory is that calibration is performed on the patient in a comfortable standing position rather than the anatomical position. For the complete detailed steps in arriving at the joint kinetics in an operational clinical laboratory, the reader is referred to Davis et al. (1991) and Õunpuu et al. (1996).

In Figure 7.2, we see two matrix rotations. [G to M] is a 3 × 3 rotation matrix that rotates from the GRS to the tracking marker axes, x_m–y_m–z_m. This is a time-varying matrix because the tracking marker axes will be continuously changing relative to the GRS. [M to A] is a 3 × 3 matrix that rotates from the tracking marker axes to the anatomical axes. This matrix is assumed to be constant and results from the calibration protocol. The combination of these two rotation matrices gives us the [G to A] rotation matrix, which, when solved for a selected angle sequence, yields the three time-varying rotation angles, θ_1, θ_2, θ_3. With this final matrix, we can get the orientation of the anatomical axes directly from the tracking marker coordinates that are collected in the GRS.

However, from Figure 7.2, we are not yet finished. We also have to find a translational transformation to track the 3D coordinates of the segment COM, **c**, over time. The location of **c** is defined by the vector, $\mathbf{R}_c$, which is a vector sum of $\mathbf{R}_m + \mathbf{c}$. The vector $\mathbf{R}_m$ is the GRS coordinates of the tracking marker, m_{T1}, while **c** is a constant vector joining m to c, as defined earlier.

7.2.1 Markerset Design

The same type of markers is typically used for both tracking and calibration. These markers are usually spherical, though cubic markers have been used in the past, and can vary in size from 1 mm to 5 cm in diameter. The appropriate marker size is determined by: (1) proximity of the subject to the camera and (2) the resolution of the camera. In the past, camera resolution was much lower than modern cameras requiring the large 5 cm markers. Current cameras have higher resolution and therefore much smaller markers can be used. For most applications, a marker that is 10 mm in diameter is appropriate. Markers under 5 mm are commonly used when the camera is in close proximity to the participant and a large number of markers are required, such as in facial expression tracking.

When selecting a marker size, marker density needs to be considered. Each marker needs to be seen as a distinctive point and caution should be taken when placing a marker close to another. Depending on the camera's view, the two markers may be detected as a single marker. Motion capture cameras work by detecting reflected near-infrared light. When a reflection is seen, it finds the centroid of that reflection and uses that location as the center of the marker, as shown in Figure 7.3a. If multiple markers are overlapping in the camera view, the computer will see the markers as one large reflection and find the centroid of

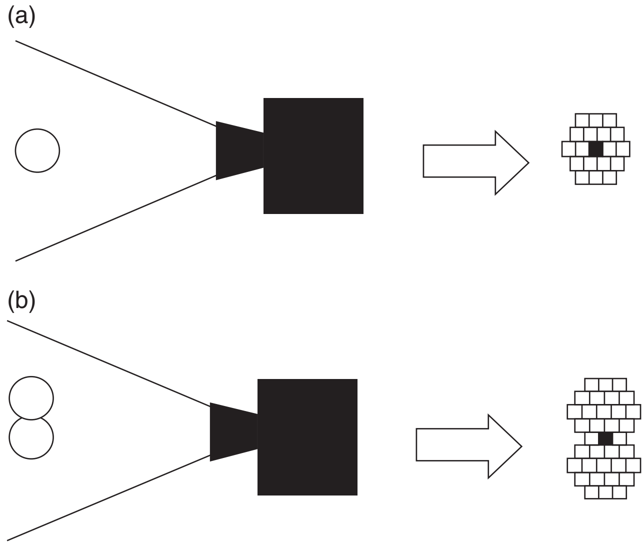

Figure 7.3 A pictorial representation of how a camera sees and interprets a (a) single and (b) overlapping marker. The black square in each is the interpreted centroid of the viewed reflection.

marker, as shown in Figure 7.3b, resulting in neither marker having its location correct and a marker missing from the markerset. The closer a marker is to the camera, the more pixels are dedicated to its reflection and, conversely, the farther away it is, the fewer pixels are dedicated to it. Therefore, choosing a marker that will be represented by an adequate number of pixels, but not overlap with other markers, is important to the markerset selection process.

In addition to size, there are other considerations when selecting an appropriate marker for motion capture. One of the biggest considerations is how the marker will be attached to the subject. Commercially sold markers generally come with a flat base attached to the sphere and many researchers use hoop-and-loop fasteners (like Velcro) to easily attach and remove the markers from subjects. In scenarios where the markers will always be in the same location, such as on a piece of sporting equipment, the markers can also be glued to the location. The markers must be reflective to the radiation emitted by the cameras (typically near infrared) and capable of withstanding wear and tear. When testing high-speed motions that are often observed in athletes, markers can collide with objects or become detached and fly off the participant. In these situations, double-sided tape or strong adhesive tape is recommended to surround the marker base to prevent detachment even when the athlete begins sweating. The marker coatings must be periodically inspected for damage to the reflective surface, such as nicks and tears. When this damage is found, the markers should be repaired with new reflective coating or replaced because breakdown of the reflective surface inhibits the ability of the marker to be properly tracked.

Over time some standardized markersets have been created and commonly used or modified to collect biomechanics data. The earliest and most well-known markersets are the Helen-Heyes and Cleveland Clinic markersets, which are full-body tracking markersets. The Helen-Heyes markerset uses a minimal number of markers to define and track the segments. However, because of the minimal nature of the markerset, it has more constraints on its degrees of freedom available for calculation and a higher error rate in joint angle calculations (Collins et al., 2009). Many research articles have used this markerset or a modified version in defining the body segments. The Cleveland Clinic markerset uses clusters on the lower body segments for tracking in order to better display data in the transverse plane. In response to the error issues with the Helen-Heyes and Cleveland Clinic markersets, a six degrees of freedom (6DOF) markerset was developed, combining the strong elements of the previous two markersets, clusters on the thigh and shank segments in order to better track motion in all planes (Collins et al., 2009). In many labs, researchers use these markersets as a starting point to create a markerset that best captures the data important to the research question under investigation and one of the strengths of markersets is the flexibility they provide in researching movement.

When placing the markers or collecting data, issues can arise that researchers need to be cognizant of and know how to fix. One of the most common problems is the medial markers, especially on the legs, colliding with each other and falling off. This can be fixed by ensuring there are enough tracking markers (at least 3) on each segment so that the medial markers are only needed in calibration and can be removed for dynamic collection. When using marker clusters on a solid backing, the clusters are frequently affixed to cloth wrapped around the segment. Attention needs to be paid to these cloth wraps to make sure they are not sliding down the limb as time progresses, which will create significant errors as the markers shift from the calibration reference position.

7.2.2 Event Detection Methods for Gait

One of the most common activities researched in biomechanics is gait. Gait data is often broken down into the stance phase and swing phase and the events that define the start and end of these phases are (usually) heel contact (HC) and toe-off (TO). These events are used as temporal markers to align data and compare biomechanical measures across trials and participants, such as graphing the knee angle during the stance phase (from HC to TO). Most often, these gait events are determined using force plates in conjunction with motion capture data. HC and TO can be defined as the time point in which the force measurement rises above and falls below a threshold value, respectively. This threshold is often set at 20 N to avoid events being placed due to noise or the foot brushing the force plate in a low clearance swing. This technique works well in healthy participants with typical HC and TO kinematics. However, in participants, such as those with cerebral palsy, with atypical HC and TO kinematics, this technique often fails.

Therefore, computing HC and TO in participants with atypical HC and TO kinematics or when kinetic data is not available can be challenging. One method that has been shown to work consistently is to find when the maximal anterior/posterior displacement between the heel marker and the sacrum marker occurs, which will determine HC. TO will be determined when maximal anterior/posterior displacement between the toe marker and the sacrum marker occur (Zeni et al., 2008). Another method, if the subject is on a treadmill, is to use the anterior/posterior velocity of the toe or heel marker. The instant the marker changes from positive to negative is when HC occurs and the instant the marker changes from negative to positive is when TO occurs (Zeni et al., 2008). While these algorithms work well for gait at normal walking speeds, they may not be accurate during running analyses.

When collecting data that will be using these methods to determine gait events, certain attention needs to be paid in collection. When using kinetic data, careful observation of foot placement is required. Because a force threshold is used as the trigger for an event to be created, the researcher must make sure that the foot was fully on one force plate exclusively and no part of the foot was in contact with any other surface at any point during stance. Additionally, no other forces can be applied to the force plate, such as objects, other feet, canes, etc. Attention needs to be paid during collections to the force plates to ensure they remain properly tared so that electrical noise or drift does not create gait events that are not real. The kinematic data gait event methods are also especially vulnerable to nonstandard gait patterns and may not be valid if the subject has gait deviations or other disabilities that alter standard biomechanics.

Once data has been collected for a participant, it is often compared to others to determine differences in biomechanical variables. First, the kinematic and kinetic data should be time normalized to 100% of the stance and/or swing phase so that the variables are temporally aligned. In order to control for varying body size, normalization techniques should be applied to kinetic data as well. Each kinetic variable may have different considerations; therefore, determination of appropriate normalization techniques should be done on a normalize kinetic data are (1) divide by the mass of the participant, (2) by the height of the participant, or (3) by the velocity of the participant. Examining standard practice among fields of study is a good way to determine the best to way to normalize the kinetic data for comparison.

7.2.3 Event Detection Methods for Other Activities

Although gait is the most common motion studied in biomechanics, there are many other motions and activities that are of interest, including jumping/landing and squatting. Jumping and landing events often divide the motion into jump initiation, liftoff, touchdown, and completion of landing. Jump initiation and landing completion can be detected when the participant deviates from a standing posture and returns to a standing posture, respectively. This is often done by setting a time event when a marker, usually on the pelvis, first appreciably moves in the vertical direction and returns to the full standing height after the touchdown event. The liftoff and touchdown time points are easiest to determine with the use of force thresholding using the vertical force component of the force plate data.

Squatting is another activity that is frequently examined in research. It is often divided into the lowering and standing phase. These events are set similarly to the jump initiation and landing completion events from jumping, where the maximum and minimum vertical displacement of the pelvic marker determines the lowering and standing phases of the squat.

While the above provides basic examples of event detection techniques, it is by no means exhaustive. The events are often movement specific and need to be determined based on the specific analysis considering the markerset used, camera set-up, and equipment.

7.2.4 Considerations for Applications with Implements

In biomechanics research, especially in the context of sports or for individuals who need assistance walking, extra implements may be included in the data collection. These implements could include balls, golf clubs, lacrosse sticks, canes, or walkers. When these implements are included, it is generally either because, it is a requirement of the sport or activity, the movement could not be performed without them or the subject is too unstable. For example, to study the biomechanics of a golf swing it is important that the athlete is swinging the club since the inertial properties of the club will influence both the kinematics and kinetics of the swing. Therefore, it would be important to track the golf club during the swing, to calculate the inertial influence on the body. If it is determined that the implement is important, it should have tracking markers placed on it in such a way that the movement in all planes is detectable, very similar to the principals involved in tracking limb segments. While tracking a relevant object may be straightforward, getting useful data from it may not. Each body segment's inertial properties have been determined from cadaveric studies. (How these inertial properties will be used in calculations is discussed later in this chapter.) This allows for quick computation because the inertial properties can be inputted in motion tracking analysis software. Objects like a baseball or basketball are not very challenging to analyze because they are uniform spheres with uniformly distributed mass and have easily calculated inertial properties. Unfortunately, the inertial properties of something like a golf club would need to be determined before calculations can be performed on it and is much more complex. Golf clubs, and similar objects, with very irregular shapes have a large portion of mass located at the extreme end, making the calculation of its inertial properties challenging. Before conducting research using such objects, its properties will need to be determined in order for it to be integrated into the model.

One additional thing to consider when using implements is their contact with the floor, especially if using force plates. The implements can move the distribution of weight of the subject onto the object and affect the location of the center of pressure of the force on the force plate. This is prevalent in assistive devices like canes and how force distribution and biomechanics changes when using a device, it is recommended to use alternative methods of support, like a bodyweight harness or gait belt.

7.2.5 Example of a Kinematic Data Set

7.2.5.1 Calibration – Calculation of [Marker to Anatomical] Matrix. Let us now look at an example of numerical data to see how the various transformations are calculated. The leg segment in Figure 7.2 will be used as an example. Recall that there are three tracking markers on this segment plus two calibration markers whose coordinates were digitized during the calibration period when the subject stood steady in the anatomical position. In Figure 7.2, the subject would be facing the +ve X direction, and the left leg was being analyzed. Table 7.1 gives the X, Y, and Z coordinates in the GRS, averaged over one second.

In this calibration position, the ankle = $(m_{T2} + m_{C1})/2$, $X_a = 2.815$, $Y_a = 22.685$, $Z_a = 10.160$. The coordinates of the knee = $(m_{T3} + m_{C2})/2$, $X_k = 6.670$, $Y_k = 20.965$, $Z_k = 41.890$. The COM of the leg = 0.567 × knee + 0.433 × ankle, $X_c = 5.001$, $Y_c = 21.710$, $Z_c = 28.151$.

Now we have to locate the anatomical x, y, and z-axes. Let the line joining the ankle to the knee be the z axis, and the line joining the lateral malleolus to the medial malleolus be an interim y axis (because it is not exactly normal to the y axis but will be corrected later). These two axes now form a plane, and the x axis, by definition, is normal to the yz plane and, therefore, is the "cross product" of y and z, or $x_{an} = (y_{an} \times z_{an})$. Using the subscript "an" to indicate anatomical axes:

$$z_{an} = (\text{knee} - \text{ankle}): \quad x_z = 3.855, \quad y_z = -1.720, \quad z_z = 31.730$$

$$y_{an} = (m_{C1} - m_{T2}): \quad x_y = -0.210, \quad y_y = 7.670, \quad z_y = 0.120$$

$$x_{an} = (y_{an} \times z_{an}): \quad x_x = 246.653, \quad y_x = 7.126, \quad z_x = -29.581$$

One final correction must be done to our anatomical model. y_{an} is the line joining the medial to lateral malleoli and is approximately 90° from the long axis z_{an}. To ensure that all three anatomical axes are at right angles to each other, y_{an} must be corrected: $y_{an} = (x_{an} \times z_{an})$: $x_y = -175.229$, $y_y = -7940.334$, $z_y = -451.714$. Note that none of these vectors is unity length, and they are normally reported as a unit vector. For example, the length of x_{an} is 245.423 cm; thus, dividing each coordinate by the length of x_{an} yields a unity vector for x_{an}: $x_x = 0.9925$, $y_x = -0.1190$, $z_x = 0.0290$. Similarly, as a unit vector, the corrected y_{an} is: $x_y = -0.0274$, $y_y = 0.9995$, $z_y = 0.0156$. Similarly, for z_{an} as a unit vector, $x_z = 0.1204$, $y_z = -0.0537$, $z_z = 0.9913$. We can now construct the leg anatomical-to-global matrix [LA to G] and it is:

$$\begin{bmatrix} 0.9925 & -0.0274 & 0.1204 \\ -0.1190 & 0.9995 & -0.0537 \\ 0.0290 & 0.0156 & 0.9913 \end{bmatrix}$$

TABLE 7.1 Marker Coordinates During Standing Calibration

Marker	Location	X (cm)	Y (cm)	Z (cm)
m_{T1}	Midleg	9.39	21.90	30.02
m_{T2}	Lateral malleolus	2.92	18.85	10.10
m_{T3}	Fibular head	5.05	15.41	41.90
m_{C1}	Medial malleolus	2.71	26.52	10.22
m_{C2}	Medial condyle	8.29	26.52	41.88

Note that the diagonal values are almost = 1, indicating that the subject stood in the calibration position with his three anatomical axes almost perfectly lined up with the global axes. A more useful transformation matrix is the leg global-to-anatomical [LG to A], which is the transpose of [LA to G]:

$$\begin{bmatrix} 0.9925 & -0.1190 & 0.0290 \\ -0.0274 & 0.9995 & 0.0156 \\ 0.1204 & -0.0537 & 0.9913 \end{bmatrix}$$

These anatomical axes have their origins at the ankle joint. However, from our inverse dynamics, it is more convenient that the origin of the anatomical axes (see Figure 7.2) be located at the COM of the leg segment with coordinates 0, 0, 0. We must now establish the local coordinates of the ankle and knee joints and the three tracking markers relative to this new origin at the COM.

From the COM to the ankle vector = (global ankle − global COM),

$$\begin{aligned} x_{al} &= X_a - X_C = 2.815 - 5.001 = -2.186, \\ y_{al} &= Y_a - Y_C = 22.685 - 21.710 = 0.975, \\ z_{al} &= Z_a - Z_C = 10.160 - 28.151 = -17.991 \end{aligned}$$

The anatomical ankle vector is the product of [LG to A][ankle vector]:

$$\begin{bmatrix} 0.9925 & -0.1190 & 0.0290 \\ -0.0274 & 0.9995 & 0.0156 \\ 0.1204 & -0.0537 & 0.9913 \end{bmatrix} \begin{bmatrix} -2.186 \\ 0.975 \\ -17.991 \end{bmatrix} = \begin{bmatrix} -2.81 \\ 0.75 \\ -18.15 \end{bmatrix}$$

This anatomical ankle vector lies along the line joining the ankle to knee and the ankle is 18.15 cm distal of the COM. If we were to repeat this procedure for the anatomical knee vector and for the anatomical vectors of the three tracking markers m_{T1}, m_{T2}, and m_{T3}, we would calculate the following:

Anatomical knee vector	Anatomical m_{T1} vector	Anatomical m_{T2} vector	Anatomical m_{T3} vector
$\begin{bmatrix} 2.14 \\ -0.57 \\ 19.86 \end{bmatrix}$	$\begin{bmatrix} 4.39 \\ 0.10 \\ 2.37 \end{bmatrix}$	$\begin{bmatrix} -2.25 \\ -3.08 \\ -17.99 \end{bmatrix}$	$\begin{bmatrix} 1.20 \\ -6.08 \\ 13.97 \end{bmatrix}$

We are now ready to calculate the constant marker-to-anatomical matrix ([M to A] in Figure 7.2). The three tracking markers form a plane in the GRS, and we can now define our maker axes in that plane. m_{T1} is chosen as the origin of the marker plane, and the line joining m_{T1} to m_{T3} is chosen to be the z axis, labeled z_m. The line joining m_{T2} to m_{T1} is a vector labeled A (an interim vector to allow us to calculate y_m and x_m). y_m is normal to the plane defined by z_m and A and x_m, is normal to the plane defined by y_m and z_m.

$$z_m = \text{anatomical } m_{T3} - \text{anatomical } m_{T2}\text{: } [3.45, -3.00, 31.96]$$

$$\text{A vector} = \text{anatomical } m_{T1} - \text{anatomical } m_{T2}\text{: } [6.64, 3.18, 20.36]$$

$$y_m = (z_m \times \text{A vector})\text{: } [-162.71, 141.97, 30\ 89]$$

$$x_m = (y_m \times z_m): [4630.11, 5306.88, -1.67]$$

The normalized axis for this leg anatomical-to-marker matrix [LA to M] is:

$$\begin{bmatrix} 0.6574 & 0.7535 & -0.0002 \\ -0.7459 & 0.6508 & 0.1416 \\ 0.1069 & -0.0929 & 0.9899 \end{bmatrix}$$

The fixed leg marker-to-anatomical matrix [M to A] is the transpose of [LA to M]:

$$\begin{bmatrix} 0.6574 & -0.7459 & 0.1069 \\ 0.7535 & 0.6508 & -0.0929 \\ -0.0002 & 0.1416 & 0.9899 \end{bmatrix}$$

7.2.5.2 Tracking Markers – Calculation of [Global to Marker] Matrix. We are now ready to calculate the [G to M] matrix in Figure 7.2. Table 7.2 lists representative GRS coordinates for the leg segment for three successive frames of walking taken during the swing phase. The procedure to calculate this [G to M] matrix is exactly the same as the latter part of the calculation of the [M to A] matrix. Consider the coordinates for frame 6:

$$z_m = (m_{T3} - m_{T2}): [24.34, -3.64, 19.99]$$

$$\text{A vector} = m_{T1} - m_{T2}: [18.86, 4.09, 9.15]$$

$$y_m = (z_m \times \text{A vector}): [-115.07, 154.30, 168.20]$$

$$x_m = (y_m \times z_m): [3696.72, 6394.16, -3336.83]$$

The normalized axis for this leg [G to M] matrix for frame 6 is:

$$\begin{bmatrix} 0.4561 & -0.4502 & 0.7676 \\ 0.7889 & 0.6036 & -0.1148 \\ -0.4117 & 0.6580 & 0.6304 \end{bmatrix}$$

TABLE 7.2 Tracking Markers During Walking

	m_{T1}			m_{T2}			m_{T3}		
Frame	X	Y	Z	X	Y	Z	X	Y	Z
5	20.65	35.95	33.87	1.30	32.14	25.74	26.52	28.10	44.43
6	25.46	35.95	34.47	6.60	31.86	25.32	30.94	28.22	45.31
7	30.18	35.94	34.97	11.98	31.60	24.64	35.08	28.36	46.10

7.2.5.3 Calculation of [Global to Anatomical] Matrix. From Figure 7.2, the final step is to calculate the [G to A] matrix that is the product of the fixed [M to A] matrix and the variable [G to M] matrix; for frame 6 this product is:

$$\begin{bmatrix} 0.6574 & -0.7459 & 0.1069 \\ 0.7535 & 0.6508 & -0.0929 \\ -0.0002 & 0.1416 & 0.9899 \end{bmatrix} \begin{bmatrix} 0.4561 & -0.4502 & 0.7676 \\ 0.7889 & 0.6036 & -0.1148 \\ -0.4117 & 0.6580 & 0.6304 \end{bmatrix}$$

$$= \begin{bmatrix} -0.3326 & -0.6758 & 0.6576 \\ 0.8953 & -0.0075 & 0.4451 \\ -0.2959 & 0.7369 & 0.6076 \end{bmatrix}$$

From Equation (7.5), this [G to A] matrix is equal to:

$$\begin{bmatrix} c_2c_3 & s_3c_1 + s_1s_2c_3 & s_1s_3 - c_1s_2c_3 \\ -c_2s_3 & c_1c_3 - s_1s_2s_3 & s_1c_3 + c_1s_2s_3 \\ s_2 & -s_1c_2 & c_1c_2 \end{bmatrix}$$

We now solve this matrix to get θ_1, θ_2, and θ_3. Equating the three terms in the bottom row: $s_2 = -0.2959$, $-s_1c_2 = 0.7369$, $c_1c_2 = 0.6304$. Therefore $\theta_2 = -17.21°$ or $162.79°$; assuming $\theta_2 = -17.21°$, $c_2 = 0.95522$ or -0.9522. Because $c_1c_2 = 0.6304$, $c_1 = 0.6304/0.95522 = 0.6600$, and $\theta_1 = 48.7°$ or $-48.7°$. Since $-s_1c_2 = 0.7369$ and $c_2 = 0.95522$, $s_1 = 0.7369/-0.95522 = 0.7714$ or -0.7714. We now validate that $\theta_1 = -48.7°$ because $-s_1c_2 \approx 0.7714$. We now use the first two terms in the first column to calculate and validate θ_3: $c_2c_3 = -0.3326$, $-c_2s_3 = 0.7535$. From this, $c_3 = -0.3326/0.95522 = 0.3482$, $\theta_3 = 69.62°$ or $-69.62°$, and $s_3 = 0.7535/-0.95522 = 0.7888$ or -0.7888. The only valid solution is $\theta_3 = -69.62°$ because $-c_2s_3 \approx 0.7500$. To summarize the results of these three rotations (see Figure 7.1), to bring the global axes in line with the anatomical axes requires an initial rotation about the global X axis of $-48.7°$. This will create new Y' and Z' axes and will be followed by a rotation of $-17.21°$ about the Y' axis. This rotation creates new X'' and Z'' axes. The final rotation is the largest (because we are analyzing the leg segment during swing), and it is $-69.62°$, which creates the final X''', Y''', and Z''' axes. These final axes are the anatomical x–y–z axes shown in Figure 7.2.

Finally, to get the COM of the segment, we must calculate **c** in GRS coordinates. We have **c** in the leg anatomical reference, and it is composed of [A to G] = [G to A]^T and [Anatomical m_{T2} vector] = [−2.25, −3.08, −17.99]. In the GRS,

$$\mathbf{c} = [\text{A to G}][-2.25, -3.08, -17.99]$$

$$= \begin{bmatrix} -0.3326 & -0.6758 & 0.6576 \\ 0.8953 & -0.0075 & 0.4451 \\ -0.2959 & 0.7369 & 0.6076 \end{bmatrix} \begin{bmatrix} -2.25 \\ -3.08 \\ -17.99 \end{bmatrix} = \begin{bmatrix} 3.314 \\ -11.713 \\ -13.781 \end{bmatrix}$$

From Figure 7.2, the global vector (remember that R_m is equal to the GRS coordinate of m_{T1})

$$R_c = R_m + c = [28.77, 24.24, 20.69].$$

7.2.5.4 ISB Recommendations. When creating segments for biomechanical analysis, it is important to define the coordinate systems in a standardized format so future studies can compare data and to reduce mathematical error. The International Society of Biomechanics has published recommendations on how to define joint segments and the Euler rotation sequence to use in analysis (Table 7.3).

The ISB also created recommendations for joint segment Euler rotation sequences specifically for the upper body (Table 7.4). The upper body can often be challenging due to a greater overall range of motion in multiple planes, which make selection of the rotation sequence less straightforward than the lower extremity. When certain rotations are used in the upper body the extremes of motion can lead to mathematical errors such as gimbal lock. Gimbal lock occurs when two axes of two segments are oriented parallel to each other thereby being unable to determine the 3D joint position. For the lower body segments, the recommended rotation order is *X–Y–Z*.

7.3 DETERMINATION OF SEGMENT ANGULAR VELOCITIES AND ACCELERATIONS

Recall from Section 7.1.2 and Figure 7.2 that we had to determine three time-varying rotation angles, θ_1, θ_2, and θ_3, prior to transforming from the GRS to the anatomical axes. The first time derivative of these transformation angles yields the components of the segment angular velocities:

$$\omega = d\theta_1/dt \cdot \mathbf{e}_x + d\theta_2/dt \cdot \mathbf{e}_{y'} + d\theta_3/dt \cdot \mathbf{e}_{z''} \tag{7.7a}$$

where $\mathbf{e}_x$, $\mathbf{e}_{y'}$ and $\mathbf{e}_{z''}$ denote the unit vectors of the three rotation axes x, y', and z'' shown in Figure 7.1. Consider an angular velocity, ω', about axis x; here, $\omega' = d\theta_1/dt \cdot \mathbf{e}_x$ and there is no rotation of θ_2 or θ_3. This angular velocity can be expressed as:

$$\omega' = \begin{bmatrix} \dot{\theta}_1 \\ 0 \\ 0 \end{bmatrix}$$

The second angular velocity $\omega'' = d\theta_2/dt \cdot \mathbf{e}_{y'}$, plus the component of ω' that is transformed by $[\Phi_2]$ in Equation (7.2), can be expressed as:

$$\omega'' = \begin{bmatrix} 0 \\ \dot{\theta}_2 \\ 0 \end{bmatrix} + \begin{bmatrix} c_2 & 0 & -s_2 \\ 0 & 1 & 0 \\ s_2 & 0 & c_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \dot{\theta}_2 \\ 0 \end{bmatrix} + \begin{bmatrix} c_2\dot{\theta}_1 \\ 0 \\ s_2\dot{\theta}_1 \end{bmatrix} = \begin{bmatrix} c_2\dot{\theta}_1 \\ \dot{\theta}_2 \\ s_2\dot{\theta}_1 \end{bmatrix}$$

TABLE 7.3 ISB Recommendations on Joint Segment Coordinate Systems

Segment	Origin	x axis	y axis	z axis
Tibia/ Fibula	The point halfway between lateral and medial malleolus	The anterior pointing line perpendicular to the frontal plane of the tibia/fibula	The line perpendicular to both x and z axis	The line connecting the medial and lateral malleolus, pointing right
Calcaneus	The point halfway between lateral and medial malleolus	The anterior pointing line perpendicular to the frontal plane of the tibia/fibula	The line pointing cranially along the long axis of the tibia/fibula	The line perpendicular to both x and y axis
Pelvis	The center of rotation of the right hip	The anterior pointing line parallel to the plane created by the two anterior superior iliac spine and the midpoint of the posterior superior iliac spine	The line perpendicular to both x and z axis	The line parallel to the line connecting the right and left anterior superior iliac spine, pointing right
Femur	The center of rotation of the right (or left) hip	The line perpendicular to both the y and z axis	The line pointing cranially through the midpoint of the lateral and medial femoral epicondyles and the origin	The line perpendicular to the y axis along the plane created by the lateral and medial femoral epicondyles and origin, pointing to the right
Thorax	The point of the suprasternal notch	The line perpendicular to both the y and z axis, pointing forward	The line pointing up connecting the midpoint of the xiphoid process and spinal process of the eighth thoracic vertebra to the midpoint of the suprasternal notch and spinal process of the seventh cervical vertebra	The line perpendicular to the plane formed by the suprasternal notch, the spinal process of the seventh cervical vertebra and the midpoint of the xiphoid process and the spinal process of the eighth thoracic vertebra, pointing right
Clavicle	The most ventral point of the sternoclavicular joint	The line perpendicular to both the y axis of the thorax and the z axis of the clavicle	The line perpendicular to both x and z axis	The line connecting the most ventral point on the sternoclavicular joint and the most dorsal point on the acromioclavicular joint, pointing toward the most dorsal point on the acromioclavicular joint
Scapula	The most laterodorsal point of the scapula	The forward pointing line perpendicular to the plane formed by the most caudal point of the scapula, the most laterodorsal point of the scapula, and the trigonum spinae scapulae	The line perpendicular to both x and z axis	The line connecting the trigonum spinae scapulae and the most laterodorsal point of the scapula, pointing toward the most laterodorsal point of the scapula

(*continued*)

TABLE 7.3 (*Continued*)

Segment	Origin	x axis	y axis	z axis
Humerus (option 1)	The glenohumeral rotation center	The forward pointing line perpendicular to the plane formed by the lateral and medial humeral epicondyle and the glenohumeral rotation center	The line connecting the glenohumeral rotation center and the midpoint of the lateral and medial humeral epicondyle, pointing toward the glenohumeral rotation center	The line perpendicular to both x and y axis, pointing right
Humerus (option 2)	The glenohumeral rotation center	The line perpendicular to both the y and z axis, pointing forward	The line connecting the glenohumeral rotation center and the midpoint of the lateral and medial humeral epicondyle, pointing toward the glenohumeral rotation center	The line perpendicular to the plane formed by the y axis of the humerus and the y axis of the forearm, pointing right
Forearm	The most caudal-medial point of the ulnar styloid	The forward pointing line perpendicular to the plane formed by the ulnar styloid, radial styloid and midpoint of the lateral, and medial humeral epicondyle	The proximal pointing line connecting the ulnar styloid and the midpoint of the lateral and medial humeral epicondyle	The line perpendicular to both x and y axis, pointing right
Hand	The most caudal-medial point of the ulnar styloid	The line perpendicular to both the y and z axis	The proximal pointing line along the long axis of the forearm	The line connecting the ulnar and radial styloid, pointing right

These data are from the ISB recommendations parts 1 and 2 (Wu et al., 2002, 2005).

TABLE 7.4 ISB Recommendations on Joint Segment Euler Rotation Sequences and Movement Directions

Segment	Rotation Sequence	x axis	y axis	z axis
Thorax relative to GCS	Z–X–Y	Lateral flexion rotation right(+)/left(−)	Axial rotation left(+)/right(−)	Flexion(−)/ extension(+)
Clavicle relative to thorax	Y–X–Z	Elevation(−)/ depression(+)	Retraction(−)/ protraction(+)	Axial rotation backward(+)/ forward(−)
Scapula relative to clavicle	Y–X–Z	Acromioclavicular lateral(−)/ medial(+) rotation	Acromioclavicular retraction(−)/ protraction(+)	Acromioclavicular anterior(−)/ posterior(+) tilt
Humerus relative to scapula	Y(*scapula*)–X–Y	Glenohumeral elevation	Glenohumeral internal(+)/ external(−) rotation	(*Scapula y axis*) Glenohumeral plane elevation
Scapula relative to thorax	Y–X–Z	Lateral(−)/ medial(+) rotation	Retraction(−)/ protraction(+)	Anterior(−)/ posterior(+) tilt
Humerus relative to thorax	Y–X–Y	Elevation	Internal(+)/ external(−) rotation	(*Thorax y axis*) Plane of elevation, 0° abduction/ 90° forward flexion
Forearm relative to humerus	Z–X–Y	Carrying angle	Pronation(+)/ supination(−)	Flexion(+)/ hyperextension(−)

These data are from the ISB recommendations parts 1 and 2 (Wu et al., 2002, 2005).

Similarly, the third angular velocity, $\omega''' = d\theta_3/dt \cdot \mathbf{e}_{z''}$, plus the contribution from ω'' that is transformed by $[\Phi_3]$ in Equation (7.3), gives us:

$$\omega''' = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix} + \begin{bmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2\dot{\theta}_1 \\ 0 \\ s_2\dot{\theta}_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix} + \begin{bmatrix} c_3c_2\dot{\theta}_1 + s_3\dot{\theta}_2 \\ -s_3c_2\dot{\theta}_1 + c_3\dot{\theta}_2 \\ s_2\dot{\theta}_1 \end{bmatrix}$$

$$= \begin{bmatrix} c_3c_2\dot{\theta}_1 + s_3\dot{\theta}_2 \\ -s_3c_2\dot{\theta}_1 + c_3\dot{\theta}_2 \\ s_2\dot{\theta}_1 + \dot{\theta}_3 \end{bmatrix}$$

Decomposing ω''' into its three components along the three anatomical axes:

$$\omega = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} c_2c_3 & s_3 & 0 \\ -c_2s_3 & c_3 & 0 \\ s_2 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} \tag{7.7b}$$

We can now calculate the three-segment angular velocities, ω_x, ω_y, and ω_z, that are necessary to solve the 3D inverse dynamics equations developed in the next section. Recall that the time-varying θ_1, θ_2, and θ_3 are calculated from Equation (7.5), and the time derivatives of these angles are individually calculated using the same finite difference technique used in two dimensions. The three-segment angular accelerations, α_x, α_y, and α_z, can now
difference. We now have all the kinematic variables

7.4 KINETIC ANALYSIS OF REACTION FORCES AND MOMENTS

Having developed the transformation matrices from global to anatomical and from anatomical to global, we are now in a position to begin calculating the reaction forces and moments at each of the joints. Because the ground reaction forces are measured in the GRS and the moments of inertia are known in the anatomical axes, these previously determined transformation matrices are used extensively in the kinetic calculations. All joint reaction forces are initially calculated in the GRS, and all joint moments are calculated in the anatomical axes.

7.4.1 Newtonian Three-Dimensional Equations of Motion for a Segment

All reaction forces are calculated in the GRS and, because the gravitational forces and the segment COM accelerations are readily available in the GRS, it is convenient to calculate all segment joint reaction forces in the GRS. Students are referred to the 2D link-segment equations and free-body diagram equations in Section 5.1. Figure 7.4 is now presented to demonstrate the steps required to calculate kinetics for this segment. The only addition is the third dimension, z. We are given the three distal reaction forces either as measures from a force plate or from the

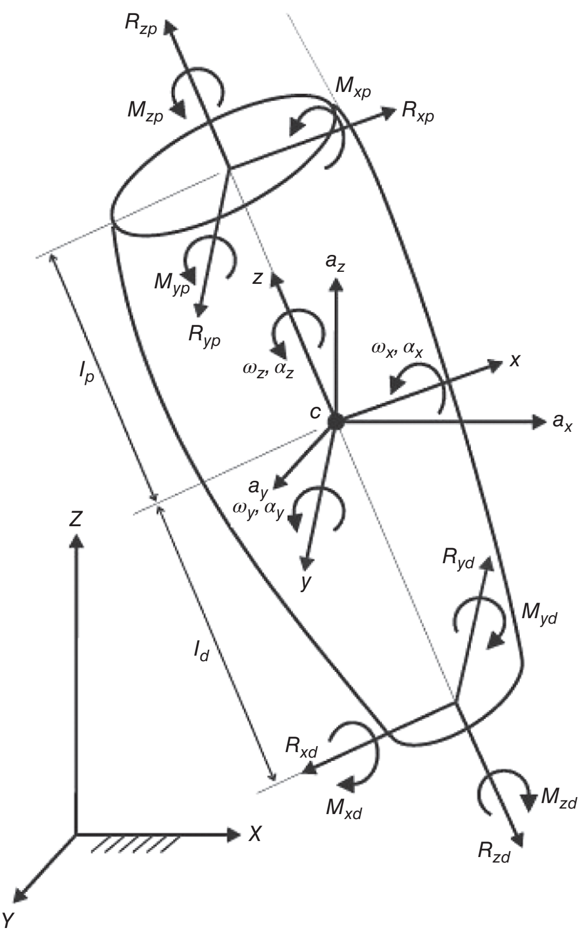

Figure 7.4 3D free-body diagram for solution of the inverse dynamics equations. Known are the distal reaction forces and moments, the COM linear accelerations, and the segment's angular velocities and accelerations. Using the kinetic Equations (7.7a), (7.7b), and (7.8a)–(7.8c), we calculate the

analysis of the adjacent distal segment. It should be noted that the reaction forces and moments at the distal end are in the reverse direction from those at the proximal end, the same convention as was used in Section 5.1.

Step 1: Calculate the reaction forces at the proximal end of the segment in the GRS:

$$\Sigma F_X = ma_X \quad \text{or} \quad R_{XP} - R_{XD} = ma_X$$

$$\Sigma F_Y = ma_Y \quad \text{or} \quad R_{YP} - R_{YD} - mg = ma_Y$$

$$\Sigma F_Z = ma_Z \quad \text{or} \quad R_{ZP} - R_{ZD} = ma_Z$$

where a_X, a_Y, a_Z are the segment COM accelerations in the X, Y, Z GRS directions and R_{XP}, R_{XD}, R_{YP}, R_{YD}, R_{ZP}, R_{ZD} are the proximal and distal reaction forces in the X, Y, and Z axes.

Step 2: Transform both proximal and distal reaction forces into the anatomical axes using the [G to A] matrix transformation based on θ_1, θ_2, and θ_3 (see Equation (7.5)). We will now have the proximal and distal reaction forces in the anatomical axes x, y, z: R_{xp}, R_{xd}, R_{yp}, R_{yd}, R_{zp}, R_{zd}.

Step 3: Transform the distal moments from those previously calculated in the GRS using the [G to A] matrix to the anatomical axes, as before, based on θ_1, θ_2, and θ_3: M_{xd}, M_{yd}, M_{zd}. We now have all the variables necessary to calculate the proximal moments in the anatomical axes.

7.4.2 Euler's Three-Dimensional Equations of Motion for a Segment

The equations of motion for the 3D kinetic analyses are the Euler equations. Considerable simplification can be made in the rotational equations of motion if these equations are written with respect to the principal (anatomical) axes of the segment with their origin at the COM of the segment. Thus, the x–y–z axes of the segment in Figure 7.4 satisfy those conditions. The angular velocity of the segment in its coordinate system is ω. The rotational equations of motion are:

$$I_x\alpha_x + \left(I_z - I_y\right)\omega_y\omega_z = \Sigma M_x = R_{zd}l_d + R_{zp}l_p + M_{xp} - M_{xd} \tag{7.8a}$$

$$I_y\alpha_y + (I_x - I_z)\omega_x\omega_z = \Sigma M_y = M_{yp} - M_{yd} \tag{7.8b}$$

$$I_z\alpha_z + \left(I_y - I_x\right)\omega_x\omega_y = \Sigma M_z = -R_{xd}l_d - R_{xp}l_p + M_{zp} - M_{zd} \tag{7.8c}$$

where

I_x, I_y, I_z	= moments of inertia about x–y–z axes
ω_x, ω_y, and ω_z	= components of angular velocity ω about x–y–z axes
α_x, α_y, α_z	= components of angular acceleration about x–y–z axes
M_{xd}, M_{yd}, M_{zd}	= previously transformed distal moments (not shown in Figure 7.4) about x–y–z axes
R_{xd}, R_{xp}, R_{yd}, R_{yp}, R_{zd}, R_{zp}	= previously transformed joint reaction forces about x–y–z axes
l_p, l_d	= distances from COM to proximal and distal joints.

The unknowns in these three equations are the three moments (M_{xp}, M_{yp}, and M_{zp}) about the proximal x, y, z axes. Note that Equations (7.8a)–(7.8c) are in the same form as the 2D Equation with the additional term $(I_1 - I_2)\ \omega_1\omega_2$ to account for the interaction of the angular velocities in the other two axes. Also note that the moments do not involve the proximal and distal reaction forces because these forces have zero moment arms about that axis.

7.4.3 Example of a Kinetic Data Set

The set of kinematic and kinetic data shown in Tables 7.5 and 7.6 is taken from stance phase of walking, and we will confine our analysis to the leg segment (see Figure 7.4). The following anthropometric measures apply $m = 3.22\,\text{kg}$, $I_x = 0.0138\,\text{kg} \cdot \text{m}^2$, $I_y = 0.0024\,\text{kg} \cdot \text{m}^2$, $I_z = 0.0138\,\text{kg} \cdot \text{m}^2$, $l_d = 13.86\ \text{cm} = 0.1386\ \text{m}$, $I_p = 18.15\ \text{cm} = 0.1815\ \text{m}$ $l_d = 13.86\ \text{cm} = 0.1386\ \text{m}$, $I_p = 18.15\ \text{cm} = 0.1815\ \text{m}$, frame rate = 60 Hz.

From the kinematic and kinetic measures presented in Tables 7.5 and 7.6, we complete the analysis of frame 6: $a_X = 7.029\ \text{m/s}^2$, $a_Y = 1.450\ \text{m/s}^2$, $a_Z = -0.348\ \text{m/s}^2$ $a_X = 7.029\ \text{m/s}^2$, $a_Y = 1.450\ \text{m/s}^2$, $a_Z = -0.348\ \text{m/s}^2$. Refer to the three steps in Section 7.4.1.

Step 1

$$R_{XP} - R_{XD} = ma_x, R_{XP} = ma_X + R_{XD}, R_{XP} = 3.22 \times 7.029 + (-119.04) = -96.41\,\text{N}$$

$$R_{YP} - R_{YD} - mg = ma_Y, R_{YP} = ma_Y + R_{YD} + mg, R_{YP} = 3.22 \times 1.450 + (-12.03) + 3.22 \times 9.814 = 24.24\,\text{N}$$

$$R_{ZP} - R_{ZD} = ma_Z, R_{ZP} = ma_z + R_{ZD}, R_{ZP} = 3.22 \times (-0.348) + (-791.44) = -792.56\,\text{N}$$

Step 2

$$\theta_1 = -0.14781\,\text{rad} = -8.46889,$$
$$\theta_2 = -0.02501\,\text{rad} = -1.43297,$$
$$\theta_3 = -0.46053\,\text{rad} = -26.38643$$

$$\cos\theta_1 = 0.9891,\ \sin\theta_1 = -0.1476$$
$$\cos\theta_2 = 0.9997,\ \sin\theta_2 = -0.025$$
$$\cos\theta_3 = 0.8958,\ \sin\theta_3 = -0.4444$$

TABLE 7.5 Ankle Reaction Forces (N) and Moments (N · m) in GRS

Frame	R_{XD}	R_{YD}	R_{ZD}	M_{XD}	M_{YD}	M_{ZD}
4	−86.55	−13.81	−766.65	10.16	−97.55	6.74
5	−102.71	−12.12	−790.27	11.94	−102.50	4.65
6	−119.04	−12.03	−791.44	14.29	−103.85	2.11
7	−134.33	−10.26	−763.38	16.13	−101.32	−0.49
8	−146.37	−5.26	−704.00	16.64	−94.59	−2.88

TABLE 7.6 Leg Angular Displacements (rad) and Velocities (rad/s)

Frame	θ_1	θ_2	θ_3	$\dot{\theta}_1$	$\dot{\theta}_2$	$\dot{\theta}_3$
4	−0.14246	−0.02646	−0.37248	−0.1752	0.2459	−2.358
5	−0.14512	−0.02362	−0.41396	−0.1607	0.0434	−2.641
6	−0.14781	−0.02501	−0.46053	−0.0911	−0.1048	−2.909
7	−0.14815	−0.02705	−0.51094	−0.1139	−0.0923	−3.209
8	−0.15161	−0.02809	−0.56749	−0.1109	0.2175	−3.472

Substituting these values in the [G to A] matrix (Equation (7.5)), we get:

$$\begin{bmatrix} c_2c_3 & s_3c_1+s_1s_2c_3 & s_1s_3-c_1s_2c_3 \\ -c_2s_3 & c_1c_3-s_1s_2s_3 & s_1c_3+c_1s_2s_3 \\ s_2 & -s_1c_2 & c_1c_2 \end{bmatrix} = \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix}$$

We can now transform the reaction forces from the global to the anatomical axes system:

$$\begin{bmatrix} R_{xd} \\ R_{yd} \\ R_{zd} \end{bmatrix} = \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix} \begin{bmatrix} R_{XD} \\ R_{YD} \\ R_{ZD} \end{bmatrix}$$
$$= \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix} \begin{bmatrix} -119.04 \\ -12.03 \\ -791.44 \end{bmatrix} = \begin{bmatrix} -170.603 \\ 32.5915 \\ -781.336 \end{bmatrix}$$

$$\begin{bmatrix} R_{xp} \\ R_{yp} \\ R_{zp} \end{bmatrix} = \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix} \begin{bmatrix} R_{XP} \\ R_{YP} \\ R_{ZP} \end{bmatrix}$$
$$= \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix} \begin{bmatrix} -96.41 \\ 24.24 \\ -792.56 \end{bmatrix} = \begin{bmatrix} -166.26 \\ 74.9789 \\ -777.674 \end{bmatrix}$$

Step 3: In a similar manner, we transform the ankle moments from the global to the anatomical axes system:

$$\begin{bmatrix} M_{xd} \\ M_{yd} \\ M_{zd} \end{bmatrix} = \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix} \begin{bmatrix} M_{XD} \\ M_{YD} \\ M_{ZD} \end{bmatrix}$$
$$= \begin{bmatrix} 0.8955 & -0.4363 & 0.0875 \\ 0.4443 & 0.8877 & -0.1215 \\ -0.0253 & 0.1473 & 0.9888 \end{bmatrix} \begin{bmatrix} 14.29 \\ -103.85 \\ 2.11 \end{bmatrix} = \begin{bmatrix} 58.2911 \\ -86.095 \\ -13.5723 \end{bmatrix}$$

Using Equation (7.7b), we calculate the angular velocities and accelerations required for the solution of Euler's kinetic Equations (7.8a)–(7.8c). As previously calculated for frame 6, $c_2 = 0.9997$, $c_3 = 0.8958$, $s_2 = -0.025$, $s_3 = -0.4444$.

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} c_2c_3 & s_3 & 0 \\ -c_2s_3 & c_3 & 0 \\ s_2 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} = \begin{bmatrix} 0.8955 & -0.4444 & 0 \\ 0.4443 & 0.8958 & 0 \\ -0.0253 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.0911 \\ -0.1048 \\ -2\ 909 \end{bmatrix} = \begin{bmatrix} -0.0350 \\ -0.1344 \\ -2\ 7067 \end{bmatrix}$$

Similarly, for frame 5,

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} -0.1632 \\ -0.0325 \\ -2.6369 \end{bmatrix}$$

and for frame 7,

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} -0.0610 \\ -0.1333 \\ -3.2061 \end{bmatrix}$$

We now calculate the angular accelerations of the segment where Δt is the sampling period:

$$\alpha_x(\text{fr.}6) = [\omega_x(\text{fr.}7) - \omega_x(\text{fr.}5)]/2\Delta t = [-0.0610 - (-0.1632)]/(2 \times 1/60) = 3.066\ \text{rad/s}^2$$

$$\alpha_y(\text{fr.}6) = [\omega_y(\text{fr.}7) - \omega_y(\text{fr.}5)]/2\Delta t = [-0.1333 - (-0.0325)]/(2 \times 1/60) = -3.024\ \text{rad/s}^2$$

$$\alpha_z(\text{fr.}6) = [\omega_z(\text{fr.}7) - \omega_z(\text{fr.}5)]/2\Delta t = [-3.2061 - (-2.6369)]/(2 \times 1/60) = -17.076\ \text{rad/s}^2$$

We are now ready to solve Euler's Equations (7.8a)–(7.8c) for the proximal moments:

$$I_x\alpha_x + (I_z - I_y)\omega_y\omega_z = \Sigma M_x = R_{zd}l_d + R_{zp}l_p + M_{xp} - M_{xd}$$

$$\begin{aligned} M_{xp} &= I_x\alpha_x + (I_z - I_y)\omega_y\omega_z - R_{zd}l_d - R_{zp}l_p + M_{xd} \\ &= 0.0138 \times 3.066 + (0.0138 - 0.0024)(-0.1344 \times -2.7067) \\ &\quad - (-781.336 \times 0.1386) - (-777.674 \times 0.1815) + 58.2911 \\ &= 307.779\ \text{N} \cdot \text{m} \end{aligned}$$

$$I_y\alpha_y + (I_x - I_z)\omega_x\omega_z = \Sigma M_y = M_{yp} - M_{yd}$$

$$\begin{aligned} M_{yp} &= I_y\alpha_y + (I_x - I_z)\omega_x\omega_z + M_{yd} \\ &= 0.0024 \times (-3.024) + (0.0138 - 0.0138)(-0.0350 \times -2.7067) \\ &\quad + (-103.85) = -103.857\ \text{N} \cdot \text{m} \end{aligned}$$

$$I_z\alpha_z + (I_y - I_x)\omega_x\omega_y = \Sigma M_z = -R_{xd}l_d - R_{xp}l_p + M_{zp} - M_{zd}$$

$$\begin{aligned} M_{zp} &= I_z\alpha_z + (I_y - I_x)\omega_x\omega_y + R_{xd}l_d + R_{xp}l_p + M_{zd} \\ &= 0.0138 \times (-17.076) + (0.0024 - 0.0138)(-0.0350 \times -2.7067) \\ &\quad + (-170.603 \times 0.1386) + (-166.26 \times 0.1815) + (-13.5723) \\ &= -67.631\ \text{N} \cdot \text{m} \end{aligned}$$

The interpretation of these moments for the left knee as the subject bears weight during single support is as follows. M_{xp} is +ve; thus, it is a counterclockwise moment and hence an abductor moment acting at the knee to counter the large gravitational load of the upper body acting downward and medial of the support limb. M_{zp} is the axial moment acting along the long axis of the leg and reflects the action of the left hip internal rotators actively rotating the pelvis, upper body, and right limb in a forward direction to gain extra step length. M_{yp} is −ve, indicating a clockwise (flexor) knee moment in the sagittal plane that would assist in starting the knee to flex late in stance shortly before TO.

The next stage of the kinetic analysis is to transform
(which are in the leg anatomical axis system), to the global

analysis can continue for the thigh segment. This transformation is accomplished by the [A to G] matrix for the leg and yields a new set of distal moments, M_{XD}, M_{YD}, and M_{ZD}, for the thigh analysis.

As an exercise, students may wish to repeat the preceding calculations for frame 5 or 7. For frame 5: $a_X = 5.89\ \text{m/s}^2$, $a_Y = 1.30\ \text{m/s}^2$, $a_Z = -1.66\ \text{m/s}^2$; for frame 7: $a_X = 9.60\ \text{m/s}^2$, $a_Y = 1.94\ \text{m/s}^2$, $a_Z = -0.020\ \text{m/s}^2$.

7.4.4 Joint Mechanical Powers

The joint mechanical power generated or absorbed at the distal and proximal joints, P_d and P_p, can now be calculated using Equation (5.5) for each of three-moment components and their respective angular velocities:

$$P_d = M_{xd}\omega_{xd} + M_{yd}\omega_{yd} + M_{zd}\omega_{zd} \tag{7.9a}$$

$$P_p = M_{xp}\omega_{xp} + M_{yp}\omega_{yp} + M_{zp}\omega_{zp} \tag{7.9b}$$

where the moments are previously defined and ω_{xd}, ω_{yd}, ω_{zd}, ω_{xp}, ω_{yp}, and ω_{zp} are the joint angular velocities at the distal and proximal ends (not shown in Figure 7.4). The angular velocities are in rad/s, the moments are in N · m, and the powers are in W. If these products were +ve the muscle group concerned would be generating energy, and if they were −ve, the muscle group would be absorbing energy.

7.4.5 Induced Acceleration Analysis

Thus far in this chapter, we have focused on calculating the position, velocity, acceleration, moment, and power that is seen on a joint-by-joint basis. However, it is important to recognize and remember that there are systemic effects to any movement. Any time a moment is applied to one joint of the body, it will induce an acceleration to any attached segment, no matter how distant. The examining of this effect is called Induced Acceleration Analysis. For example, consider a position where you start with your upper arm at your side and your forearm at 90° in front of you. When you swing your upper arm forward in flexion (applying a moment at the shoulder), your forearm must follow the upper arm. Because the forearm has inertial properties, it will want to stay in its original position. Therefore, unless active muscle contraction is applied, as the upper arm flexes, the forearm will extend at the elbow. This would be the upper arm-inducing acceleration on the forearm. If, in the above example, you suddenly stopped moving the upper arm in flexion, the inertial properties of the forearm will cause it to continue moving and, especially when it hits the end of range of motion (fully extended), will act to apply an extension moment on the upper arm. This would be the forearm-inducing acceleration on the upper arm.

As we can see, any start or stop to one segment has interactions with all the other systemic segments to which it is attached. While the most obvious effects are seen in adjacent segments, the movements do affect segments further away in the chain as well, such as an ankle plantarflexion moment inducing an acceleration of the head. These interactions can be seen with the mathematical equations of motion for the segments. In the equations, we can see that the angular acceleration for a segment is determined by both that segment's moment, angle, angular velocity, and moment applied by gravity and the moment, angle, angular velocity, and moment applied by gravity of the attached segments in the system.

It is important to keep this concept in mind when studying the movements of a body. When segments are moving at high speeds and with large moments the explanation may be more clearly explained by the movements induced by other parts of the body than by focusing on single segments. A great example of this is in a baseball pitch. If you looked at the moment generated by each segment's muscles individually, it is not likely to be able to explain the speed and power that is delivered to a baseball. Only when you look at how the each other; can you see how the interactions create a cumulative effect that is greater than its constituent parts.

7.4.6 Sample Moment and Power Curves

Figure 7.5 presents a set of intersubject averaged 3D joint moments at the ankle, knee, and hip during one walking stride (Eng and Winter, 1995). HC is at 0% stride and TO is at 60%. Figure 7.6 shows the 3D power curves for these intersubject averages (Eng and Winter, 1995). The curves are normalized relative to body mass; the moments are reported in N · m/kg and the powers in W/kg. Convention plots extensor moments in the sagittal plane as positive; in the transverse plane, external rotation moments are positive, and in the frontal plane, evertor moments are positive. A detailed explanation of the specific function of each of these moments is beyond the scope of this text; however, a few comments will be made on the larger or functionally more important moments.

1. The largest moment during walking is seen at the ankle in the sagittal plane. Immediately at HC, there is a small dorsiflexor moment to lower the foot to the ground; this is followed by a large increase in plantarflexor moment reaching a peak at about 50% of stride to cause the ankle to rapidly plantarflex and achieve an upward and forward "push off" of the lower limb as the subject starts swing at TO (see A2-S power generation burst in Figure 7.6).
2. The knee extensors are active at 8–25% of stride to control knee flexion as the limb accepts weight (K1-S absorption phase), then the moment reverses to a flexor pattern as a by-product of the gastrocnemic's contribution to the increasing ankle plantarflexor moment.

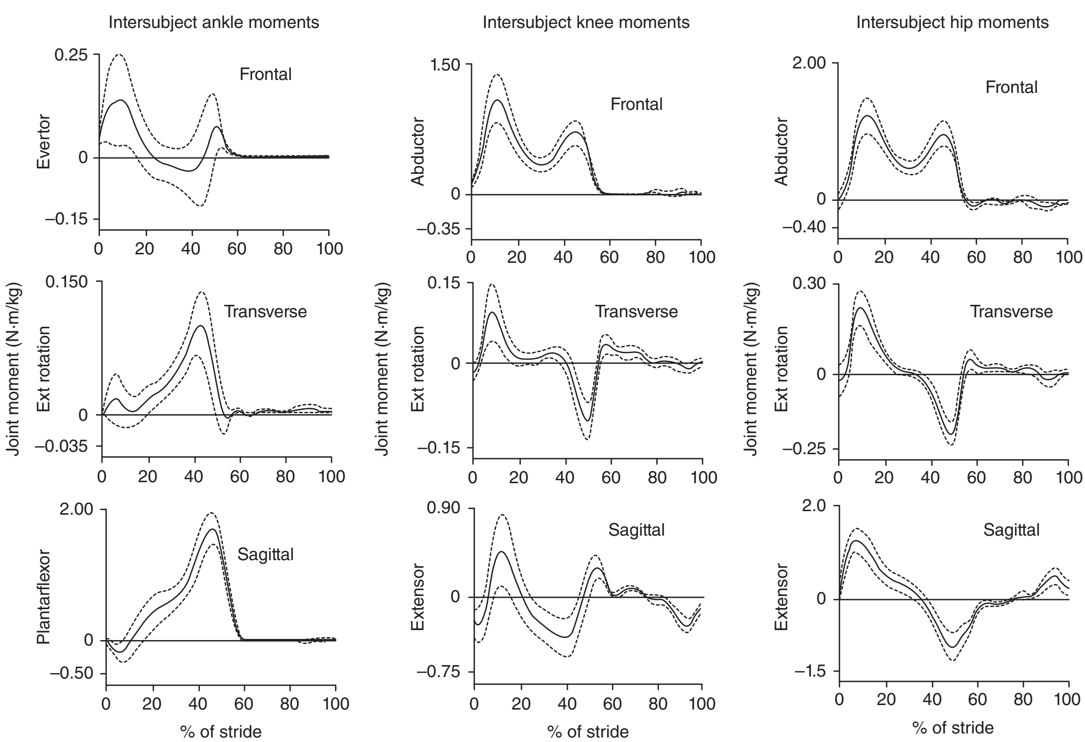

Figure 7.5 Typical 3D moments at the ankle, knee, and hip during one stride of walking (heel contact at 0 and 100%). Profiles are intersubject averages; solid line is the average curve with one standard deviation plotted as a dotted line. For interpretation of the more important profiles, see the

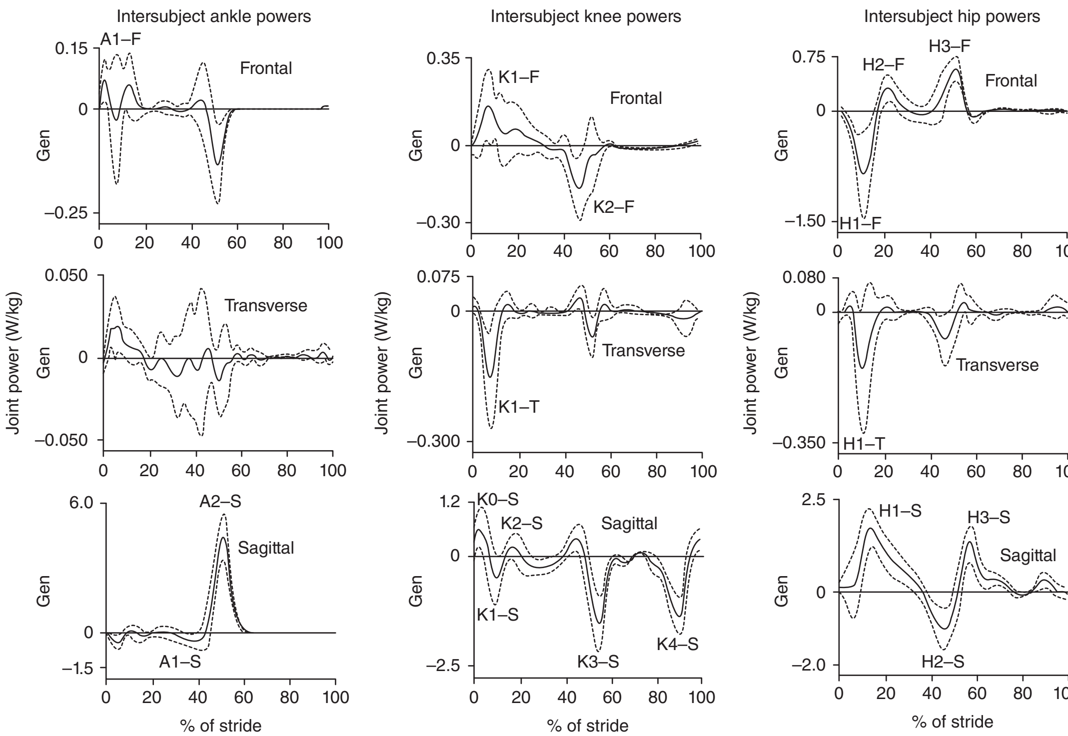

Figure 7.6 Mechanical power generation and absorption at the ankle, knee, and hip for the same averaged trials as in Figure 7.5. Generation power is +ve and absorption power is −ve. For interpretation of selected profiles, see the text.

Then, just before and after TO, a small knee extensor moment acts to limit the amount of knee flexion in late stance and early swing (K3-S absorption phase). The final burst of flexor activity just before HC is to decelerate the swinging leg prior to HC (K4-S absorption phase).

3. The hip pattern is characterized by an extensor moment for the first half of stance, followed by a flexor for the latter half. During the first half, the extensors stabilize the posture of the trunk by preventing it from flexing forward under the influence of a large posterior reaction force at the hip; this also assists the knee extensors in preventing collapse of the knee joint and, in addition, contributes to forward propulsion by what has been described as a "push from behind" (H1-S generation burst). The flexor moment during the second half of stance serves two functions: first, to stabilize the trunk posture by preventing it from flexing backward under the influence of the forward reaction force at the hip; second, in the last phase of stance and early swing (50–75% of stride), to achieve a "pull-off" of the thigh into swing (H3-S generation burst).
4. The major transverse activity is at the hip. During the first half of stance, the external rotators of the stance limb act to decelerate the horizontal rotation of the pelvis (and trunk) over the stance limb (H1-T absorption phase); then, during the second half, the internal rotators are active to stabilize the forward rotation of the pelvis and swing limb.
5. The frontal plane moments at the stance hip are a strong abductor pattern to prevent drop of the pelvis (and entire upper body) against the forces of gravity, which are acting about 10 cm medial of the stance hip (H1-F absorption phase as the pelvis drops, followed by H2-F and H3-F generation phases as the pelvis, trunk, and swing limb are lifted to help achieve a safe toe clearance during swing). It is

at the stance knee, but this is not as a result of muscle activity. Rather, it is the response of the weight-bearing knee to the large gravitational load; the knee tries to invert but the passive loading of the medial condyles and unloading of the lateral condyles creates an internal abductor moment. Such an example illustrates the fact that internal skeletal and ligament structures can aid (or in some cases hinder) the activity of the muscles. Rehabilitation engineers involved in prothesis design must be aware of the internal moments created by the springs, dampers, and mechanical stops in their design.

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8

MUSCLE MECHANICS

Stephen J. Thomas[1], Joseph A. Zeni[2], and David A. Winters[3,†]
[1]Thomas Jefferson University, Philadelphia, PA, USA
[2]Rutgers University, Newark, NJ, USA
[3]University of Waterloo, Waterloo, Ontario, Canada

8.0 INTRODUCTION

The most intriguing and challenging area of study in biomechanics is probably the muscle itself. It is the "living" part of the system. The neural control, metabolism, and biomechanical characteristics of muscle are subjects of continuing research. The purpose of this chapter is to report the state of knowledge with regard to the biophysical characteristics of individual motor units (MUs), connective tissue, and the total muscle itself. Each of the biophysical characteristics is described in detail, and it is shown how these characteristics influence the biomechanical function of the overall muscle.

8.0.1 The Motor Unit

The smallest subunit that can be controlled is called a *MU* because it is innervated separately by a motor axon. Neurologically the MU consists of a synaptic junction in the ventral root of the spinal cord, a motor axon, and a motor end plate in the muscle fibers. Under the control of the MU are as few as three muscle fibers or as many as 2000, depending on the fineness of the control required (Feinstein et al., 1955). Muscles of the fingers, face, and eyes have a small number of shorter fibers in a MU, while the large muscles of the leg have a large number of long fibers in their MUs. A muscle fiber is about 100 μm in diameter and consists of fibrils about 1 μm in diameter. Fibrils, in turn, consist of filaments about 100 Å in diameter. Electron micrographs of fibrils show the basic mechanical structure of the interacting actin and myosin filaments. In the schematic diagram of Figure 8.1, the darker and wider myosin protein bands are interlaced with the lighter and smaller actin protein bands. The space between them consists of a cross-bridge structure, and it is here that the tension is created and the shortening or lengthening takes place. The term *contractile element* is used to describe the part of the muscle that generates the tension, and it is this part that shortens and lengthens as positive or negative work is done. The basic length of the

†Deceased

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

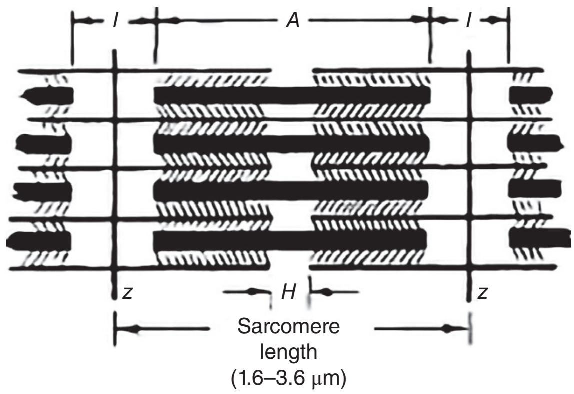

Figure 8.1 Basic structure of the muscle contractile element showing the *Z* lines and sarcomere length. Wider dark myosin filament interacts across cross-bridges (cross-hatched lines) with the narrower actin filament. Darker and lighter bands (*A*, *H*, and *I*) are shown.

myofibril is the distance between the *Z* lines and is called the *sarcomere length*. It can vary from 1.5 μm at full shortening to 2.5 μm at resting length to about 4.0 μm at full lengthening.

The structure of the muscle is such that many filaments are in parallel and many sarcomere elements are in series to make up a single contractile element. Consider a MU of a cross-sectional area of 0.1 cm^2 and a resting length of 10 cm. The number of sarcomere contractile elements in series would be 10 cm /2.5 μm = 40, 000 and the number of filaments (each with an area of 10^{-8} cm^2) in parallel would be $0.1/10^{-8} = 10^7$. Thus, the number of contractile elements of sarcomere length packed into this MU would be 4×10^{11}.

The active contractile elements are contained within another fibrous structure of connective tissue called *fascia*. These connective tissue coverings enclose each of the muscle's subscales. Individual muscle fibers are enclosed by endomysium. A group of fibers or fascicles are then enclosed by perimysium. Lastly, a group of fascicles or whole muscles are enclosed by epimysium. Ultimately, these connective tissue layers converge at both ends to form tendons. The mechanical characteristics of connective tissue are important in the overall biomechanics of the muscle. Some of the connective tissue is in series with the contractile element, while some is in parallel. The effect of this connective tissue has been modeled as springs and viscous dampers.

8.0.2 Recruitment of Motor Units

Each muscle has a finite number of MUs, each of which is controlled by a separate nerve ending. Excitation of each unit is an all-or-nothing event. The electrical indication is a MU action potential; the mechanical result is a twitch of tension. An increase in tension within a single muscle can, therefore, be accomplished in two ways: by an increase in the stimulation rate for that MU and/or by the excitation (recruitment) of an additional MU. Figure 8.2 shows the electromyogram (EMG) of a needle electrode in a muscle as the tension was gradually increased. The upper tracing shows one MU firing, the middle trace, two MUs, and the lower trace three units. Initially, muscle tension increases as the firing rate of the first MU increases. Each unit has a maximum firing rate, and this rate is reached well after the next unit is recruited (Erim et al., 1996). When the tension is reduced, the reverse process occurs. The firing rates of all recruited units decrease until the last-recruited unit drops out at a rate usually less than when it was originally recruited. As the tension further decreases, the remainder of the units drops out in the reverse order from that in which they were recruited. The firing rate of individual units increases monotonically with force; however, the collective firing rates of all active MUs increase nonlinearly as force increases from 0 to 100% maximum voluntary contraction (MVC) (Erim et al.,

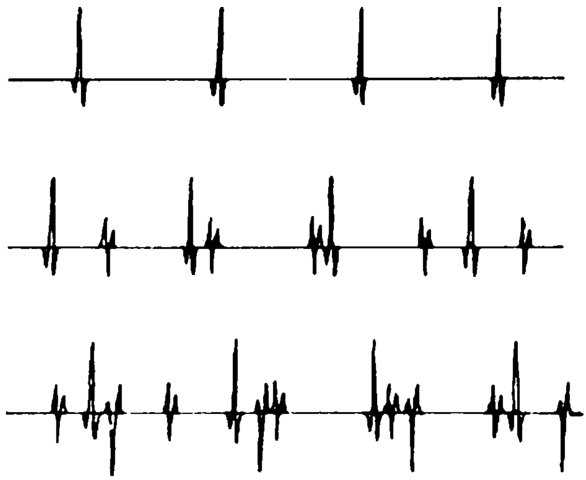

Figure 8.2 EMG from an indwelling electrode in a muscle as it begins to develop tension. The smallest motor unit is first recruited, and as its rate increases, a second, then a third motor unit are recruited. Each motor action potential has a characteristic shape at a given electrode, which depends on the size of the motor unit and the distance from the electrode to the fibers of the motor unit (see Chapter 9).

firing pattern of MUs within a single muscle. This strategy is thought to prevent fatigue of the larger MUs. Mathematical models describing recruitment have been reported by Wani and Guha (1975), Milner-Brown and Stein (1975), Fuglevand et al. (1993), and Erim et al. (1996).

8.0.3 Size Principle

Considerable research and controversy have taken place over the past several decades over how the MUs are recruited. Which ones are recruited first? Are they always recruited in the same order? It is now generally accepted that they are recruited according to the *size principle* (Henneman, 1974*a*), which states that the size of the newly recruited MU increases with the tension level at which it is recruited. This means that the smallest unit is recruited first and the largest unit last. In this manner, low-tension movements can be achieved in finely graded steps. Conversely, those movements requiring high forces but not needing fine control are accomplished by recruiting the larger MUs. Figure 8.3 depicts a hypothetical tension curve resulting from successive recruitment of several MUs. The smallest MU (MU 1) is recruited first, usually at an initial frequency ranging from about 5 to 13 Hz. Tension increases as MU 1 fires more rapidly, until a certain tension is reached at which MU 2 is recruited. Here, MU 2 starts firing at its initial low rate, and further tension is achieved by the increased firing of both MU 1 and 2. At a certain tension, MU 1 will reach its maximum firing rate (15–60 Hz) and will, therefore, be generating its maximum tension. This process of increasing tension, reaching new thresholds, and recruiting another larger MU continues until MVC is reached (not shown in Figure 8.3). At that point, all MUs will be firing at their maximum frequencies but in an asynchronous pattern to prevent fatigue. Tension is reduced by the reverse process: successive reduction of firing rates and dropping out of the larger units first (DeLuca et al., 1982). During rapid ballistic movements, the firing rates of individual MUs have been estimated to reach 120 Hz (Desmedt and Godaux, 1978).

The MU action potential increases with the size of the MU with which it is associated (Milner-Brown and Stein, 1975). The reason for this appears to be twofold. The larger the MU, the larger the motor neuron that innervates it, and the greater the depolarization potentials seen at the motor end plate. Second, the greater the mass of the MU, the greater the voltage changes seen at a given electrode. However, it is never possible from a given

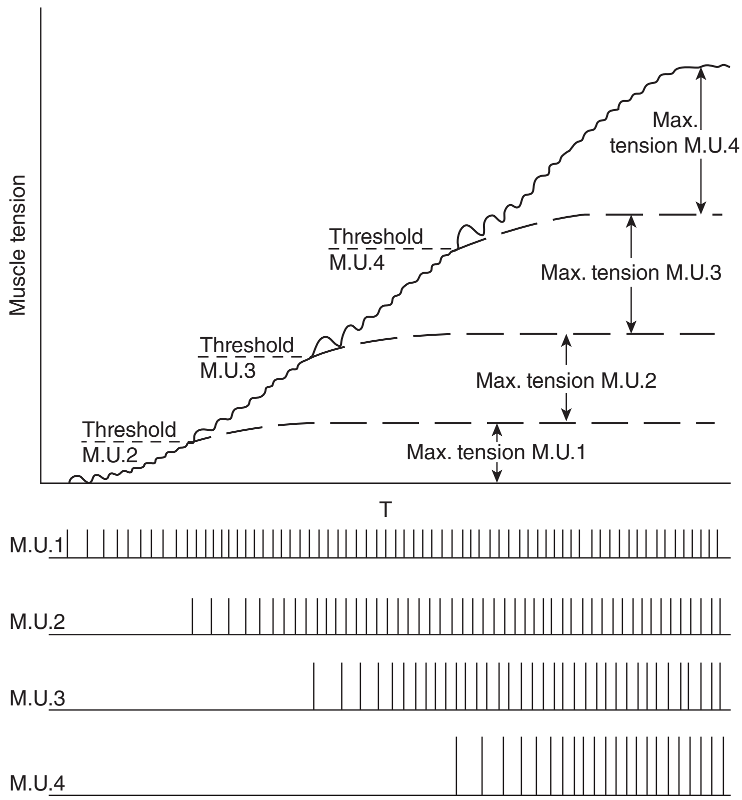

Figure 8.3 Size principle of recruitment of motor units. Smaller motor units are recruited first; successively larger units begin firing at increasing tension levels. In all cases, the newly recruited unit fires at a base frequency then increases to a maximum.

because its action potential also decreases with the distance between the recording site and the electrode. Thus, a large MU located a distance from the electrode may produce a smaller action potential than that produced by a small MU directly beneath the electrode.

8.0.4 Types of Motor Units – Fast- and Slow-Twitch Classification

There have been many criteria and varying terminologies associated with the types of MUs present in any muscle (Henneman, 1974*b*). Biochemists have used metabolic or staining measures to categorize the fiber types. Biomechanics researchers have used force (twitch) measures (Milner-Brown et al., 1973*b*), and electrophysiologists have used EMG indicators (Wormolts and Engel, 1973; Milner-Brown and Stein, 1975). The smaller slow-twitch MUs have been called *tonic units*. Histochemically they are the smaller units (type I), and metabolically they have fibers rich in mitochondria, are highly vascularized, and therefore have a high capacity for aerobic metabolism. Mechanically they produce twitches with a low peak tension and a long time to peak (60–120 ms). The larger fast-twitch MUs are called *phasic units* (type II). They have less mitochondria, are poorly vascularized, and therefore rely primarily on anaerobic metabolism. The twitches have larger peak tensions in a shorter time (10–50 ms). Figure 8.4 is a typical histochemical stain of fibers of a muscle that contains both slow- and fast-twitch fibers. An adenosine triphosphatase (ATPase)-type stain was used here, so slow-twitch fibers appear dark and fast-twitch fibers appear light. If an indwelling microelectrode were present in the area of these fibers, the muscle action potential from these darker slow-twitch fibers would be smaller than that from the lighter stained fast-twitch fibers.

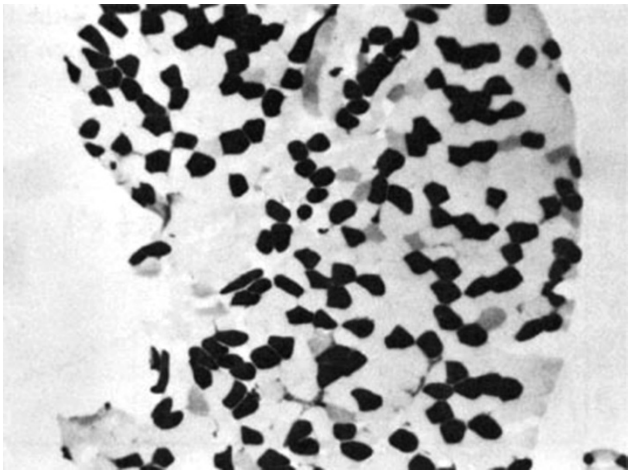

Figure 8.4 Histochemical stain showing dark slow-twitch fibers and light fast-twitch fibers. A myofibrillar ATPase stain, pH 4.3, was used to stain the vastus lateralis of a female volleyball player. (Reproduced by permission of Professor J. A. Thomson, University of Waterloo, Waterloo, Ont., Canada.)

In spite of the histochemical classification described, there is growing biophysical evidence (Milner-Brown et al., 1973b) that the MUs controlled by any motor neuron pool form a continuous spectrum of sizes and excitabilities.

8.0.5 The Muscle Twitch

Thus far, we have said very little about an individual MU twitch. As was described in Section 8.0.4, each MU has its unique time course of tension. Although there are individual differences in each newly recruited MU, they all have the same characteristic shape. The time-course curve follows closely with that of the impulse response of a critically damped second-order system (Milner-Brown et al., 1973*a*). The electrical stimulus of a MU, as indicated by this action potential, is of short duration and can be considered an impulse. The mechanical response to this impulse is the much longer duration twitch. The general expression for a second-order critically damped impulse response is

$$F(t) = F_0 \frac{t}{T} e^{-t/T} \tag{8.1}$$

For the curve plotted in Figure 8.5, the twitch time, T, is the time for the tension to reach a maximum, and F_0 is a constant for that given MU. T is the contraction time and is larger for the slow-twitch fibers than for the fast-twitch MUs, while F_0 increases for the larger fast-twitch units. Muscles tested by Buchthal and Schmalbruch (1970) using submaximal stimulations showed a wide range of contraction times. Muscles of the upper limbs generally had short T values compared with the leg muscles. Typical mean values of T and their range were:

Triceps brachii	44.5 ms (16–68 ms)
Biceps brachii	52.0 ms (16–85 ms)
Tibialis anterior	58.0 ms (38–80 ms)
Soleus	74.0 ms (52–100 ms)
Medial gastrocnemius	79.0 ms (40–110 ms)

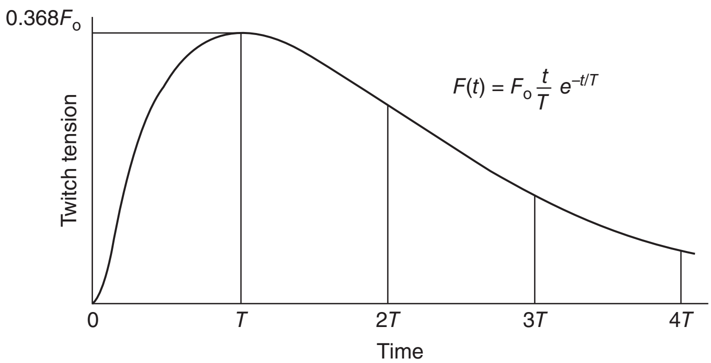

Figure 8.5 Time course of a muscle twitch, modeled as the impulse response of a second-order critically damped system. Contraction time T varies with each motor unit type, from about 20 to 100 ms. Effective tension lasts until about 4 T.

These same researchers found that T increased in all muscles as they were cooled; for example, the biceps brachialis had a contraction time that increased from 54 ms at 37°C to 124 ms at 23°C. It is evident that the cause of the delayed build-up of tension is the slower metabolic rates and increased muscle viscosity seen at lower temperatures.

Many other researchers have tested other muscles using different techniques. Milner-Brown et al. (1973b) looked at the first dorsal interossei during voluntary contractions and measured T to average 55 ms (±12 ms), while Desmedt and Godaux (1978) calculated a mean of 60 ms on the same muscle but used supramaximal stimulations. These latter researchers, using the same technique, reported T for the tibialis anterior as 84 ms and for the soleus as 100 ms. Bellemare et al. (1983), also using supramaximal stimulations, reported soleus T = 116 ms (±9 ms) and biceps T = 66 ms (±9 ms). It appears that the contraction time varies considerably depending on the muscle, the person, and the experimental technique employed. It is important to note that in these studies participants were asked to maintain a certain level of contraction prior to stimulations of specific MUs thereby eliminating the time required to take the slack out of the passive connective tissue at rest.

8.0.6 Shape of Graded Contractions

The shape of a voluntary tension curve depends to a certain extent on the shape of the individual muscle twitches. For example, if we elicit a maximum contraction in a certain muscle, the rate of increase of tension depends on the individual MUs and how they are recruited. Even if all MUs were turned on at the same instant and fired at their maximum rate, maximum tension could never be achieved in less than the average contraction time for that muscle. However, as described in Figure 8.2 and depicted in Figure 8.3, the recruitment of MUs does not take place all at once during a voluntary contraction. The smaller slow-twitch units are recruited first in accordance with the size principle, with the largest units not being recruited until tension has built up in the smaller units. Thus, it can take several hundred milliseconds to reach maximum tension. During voluntary relaxation of the same muscle, the drop in tension is governed by the shape of the trailing edge of the twitch curve. This delay in the drop of tension is even more pronounced than the rise. This delay, combined with the delay in dropping out of the MUs themselves according to the size principle, means that a muscle takes longer to turn off than to turn on. Typical turn-on times are 200 ms and turn-off times 300 ms, as shown in Figure 8.6, where maximum rapid turn-on and turn-off are plotted showing the tension curve and the associated EMG.

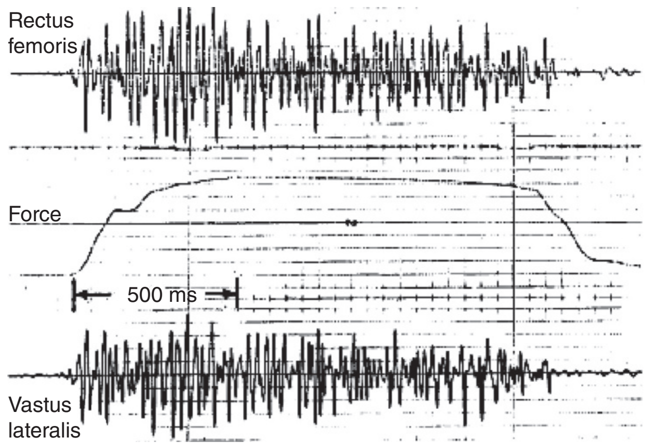

Figure 8.6 Tension build-up and decrease during a rapid maximum voluntary contraction and relaxation. The time to peak tension can be 200 ms or longer, mainly because of the recruitment according to the size principle and because of delay between each motor unit action potential and twitch tension. Note the presence of tension for about 150 ms after the cessation of EMG activity.

8.1 FORCE–LENGTH CHARACTERISTICS OF MUSCLES

As indicated in Section 8.0.1, the muscle consists of an active element, called the *contractile element*, and passive connective tissue. The net force–length characteristics of a muscle are a combination of the force–length characteristics of both active and passive elements.

8.1.1 Force–Length Curve of the Contractile Element

The key to the shape of the force–length curve is the changes of the structure of the myofibril at the sarcomere level (Gordon et al., 1966). Figure 8.7 is a schematic representation of the myosin

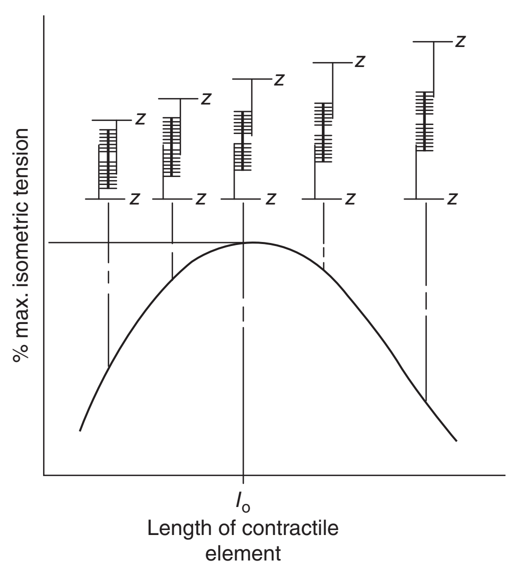

Figure 8.7 Tension produced by a muscle as it changes length about its resting length, l0. A drop in tension on either side of maximum can be explained by the interactions of

and actin cross-bridge arrangement. At resting length, about 2.5 μm, there are a maximum number of cross-bridges between the filaments and, therefore, a maximum tension is possible. As the muscle lengthens, the filaments are pulled apart, the number of cross-bridges reduces, and ultimate force decreases. At full length, about 4.0 μm, there are no cross-bridges and the force reduces to zero. As the muscle shortens to less than resting length, there is an overlapping of the cross-bridges and interference takes place. This results in a reduction of force that continues until a full overlap occurs, at about 1.5 μm. The force does not drop to zero but is drastically reduced by these interfering elements. It is important to note that these muscle lengths are not referencing joint positions instead the lengths of muscle if they were excised from the body. Each muscle crossing a joint in the body will have varying length positions for a given joint position and therefore will have altering force-producing capabilities.

8.1.2 Influence of Parallel Connective Tissue

The connective tissue that surrounds the contractile element influences the force–length curve. It is called the *parallel elastic component*, and it acts much like an elastic band. When the muscle is at resting length or less, the parallel elastic component is in a slack state with no tension. As the muscle lengthens beyond the resting length, the parallel element is no longer loose, so tension begins to build up, slowly at first and then more rapidly. Unlike most springs, which have a linear force–length relationship, the parallel element is quite nonlinear. In Figure 8.8, we see the force–length curve of this element, F_p, combined with that of the overall contractile component F_c. If we sum the forces from both elements, we see the overall muscle force–length characteristic, F_t. The force–length curve typically presented is usually for a maximum contraction. The passive force, F_p, of the parallel element is always present, but the amount of active tension in the contractile element at any given length is under voluntary control. Thus, the overall force–length characteristics are a function of the percentage of excitation, as seen in Figure 8.9.

8.1.3 Series Elastic Tissue

All connective tissue in series with the contractile component, including the tendon, is called the *series elastic element*. Under isometric contractions, it will lengthen slightly as tension increases.

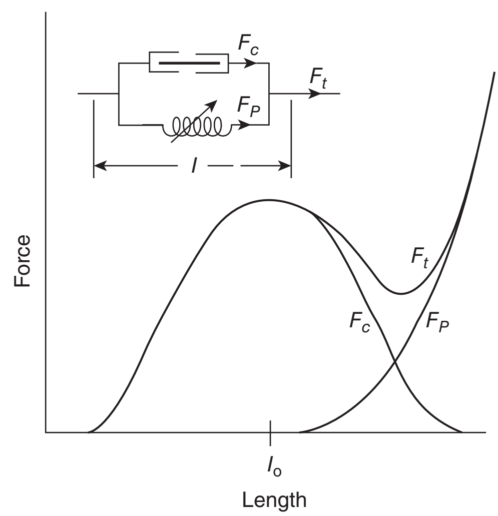

Figure 8.8 Contractile element producing maximum tension F_c along with the tension from F_p from the parallel elastic element. Tendon tension is $F_t = F_c + F_p$.

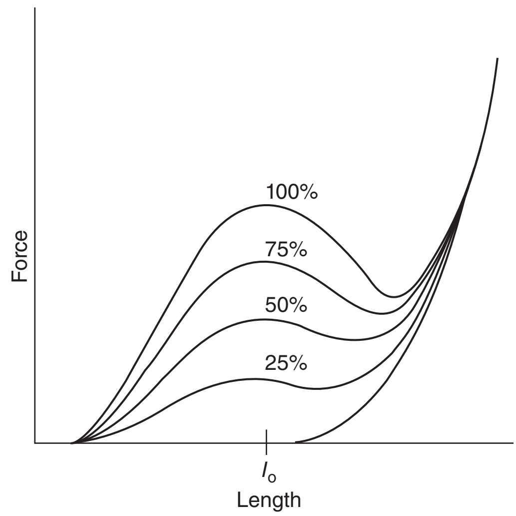

Figure 8.9 Tendon tension resulting from various levels of muscle activation. Parallel elastic element generates tension independent of the activation of the contractile element.

However, during dynamic situations, the series elastic element, in conjunction with viscous components, does influence the time course of the muscle tension.

During isometric contractions, the series elastic component is under tension and, therefore, is stretched a finite amount. Because the overall length of the muscle is kept constant, the stretching of the series elastic element can only occur if there is an equal shortening of the contractile element itself. This is described as internal shortening. Figure 8.10 illustrates this point at several contraction levels. Although the external muscle length L is kept constant, the increased tension from the contractile element causes the series elastic element to lengthen by the same amount as the contractile element shortens internally. The amount of internal shortening from rest to maximum tension is only a few percent of the resting length in most muscles (van Ingen Schenau, 1984) but has been shown to be as high as 7% in others (Bahler, 1967). It is widely inferred that these series elastic elements store large amounts of energy as muscles are stretched prior to an explosive shortening in athletic movements. However, van Ingen Schenau (1984) has shown that the elastic capacity of these series elements is far too small to explain improved performances resulting from pre-stretch. It is possible that a combination of the stored elastic energy in the series elastic components, enhanced sensitivity of muscle spindles, and desensitivity of Golgi tendon organs result in the improved performance seen following a pre-stretch. This can be achieved through plyometric training programs.

Experiments to determine the force–length characteristics of the series element can only be done on isolated muscle and will require dynamic changes of force or length. A typical experimental setup is shown in Figure 8.11. The muscle is stimulated to a certain level of tension while held at a certain isometric length. The load (tension) is suddenly dropped to zero by releasing one end of the muscle. With the force suddenly removed, the series elastic element, which is considered to have no mass, suddenly shortens to its relaxed length. This sudden shortening can be recorded, and the experiment repeated for another force level until a force-shortening curve can be plotted. During the rapid shortening period (about 2 ms), it is assumed that the contractile element has not had time to change length, even though it is still generating tension. Two other precautions must be observed when conducting this experiment. First, the isometric length prior to the release must be such that there is no tension in the parallel elastic element. This will be so if the muscle is at resting length or shorter. Second, the system that records the sudden shortening of the series elastic

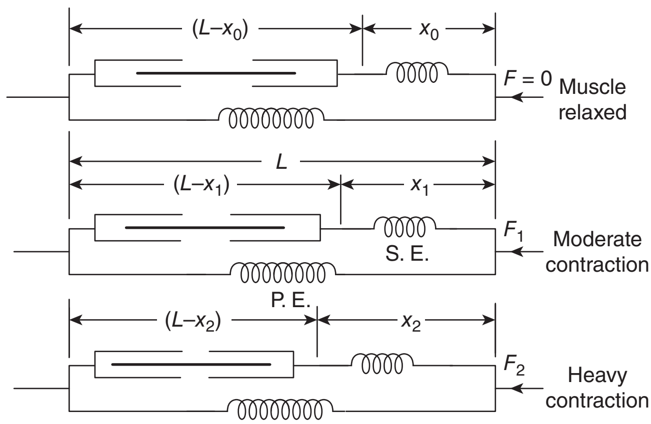

Figure 8.10 Introduction of the series elastic (S.E.) element. During isometric contractions, the tendon tension reflects a lengthening of the series element and an internal shortening of the contractile element. During most human movement, the presence of the series elastic element is not too significant, but during high-performance movements, such as jumping, it is responsible for storage of energy as a muscle lengthens immediately prior to rapid shortening.

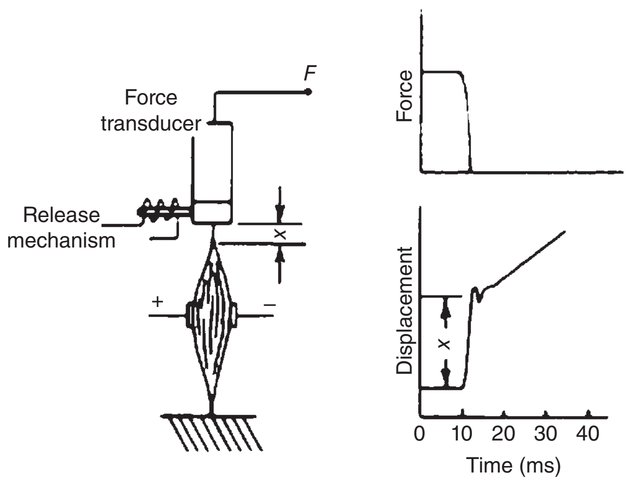

Figure 8.11 Experimental arrangement to determine the spring constant of the series elastic element. Muscle is stimulated; after the tension builds up in the muscle, the release mechanism activates, allowing an almost instantaneous shortening, x, while the force change is measured on a force transducer.

viscosity. Because of the high acceleration and velocity associated with the rapid shortening, the resisting force of the attached recording transducer should be negligible. If such conditions are not possible, a correction should be made to account for the mass or viscosity of the transducer.

8.1.4 In Vivo Force–Length Measures

Length–tension relationships of striated muscle have been well researched in mammalian and amphibian species, but in humans, the studies have been limited to changes in joint moments during MVC as the joint angle is altered. Two problems arise in such studies. First, it is usually impossible to generate an MVC for a single agonist without activating the remaining agonists. Thus, the joint moment is the sum of the moments generated by several agonists, each of which is likely to be operating at a different point in its force–length curve. Also, the moment generated by any muscle is a product of its force and its moment arm length, both of which change as the joint angle changes. Thus, any change in joint moment with joint angle cannot be attributed uniquely to the length–tension characteristics of one specific muscle.

One in vivo study that did yield some meaningful results was reported by Sale et al. (1982), who quantified the plantarflexor moment changes as the ankle angle was positioned over a range of 50° (30° plantarflexion to 20° dorsiflexion). The contribution of the gastrocnemius was reduced by having the knee flexed to 90°, thus causing it to be slack. Thus, the contribution of the soleus was considered to be dominant. Also, the range of movement of the ankle was considered to have limited influence on the moment arm length of the soleus. Thus, the moment–ankle angle curve was considered to reflect the force–length characteristics of the soleus muscle. The results of the MVC, peak twitch tensions, and a 10-Hz tetanic stimulation were almost the same: the peak tension was seen at 15° dorsiflexion and dropped off linearly to near zero as the ankle plantarflexed to 30°. Such a curve agreed qualitatively with a muscle whose resting length was at 15° dorsiflexion and whose cross-bridges reached a full overlap of 30° plantarflexion.

8.2 FORCE–VELOCITY CHARACTERISTICS

The previous section was concerned primarily with isometric contractions, and most physiological experiments are conducted under such conditions. However, movement cannot be accomplished without a change in muscle length. Alternate shortening and lengthening occur regularly during any given movement and the velocity associated with the shortening and lengthening will relate to the overall muscle tension. In the next section, we will discuss the velocity-based mechanisms to muscle tension.

8.2.1 Concentric Contractions

The tension in a muscle decreases as its shortening velocity increases. The characteristic curve that describes this effect is called a *force–velocity curve* and is shown in Figure 8.12. The usual curve is plotted for a maximum (100%) contraction. However, this condition is rarely seen except in athletic events, and then only for short bursts of time. In Figure 8.12 the curves for 75, 50, and 25% contractions are shown as well. Isometric contractions lie along the zero-velocity axis of this graph and should be considered as nothing more than a special case within the whole range of possible velocities. It should be noted that this curve represents the characteristics at a certain muscle length. To incorporate length as a variable as well as velocity requires a three-dimensional plot, as is discussed in Section 8.2.3.

The decrease of tension as the shortening velocity increases has been attributed to two main causes. A major reason appears to be the loss in tension as the cross-bridges in the contractile element break and then reform in a shortened condition. A second cause appears to be the fluid viscosity in both the contractile element and the connective tissue. Such viscosity results in friction that requires an internal force to overcome and, therefore, results in a reduced tendon force. Whatever the mechanism of lost tension, it is clear that the total effect is similar to that of viscous friction in a mechanical system and can, therefore, be modeled as some form of a fluid damper. The loss of tension related to a combination of the number of cross-bridges breaking and reforming and passive viscosity makes the force–velocity curve somewhat complicated to describe. If all the viscosity were passive, the slope of the force–velocity curve would be independent of activation. Conversely, if all the viscosity were related to the number of active cross-bridges, the slope would be proportional to the activation. Green (1969) has analyzed families of force–velocity curves to determine the relative contribution of each of these mechanisms.

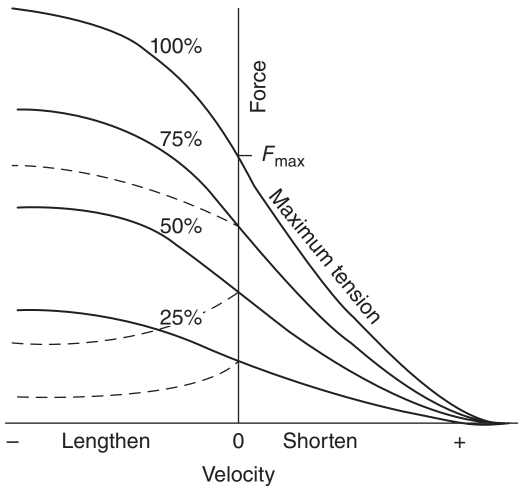

Figure 8.12 Force–velocity characteristics of skeletal muscle for different levels of muscle activation; shown are 25, 50, 75, and 100% levels of activation. All such measures must be taken as the muscle shortens or lengthens at a given length, and the length must be reported. During shortening, the curves follow the hyperbolic Hill model, but during lengthening, the curves depend on the experimental protocol; the solid lines are for isotonic activity, while the dashed lines are for isovelocity activity.

A curve fit of the force–velocity curve was demonstrated by Fenn and Marsh (1935), who used the equation:

$$V = V_0 e^{P/B} - KP \tag{8.2}$$

where

V = shortening velocity at any force
V_0 = shortening velocity of unloaded muscle
P = force
B, K = constants.

A few years later, Hill (1938) proposed a different mathematical relationship that bore some meaning with regard to the internal thermodynamics. Hill's curve was in the form of a hyperbola,

$$(P + a)(V + b) = (P_0 + a)b \tag{8.3}$$

where

P_0 = maximum isometric tension
a = coefficient of shortening heat
$b = a \cdot V_0/P_0$
V_0 = maximum velocity (when $P = 0$).

More recently, this hyperbolic form has been found to fit the empirical constant only during isotonic contractions near resting length.

The maximum velocity of shortening is often expressed as a function of l_0, the resting fiber length of the contractile elements of the muscle. Animal experiments have shown the maximum shortening velocity to be about $6l_0$/s (Close, 1965; Faulkner et al., 1980). However, it appears that this must be low for some human activity. Woittiez (1984) has calculated the shortening velocity in soleus, for example, to be above $10l_0$/s (based on plantarflexor velocity in excess of 8 rad/s and a moment arm length of 5 cm).

8.2.2 Eccentric Contractions

The vast majority of research done on isolated muscle during in vivo experiments has involved concentric contractions. As a result, there is relatively limited knowledge about the details of the force–velocity curve as the muscle lengthens. The curve certainly does not follow the detailed mathematical relationships that have been developed for concentric contractions.

This lack of information about eccentric contractions is unfortunate because normal human movement usually involves as much eccentric as concentric activity. If we neglect air and ground friction, level walking involves equal amounts of positive and negative work, and in downhill gait, negative work dominates. Figure 8.12 shows the general shape of the force–velocity curve during eccentric contractions. It can be seen that this curve is an extension of the concentric curve. During isotonic eccentric action, the solid line curves apply (Winters, 1990), but during isovelocity eccentric activity, the relationship follows the dotted lines shown in Figure 8.12 (Zahalak, 1990; Sutarno and McGill, 1995). If the maximum isometric force is $F_{\max}$, the plateau reached in the eccentric phase varies from 1.1 $F_{\max}$ to 1.8 $F_{\max}$ (Winters, 1990). The reason was given for the force increasing as the lengthening velocity increases was that within the cross-bridges the force required to reverse pivot the myosin heads and break the bonds is greater than that required to hold it at its isometric length. The plateau is reached at higher velocities when the cross-bridge links simply "give way" to produce no further increase in force. Also, there is viscous friction in the fluid surrounding the muscle fibers, and this friction force must be overcome as well. This is similar to opening a screen door (with a pneumatic door closer) very quickly. Air pressure is built-up within the pneumatic closer since air cannot escape at that rate through the small hole. During eccentric muscle contractions, the connective tissue element will undergo tension and water will be forced out of the tissue. When lengthened quickly the water cannot escape at that rate and water pressure will increase, creating viscous friction. However, the distributed moment (DM) state variable model that approximates the Huxley-type cross-bridge theories (Zahalak and Ma, 1990) predicted the drop in tension reported in the eccentric region for constant velocity stretch.

Experimentally, it is somewhat more difficult to conduct experiments involving eccentric work because an external device must be available to do the work on the human muscle. Such a requirement means that a motor is needed to provide an external force that will always exceed that of the muscle. Experiments on isolated muscle are safe to conduct, but in vivo experiments on humans are difficult because such a machine could cause lengthening even past the safe range of movement of a joint. The excessive force could tear the limb apart at the joint. Foolproof safety mechanisms have to be installed to prevent such an occurrence. In addition to safety concerns, it needs to be highlighted that joints have several muscles combining to create the internal moment which will counteract the external load. Therefore, it is impossible to isolate muscles during such experiments as these devices can only apply external force to a joint.

8.2.3 Combination of Length and Velocity Versus Force

In Section 8.1 and in this section, it is evident that the tendon force is a function of both length and velocity. Therefore, a proper representation of both these effects requires a three-dimensional plot like that shown in Figure 8.13. The resultant curve is actually a surface, which here is represented only for the maximum contraction condition. The more normal contractions are at a fraction of this maximum, so that surface plots would be required for each level of muscle activation, say at 75, 50, and 25%.

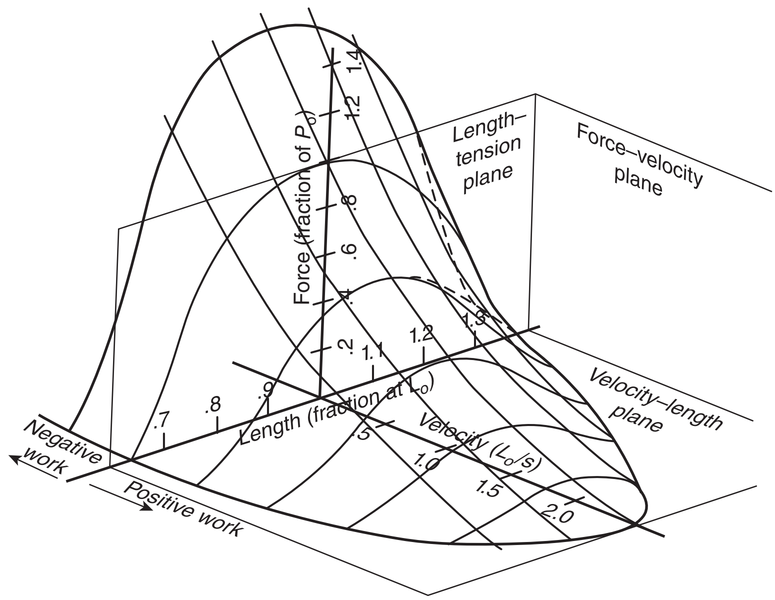

Figure 8.13 Three-dimensional plot showing the change in contractile element tension as a function of both velocity and length. Surface shown is for maximum muscle activation; a new "surface" will be needed to describe each level of activation. Influence of parallel elastic element is not shown.

8.2.4 Combining Muscle Characteristics with Load Characteristics: Equilibrium

Muscles are the only motors in the human system, and when they are active, they must be in equilibrium with their load. The load can be static, for example, when holding a weight against gravity or applying a static force against a fixed object (e.g. a wall, the floor), or in an isometric co-contraction, where one muscle provides the load of the other. Or the load may be dynamic and the muscles are accelerating or decelerating an inertial load or overcoming the friction of a viscous load. In the vast majority of voluntary movements, there is a mixture of static and dynamic load. The one condition that is satisfied during the entire movement is that of equilibrium: at any given point in time, the operating point will be the intersection of the muscle and load characteristics. In a dynamic movement, this operating or equilibrium point will be continuously changing. In Chapter 5, an equilibrium was recognized every time the equations of motion were written. However, such equations recognized the net effect of all muscles acting at each joint and represented them as a torque motor without any concern about the internal characteristics of the muscles themselves. These are now examined.

Consider a muscle that is contracting against a spring-like load that can have a linear or a nonlinear force–length characteristic. Figure 8.14a plots the force–length characteristics of a muscle at four different levels of activation, along with the linear and nonlinear characteristics of the spring loads. Assume that the muscle is at a length l_1 longer than the resting length, l_0 when the spring is at rest. The linear spring is compressed with a 50% muscle contraction, the equilibrium point *a* is reached, and the muscle will shorten from l_1 to l_2. When the nonlinear spring is compressed with a 50% contraction, the muscle shortens from l_1 to l_3 and the equilibrium point *b* is reached. If a gravitational load is lifted by the elbow flexors, the flexor muscles have the characteristics shown in Figure 8.14b. Starting with the forearm lowered to the side so that it is vertical, the initial equilibrium point is *a*. Then, as the flexors contract and shorten, equilibrium points *b* through *e* are reached. Finally, as the muscles are activated 100%, the equilibrium point, *e*, is reached. During the transient conditions en route from

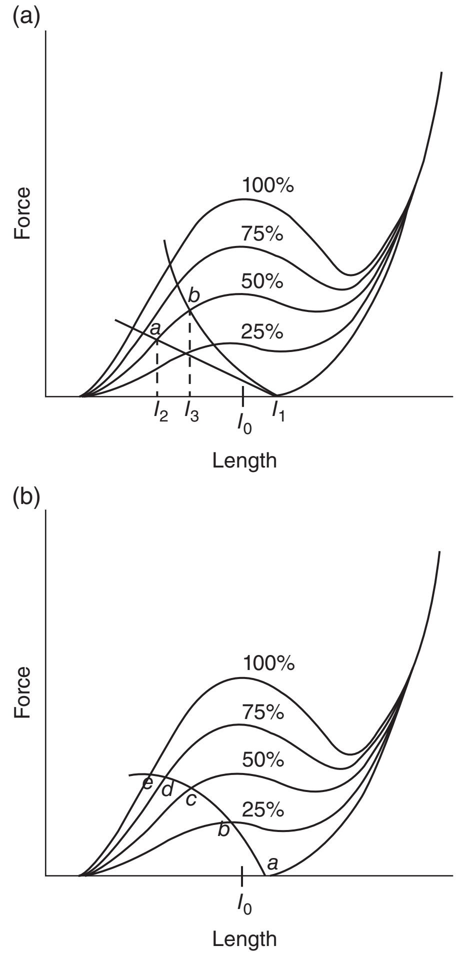

Figure 8.14 (a) Force–length characteristics of a muscle at four different levels of activation along with the characteristics of a linear and nonlinear spring. Intersection of spring (load) and muscle characteristics is the equilibrium point. (b) Equilibrium points of a load and elbow flexor muscle as the elbow is flexed against a gravitational load.

move about on the force–velocity characteristics, tracing a three-dimensional locus. The concept of dynamic equilibrium is now addressed.

We use the data from a typical walking stride to plot the time course of the contractions of the muscles about the ankle. Because the angular changes of the ankle are relatively small, we can consider that the muscle lengths are proportional to the ankle angle, and these length changes are also small. Thus, we can plot the event on a two-dimensional force–velocity curve. The lengthening or shortening velocity can be considered proportional to the angular velocity, and the tendon forces are considered proportional to the muscle moment. Thus, using in vivo data, the traditional force–velocity curve becomes a moment–angular velocity curve. Figure 8.15 is the resultant plot for the period of time from heel contact to toe-off.

It can be seen from this time course of moment and angular velocity that this common movement is, in fact, quite complex. Contrary to what might be implied by force–velocity curves, a muscle does not operate along any simple curve, but actually goes through a complex combination of force and velocity changes. Initially, the dorsiflexor muscles are active as the foot plantarflexes between the time of heel contact and flat foot (when the ground reaction force lies behind the ankle joint). Negative work is being done by the dorsiflexors as they lower the foot to the ground. After flat foot, the plantarflexors dominate and

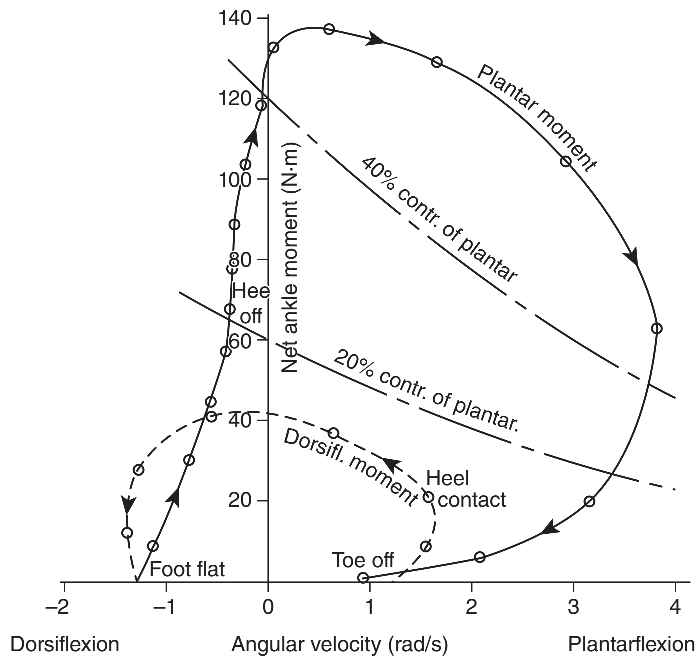

Figure 8.15 Time history of muscle moment–velocity during stance phase of walking. Dorsiflexor moment is shown as a dashed line; plantarflexor moment is a solid line. Stance begins with negative work, then alternating positive and negative, and finally, a major positive work burst later in push-off, ending at toe-off.

the leg as it rotates over the foot, which is now fixed on the floor. Again, this is negative work. A few frames after heel-off, positive work begins, as indicated by the simultaneous plantarflexor muscle moment and plantarflexor velocity. This period is the active push-off phase, when most of the "new" energy is put back into the body.

8.3 TECHNIQUE TO MEASURE IN VIVO TENDON MECHANICAL PROPERTIES

Methodologies have been developed and tested to calculate tendon stiffness and force during isometric contractions. The majority of the literature is focused on the Achilles tendon; however, similar techniques have been performed at other tendons, such as the patellar tendon. These techniques are useful to examine the effectiveness of surgical and rehabilitation approaches. The following section will describe the detailed steps to accurately calculate Achilles tendon stiffness and force.

8.3.1 Ankle Joint Moment

The first step in determining Achilles tendon stiffness is to measure ankle joint moments while performing a series of targeted isometric contractions based on a maximal voluntary isometric contraction (MVIC) using a dynamometer. Participants lie prone on the dynamometer table with their knee in 0° of flexion, and their ankle in 20° of plantarflexion. This lower extremity position is important to minimize the passive torque from the connective tissue structures. While other lower extremity positions can also minimize the passive connective tissue torque, it is important to keep the position standardized across participants. Care should be taken to align the ankle axis of rotation with the dynamometer's axis of rotation. Even with proper alignment and fixation, the axis of rotation of the ankle will shift during an isometric
the calculation of the joint moment and tendon elongation.

ankle and the dynamometer attachment need to be included to correct joint moments and tendon elongations (Magnusson et al., 2001). In addition, EMG is often used to correct for co-activation of the tibialis anterior muscle during a plantarflexion isometric contraction (Maganaris et al., 1998). During testing, it is necessary to collect passive plantar and dorsiflexion trials to correct for gravitational forces. In addition, a maximal dorsiflexion contraction is performed to later estimate the ankle dorsiflexion moment (antagonist co-contraction) occurring during a plantarflexion contraction (Arampatzis et al., 2005).

8.3.2 Tendon Mechanical Properties

Diagnostic ultrasound is used to measure the displacement of the myotendinous junction (MTJ) of the medial gastrocnemius muscle during isometric plantarflexion. To account for MTJ displacement resulting from joint rotation during the isometric contractions, ultrasound should also be used during the passive plantarflexion and dorsiflexion trials. To correct the displacement, a simple difference was taken between active and passive trials. Next, the resting length of the Achilles tendon needs to be measured in 20° of plantarflexion to minimize any torque to the connective tissue structures. The resting length is measured from the MTJ along the curved path to the insertion on the calcaneal tuberosity (Arampatzis et al., 2008). Achilles tendon moment arms are typically estimated based on the relationship between MTJ elongation and ankle joint rotation (Fath et al., 2010). To calculate tendon force, the ankle joint moment is divided by the ankle joint moment arms. Tendon stiffness will then be calculated as the slope of the tendon force/elongation curve. This curve will typically be created using a series of targeted isometric contraction levels based on the MVIC.

Calculating tendon properties in vivo can be very useful to determine the effectiveness of interventions designed to improve healing following surgery or rehabilitation protocols. Although the most established methods have been developed and tested at the Achilles tendon, other tendons are possible. When attempting these calculations at other joints, it is important to consider the previously described variables (gravitational forces, moment arm changes, axis of rotation misalignment, joint rotation, etc.) in addition to isolating the contribution of a single muscle/tendon unit.

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9

KINESIOLOGICAL ELECTROMYOGRAPHY

JOSEPH A. ZENI[1], STEPHEN J. THOMAS[2], AND DAVID A. WINTERS[3,†]
[1]*Rutgers University, Newark, NJ, USA*
[2]*Thomas Jefferson University, Philadelphia, PA, USA*
[3]*University of Waterloo, Waterloo, Ontario, Canada*

9.0 INTRODUCTION

The electrical signal associated with the contraction of a muscle is called an *electromyogram*, or EMG. The study of EMGs, called *electromyography*, has revealed some basic information; however, much remains to be learned. Voluntary muscular activity results in an EMG that increases in magnitude with the tension. However, there are many variables that can influence the signal at any given time: velocity of shortening or lengthening of the muscle, rate of tension build-up, fatigue, and reflex activity. An understanding of the electrophysiology and the technology of recording is essential to the appreciation of the biomechanical relationships that follow.

9.1 ELECTROPHYSIOLOGY OF MUSCLE CONTRACTION

It is important to realize that muscle tissue conducts electrical potentials somewhat similarly to the way axons transmit action potentials. The name given to this special electrical signal generated in the muscle fibers as a result of the recruitment of a motor unit is a *motor unit action potential* (m.u.a.p.). Electrodes placed on the surface of a muscle or inside the muscle tissue (indwelling electrodes) will record the algebraic sum of all m.u.a.p.'s being transmitted along the muscle fibers at that point in time. Those motor units far away from the electrode site will result in a smaller m.u.a.p. than those of similar size near the electrode.

9.1.1 Motor End Plate

For a given muscle, there can be a variable number of motor units, each controlled by a motor neuron through special synaptic junctions called *motor end plates*. An action potential transmitted down the motoneuron (sometimes called the *final common pathway*) arrives at the motor end

†Deceased

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

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plate and triggers a sequence of electrochemical events. A quantum of Acetylcholine (ACh) is released; it crosses the synaptic gap (200–500 Å wide) and causes a depolarization of the postsynaptic membrane. Such a depolarization can be recorded by a suitable microelectrode and is called an *end plate potential* (EPP). In normal circumstances, the EPP is large enough to reach a threshold level, and an action potential is initiated in the adjacent muscle fiber membrane. In disorders of neuromuscular transmission (e.g. depletion of ACh), there may not be a one-to-one relationship between each motor nerve action potential and a m.u.a.p. The end plate block may be complete, or it may occur only at high stimulation rates or intermittently.

9.1.2 Sequence of Chemical Events Leading to a Twitch

The beginning of the m.u.a.p. starts at the *Z* line of the contractile element (see Chapter 8) by means of an inward spread of the stimulus along the transverse tubular system. This results in a release of Ca^{2+} in the sarcoplasmic reticulum. Ca^{2+} rapidly diffuses to the contractile filaments of actin and myosin, where adenosine triphosphate (ATP) is hydrolyzed to produce adenosine diphosphate (ADP) plus heat plus mechanical energy (tension). The mechanical energy manifests itself as an impulsive force at the cross-bridges of the contractile element. The time course of the contractile element's force has been the subject of considerable speculation and has been modeled mathematically, as described in Chapter 8.

9.1.3 Generation of a Muscle Action Potential

The depolarization of the transverse tubular system and the sarcoplasmic reticulum results in a depolarization "wave" along the direction of the muscle fibers. It is this depolarization wave front and the subsequent repolarization wave that are "seen" by the recording electrodes.

Many types of EMG electrodes have developed over the years, but generally, they can be divided into two groups: surface and indwelling (intramuscular). The seminal paper by Basmajian (1973) gives a detailed review of the use of different types along with their connectors. Surface electrodes consist of disks of metal, usually silver/silver chloride, of about 1 cm in diameter. These electrodes detect the average activity of superficial muscles and give more reproducible results than do indwelling types (Komi and Buskirk, 1970; Kadaba et al., 1985). Smaller disks can be used for smaller muscles. Indwelling electrodes are required, however, for the assessment of fine movements or to record from deep muscles. A needle electrode is a fine hypodermic needle with an insulated conductor located inside and bared to the muscle tissue at the open end of the needle; the needle itself forms the other conductor (Figure 9.1). For research purposes, multielectrode types have been developed to investigate the "territory" of a motor unit (Buchthal et al., 1959), which has been found to vary from 2 to 15 mm in diameter. Fine-wire electrodes, which

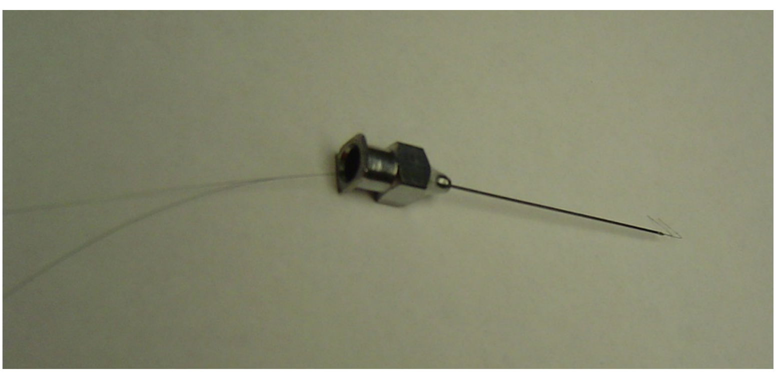


Figure 9.1 Intramuscular EMG electrode with fine wire passing hooked ends for fixation within muscle tissue.

have a diameter about that of human hairs, are now widely used. They require a hypodermic needle to insert. After removal of the needle, the fine wires with their uninsulated tips remain inside in contact with the muscle tissue. A comparison of this experimental investigation of motor unit territory was seen to agree with theoretical predictions (Boyd et al., 1978).

Indwelling electrodes are influenced not only by waves that actually pass by their conducting surfaces but also by waves that pass within a few millimeters of the bare conductor. The same is true for surface electrodes. The field equations that describe the electrode potential were originally derived by Lorente de No (1947) and were extended in rigorous formulations by Plonsey (1964, 1974) and Rosenfalck (1969). These equations were complicated by the formulation of current density functions to describe the temporal and spatial depolarization and repolarization of the muscle membrane. Thus, simplification of the current density to a dipole or tripole (Rosenfalck, 1969) yielded a reasonable approximation when the active fiber was more than 1 mm from the electrode surface (Andreassen and Rosenfalck, 1981).

In the dipole model (see Figure 9.2), the current is assumed to be concentrated at two points along the fiber: a source of current, I, representing the depolarization, and a sink of current, $-I$, representing the repolarization, both separated by a distance b. The potential Φ, recorded at a point electrode at a distance r from the current source, is given by:

$$\Phi = \frac{I}{4\Pi\sigma} \cdot \frac{1}{r} \tag{9.1}$$

where σ is the conductivity of the medium, which is assumed to be isotropic (uniform in all spatial directions).

The net potential recorded at the point electrode from both source and sink currents is:

$$\begin{aligned} \Phi &= \frac{I}{4\Pi\sigma} \cdot \frac{1}{r_1} - \frac{I}{4\Pi\sigma} \cdot \frac{1}{r_2} \\ &= \frac{I}{4\Pi\sigma}\left(\frac{1}{r_1} - \frac{1}{r_2}\right) \end{aligned} \tag{9.2}$$

where r_1 and r_2 are the distances to the source and sink currents.

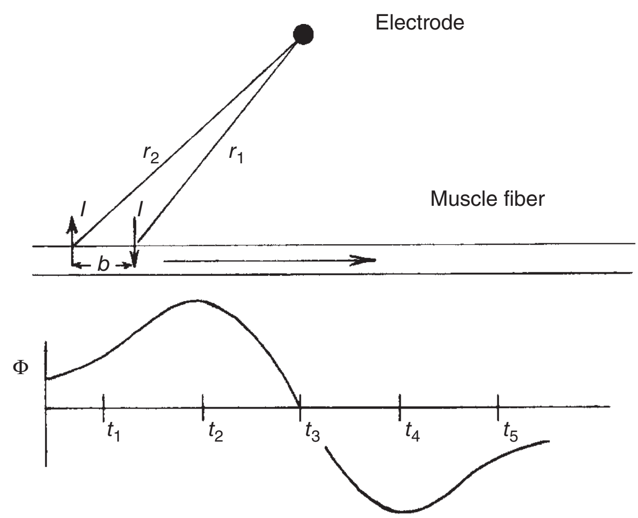


Figure 9.2 Propagation of motor unit action potential wave front as it passes beneath a recording electrode on the skin surface. The electrode voltage is a function of the magnitude of the
electrode to the depolarizing and repolarizing currents.

The time history of the action potential then depends on r_1 and r_2, which vary with time as the wave propagates along the muscle fiber. At t_1, as the wave approaches the electrode ($r_1 < r_2$), the potential will thus be positive and will increase. It will reach a maximum at t_2; then as r_1 becomes nearly equal to r_2, the amplitude suddenly decreases and passes through zero at t_3 when the dipole is directly beneath the electrode ($r_1 = r_2$). Then it becomes negative as the dipole propagates away from the electrode ($r_1 > r_2$). Thus, a biphasic wave is recorded by a single electrode.

A number of recording and biological factors affect the magnitude and shape of the biphasic signal. The duration of each phase is a function of the propagation velocity, the distance b (which varies from 0.5 to 2.0 mm) between source and sink, the depth of the fiber, and the electrode surface area. Equation (10.2) assumes a point electrode. Typical surface electrodes are not point electrodes but have a finite surface area, and each point on the surface can be considered an area of point sources; the potential on the surface is the average of all point-source potentials. Figure 9.3 is presented from a study by Fuglevand et al. (1992) to demonstrate the influence of the electrode size and shape on the electrode potential. Two different orientations of strip electrodes are shown: (a) along an axis parallel to the muscle fiber – and (d) at right angles to the muscle fiber. (b) and (e) show the location of a strip of point-source electrodes to represent these two strip electrodes, each 10 mm long with 10 point sources; (c) and (f) show the action potentials at each point source. For the strip electrode in (a), the distance from the points on the electrode to the fiber, r_f, is constant; thus, each action potential has the same amplitude and shape. However, there is a phase difference between each of the point-source action potentials such that the average action potential is lower and longer than the individual action potentials. For the strip electrode in (d), the distance r_f is different for every point source; thus, each action potential decreases as r_f increases, as is seen in (f). The phase of each point-source action potential has the same zero-crossing time, but the smaller action potentials from the further point sources are slightly longer in duration. Finally, (g) compares the net action potentials from a point source, strip (a), strip (d), and a square electrode $10 \times 10\ \text{mm}^2$. As is evident from these action potentials, it is desirable to use electrodes with as small a surface area as possible. Also, as will be seen in Section 9.2.5, the smaller surface area electrodes are less influenced by cross-talk.

The zone of pick-up for individual fibers by an electrode depends on two physiological variables, I and r. Larger muscle fibers have larger dipole currents, which increase with the diameter of the fiber. For an m.u.a.p., the number of fibers innervated must also be considered. The potential detected from the activation of a motor unit will be equivalent to the sum of the constituent fiber potentials. Therefore, the pick-up zone for a motor unit that innervates a large number of fibers will be greater than that for a unit with fewer fibers. An estimate of the detectable pick-up distance for small motor units (50 fibers) is about 0.5 cm and, for the largest motor units (2500 fibers), about 1.5 cm (Fuglevand et al., 1992). Thus, surface electrodes are limited to detecting m.u.a.p.'s from those fibers quite close to the electrode site and are not prone to pick up from adjacent muscles (the phenomenon known as *cross-talk*) unless the muscle being recorded from is very small.

Most EMGs require two electrodes over the muscle site, so the voltage waveform that is recorded is the difference in potential between the two electrodes. Figure 9.4 shows that the voltage waveform at each electrode is almost the same but is shifted slightly in time. Thus, a biphasic potential waveform at one electrode usually results in a different signal that has three phases. The closer the spacing between the two electrodes, the more the difference signal looks like a time differentiation of that signal recorded at a single electrode (Kadefors, 1973). Thus, closely spaced electrodes result in the spectrum of the EMG having higher-frequency components than those recorded from widely spaced electrodes.

9.1.4 Duration of the Motor Unit Action Potential

As indicated in the previous section, the larger the surface area, the longer the duration of the m.u.a.p. Thus, surface electrodes automatically record

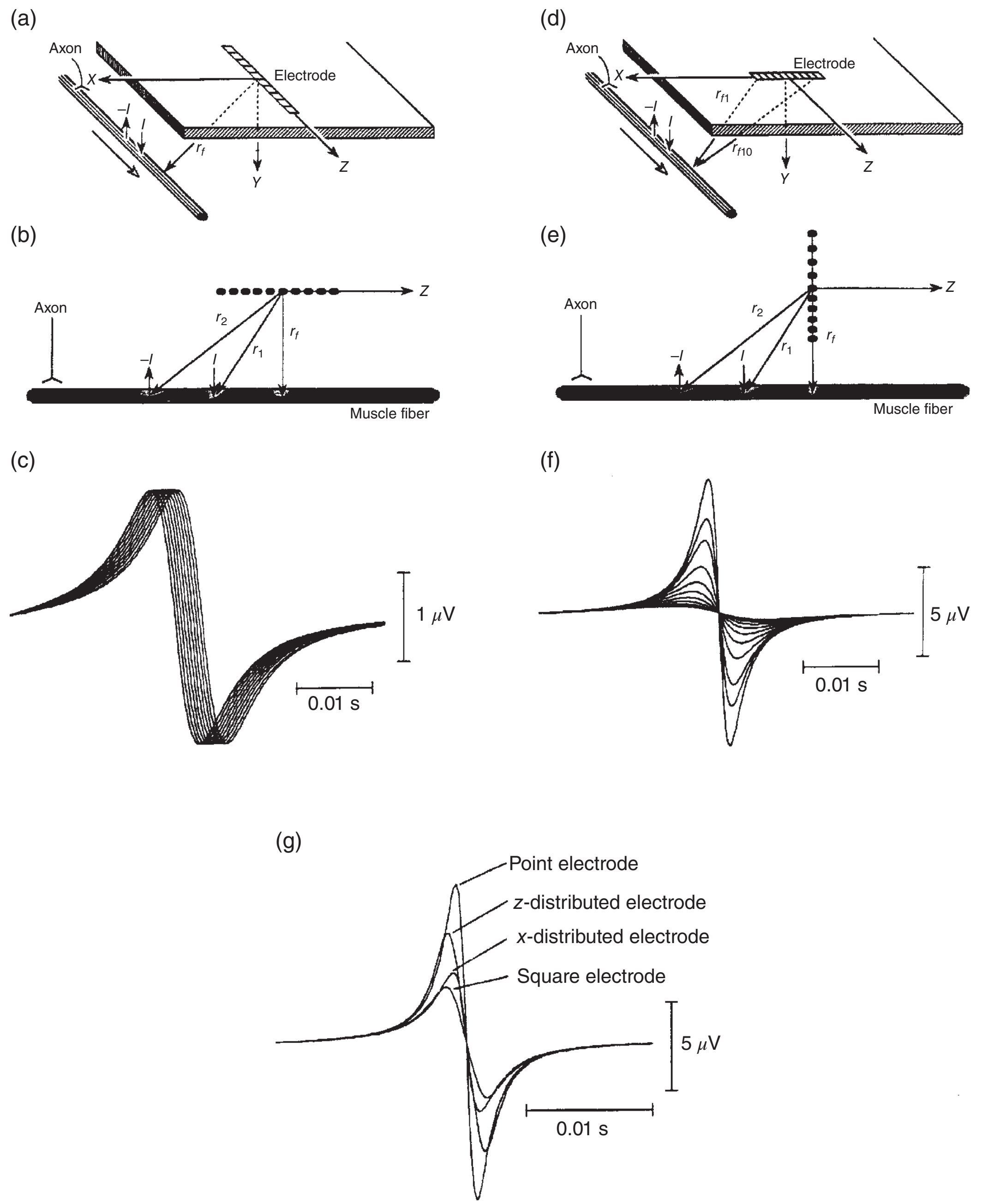


Figure 9.3 Influence of electrode geometry on the predicted action potentials from a single electrode with a variety of shapes and sizes. The potential was computed as the algebraic mean of the potentials detected by an array of point-source electrodes. See the text for detailed discussion. (Reproduced by permission from *Biological Cybernetics*, Fuglevand et al. (1992). With kind permission of Springer Science + Business Media.)

electrodes (Kadefors, 1973; Basmajian, 1973). Needle electrodes record durations of 3–20 ms, while surface electrodes record durations about twice that long. However, for a given set of electrodes, the duration of the m.u.a.p.'s is a function of the velocity of the propagating wave front and the depth of the motor unit below the electrode's surface. The velocity of propagation of the m.u.a.p. in normals has been found to be about 4 m/s

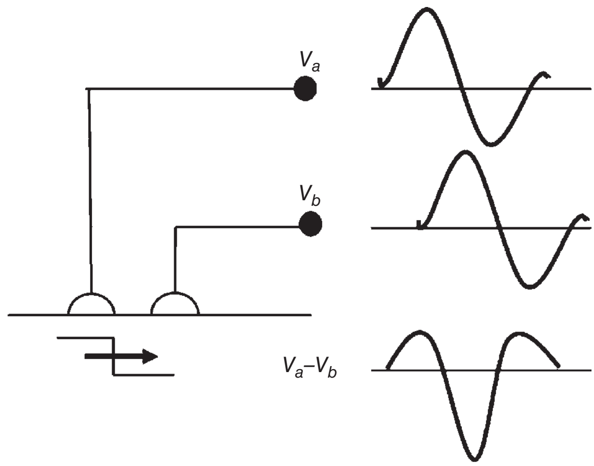


Figure 9.4 Voltage waveform present at two electrodes because of a single propagating wave. Voltage recorded is the difference voltage $V_a - V_b$, which is triphasic in comparison with the biphasic waveform seen at a single electrode.

velocity, the shorter the duration of the m.u.a.p. Such a relationship has been used to a limited extent in the detection of velocity changes. In fatigue and in certain myopathies (muscle pathologies), the average velocity of the m.u.a.p.'s that are recruited is reduced; thus, the duration of the m.u.a.p.'s increases (Johansson et al., 1970; Gersten et al., 1965; Kadefors, 1973). Because the peak amplitude of each phase of the m.u.a.p. remains the same, the area under each phase will increase. Thus, when we measure the average amplitude of the EMG (from the full-wave rectified waveform), it will appear to increase (Fuglevand, et al., 1992). During voluntary contractions, it has been possible, under special laboratory conditions (Milner-Brown and Stein, 1975), to detect the duration and amplitude of muscle action potentials directly from the EMG. However, during unconstrained movements, a computer analysis of the total EMG would be necessary to detect a shift in the frequency spectrum (Kwatny et al., 1970), or an autocorrelation analysis can yield the average m.u.a.p. duration (Person and Mishin, 1964).

The distance between the motor unit and the surface of the electrode severely influences the amplitude of the action potential, as is predicted in Equation (10.2), and also influences the duration of the action potential. Figure 9.5, taken from Fuglevand et al. (1992), predicts the duration and amplitude of the action potential from a 50-fiber motor unit with electrode-unit distances out to 20 mm. It is evident that the duration of the action potential increases as the amplitude decreases with distance. Thus, the frequency content of the m.u.a.p.'s from the more distant sites decreases. Fuglevand et al. (1992) showed that the mean power frequency (MPF) of the m.u.a.p. would decrease from 160 Hz at an electrode-motor unit distance of 1 mm to 25 Hz at an electrode-motor unit distance of 20 mm.

9.1.5 Detection of Motor Unit Action Potentials from Electromyogram During Graded Contractions

Using recordings from multiple indwelling electrodes, DeLuca et al. (1982) pioneered techniques to identify individual m.u.a.p.'s during a low-level graded contraction. More recent techniques use a quadrifilar needle electrode recording of three channels followed by decomposition algorithms involving template matching, template updating, and motor unit firing statistics (DeLuca, 1993). As many as four motor units were detected and their firing rates were tracked from 0 to 100% of maximum voluntary contraction (MVC) of the tibialis anterior (TA) muscle (Erim et al., 1996). It should be noted that only a fraction of all active

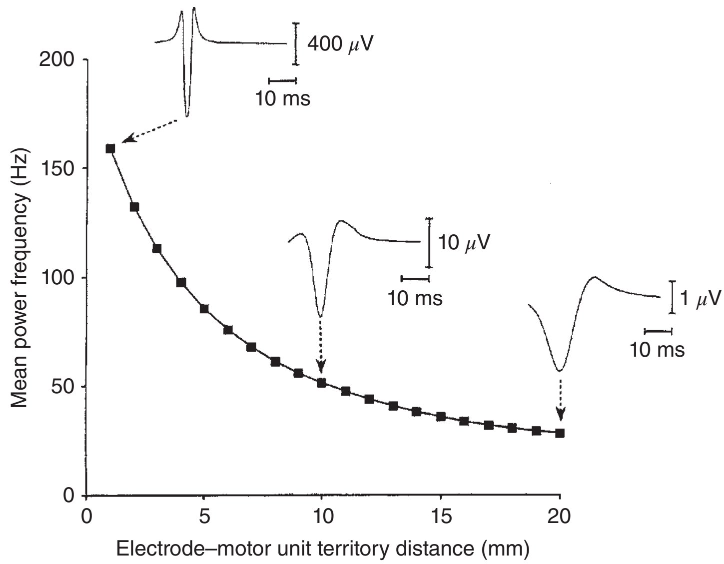


Figure 9.5 Variation in the amplitude and frequency content of the motor unit action potential with increased electrode-motor unit distance. Predictions were for a small motor unit (50 fibers) with 4-mm^2 bipolar electrodes with an interelectrode spacing of 11 mm. Shown are the m.u.a.p.'s at 1-, 10-, and 20-mm distances. (Reproduced by permission from *Biological Cybernetics*, Fuglevand et al. (1992). With kind permission of Springer Science + Business Media.)

within the pick-up region of the electrode array. EMG recordings from surface electrodes have been decomposed in order to determine the firing profile of those motor units detected (McGill et al., 1987).

9.2 RECORDING OF THE ELECTROMYOGRAM

A biological amplifier of certain specifications is required for the recording of the EMG, whether from surface or from indwelling electrodes. It is valuable to discuss the reasons behind these specifications with respect to considerable problems in getting a "clean" EMG signal. Such a signal is the summation of m.u.a.ps and should be undistorted and free of noise or artifacts. Undistorted means that the signal has been amplified linearly over the range of the amplifier and recording system. The larger signals (up to 5 mV) have been amplified as much as the smaller signals (100 μV and below). The most common distortion is overdriving of the amplifier system such that the larger signals appear to be clipped off. Every amplifier has a dynamic range and should be such that the largest EMG signal expected will not exceed that range. Noise can be introduced from sources other than the muscle and can be biological in origin or man-made. An electrocardiogram (ECG) signal picked up by EMG electrodes on the thoracic muscles can be considered unwanted biological noise. Man-made noise usually comes from power lines (60 Hz in the United States or 50 Hz in Europe) or from electronics or is generated within the components of the amplifier. Motion artifacts generally refer to false signals generated by the electrodes themselves or the cabling system. Therefore, care should be taken to secure wires and electrodes to the participant so that movement near the electrode detection sites would not be affected (Figure 9.6).

The major considerations to be made when specifying the EMG amplifier are:

1. Gain and dynamic range
2. Input impedance

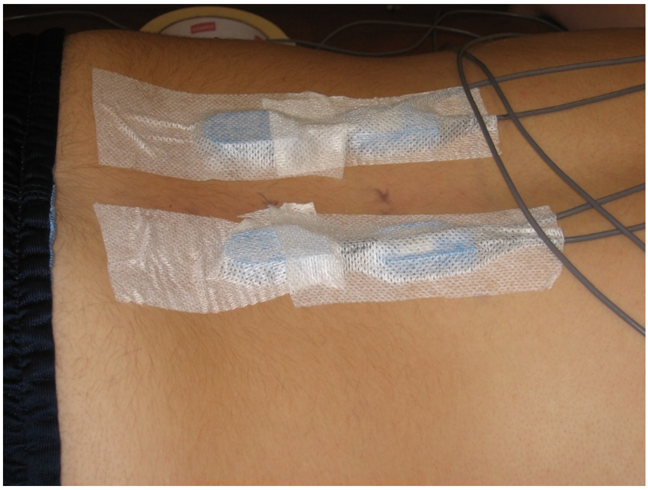


Figure 9.6 Surface EMG electrodes secured with adhesive material to a patient to prevent movement artifact of the wires and electrode.

3. Frequency response
4. Common-mode rejection

9.2.1 Amplifier Gain

Surface EMGs have a maximum amplitude of 5 mV peak to peak, as recorded during a MVC. Indwelling electrodes can have a larger amplitude, up to 10 mV. A single m.u.a.p. has an amplitude of about 100 μV. The noise level of the amplifier is the amplitude of the higher-frequency random signal seen when the electrodes are shorted together, and should not exceed 50 μV, preferably 20 μV. The gain of the amplifier is defined as the ratio of the output voltage to the input voltage. For a 2-mV input and a gain of 1000, the output will be 2 V. The exact gain chosen for any given situation will depend on what is to be done with the output signal. EMG had traditionally been viewed and recorded on a pen recorder, but nearly all modern units will transfer the data into digital format and be stored on a computer. The amplified EMG should not exceed the range of input signals expected by the recording equipment. Fortunately, most of these recording systems have internal amplifiers that can be adjusted to accommodate a wide range of input signals. In general, a good bioamplifier should have a range of gains selectable from 100 to 10,000. Independent of the amplifier gain, the amplitude of the signal should be reported as it appears at the electrodes, in millivolts.

9.2.2 Input Impedance

The input impedance (resistance) of a biological amplifier must be sufficiently high so as not to attenuate the EMG signal as it is connected to the input terminals of the amplifier. Consider the amplifier represented in Figure 9.7a. The active input terminals are 1 and 2, with a common terminal *c*. The need for a three-input terminal amplifier (differential amplifier) is explained in Section 9.2.4.

Each electrode-skin interface has a finite impedance that depends on many factors: thickness of the skin layer, cleaning of the skin prior to the attachment of the electrodes, area of the electrode surface, and temperature of the electrode paste (it warms up from room temperature after

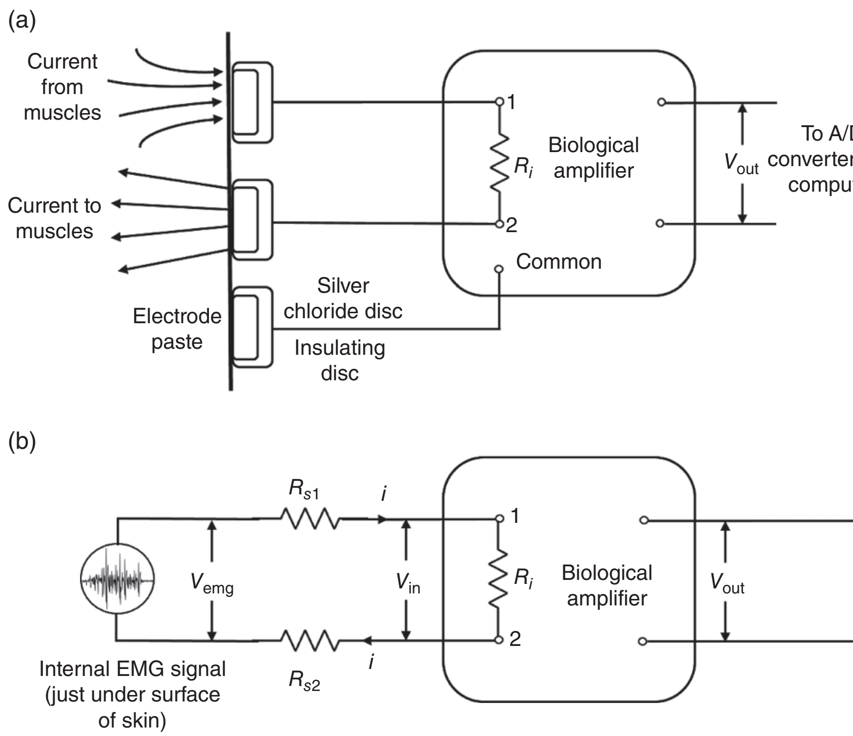


Figure 9.7 Biological amplifier for recording electrode potentials. (a) Current resulting from muscle action potentials flows across skin – electrode interface to develop a voltage V_{in} at the input terminals of the amplifier. A third, common electrode is normally required because the amplifier is a differential amplifier. (b) Equivalent circuit showing electrodes replaced by series resistors R_{s1} and $R_{s2}$$\cdot$$V_{in}$ will be nearly equal to V_{EMG} if $R_i \gg R_s$.

attachment). Indwelling electrodes have a higher impedance because of the small surface area of bare wire that is in contact with the muscle tissue.

In Figure 9.7b, the electrode-skin interface has been replaced with an equivalent resistance. This is a simplification of the actual situation. A correct representation is a more complex impedance to include the capacitance effect between the electrode and the skin. As soon as the amplifier is connected to the electrodes, the minute EMG signal will cause current to flow through the electrode resistances R_{s1} and R_{s2} to the input impedance of the amplifier R_i. The current flow through the electrode resistance will cause a voltage drop so that the voltage at the input terminals V_{in} will be less than the desired signal V_{EMG}. For example, if $R_{s1} = R_{s2} = 10{,}000\ \Omega$ and $R_i = 80{,}000\ \Omega$, a 2-mV EMG signal will be reduced to 1.6 mV at V_{in}. A voltage loss of 0.2 mV occurs across each of the electrodes. If R_{s1} and R_{s2} were decreased by better skin preparation to 1000 Ω, and R_i was increased to 1 M Ω, the 2-mV EMG signal would be reduced only slightly, to 1.998 mV. Thus, it is desirable to have input impedances of 1 M Ω or higher and to prepare the skin to reduce the impedance to 1000 Ω or less. For indwelling electrodes, the electrode impedance can be as high as 50, 000 Ω, so an amplifier with at least 5 M Ω input impedance should be used.

9.2.3 Frequency Response

The frequency bandwidth of an EMG amplifier should be such as to amplify, without attenuation, all frequencies present in the EMG. The bandwidth of any amplifier, as shown in Figure 9.8, is the difference between the upper cutoff frequency f_2 and the lower cutoff frequency f_1. The gain of

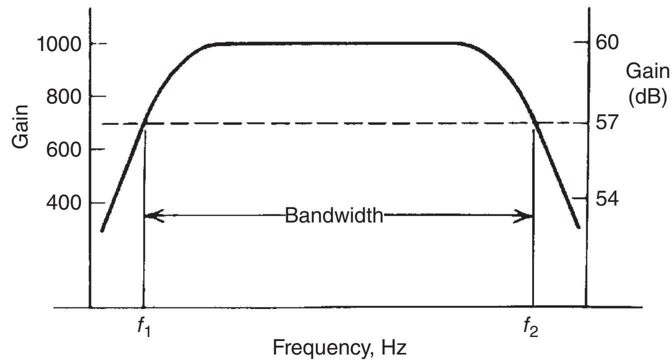


Figure 9.8 Frequency response of the biological amplifier showing a gain of 1000 (60 dB), and lower and upper cutoff frequencies f_1 and f_2.

the amplifier at these cutoff frequencies is 0.707 of the gain in the midfrequency region. If we express the midfrequency gain as 100%, the gain at the cutoff frequencies has dropped to 70.7%, or the power has dropped to $(0.707)^2 = 0.5$. These are also referred to as the half-power points. Often amplifier gain is expressed in logarithmic form and expressed in decibels,

$$\text{gain (dB)} = 20 \log_{10} (\text{linear gain}) \tag{9.3}$$

If the linear gain were 1000, the gain in decibels would be 60, and the gain at the cutoff frequency would be 57 dB (3 dB less than that at midfrequency).

In a high-fidelity amplifier used for music reproduction, f_1 and f_2 are designed to accommodate the range of human hearing, from 50 to 20,000 Hz. All frequencies present in the music will be amplified equally, producing a true undistorted sound at the loudspeakers. Similarly, the EMG should have all its frequencies amplified equally. The spectrum of the EMG has been widely reported in the literature, with a range from 5 Hz at the low end to 2000 Hz at the upper end. For surface electrodes, the m.u.a.p.'s are longer in duration and, thus, have negligible power beyond 1000 Hz. A recommended range for surface EMG is 10–1000 Hz and 20–2000 Hz for indwelling electrodes. If computer pattern recognition of individual m.u.a.p.'s is being done, the upper cutoff frequencies should be increased to 5 and 10 kHz, respectively. Figure 9.9 shows a typical EMG spectrum, and it can be seen that most of the signal is concentrated in the band between 20 and 200 Hz, with a lesser component extending up to 1000 Hz.

The spectra of other physiological and noise signals must also be considered. ECG signals contain power out to 100 Hz, so it may not be possible to eliminate such interference, especially

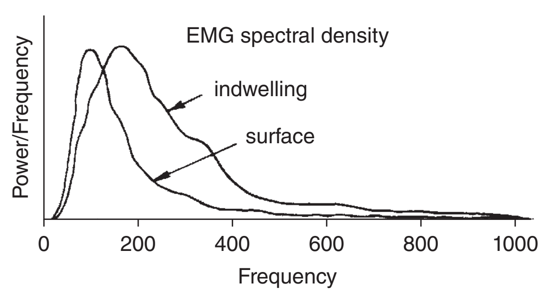


Figure 9.9 Frequency spectrum of EMG as recorded via surface and indwelling electrodes. The higher-frequency content of indwelling electrodes is the result of the closer spacing between electrodes and their closer proximity to the active muscle fibers.

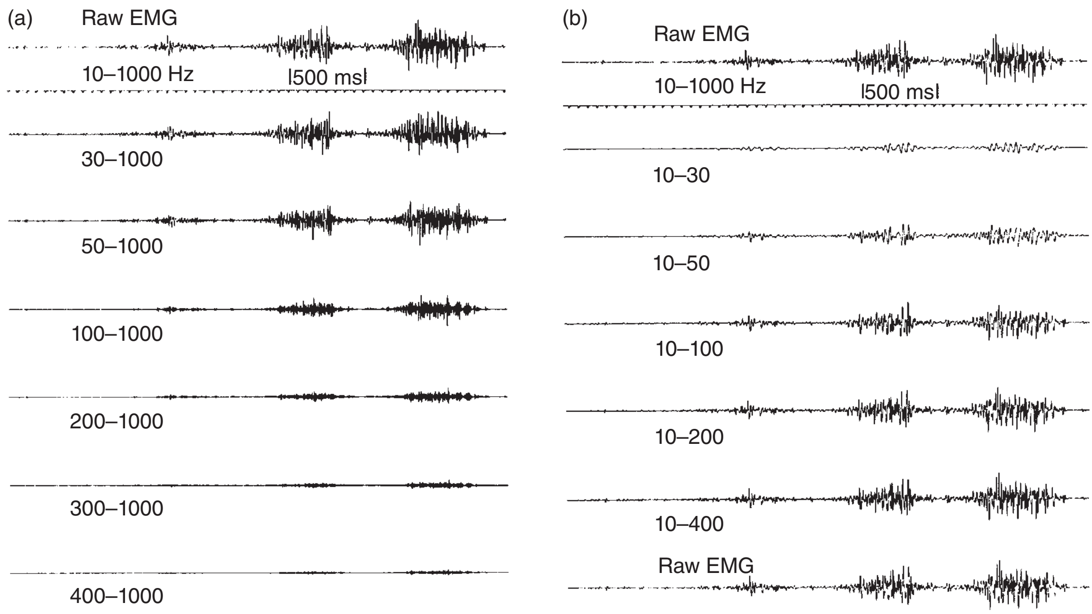


Figure 9.10 Surface EMG filtered with varying cutoff frequencies. (a) With lower cutoff frequency varied from 30 to 400 Hz, showing the effect of rejecting lower frequencies. (b) With upper cutoff frequency varied from 30 to 400 Hz, showing the loss of higher frequencies.

when monitoring muscle activity around the thoracic region. The major interference comes from power line hum: in North America it is 60 Hz, in Europe 50 Hz. A notch filter can be applied to remove that source of noise, however physiologic EMG at the frequency will also be removed. Movement artifacts, fortunately, lie in the 0–10 Hz range and normally should not cause problems. Unfortunately, some of the lower-quality cabling systems can generate large low-frequency artifacts that can seriously interfere with the baseline of the EMG recording. Usually, such artifacts can be eliminated by good low-frequency filtering, by setting f_1 to about 20 Hz. If this fails, the only solution is to replace the cabling or use an EMG system that includes microamplifiers at the skin surface.

It is valuable to see the EMG signal as it is filtered using a wide range of bandwidths. Figure 9.10 shows the results of such filtering and the obvious distortion of the signal when f_1 and f_2 are not set properly.

9.2.4 Common-Mode Rejection

The human body is a good conductor and, therefore, will act as an antenna to pick up any electromagnetic radiation that is present. The most common radiation comes from domestic power: power cords, fluorescent lighting, and electric machinery. The resulting interference may be so large as to prevent recording of the EMG. If we were to use an amplifier with a single-ended input, we would see the magnitude of this interference. Figure 9.11 depicts hum interference on the active electrode. It appears as a sinusoidal signal, and if the muscle is contracting, its EMG is added. However, hum could be 100 mV and would drown out even the largest EMG signal.

If we replace the single-ended amplifier with a differential amplifier (see Figure 9.12), we can possibly eliminate most of the hum. Such an amplifier takes the difference between the signals on the active terminals. As can be seen, this hum interference appears as an equal amplitude on both active terminals. Because the body acts as an antenna, all

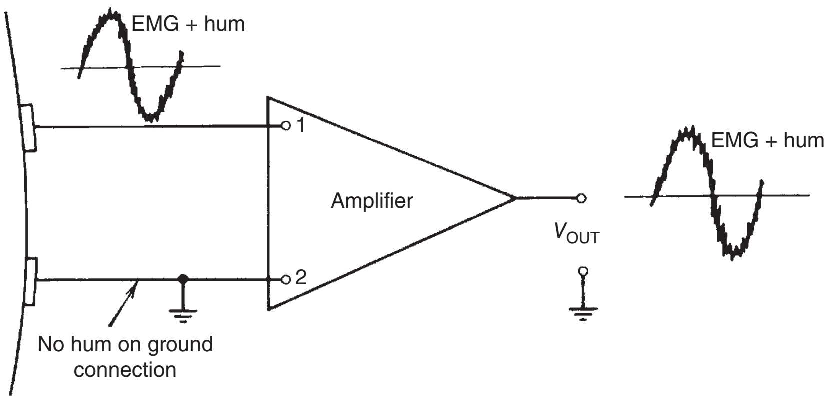


Figure 9.11 Single-ended amplifier showing lack of rejection of hum present on ungrounded active terminal.

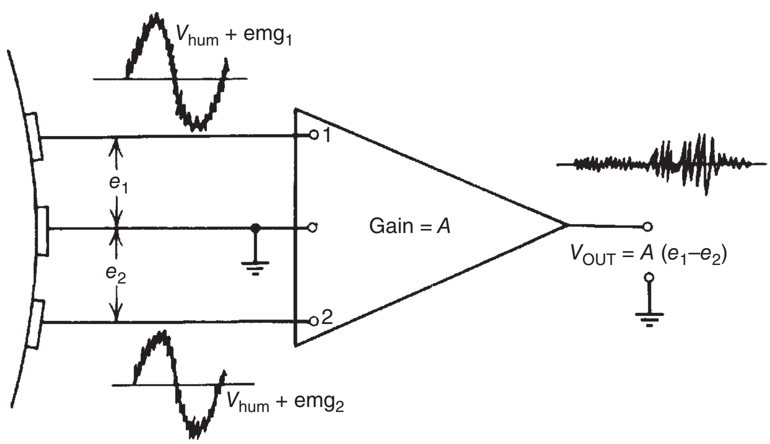


Figure 9.12 Biological amplifier showing how the differential amplifier rejects the common (hum) signal by subtracting the hum signal that is present at an equal amplitude at each active terminal. Different EMG signals are present at each electrode; thus, the subtraction does not result in a cancellation.

Because this unwanted signal is common to both active terminals, it is called a *common-mode signal*. At terminal 1, the net signal is $V_{\text{hum}} + \text{emg}_1$; at terminal 2 it is $V_{\text{hum}} + \text{emg}_2$. The amplifier has a gain of A. Therefore, the ideal output signal is:

$$\begin{aligned} e_o &= A(e_1 - e_2) \\ &= A(V_{\text{hum}} + \text{emg}_1 - V_{\text{hum}} - \text{emg}_2) \qquad (9.4) \\ &= A(\text{emg}_1 - \text{emg}_2) \end{aligned}$$

The output e_o is an amplified version of the difference between the EMGs on electrodes 1 and 2. No matter how much hum is present at the individual electrodes, it has been removed by a perfect subtraction within the differential amplifier. Unfortunately, a perfect subtraction never occurs, and the measure of how successfully this has been done is given by the common-mode rejection ratio (CMRR). If CMRR is 1000 : 1, then all but 1/1000 of the hum will be rejected. Thus, the hum at the output is given by:

$$V_o(\text{hum}) = \frac{A \times V_{\text{hum}}}{\text{CMRR}} \tag{9.5}$$

Example 9.1. Consider the EMG on the skin to be 2 mV in the presence of hum of 500 mV. CMRR is 10,000:1 and the gain is 2000. Calculate the output EMG and hum.

$$\begin{aligned} e_o &= A(\text{emg}_1 - \text{emg}_2) \\ &= 2000 \times 2\,\text{MV} = 4\ \ \text{V} \\ \text{output hum} &= 2000 \times 500\,\text{mV} \div 10{,}000 = 100\,\text{mV} \end{aligned}$$

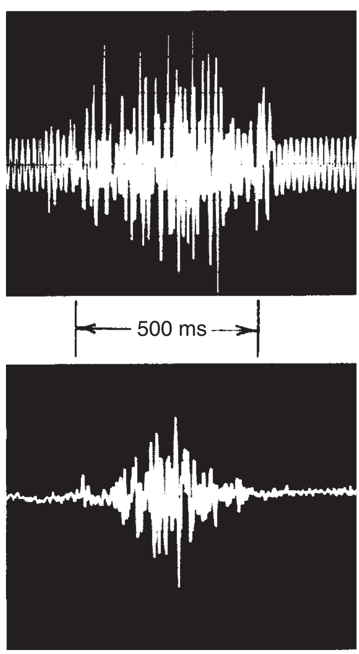
Hum is always present to some extent unless the EMG is being recorded with battery-powered equipment not in the presence of domestic power. Its magnitude can be seen in the baseline when no EMG is present. Figure 9.13 shows two EMG records, the first with hum quite evident, the latter with negligible hum.

(a) With hum present.
(b) Without hum.

CMRR is often expressed as a logarithmic ratio rather than a linear ratio. The units of this ratio are decibels,

$$\text{CMRR (dB)} = 20\log_{10}\ \text{CMRR(linear)} \tag{9.6}$$

(a)



(b)

Figure 9.13 Recordings of an EMG signal.

If CMRR = 10, 000 : 1, then CMRR (dB) = $20\log_{10}10,000$ = 80 dB. In good-quality biological amplifiers, CMRR should be 80 dB or higher.

9.2.5 Cross-Talk in Surface Electromyograms

The detectable pick-up distance was estimated to be about 0.5 cm for small motor units and about 1.5 cm for the largest units (Fuglevand et al., 1992). In Figure 9.14, for example, we see a number of surface electrodes over several thigh muscles in the midthigh region. The range of pick-up is shown as an arc beneath each electrode. Those m.u.a.p.'s whose fibers are close to each electrode are not subject to cross-talk; however, there is an overlapping zone of pick-up where both electrodes detect m.u.a.p.'s from the same active motor units. This common pick-up is called cross-talk.

Recommendations for the placement of surface electrodes have been established and published through the project "Surface ElectroMyoGraphy for the Non-Invasive Assessment of Muscles" (SENIAM). These sites were chosen to maximize signal amplitude from the desired muscle while reducing cross-talk from adjacent muscles. However, because of close proximity of the desired muscles in many research applications, it may be necessary to check that cross-talk is minimal. The most common way to test for cross-talk is to conduct a manual resistance test prior to any given experiment. Consider two adjacent muscles A and B that, during the course of a given movement, have common periods of activity. The purpose of the manual resistance tests is to get a contraction on muscle A without any activity on B and vice versa. Consider electrodes placed on the leg on three muscles that are fairly close together: TA, peroneus longus (PER), and

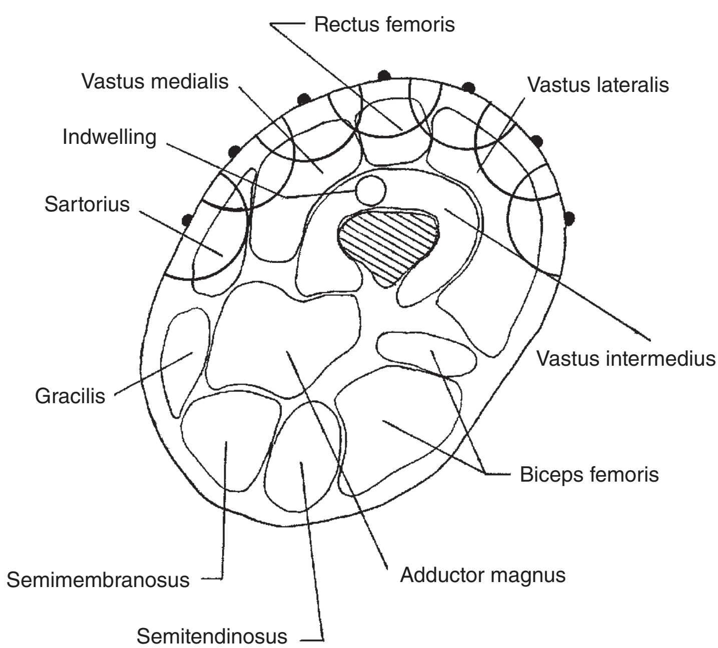


Figure 9.14 Location of seven surface electrodes placed across the midthigh showing the region of pick-up for each site relative to the underlying muscles. The zone of pick-up for each is shown as an arc beneath electrode; there is an overlapping zone between adjacent electrodes that will result in some cross-talk. Deeper muscles, such as the vastus intermedius and adductor magnus, require indwelling electrodes whose zone of pick-up is small because of the small surface area and close spacing of the electrode tips.

lateral gastrocnemius (LG). Functionally, the TA is a dorsiflexor and invertor, PER is an evertor and plantarflexor, and LG is a plantarflexor. The results of a functional test for cross-talk from the PER to the TA and LG is presented in Figure 9.15. Because it is difficult for many subjects to voluntarily create a unique contraction (such as eversion for the PER), it is very helpful for the subject to view the EMG recording and use visual feedback to assist in the test. Here, we see two very distinct PER contractions. The first was eversion with a small amount of dorsiflexion: note the minor activity on TA and negligible amount on LG. The second contraction was eversion with a minor amount of plantarflexion: note the minor activity on the LG and none on TA. It is not critical to be successful on both tests simultaneously, or even on the very first attempt. The student is referred to Winter et al. (1994) for more details and examples of manual resistance tests.

However, there are many situations when a manual resistance test cannot be done, and there is a good chance that adjacent channels may have some common EMG signal. The question is how much do they have in common? This question was initially addressed by Winter et al. (1994) using the well-recognized signal-processing technique called cross-correlation (see Section 2.1). The adjacent channels, $x(t)$ and $y(t)$, are isometrically contracted for about 10

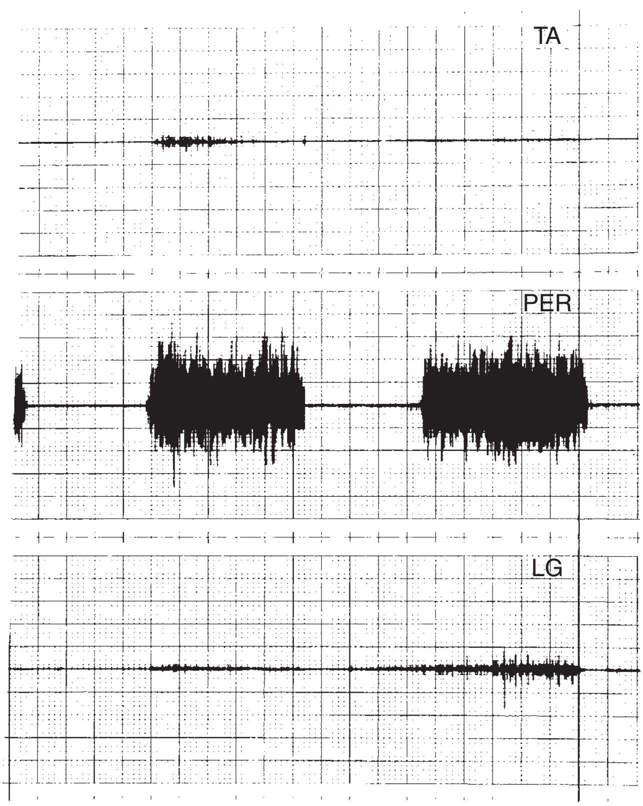


Figure 9.15 Results of a manual resistance test to ensure no cross-talk between the peroneii (PER) muscle and the tibialis anterior (TA) and lateral gastrocnemius (LG) muscles. The first trial showed negligible cross-talk between PER and TA; the second contraction showed negligible cross-talk between PER and LG. See text for further details. (Reprinted from the *Journal of Electromyography and Kinesiology*, Winter et al. (1994), with permission of Elsevier.)

seconds at a moderate contraction level. The general equation for cross-correlation, $R_{xy}(\tau)$, between signals $x(t)$ and $y(t)$ is:

$$R_{xy}(\tau) = \frac{\frac{1}{T}\int_0^T x(t) \cdot y(t-\tau)dt}{\sqrt{R_{xx}(0) \cdot R_{yy}(0)}} \tag{9.7}$$

where $x(t)$ and $y(t)$, respectively, are at a phase shift of 0. $R_{xx}(0)$ and $R_{yy}(0)$ are the mean squares of $x(t)$ and $y(t)$, respectively, over the time T. τ is normally varied ±30 ms, and a typical cross-correlation for two electrode pairs located 2 cm apart is presented in Figure 9.16. Note that the peak correlation is 0.6 at zero phase shift. Because the electrode pairs were placed parallel with the muscle fibers, each pair was the same distance from the motor point; thus, any common m.u.a.p.'s will be virtually in phase. With the peak $R_{xy} = 0.6$, the net correlation, $R_{xy}^2 = 0.36$, tells us the two EMG had 36% of their signals in common. R_{xy} drops off sharply as the distance between electrode pairs is increased (Winter et al., 1994). At 2.5 cm, $R_{xy} = 0.48$ (23% cross-talk); at 5.0 cm, $R_{xy} = 0.24$ (6% cross-talk); and at 7.5 cm, $R_{xy} = 0.14$ (2% cross-talk). Reducing the size of the electrodes reduces the pick-up zone of the electrode, and thus the cross-talk. The motor units at a greater distance from the electrodes contribute most to the cross-talk, and the m.u.a.p.'s from those units are longer duration (lower frequency). Thus, by processing the EMG through a differentiator, the higher-frequency (closer) m.u.a.p.'s are emphasized, while the lower-frequency (further) m.u.a.p.'s are attenuated.

The double differentiation technique can be used to either remove or identify if cross-talk is occurring during an EMG recording (DeLuca, 1997). The use of an electrode with 3 detection surfaces is required to perform this technique which differentiations the EMG signal between the 1 and 2 detection surfaces and the 2 and 3 detection surfaces to cancel out very distant signals likely coming from further away muscles. Although useful this approach is not commonly used in practice. Students should spend time perfecting electrode placement within the mid muscle belly to avoid cross-talk and accurate and reliable EMG signals. Ultrasound can be used during training to help with muscle identification and electrode placement.

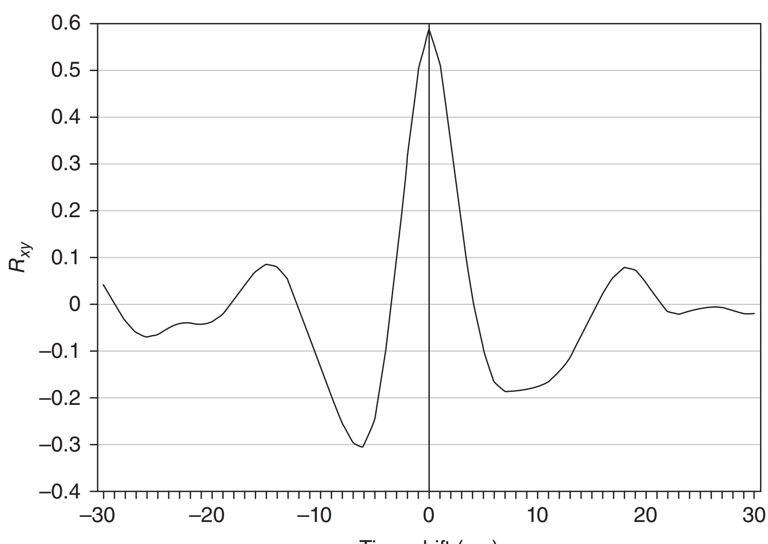


Figure 9.16 Results of a cross-correlation, *Rxy*, between two adjacent electrodes in Figure 9.13. Here, $R_{xy} = 0.6$, indicating a common pick-up of $R_{xy}^2 = 0.36$, or 36% cross-talk. (Reprinted from the *Journal of Electromyography and Kinesiology*, Winter et al. (1994), with permission of Elsevier.)

9.2.6 Recommendations for Surface Electromyogram Reporting and Electrode Placement Procedures

Over the past 30 years, the availability of EMG recording equipment and the number of laboratories using such equipment has exploded dramatically. Most labs have developed their own protocols regarding the details needed for reporting results and for the selection of the electrode sites over the muscles. The first attempt at developing recommended standards was done by the International Society of Electrophysiological Kinesiology, which resulted in an Ad Hoc Committee Report (Winter et al., 1980). More recently, a considerably more in-depth report was published with the support of the European Union and was called SENIAM (Surface EMG for a Non-Invasive Assessment of Muscles). During 1996–1999, the report was developed and reviewed by over 100 laboratories (Hermens et al., 2000), and the final report was published as a booklet and as a CD-ROM: SENIAM 8: European Recommendations for Surface Electromyography, 1999.

The SENIAM recommendations as to information to be reported are as follows:

1. Electrode size, shape (square, circular, etc.), and material (Ag, AgCl, Ag/AgCl, etc.)
2. Electrode type (monopolar, bipolar, one- or two-dimensional array)
3. Skin preparation, interelectrode distance
4. Position and orientation on the muscle (recommendations for 27 muscles are included)

The location of electrode placement can have a dramatic effect of the recorded EMG signal. Electrodes should be placed at the mid muscle belly (transverse and sagittal plane). When a muscle contracts the distal and proximal attachment will be brought to the middle of the muscle.

Therefore, by placing the electrodes at the mid muscle belly there will be minimal movement of the motor units being recorded under the electrodes. Recording a similar group of motor units throughout a range of motion will provide a valid amplitude and frequency profile, which can be used with confidence to identify neuromuscular characteristics of a given muscle. If an electrode is placed closer to either tendon end there will be a smaller pool of motor units represented in the EMG signal and that pool will change during the contraction of the muscle leading to inaccurate conclusions about the neuromuscular behavior of the muscle. If the electrode is placed medial or lateral to the sagittal plane of the muscle there is potential for cross-talk to occur as described previously. Another consideration for electrode placement is that the detection surfaces are oriented perpendicular to the direction of muscle fibers. If the detection surfaces are off angle to the muscle fibers the conduction velocity will be affected and the amplitude and frequency of the signal profile will be altered. For these reasons, it is important to have a strong knowledge of muscular anatomy prior to performing electrode placements. In addition, palpation techniques of surface anatomy should be practiced to enhance electrode placement accuracy.

9.3 PROCESSING OF THE ELECTROMYOGRAM

Once the EMG signal has been amplified, it can be processed for comparison or correlation with other physiological or biomechanical signals. Given the high frequency nature of the raw EMG and typical mean value of zero, additional processing makes it better suited for quantitative clinical and research applications.

The more common types of online processing are:

1. Half- or full-wave rectification (the latter also called absolute value)
2. Linear envelope detector (half- or full-wave rectifier

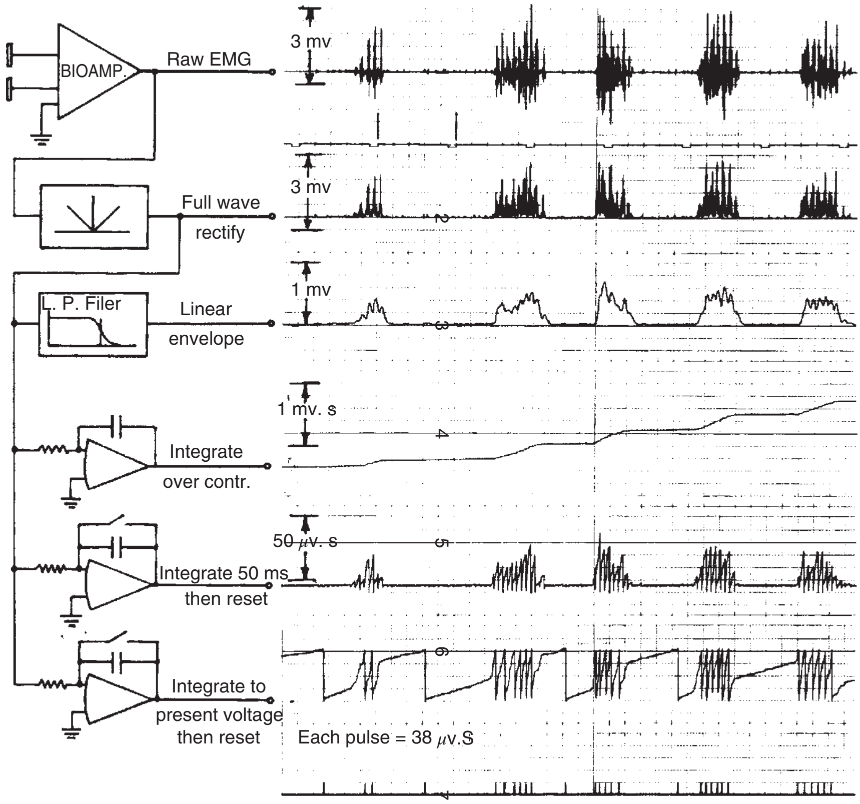


Figure 9.17 Schematic diagram of several common EMG processing systems and the results of simultaneous processing of EMG through these systems. See the text for details.

3. Integration of the full-wave rectified signal over the entire period of muscle contraction
4. Integration of the full-wave rectified signal for a fixed time, reset to zero, then integration cycle repeated
5. Integration of the full-wave rectified signal to a preset level, reset to zero, then integration repeated

These various processing methods are shown schematically in Figure 9.17, together with a sample record of a typical EMG processed in all five ways. This is now discussed in detail.

9.3.1 Full-Wave Rectification

The full-wave rectifier generates the absolute value of the EMG, usually with a positive polarity. The original raw EMG has a mean value of zero because it is recorded with a biological amplifier with a low-frequency cutoff around 10 Hz. However, the full-wave rectified signal does not cross through zero and, therefore, has an average or bias level that fluctuates with the strength of the muscle contraction. The quantitative use of the full-wave rectified signal by itself is somewhat limited; it serves as an input to the other processing schemes. The main application of the full-wave rectified signal is in semi-quantitative assessments of the phasic activity of various muscle groups. A visual examination of the amplitude changes of the full-wave rectified signal gives a good indication of the changing contraction level of the muscle. The proper unit for the amplitude of the rectified signal is the millivolt, the same as for the original EMG.

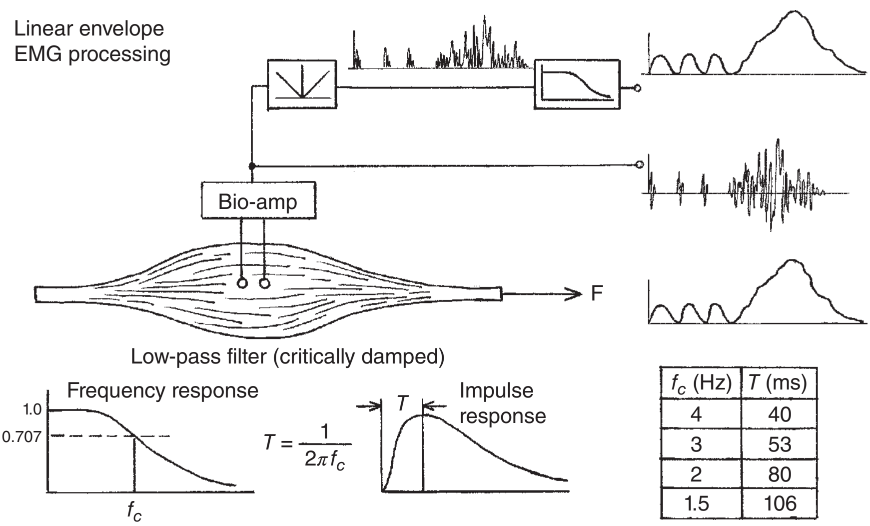
9.3.2 Linear Envelope

If we filter the full-wave rectified signal with a low-pass filter, we have what is called the *linear envelope*. It can be described best as a moving average because it follows the trend of the EMG and quite closely resembles the shape of the tension curve. It is reported in millivolts. There is considerable confusion concerning the proper name for this signal. Many researchers call it an *integrated EMG* (IEMG). Such a term is quite wrong because it can be confused with the mathematical term *integrated*, which is a different form of processing.

There is a need to process the signal to provide the assessor with a pattern that can be justified on some biophysical basis. Some researchers have full-wave rectified the raw EMG and low-pass filtered at a high frequency but with no physiological basis (Forssberg, 1985; Murray et al., 1985). If it is desired that the linear envelope bear some relationship to the muscle force or the joint moment of force, the processing should model the biomechanics of muscle tension generation. The basic unit of muscle tension is the muscle twitch, and the summation of muscle twitches as a result of recruitment is matched by a superposition of m.u.a.p.'s. There is an inherent delay between the m.u.a.p. and the resultant twitch waveform. If we consider the full-wave rectified signal to be an impulse and the twitch to be the response, we can define the transfer function of the desired processing. The duration of the full-wave rectified m.u.a.p. is about 10 ms, while the twitch waveform peaks at 50–110 ms and lasts up to 300 ms, so this impulse/response relationship is close. Twitch waveforms have been analyzed and have been found to be a second-order system, critically damped or slightly overdamped (Crosby, 1978; Milner-Brown et al., 1973). The cutoff frequency of these second-order responses ranged from 2.3 to 7.8 Hz. Figure 9.18 is presented to show the linear envelope processing of the EMG to model the relationship between the m.u.a.p. and the twitch. A critically damped second-order low-pass filter has a cutoff frequency f_c, which is related to the twitch time T as follows:

$$f_c = 1/2\pi T \tag{9.8}$$

Thus, the table in Figure 9.18 shows the relationship between f_c and T for the range of twitch times reported in the literature. The soleus muscle with a twitch time ≈106 ms would require a



f_c (Hz)	T (ms)
4	40
3	53
2	80
1.5	106

Figure 9.18 Linear envelope processing of the EMG with a critically damped low-pass filter to match the impulse response of the muscle being recorded. The full-wave rectified EMG acts as a series of impulses that, when filtered, mimic the twitch response of the muscle and, in a graded contraction, mimic the superposition of muscle twitches.

filter with $f_c = 1.5$ Hz. The reader is referred back to Section 8.0.5 for information on the twitch shape and times.

Good correlations have been reported between the muscle force and linear envelope waveforms during isometric anisotonic contractions (Calvert and Chapman, 1977; Crosby, 1978). Muscle contraction can also be modeled as a mass-spring-damper system that was critically damped. The critically damped low-pass filters described earlier have exactly the same response as this previously mechanical damper system. Thus, the matching of the muscle force waveform with that from the model is exactly the same as would be achieved with a linear envelope detector with the same cutoff frequency as used in the mechanical model.

9.3.3 True Mathematical Integrators

There are several different forms of mathematical integrators, as shown in Figure 9.17. The purpose of an integrator is to measure the "area under the curve." Thus, the integration of the full-wave rectified signal will always increase as long as any EMG activity is present. The simplest form starts its integration at some preset time and continues during the time of the muscle activity. At the desired time, which could be a single contraction or a series of contractions, the integrated value can be recorded. The unit of a properly integrated signal is millivolt-seconds (mV s). The only true way to find the average EMG during a given contraction is to divide the integrated value by the time of the contraction; this will yield a value in millivolts.

A second form of integrator involves a resetting of the integrated signal to zero at regular intervals of time (40–200 ms). Such a scheme yields a series of peaks that represent the trend of the EMG amplitude with time. Each peak represents the average EMG over the previous time interval, and the series of peaks is a "moving" average. Each peak has units of millivolt-seconds so that the sum of all the peaks during a given contraction yields the total integrated signal, as described in the previous paragraph. There is a close similarity between this reset, integrated curve, and the linear envelope. Both follow the trend of muscle activity. If the reset time is too high, it will not be able to follow rapid fluctuations of EMG activity, and if it is reset too frequently, noise will be present in the trendline. If the integrated peaks are divided by the integration time, the amplitude of the signal can again be reported in millivolts.

A third common form of integration uses a voltage level reset. The integration begins before the contraction. If the muscle activity is high, the integrator will rapidly charge up to the reset level; if the activity is low, it will take longer to reach rest. Thus, the strength of the muscle contraction is measured by the frequency of the resets. A high frequency of reset pulses reflects high muscle activity; a low frequency represents low muscle activity. Intuitively, such a relationship is attractive to neurophysiologists because it has a similarity to the action potential rate present in the neural system. The total number of counts over a given time is proportional to the EMG activity. Thus, if the threshold voltage level and the gain of the integrater are known, the total EMG activity (millivolt-seconds) can be determined.

9.4 RELATIONSHIP BETWEEN ELECTROMYOGRAM AND BIOMECHANICAL VARIABLES

The major reason for processing the basic EMG is to derive a relationship between it and some measure of muscle function. A question that has been posed for years is: "How valuable is the EMG in predicting muscle tension?" Such a relationship is very attractive because it would give an inexpensive and noninvasive way of monitoring muscle tension. Also, there may be information in the EMG concerning muscle metabolism, power, fatigue state, or contractile elements recruited.

9.4.1 Electromyogram Versus Isometric Tension

Bouisset (1973) has a seminal review of the state of knowledge regarding the EMG and muscle tension in normal isometric contractions. The EMG processed through a linear envelope detector has been widely used to compare the EMG-tension relationship, especially if the tension is changing with time. If constant tension experiments are done, it is sufficient to calculate the average of the full-wave rectified signal, which is the same as that derived from a long time-constant linear envelope circuit. Both linear and nonlinear relationships between EMG amplitude and tension have been discovered. Typical of the work reporting linear relationships is an early study by Lippold (1952) on the calf muscles of humans. Zuniga and Simons (1969) and Vredenbregt and Rau (1973), on the other hand, found quite nonlinear relationships between tension and EMG in the elbow flexors over a wide range of joint angles. Both these studies were, in effect, static calibrations of the muscle under certain length conditions; a reproduction of their results is presented in Figure 9.19.

Another way of representing the level of EMG activity is to count the action potentials over a given period of time. Close et al. (1960) showed a linear relationship between the count rate and the IEMG, so it was not surprising that the count rate increased with muscle tension in an almost linear fashion.

The relationship between force and linear envelope EMG also holds during dynamic changes of tension. Inman et al. (1952) first demonstrated this by a series of force transducer signals that were closely matched by the envelope EMG from the muscle generating the isometric force. Gottlieb and Agarwal (1971) mathematically modeled this relationship with a second-order low-pass system (e.g. a low-pass filter). Under dynamic contraction conditions, the tension is seen to lag behind the EMG signal, as shown in Figure 9.20. The delay is caused by the fact that the twitch corresponding to each m.u.a.p. reaches its peak 40–100 ms afterward. Thus, as each motor unit is recruited, the resulting summation of twitch forces will also have a similar delay behind the EMG.

(a) During a gradual build-up and rapid relaxation.
(b) During a short 400-ms contraction.

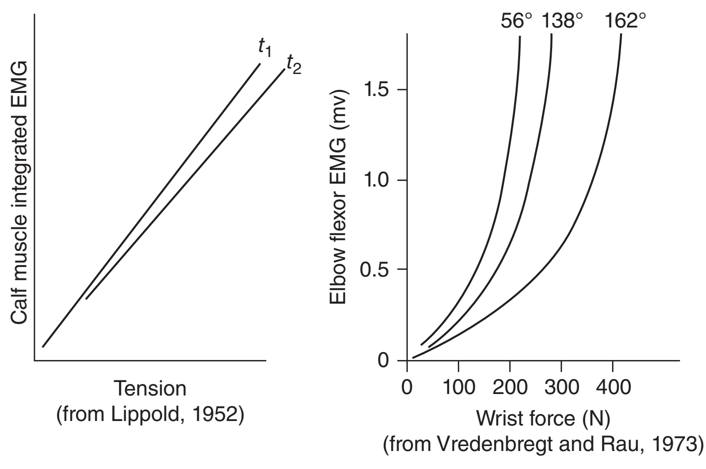


Figure 9.19 Relationship between the average amplitude of EMG and the tension in the muscle in isometric contraction. Linear relationships have been shown by some researchers, while others report the EMG amplitude to increase more rapidly than the tension.

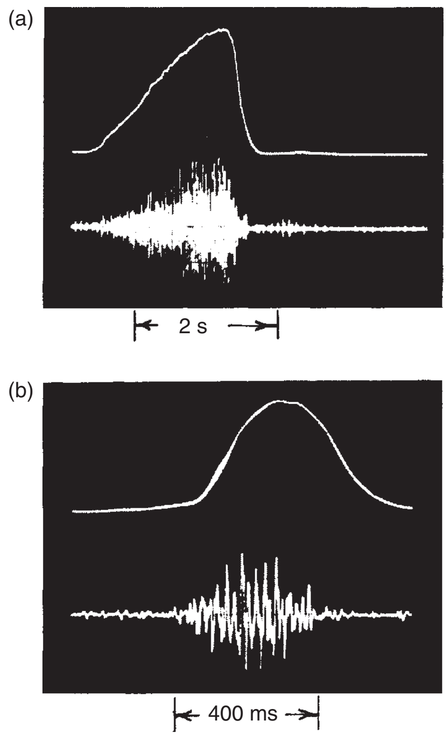


Figure 9.20 EMG and muscle tension recordings during varying isometric contractions of the biceps muscles. Note the delay between the EMG and the initial build-up of tension, time to reach maximum tension, and drop of tension after the EMG has ceased.

In spite of the reasonably reproducible relationships, the question still arises as to how valid these relationships are for dynamic conditions when many muscles act across the same joint:

1. How does the relationship change with length? Does the length change merely alter the mechanical advantage of the muscle, or does the changing overlap of the muscle fibers affect the EMG itself (Vredenbregt and Rau, 1973)?
2. How do other agonist muscles share the load at that joint, especially if some of the muscles have more than one function (Vredenbregt and Rau, 1973)?
3. In many movements, there is antagonist activity. How much does this alter the force being predicted by creating an extra unknown force?

With the present state of knowledge, it appears that a suitably calibrated linear envelope EMG can be used as a coarse predictor of muscle tension for muscles whose length is not changing rapidly.

9.4.2 Electromyogram During Muscle Shortening and Lengthening

In order for a muscle to do positive or negative work, it must also undergo length changes while it is creating tension. Thus, it is important to see how well the EMG can predict tension under these more realistic conditions. A major study has been reported by Komi (1973). In it, the subject did both positive and negative work on an isokinetic muscle-testing machine. The subject was asked to generate maximum tension while the muscle lengthened or shortened at controlled velocities. The basic finding was that the EMG amplitude remained tension during shortening and increased tension during eccentric contractions (see Section 8.2.1). Such results support the theory that the EMG amplitude indicates the state of activation of the contractile element, which is quite different from the tension recorded at the tendon. Also, these results combined with later results (Komi et al., 1987) indicate that the EMG amplitude associated with negative work is considerably less than that associated with the same amount of positive work. Thus, if the EMG amplitude is a relative measure of muscle metabolism, such a finding supports the experiments that found negative work to have somewhat less metabolic cost than positive work.

9.4.3 Electromyogram Changes During Fatigue

Muscle fatigue occurs when the muscle tissue cannot supply the metabolism at the contractile element, because of either ischemia (insufficient oxygen) or local depletion of any of the metabolic substrates. Mechanically, fatigue manifests itself by decreased tension, assuming that the muscle activation remains constant, as indicated by a constant surface EMG or stimulation rate. Conversely, the maintenance of a constant tension after onset of fatigue requires increased motor unit recruitment of new motor units to compensate for a decreased firing rate of the already recruited units (Vredenbregt and Rau, 1973). Such findings also indicate that all or some of the motor units are decreasing their peak twitch tensions but are also increasing their contraction times. The net result of these changes is a decrease in tension.

Fatigue not only reduces the muscle force but also may alter the shape of the motor action potentials. It is not possible to see the changes of shape of the individual m.u.a.p.'s in a heavy voluntary contraction. However, an autocorrelation shows an increase in the average duration of the recruited m.u.a.p. (see Section 9.1.4). Also, the EMG spectrum shifts to reflect these duration changes; Kadefors et al. (1973) found that the higher-frequency components decreased. The net result is a decrease in the EMG frequency spectrum, which has been attributed to the following:

1. Lower conduction velocity of the action potentials along the muscle fibers below the non-fatigued velocity of 4.5 m/s (Mortimer et al., 1970; Krogh-Lund and Jørgensen, 1991).
2. Some of the larger and faster motor units with shorter duration m.u.a.p.'s dropping out.
3. A tendency for the motor units to fire synchronously, which increases the amplitude of the EMG. Normally, each motor unit fires independently of others in the same muscle so that the EMG can be considered to be the summation of a number of randomly timed m.u.a.p.'s. However, during fatigue, a tremor (8–10 Hz) is evident in the EMG and force records. These fluctuations are neurological in origin and are caused by motor units firing in synchronized bursts.

The major EMG measures of fatigue are an increase in the amplitude of the EMG and a decrease in its frequency spectrum. One of those measures is median power frequency, f_m, which is the frequency of the power spectral density function below which half the power lies and above which the other half of the power lies:

$$\int_0^{f_m} X^2(f)df = \int_{f_m}^{\infty} X^2(f)df = \frac{1}{2}\int_0^{\infty} X^2(f)df \tag{9.9}$$

where $X(f)$ is the amplitude of the harmonic at frequency f, and $X^2(f)$ is the power at frequency f.

A representative paper that shows the majority of these measures (Krogh-Lund and Jørgensen, 1991) reports a 10% decrease in conduction velocity, a 45% decrease in the median power frequency, and a 250% increase in the root mean square (rms) amplitude. The median power frequency has also been shown to be highly correlated with bandwidth (15–45 Hz) and the ratio of the high-frequency bandwidth (>95 Hz) to the low-frequency bandwidth (Allison and Fujiwara, 2002). An alternate and very common statistical measure (Öberg et al., 1994) is MPF:

$$\text{MPF} = \frac{\int_0^F f \cdot X^2(f)df}{\int_0^F X^2(f)df} \text{ Hz} \tag{9.10}$$

where F is the maximum frequency analyzed.

There is virtually no difference in these two frequency measures and their correlations with independent measures of fatigue (Kerr and Callaghan, 1999).

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10

MODELING OF HUMAN MOVEMENT

Brian A. Knarr[1], Todd J. Leutzinger[1], and Namwoong Kim[2]

[1]Department of Biomechanics, College of Education, Health, and Human Sciences, University of Nebraska at Omaha, Omaha, NE, USA

[2]Division of Health & Kinesiology, Incheon National University, Incheon, Republic of Korea

10.0 INTRODUCTION

The analysis of human movement has been expanding and improving for several decades. Previously, joint kinetics during human movement have been analyzed through inverse dynamics solutions. Detailed in Chapters 5 and 7, inverse dynamics solutions utilize the body's measured kinematics and external forces applied to the body (i.e. ground reaction forces) to estimate internal joint reaction forces and joint moments. A link-segment model uses previously calculated joint reaction forces and moments to predict the forces that caused a given movement. However, inverse dynamics is the opposite of how our bodies actually move in real life. The true sequence that occurs in our bodies begins with the central nervous system (CNS) utilizing a top-down approach sending signals to the muscles through the peripheral nervous system. If the signal to the muscles is strong enough, this results in the recruitment of agonist and antagonist muscles. The net effect of muscle forces results in a moment acting about that joint. This moment leads to the acceleration of adjacent body segments and ultimately causes a person to move.

Computer modeling that implements the top-down approach utilized by our bodies is referred to as a forward solution. Unfortunately, the necessary information and constraints required to perform a forward solution are considerably greater than performing an inverse dynamics solution. For example, if we wish to calculate the ankle and knee moments on one limb utilizing the inverse dynamics approach, we only need data from the adjacent segments involved in the joint moment calculation (in this case, the foot and the shank of the respective leg). On the other hand, to calculate a forward solution, we must model all parts of the body including the thighs, hips, and trunk. This is a direct result of the interlimb coupling of forces that occurs in the body. Any given joint moment acts directly on the two adjacent segments that make up the joint. A sort of ripple effect occurs and these two segments create reaction forces on segments further away from the original joint. In human gait, the plantarflexor muscles create a plantarflexion moment about the ankle. Based on the magnitude of the plantarflexion moment, the ankle will plantarflex at a

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

particular speed and with a certain amount of torque. Due to the resulting accelerations experienced, the horizontal and vertical reaction forces at the knee will be altered. This in turn has a direct effect on the acceleration of the thigh which will alter the reaction forces at the hip, contralateral hip, and trunk. In this way, the accelerations of multiple body segments are affected by the plantar flexor moment.

Not only must we model each segment of the body during a forward solution, but it is crucial that our model be anatomically correct. If we have errors in our anatomical model, even if it is not directly related to the joint in question, we will have errors in our model predictions. For example, if a body segment has a disproportionately inaccurate mass, a joint has a missing constraint, or the constraints placed on a joint are unrealistic, the entire link system will start to generate displacement errors and these errors will continue to multiply over time at increasing rates. Therefore, despite the fact forward solutions utilize a top-down approach similar to how our bodies interact with the world, accurate anatomical information is crucial when running forward simulations given they present more opportunities for errors to be introduced.

10.1 REVIEW OF FORWARD SOLUTION MODELS

Human locomotion and movement mechanics have been the most intriguing topics for researchers performing modeling simulations. Unfortunately, due to the complexities of both human movement and the link-segment model required to perform these simulations, biomechanical simplifications have been made and many of the simulations were confined to short periods of the movement. The first true forward simulation model is known as the inverted pendulum model (Cavagna et al., 1963). This model used a simple inverted pendulum illustration to describe the movement of the body's center of mass throughout the gait cycle. In 1980, McMahon and Mochon made further additions to the inverted pendulum model that utilized the law of conservation of energy (Mochon and McMahon, 1980). The model used the assumptions that the body exchanges kinetic energy for potential energy based on stance phase and swing phase and the vertical displacement of the body's center of mass. This model is known as the ballistic model. Hemami proposed a three-segment three-dimensional model with rigid legs and no feet (Hemami, 1980), and Pandy and Berme simulated single support only, using a five-segment planar model with no feet (Pandy and Berme, 1988). Unfortunately, such serious simplifications could not produce valid answers. Even with more complete models, many researchers constrained parts of their models kinematically by assuming sinusoidal trajectories of the trunk or pelvic segments (Beckett and Chang, 1968; Chao and Rim, 1973; Townsend, 1981). Such constraints violate one of the major requirements of a true simulation. Onyshko and Winter modeled the body as a seven-segment system (two feet, two legs, two thighs, and a head, arm, and trunk (HAT) segment), but the model did not satisfy the requirements of internal validity because of certain anatomical constraints (sagittal plane movement only at all three joints plus a rigid foot segment) (Onyshko and Winter, 1980). The Onyshko and Winter model was eventually made to walk but only by altering the moment patterns. Such results should alert researchers that two wrongs can make a right, but this does not necessarily make it correct. An incomplete model will result in a valid movement only if faulty motor patterns are used.

In terms of modeling simulations, simple motions have been modeled and deemed reasonably successfully. Phillips et al. modeled the swinging action of a human limb using the accelerations and moments of the hip during the swing phase of running (Phillips et al., 1983). Hemami et al. modeled the sway of the body in the coronal plane with locked out knees using adductor/abductor actuators at the hip and ankle as input to define the stability of the total system (Hemami et al., 1982).

More recent modeling of three-dimensional (3D) gait has been somewhat more successful. One of the major problem points in previous attempts was the modeling of initial foot contact with the ground. As a solution, springs were used to demonstrated by the bottom of the human foot and shoe. Unfortunately, the use of these springs resulted in extremely large accelerations of the foot segment and similarly large spikes in the ground reaction forces. A viscoelastic model of the foot with an array of parallel springs and dampers under the rigid foot segment was used to address this issue (Gilchrist and Winter, 1996).

In 1997, a 3D nine-segment model of walking that used ADAMS software (discussed in Section 8.2) demonstrated some success using the inverse dynamic joint moments as inputs (Gilchrist and Winter, 1997). Nonlinear springs at the knee, ankle, and metatarsal phalangeal joints constrained those joints to their anatomical range. Linear springs were used at the hips, and dampers were used at all joints to ensure a smooth motion. However, it became apparent that any small error in the joint moments after about 500 ms resulted in accumulating kinematic errors that ultimately became too large. This resulted in the model either becoming unbalanced or ultimately collapsing. The build-up of these errors is an inherent characteristic of any forward solution. The double integration of the segment accelerations caused by the input and reaction moments and forces result in displacement errors that increase over time and can only be corrected by continuous fine-tuning of the input joint moments. Thus, we are forced to violate some of the constraints of forward solution modeling as listed in Section 10.1.1.

10.1.1 Assumptions and Constraints of Forward Solution Models

1. The link-segment model has the same assumptions as those presented in Section 5.0.1 for the inverse dynamic solutions.
2. No kinematic constraints should be applied to the model whatsoever. The model must be permitted to fall over, jump, or collapse as dictated by the motor inputs.
3. The initial conditions must include the position and velocity of every model segment.
4. The only inputs to the model are externally applied forces and internally generated muscle forces or moments.
5. The model must incorporate all important degrees of freedom and constraints.
6. External reaction forces must be calculated. For example, the ground reaction forces would be equal to the algebraic summation of the mass-acceleration products of all segments when the feet are on the ground. Partitioning of the reaction forces when two or more points of the body are in contact with external objects is a separate and possibly major problem.

10.1.2 Potential of Forward Solution Simulations

The sky is the limit in regards to the potential for groundbreaking research and discovery when utilizing simulations. Unfortunately, due to the severe constraints associated with forward dynamics, the full potential of using simulations has not been realized. However, before we can use simulations for research, we must first make sure the link-segment representation of our model is anatomically correct and valid as indicated in Section 10.1.1. Researchers have been working on testing the internal validity of forward simulation models, but more work is still needed to ensure the validity of simulation results (Hicks et al., 2015; Fregly et al., 2012; Taylor et al., 2017).

Traditionally, testing a model's internal validity requires an inverse dynamics solution to calculate the moments at each joint in each plane of movement (sagittal, frontal, and coronal). Then, by using these motor patterns as inputs along with measured initial conditions, the forward solution should reproduce the originally measured kinematics from the inverse dynamics solution. If the model does not pass this test, it is counterproductive to use the model to answer any functionally related questions. All that will result will be an erroneous set of motor patterns overlaid on an erroneous biomechanical model. Fortunately, recent advancements in the field of instrumented prostheses have greatly progressed the validation process for musculoskeletal models and will be highlighted in later sections of this chapter.

10.2 MUSCLE-ACTUATED SIMULATION OF MOVEMENT

The human body is an extremely complex dynamic system composed of ligaments, tendons, bones, muscles, and other soft tissues that interact with the world around it. All these components within the body work together to generate forces and create movement. Muscle-actuated simulations provide the opportunity to study all these factors together and allow for a better understanding of human movement.

There are two main outcomes of musculoskeletal simulations that aid in our understanding of human movement: The first is the estimation of internal joint forces during dynamic movements. Previously, internal joint forces were only able to be measured through in vivo measurements. Additionally, if surrogate measures were used, such as an inverse dynamics approach, muscle forces were left unaccounted for. Through the use of musculoskeletal simulations, we can now estimate forces occurring inside the joint while adequately accounting for the muscle forces that contribute to this load. The second main outcome is the capability of performing "what if" studies to establish cause–effect relationships for phenomena that occur in the body. Traditional methods used to analyze human movement, such as motion capture, have been limited mainly to joint angles, moments, and powers. However, with musculoskeletal simulation, we can reasonably estimate the forces occurring inside a joint (Knarr and Higginson, 2015), determine the effects of surgery prior to the procedure (Homayouni et al., 2015), analyze landing mechanics on surfaces deemed unsafe for human subjects (DeMers et al., 2017), investigate the effectiveness of assistive devices (Rosenberg and Steele, 2017), and so much more.

In this chapter, we will discuss the key components of musculoskeletal simulations to better understand how this tool can be used to answer a variety of different research questions.

10.2.1 Musculoskeletal Modeling

Developing a musculoskeletal model is the first step needed to investigate the underlying mechanisms responsible for movement disorders that would otherwise be impossible due to the complexity of the human body. The basic component for building a musculoskeletal model is the geometry of the musculoskeletal system. Musculoskeletal geometry includes skeletal geometry, joint structure, structure of passive objects, muscle attachment sites, and muscle moment arms.

10.2.1.1 Skeletal Geometry. Accurate skeletal geometry is essential in creating a valid and reliable musculoskeletal model. Inaccurate skeletal geometry can lead to the false representation of muscle origins, insertion points, and muscle lines of action, which ultimately will lead to inaccurate estimations of muscle forces. Individual subject-specific skeletal geometry is often neglected due to the considerable investment of time and effort that is required.

The initial steps in building the geometry of a musculoskeletal model begin with the construction of individual rigid bodies that make up the skeletal system of the model. Synonymous to the human body, one rigid body (parent body) is connected to and able to manipulate another rigid body (child body) through the connection at a joint. A joint is an articulation between two rigid bodies in the model. In a model, a joint can allow anywhere from zero degrees of freedom to six degrees of freedom. Unfortunately, there is no perfect joint definition for a model that describes the human joint. For example, the knee joint is often defined as a hinge joint with one degree of freedom permitted in the sagittal plane. However, in the human body, the knee joint is able to translate anteriorly/posteriorly, abduct/adduct, and internally/externally rotate as well. Because these other movements are not as prevalent as flexion/extension in the knee joint, they are often omitted. The level of complexity of a joint in a model largely varies depending on the model and the research question being asked. If a model is too complex it becomes difficult to validate and computationally expensive to

10.2.1.2 Passive Structures (Ligament, Cartilage, Fascia, etc.). Skeletal muscles are the primary actuators, driving forces in a model that act as muscles, that generate joint motion in the body. However, passive structures such as ligament, cartilage, and fascia have mechanical properties that influence muscle force generation and can be influential toward the forces and moments occurring at a joint during dynamic movements. For example, when running, the plantarflexor muscles are the primary drivers that allow an individual to move their legs and produce power. However, elastic energy stored and released in the Achilles tendon during certain phases of the gait cycle plays a significant role in the ability to run. It is believed a major reason why kangaroos are able to run so fast is because of their Achilles tendon (Morgan et al., 1978). Additionally, tendon slack length (resting tendon length) is an important parameter included in the musculoskeletal model. Because the tendon acts as nonlinear spring, tendon slack length substantially affects muscle force production both in timing and magnitude.

10.2.1.3 Muscle Moment Arms. A muscle generating force vector at a perpendicular distance from a joint center of rotation creates a moment about that joint.

$$T = r \times F \tag{10.1}$$

In Equation 10.1, the moment about a joint is the result of the vector cross product between the force of the muscle crossing that joint (F) and the muscle moment arm (r). The muscle moment arm is the perpendicular distance between the joint center of rotation and the muscle force vector. The muscle moment arm carries just as much weight in the calculation of joint moments as the muscle force itself. Therefore, it is important to measure the moment arm about a joint accurately. The use of magnetic resonance imaging (MRI) imaging is helpful in measuring this perpendicular distance between the muscle force vector and the joint center of rotation, but it is important to note this measurement cannot just be taken once. The moment arm must be the perpendicular distance which means it changes as the joint angle changes. Therefore, to accurately calculate the moment about a joint, one needs to accurately measure the length of the moment arm at each point of the joint's range of motion. The use of an MRI machine to perform this task would be extremely tedious and expensive which begs the question "is there a better way?".

Another way to define the moment arm is the tendon excursion definition. This definition allows for the computational discovery of a moment arm with respect to the joint angle without having to measure it "by hand" with an MRI. By simultaneously measuring the position of a muscle-tendon unit and the joint angle, one can then compute the derivative of the muscle-tendon unit length with respect to joint angle. This results in the ability to calculate the muscle moment arm as a function of joint angle throughout the range of motion of the joint (An, 2007). A much simpler way than taking MRIs. Using this definition, the moment arm of a muscle about a joint is a function of both the joint angle and muscle-tendon length. Both of which can be affected by the origin and insertion of the muscle, i.e. the muscle attachment sites making this an important factor to accurately represent within the model.

10.2.1.4 Muscle Attachment Sites. As discussed previously, the moment created about a specific joint is dependent on the moment arm. That being said, it is important to accurately place the origin and insertion point for each muscle-tendon unit when building a musculoskeletal model. In a musculoskeletal model, muscles can be attached to bone using a variety of different methods depending on the type of muscle, its location on the body, and its overall range of motion. The first and simplest form of creating muscle attachments is through the use of fixed points. When using fixed points, the muscle path is set in a straight line and is defined by a series of attachment points that are fixed to the body. The points where the muscle attaches to the bone are called fixed points. The second form of muscle attachment used in musculoskeletal modeling is called via points. The use of via points is similar to using fixed are fixed to the body. However, they differ from fixed points because they are only fixed at the origin and insertion points of the muscle. This allows via points to be used in simple cases of wrapping when a joint moves to a certain angle and can assist with accurately portraying the muscle's line of action. For example, via points can be used on the ulna when the elbow flexes beyond a specific angle. Wrap points are attachment points that are automatically calculated by OpenSim to traverse over the surface of wrap objects. This is done to constrain the path of the muscle and guide it to avoid interference with other geometric structures in the model. Lastly, moving muscle points are used when the muscle paths are supposed to move with the joint and occur when the XYZ offsets in a body act as functions of the joint motion rather than a constant as seen with fixed points.

10.2.1.5 Muscle-Tendon Parameters. Skeletal muscles have certain mechanical parameters that are common across all muscles. Specifically, in musculoskeletal modeling, there are four important parameters that are included in the Hill-type muscle model: optimal fiber length, pennation angle at optimal fiber length, maximum isometric force, and maximum contraction velocity. The Hill-type muscle model has been widely used and is one of the common and basic muscle models. Optimal fiber length is the sarcomere length that can generate maximum isometric force within the muscle. Optimal muscle fiber length can be determined using the following equation assuming that all sarcomeres have the same length in a muscle fiber.

$$l_O^M = n * l_O^S \tag{10.2}$$

In Equation (10.2), l_O^M is the optimal fiber length, l_O^S is the optimal sarcomere length, and n is the number of sarcomeres in series. Optimal muscle fiber length affects the force–length and force–velocity curve, which in turn affects muscle fiber force production.

The angle of the fibers within a muscle also plays a role in the force production of a muscle. This is known as the pennation angle and is the angle at which the muscle fibers are arranged within the muscle (see Chapter 8 for more detail on pennation angles of muscle fibers). The way a muscle contracts, the number of fibers in a muscle, and the amount of force a muscle can produce are all affected by its pennation angle.

An isometric contraction occurs when the muscle generates force but the joint angle does not change and the length of the muscle remains the same. A muscle's maximum isometric force is the maximum amount of force generated from a muscle when the muscle fibers are at their optimal length and do not change. This maximum isometric force is a critical measurement in terms of measuring a muscle's ultimate force production. Unfortunately, maximum isometric force production for each individual muscle is difficult to obtain because it is hard to truly isolate a single muscle due to the body's redundancy. Therefore, physiological cross-sectional area (PCSA) is a viable alternative that is often used to estimate a muscle's maximum isometric force. It should be noted however, a muscle's anatomical cross-sectional area and its PCSA are not always the same thing. The PCSA of a muscle is the cross-section of the muscle taken perpendicular to the muscle fibers. Therefore, if we take a muscle's PCSA and multiply it by its specific tension (the force per unit area of a muscle) we are able to calculate the maximum isometric force for any muscle. Values for the specific tension of a muscle can vary and should be considered when developing a model (Dickerson et al., 2007; Buchanan et al., 1998; Arnold et al., 2010).

Muscle force production is also influenced by muscle contraction velocity. Each muscle contains two different muscle fiber types: fast-twitch and slow-twitch fibers. Slow-twitch muscle fibers (also called type 1) are more aerobic in nature, less resistant to fatigue, and generate force at a slow rate. Conversely, fast-twitch muscle fibers (type 2) are anaerobic in nature, fatigue quickly, and generate force at fast rates. All skeletal muscle in the body contains both fast and slow-twitch muscle fibers, but the ratio of fibers depends on each individual muscle and even each individual person. For example, a sprinter will have a fibers compared to a marathon runner who will mainly have muscle built up of slow-twitch fibers. Training can alter fiber type within a muscle to a certain degree but this make-up of fast-twitch versus slow-twitch muscle fibers is largely determined in the first few years of life and can be highly genetic (Kenney et al., 2015). The muscle contraction velocity of a muscle is important as it influences the force–velocity profile. According to the force–velocity relationship for muscles, the slower a concentric contraction occurs, the more force the muscle is able to produce (see Chapter 8 for muscular force–velocity relationships). Although fast-twitch muscles may be able to generate force at higher speeds, they are unable to optimize their force output at these higher speeds. Therefore, muscle contraction velocity and muscle fiber type are important to consider when modeling the musculoskeletal system.

Tendons behave similarly to nonlinear springs and like skeletal muscle, they too have a force–length relationship. The amount of force that can be transmitted from muscle to bone by a tendon is dependent on the length of the tendon. The tendon force–length relationship is not linear and the slope of the force–length curve is the stiffness of the tendon. The slope can be divided into three different parts on the curve and represents three different areas where the tendon elicits three distinctly different properties. In the first three percent of the curve when the tendon has not been stretched, the tendon is very flexible but produces very little force. Similar to stretching a rubber band to fling it across the room, if the rubber band has only been stretched very little it will go a very short distance when released. In the second part of the curve, as the tendon is stretched more and more, the amount of force that can be produced increases at a linear rate. There is a point where you can increasingly stretch the rubber band and it will go incrementally further across the room. Lastly, the last ten percent of the curve is where the tendon is pushed past its mechanical limit and plastic deformation occurs resulting in injury to the tendon. This is similar to when the rubber band becomes overstretched and is in danger of snapping. Therefore, in order to produce maximal force from a muscle, it is crucial the tendon length is also optimal. This is crucial to avoid injury while still stretching the tendon enough to produce ample power from the muscle. This force–length relationship of the tendon emphasizes the importance of properly modeling tendon properties when performing musculoskeletal simulations.

10.2.2 Control

Because the human musculoskeletal system is a redundant system (there are more unknown variables than equations), it is not well understood how the CNS controls the human body to adapt to its ever-changing environment. For example, there are multiple ways to grab a coffee cup in front of you, and it is assumed the CNS chooses the most effective movement based on certain criteria. In this case, if your musculoskeletal model needs to produce a desired motion, what control strategy should be used to produce muscle force for the desired movement?

The most common control models include (i) optimization approaches and (ii) EMG-driven approaches. Optimization approaches are further divided into static optimization (SO) and dynamic optimization.

10.2.2.1 Optimization.

10.2.2.1.1 Static Optimization of Muscle Forces As discussed in previous chapters, a net joint moment can be estimated using an inverse dynamics approach from motion capture data. However, there are many plausible solutions that could possibly generate the same calculated net joint moment. To solve this problem of redundancy, an optimization approach, utilizing certain basic criteria (e.g. Minimizing or maximizing cost function) and constraints, can decompose the net joint moment into individual muscle forces at a single instant in time. This decomposition provides greater insight into the forces occurring around a specific joint and removes the guesswork in interpreting the cause of certain joint moments. Static equilibrium problems ignore time-dependent characteristics of human movement and have modeling because it is computationally efficient and able to estimate muscle forces during dynamic movement.

SO has been used for a variety of different clinical initiatives to help advance the field of rehabilitative care. For example, Steele et al. utilized SO to investigate the change in muscle forces that occur in individuals with cerebral palsy who suffer from crouch gait (Steele et al., 2012). This estimation of muscle forces allowed researchers to investigate knee joint contact forces present in this population as many individuals with crouch gait also struggle with knee pain. Additionally, Arnold et al. investigated muscle activations of the lower extremities during walking to rank the potential of each muscle to alter lower joints (Arnold et al., 2005). This was done in an attempt to determine the cause of crouch gait. Wesseling et al. utilized two different SO techniques to investigate how different estimates of muscle force would affect the calculation of hip joint contact forces (Wesseling et al., 2015). Rosenberg and Steele performed simulations to investigate how powered and passive ankle-foot orthoses would affect muscle demand and recruitment in individuals with cerebral palsy (Rosenberg and Steele, 2017). Lastly, Kian et al. used SO and musculoskeletal modeling to estimate the muscle and joint forces occurring in the glenohumeral joint and compare their findings to an EMG-driven model of identical structure (Kian et al., 2019). These are just a few examples of how SO is used in the literature to investigate muscle forces and improve the world of clinical biomechanics.

10.2.2.1.2 Dynamic Optimization Dynamic optimization is fundamentally distinguished from SO for two reasons. First, dynamic optimization employs forward dynamics. Therefore, the dynamic optimization approach often calculates muscle excitation patterns to control the musculoskeletal model. Then, these muscle excitation patterns are used to estimate muscle forces, moments, joint angular accelerations, joint angular velocities, and joint angles to ultimately calculate the positions of the musculoskeletal model. Second, the goal of motor tasks can be included in the formulation of the problem for the whole simulation of human movement. Because of the time-dependent characteristic of forward simulation, we can calculate time histories of muscle forces for the whole movement. This is distinguishably different from SO that calculates muscle forces at a single instant in time. One of the downfalls associated with SO is the abrupt activation and deactivation of muscles. Because of the time independence of SO, the estimation of muscle forces can be quite different for each time step. In other words, SO chooses a different set of muscle groups and a different magnitude of muscle forces for each time step resulting in a lack of continuity for muscle activations throughout a movement. Contrary to SO, dynamic optimization is designed to determine muscle excitation in conjunction with forward dynamics to find optimal motion. For example, we can find a set of muscle excitation patterns that maximize jump height or minimize energy expenditure during walking with additional weight. This is advantageous because it allows for the prediction of complex human movements without the need for experimental data.

Dynamic optimization often requires more detailed musculoskeletal models and computation time compared to SO. To decrease computational cost, hybrid optimization approaches such as computed muscle control (CMC) and direct collocation have been used with tracking experimental data. CMC is an algorithm to solve tracking problems with forward dynamics along with SO. Kinematics and muscle excitation levels from SO are used as feedback to drive forward dynamic simulations that track experimental data. By doing so, CMC can significantly reduce computation time. However, CMC involving SO still solves an optimization problem at each single time point, which may limit the benefits of dynamic optimization.

Direct collocation, a recently developed method for dynamic optimization, converts dynamic equations of motion into algebraic constraints. Both the controls (muscle excitations) and the states (positions and velocities) are parameterized while solving an optimization problem resulting in efficient computation. OpenSim Moco (Musculoskeletal Optimal Control) provides a way for researchers to utilize direct collocation to answer their scientific questions (Dembia et al., 2020). With computationally efficient direct collocation and to solve various kinds of problems including motion tracking problems, predictive dynamic optimization, EMG-driven simulation, model parameter optimization, and device design.

Similar to SO, dynamic optimization has been used in the literature to help advance the field of clinical biomechanics. For example, Davy and Audu utilized a dynamic optimization solution to predict muscle forces in the swing phase of gait (Davy and Audu, 1987). Additionally, McLean et al. investigated knee forces in the sagittal plane to determine if these shear forces could rupture the anterior cruciate ligament (ACL) during sidestepping to aid in athlete injury prevention (McLean et al., 2004). Kuzelicki et al. used a dynamic optimization technique to compute the trajectories of a sit-to-stand movement in amputees as a step in the development of assistive devices to aid in the sit-to-stand transition for this population (Kuzelicki et al., 2005). Eskinazi and Fregly used a dynamic optimization approach to calculate muscle forces, joint contact forces, and body motion simultaneously, to speed up the computation process and help facilitate model-based efforts to improve surgical outcomes and rehabilitative interventions (Eskinazi and Fregly, 2018). These are just a few examples of how dynamic optimization solutions are used to investigate human movement and improve the lives of many from a clinical perspective.

10.2.3 OpenSim

One key component in the field of dynamic simulations and musculoskeletal modeling is OpenSim simulation software (Delp et al., 2007). OpenSim is an open-source software that allows users to both develop musculoskeletal models and run dynamic simulations for a variety of different movements. OpenSim encourages collaboration between labs and institutions as a way for the scientific community to build an archive of simulations that can be exchanged, verified, analyzed, and improved by multiple different researchers. Previously, researchers spent considerable effort replicating simulations from other labs because the software used by each lab was customized or simply unavailable. By being open source, OpenSim has sped up the development and sharing of musculoskeletal models and simulation technology tremendously and since has allowed the field of biomechanics to make leaps and bounds in the optimization of musculoskeletal simulations.

10.2.3.1 Assumptions and Constraints. Like all models and dynamic simulations, there are many assumptions and constraints with the OpenSim software. Each individual model has its own constraints and assumptions that dictate its motion. For example, many of the joints within an OpenSim model are constrained to reduced degrees of freedom compared to the real world with some models even constraining joints to no motion at all. Additionally, all models have limitations or make assumptions which can cause dynamic inconsistencies in the simulations. Some common assumptions and limitations include models not accounting for ligaments, tendons, cartilage, and other soft tissues; models assuming separate tendons exist for each muscle; oversimplified geometry; and the ability for muscles to actually go through bones. Additionally, OpenSim assumes muscles to be massless and disregards muscle fatigue during motion. Fiber type distribution within a muscle is often ignored and fiber lengths and velocities are assumed to be constant throughout the muscle. These limitations, assumptions, and constraints within a model can create inconsistencies in the simulated movement and should be made known to the reader indicating the potential for inaccuracies in the final simulated movement.

10.2.3.2 How It Works. OpenSim follows a logical progression that begins with creating and scaling a model and advances to finding a set of muscle excitations that properly fit the experimental motion performed in the lab (Figure 10.1).

10.2.3.3 Scaling. First, each model is scaled to the anthropometrics of the study participant by utilizing the relationship between the marker locations of the experimental motion capture data and the virtual marker locations on the OpenSim

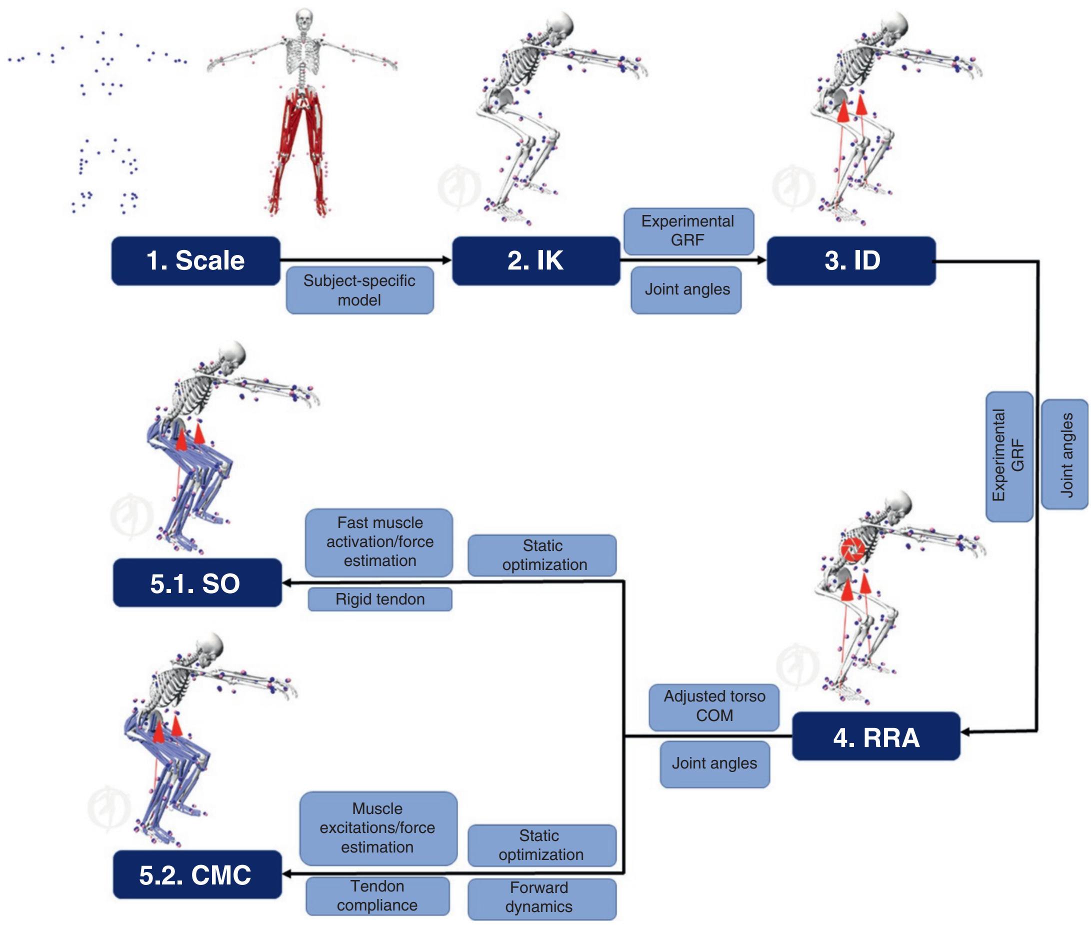

Figure 10.1 General musculoskeletal modeling workflow. *IK, inverse kinematics; ID, inverse dynamics; SO, static optimization; RRA, residual reduction algorithm; CMC, computed muscle control.

frame and mass center locations along with the force application and muscle attachment points. This is all based on scale factors that are computed in the scaling process. Next, the mass and inertial properties of each body segment are scaled proportionally based on the mass of the participant using generalized ratios from human or cadaveric work. Unfortunately, these data often come from single cadaver studies or cadavers of older individuals making them an inaccurate representation of healthy young individuals. Additionally, individual properties of these mass ratios can vary a great deal across age, sex, and disease state making this a limitation within the scaling process. After scaling the mass and inertial properties of each body segment, muscles and other model components that are dependent on length such as ligaments and muscle actuators are scaled. A ratio of the length of the model component before scaling to the length of the model component after scaling is used as the scale factor to scale the component's length-dependent properties. Lastly, the final step is to move the virtual marker locations of the model to match the marker locations of the motion capture data. Once the virtual marker locations of the model are aligned with the experimental marker data, the model is considered to be scaled. Unfortunately, the virtual and experimental markers will never align perfectly. OpenSim's best practices for marker errors should be consulted to assess simulation quality.

10.2.3.4 Inverse Kinematics. Once the model has been scaled to match the study participant anthropometrics, an inverse kinematics (IK) solution is performed to determine the joint angles of the model that most accurately reproduce the experimental kinematics from the motion capture data. This is done through a least-squares calculation in which the difference between the experimental marker data and the virtual marker data from the model are minimized throughout the motion of the simulation.

10.2.3.5 Residual Reduction Algorithm. Once the virtual markers are adequately aligned with the experimental motion capture data, the motion goes through the residual reduction algorithm. The residual reduction algorithm, commonly known as RRA, seeks to make the joint angles and translations of the model dynamically consistent with the experimental motion performed in the lab. This is done by altering the mass center of a model's torso to make the motion dynamically consistent with the ground reaction force data of the movement. By adjusting the mass of the body segments in the model and making small perturbations to the motion trajectories, OpenSim is able to decrease the inconsistency between the experimental motion and the model. This will result in lower "hand of god" residual forces required for the simulation to run. The residuals are calculated and averaged throughout the entire movement providing suggestions for adjustments to both the model translations and body mass segments. A performance standard is used to calculate tracking errors across the joint angles and is repeated at each instant in time throughout the simulation until the end of the desired movement.

10.2.3.6 Static Optimization. The fourth step of the simulation progression involves solving for the model's muscle activations along with other forces that may be responsible for the studied movement pattern. This is done using one of two optimization techniques: CMC or SO. SO is an extension of inverse dynamics, and utilizes the known positions, velocities, and accelerations of a model to find the muscle activations and other forces responsible for the given motion. This is done using a performance criterion that minimizes the sum of the muscle activations.

$$J = \sum_{m=1}^{n} (a_m)^p \tag{10.3}$$

In Equation 10.3, n is the number of muscles in the model, a_m is the activation level of a given muscle m at a particular point in time and p is the user-defined constraint.

10.2.3.7 Computed Muscle Control. On the other hand, CMC calculates coordinated muscle excitations that drive the simulated movement while tracking the desired kinematics through a forward dynamics solution. CMC first calculates the desired accelerations that will push the model coordinates to be more like the experimental movement. Then CMC calculates the actuator controls that distribute forces across muscles in an attempt to achieve the desired accelerations that will match the tracked kinematics. This process is performed for each instant in time throughout the simulation. Lastly, the CMC algorithm uses these actuator controls to produce a forward dynamic simulation of the experimental movement. This process is repeated at each instant in time throughout the simulation until the end of the desired movement. The benefit of using CMC is that it is computationally efficient like SO while performing a harmonious forward dynamic solution.

Upon the completion of CMC, a variety of analysis plug-ins can be performed on the model that are able to calculate joint contact forces, muscle-induced accelerations, and muscle power among others.

10.2.3.7.1 Review of Literature Since its inception, OpenSim has been used by a variety of different researchers to investigate a wide range of outcomes through a variety of different methodologies. For example, both SO and CMC are used to estimate muscle activations and muscle forces. Both are available for investigators to utilize and both have their advantages and disadvantages, but it is important to know the differences between the two methods before using them in a simulation. Previous literature has investigated the differences between SO and CMC and found CMC often overestimates muscle activations and muscle forces because it accounts for passive muscle dynamics and activation-contraction dynamics. However, a study by Roelker et al. investigated differences in muscle activation between SO and CMC using two different models and found that the agreement between EMG readings and SO versus CMC was both muscle-dependent and model-dependent rather than a fixed characteristic of the optimization technique (Roelker et al., 2020). Regardless of which method is used, both allow for the investigation of a variety of different outcome variables such as muscle activation dynamics, mass center accelerations, joint contact forces, and more. For example, Simpson et al. investigated the feasible muscle activation ranges for a variety of different muscles during a full gait cycle (Simpson et al., 2015). Hamner et al. used induced acceleration analyses to investigate individual muscle contributions to the center of mass accelerations while running at different speeds (Hamner et al., 2010). Lastly, Knarr et al. investigated the knee joint contact forces that occurred during walking utilizing four different methods varying in the model's specificity to the study participant (Knarr and Higginson, 2015). As demonstrated here, OpenSim has a wide variety of applications and although we mention the general linear progression one takes when using OpenSim, it is not uncommon for researchers to expand on that line to continue to progress the field forward.

10.2.3.7.2 Model Applications One of the main outcomes the creators of OpenSim desired was for collaboration across labs to help improve the field of musculoskeletal modeling. In an attempt to promote collaboration and field advancement, the creators of OpenSim sponsored "the grand knee challenge" at the 2010 ASME Summer Bioengineering Conference. The grand knee challenge datasets were collected from individuals with an instrumented knee implants (Fregly et al., 2012). The goal of this challenge was to encourage scientists to develop musculoskeletal models that would accurately predict muscle forces and ultimately knee joint contact forces. The CAMS knee data set is another example of using instrumented implants to better understand knee joint function. The CAMS knee data set was developed to better understand how internal movement of the knee relates to internal knee joint loading (Taylor et al., 2017).

Additionally, OpenSim has been used for a variety of different clinical applications. For example, Homayouni et al. utilized OpenSim to investigate passive mechanical implants used to improve an individual's ability to grasp and pinch objects following hand tendon-transfer surgery (Homayouni et al., 2015). Additionally, muscle-actuated forward dynamic simulations allow for the prediction of underlying muscular function, which drives an individual's movement. This is critical when investigating knee joint contact forces given muscle forces are a significant contributor to knee joint loading. Using modeling techniques, Sasaki and coworker demonstrated the vastii muscles are the primary contributors to the axial knee joint contact force in early stance with additional contributions from the rectus femoris, and biceps femoris short head (Sasaki and Neptune, 2010). The gluteus maximus also contributed to the axial knee joint contact force during early stance despite it not crossing the knee joint through its contribution to the ground reaction force vector. During late stance, the triceps surae muscles were the main contributors to the axial forces occurring at the knee. Shelburne et al. investigated the contributions of muscles, ligaments, and ground reaction forces to medial and lateral knee joint loading (Shelburne et al., 2006). The authors demonstrated that medial compartment loading of the tibiofemoral joint was mainly attributed to the orientation of the ground reaction force vector whereas the quadriceps and gastrocnemius muscles provided a substantial contribution to lateral joint loading. Lastly, Myers et al. utilized OpenSim to investigate the effect of strengthening the hip abductors following a total hip arthroplasty on joint contact forces at the hip, knee, ankle, and low back (Myers et al., 2019). These are just a few examples of how OpenSim has been used to investigate human movement and improve our understanding of the human body to improve the lives of many from a clinical perspective.

10.2.4 EMG-Driven Modeling

Electromyography, or EMG, has been widely used in biomechanics research to measure muscle activations and investigate CNS control. For this reason, EMG data have been used as a control signal to drive musculoskeletal models. Specifically, EMG data have been used to generate muscle-tendon unit excitations to drive forward dynamic solutions. If you recall, a forward dynamic simulation begins with a neural command, which is comparable to how our bodies perform movement tasks in the real world. When using EMG-driven modeling, this neural command is represented as the amplitude of the EMG signal (Figure 10.2). The EMG signal is then processed to represent the magnitude of the muscle activation patterns. Once we have solved for the muscle activation, we must determine the muscle forces that result from these activations through the use of a muscle model. Many muscle models that focus on characterizing the physiological aspects of muscle are very complex and become computationally expensive when modeling forces in multiple muscles. The Hill-type muscle model (referred to as phenomenological) is simpler than most other models such as the Huxley (Huxley, 1957) and distributed moment models (Gielen et al., 2000) because it focuses on modeling the function of the muscle and is less concerned with the actual muscle physiology. Because of this, the Hill-type muscle model is often used with EMG-driven modeling. A Hill-type muscle model places the contractile element of the muscle fibers in series with the viscoelastic component of the tendon to estimate the force generated by the muscle-tendon unit. However, as mentioned previously, each muscle has its own optimal fiber length, maximum contraction velocity, pennation angle, and other factors which dictate the muscle's ability to generate force effectively. Therefore, it is important to account for these specific muscle parameters within the Hill-type model when modeling muscle forces. Unfortunately, the forces produced by muscle-tendon units are nonlinear due to the changes in muscle fiber length as the muscle contracts. Therefore, one must find the muscle-tendon force at each point of a time series to accurately measure muscle forces throughout the duration of the movement being modeled.

As discussed previously, muscle-tendon unit anatomy plays a vital role in ensuring accurate calculations of the moment created about a joint. Therefore, it is crucial to accurately determine these variables before they are inserted in the model. The muscle-tendon moment arm is directly related to the calculation for determining joint moments, but can be difficult to find because it is constantly changing with the angle of the joint. However, if we know the muscle-tendon unit's length and the angle of the joint, the moment arm can be found rather easily using the tendon displacement method (An, 2007). Once muscle-tendon unit forces and moment arms are calculated, we can calculate joint moments through the multiplication of force and the moment arm. If we know these two parameters for every muscle, we can calculate the net moment about a joint. Models can be validated through the use of a nonlinear optimization technique that aims to reduce the squared error between the model predicted joint moment and the measured joint moment. Using this technique, model coefficients can be refined and tuned for individual participants. However, one should be wary of allowing too many variables to be altered because models containing a lot of varying parameters tend to be inaccurate and display poor predictive power for novel data. Therefore, it is a delicate balance between having a model that accurately replicates the anatomy of the human body without having too many parameters that need to be adjusted at each step of a time series. Similar to the Hill-type model, musculoskeletal models should accurately describe the human body without being overly complex.

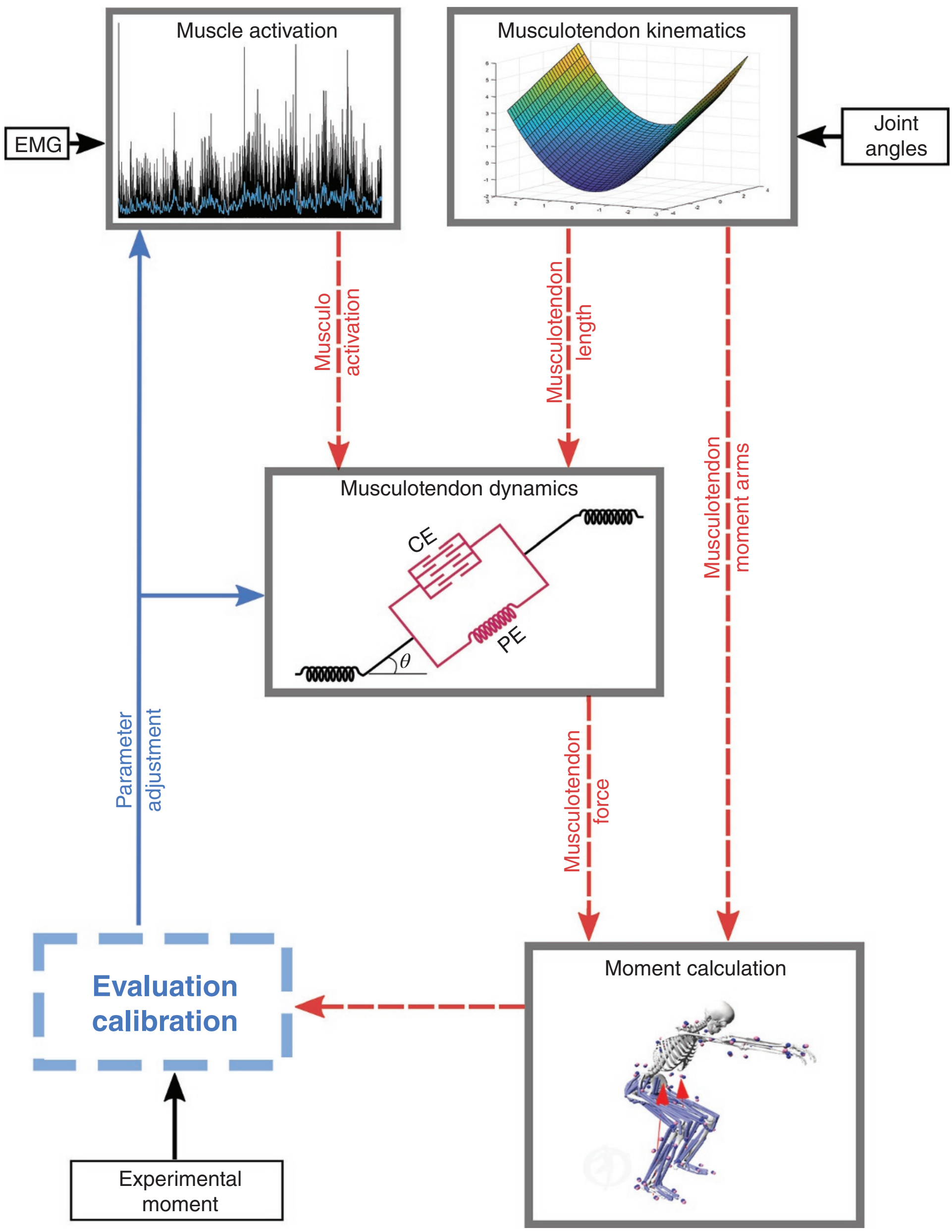

Figure 10.2 Schematic depiction of the EMG-driven modeling.

An EMG-driven modeling approach has been used extensively in the field of biomechanics to provide insights for injury prevention, motor control, surgical intervention, and pathological disorders. One strength the EMG-driven modeling approach has over the optimization approach is that EMG-driven modeling can account for individual CNS control rather than adopting a standardized performance criterion. Additionally, the EMG-driven approach is computationally more efficient than dynamic optimization because this approach does not require the estimation of muscle excitations through dynamic optimization. However, EMG-driven modeling has some limitations. One limitation is that muscular activity from deep musculature is not measurable and cannot be accounted for in this approach. Indwelling EMG can measure deep muscle activity, but it is invasive and can alter one's natural behavior during a dynamic task. Additionally, inherent technical challenges raise the question as to whether EMG data can truly represent CNS control. However, the EMG-driven modeling approach appears to be more suitable for individuals who have a disability or history of injury as their movement strategies may be adapted to accommodate their disabilities or previous injuries. Lastly, EMG-driven modeling currently requires calibration procedures to determine physiological parameters such as tendon slack length, optimal fiber length, EMG-to-activation recursive filter coefficients, maximum isometric force, and nonlinear shape factor (Lloyd and Besier, 2003; Buchanan et al., 2004; Manal and Buchanan, 2013). The calibrated parameters need to be applied to a musculoskeletal model to predict muscle forces using the EMG-driven modeling approach.

EMG-driven modeling is very beneficial for understanding muscle dynamics during movement, but unfortunately, it can become overly complex when multiple segments are involved making the modeling procedure computationally expensive. A hybrid method is available that combines both forward and inverse dynamic solutions. The hybrid scheme is beneficial because it allows for a cross-validation measure of the joint moments calculated from the forward dynamics method within the error of the inverse dynamics method.

10.2.4.1 Model Applications. Previous researchers have utilized EMG-driven modeling to investigate the mechanics of the human body through the use of a forward dynamics solution (McGill and Norman, 1986; McGill, 1992; Cholewicki et al., 1995; van Dieën and Visser, 1999). Unfortunately, to perform a forward dynamics solution the estimation of muscle activations from EMG signals and the transformation of muscle activation to muscle force is not a trivial task. The use of cost functions can be utilized to bypass the task of determining muscle force from EMG data, but the question of "which cost function to use?" is highly debated. Additionally, it does not seem wise to replace the entire CNS with a simple, unverified equation. However, in the last several decades, EMG-driven *neuro*musculoskeletal modeling (Buchanan et al., 2004) has been used to investigate human body function from the body's neural signals. For example, Kumar et al. utilized EMG-driven modeling to estimate muscle forces and joint loading in individuals with knee osteoarthritis (Kumar et al., 2012). Knee joint loading is altered during the onset of knee osteoarthritis and throughout its progression. Additionally, muscle forces are an important factor in determining knee joint loading. EMG-driven modeling allowed Kumar et al. to utilize subject-specific muscle activation patterns which provided greater accuracy in their estimation of muscle forces and ultimately provided a better estimation of the loads occurring at the knee joint (Kumar et al., 2012). Shao et al. utilized an EMG-driven musculoskeletal model to estimate the muscle forces and joint moments occurring about the ankle in individuals who suffered from a stroke (Shao et al., 2009). Especially during complex tasks like walking, muscle activation and movement patterns are significantly altered in individuals post stroke. By utilizing an EMG-driven musculoskeletal model, Shao et al. sought to better understand these changes and develop improved rehabilitation techniques for this population. Manal et al. developed a real-time EMG-driven model of the ankle with the intention of monitoring muscle forces during rehabilitation of injuries such as an Achilles tendon tear (Manal et al., 2011). Achilles tendon tears are the kiss of death for many athletes, and it is important that patients and rehabilitation specialists alike are able to see the muscle forces produced by the ankle to determine if the patient is working hard enough to produce improvement, but not too hard that they would risk reinjury. The authors believe this model is an integral step in improving the rehabilitation process for Achilles tendon injuries allowing for athletes and laypersons alike to return to function sooner. Nikooyan et al. developed an EMG-driven model of the shoulder that was able to account for co-contractions that occur at the shoulder (Nikooyan et al., 2012). By developing an EMG-driven musculoskeletal model of this caliber, more reliable measurements of the forces occurring at the shoulder joint can be obtained. This is especially crucial for individuals who have rotator cuff tears. Lastly, Koo and Mak demonstrated the ability to use an EMG-driven model to predict individual muscle forces at the elbow during dynamic movements (Koo and Mak, 2005). EMG-driven modeling truly gives researchers greater insight into the muscle forces occurring at a variety of different joints and helps increase the knowledge and understanding of the movements at these joints to ultimately help improve the quality of life for a variety of different clinical populations.

10.3 MODEL VALIDATION

Musculoskeletal modeling is an excellent way to test hypotheses that are not safe or feasible for human subjects to perform and many great scientific discoveries have come from modeling and simulations. However, we must admit that there is no perfect model or control algorithm to represent all variables related to human movement and physiological characteristics. Therefore, before using a model and performing a simulation it is important to verify the model to ensure it is valid, accurate, and reliable. According to the American Society of Mechanical Engineers (ASME), validation of a model is “the process of determining the degree to which a model is an accurate representation of the real world from the perspective of the intended uses of the model” and verification of a model is “the process of determining that a computational model accurately represents the underlying mathematical model and its solution” (Hicks et al., 2015).

The process of verifying and validating a model takes a variety of different steps.

1. Devise a research question that a modeling simulation can answer. In this step, it is important to formulate a research question that will not only advance the field forward, but it is also imperative to make sure modeling and simulations are both necessary and capable of answering the research question.
2. Design methods for the study and create a plan for model validation and verification to add credibility to the results.
3. Before any data is collected, it is important to verify the software that is being used to ensure the model and algorithms used in the simulation are correct.
4. Compare the model used and validate the study results to other experiments in the literature.
5. Test the robustness of the model. This is often done by simulating other scenarios not tested in the experiment.
6. Document the modeling and simulation methods used in the study and sharing the model and simulations with other researchers to allow them to test the model.
7. Generate other hypotheses that do not need modeling simulations but are testable in the real world.

One of the most common validation approaches used in the current literature is to compare simulated muscle activations to muscle activations from EMG sensors. It is important to compare temporal agreement between simulated muscle activation and muscle activations from EMG sensors because muscle activation timing significantly impacts muscle force production. When comparing simulated muscle activations to EMG data, it is important to consider electromechanical delay from EMG recordings. Additionally, when filtering EMG data different cutoff frequencies have been shown to alter the electromechanical delay recorded by the EMGs (Manal et al., 2002; Lloyd and Besier, 2003). As a result of these findings, investigators need to be cognizant of the filtration methods that should be used for their data based on the specific aims of the study (see Chapter 2 for signal processing methods). Unfortunately, there is no established standard for validating a model using EMG data, but visual confirmation for temporal agreement between simulated muscle activation and EMG activation is often used.

It is important to understand that a good agreement between experimental data and simulated outcomes does not always mean that the model is correct. In complex models and movements, it is challenging to detect errors from various sources such as simplified models, control algorithms, and imperfect experimental data in complex models and movements. Therefore, there is a risk that a combination of errors may mask or offset errors making simulated results seemingly correct. To reduce uncertainty from a model, sensitivity analysis is often used to investigate the impact of any particular parameters on key variables. Creating and validating models requires extensive effort and time, researchers are encouraged to share their knowledge, models, and report errors to accelerate musculoskeletal simulation studies.

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11

STATIC AND DYNAMIC BALANCE

STEPHEN J. THOMAS[1], JOSEPH A. ZENI[2], AND DAVID A. WINTERS[3,†]
[1]*Thomas Jefferson University, Philadelphia, PA, USA*
[2]*Rutgers University, Newark, NJ, USA*
[3]*University of Waterloo, Waterloo, Ontario, Canada*

11.0 INTRODUCTION

The complex interconnectivity of the neuromuscular system makes it extremely challenging to interpret the function of a single variable at any single joint during the time course of any total body movement. The various levels of integration were described in Section 1.2 where the first three levels were involved with neuromuscular integration to generate the moment profile at a given joint. However, the reason for this final motor profile is not evident until we look at the movement task to see how the muscles at all joints contribute to the final goal. Also, during any given movement, an individual group of muscles may also have more than one simultaneous subtask to accomplish.

Biomechanics continues to evolve the methodologies and technologies used to measure and analyze the total body in 3D space and, therefore, is capable of identifying total body movement synergies either during normal everyday tasks or in response to perturbations, either internal or external in origin. Some very quick examples will be referenced to illustrate some of these total body analyses.

1. MacKinnon and Winter (1993) reported the total balance control in the frontal plane during normal walking. The hip abductors responded proactively to the gravitational and inertial forces acting on the head, arms, and trunk (HAT) segment to keep it nearly vertical during single support.
2. Eng et al. (1992) identified the total body responses to total arm voluntary movements in the sagittal plane: the hip, knee, and ankle moments responded in anticipation with appropriate polarities to the polarity of the shoulder moment. A flexor shoulder moment resulted in a posterior postural response (hip extensors, knee flexors, and ankle plantarflexors), while an extensor shoulder moment resulted in an anterior postural response (hip flexors, knee extensors, and ankle dorsiflexors).

[†]Deceased

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

3. Rietdyk et al. (1999) identified the total body balance recovery mechanisms from mediolateral external perturbations of the upper body during standing.
4. Scholz and Schöner (1999) established the uncontrolled manifold concept as a method to examine the relationship between segmental coordination and stability of the bodies center of mass in 3-D space during a sit-to-stand task. Yamagata et al. (2021) also demonstrated its validity in predicting falls during walking in the elderly.

Human gait is a complex bipedal movement with many subtasks that must be simultaneously satisfied and that are continuously changing over the stride period. These tasks may be complementary or competitive (Winter, 1991): muscles generate and absorb energy at the same time as they are also responsible for the control of balance and vertical collapse of the body. This chapter is presented to detail major examples of synergistic motor patterns and demonstrates how kinetic and electromyographic (EMG) profiles aid in identifying these movement synergies.

The term synergy is defined here as muscles collaborating toward a common goal and therefore to identify such synergies, it is critical to clarify the goal and over what period of time the muscle groups collaborate.

11.1 THE SUPPORT MOMENT SYNERGY

In Chapter 5, a detailed description of the three lower limb moments over stance was presented. Also introduced was the concept of a total limb extensor pattern called the support moment $M_s = M_k - M_a - M_h$ (Winter, 1980). Considerably more information about this synergy becomes evident when we analyze repeat trials on the same subject and analyze the considerable variability at each of the joints (Winter, 1984, 1991). Figure 11.1 presents the ensemble-averaged profiles for the same subject over nine days, where the subject was instructed to walk with her natural cadence. Note that the convention for this plot has changed, with extensor moments at each joint being plotted +ve, thus $M_s = M_h + M_k + M_a$.

The joint angle kinematics over these nine trials was very consistent: the rms standard deviation over the stride period was 1.5° at the ankle, 1.9° at the knee, and 1.8° at the hip. The cadence remained within 2%. As can be seen, M_h and M_k show negligible variability over swing but considerable variability over the stance period (CV_h = 68%, CV_k = 60%), while $M_h + M_k$ shows considerably reduced variability (CV_{h+k} = 21%). Also, M_s, which is the sum of all three moments, has CV_s = 20%. Thus, on a trial-to-trial basis, there is some "trading off" between the hip and knee; on a given day, the hip becomes more extensor, while the knee becomes more flexor, and vice versa on subsequent days. We can quantify these day-to-day interactions by a covariance analysis. Figure 11.2 presents the variances and covariances for these repeat day-to-day trials along with 10 repeat trials done on a second subject done minutes apart. The hip and knee variances and covariances are related as follows with all units in $(\mathrm{N \cdot m})^2$

$$\sigma^2_{hk} = \sigma^2_h + \sigma^2_k - \sigma^2_{h+k} \tag{11.1}$$

where

σ^2_h and σ^2_k are the mean hip and knee variances over stance
σ^2_{+k} is the mean variance of the hip + knee profile over stance
σ^2_{hk} is the mean covariance between hip and knee over stance

The term σ^2_{hk} can be expressed as a percent of the maximum possible, which would be 100% if $\sigma^2_{h+k} = 0$, meaning that day-to-day changes in M_h and M_k were completely out of phase and were

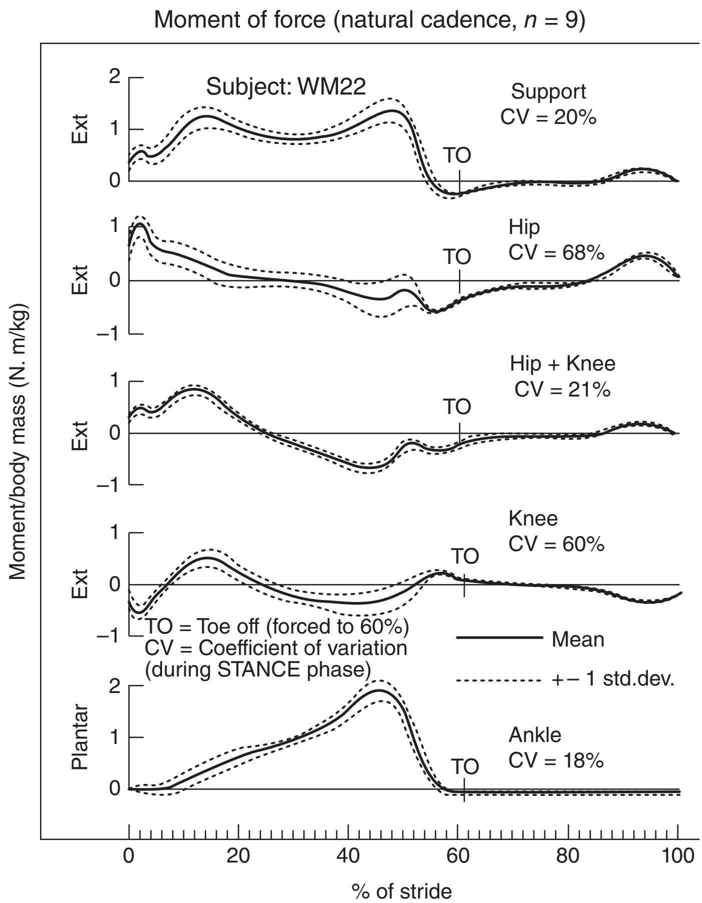

Figure 11.1 Ensemble averaged moment profiles for repeat trials from the same subject over nine days. The support moment, $M_s = M_h + M_k + M_a$ with extensor moment at each joint plotted +ve. The hip + knee moment, $M_h + k = M_h + M_k$ is plotted to demonstrate the dramatic reduction in variability because of day-to-day trade-offs between these two joints. See the text for a detailed discussion. (Reproduced with permission from Winter (1991).)

canceling each other out completely. Thus, the maximum possible $\sigma_{hk}^2 = \sigma_h^2 + \sigma_k^2$, and the % covariance is:

$$\mathrm{COV} = \sigma_{hk}^2 / \left(\sigma_h^2 + \sigma_k^2\right) \times 100\% \tag{11.2}$$

As is evident from Figure 11.2 over the nine days, the $\sigma_{hk}^2 = 19.1\ \mathrm{N{\cdot}m^2}$, which is 89% of the maximum; even σ_{ak}^2 is 76% of the maximum. The trial-to-trial covariances are somewhat reduced, but this is because the individual joint variances were drastically smaller over repeat trials minutes apart compared to repeat trials over days.

11.1.1 Relationship Between M_s and the Vertical Ground Reaction Force

As the support moment represents a summation of the extensor moments at all three joints, it quantifies how much the total limb is pushing away from the ground. The profile of M_s has the characteristic "double-hump" shape seen in the vertical ground reaction force, F_y. To check this out, a linear correlation was done between the averaged M_s and F_y profiles for three groups of

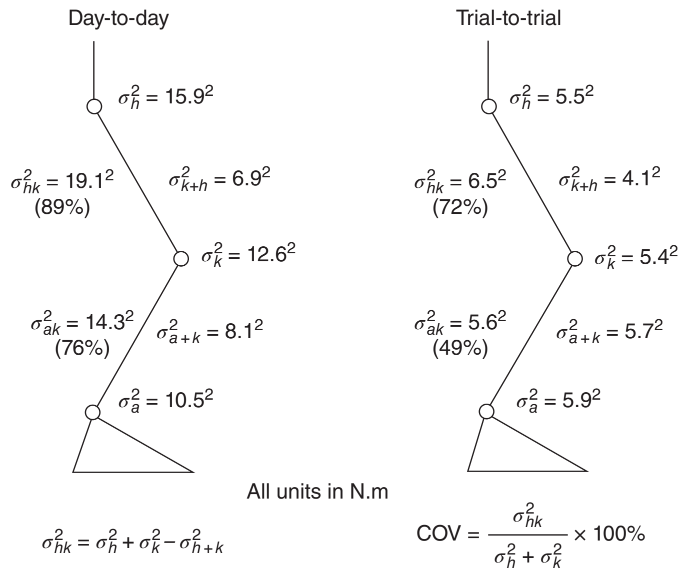

Figure 11.2 Variance and covariance analyses of the joint moments of the subject shown in Figure 11.1 plus a similar analysis of 10 repeat trials of a second subject done minutes apart. There is a very high covariance between the hip and knee moments (89%) and a moderate covariance between the knee and ankle moments (75%). The trial-to-trial covariances are still moderately high but are reduced because the individual joint variances are considerably smaller. See the text for detailed calculations. (Reproduced with permission from Winter (1991).)

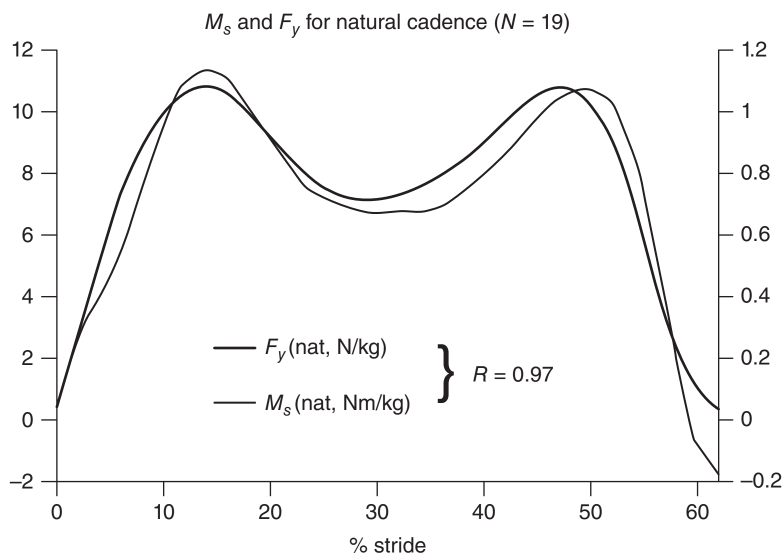

Figure 11.3 Average vertical ground reaction force, F_y, and the average support moment, M_s, for 19 adult subjects walking with their natural cadence. The close similarity in these two profiles yielded r = 0.97 from a linear regression. (Reproduced with permission from Winter (1991).)

walking adults (Winter, 1991): those walking their natural cadence, fast cadence (natural + 20), and slow cadence (natural − 20). Figure 11.3 is a plot of M_s and F_y averaged profiles for the 19 subjects walking with their natural cadence. Here, we see an almost perfect match between the two profiles, and this is reinforced by the

$r = 0.95$ and for 17 slow walkers, $r = 0.90$. Even the atypical moment patterns seen in pathological gait yielded $r = 0.96$ for two walking trials for a 69-year-old female total knee replacement patient and $r = 0.92$ for two trials for a 69-year-old male knee replacement patient.

11.2 MEDIAL/LATERAL AND ANTERIOR/POSTERIOR BALANCE IN STANDING

11.2.1 Quiet Standing

Standing has been the subject of considerable research with posture and balance being the dominant tasks in both medial–lateral (M/L) and anterior–posterior (A/P) directions (Horak and Nashner, 1986; Winter et al., 1996; Gage et al., 2003). The primary outcome measures are center-of-pressure (COP) and center-of-mass (COM), and it has been shown that as an inverted pendulum model (see Chapter 5) that for sway angles of less than 8° the COP and COM are related to the horizontal acceleration of the COM in either the A/P or M/L directions (Winter et al., 1996):

$$\text{COP} - \text{COM} = -I\ddot{x}/Wd = -K\ddot{x} \tag{11.3}$$

where
- I is the moment of inertia of the total body about the ankle in the direction of interest
- $\ddot{x}$ is the horizontal acceleration of the COM in the direction of interest
- d is the vertical distance from the ankle joints to the COM
- W is the total body weight above the ankle joints.

Thus, we can think of COP–COM as an error signal in the balance control system for controlling the horizontal acceleration of the COM, and it forces us to focus on how the central nervous system (CNS) controls the COP to achieve a stable balance. In quiet standing in the A/P direction, COP is controlled by the ankle dorsiflexors/plantorflexors (Horak and Nashner, 1986), whereas in the M/L direction, COP is controlled by the hip abductors/adductors in what has been described as a "load/unload" mechanism (Winter et al., 1996). Figure 11.4 summarizes this mechanism during quiet standing and shows when two force platforms are used the right and the left vertical ground reaction forces along with the M/L COP. Note that these vertical ground reaction forces oscillate about 50% of body weight and the fluctuations about this 50% are virtually equal in magnitude and also exactly out of phase. The M/L COP (cm) is a weighted average of the left and right vertical reaction forces and with the convention shown is in phase with the right vertical reaction force. The horizontal ground reaction forces are negligible; thus, from our inverse dynamics, the left and right hip moments are respectively in phase with the left and right vertical reaction forces. Thus, this "load/unload" mechanism is accomplished by hip abductor/adductor moments that are exactly equal in magnitude and also 180° out of phase. A detailed analysis of the COP and COM waveforms has attributed these motor patterns to a simple stiffness control (Winter et al., 1998). This stiffness control is not reactive because the COP is virtually in phase with the COM except the COP oscillates with larger amplitude on either side of the COM and the COP–COM error signal [Equation (11.3)] keeps accelerating the COM toward a central position. This mechanism can be considered as a bilateral synergy because the CNS has to keep the left and right hip abductors/adductors at a small level of muscle tone so that the miniscule M/L sway results in small synchronized fluctuations in hip frontal-plane moments sufficiently large to maintain M/L balance. The fact that this balance mechanism does not involve continuous reactive control allows all the reactive sensors to be on "standby" ready to react to an unexpected perturbation. One final question relates to what causes the fluctuations in the COM. The COM in these studies was estimated from a 14-segment total body model (see Chapter 4), which included four trunk segments; this was necessary because of internal mass shifts mainly from the lungs and heart (Winter and Lodge, 1966; Hunter and Kearney,

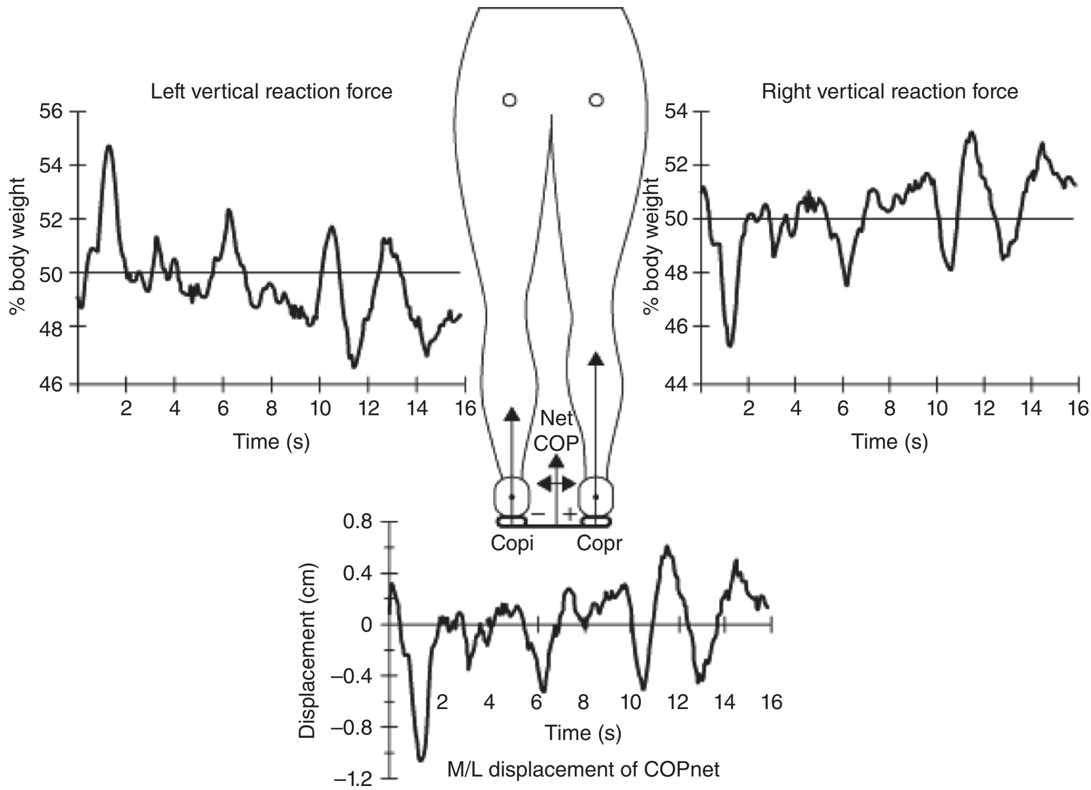

Figure 11.4 Left and right vertical ground reaction forces from two force platforms from a subject standing quietly. Note that the left and right vertical reaction forces oscillate about 50% of body weight and that these oscillations are virtually equal in amplitude and exactly out of phase. The M/L COP is a weighted average of these two reaction forces and with the convention shown is in phase with the right vertical reaction force. See the text for details of this load/unload mechanism as a synergy to maintain balance during quiet standing. (Reproduced with permission from Winter (1991).)

11.2.2 Medial Lateral Balance Control During Workplace Tasks

When standing on both feet human beings must maintain their balance whether they are standing quietly or whether they are performing manual tasks with their upper extremities. A/P balance is controlled by the plantarflexors/dorsiflexors (Horak and Nashner, 1986), whereas M/L balance is controlled by the hip abductors/adductors (Winter et al., 1996). M/L balance was identified as a "load/unload" mechanism, where increased abductor forces on one side are accompanied by decreased abductor forces on the contralateral side. Such a pattern attempts to lift the pelvis/HAT mass, thus loading the ipsilateral side, while simultaneously unloading the contralateral side. This loading/unloading moves the COP toward the ipsilateral foot, thus causing an acceleration of the COM of the body toward the contralateral side.

This synergistic pattern was examined in an ergonomics task, where subjects stood in front of a table carrying out a variety of manual tasks for a period of two hours (Nelson-Wong et al., 2008). Surface EMG records of right and left gluteus medius muscles were recorded to quantify this synergistic pattern during this fatiguing event. A cross-correlation analysis (see Chapter 2) of left and right gluteal activity quantified this synergy: a negative correlation resulted when the right activity increased at the same time as the left activity decreased, and vice versa (for a typical subject, see Figure 11.5). For this subject, $R_{xy}(\tau)$ had a peak value of −0.677 during a 15-minute block of the 2-hour trial. The $R_{xy}(\tau)$ is plotted for $\tau = \pm 1$ s and is fairly flat over that range because of the

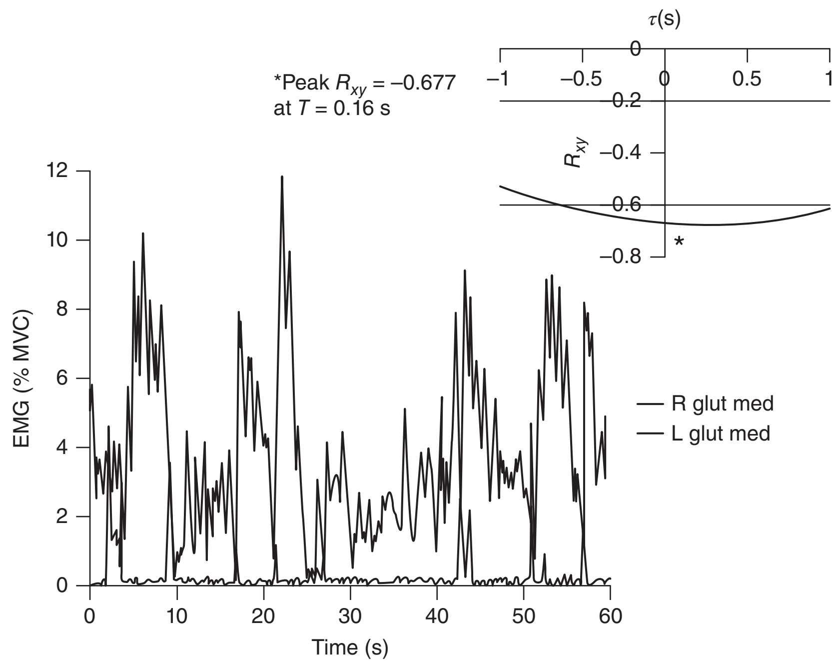

Figure 11.5 A 60-seconds linear envelope record of the left and right gluteus medius muscles showing a synergistic reciprocal pattern. R_{xy} (upper right) is plotted for $\tau = \pm 1$ second and showed a peak of −0.677 at $\tau = 0.16$ second showed these left and right muscles were virtually out of phase, indicating a synergistic load/unload pattern. See the text for more details on the reported low back pain of those subjects exhibiting this pattern.

fairly long duration of the left and right gluteal bursts of activity. Thus, the peak of R_{xy} at $\tau = 0.16$ s reflects what we see from these bursts of activity are virtually 180° out of phase.

Conversely, when the load/unload synergy was not present, there was a positive correlation [$R_{xy}(\tau) = 0.766$ during the 15-minute block]; both right and left activity were increasing and decreasing together, indicating an inefficient co-contraction (see Figure 11.6). Again, the peak of the $R_{xy}(\tau)$ was at $\tau = 0.06$ s, indicating on this 15-minute record that the left and right gluteal activity was virtually in phase, and therefore the right and left abductors were fighting each other to maintain M/L balance.

In this 2-hour workplace study (Nelson-Wong et al., 2008), 23 subjects reported their pain levels for each 15-minute work task on a visual analog scale, scored from 0 to 40. Fifteen of the subjects who exhibited this inefficient co-contraction pattern reported a pain score increasing from an average of 4 in the first work task to 32 in the last task. The remaining eight subjects exhibited the efficient load/unload pattern and reported an average pain score of 1 in the first task, increasing slowly to 8 in the last task. There was a significant main effect between these two groups with a $p < 0.0005$.

11.3 DYNAMIC BALANCE DURING WALKING

11.3.1 The Human Inverted Pendulum in Steady State Walking

During the gait cycle, there are two periods of single support (each about 40% of the gait cycle) and two short periods of double-support when both feet are never flat on the ground: the heel contact foot is about 20° from the horizontal and is plantarflexing rapidly toward a flat foot position, while the push off heel is well off the ground and weight is entirely on the metatarsals

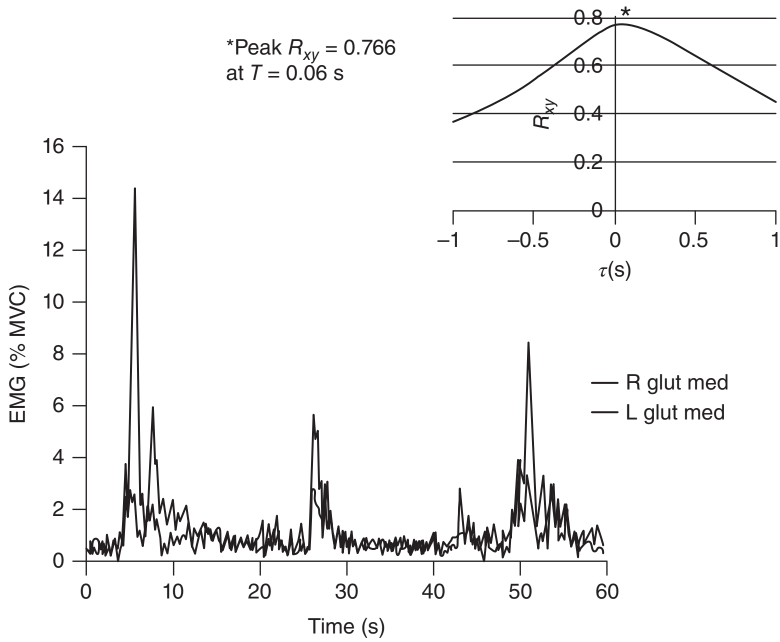

Figure 11.6 A 60-seconds linear envelope record of the left and right gluteus medius muscles showing an antagonistic co-contraction pattern. R_{xy} (upper right) is plotted for $\tau = \pm 1$ second and shows a peak of 0.766 at $\tau = 0.06$ second; this shows that these left and right muscles were virtually in phase, indicating an inefficient coactivation. See the text for more details on the reported low back pain of those subjects exhibiting this pattern.

and toes. When we examine the trajectories of the COP under the feet and the COM of the total body, we see the challenge to the human control system and more specifically the human as an inverted pendulum. Figure 11.7 plots the COP and COM of the body during two steps to illustrate this challenge. The first observation is that the COM never passes within the base of the foot; rather, it moves forward passing just medial of the inside of each foot. Thus, during each 40% single-support period (LTO to LHC and RTO to RHC) the body is a single-support inverted pendulum and its horizontal acceleration is decided by the vector joining the COP to the COM (review the inverted pendulum equations in Chapter 5). Thus, from LTO the M/L trajectory of the COM is headed toward the right foot, but from the curvature of this trajectory, it is evident that it is being accelerated medially toward the future position of the left foot. In the plane of progression (shown by the centerline), there is a deceleration of the COM while it is posterior of the COP and an acceleration of the COM after the COM moves forward along the foot and is anterior of the COP. This would be seen in the velocity of the COM, which is decreasing during the first half of stance and increasing during the latter half of stance. The challenge to the CNS is that the human is never more than about 400 ms away from falling, and it is the trajectory of the swinging foot that decides its future position and, therefore, its stability for the next single-support period.

11.3.2 Initiation of Gait

The complete collaboration of both the A/P and M/L muscle groups is never more evident in the initiation of gait. Going from a very stable balance condition during quiet standing to a walking state in about two steps requires coordination of the A/P muscles (plantarflexors/dorsiflexors) and the M/L control (hip abductors/adductors). The goal of these motor patterns during initiation is to change from the quiet standing pattern to the steady state COP/COM patterns, as described in Section 11.3.1, in as short a time as possible.

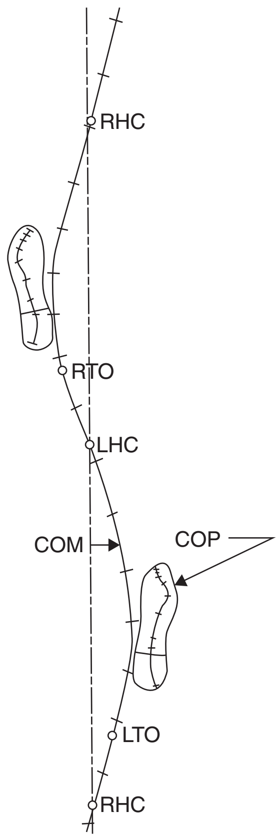

Figure 11.7 Trajectories of the COM and COP for two steps during steady state walking. Plotted on the COM trajectory is shown when key stance events occur: LTO, left toe off; LHC, left heel contact; RTO, right toe off; RHC, right heel contact. Note that the COM never passes within the base of support of either foot. See the text for details of the balance challenges to the inverted pendulum model. (Reproduced with permission from Winter (1991).)

The first major study to track COP/COM profiles during gait initiation yielded the patterns shown in Figure 11.8 (Jian et al., 1993). Justification for the inverted pendulum model during this study [Equation (11.3)] yielded correlations that averaged −0.94 in both the A/P and M/L directions. The origin of both axes is located over the quiet standing position. The initial task of gait initiation is to accelerate the COM forward and toward the stance limb. Thus, the first small movement of the COP must be in the opposite direction: posteriorly and toward the swing limb. This posterior shift is accomplished by a sudden decrease in plantarflexor activity; the lateral shift toward the swing limb results from a momentary loading of the swing limb via increased activity of the swing limb abductors and decreased activity in the stance limb abductors. As predicted by Equation (11.3), this momentary shift of the COP accelerates the COM in the desired direction forward and toward the stance limb. This constitutes the release phase. Interestingly, this initial increase in swing limb ground reaction force and decrease in stance limb force appeared in graphs considerably earlier (Herman et al., 1973) but no comment was made. Then the COP moves rapidly laterally toward the stance limb as the stance limb abductors turn on, the swing limb abductors turn off, and the swing limb hip flexors/knee extensors accelerate the swing limb upward and forward. Then, at RTO, the COP is under the left foot and is controlled by the left leg plantarflexors. The start of this single-support phase is labeled 0%. At this time, the COM has moved forward about 6 cm, but from its curvature, it is continuing to be accelerated forward

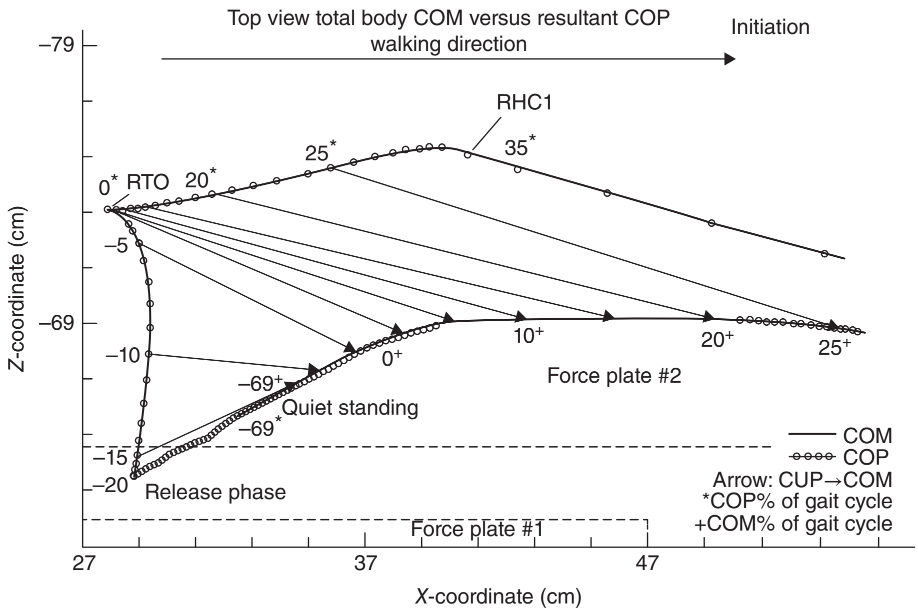

Figure 11.8 Trajectories of the COM and COP during initiation of gait. Subject was standing quietly with the left foot on force plate # 2 and the right foot on force plate #1. From the quiet standing position (−69% of the initial gait cycle), the COP moves posteriorly and laterally toward the swing limb, reaching the release position at −20% of the gait cycle, then moving rapidly laterally toward the stance foot, reaching it at right toe-off (RTO). The arrows from the COP to the COM are acceleration vectors showing the acceleration of the COM as predicted by the inverted pendulum model. See the text for details of the motor control of the COP to achieve the desired COM trajectory.

and now away from the stance foot. During this single-support phase, the stance limb plantarflexors increase their activity, causing the COP to move rapidly forward to an initial "toe-off." Simultaneously, the right swing limb is swinging forward and RHC1 occurs at 35% of this first gait cycle. Then, during the double-support phase, the COP moves very rapidly forward toward the right foot. The trajectory of the COM has now moved forward about 25 cm and (not shown here) is headed to pass forward along the medial border of the right stance foot (Jian et al., 1993). Thus, by the end of the first step, the trajectory of the COM has already reached the steady state walking pattern, as shown previously in Figure 11.7.

The initiation of gait in the young, the elderly, and Parkinson's patients was analyzed in considerable detail (Halliday et al., 1998). The basic finding in virtually all the kinematic and kinetic variables was that the temporal patterns were the same but that they were altered by a scaling factor related to their final steady state walking velocity. The young adults had the highest velocity; the fit and healthy elderly were slower, and the Parkinson's patients were slowest. All of the several significant differences disappeared after the variables were divided by the subjects' walking velocity. Thus, we can conclude that the synergistic patterns are still present but are reduced by some tonic gain control that decreases with aging and disease.

11.3.3 Gait Termination

The challenge to the balance control system during termination of gait is even more critical during termination of steady state walking. The forward momentum of the body must be removed within the last two steps, and the COP must be controlled to a position slightly ahead of the COM trajectory as the COM comes to a near stop. Figure 11.9 depicts the COM and COP trajectories during the contact of the right foot on force plate #2 and then the left foot on force plate #1;

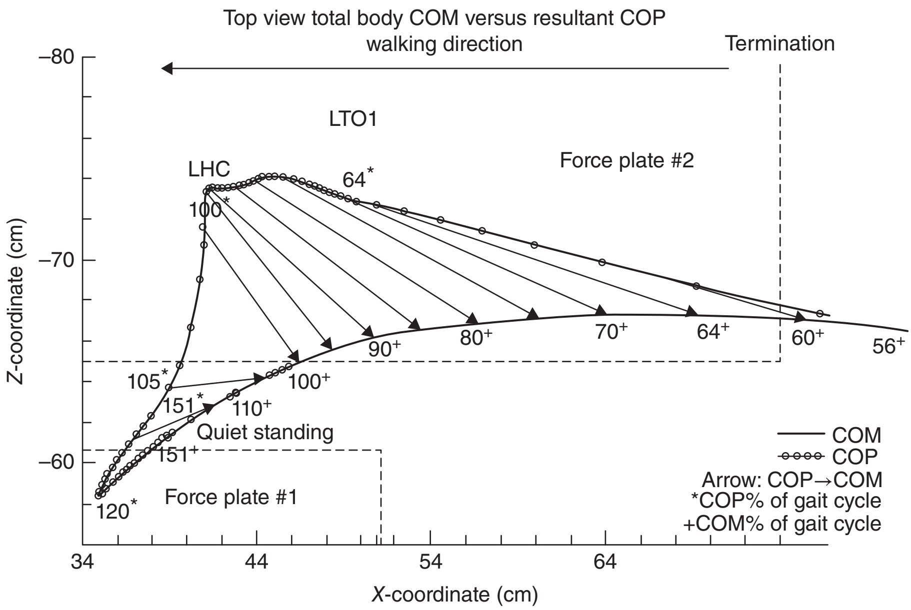

Figure 11.9 Trajectories of the COP and COM during termination of gait. The final two steps are shown; the right foot on force plate #2 and the left foot on force plate #1. The last gait cycle, from left heel contact (LHC) to LHC, was 0–100%, and shown here is 56–100%. From 56 to 64% is double-support, from 64 to 100% is right single support, and 100–151% is the final double-support. The COP trajectory moved rapidly forward toward the right foot during double-support and continues forward under the right foot until LHC, when the left foot begins to load, moving the COP rapidly laterally until it reaches 120%, when it stops ahead of the COM trajectory. See the text for the synergistic motor patterns to achieve the desired COM trajectory.

the command to stop was made by a flashing light when the previous left heel made contact with force plate #3 (not shown). The percent of that stride period began at that heel contact and at 64% of that stride was LTO1 when 100% of body weight is supported by the right foot. Prior to this 64% point was the double-support period, and the trajectory of the COP moved rapidly forward from the left foot to the right foot (shown is the 56–64% period). During this double-support period, the COP moves rapidly forward so that the COP–COM vectors (shown as arrows) indicate a rapid deceleration of COM. Then, during single support of the right foot (64–100%), the right plantarflexors' activity increases dramatically, causing COP to move forward to increase the COP–COM deceleration vector, and by 100%, the COM forward velocity is reduced by about 70%. During this right single-support period, the left swing limb is rapidly decelerated by its hip extensors/knee flexors, resulting in a step length about half of normal. The rapid loading of the left foot after LHC causes a rapid lateral and forward shift of the COP until at 120% the COP stops directly ahead of the trajectory of the COM. This loading of the left foot was achieved simultaneously by increased left hip abductor activity and unloading of the right foot by decreased right hip abductor activity. The final phase, from 120 to 151%, sees the final deceleration of the COM as the COP moves posteriorly (by reduced plantarflexors of both limbs) and slightly to the right (by increased right hip abductor and decreased left hip abductor activity) to come to rest at quiet standing. The most significant aspect of this termination synergy is the proactive control of the both hip abductors to arrest the later trajectory of the COP at a point directly ahead of the COM trajectory. About 85% of the forward velocity was arrested by the right plantarflexors and 15% by the left plantarflexors.

Interestingly, on two unpublished termination trials, one on a patient with peripheral sensory loss and one on subject with ischemic leg block, the COP trajectory in this final stage when both feet were loaded was very erratic: the COP trajectory overshot the COM trajectory in both the M/L and A/P directions. This indicates that the peripheral sensory system plays a major role in monitoring and feedback for both the COP and COM positions.

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12

CENTRAL NERVOUS SYSTEM'S ROLE IN BIOMECHANICS

ALAN R. NEEDLE[1,2] AND CHRISTOPHER J. BURCAL[3]

[1]*Department of Public Health & Exercise Science, Appalachian State University, Boone, NC, USA*
[2]*Department of Rehabilitation Science, Appalachian State University, Boone, NC, USA*
[3]*School of Health and Kinesiology, University of Nebraska at Omaha, Omaha, NE, USA*

12.0 INTRODUCTION

Charles Sherrington's (1904) work, "Correlation of Reflexes and the Principle of the Common Path," established the concept of the motor unit – a lower alpha motor neuron and the muscle fibers it innervates – as the final common output of the nervous system (Sherrington, 1904). It is this finding that spurred decades of research whereby electromyography served as the primary mechanistic measure of the processes of the nervous system during human movement (see Chapter 9). However, the processes by which the final common output is generated have recently sparked interest in biomechanists across a variety of contexts. This chapter will focus on these areas of interest and describe the methodologies used to collect data.

12.1 CENTRAL NERVOUS SYSTEM AND VOLITIONAL CONTROL OF MOVEMENT

12.1.1 Key Structures for Movement

The sensorimotor system originates at the environmental level (e.g. somatosensory, visual, vestibular afferent information), is integrated within the central nervous system, and then generates an output along the final common pathway (i.e. motor units). While the anatomy of the central nervous system is extensive, we can focus on key structures for the production of movement that offer the potential for measurement or intervention (Figure 12.1). The cerebral cortex, largely responsible for conscious control of movement offers several important areas. The *somatosensory cortex* (Brodmann's area 3, 1, 2), located in the postcentral gyrus is the terminus for sensory information from somatic receptors. It is located adjacent to the *primary motor cortex* (Brodmann's area 4) in the precentral gyrus, which serves as the origin for motor signals in the conscious control of movement. The primary motor and somatosensory cortices receive input from many different cortical areas. Some key cortical areas that interact with the sensorimotor

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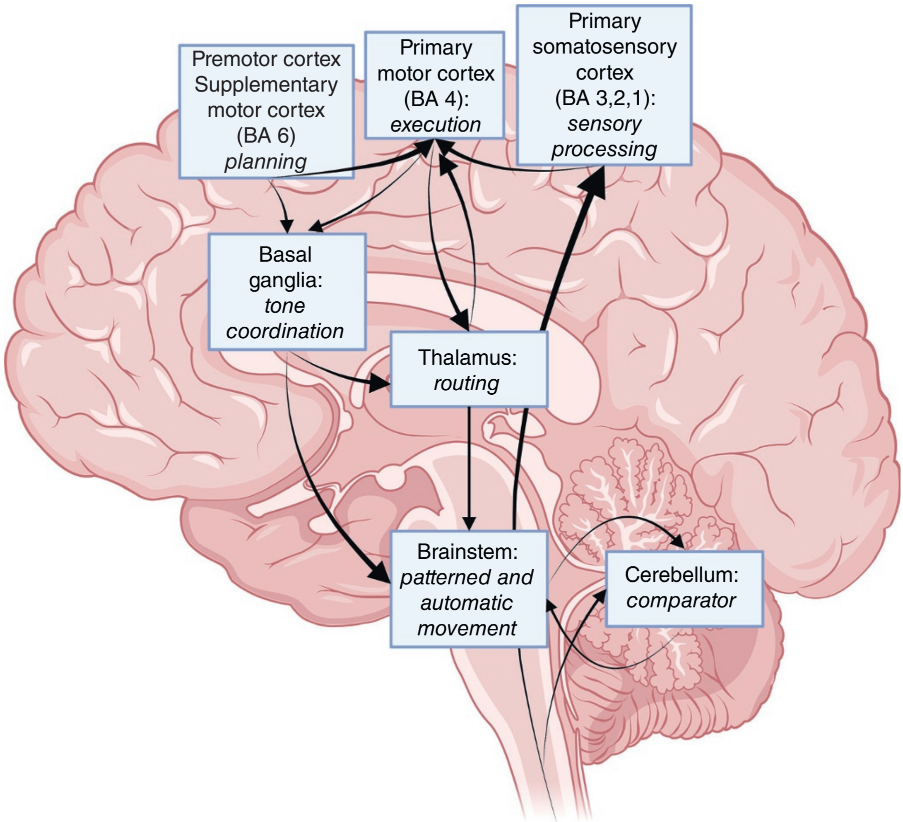

Figure 12.1 Schematic of relevant cortical and subcortical areas for sensorimotor function, including relationships between these areas. (Created with BioRender.com.)

cortices include the *Premotor Area* and the *Supplementary Motor Area* (Brodmann's area 6) that will work in a coordinated system with subcortical structures to plan, modify, and execute motor skills.

It is important to consider that the majority of observed movement happens without significant conscious control (e.g. gait), with much of this movement being regulated by subcortical areas. The *basal ganglia* are a series of nuclei located in cerebral white matter, which include the caudate nucleus, putamen, globus pallidus, substantia nigra, and subthalamic nucleus. These nuclei combine to initiate complex, coordinated movement, largely through the modification of *muscle tone* and posture. The *cerebellum*, while largely associated with vestibular function and control of balance, offers a key role in movement as the comparator of the nervous system. That is comparing somatosensory information to an efference copy of the motor action to ensure that the executed movement matches the planned movement, and offering direct corrections to these movement patterns. Lastly, the brainstem, consisting of the *midbrain, pons*, and *medulla*, contains a series of direct connections with functional groupings of muscles through descending tracts capable of controlling posture and muscle tone. The brainstem is the main location for *central pattern generators (CPGs)* that coordinate patterned movement (e.g. walking, cycling) through the rhythmic activation of functional groupings of flexors and extensors. Lastly, the spinal cord provides more than just the pathway for these signals to reach lower motor neurons and effector organs. The spinal cord contains dedicated tracts connecting cortical and subcortical structures with both the somatic receptors sensing environmental stimuli and motor neurons to execute (or inhibit movement). Lastly, the spinal cord gray matter is also the home of a large number of interneurons capable of integrating sensory stimuli into excitatory or inhibitory reflexive responses without cortical or subcortical involvement.

12.1.2 Synapses and Neurotransmitters

Neurons are the primary functional unit of the nervous system, with all neurons consisting of a cell body (soma), axon, and one or more dendrites. Impulses along a neuron travel in a single direction, with the presynaptic membranes of the dendrites receiving an impulse, transmitting it to the soma, and then going along an axon to the axon terminal(s). The interaction between the presynaptic membrane and axon terminals forms the *synapse* and may be electrical or chemical in nature. Electrical synapses will transmit voltage across a small gap junction, whereas chemical synapses rely on the transmission of neurotransmitters or neuromodulators across a synaptic cleft.

These synapses can be convergent, whereby multiple axon terminals form a synapse at the same dendrite; or divergent, where the same axon terminal transmits a signal to multiple dendrites. With regard to motor function, the most relevant neurotransmitters within the central nervous system include *glutamate*, *gamma-aminobutyric acid (GABA)*, and *glycine*. Glutamate is the primary excitatory neurotransmitter within the central nervous system, whereas GABA is the primary inhibitory neurotransmitter within the cortex, with glycine being a prevalent inhibitory neurotransmitter in the spinal cord. Activation of neurons, whether in an electrical or chemical synapse relies on the generation of an action potential. In the case of chemical synapses, those neurotransmitters result in an increase or decrease in the postsynaptic voltage, creating excitatory postsynaptic potentials (EPSPs) or inhibitory postsynaptic potentials (IPSPs) that enable the propagation of an action potential, constituting a key element of decision making.

12.1.3 CNS Adaptations

A theory posited by Donald Hebb describes the associative learning of neurons that is often described as, “cells that fire together wire together.” This Hebbian plasticity describes the nature of synapses to form or increase strength with continued stimuli, or to weaken with the lack of stimuli (Hebb, 1949). However, there are a number of ways by which synapses adapt that vary in their speed and durability. Many forms of synaptic facilitation and depression are dependent on the availability or depletion of calcium ions surrounding the synapse, respectively. The presence of extracellular calcium enables the transmission of glutamate through higher conductance *N*-methyl-d-aspartate (NMDA) channels. This is true for homosynaptic facilitation or depression, where a previous EPSP either facilitates or diminishes (depresses) a second EPSP, typically within 200 ms of the initial EPSP. This can further happen over multiple EPSPs, termed *post-tetanic potentiation* that can facilitate subsequent potentials for several minutes. Alternately, a secondary neuron may act on the presynaptic neuron to facilitate or inhibit the release of a neurotransmitter and calcium ions from the presynaptic neuron, thereby strengthening (presynaptic facilitation, Figure 12.2), or weakening subsequent responses (presynaptic inhibition, Figure 12.2). The net result of these forms of temporary plasticity are transient changes in the ability of a neuron to activate – or in the case of motor function – greater (or lesser in the case of inhibition) muscle activation.

Most forms of synaptic plasticity listed above last from the range of milliseconds to minutes; however, the goal of most learning is near-permanent changes. This type of learning is facilitated through long-term potentiation (LTP), or its negative counterpart, long-term depression (LTD). Our understanding of these processes is best understood as it relates to LTP within the

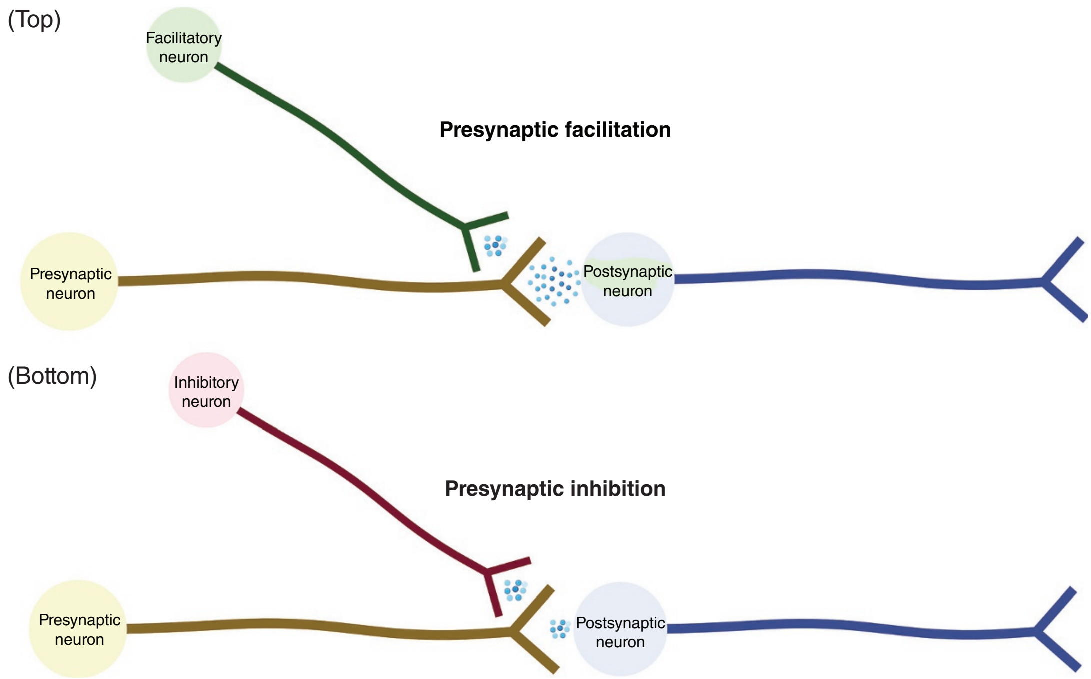

Figure 12.2 (Top) Presynaptic facilitation. A third neuron acts on a synapse, providing input to the presynaptic neuron that will increase neurotransmitter release and subsequently the likelihood of a subsequent action potential. (Bottom) Presynaptic inhibition. A third neuron acts on a synapse, providing input to the presynaptic neuron that will decrease neurotransmitter release and subsequently the likelihood of a subsequent action potential. (Created with BioRender.com.)

hippocampus; however, the general goal of LTP is to increase the number of NMDA-type glutamate receptors on a postsynaptic neuron (Figure 12.3). Neurons will have AMPA-type glutamate receptors that are permeable to sodium and potassium ions; however, NMDA-type receptors are permeable to those ions as well as stronger calcium ions. In order for these NMDA-type receptors to be activated, it requires the removal of a magnesium ion through internal cellular signaling. Throughout the process of LTP, additional AMPA and NMDA receptors are created to facilitate action potential transmission from presynaptic to postsynaptic neuron. This process is the primary mechanism by which new synapses are believed to be formed and strengthened.

12.2 PERIPHERAL NERVOUS SYSTEM AND REFLEXIVE CONTROL OF MOVEMENT

The role of central nervous system (CNS) structures in generating volitional movement is described above; however, the peripheral nervous system is also able to independently control movement through some modulation of reflexive control. While it may be considered *peripheral nervous system* control of movement, this is a misnomer as all reflexes require an integrating center that exists within the CNS (often the spinal cord or brainstem). In fact, all reflexes will have five components: a sensory receptor, a sensory neuron, an integrating center, a motor neuron, and an effector (e.g. muscle). These differ from volitional control of movement as they are largely automatic and lack volitional control; however, conscious control can modify reflex actions, as they are not "hard-wired." The most basic example of reflexive control comes in the stretch reflex – a rare monosynaptic reflex. A stretch to

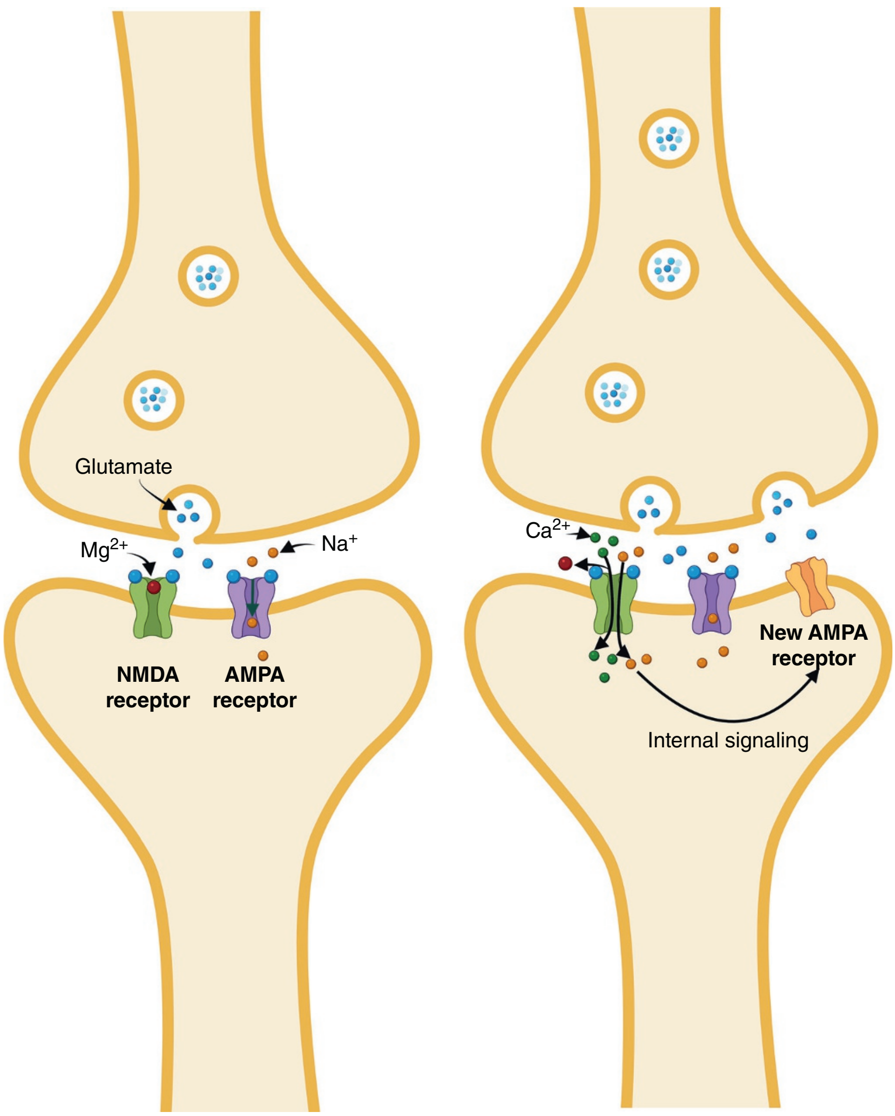

Figure 12.3 Long-term potentiation. Consistent firing at a synapse will lead to higher concentrations of calcium ions and maintain open channels such that internal cell signaling leads to increased NMDA receptors in the postsynaptic neuron. (Created with BioRender.com.)

(sensory receptor), which transmits its signal along Type Ia/II afferent neurons, is integrated in the spinal cord with a direct synapse from the sensory neuron to a motor neuron, generates a response along the α-motor neuron, and causes a contraction of the homonymous muscle. However, this same stimulus causes multiple responses as the same sensory stimulus gets transmitted along multiple interneurons within the spinal cord to inhibit the antagonist muscle, constituting a polysynaptic reflex.

As the stretch reflex is considered the only monosynaptic reflex, it stands to reason that the remainder of reflexes throughout the nervous system are polysynaptic, with multiple interneurons – those converging and diverging, ascending and descending – generating responses. While reflexes are trainable and adaptable to various long-term stimuli, there are some common ones described. For instance, the withdrawal reflex glass) that causes contraction of a withdrawal muscle (e.g. hamstring), inhibition of the antagonist muscle (e.g. quadriceps), and potentially a stabilizing contraction of the contralateral extremity.

A number of references above describe *control* of these reflexes that are not hard-wired. Among the interneuronal responses to these reflexes include those transmitting descending input from cortical, cerebellar, and brainstem structures. This descending control will offer the ability to facilitate or inhibit responses (through presynaptic mechanisms), allowing for reflexes to be tuned to the specific task. For instance, sitting still and generating a rapid stretch to the Achilles may facilitate a contraction from the triceps surae; however, if actively stretching the calf, descending control can inhibit the reflexive response and allow the muscle to remain relaxed. These task-specific adjustments to reflex sensitivity are termed *state-dependent reflex reversal*, whereby the same stimulus can generate separate responses based on descending input from the cortex and other higher structures. This descending control of reflexes forms the basis for the ability to produce patterned movement (e.g. gait, cycling, swimming) with minimal volitional input. Within the brainstem and spinal cord are *CPG*, circuits that when activated pattern movement through reflex modulation. For instance, during gait, a CPG may sense plantar sensation (stance phase) and recognize that a stretch of the triceps surae should trigger a homonymous response (push-off). Meanwhile, with no plantar sensations (swing phase), a similar stretch of the triceps surae triggers no such response; however, reflexive activation of the tibialis anterior may occur to facilitate foot clearance.

12.2.1 Sensory Receptors and Motor Units

To understand reflexive control of movement, we must briefly address the types of sensory input and motor responses in the peripheral nervous system. Rather than a detailed description of all sensory receptors, we can consider the types of information that are conveyed by these receptors. We can generally divide the sensory information from receptors into three modalities: proprioception, pressure and touch, and pain. Proprioception consists of the sense of joint position, movement, and load throughout the body and is largely provided by the muscle spindle (muscle length, movement), Golgi tendon organ (load), Ruffini endings (ligament stretch), and Pacinian corpuscles (ligament pressure). These structures will provide some of the most vital information as it relates to generating and regulating human movement. A number of receptors offer more crude sensations, including touch and pressure, with these receptors located within the skin and connective tissue. Examples of these receptors include Merkel disks, Meissner's corpuscles, and Pacinian corpuscles. Their specialization comes largely from their ability to discriminate between fine versus crude touch and texture, and their ability to rapidly adapt or slowly adapt to changes. Lastly, pain is largely transmitted by free nerve endings throughout the body.

These different modalities should be considered with regard to the measurement of joint sensation. Although the focus of this chapter is techniques to directly measure the CNS, a number of techniques are used to quantify sensation within biomechanical contexts. Given that proprioception includes a sense of joint position, movement, and load; to fully understand proprioception these should each be given consideration. Joint position sense is often quantified through a subject's active or passive ability to replicate a given joint angle and quantifying the error in angle replication (Strong et al., 2021). Movement sense is often quantified as the threshold-to-detect passive motion, where a device provides a slow movement (~0.1°/second), and participants hit a hand switch to indicate they sense movement, with the amount moved prior to sensing movement used as an outcome measure (Konradsen, 2002). Lastly, force sense can be determined through force matching tasks, where participants blindly attempt to replicate a given force level, quantifying the error or variability as an outcome measure (Wright and Arnold, 2011). For each of the items above, the outcome measure is a form of "error," considering the degree to which angle replication is off, the movement offset to detect passive motion, and the difference between the target load
all of these measurement techniques is that they all

outcome measure (e.g. depression of a hand-switch, actively generating a joint angle, actively producing a contraction), and therefore does not measure the pure sensory modality of proprioception (Gandevia et al., 1992, 2006).

Discussions of lower motor neurons as it pertains to electromyographic assessment can be found in Chapter 9. The focus of that chapter is on α-motor neurons; however, consideration could be given to γ- or even β-motor neurons. γ-Motor neurons are responsible for the innervation of intrafusal muscle, from which the muscle spindle is receiving its sensation. It therefore is able to modify the sensory input to the CNS by increasing or decreasing the sensitivity of the muscle spindle (Johansson et al., 1991). β-Motor neurons function similarly to γ-motor neurons but innervate both extrafusal and intrafusal muscle. Electromyographic assessment collects ensemble activation from all of these motor neurons, with α-motor neurons being the most prominent. There are very limited in vivo methods to understand the function of these other motor neurons, although consideration of joint stiffness and muscle tone could be considered. Since γ-motor neurons have a more consistent level of activation, they have the ability to modify muscle tone, or the baseline tension in muscles. A number of joint stiffness measurement techniques could quantify this, but caution should be used as the property of muscle tone would be sensitive to the speed of stretch (Needle et al., 2014).

12.3 METHODOLOGIES TO UNDERSTAND CENTRAL NERVOUS SYSTEM FUNCTION

Measurement of CNS function in biomechanics research has grown largely over the last several decades. Understanding the process of sensory integration and movement production has been a crucial step to advance the scientific understanding of movement and to generate new rehabilitation options for a range of musculoskeletal and neurological pathologies. However, there is often a trade-off between the recording accuracy and the functionality of the measurement techniques. For instance, functional magnetic resonance imaging (fMRI) provides unparalleled spatial resolution to observe CNS function; however, gross joint motion can lead to artifact that is disruptive to the MRI technology, and temporal accuracy is limited to the length of the scan. Alternately, electroencephalography (EEG) drastically improves temporal resolution and allows for some functional movement; however, generally at the cost of spatial resolution. Technological innovations are working toward improving the spatial and temporal resolution, especially during functional tasks. Therefore, consider the following section as a guide for the broad use of these technologies, in the context of the current evidence.

12.3.1 Functional Magnetic Resonance Imaging (fMRI)

12.3.1.1 Overview of Imaging Techniques. Generally speaking, the gold standard for assessment of cortex activation comes from magnetic resonance imaging technology, in the form of fMRI. While using magnetic fields to reorient and subsequently measure energy release from hydrogen ions similar to typical magnetic resonance imaging, fMRI quantifies changes in cerebral hemodynamics between resting and functional states. This will quantify the hemodynamic response function (HRF) using a blood-oxygen-level-dependent (BOLD) contrast. Therefore, the "cortical activation" described through fMRI represents changes in cerebral blood flow, which is a correlate of the electrical firing of neurons.

Because of the demands of sugar and oxygen during neuronal firing, a hemodynamic response allows for an understanding of cortical activation. As oxygenated and deoxygenated hemoglobin have differing magnetic properties, their response to magnetic field alignment can be detected using MRI, most commonly using T2* weighted acquisitions (Buxton, 2009). Therefore, the BOLD signal increases as a function of blood flow, blood volume, and blood oxygenation throughout the cortex in relation to neuronal activity.
in the signal's acquisition that must be addressed in

fMRI offers a high level of spatial resolution that allows researchers to identify activation within specific regions and structures of the cortex. However, this comes with two notable limitations. First, temporal resolution is limited, as full scans last one to several minutes long and detect the peak HRF seconds after it actually occurs. Therefore, average activation during a task is explored relative to a control, resting period. Second, in line with the equipment demands, there are limitations to the level of functional tasks that can be implemented. The head must remain still throughout the duration of the task, limiting the amount of movement that can be performed, and any implements used to provide stimuli must be nonmetallic, due to the use of magnetic fields.

12.3.1.2 Experimental Design and Data Collection. While fMRI acquisition will differ based on the stimulus type and movement paradigm incorporated, there are some standard features. As with other physiological techniques, efforts must be made to maximize a signal-to-noise ratio. In the case of neuroimaging, signal strength can be increased with equipment considerations. The majority of scanners implemented in research to date have utilized a scanner magnet between 1.5 to 3 T in strength, with scanners up to 7 T currently available. A stronger magnetic field is able to increase the BOLD signal, though at the cost of possibly increased artifact. Further, radio frequency (RF) coils, responsible for detecting the energy release, have an impact on image quality, with more channels yielding higher quality through both spatial and temporal resolution. Most commonly, 20-, 32-, or 64-channel RF coils are utilized. While these equipment considerations will increase the signal quality, there are numerous considerations to minimize the amount of noise. While several processing steps will serve this function (described below), stabilization of the head is generally seen as the key aspect in decreasing extraneous noise, such that signal quality is improved without changing the MRI hardware itself.

Prior to considering technical configurations and settings of the scan itself, thought must be put into the task and experimental design. While resting-state fMRI (rs-fMRI) can be collected to study functional connectivity between cortical regions at rest, most fMRIs used in biomechanical research are task-related in nature. These tasks could be structured into blocks, in which a task is performed continuously for 15–30 seconds, with a period of rest between blocks (Figure 12.4a) (Soares et al., 2016). These blocked approaches represent the most frequently seen design in biomechanical research, as they offer robust power to detect changes. Limitations of this approach include the ability of a participant to habituate to the task, and a limited ability to link changes to specific aspects of the task. For instance, in a knee flexion and extension task in which a participant straightens and flexes their knee 10 times in a block, it is unclear whether the cortical activation observed is tied to the knee extension or flexion component. An alternate option is event-related designs, in which a brief (0.5–8 seconds) stimulus (e.g. sensory pulse, passive movement) is provided, separated by interstimulus intervals of rest, with analysis attempting to track the HRF over time (Figure 12.4b). While this increases the temporal resolution of fMRI, there is a smaller signal-to-noise ratio, and a higher number of trials are needed to adequately power and detect task-related changes (Soares et al., 2016).

The selection of blocked or event-related designs is largely dependent on the type of task or stimulus being investigated. Within biomechanics research, there is a challenge to minimize head movement, while having a participant generate movement and/or providing a sensory stimulus. There are three common paradigms used to understand cortical activation during movement tasks in biomechanics research using fMRI. First, simple, isolated movement – often in blocked designs – can determine cortical activation patterns. This approach requires conscious and careful stabilization of the head as even ankle movement can be associated with movement artifact at the head. Thus, a higher number of trials might be needed to ensure enough volumes of high-quality images are obtained. Second, the use of a projector in combination with mirrors or specialized goggles can be used to present images or videos to participants depicting movement and measuring response. Often, these can be used to provide first-person presentation of joint movement, similar to virtual reality. However, as this approach does not have participants initiating motion, the signals recorded are largely

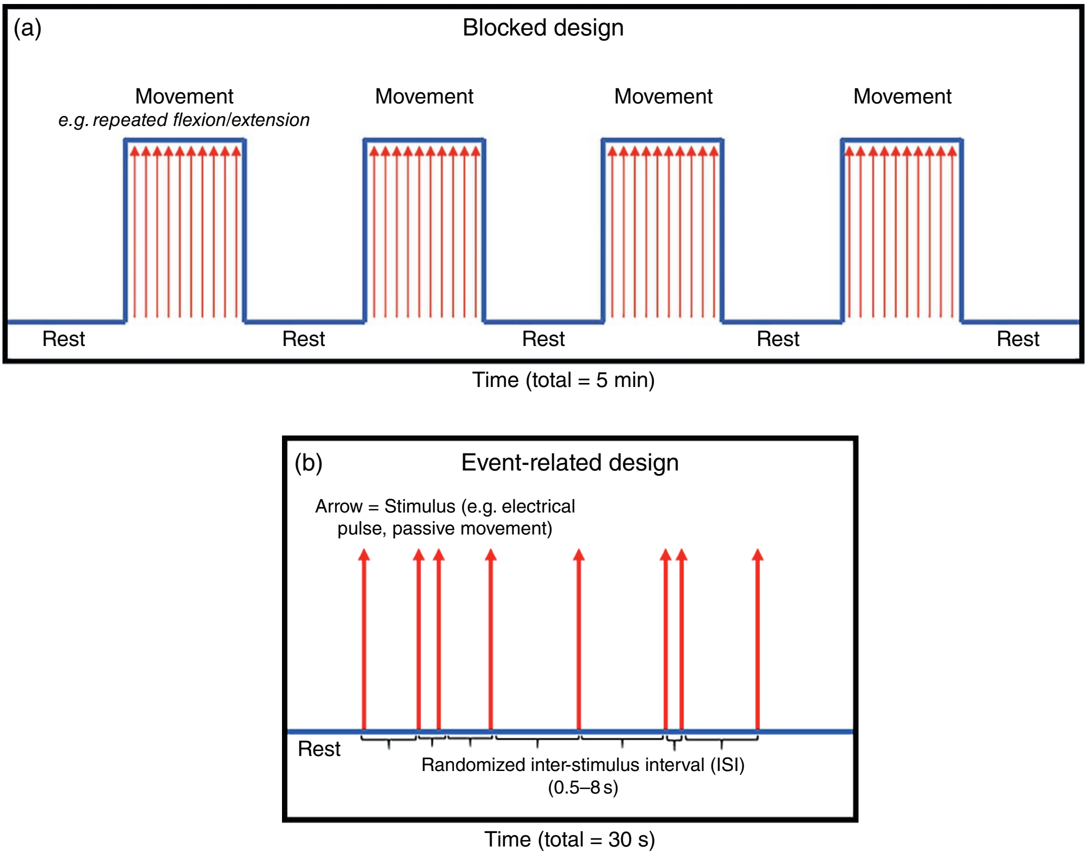

Figure 12.4 (a) Example of blocked research design. Stimulus is provided or movement is performed continuously across a block of time and compared to a rest period. (b) Example of an event-related research design. Stimuli are provided with varying inter-stimulus intervals (ISIs).

(i.e. mirror neurons) and not direct pathways. Additionally, these approaches are largely dependent on ensuring a participant focuses on the images or video presented. A final approach utilized is motor imagery, where participants are guided to picture themselves producing movement (or performing a sport-specific task). The limitations of this approach are similar to the use of image projection, as recordings are dependent on the motor resonance system and require participant focus, with the additional consideration of standardization of imagery instructions.

With equipment and study design determined, the final considerations are technical configurations and settings of the scan. Gradient- or spin-echo planar imaging using $T2^*$ weighted MRI acquisitions is the most common for obtaining whole brain acquisition of cortical activation, though other options are available (Soares et al., 2016). Each fMRI image consists of more than 100,000 voxels that cover a 2.8–3.5 mm^3 area, with a series of 2D images combined to create a volume (whole brain). Therefore, a number of volumes are collected over time to create a time series, with the time between volumes referred to as the *repetition time* (TR). With this understanding, we can consider some key decisions, such as the gap between each slice within a volume, the sequence in which slices may be obtained, and determining an appropriate TR. While the spatial resolution is improved with a smaller gap between slices, a higher number of slices can lead to interference and spin history effects that contribute to signal noise. Shorter TRs have the potential to improve temporal resolution and could be preferred in event-related designs; however, these shorter times can increase the sensitivity to movement artifact and therefore decrease signal-to-noise ratio. One final, and key constraint toward

TABLE 12.1 List of Common Steps Performed During fMRI Analysis

Step	Description
Image stabilization	Eliminating the first several seconds of a scan to remove inconsistent magnetic gradients
Slice-timing	Corrects for repetition time (TR) across images through interpolation to the first slice or main image
Motion correction	Realignment of each slice to a reference volume, including translations and rotations
Spatial transformation	Use of linear and nonlinear methods to align images from individual space to a referenced standard
Spatial smoothing	Spatial smoothing across a 5–10 mm space
Temporal filtering	High pass filtering at 0.008–0.01 Hz
First-level analysis	Locating voxels that differ between the rest periods and task periods for an individual subject
Second-level analysis	Locating voxels that differ between the rest periods and task periods across a sample

reflecting the amount of time between the excitation of an ion by magnetic fields and the detection of that signal, and will typically range from 25 to 40 ms.

12.3.1.3 Processing and Analysis. While a number of sources of artifact can be addressed through data collection considerations described above, a number of steps must be taken during preprocessing to optimize the signal-to-noise ratio (Table 12.1). A number of software pipelines and workflows are available, with the steps described subsequently serving as a guide for what may be considered, and further reading is recommended to better understand the parameters of each step (Poldrack et al., 2011; Soares et al., 2016). *Image stabilization* will often be done by eliminating the first 10 seconds of any scan, as magnetic gradients may be inconsistent during this time period. *Slice-timing* is a technique to normalize the potential offset of TR between images and will interpolate each image to either the first image or a mean slice. *Motion correction* will consist of the realignment of each slice to a reference volume. This may consist of linear steps including x, y, and z translations and rotations, as well as the process of Scrubbing, where structured noise can be assessed and removed from the analysis. While translations and rotations are part of motion correction, these will be further done during a *spatial transformation* step of fMRI processing. Spatial transformation will use both linear (translation, rotation, scaling) and nonlinear (warping and distortion maps) methods to align the images from an individual's space to a standard reference template. For instance, Talairach space, or the Montreal Neurological Institute (MNI) template represents standardized models, associated with atlases to identify notable structures (e.g. automated anatomic labeling, AAL; Harvard-Oxford Atlas). These spatial transformations are therefore necessary for standardizing across patients and studies and allowing for detected differences to be associated with a specific cortical region. Finally, a degree of filtering is warranted, both spatially and temporally. Spatial smoothing might occur across a 5–10 mm space for group analyses, while temporal filtering might consist of a high pass filter at approximately 0.008 to 0.01 Hz.

Once processed, data must undergo two levels of analysis. The first-level analysis is used to determine activated areas of the cortex during the task, when compared to resting blocks, within an individual subject. The second-level analysis consists of compiling data across subjects, in order to determine the task-related changes that occur across all subjects. Caution must be used for these analyses, as more than 100,000 voxels of data may be compared, raising the risk of Type I error regarding findings (Lindquist, 2008). There are several strategies to subsequently decrease this risk. For instance, many studies may look at activation across regions of interest (ROIs), which would compile activation of voxels within an the number of comparisons. However, this is reliant on a series of a priori ROIs that would be targeted. Alternately, correction to the error rate may occur, including a family-wise error rate (FWER), representing a conservative option akin to a Bonferroni correction; or by correcting the false discovery rate (FDR) that corrects *p*-values to correct for the risk of false positives, which is considered less conservative.

A large number of software packages are available that can aid investigators with data processing and first-level analyses. Analysis of Functional NeuroImages (AFNI) is a tool supported by the National Institutes of Health that provides tools for smoothing, normalization, and mapping of data, as well as statistical analyses. The FMRIB Software Library (FSL) is a publicly available toolkit that provides visualization, normalization, and mapping tools, and is well equipped for SEM-based analyses. A third, though not final, software package is statistical parametric mapping (SPM), a tool that facilitates SPM-based whole-brain analyses of fMRI.

A final step to fMRI analysis includes comparison across groups or time points in a research study. Though *t*-tests, analyses of variance, and multiple regression are frequently implemented, caution is often utilized to minimize the chance of type I error, particularly when performing voxel-level analyses. At the minimum the level of significance is often lowered to 0.001; however, this can still lead to 100 voxels providing erroneous data. Therefore, voxel-based thresholding or cluster-extent-based thresholding are often used to lower the rate of false discovery, with the former considered more conservative (in line with a Bonferroni correction) (Lindquist, 2008; Soares et al., 2016).

12.3.1.4 Other Imaging Techniques. fMRI represents only one form of advanced imaging technique for understanding activation of the cortex. While serving as a gold standard in providing spatial resolution of cortical activation, other MR-based technologies can offer insight into CNS function and are finding their way into biomechanics research. Diffusion tensor imaging (DTI) uses MRI technology to track water molecules within the central nervous system over a period of time (5–20 minutes). These data can then be analyzed with fiber tractography to generate a structural map of white matter within the brain. The width can provide a basis for connectivity within the cortex, and the atrophy of white matter may be associated with neurological diseases and even musculoskeletal injury (Soares et al., 2013).

MR spectrography (MRS) determines the chemical composition within a single voxel, allowing investigators to determine the neurometabolites (Rhodes, 2017). MRS is typically a static scan studying single voxels; however, whole-brain imaging can occur using chemical shift imaging (CSI), allowing for investigation into ROIs in a relatively quick time. MRS can also occur during functional tasks (fMRS) and allow for tracking of metabolites over time (as fast as ~three seconds). These imaging techniques have seen limited to no use in biomechanics, but the ability of MRS, CSI, and fMRS to determine changes in neurotransmitter activity could have notable benefits for studies implementing motor tasks, and changes throughout learning.

12.3.2 Electroencephalography (EEG)

12.3.2.1 Overview of Electroencephalography. EEG is a method of recording the electrical potentials generated by populations of neurons from the cerebral cortex. These local potentials may result from action potentials used for transmission within neurons or postsynaptic potentials, which reflect the relatively slower process of ion channels opening or closing. Due to the rapid time scale and propagation of individual action potentials, it is difficult to record this process from scalp electrodes. Since postsynaptic potentials lead to relatively slower changes in local voltage, it is possible to record these changes in voltage when large populations of neurons are active and there is a summation of their individual postsynaptic potentials. It is important to remember that EEG is recording the electrical activity of tens of billions of cortical neurons which are constantly communicating locally as well as across regions of the central nervous system. The voltages recorded from the EEG are small, typically measured in

Positive or negative fluctuations in voltage that are measured at the scalp electrode represent activity that: (1) comes from large populations of neurons that are simultaneously active, (2) is oriented in a similar direction with respect to the measuring electrode, (3) comes from post-synaptic potentials arising from the same part of the neuron such as the dendrites or cell body, and (4) the net current flow (e.g. positive or negative) is sufficient enough to overcome opposing currents (Cohen, 2014; Luck, 2014). These criteria are met by many, but not all, processes that occur within the brain. Since EEG is recording electrical potentials generated by neurons, it is one of the few neuroimaging modalities that directly measures cortical activity as opposed to fMRI which measures change in the local uptake of oxygen.

The summed electrical potentials that makeup EEG combine to produce multidimensional data. For example, when cortical neurons respond to an external stimulus such as a perturbation, electrical activity may evolve over *time*. There may be shifts in *frequency*, where an oscillatory brain rhythm becomes more or less dominant in the EEG such as alpha (8–12 Hz) during periods of alertness or rest. There is also *space*, as the EEG recorded from each electrode will differ owing to the proximity of the electrode to the cortical sources and vectors of the electrical potentials. This phenomenon is best observed when eye blink artifact is present in the EEG as it will have a greater magnitude in more rostral electrodes closer to the eyes and the magnitude will decrease in more caudal channels. Information is also contained in the *power* and *phase* components of the overall electrical activity that can be extracted through time-frequency decomposition. This data can be leveraged to test hypotheses pertaining to dynamic brain activity. Accompanying this multidimensionality are many analytic approaches that can be applied to EEG data. It is therefore critically important to have clear scientific objectives when beginning any research project.

12.3.2.2 Experimental Setup. A basic EEG system consists of a recording computer, an amplifier, electrodes, and an EEG cap which houses the electrodes. The electrodes can be either active electrodes or passive electrodes, with active electrodes preamplifying the cortical signal before reaching the digital amplifier and therefore helping to minimize certain artifacts. EEG caps should be fitted to the participant based on their head circumference to ensure a snug, but not uncomfortable fit. Most EEG caps have precise electrode spacing according to the international 10-20, 10-10, or 10-5 system of electrode spacing (Figure 12.5). This allows for consistency across participants during analysis efforts focusing on an individual or array of electrodes. Efforts should be taken to minimize electromagnetic noise (e.g. cell phones, wireless routers) and avoid signal contamination (similar to EMG), which can be accomplished by collecting data within a Faraday cage. The laboratory climate (temperature and relative humidity) should also be closely controlled to minimize the occurrence of static electricity discharge which may impart "pops" or square waveforms into the recorded EEG. The air should not be so dry as to promote the buildup of static electricity and not so warm that it leads to increased scalp perspiration. Attention should be paid to minimizing any environmental distractions both within the laboratory as well as the immediate surroundings when possible as these events can further contaminate the signal. By minimizing environmental sources of noise/data contamination as well as limiting any extraneous sensory, cognitive, or perceptual processes for the participant, one gains more confidence that the cortical activity being recorded is the result of experimental conditions and/or manipulations.

One of the most important factors for EEG data analysis is synchronization. Changes in neural activity without context (e.g. movement performance) lead to speculative conclusions that are not fully supported by the data. Therefore, it is critical to ensure that the EEG is accurately synchronized with stimuli (auditory, mechanical, visual) or other types of data (EMG, inertial measurement unit (IMUs), kinematics, etc.). Most EEG amplifiers allow for both analog and digital inputs, enabling researchers to use either custom-made synchronization devices or built-in synchronization output signals from their hardware. Software solutions exist as well, with Lab Streaming Layer, an open-source software package that
and synchronization across many common hardware

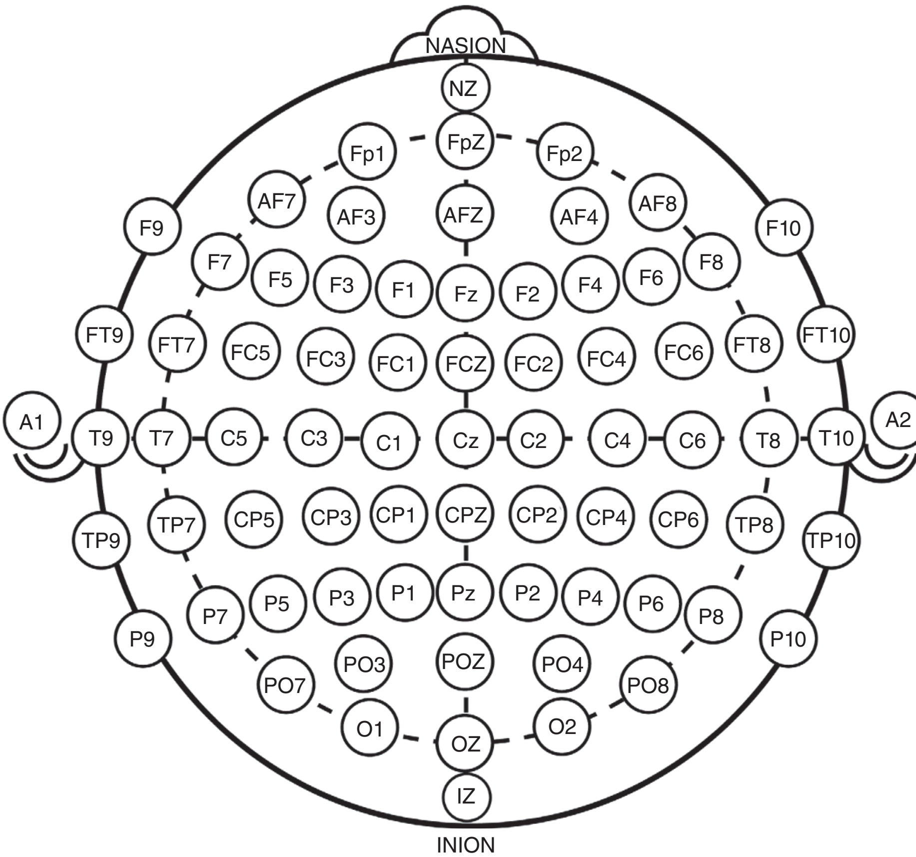

Figure 12.5 The 10-10 system is an international labeling system for describing electrode locations during EEG research. It allows for standardized electrode locations based on the distance between electrodes. The distance is measured between the nasion and inion, as well as the distance between the ears at the tragus. Electrodes are spaced based on 10% of that distance, with the Cz electrode at the midpoint of those two lines. The letters refer to the lobe of the cortex, and reference channels are often placed on the mastoid process, earlobes, or nose and are labeled as "A" or "M." The 10-20 and 10-5 systems may also be adapted for lower and higher density electrode arrays with 20 and 5% distances between electrodes, respectively.

12.3.2.3 Processing and Managing Electroencephalographic Data. It is very unlikely that an experiment will be conducted that collects EEG data without artifacts. The nature of EEG recordings makes them very susceptible to contamination with movement artifacts, line noise with equipment, perspiration, and many more. Artifacts can even occur from processing unexpected environmental stimuli (startle) or natural bodily functions like swallowing saliva. Taking the time to pilot test helps to identify potential sources of noise and minimize them. While a number of analysis techniques will aid in the reduction of extraneous noise, the less noise in the originating signal will lead to ultimately cleaner data, and allow an adequate number of trials to be maintained. Since many biomechanics experiments inevitably record movement, artifacts resulting from movement will be one of the largest causes of contaminated or even unstable data. To ensure the highest signal-to-noise ratio, artifacts in the data need to be addressed, through filtering, manual or automated artifact rejection, or artifact correction. Each experiment will likely use a unique approach to offline EEG processing; therefore, it is critical that the exact steps and order of operations are reported clearly when disseminating the results of an experiment.

Filtering is useful for attenuating artifacts in the data that
Processing the data with a 50 Hz or 60 Hz notch filter can

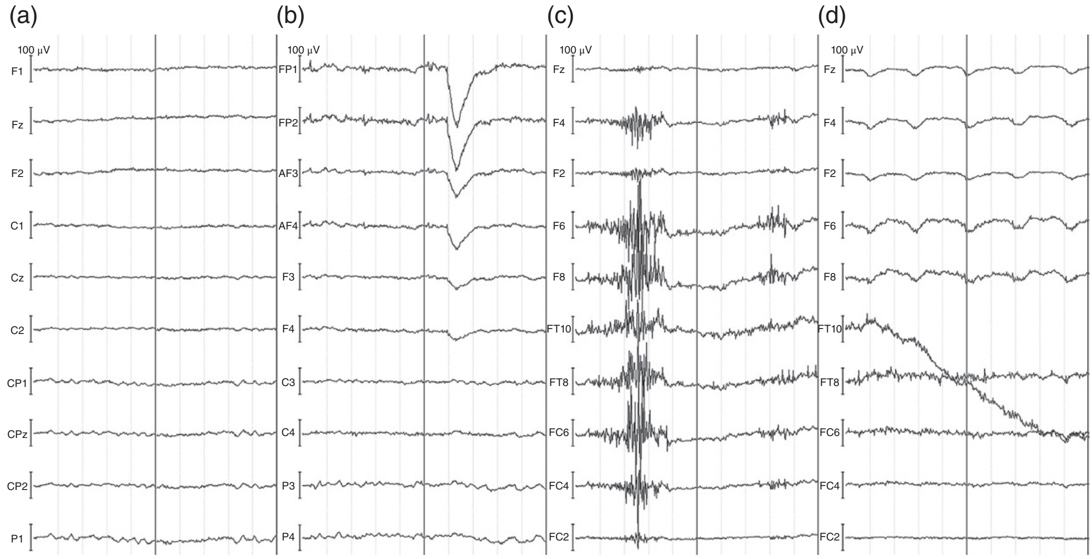

Figure 12.6 This figure shows three common artifacts that may be encountered when recording EEG. All panels and EEG channels have a scale of ±50 μV. Panel (a) shows normal EEG clean of artifact. Panel (b) shows an artifact caused by an eye blink and demonstrates how the amplitude of this artifact decreases as channels move further from the source. Panel (c) shows EMG artifact caused by neck muscle activity. Panel (d) shows drift that can occur due to local perspiration under an individual electrode.

high-frequency artifact. High-pass filtering is a useful technique for addressing signal drift that may occur over time within the amplifier, or because of local skin potentials due to perspiration or changes in scalp conductance. Band-pass filtering is commonly used in EEG postprocessing prior to running automated or semi-automated artifact identification, rejection, or correction. Band-pass filtering also allows for specific bands of activity to be extracted for further analysis such as event-related (de)synchronization (ERD/S), or spectral power density. Since filters create an edge artifact due to the transformation applied to attenuate the nondesired frequencies, the beginning, and end of epochs may render the data unusable. Recording an additional 30–60 seconds before a testing block is recommended to limit this edge artifact.

Filtering struggles to attenuate the impact that transient, stereotyped artifacts have on the EEG (Figure 12.6). Once common artifacts have been identified (e.g. eyeblinks), the next step is to find a way to more directly measure this artifact. Many EEG amplifiers allow for simultaneous collection of online digital or analog auxiliary channels. The electrooculogram (EOG) is used to measure and monitor the potentials created by eye movement and blinks; accelerometers can be used to monitor perturbations that can create microscopic movements of the EEG electrodes, and EMG can be used to monitor muscular activity of the head and neck. Most EEG analysis software packages include several options for manual, semi-automated, and fully automated artifact detection. Once an artifact is identified there are several options. Artifacts can be rejected, where the segment of data or epoch containing the artifact is removed from the analysis. Artifacts can be corrected, using one of many different algorithmic approaches that remove the artifact and attempt to reconstruct the underlying EEG. It is important to remember that each corrected artifact is a transformation of the original EEG data; the signal to noise ratio is being altered and it is possible that endogenous EEG activity is being removed any time an artifact is corrected. A shift toward independent component analysis (ICA)-driven artifact correction has led to less contamination of the data during artifact correction and results in improved data quality. Pairing ICA with artifact-specific sensors such as EOG allows for a much more accurate removal of artifacts with less risk of removing endogenous brain activity

TABLE 12.2 Overview of Common EEG Analysis Techniques and their Neurophysiologic Mechanisms

EEG Technique	Measurement	Neurophysiologic Mechanism
Event-related potential	Activation from a single or series of electrodes in response to a stimulus (e.g. postural perturbation, initiation of movement)	Stimulus ascends through spinal tracts to various areas of the cortex (stimulus specific) and creates a patterned response of activity observed when many trials are averaged together
Power Spectral Density	EEG is measured continuously through a task (e.g. balance, force matching, etc.). Spectral decomposition is used to determine the spectral power of specific frequencies at each electrode	Delta (0–4 Hz), theta (4–7 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30 80 Hz) oscillations are associated with varying degrees of cortical activation and deactivation. For instance, alpha frequencies represent inhibitory thalamocortical activity, whereas delta frequencies reflect excitatory cortical activation
Event-related (De) synchronization	Continuous EEG recordings are measured relative to an event (e.g. applied joint load, initiation of movement, etc.). Time- and phase-locked changes allow investigators to determine frequency-specific changes relative to a stimulus	Combining the above techniques, specific frequency bands are assessed relative to a specific event. An increase (ERS) or decrease (ERD) in that band's power relative to a baseline period prior to the event determine event-related excitatory or inhibitory changes

12.3.2.4 Time-Domain Analysis: Event-Related Potentials. One of the most common techniques in EEG research is measuring the event-related potential, or ERP (Table 12.2). This analytic technique is recommended for testing hypotheses relating to differences in cortical processing in response to experimentally manipulated stimuli. The high temporal resolution of EEG can be leveraged to evaluate how cortical activity unfolds as it relates to a time-locked event. ERPs are often measured using outcomes that reflect the magnitude or timing/latency of a potential. For example, you may be interested in determining if there is a difference in the magnitude of a motor potential prior to a preplanned movement with varying levels of complexity or effort and the peak value could be compared between conditions. An averaged ERP also often has multiple components that are named due to their polarity or latency with respect to the stimulus or order of components (Figure 12.7).

Processing EEG data to measure an ERP uses the simple analytic approach of averaging together many trials to increase the signal-to-noise ratio. Since the recorded EEG represents the sum of all cortical activity, it is very difficult to extract meaningful information from an individual trial. The signal (the induced cortical activity as the result of an action or stimulus) is masked by other ongoing processes. By using a precisely measured temporal event code and averaging together dozens (sometimes hundreds) of trials, the ERP will be revealed. Therefore, it is important to ensure that consistent and reliable criteria for creating event markers are used. Events may be software-driven or hardware-driven, based on input or criteria from other measurement techniques like a force plate or EMG. Since proper synchronization is important in this method, the accuracy, precision, and reliability of the other measurement modalities should be taken into consideration when planning an experiment and identifying an appropriate event trigger (Luck, 2014).

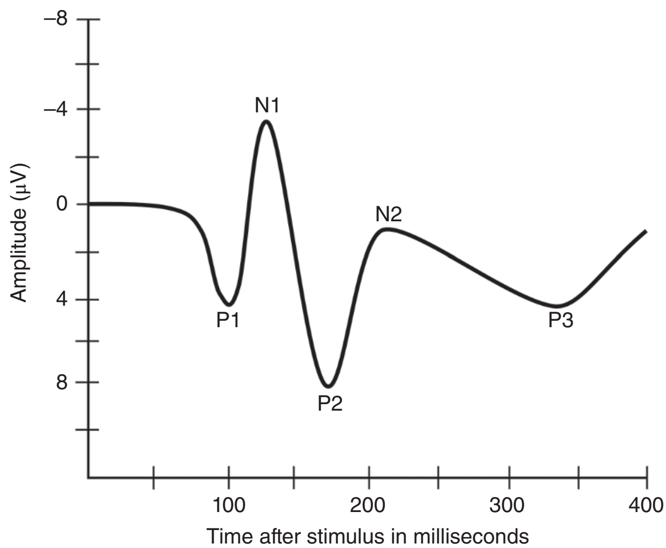

Figure 12.7 An example ERP trace reflecting five individual peaks, or components. ERP components are often named for their directional polarity as well as the order they occur, or the latency at which they are observed. In this case, P1 could also be named P100. ERPs are conventionally posted with negative values upward.

12.3.2.5 Time-Frequency Analysis Approaches. Decomposing any signal such as EEG into the time-frequency domain provides researchers with an ability to explore many complex phenomena in the cerebral cortex. For more information on time-frequency decomposition methods such as Fourier and wavelet transforms, see Chapter 2. The decomposition method is an important factor that can significantly impact both the temporal and frequency resolution of your analysis. In the following section, we will focus on analyses that utilize spectral power. Each analysis method on its own, much like the decomposition methods, is complex and may have analysis-specific parameters to be considered (Cohen, 2014).

12.3.2.6 Power Spectral Density. Spectral power provides an overview of the summed magnitude of activity at a given frequency (or range of frequencies) within a window of time. Individual frequencies are not often assessed in EEG research, primarily due to differences between individuals in how their brain processes information. Therefore, a band-based approach is often used, sometimes called waves (e.g. delta band/waves), wherein a range of frequencies is assessed in order to determine changes in activity. It is important to clearly define the upper and lower frequency limits that are being analyzed in the study, as these may differ from research group to research group. Examining differences in signal power is useful for determining if frequency band-specific activity is different between conditions of an independent variable. A limitation to using spectral power to test broader hypotheses in biomechanics comes back to a core principle of EEG: volume conduction. The EEG signal will contain all the ongoing sensory, motor, perceptual, and cognitive processes, therefore clean data that has minimal artifacts is critical. While it may be possible to mitigate these influences with a high signal-to-noise ratio and subtracting baseline power from a reference period (see below, event-related synchronization), it is difficult to draw concrete conclusions using spectral power alone.

12.3.2.7 Event-Related Synchronization. Determining changes in event-related signal power is similar to the ERP; however, it requires the signal to be decomposed in order to integrate frequency information. This analytic approach has several common names, commonly referred to as event-related spectral perturbations, or ERD/S. When can generate postsynaptic potentials, which can add to the magnitude of activity at a given frequency. Neurons also communicate information through discharge rate (e.g. frequency), and thus these two components can combine and reveal information about changes in power at a given frequency (or band). These changes in power are often used to reflect an increase or decrease in synchrony among neural populations. Therefore, we can use ERD/S outcomes to test for changes in cortical activity with respect to an event or stimulus. The practical considerations for calculating ERD/S are very similar to those when designing an experiment that measures an event-related potential. ERD/S is a referential outcome, meaning it reflects how change in power unfolds in a window of time across the given frequencies with respect to a baseline period. Since changes in oscillation reflect a change in a dynamic signal, there may be a longer refractory period required for the brain to return to its original state. For example, there is a prolonged change in oscillations that occurs following the completion of a simple discrete movement task; if one were to repeat multiple trials of the same movement over and over, a short interval between trials would contaminate a referential baseline period with postmovement neural activity. Since the baseline period is also being decomposed, you should use the same parameters for both the baseline and period of interest. The signal-to-noise ratio also has implications for ERD/S analyses; however, since the temporal resolution is relatively lower for time-frequency analyses than event-related potentials, you often do not need as many trials. Many studies will report averaging between 50 and 100 trials per condition, and much like conducting ERP studies, you should a priori set the required number of trials/epochs and report the number of trials/epochs averaged (Pfurtscheller and Lopes da Silva, 1999).

Careful attention should be given to the part of the experiment that is used for the baseline/reference period. This is important when calculating ERD/S, or comparing power spectral density between conditions of a continuous task such as balance. For instance, if an experiment manipulates variables during different gait conditions then it would make more sense to use a baseline gait condition to capture the normal spectral activity that occurs during gait; this would allow for differences in cortical activity between conditions to be better evaluated. If using a reference period from a quiet, seated baseline period prior to the gait conditions, then the change in spectral activity would reflect both the difference in activity between sitting and gait, as well as any experimental conditions that were manipulated. Therefore, when designing an experiment, it is important to factor in comparative conditions and allow for ample time to get a good signal-to-noise ratio for both the experimental manipulations of interest as well as any baseline/reference periods.

12.3.2.8 Source Localization with Electroencephalography. A difficult question to answer with EEG is identifying the brain region that is generating the signal being recorded. To answer this question of source localization, the head is first modeled and then the EEG signal is decomposed. Once the head is appropriately modeled, the *forward solution* is calculated; the forward solution refers to what the scalp EEG would look like given a known cortical source. The forward solution is critical for the next step in the processing pipeline when the signal is decomposed to try to identify the sources generating the potentials recorded in the EEG. This is referred to as the inverse solution, which aims to identify the most likely cortical source of a signal based on the EEG. Finding the *inverse solution* is a computationally intensive process. Many factors can dictate how much each active cortical region contributes to the scalp EEG being recorded, including current strength, number of neurons active, orientation of the dipole moment, conductivity of tissues between the source and electrode, and many more that are covered in more detail in a comprehensive guide by Michel and Brunet (2019).

There are several approaches that can be used to estimate the brain region that is most likely responsible for the measured scalp EEG. Dipole localization was one of the first methods described; this method relies on nonlinear optimization equations to identify the best mathematical location of a single or a small number of dipole moments that are influencing the EEG. This method is often used with time-locked changes in cortical activity such as ERPs or ERD/S. A major assumption with this method is that there is only a given time, which may not be the case during most biomechanical applications. The other two main methods are nonadaptive and adaptive distributed source imaging. These methods differ from the earlier dipole localization methods in that they do not assume a small number of sources, instead relying on thousands of potential sources in their forward solution model. In both methods, each location and electrode have a weighting factor that is used to determine the likelihood that the EEG signal originated from a given source in this 3-dimensional space. Nonadaptive methods such as low-resolution electromagnetic tomography (LORETA) can offer rapid calculation of the likely distribution of cortical sources at a given point in time. One drawback to using nonadaptive methods, such as LORETA, is these methods have relatively lower spatial resolution compared to other methods, where the data may appear to look like a blurry fMRI (Pascual-Marqui et al., 1994). This may make it difficult to precisely identify sources if that is the primary goal of the analysis. Adaptive distributed source imaging methods differ in that these models assign weights to the EEG data itself in addition to the weights of the electrode positions. Such methods rely on ICA or principal component analysis (PCA) decomposition to determine the relative weight that each component contributes to the signal. This additional factor is used by these models to solve the inverse solution and identify sources. One common method is called beamforming, which shares the resolution issues of LORETA, but is more sensitive to the dynamics of the EEG signal due to the integration of signal weights. This added information could in theory improve the ability to detect components or ERPs of interest that have a smaller magnitude. A drawback to using adaptive methods such as beamforming is the large number of parameters that can significantly impact the results of your analysis (Cohen, 2014; Michel and Brunet, 2019).

Caution should be taken when evaluating results from an investigation using source localization due to many of the same factors that affect EEG data collection in the first place. There are many steps required when conducting these analyses (Michel and Brunet, 2019). While it is theoretically possible to achieve high spatial resolution, these would occur under situations where individual participant-level MRI scans are known, electrode locations are precisely recorded, head and electrode movement do not occur, the scalp and skull evenly conduct potentials, and many more. Using a high-density (>32 electrodes) montage is recommended as a minimum to facilitate the decomposition of the EEG signal, which may improve the ability of your software to resolve the likely cortical sources and/or dipole locations.

12.3.2.9 Other Analysis Methods for Electroencephalographic Data. There are methods for EEG analysis that are not covered in this text. Coherence is a common measure that can be used to determine the relative relationship or consistency between two signals in time and/or phase. Coherence can be applied to activity occurring at ROI, to look at connectivity, or even to peripheral measures to determine the relationship of cortical activity to peripheral efferent signals. Corticomotor coherence is an example that can show the strength of coupling between motor unit output/recruitment and cortical rhythms. Coherence can also be used to provide estimates of brain connectivity. By evaluating the phase-locking properties of signals from different regions and the evolution of the EEG signal in the time domain, it is possible to use EEG to estimate the dynamic coordination of brain regions. There are several other methods for determining network connectivity including Granger causality that uses the ability of one signal at predicting the next signal, mutual information uses linear and nonlinear approaches to detect coupling between regions or electrodes and is an area of continued algorithmic development and validation. There are dozens of approaches that can be used when analyzing electrophysiological data, especially with EEG. Careful thought needs to be applied when determining the best analysis approach to test your hypothesis, and not fall victim to post hoc over-analysis of your data. It is always best practice to pilot test both your data collection protocols, as well as, your data analysis procedures and approach.

12.3.3 Neural Excitability

12.3.3.1 Overview of Neural Excitability. Neural excitability describes the recruitment and depolarization properties of motor neurons in response to external stimuli. Whereas EEG and fMRI are generally a trade-off between spatial and temporal resolution of cortical activation during a task, measurements of neural excitability describe a largely separate component: the amount of energy needed to elicit a depolarization of a motor neuron, which is indicative of the resting membrane potentials of that neuron. Further, excitability can also determine the increases in size of motor responses in relation to the size of the stimulus, often providing a measure of maximal excitability that indicates the total amount of the neuronal pool available. Clinically, neurons that have higher resting membrane potentials (higher excitability) would be easier to recruit quickly and possibly effectively during functional tasks. Similarly, neurons that have larger available neuron pools and better recruitment can lead to larger and stronger muscle contractions during activity.

The counterpart of neural excitability is neural inhibition, or in the case of some of these measurement techniques, intracortical inhibition. Generally speaking, the application of external energy will activate neurons, leading to a measurable response in the periphery. However, in the case of the cortex, while stimuli will activate descending excitatory neurons, it will also activate inhibitory neurons in the cortex. These inhibitory neurons could potentially decrease the excitability observed through the motor evoked potential (MEP); but alternately can be better understood through techniques that measure the relative silence following a magnetic pulse, or the effects of precisely timed conditioning pulses on the MEP. The clinical impact of intracortical inhibition is less clear, although some relationships with levels of resting muscle tone have been observed.

The implications of neural excitability are not quite as clear as the previous techniques, where brain activation is recorded during functional tasks. Neural excitability is most often collected at rest (though not exclusively), and instead informs the researcher about properties of the nervous system that may translate to movement.

12.3.3.2 Methodologies to Assess Neural Excitability. Assessment of cortical (or corticospinal) excitability is typically performed using *transcranial magnetic stimulation* (TMS). These devices utilize dual magnetic fields applied through a double-coil that is placed over the skull. The magnetic energy from these coils (often up to 2.0 T) is able to painlessly travel through the scalp and skull and generate a response in a target muscle. There are several types of TMS seen in the literature, with the focus of this chapter being single-pulse and dual-pulse TMS, often considered *diagnostic* TMS. Also observed is repetitive TMS (rTMS) or deep TMS (dTMS), which differ in the frequency of pulses and coil types and are used for *therapeutic* purposes rather than for understanding neural excitability (Groppa et al., 2012).

For both single- and dual-pulse TMS paradigms, protocols will follow a similar procedure. Outcome measures will be tracked with electromyography electrodes placed over the target muscle(s). Participants should be familiarized with TMS and the sensations it provides by placing the coil in the approximate location over the skull, and slowly increasing the intensity of the coil to the desired intensity, or until a motor response is visible. For investigations into lower extremity function, the coil should be placed just anterior and lateral to the vertex of the skull (CZ or C1/C2 of the 10:20 system, See Figure 12.5), while recordings from the upper extremities should be placed at approximately the C3/C4 or C5/C6 locations contralateral to the target muscle. The specific hotspot of the target muscle should be located by providing stimuli in a 2.5–5 cm^2 radius while observing recordings of electromyography from the targeted muscle. These responses, termed MEPs, represent compound muscle action potentials in the target muscle occurring 20–100 ms after the stimulus onset. The hotspot should be identified as the location that provides the largest MEP at the lowest intensity possible. There are newer methods to identify the hotspot; however, they require additional technology. For instance, systems can combine motion analysis cameras with a headband worn by the participant, and an image of the subject's brain (from MRI) or a standard reference patient. While requiring more equipment and expertise, these methods provide a greater amount of intersession reliability.

Once the hotspot is identified, protocols will vary based on the outcome measure being targeted. Often, investigators will attempt to identify the resting motor threshold (RMT) or active motor threshold (AMT). While several methods exist to measure this, the difference between RMT and AMT is whether the participant is relaxed (RMT) as pulses are applied or whether there is a level of muscle facilitation (AMT), with 10–20% facilitation being common. Common options for determining motor thresholds include the "5 of 10"' approach, whereby 10 pulses are administered at gradually increasing intensities until a visible MEP is observed on 5 of 10 pulses. Alternately, a range of pulses may be applied across a range of intensities, and the inflection point where motor responses are observed could be inferred as the motor threshold (Groppa et al., 2012; Hallett, 2007; Needle et al., 2017).

While the RMT and AMT provide information about the depolarization threshold of intracortical neurons, several additional measures can provide additional insight regarding the recruitment of additional neurons. MEP size can be quantified at a given percentage of the RMT or AMT, with or without muscle facilitation. Further, a stimulus-response curve can be obtained by providing pulses across a range of intensities ranging from below the RMT to above a maximal response (Figure 12.8). Multiple pulses should be applied at each intensity, with an increasing number of pulses leading to a more accurate and reliable curve. The curve of MEP size to stimulus intensity can then be fitted to a modified Boltzmann equation using a Levenberg–Marquardt algorithm, where y represents the MEP size and *x* represents stimulus intensity:

$$y = \mathrm{MEP}_{\min} + \frac{(\mathrm{MEP}_{\max} - \mathrm{MEP}_{\min})}{1 + e^{m(I_{50} - x)}} \tag{12.1}$$

Equation 12.1 above allows for multiple potential dependent measures. While minimum MEP ($\mathrm{MEP}_{\min}$) represents potential EMG noise or background activation, maximum MEP ($\mathrm{MEP}_{\max}$) provides a maximum amount of the muscle that can be recruited through cortical activation and can be normalized/compared to a maximal voluntary contraction or direct nerve or muscle stimulation. Additional parameters include the 50% intensity (I_{50}), indicating the stimulus intensity where MEP size is 50% of the maximum and the peak slope (m), which represents a parameter that increases or decreases proportionally to the slope of the curve. The variable e represents Euler's number. The I_{50} and slope together provide valuable information regarding the additional recruitment of neurons throughout the curve, although the lowest reliability is noted for the slope parameter. Further single-pulse TMS measures that inform cortical excitability include the central motor conduction time, representing the conducting delay. This measure should be compared to the delay from peripheral nerve stimulation, to determine the delay from cortical activation and no other factors such as electromechanical delay (Devanne et al., 1997).

While TMS is typically not utilized to map brain function, cortical mapping has been done to determine motor function (Lüdemann-Podubecká and Nowak, 2016). In these protocols, the coil is systematically moved along a grid while the same sized pulse is provided. The area and intensity of responses across an area can be assessed to understand how brain function changes in response to a medical condition or intervention. To increase the accuracy of this measure, the use of technology that incorporates motion analysis with the coil is encouraged, as it allows for more systematic movements of the coil.

Lastly, single-pulse TMS can be utilized to investigate one measure of intracortical inhibition: the *cortical silent period* (CSP). This requires the collection of MEPs during facilitated contraction (typically 10–20% of maximum effort). As the single-pulse stimulates the excitatory neurons (generating a MEP), inhibitory neurons are similarly activated, and those mediated by $\mathrm{GABA_B}$ as a neurotransmitter can be observed as a period of relative silence following the MEP, lasting between 150 and 400 ms, termed the CSP (Kimberley et al.,
of more intracortical inhibition that is mediated by GABA

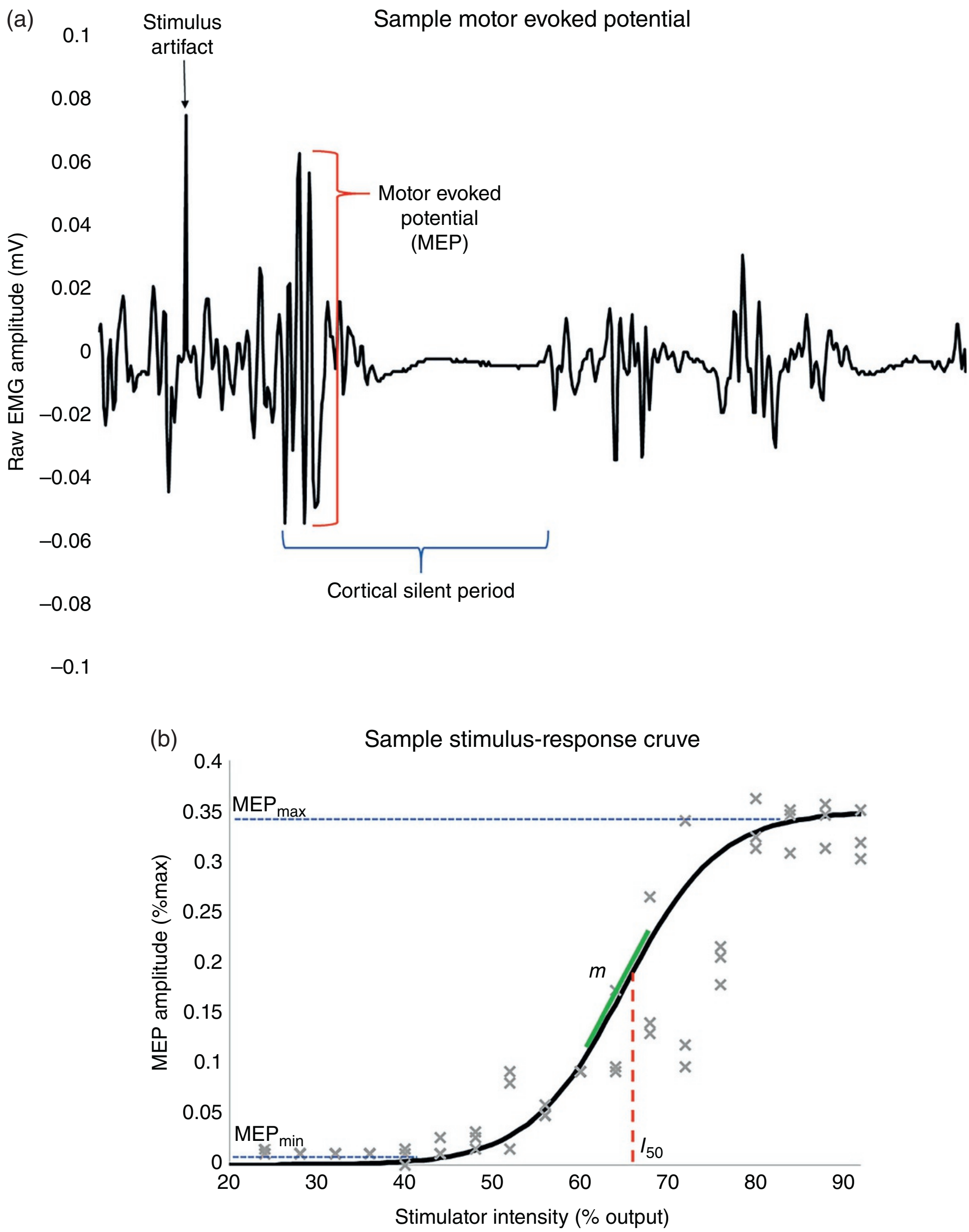

Figure 12.8 (a) Sample motor evoked potential (MEP) during a facilitated trial. (Stirling et al. (2018)/With permission of Elsevier.) (b) Stimulus-response curve of MEP size versus stimulus intensity. The curve can be used to extract maximal MEP size (MEP_{max}), minimal MEP size (MEP_{min}), slope (*m*), and the intensity at 50% of the peak (I50). (Adapted from Needle et al. (2013).)

Dual-pulse TMS differs from single-pulse TMS in that it utilizes dual capacitors, allowing for customized timing from two pulses provided through the same coil. There are three measures typically extracted using this technology: short-interval intracortical inhibition (SICI), long-interval intracortical inhibition (LICI), and intracortical facilitation (ICF). All of these paradigms utilize a conditioning pulse, followed by a precisely
informing intracortical inhibition from $GABA_A$ (SICI),

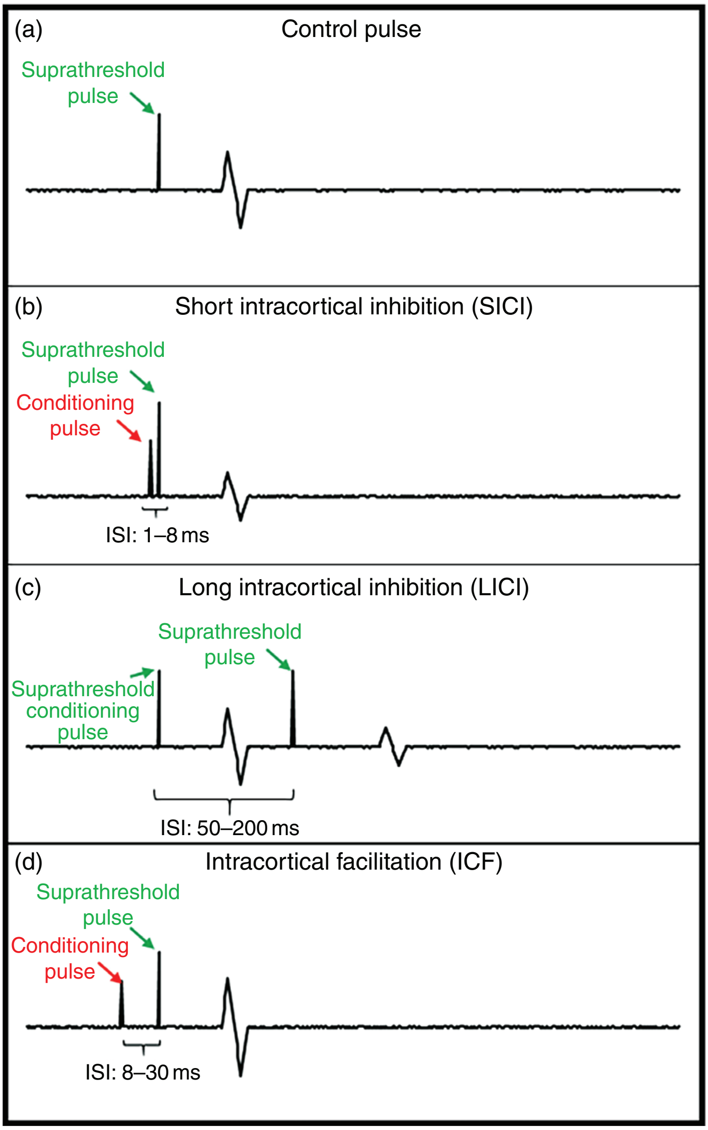

Figure 12.9 (a) Example of a normal motor evoked potential (MEP) from a single-pulse of transcranial magnetic stimulation. (b) Short-interval intracortical inhibition. A sub-threshold conditioning pulse is provided 1–8 ms before the test stimulus. The relative decrease in MEP size compared to single-pulse is a measure of inhibitory activity. (c) Long-interval intracortical inhibition. A suprathreshold conditioning pulse is provided 50–200 ms prior to the test pulse. The decrease in MEP amplitude from the first to second pulse is a measure of inhibitory activity. (d) Intracortical facilitation. A sub-threshold conditioning pulse is provided 8–30 ms prior to a test pulse. The increase in MEP amplitude compared to single-pulse is a measure of intracortical facilitation.

NMDA receptors (ICF). In the case of SICI, a subthreshold conditioning pulse (70–80% RMT) is applied 1–6 ms prior to a suprathreshold test stimulus with the reduction in MEP size describing the inhibition occurring. In the case of LICI, a suprathreshold conditioning pulse is provided 50–200 ms prior to a suprathreshold test stimulus, with the reduction in MEP size describing the inhibition occurring. Finally, ICF is quantified by measuring the *increase* in MEP size with a subthreshold conditioning pulse applied 8–30 ms prior to a suprathreshold test stimulus (Figure 12.9) (Chen, 2004; Ridding et al., 1995).

TMS is used to understand the connection between the cortex and muscle, and measures are often described as corticospinal excitability since the stimulus must travel down the corticospinal tracts to elicit a response at the measurement site. Therefore, it is often done in conjunction with peripheral electrical stimulation, often in the form of the Hoffmann reflex (H-reflex). The H-reflex involves stimulation to a peripheral nerve with a similar measurement along a muscle through electromyography. The nerve should be located by providing supra-threshold pulses (1 ms width, square wave), and observing where the largest response can be observed at the lowest stimulation intensity. Pulses should then be applied at that location with gradually increasing intensities. The electromyographic response from these stimuli can be divided into two categories: a direct response occurring 5–25 ms after the stimulus (M-wave), and the reflexive response observed 25–70 ms after the stimulus (H-wave) (Hoffman et al., 2003; Hoffman, 1922). During a typical collection, low-intensity pulses will yield an increase in H-wave amplitude as sensory neurons that transmit muscle spindle activation are larger in diameter and will trigger a reflex, akin to the stretch reflex. As intensity increases, M-wave amplitude will increase as the stimulation threshold for motor neurons is reached, while H-reflex amplitude will decrease due to antidromic inhibition. Stimuli are provided every 10 s until a maximal response is observed on the M-wave, as indicated by a plateau in the sigmoidal stimulus-response curve (Figure 12.10). The most common outcome from these measures is the ratio of maximum H-wave to maximum M-wave (H_{max}:M_{max}), which is considered a measure of the excitability of the reflexive motor neuron pool (Hoffman, 1922).

Additional outcome measures can be assessed using H-reflex methodology that could provide insight into segmental mechanisms. Similar to TMS, investigating the size of motor responses following the application of a precisely timed conditioning pulse is the key to understanding these mechanisms. Paired-reflex depression (PRD) uses a conditioning pulse and test stimulus of 15–25% of M_{max} intensity, spaced 80 ms apart. Alternately, recurrent inhibition, proportionate to activation of Renshaw cells within the spinal cord, can be obtained using a conditioning pulse at 25% of M_{max} intensity, and a test stimulus at the M_{max} intensity, with pulses provided at both 10 and 80 ms inter-stimulus intervals (ISIs), with the difference between these conditions indicating recurrent inhibition (Earles et al., 2002; Oza et al., 2017).

Collectively, measures of cortical and reflexive excitability and inhibition can offer insight into the mechanisms that may be contributing to altered motor output, and/or provide mechanistic data regarding the effects of various interventions on motor function. However, caution should be used when exploring these components in isolation, as large individual variability may be observed across populations, and can vary relative to arousal level, recent exercise, or other extrinsic factors.

12.4 PERIPHERAL NERVOUS SYSTEM MEASUREMENT TECHNIQUES

To fully understand the role of the CNS during movement, one must also consider peripheral nervous system activity. Much is understood about motor activation within the central nervous system, as electromyography is the gold standard for understanding alpha motor neuron activation (See Chapter 9). However, there are a number of additional techniques to understand both sensory and motor peripheral nervous system function.

12.4.1 Nerve Conduction Studies

As opposed to electromyography that quantifies the amplitude and onset of activation within the muscle, nerve conduction studies can be used to determine the timing and activation along the path of a peripheral nerve. As such, these can be investigated both in the context of sensory or motor stimuli. Similar to electromyography, surface (or sometimes needle) electrodes are placed along the pathway of a nerve. In the case of a motor study,

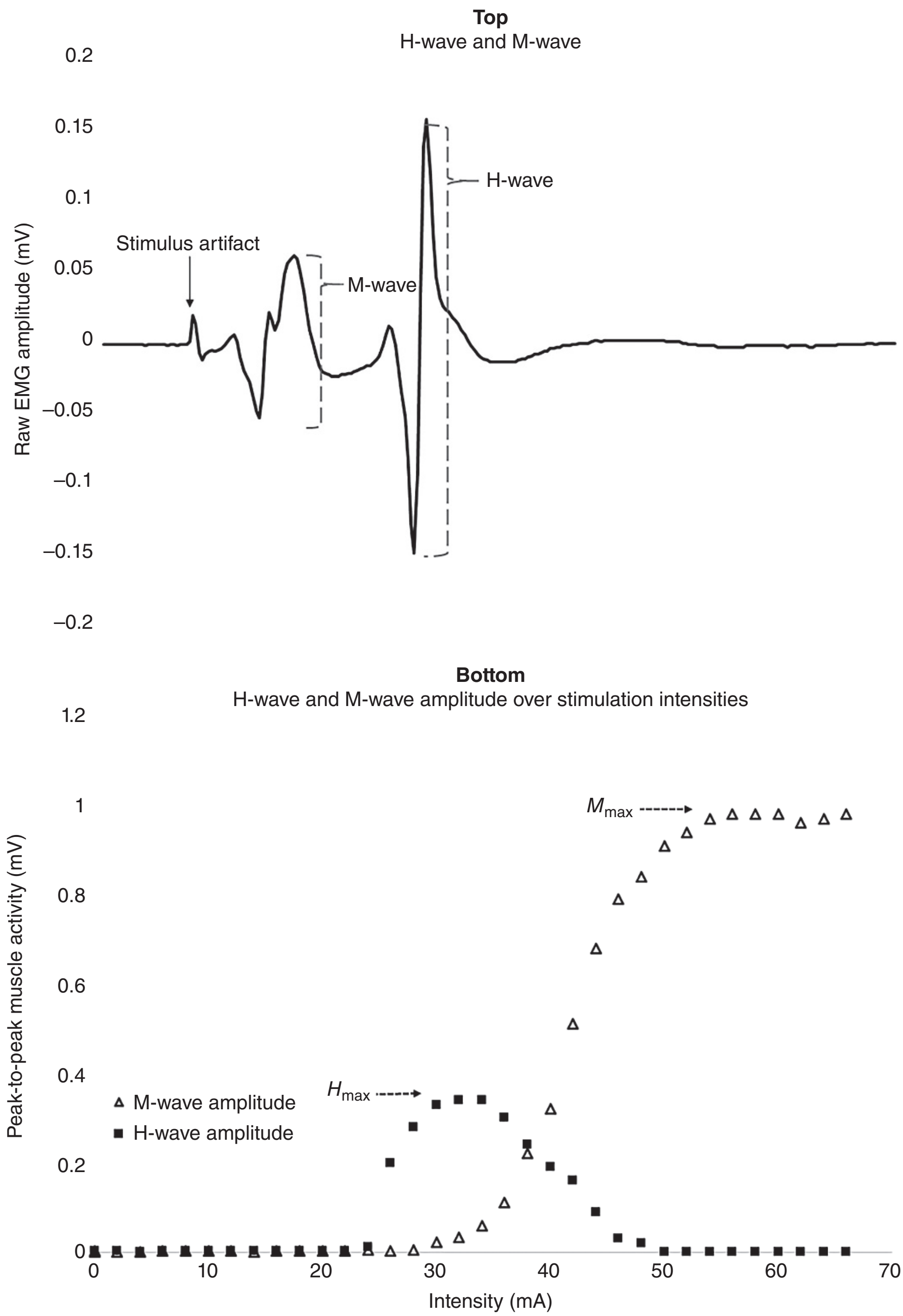

Figure 12.10 Top: Sample response from peripheral nerve stimulation. The M-wave indicates the direct response from the muscle as the electrical pulse stimulates the motor nerve. The H-wave indicates the reflexive response as the afferent nerve is stimulated and initiates a reflexive contraction of the muscle. Bottom: A sample response of H-wave and M-wave amplitudes as stimulus intensity increases. (Stirling et al. (2018)/With permission of Elsevier.)

brief electrical pulse, similar to that of the H-reflex, while *compound motor action potentials* (CMAPs) are collected from sites along the nerve path, quantifying the amplitude and onset of the response. Stimuli are often applied across a number
to understand where a potential decrease in amplitude or

sensory nerve conduction studies rely on stimulation at the distal end of the nerve distribution, while measuring a similar response, termed the *sensory nerve action potential* (SNAP) across a series of sites along the nerve (Mallik and Weir, 2005).

Deficits in CMAP or SNAP amplitude and timing are mostly associated with nervous system disorders (e.g. spinal cord lesion, multiple sclerosis) but can offer additional insight into the potential sources of delayed or decreased activation that may be observed through broad EMG testing, or understanding proprioceptive deficits.

12.4.2 Microneurography

A limitation of understanding sensory function in individuals is that most proprioceptive tests (e.g. joint position sense, kinesthetic awareness, force matching) rely on motor responses to infer changes in sensory traffic. One of the few methods to directly understand sensory nerve activity is through microneurography, consisting of direct recordings of the nerve through a microelectrode placed directly into the nerve fascicle. This technique is often utilized to understand sympathetic outflow and its relationship to cardiovascular function but offers several advantages to understanding sensory neuron activity (Hagbarth and Burke, 1977).

The microneurographic technique first utilizes ultrasound imaging or electrical stimulation to locate the approximate location of a nerve. Then, two tungsten microelectrodes are inserted into the subject – a reference electrode in nearby skin and an active electrode at the nerve location. The electrodes are connected to a preamplifier and subsequently a nerve traffic analyzer to amplify the signal. Once inserted, the active electrode will be manipulated by the investigator until it is situated within the nerve (noted through a discharge signal) and is recorded from the target population of neurons. Muscle spindle afferents (i.e. Type Ia, II) are the most common neurons studied using this technique, although some investigations have explored Golgi tendon organ afferents (Type Ib) and some have even been performed in Type III/IV afferents. The afferent signal must then be confirmed to be originating from its source. For muscle spindle afferents, this consists of (1) responding to passive stretch (2) silence during shortening/muscle twitch, (3) responding to squeezes of the tendon/muscle belly but not skin brushes, and (4) responding to vibratory stimuli (Mano et al., 2006). This signal may not only be collected in its raw form, with wavelet decomposition utilized to extract individual fiber recordings; but also be collected as an ensemble, where filtering similar to EMG can help to determine the amplitude and timing of responses (Mano et al., 2006; Needle et al., 2013).

The challenge with the utilization of microneurography in the context of biomechanics research is the difficulty of coordinating this signal with sensory stimuli and/or movement. The inserted electrode is often very easy to dislodge, even with the stretch of the skin, and therefore only small, controlled motions can typically be implemented. A further methodological implementation is the potential difficulty in obtaining the desired signal, as the more time spent searching for the appropriate signal raises the risk of experiencing residual paresthesia following the protocol.

12.5 METHODOLOGIES TO UNDERSTAND CENTRAL NERVOUS SYSTEM BEHAVIOR AND ENVIRONMENTAL INTERACTIONS

12.5.1 Virtual Reality

As described in the above sections, many of the techniques for assessing CNS activation in biomechanics research suffer a trade-off between functional movement and accuracy of recording, as minimizing movement is required to obtain high-quality signals. Virtual reality, often abbreviated as VR, is a broad term used to reflect computer-generated immersive experiences that represents a way to improve our ability to understand how the CNS interacts with the environment. In research, virtual reality is a strong tool for the motor and perceptual consequences of these visuomotor transformations. For example, if someone stands still while observing a roller coaster in VR, they may unintentionally begin leaning backward to account for their perceived forward momentum. Due to the manipulation of optic flow in the above scenario, our brain perceives motion and tries to counteract that movement. This is the foundation of most VR-focused experiments in biomechanics: leveraging the immersive capabilities and temporal resolution of stimulus presentation to observe changes in motor output.

12.5.1.1 Types of Virtual Reality. VR is most often associated with video games and head-mounted screens that fully immerse the user in a 3D-rendered scene. Most experiments in VR will require a custom-coded environment; therefore, it is recommended to consult or include a software programmer in the research team. There are several types of VR that can be used in research. Head-mounted VR relies on a headset with precisely shaped lenses that refract the light from individual computer displays in a manner that creates a three-dimensional projection of the image that the computer is generating (Figure 12.11). When using a head-mounted VR headset, the participant will be fully immersed in the scene portrayed on the screen and will not have any visual cues from their immediate environment to rely on, making them useful if the goal is to provide a focused visual stimulus. For research purposes, most head-mounted VR headsets (Figure 12.11) will require a powerful computer capable of rendering the 3D environment in real-time with minimal latency, often coded using the Unity. One drawback to these methods is that the individual screens used to generate the 3D environment lead to a limited vertical and horizontal field of view, which may limit the external validity of the findings of a study. Normal human vision has 200–220° of horizontal field of view compared to 75–110° in head-mounted VR headsets; vertical field of view is also limited, with normal vision approximately 130° compared to 90–100° when using a VR headset. It is also possible to use multiple projectors (at least 2) and large screens to create a virtual environment. This method has a few advantages to head-mounted VR for biomechanics experiments and is the most used approach for gait studies that manipulate visual inputs. The common setup for projection-based VR is a treadmill that is placed a few meters in front of a curved projection screen or series of flat projection screens (Figure 12.12). Projection-based VR does also limit the field of view of the participant, but unlike head-mounted VR, the immersive field of view is limited by the height and width of the projectors and their distance from the participant. Despite the limitations in how much of the field of view can be rendered virtually, this setup allows the

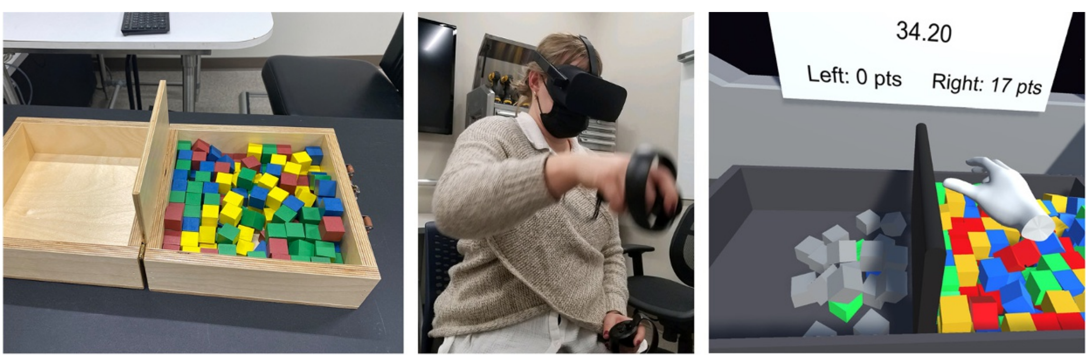

Figure 12.11 Head-mounted virtual reality can be used to recreate a physical environment. The left panel shows the physical task that requires moving blocks over a barrier from one side to the other. The middle panel shows the seated participant interacting with the virtual environment. The right panel shows the participant performing the same task as the physical task in virtual reality.

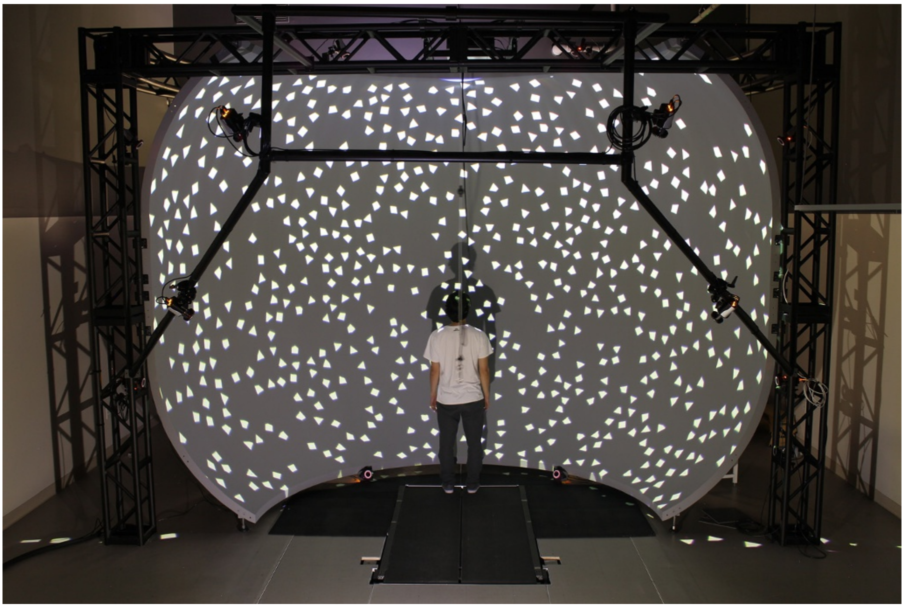

Figure 12.12 Projection-based virtual reality utilizes a screen, sometimes curved, and one or more projectors to generate an interactive environment. This method is commonly combined when motion analysis is a primary measurement of interest to researchers. (Photo provided with permission from Fernando Santos of Bertec LLC. Laboratory pictured is Bertec Immersive Labs, CoBal Lab at the University of Delaware.)

participant to walk and still perceive cues from their peripheral vision for a more natural gait, as opposed to head-mounted VR which blocks out all visual input other than the screen. Another form of virtual reality that can be used for research purposes is augmented reality (AR). The major difference between AR and VR is that there is no 3D virtual environment, rather the current environment the participant is in is supplemented with a digital overlay. Therefore, AR is not an optimal choice if the goal of the research is to fully manipulate and control the visual scene. Much like head-mounted VR, a headset, or set of specialized goggles is used for AR, but the participant can clearly see through the lenses and observe their surroundings.

12.5.1.2 Technical Considerations with VR. There are several important factors to consider with research that uses VR. From a software side, it is important to ensure that the 3D-rendered environment is ecologically valid, meaning that 3D-rendered distances between objects and their digital dimensions are portrayed in a realistic manner. This will help to improve the participants' immersion and perception in the environment so they can focus on the experimental tasks. This is accomplished by working with skilled software programmers with experience in the 3D engine and pilot testing iterations. One of the most critical steps in a VR experiment is familiarizing the participant with VR. Most people have not experienced virtual reality, and the novelty of interacting with a virtual environment (projection VR) or a 3D scene responsive to their head movements (head-mounted VR) can significantly impact the focus and attention a participant is able to afford to the experiment. Familiarization also serves to identify and potentially limit several other confounding factors that will be outlined in the following sentences. Motion sickness is a common complaint in both head-mounted and projection-based VR, which can occur acutely and transiently. Due to the potential for transient motion sickness, frequent breaks should be taken during VR experiments; long-term effects can also occur such as migraines. If a participant reports any motion sickness they should be withdrawn from the study for their safety, and individuals with a history of light sensitivity, vertigo, vestibular conditions (e.g. Meniere's disease), migraines, or other conditions that may interact poorly with VR should be excluded from the study. If the experiment requires the participant to perform a specific task in VR, they should also be familiarized with the task to minimize early learning/practice effects. It is good practice to report the standardized procedures and tasks that are used in the experiment, as familiarization should be sufficient to mitigate the potential negative effects of the novelty of VR environments.

There are also a few hardware and laboratory setup factors that should be considered during VR research. Since VR is a stimulus presentation modality, experiments will likely be collecting other forms of data such as kinematics. The headsets that are used for head-mounted VR also contain accelerometers and IMUs which are used to help track head rotation so the user can interact with the virtual environment. Experiments focused on the neural control of movement may also be collecting EEG or functional near-infrared spectroscopy (fNIRS) to measure cortical activity. Since EEG and fNIRS require specialized caps that have precise electrode/optode locations, you should ensure that the straps that secure the VR headset to the user do not interfere with the ability to collect high-quality data on CNS function. Cable management is also another factor that could interfere with data collection, as additional modalities (EEG) or the head-mounted VR itself often require cables that could alter the way the participant moves. All of these factors can be addressed through careful planning and in-depth pilot testing prior to beginning an experiment.

12.6 NERVOUS SYSTEM ROLE IN MUSCLE SYNERGIES

One of the major issues the CNS faces when planning for/executing movement or responding to perturbations is how to appropriately activate muscles to generate the required torque. Muscle synergies are a way of describing neuromuscular behavior that attempts to reduce the degrees of freedom that need to be accounted for during human movement. Synergistic muscle activity describes a group of muscles working together to complete a movement. Most muscle synergies described in the literature are stereotypical patterns of muscular activity that underly a behavior, such as gait. One interpretation of muscle synergies is that these patterns are carried out by subcortical structures, allowing for the cortex to override when movement pattern flexibility is needed, such as avoiding stepping in a puddle while running or hitting a tennis forehand while running. Synergies can also be used to understand how the CNS organizes a reflex-driven response to a support perturbation such as a translation or slip. In this instance, a group of muscles works together to accomplish a common goal, such as the posterior leg muscles contracting together to generate a posterior shift of the center of mass following a support perturbation. The responses to perturbations or threats to stability are used for insight into how pathology or injury may influence feedback neuromuscular control. Although there is evidence both in support of and opposing the role of muscle synergies, this may be a contentious topic. You can read more on the role of muscle synergies by Tresch and Jarc (2009). In this section, we will outline the experimental requirements and basic analysis strategies for muscle synergy research.

12.6.1 Measurement Techniques and Experimental Setup

The technique of measuring and evaluating muscle synergies relies on EMG. Detailed information on how to prepare participants for EMG data collection and appropriate surface electrode placement can be read in Chapter 9. First, it is important to record data from a large number of muscles (12 or more), which adds dimensionality to the data. Ideally, the entire limb should be recorded, as muscles often coordinate and work synergistically across multiple joints. Both agonist and antagonist muscle groups should be recorded as well. The EMG data should be appropriately synchronized with all additional sources of data such as perturbation platforms, force platforms, 3D motion capture systems, accelerometers, or any other equipment that is being used to measure the performance of the motor task. Perturbation paradigms inherently place the participant in an unstable condition, therefore a harness should be used to ensure participant safety. Pilot testing is important in these experiments and will help you to fine-tune the magnitude and timing of perturbations. Perturbation research will also require a movable platform, which can vary greatly in cost; in general, the cost of the platform will rise as the number of controllable degrees of freedom increases. Muscle synergies are also of interest to researchers during common movements, especially stereotyped movements such as gait, weightlifting techniques, and reaching tasks. Investigating the coordination of muscular activation can be of interest when researchers are concerned with fine motor tasks that involve target manipulation or interception with the upper and lower extremities.

Experimental design is a critical factor for muscle synergy research. As explained in further detail by Tresch and Jarc (2009), it is difficult to determine if most synergies reflect neural strategies for motor control or if they simply reflect the constraints of the task the participants are performing. One solution to this issue is to test the movement or perturbation of interest under a broad range of experimental conditions and magnitudes. The ironic solution to this issue can be accomplished by manipulating the task constraints such as perturbation magnitude or direction, as well as modifying the speed or force of movement. For example, having a participant bench press several different loads as a percentage of their 1-repetition maximum would allow for ample variation in task constraints to assist with identifying the most common underlying muscle synergies. In perturbation studies, steps should also be taken to limit anticipation of support surface perturbations, including randomizing the order of trials or using a computer to randomly trigger the perturbation. Due to between- and within-subjects differences in how they may perform a movement or respond to a stimulus, it is recommended to have participants perform at least 10 trials per movement/stimulus condition as an increased number of trials help to improve the accuracy of the analysis.

12.6.2 Analysis Techniques

Analyzing EMG data to evaluate muscle synergies does not require many steps. The first two steps are collecting and preprocessing the EMG data. Researchers should take a similar approach to setup and data management that are described in Chapter 9. Preprocessing also includes defining the post-event window (i.e. perturbation, movement onset) of EMG activity that will be analyzed. The multiple channels of EMG data (i.e. muscles) need to be decomposed to identify the underlying patterns of activity. There are several approaches that can be used. The most rudimentary approach is to calculate the onset latency of the given muscles and order them with respect to a time-locking event such as a perturbation or the gait cycle. More rigorous methods include ICA/PCA and nonnegative matrix formation. The goal of these analyses is to determine the underlying patterns of activity that make up the signals being analyzed. The ICA or PCA assigns a weighting factor to each identified component of activity analyzed through the signal. This step serves to identify the muscle synergies through mathematical inference. This approach is recommended as not all synergies have a specific pattern of activation in the temporal domain, and some muscles may co-activate to generate a specific joint torque rather than activate in an ordered fashion. The next step is to determine if the EMG data can be explained using these synergies. The data should also be evaluated for variability both between- and within subjects, especially when conducting hypothesis-generating experiments and observational research. This leads to the calculation of the variance accounted for, often abbreviated VAF, for which most studies will have an a priori cut-off score set to ensure the synergy reflects the between- and within-subject variability in the data. It is also possible to extract the temporal profile of muscle synergies when using nonnegative matrix formation techniques, which is beneficial during phasic and/or continuous tasks such as gait. A critical assumption underlying muscle synergy is a strong relationship with variables that represent movement or performance such as shifts in the center of mass or joint angles. This relationship should always be confirmed once the candidate synergies have been identified and will assist when interpreting the findings of the study (Tresch et al., 2009).

12.7 THE CENTRAL NERVOUS SYSTEM AND LEARNING, AND INJURY

12.7.1 Translation of Synaptic Plasticity to Motor Learning

In traditional contexts, motor learning is often inferred from performance, with the *retention* of that performance and the *transfer* of that performance outside of the learned setting serving as the key elements to infer motor learning has occurred. Further, the timeframe of "learned" adaptations allows for the inference as to whether learning is secondary to long-term (i.e. LTP, LTD) or short-term changes (e.g. PTP, presynaptic facilitation/depression) in synaptic plasticity.

The implementation of the nervous system measurement techniques described in this chapter increase the tools for investigators to infer whether learning has occurred. Most typically, as a task is introduced, it requires a large degree of cortical activation to coordinate and recognize the task. For instance, learning a new movement pattern requires distinct areas of the motor cortex to activate, large frontal cortex (e.g. premotor areas, dorsolateral prefrontal cortex) activation to plan and coordinate muscle activation, and perhaps visual cortex activation to visually track the motion. However, as a task is learned, the amount of activation decreases, as functional groupings may allow for coordinated muscle activation through less activation, and increased intracortical connections increase the efficiency of coordinating visual, vestibular, or other relevant control areas (Seidler, 2010). These types of changes are evident both in motor learning studies (tracking individuals as they learn a task) and in case-control investigations comparing experts in a task to novices.

The central nervous system measuring techniques described in this chapter can offer a number of ways in which we can gain understanding and confirm these effects. One of the most common potential decrements is decreased BOLD activation in ROIs through fMRI and/or changes in cortical activation patterns using EEG (e.g. increases in alpha, decreased theta/beta rhythms). While these represent efficiency in the form of decreased activation, several measurements may reflect the increased connectivity between areas of the brain. Resting-state fMRI can explore the activation of various cortical areas without any movement or stimulus, indicating a degree of functional connectivity between areas that might be responsible for coordinating movement. Further, DTI and subsequent tractography can describe hypertrophy in intracortical tracts, while MRS and fMRS can be used to track synaptic strength. Measures of cortical and reflexive excitability represent some of the most interesting results regarding motor learning – while other cortical areas are expected to decrease activation – the motor cortex as targeted by TMS may be likely to increase excitability, represent stronger connections to relevant muscle groups.

12.7.2 Role of Pathology on the Central Nervous System

One final key consideration in biomechanical investigations is the manner in which pathology affects central nervous system function in the context of movement. For this discussion, we can divide pathology to include direct lesions to the central nervous system, or upper motor neuron lesions (e.g. stroke, cerebral palsy, and traumatic brain injury.); as compared to injuries to the periphery (e.g. musculoskeletal injury). Upper motor neuron lesions have been studied extensively in the context of biomechanics research.

Upper motor neuron lesions represent a constraint (e.g. infarct) directly within the central nervous system that will typically impede movement in some way, that varies depending on the area of the cortex affected. Imaging and investigation of the central nervous system will observe decreased activation and/or excitability within the areas affected. However, given the plastic nature of the nervous system, activation from surrounding areas may increase and allow individuals to maintain some function. For instance, following a stroke, an infarct may decrease the ability of the motor cortex to appropriately activate during gait as well as decreasing corticospinal excitability; however, increased activation and excitability might be observed from the contralateral motor cortex. This likely represents strengthening of non-decussating fibers that activate the impacted muscles, in an attempt to maintain some level of motor control (Palmer et al., 2016). One final (and important) consideration in the nervous system function of individuals with upper motor neuron lesions is the common observations of spasticity or rigidity. These symptoms are due to a lack of inhibitory input from the central nervous system in regulating the reflex sensitivity of the muscles surrounding a joint, leading to a persistent stretch reflex loop. Most biomechanists would observe this symptom through increased muscle activation and joint stiffness, particularly when observing stiffness relative to the speed of stretch (Katz and Rymer, 1989). Excitability measures can elucidate these symptoms further, as reflexive excitability would notably increase in these individuals, but also disinhibition may be noted within the cortex, as observed through the CSP, SICI, and/or LICI (Stinear, 2017).

As opposed to pathologies directly impacting the CNS, recent evidence has highlighted a myriad of changes in the cortex and spinal cord that occur following musculoskeletal injury (Needle et al., 2017). As opposed to upper motor neuron lesions, there is no constraint within the CNS itself that limits movement; however, the injury may present a peripheral constraint that still forces CNS remodeling. For instance, most injuries are associated with an inflammatory response that leads to pain and swelling throughout the joint. These aberrant sensations decrease the quality of proprioceptive information that reaches the CNS and forces an individual to coordinate movement with this decreased input (Figure 12.13). Similar negative effects on proprioceptive feedback can be caused through deafferentation or increased laxity following ligamentous injury. While these injuries place a sensory constraint on the CNS, pain, and inflammation also negatively affect motor function through *arthrogenic muscle inhibition*, evident through decreased reflexive excitability following musculoskeletal injury, as well as models that simulate joint effusion (Hopkins and Ingersoll, 2000; Sonnery-Cottet et al., 2019). Therefore, observations have indicated that an acute injury makes activation of the stabilizing musculature more difficult due to direct inhibition, and more challenging to coordinate movement from faulty sensory feedback. As such, these factors force the central nervous system to reorganize in such a way that makes reinjury more likely over the 2–4 weeks following injury. For instance, with motor excitability decreased, activation of muscle utilizes activation from extraneous cortical areas (e.g. contralateral motor cortex, premotor areas, visual cortex), allowing for the motion to appear normal, but through atypical processes (Grooms et al., 2016). However, when outside of a laboratory or clinic, the *cortical spread* (i.e. increased activation of extraneous areas with decreased activation of the motor cortex) means that individuals have fewer cortical resources available for movement flexibility, and become more likely to be injured when under cognitive load, fatigued, or facing other external constraints (Burcal et al., 2019).

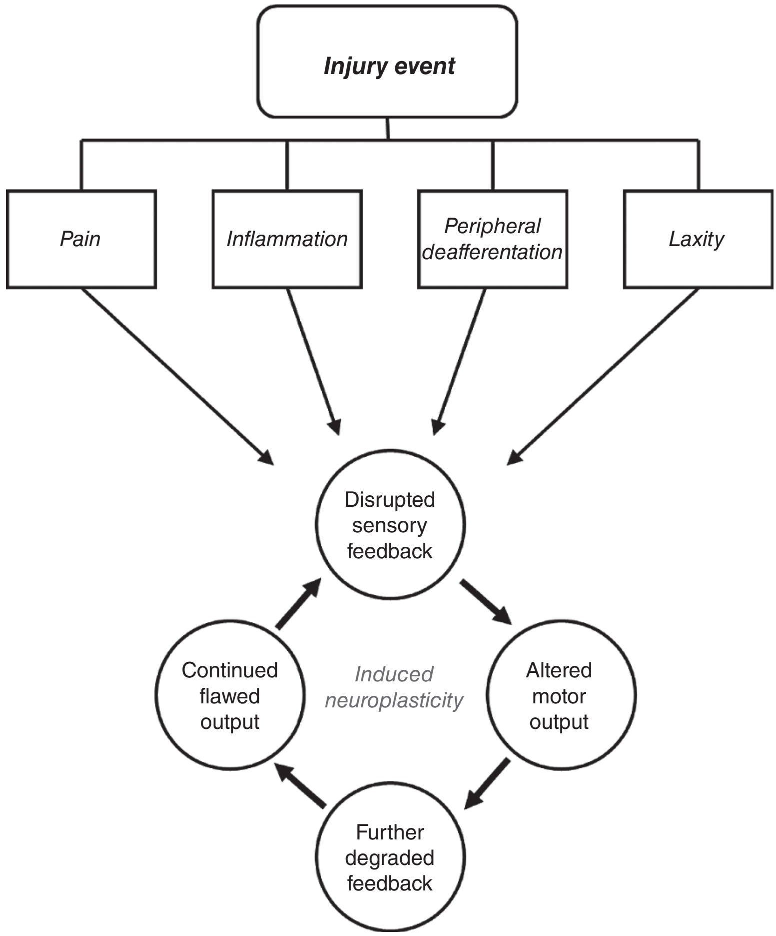

Figure 12.13 Paradigm of injury-induced maladaptive neuroplasticity introduced by Needle et al. (2017). Sensory aberrations secondary to acute and chronic injury perpetuate a cycle of flawed sensory input and motor output that contributes to subsequent instability and reinjury. (Needle et al. (2017)/With permission of Springer Nature.)

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13

A CASE-BASED APPROACH TO INTERPRETING BIOMECHANICAL DATA

Ankur Padhye[1], John D. Willson[2], Joseph A. Zeni[3], Kristen F. Nicholson[4], and Garrett S. Bullock[4]

[1]*Rehabilitation Sciences/Movement Science and Disorders Concentration, Department of Physical Therapy, East Carolina University, Greenville, NC, USA*

[2]*Department of Physical Therapy, East Carolina University, Greenville, NC, USA*

[3]*Department of Rehabilitation and Movement Sciences, Rutgers University, Newark, NJ, USA*

[4]*Department of Orthopaedic Surgery, Wake Forest School of Medicine, Winston-Salem, NC, USA*

13.0 PATELLOFEMORAL PAIN

13.0.1 Introduction

The patellofemoral joint (PFJ) includes the articulation of the patella and the trochlear groove of the femur. As a sesamoid bone within the quadriceps tendon, the patella increases the ability of the quadriceps to create knee extension torque by increasing the moment arm of the quadriceps muscles. When the knee is flexed, quadriceps contractions produce retropatellar compressive forces that can be several times a person's body weight, particularly during dynamic activities such as running. These large compressive forces are distributed across the PFJ articular surfaces as a function of knee and hip joint kinematics in the sagittal, frontal, and transverse plane. Repetitive exposure to a large compressive force, abrupt changes in training loads, or knee and hip joint kinematics that concentrate the distribution of PFJ force to a smaller area (increased PFJ stress) may contribute to patellofemoral pain (PFP), one of the most common musculoskeletal injuries among physically active populations. Unfortunately, PFP is not a self-limiting condition and symptoms frequently persist for years following the initial presentation until the proximate cause of the injury is addressed. Therefore, a thorough clinical and biomechanical evaluation of lower extremity (LE) mechanics during exacerbating activities is important to design an effective treatment protocol for PFP.

13.0.2 Case Description

The patient was a 35-year-old female with insidious onset of right anterior knee pain 18 months ago. She was an avid runner over the previous five years, progressively increasing her distance and intensity, but she was currently unable to tolerate more than a 3 mile run and her current

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

running volume was 12 miles/week. Her symptoms increased up to 7/10 with running for more than 10 minutes and increased with repetitive squatting, ascending and descending stairs, and prolonged sitting (>30 minutes). She denied experiencing instability, locking, numbness, or a history of trauma to either knee. Her Anterior Knee Pain Scale score was 73% (100% = no symptoms). She was given general LE stretching and strengthening exercises and foot orthoses by her general physician one year ago, but she experienced little benefit. Her motivation to attend physical therapy included reducing pain so that she can resume regular participation in running and compete in a half marathon that was six months away.

13.0.3 Patient Examination

The physical examination included LE joint strength and mobility testing. PFJ assessment for patellar alignment and patellar tracking in open-chain knee flexion/extension did not reveal any abnormalities. Marked contralateral pelvis drop and medial knee joint displacement were observed during a single leg step-down activity. The patient also noted apprehension to movement and anxiety during this functional closed chain activity. Such lack of confidence and kinesiophobia may adversely affect overall rehabilitation potential. In addition, a gait analysis during running was performed and used both 2D and 3D video systems.

13.0.4 Gait Analysis

The patient was asked to run on a treadmill with separate 2D cameras filming at 120 Hz covering the anterior–posterior and lateral views and a scaling rod for spatial measurements of her running mechanics. During running at the patient's preferred speed (3.2 m/s), she demonstrated a cadence of 160 steps/minute and an average stance duration of 275 ms. Sagittal plane video analysis revealed a rearfoot strike pattern well anterior (11 cm) to her center of mass, a large (12 cm) vertical excursion of her center of mass from peak height in swing phase to minimum height in mid-stance, and only five degrees of knee flexion at initial contact. Frontal plane video during running revealed increased (eight degrees) hip adduction excursion during the stance phase due to increased contralateral pelvic drop as well as medial knee displacement of the stance limb, reducing the space between both the knees during mid-stance. Peak PFJ stress and stress impulse during running, estimated using 3D motion capture and an inverse-dynamics informed musculoskeletal model, were 5.2 MPa and 0.059 MPa·s, respectively (Figure 13.1). These baseline PFJ kinetics served as a reference for the comparison of running mechanics following treatment.

13.0.5 Interpretations and Intervention

The patient demonstrated signs and symptoms consistent with PFJ pain that can be a consequence of an imbalance of mechanical load (PFJ stress) during running relative to the adaptive/reparative capabilities of the affected tissues. The intervention developed included a focus on three elements to reduce PFJ mechanical loading during running and to provide a framework for meeting her physical activity goals: (1) increase running cadence to reduce step length at her preferred training velocity (2) address hip and knee joint transverse and frontal plane rotations that reduce PFJ contact area during stance phase, and (3) provide patient education regarding self-management strategies and load management principles for running exposure progression and regression guided by symptom response.

To achieve these small alterations to running mechanics that can reduce PFJ mechanical loads during running, the patient was asked to practice running at her preferred pace while matching the audio feedback of running cadence using a metronome set at 10% above her preferred cadence (Bramah et al., 2019). Additionally, to increase awareness of hip and knee joint rotations that decrease PFJ contact area, a mirror was set in front of the

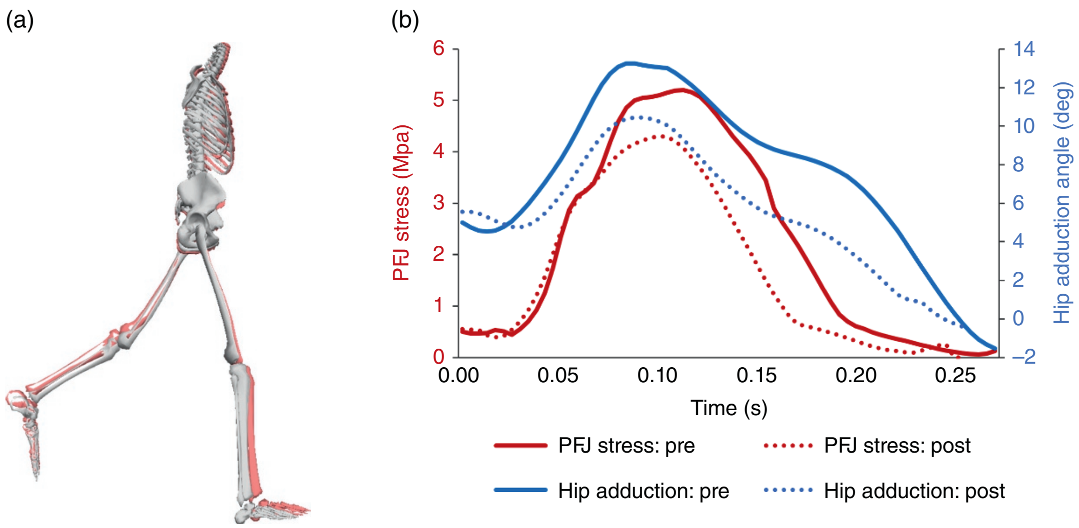

Figure 13.1 (a) Running kinematics at initial contact before (dark) and after (light) adopting increased cadence at preferred running speed. Note increased knee flexion and heel contact closer to her center of mass. (b) Within-day changes in patellofemoral joint stress (gray) and hip adduction (black) during the stance phase of running before and after treatment.

on running kinematics. Step rate feedback and verbal cues such as "increase the space between your knees" and "keep the waistband of your shorts level" were provided for the first five minutes of practice to assist the patient successfully increasing step rate and reducing stance phase hip adduction and internal rotation excursion during running. Over the next five minutes, verbal and audio cues were removed and provided only as needed. A sample of her running mechanics while adopting these technique changes was recorded for comparison with her original presentation (Figure 13.1). Taken together, these adjustments to the running technique reduced hip adduction excursion by three degrees, decreased peak patellofemoral stress by 19% to 4.2 MPa, and decreased PFJ stress impulse by 25% to 0.044 MPa·s during the stance phase.

The patient was educated regarding pain management strategies in the context of a schedule to progressively increase her running exposure with these recommended kinematic adjustments. Additionally, she was asked to use a readily available phone application to monitor her compliance with increased cadence while running outside the clinic. Follow-up appointments were scheduled once a week for two weeks, decreasing to once a month or as needed thereafter.

13.0.6 Patient Outcomes and Discussion

The patient appeared at four total visits over three months. At the end of this time, the patient reported independence with all aspects of her treatment and a significant reduction in her presenting symptoms. 2D video analysis during running at her preferred pace (3.2 m/s) at three months revealed substantial changes in running mechanics compared to her initial evaluation. In the sagittal view, compared to her initial examination, several hallmarks of a shortened step length were observed including that step rate increased 10 steps/minutes, stance time decreased to 250 ms, foot placement was 3 cm closer to her center of mass at initial contact, vertical excursion of her center of mass decreased 3 cm, and she displayed five degrees more knee flexion at initial contact while maintaining a rear foot strike pattern. In the frontal plane view, she displayed three fewer degrees of hip adduction excursion, resulting in decreased medial knee displacement during the stance phase. Her Anterior Knee Pain Scale score increased to 95% and she had carefully increased her running exposure to 24 miles/week.

The above changes in running kinematics produce kinetic effects culminating in lower per-step PFJ stress. For example, at a given running velocity, increasing cadence decreases step length, vertical excursion of the center of mass, and peak vertical ground reaction force. Additionally, landing with the foot closer to the center of mass reduces posterior ground reaction force following initial contact. Combined, the reductions in vertical and posterior ground reaction forces lower quadriceps force requirements to generate internal knee extension moment. Because stance time is also reduced when running with a shorter step length, decreased knee extension moment is produced over a shorter duration, resulting in a substantially lower angular impulse and decreased knee joint work (Heiderscheit et al., 2011). These lower quadriceps requirements during the stance phase reduce mechanical loads at the PFJ in the form of decreased peak joint contact force and joint force impulse for each step. Further, reducing hip adduction excursion may reduce iliotibial band tension and laterally directed patellar forces that can adversely affect the distribution of PFJ contact forces. This improvement in hip adduction excursion observed after treatment may be a consequence of increased attention and movement feedback training (Willy et al., 2012), increased gluteal muscle contributions during the late swing and early stance phase with increased step rate (Lenhart et al., 2014), or contributions from both mechanisms.

13.0.7 Conclusion

The patient in this case demonstrated successful resolution of symptoms and increased tolerance for running, presumably due to a treatment approach grounded in PFJ mechanics. The case underscores the value of a targeted application of biomechanical principles into practice for a person with PFJ symptoms and clinically identifiable running gait deviations as a proximate cause of her complaints.

13.0.8 References

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13.1 BIOMECHANICAL APPROACH TO MANAGE KNEE OSTEOARTHRITIS

13.1.1 Osteoarthritis and Biomechanics

Knee osteoarthritis (OA) is defined by the deterioration of articular cartilage in the tibiofemoral and/or patellofemoral compartments of the knee joint. Although cartilage loss is the primary radiographic determinant of OA, this pathology involves many other structures in and around the knee joint. Notably, there are changes in muscle morphology and function (decreased physiological cross-sectional area, increased intramuscular fat, decreased muscle force production), sclerosis of subchondral bone, and the abnormal presence of osteophytes (bone spurs) (Figure 13.2).

Figure 13.2 Frontal plane radiograph view of the tibiofemoral joint. Note that there is a lack of joint space on the medial (left) side of the image indicating loss of articular cartilage in the medial compartment. Osteophytes and subchondral sclerosis are present in the medial compartment. Collectively these findings indicate severe structural osteoarthritis disease.

Unlike many other pathologies, there is an established link between movement patterns and the risk of OA initiation and progression. For kinetics, greater adduction moments during gait have been shown to increase the risk for radiographic OA progression at follow-up (Miyazaki et al., 2002). As the adduction moment increases, there is a concurrent shift in the distribution of joint forces in the tibiofemoral interface of the knee; the greater the adduction moment the greater the load distribution in the medial compartment of the knee joint. Because direct measurement of joint force requires an implanted and instrumented prosthetic device, the adduction moment is often used as a surrogate measure of internal loads (Kutzner et al., 2013). It is postulated that this asymmetrical loading creates an environment that fosters cartilage loss, bony sclerosis, and osteophyte development in the medial compartment; hence OA progression.

From a kinematic perspective, varus and valgus thrusts in the frontal plane have also been associated with a greater risk of OA progression (Chang et al., 2004). This thrust occurs when there is a rapid change in a frontal plane joint position toward the knee varus (adduction, bow-legged position) or knee valgus (abduction, knock-knee position). Rapid changes in frontal plane position increase the rate of loading in the medial or lateral joint compartments. As these loading rates exceed the physiological levels, it may also lead to cartilage deterioration and bony changes within the joint.

Biomechanical changes are not isolated to the frontal plane. Decreased joint excursions in the sagittal plane during the loading response phase of the gait cycle (knee flexion) have also been identified as a risk factor for OA progression (Zeni et al., 2019). During the loading phase of the gait cycle, bodyweight is transitioned from double-limb support to single-limb support. The knee flexion that occurs immediately after this transition dampens the transition of force on the forward limb and is thought of as a "shock-absorbing" mechanism during stance. Individuals with knee OA often present with decreased joint excursion throughout the loading response and maintain the knee in a flexed position. This may be a consequence or driver of the increased co-contraction between the quadriceps and hamstring muscles often seen in this population.

Because these biomechanical alterations are part of the disease process or may increase the risk of OA progression, a biomechanical analysis is an important

OA. Abnormal kinetics and kinematics can be addressed through the use of movement retraining, bracing, assistive devices, or other targeted interventions.

13.1.2 Patient History

Case information provided courtesy of Emovi, Inc., Montreal, Canada. The patient in this case was a 77-year-old male who was diagnosed with "end-stage" or severe medial compartment knee OA in his left knee. His X-rays revealed a substantial loss of cartilage, subchondral sclerosis, and osteophytes. Although this patient was physically active and motivated to return to his normal recreational activities, substantial joint pain prevented him from playing tennis and walking long distances. This patient followed a typical course of medical management. He received multiple joint injections for pain relief from his orthopedic physician and participated in several months of physical therapy. During his therapy, he worked to improve his joint range of motion, muscle weakness, and swelling in the knee.

After six months of therapy, the patient had improved knee range of motion, small improvements in strength, and a reduction in acute knee pain at rest. However, joint pain during activity still prevented him from returning to tennis. He continued to report substantial pain during functional weight-bearing activities, including walking for longer than 10 minutes and ascending and descending stairs. Because of his persistent pain, it was recommended he undergoes a total knee replacement.

13.1.3 Biomechanical Assessment

In an attempt to delay or prevent the total knee replacement, the patient sought an additional assessment and opinion on care. To identify movement alterations and potential biomechanical targets for rehabilitation, the patient underwent a three-dimensional kinematic assessment of the knee (KneeKG™, Emovi Inc., Canada). As part of this examination, he walked on a treadmill at a comfortable self-selected pace while knee joint motions in all cardinal planes were evaluated.

This exam revealed several biomechanical risk factors associated with OA progression and poor function. In the frontal plane, it was found that this individual had a varus thrust, as well as a static and dynamic varus alignment during standing and walking (Figure 13.3). Peak knee adduction on the affected limb approached 14 degrees during the early portion of the stance phase, compared to less than four degrees of adduction in the non-affected knee. The magnitude of the varus thrust, measured as the difference between the local minimum and maximum values during the loading response was six degrees. In the sagittal plane, the patient was found to have limited knee flexion excursion during loading and walked with a characteristic "stiff-leg" gait pattern (Figure 13.4). This was identified as the reduced arc of motion in the affected knee (six degrees) compared to the unaffected knee (16 degrees) during the loading response phase of the gait cycle.

Based on this examination, the therapist developed a targeted home-based exercise program that focused on neuromuscular retraining to restore more normal movement patterns in the frontal and sagittal planes. Specifically, these exercises focused on improved control of the knee in the frontal plane during standing, squatting, and lunging. He was given visual and verbal feedback on the frontal plane position of the knee as he practiced transitioning weight from the unaffected to the affected knee. In addition to these exercises, the patient was also given a stabilization brace to be worn during higher-level activities to limit excessive frontal plane motion and prevent the varus thrust.

To address the limited sagittal plane joint excursion, the patient was given exercises that focused on quadriceps strengthening and control in the terminal portion of the range of motion (0–20 degrees). He performed these exercises on stable surfaces and transitioned to unstable surfaces to increase the difficulty of the task over time.

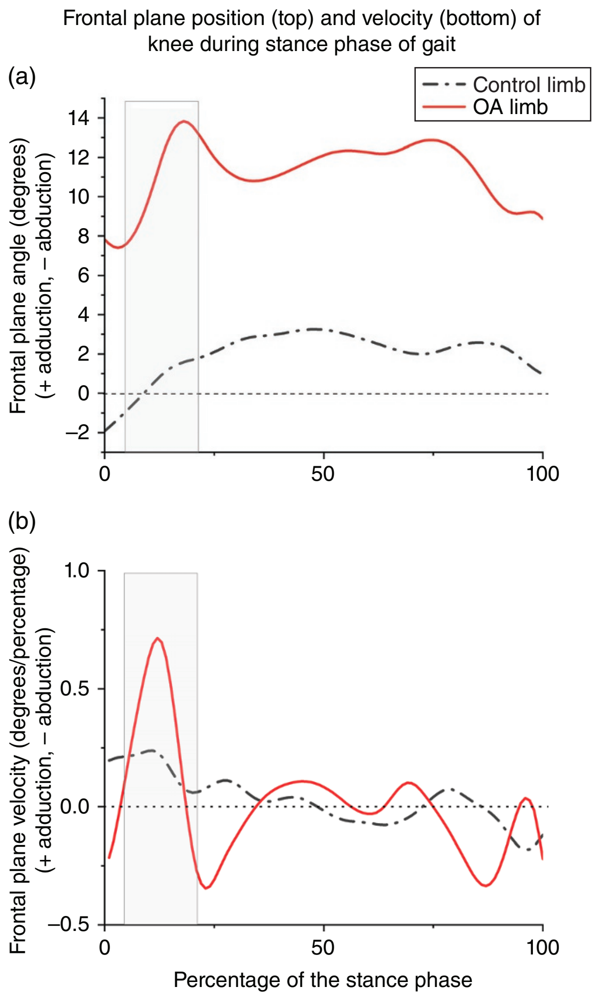

Figure 13.3 Analysis of frontal plane joint angles during walking (a) revealed the patient had considerable knee varus (adduction positioning) throughout the stance phase of gait on the involved limb. The varus thrust is identified as the rapid change in joint position during the loading response portion of the gait cycle as the limb transitions from double-limb support to single-limb support (gray box). The thrust can be seen as the change in position (a) and the larger adduction velocity (b) in the time-series curves. Note: time in the x-axis has been normalized to 100% of the stance phase of the gait cycle, so units for velocity are not given in change per second, but rather change per percentage of stance phase.

The patient returned for a follow-up movement assessment six weeks after his initial evaluation. Results from this examination revealed that the patient was able to walk without a varus thrust, even without wearing his brace. He also reduced the peak varus alignment during walking by 40% (8 degrees vs. 14 degrees at baseline) and increased his knee flexion excursion during loading by nearly 70% (10 degrees vs. 6 degrees at baseline). While these changes are substantial, the improved movement strategies also led to a reduction in knee pain and function. At a six-month follow-up, the patient was able to indefinitely postpone his knee replacement and was able to return to playing tennis with only mild discomfort.

Clinicians should consider a biomechanical assessment when working with patients with knee OA. Progression of knee OA is driven, in part, by biomechanical markers. Selecting

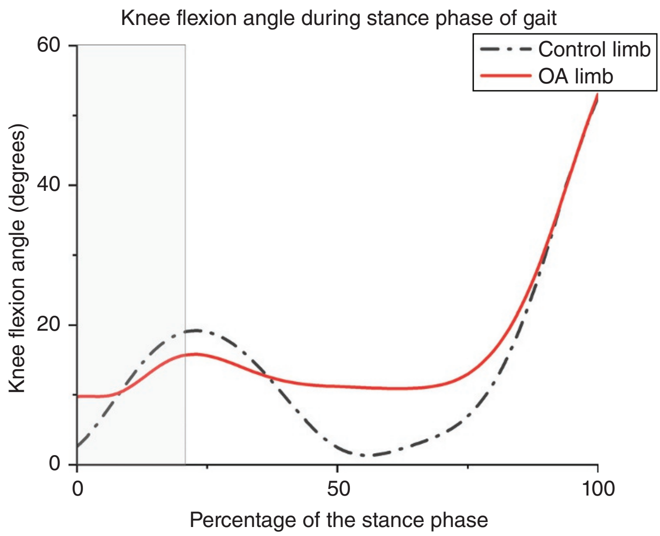

Figure 13.4 Analysis of sagittal plane joint angles during walking revealed the patient had considerably less knee flexion excursion during the loading phase and maintained the knee in a flexed position throughout the stance phase of gait.

exercises and interventions to address abnormal mechanics may lead to better long-term outcomes and preserve the structural integrity of the joint.

13.1.4 References

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13.2 ULNAR COLLATERAL LIGAMENT RECONSTRUCTION

Overuse conditions and throw-related injuries are a significant problem at the shoulder and elbow for athletes, particularly baseball pitchers (Conte et al., 2016; Hootman et al., 2007; Posner et al., 2011; Shanley et al., 2011). Elbow injuries are often attributed to excessive elbow valgus torque (Pytiak et al., 2018; Agresta et al., 2019; Anz et al., 2010), which can reach 115 N · m when pitching. The ulnar collateral ligament (UCL) is designed to reduce valgus motion at the elbow and can become injured or torn with repetitive pitching activities (Feltner and Dapena, 2016; Werner et al., 1993, 2002). Modern biomechanical pitching analysis can identify risk factors associated with elbow injury. Clinicians and researchers can use this information to reduce injury and improve performance through targeted training activities.

Three-dimensional motion capture has been used to examine how full-body kinematics influence joint stresses and pitching velocity (Agresta et al., et al., 2020). While there are hundreds of variables of interest when analyzing pitching, pitch velocity was the most influential variable, and certain kinematic variables revealed significant influence on elbow stress (Nicholson et al., 2021). Some of these variables include maximum shoulder external rotation, ground reaction force, shoulder abduction, and humeral rotation velocity. Clinicians and coaches can focus on these variables and provide feedback to change pitching mechanics and, ultimately, reduce elbow stress. Decreasing shoulder abduction at foot strike (Werner et al., 2002; Aguinaldo and Escamilla, 2019) and increasing shoulder external rotation torque (Aguinaldo et al., 2007; Sabick et al., 2004; Sgroi et al., 2015) (achievable by increasing maximum external rotation and/or external rotation velocity) have been shown to reduce elbow valgus load when pitching.

Evaluating and interpreting key biomechanical variables is an important part of pitching analysis. The purpose of this case is to explore an approach to interpreting biomechanical pitching data to limit elbow stress. While every individual is unique, this case demonstrates some of the more common deviations and variables of interest that can be identified and corrected through biomechanical analysis.

13.2.1 Player History

A 22-year-old right-handed relief pitcher in a Division I college baseball program presented for a 3D biomechanical pitching assessment. The pitcher had suffered a season-ending injury to the UCL of his elbow in 2018. He underwent UCL reconstruction and missed the 2019 season but was cleared by his sports medicine physician to return to throwing. The 3D biomechanical pitching assessment was conducted in August 2019, about 14 months after surgery. Two weeks after the assessment, the pitcher re-tore his UCL during competitive play.

Elbow valgus torque is a measure of the torque that is created in resistance to elbow valgus motion and is considered an indicator of stress in the UCL. As discussed in previous chapters, joint torques are influenced by height and body weight. To account for these anthropometric variables, elbow valgus torque is normalized by body weight times height (BW×H) during the analysis. In our experience, elbow valgus torque that exceeds 5% BW×H should be reduced to decrease UCL stress. This pitcher presented with maximum elbow valgus torques between 6 and 8% BW×H when throwing.

A cursory examination of the key performance and injury indicators at specific points in the pitching cycle (Table 13.1) indicates that the pitcher had excessive shoulder external rotation motion, while he had lower than normal values for hip–shoulder separation at footstrike, knee extension from footstrike to ball release, and shoulder internal rotation angular velocity. While some of these metrics do not directly indicate an increase in elbow stress, they warrant further exploration.

When evaluating key performance and injury indicators, a bottom-up approach evaluates these metrics from the legs first, followed by the trunk and upper extremities. When landing on the lead leg, this pitcher landed in a “closed” position where the landing leg crossed the midline. Ideally, the lead foot should land in line with the drive (rear) leg while also being slightly internally rotated (Dillman et al., 1993). Landing in the closed position reduces throwing power and places the pitcher with an unstable base of support (Figure 13.5).

There were also concerning findings in sagittal plane knee motion in the lead leg. It is recommended that the lead leg moves through at least 10 degrees of knee extension excursion from footstrike to ball release (Figure 13.6). Extending the knee at this point in the delivery allows a pitcher to use the powerful quadriceps and gluteal muscles before ball release to create a catapult effect with the knee and hip. This increases ball velocity without increasing elbow valgus torque. In this case study, the pitcher continues to flex his knee from footstrike to ball release. This may be the result of altered motor control patterns or impairments with muscular strength. Knee flexion during this period reduces the lower extremity’s contribution to velocity, which requires the pitcher to generate more energy from the upper extremity to maintain ball velocity. It is possible that this contributes to excessive torques at the elbow.

TABLE 13.1 Key Performance and Injury Indicators at Maximums, Footstrike, or Ball Release

	Breaking Ball ($n = 3$)	Fastball ($n = 3$)	Off Speed ($n = 1$)	Reference ($n = 261$)
Ball velocity	78.2 ± 0.3 mph	83.8 ± 0.7 mph	80.0 mph	76.0–85.3 mph
Stride length	74.9% *H* ± 0.4 *H*	71.2% *H* ± 0.5% *H*	72.8% *H*	69.5–86.7% *H*
Shoulder abduction at footstrike	94.4° ± 0.4°	96.0° ± 0.2°	96.9°	76.5°–100.2°
Shoulder external rotation maximum	194.8° ± 1.6°	194.2° ± 1.9°	192.5°	156.7°–179.3°
Hip–shoulder separation at footstrike	43.0° ± 1.1°	40.3° ± 1.2°	40.6°	38.3°–55.3°
Shoulder horizontal abduction at footstrike	−68.9° ± 0.5°	−67.9° ± 0.5°	−68.8°	−35.9° to −16.1°
Elbow flexion at footstrike	94.3° ± 3.6°	93.1° ± 1.8°	94.2°	73.2°–104.3°
Elbow flexion at ball release	47.3° ± 1.7°	44.4° ± 5.2°	60.8°	20.1°–36.6°
Lead knee flexion at footstrike	49.7° ± 0.8°	48.7° ± 2.2°	49.3°	38.4°–57.4°
Lead knee flexion at ball release	43.0° ± 2.8°	48.0° ± 2.2°	51.0°	28.4°–59.8°
Pelvis rotation angular velocity	650.0 ± 6.3°/s	652.4 ± 5.1°/s	661.8°/s	596.9–788.9°/s
Trunk rotation angular velocity	1156.0 ± 5.6°/s	1162.3 ± 5.2°/s	1166.5°/s	995.7–1174.3°/s
Shoulder horizontal adduction angular velocity	1009.8 ± 16.2°/s	952.4 ± 28.0°/s	923.9°/s	218.3–1090.3°/s
Shoulder internal rotation angular velocity	4604.7 ± 196.6°/s	4371.3 ± 174.4°/s	4589.8°/s	4638.7–5751.3°/s
Lead leg GRF maximum	2.1% *W* ± 0.04% *W*	2.1% *W* ± 0.01% *W*	2.1% *W*	1.9–2.5% *W*
Drive leg GRF maximum	1.2% *W* ± 0.2% *W*	1.3% *W* ± 0.2% *W*	1.3% *W*	1.1–1.5% *W*

References are lab-generated norms for collegiate baseball pitchers reported as a range of mean ± one standard deviation. Breaking balls are curveballs or sliders while off-speed pitches are changeups. Mean ± standard deviation. *H*, height; *W*, weight.

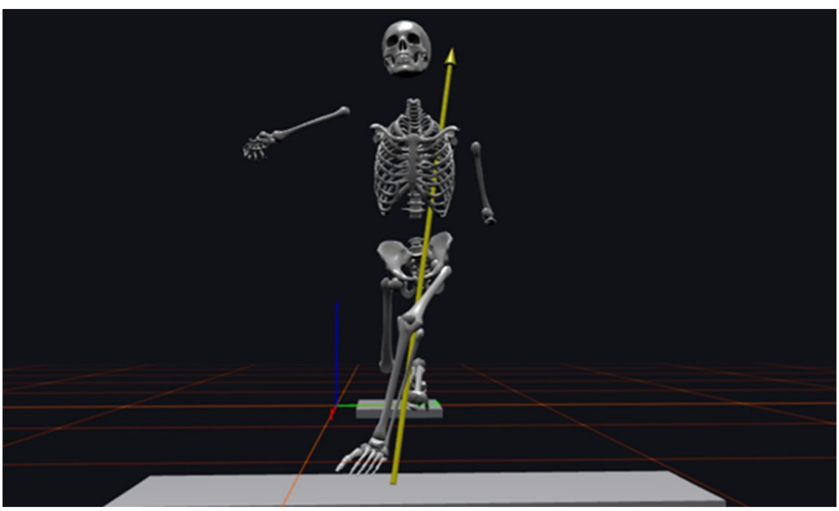

Figure 13.5 The pitcher's stride leg crosses over midline with the shank angled toward first base. This is not a powerful position, and the pitcher lacks a stable base for rotation and power generation.

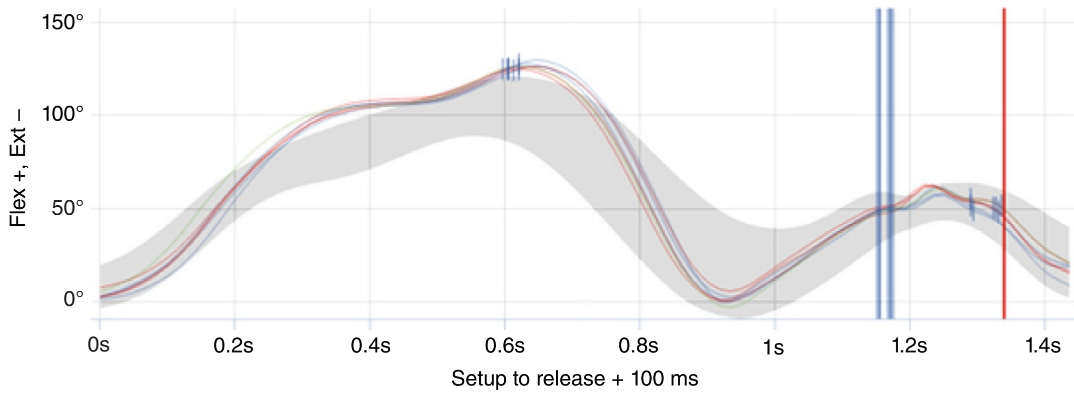

Figure 13.6 Knee flexion graph. The first vertical lines indicate footstrike. The second vertical line indicates ball release. It is recommended to have at least 10 degrees of knee extension from footstrike to ball release. Gray band represents ± one standard deviation of the reference normal. This graph represents the knee action of the pitcher in this case example. He is landing with about 50 degrees of knee flexion/bend. He flexes his knee further before achieving some extension, but by ball release, he is in a more flexed position (51 degrees) than at footstrike.

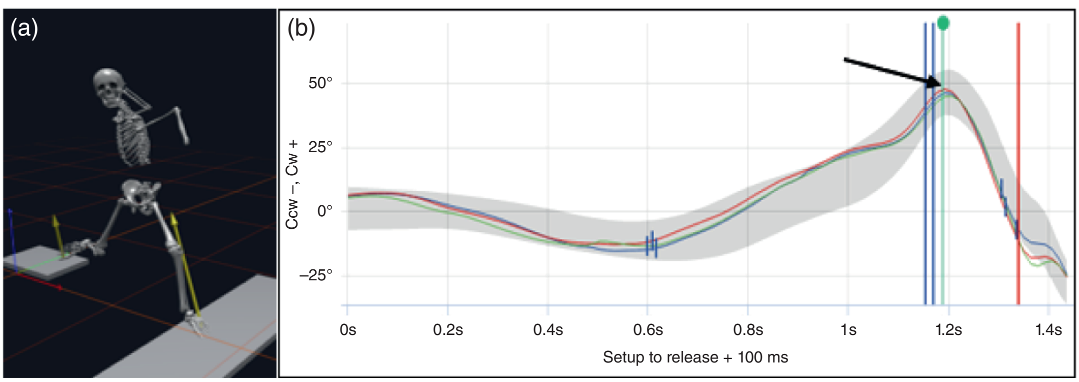

Figure 13.7 (a) Moment of maximum hip–shoulder separation. Pitchers who exhibit maximum hip–shoulder separation angles of greater than 55 degrees have pelvis facing home at maximum hip–shoulder separation. (b) Hip–shoulder separation graph. The first vertical lines indicate footstrike. The second vertical line indicates ball release. Gray band represents ± one standard deviation of the reference normal. This pitcher's maximum hip–shoulder separation is less than 55 degrees (arrow).

Hip–shoulder separation is another important metric related to elbow torque and is measured as the angle in the transverse plane created between the frontal plane alignment of the pelvis and frontal plane alignment of the shoulders. This angle is a key indicator of lower extremity loading; greater angles create a larger spring effect between the lower extremity and trunk. When evaluating throwing mechanics, a pitcher should have at least 55 degrees of hip–shoulder separation just after stride leg footstrike. The pitcher in this case had a maximum hip–shoulder separation of 45 degrees (Figure 13.7). Decreased hip–shoulder separation limits trunk rotation velocity and potentially pitch velocity (Bullock et al., 2020a). To maintain appropriate pitch velocity, a pitcher with limited hip–shoulder separation may use other centripetal force generation mechanisms at the shoulder and elbow, which can increase detrimental elbow torques (Hirashima et al., 2002). For this pitcher, he compensated with excessive shoulder horizontal abduction (Figure 13.8). This is described as "throwing out of the scapular plane" and can contribute to shoulder and elbow injuries.

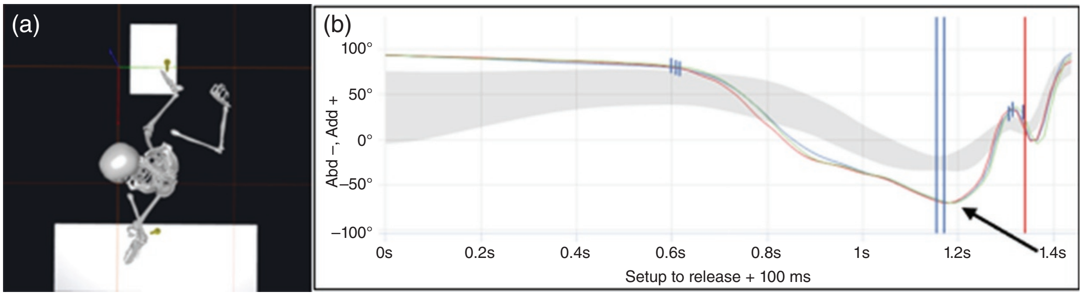

Figure 13.8 (a) Moment of maximum shoulder horizontal abduction. (b) Shoulder horizontal abduction graph. The first vertical lines indicate footstrike. The second vertical line indicates ball release. Gray band represents ± one standard deviation of the reference normal. This pitcher's shoulder horizontal abduction far exceeds the reference normal around footstrike (arrow).

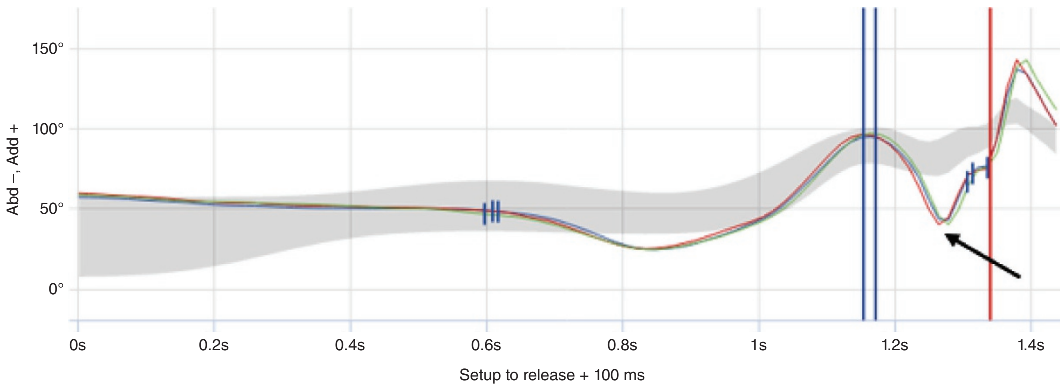

Figure 13.9 Shoulder abduction graph. The first vertical lines indicate footstrike. The second vertical line indicates ball release. Gray band represents ± one standard deviation of the reference normal. This pitcher is around 90 degrees of shoulder abduction at footstrike, but then drops his arm as he beings to externally rotate (arrow). His arm is still low at ball release.

The ideal arm position at footstrike is 90 degrees of abduction, 90 degrees of elbow flexion, and 60 degrees of shoulder external rotation. The ideal arm position at ball release is 90 degrees of abduction. The arm should follow a relatively straight path from footstrike to ball release. This pitcher has 90 degrees of abduction at footstrike (Figure 13.9), but there is a sharp decrease to 40 degrees of shoulder abduction. This is a dramatic deviation from normal and expected values. At ball release, his shoulder abduction is still lower than anticipated values. Leading with the elbow and having the elbow lower than the shoulders at ball release will increase the elbow valgus torque and the stress on the UCL.

Greater shoulder external rotation has been shown to increase elbow valgus torque (Nicholson et al., 2021). This pitcher has a maximum shoulder external rotation of 195 degrees when pitching, while the normal range for collegiate pitchers is 157–180 degrees (Table 13.1). Most pitchers release the baseball around 100 degrees of shoulder external rotation, but this pitcher released the ball between 135 and 165 degrees of shoulder external rotation. This indicates he releases the ball early in the pitch cycle. This reduces internal rotation velocity and can limit pitch velocity. As with other deficits, the pitcher may compensate with other abnormal mechanics to maintain a normal and effective velocity during competition.

In summary, this pitcher had abnormalities in sagittal, frontal, and transverse planes in the legs and trunk. Lower extremity deficits were likely the upper extremity. Specifically, he substituted excessive shoulder horizontal adduction and excessive shoulder external rotation to overcome low hip/shoulder separation angles. This led to excessive horizontal abduction motion and increased elbow valgus torque. Inefficient power transfer and early ball release further contributed to increased elbow stress and inefficient pitching mechanics. Identifying these flaws and implementing corrections may have been essential in reducing future injury. Unfortunately, the biomechanical data, although collected, was not evaluated and integrated into a training plan and this pitcher sustained a career-ending injury.

13.2.2 References

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APPENDIX A

KINEMATIC, KINETIC, AND ENERGY DATA

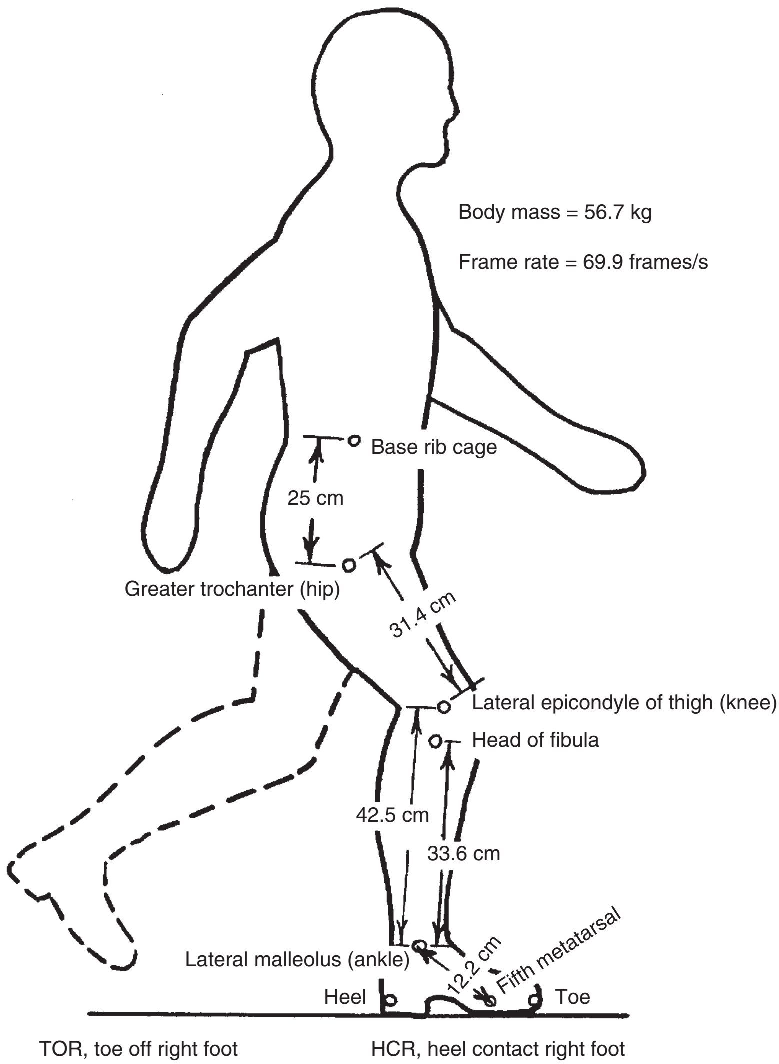

Figure A.1 Walking trial – marker locations and mass and frame rate information.

Winter's Biomechanics and Motor Control of Human Movement, Fifth Edition.
Stephen J. Thomas, Joseph A. Zeni, and David A. Winter.

Companion website:

TABLE A.1 Raw Coordinate Data (cm)

	Time	Base Rib	Cage	Right	Hip	Right	Knee	Right	Fibula	Right	Ankle	Right	Heel	Right	Metat.	Right	Toe
Frame	s	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*
1	0.000	46.98	104.41	44.94	78.58	41.00	47.40	35.91	40.53	9.31	21.44	2.95	24.24	7.53	9.35	11.73	3.63
2	0.014	49.22	104.79	47.31	78.58	45.02	46.89	40.06	40.02	12.70	22.46	7.23	26.02	10.54	10.63	13.72	4.77
3	0.029	51.10	105.17	49.57	78.71	48.68	47.27	44.23	40.15	16.49	23.73	10.64	27.30	14.20	12.03	17.63	6.43
4	0.043	53.13	105.30	51.74	79.21	52.50	47.53	48.43	40.15	20.81	24.37	14.71	27.55	18.78	12.53	22.47	6.94
5	0.057	54.86	105.43	53.33	79.09	56.13	47.91	52.70	40.78	24.96	24.24	18.72	27.42	23.17	12.66	27.12	7.32
6	0.072	56.81	106.06	55.41	79.98	59.87	48.67	56.56	41.29	29.33	24.62	23.09	27.17	28.31	12.41	32.51	6.81
7	0.086	58.25	106.32	56.73	80.49	63.34	49.44	60.29	41.80	33.57	23.73	26.95	26.02	33.44	12.03	38.02	6.81
8	0.100	60.03	106.95	58.89	81.00	66.90	50.84	64.36	42.69	38.78	23.22	31.27	25.01	39.55	11.26	44.38	6.30
9	0.114	61.56	107.08	60.79	81.12	69.96	51.09	67.79	43.20	43.11	22.21	35.73	22.84	45.01	10.24	50.36	6.04
10	0.129	63.54	107.46	62.78	82.01	73.22	51.73	71.31	43.84	48.15	21.06	40.64	20.93	51.46	8.97	57.05	5.28
11	0.143	65.20	107.85	64.69	82.40	76.27	53.00	74.74	44.86	53.23	20.04	45.73	19.02	57.56	8.21	63.54	5.15
12	0.157	66.92	107.85	66.67	82.78	79.01	53.51	77.86	45.11	58.27	18.52	51.27	16.86	63.99	7.44	70.23	5.03
13	0.172	68.77	107.59	68.65	82.65	81.62	54.15	80.73	45.49	63.68	16.86	56.56	14.44	70.30	6.30	76.79	4.52
14	0.186	70.37	107.59	70.88	83.16	83.99	54.53	83.48	45.87	69.22	15.59	62.23	12.66	76.73	5.79	83.48	4.77
15	0.200	72.43	107.59	73.19	82.90	86.56	54.78	86.30	46.00	74.72	14.44	68.23	11.01	83.63	5.66	90.12	5.54
16	0.215	74.00	107.59	74.89	83.03	88.51	54.91	88.89	46.13	80.36	13.43	74.13	9.10	89.53	5.28	96.14	5.54
17	0.229	75.68	107.59	76.95	82.90	90.57	54.91	90.95	45.87	85.86	12.66	80.01	7.57	95.53	5.79	102.40	6.55
18	0.243	77.67	107.08	79.19	82.65	93.06	54.65	93.32	46.00	91.54	12.03	86.32	6.55	101.72	5.79	108.84	7.70
19	0.257	79.55	106.95	81.20	82.40	94.56	54.02	95.45	45.62	96.85	11.64	92.02	5.41	107.79	6.17	114.16	8.97
20	0.272	81.47	107.21	83.12	82.14	96.10	54.02	96.99	45.24	101.45	11.77	97.50	5.15	113.28	7.70	119.39	10.88
21	0.286	83.53	106.45	85.69	81.50	98.16	53.25	99.18	44.73	106.56	11.77	103.13	4.77	118.78	8.21	124.38	12.15
22	0.300	85.86	105.68	87.77	80.49	99.73	51.86	101.13	43.84	110.68	11.52	108.39	4.14	123.15	9.35	128.24	13.30
23	0.315	87.74	105.81	89.91	80.61	101.36	51.73	103.14	43.71	114.85	12.15	112.43	4.90	126.94	10.88	131.77	15.33
24	0.329	90.34	105.17	92.25	80.49	103.57	51.09	105.36	43.58	118.33	12.15	116.30	5.28	130.55	11.39	135.39	16.23
25	0.343	92.25	104.79	94.41	80.10	105.23	50.96	107.26	43.46	120.75	12.28	118.84	4.65	133.22	11.64	137.80	16.73
26	0.357	94.36	104.16	97.03	79.72	107.59	50.84	109.50	42.69	122.99	12.03	121.34	4.14	135.21	11.39	139.79	16.23
27	0.372	96.57	103.77	99.37	79.98	109.93	50.58	111.58	42.95	124.31	11.90	122.14	3.75	136.27	10.37	140.85	15.21
28	0.386	98.73	103.52	101.53	79.34	112.34	50.46	113.87	41.80	124.94	10.63	122.65	3.25	137.16	8.97	141.99	13.68
29	0.400	101.40	103.26	104.45	79.60	114.89	50.33	116.28	41.80	125.83	9.61	123.03	3.50	137.66	7.95	142.88	11.90
30	0.415	103.60	103.65	106.78	79.60	116.96	50.46	118.11	41.80	127.02	9.74	123.20	4.14	138.34	6.30	144.45	10.24

31	0.429	105.94	103.65	108.86	79.60	118.53	50.33	119.81	41.55	127.82	9.48	123.12	4.01	138.77	5.41	145.13	8.59
32	0.443	108.22	103.52	111.28	80.10	120.44	50.58	121.71	42.18	128.33	9.61	123.24	4.14	139.27	4.65	145.38	6.81
33	0.458	110.75	104.03	113.80	80.49	122.84	50.96	124.24	42.44	129.07	9.35	123.60	4.14	139.51	4.14	146.38	6.30
34	0.472	112.91	104.03	115.58	81.38	125.25	51.86	126.14	42.57	129.45	9.61	123.98	4.39	139.63	3.37	146.25	5.41
35	0.486	115.20	104.28	117.36	81.89	127.54	51.86	128.05	42.44	129.83	9.74	123.98	4.26	140.01	3.63	147.14	5.41
36	0.500	116.74	104.54	118.78	81.63	128.32	51.86	128.83	42.69	129.46	9.10	123.48	4.14	139.39	3.75	146.52	4.77
37	0.515	119.05	105.43	120.96	81.63	129.87	51.86	130.13	42.95	129.74	9.35	123.89	4.65	139.80	3.63	146.54	4.90
38	0.529	121.31	105.30	123.10	82.27	130.86	51.60	131.49	42.18	130.10	9.23	124.24	4.52	140.02	3.75	147.27	4.52
39	0.543	123.28	106.19	124.93	83.03	132.18	52.36	132.31	42.82	130.53	9.61	124.55	4.90	140.45	3.75	147.33	5.28
40	0.558	125.03	106.19	126.94	82.78	133.04	51.35	132.79	42.44	130.37	9.35	124.64	4.26	140.55	3.50	147.04	4.77
41	0.572	127.21	106.83	128.61	83.03	133.58	51.73	132.81	42.44	130.27	9.61	124.67	4.77	139.94	3.63	147.19	4.52
42	0.586	128.71	106.95	130.24	83.03	134.31	51.35	133.17	43.20	129.98	8.97	123.75	4.77	139.91	3.37	146.91	4.77
43	0.601	130.90	107.34	132.04	82.90	134.72	50.96	133.70	42.57	130.39	9.61	124.54	4.14	140.19	3.50	147.31	4.52
44	0.615	132.52	107.97	133.03	83.41	135.70	52.62	134.56	43.20	130.23	9.61	124.12	4.52	139.90	4.01	146.78	4.26
45	0.629	134.48	107.85	134.86	83.54	136.26	51.86	134.73	42.95	129.89	9.48	123.91	4.77	139.82	3.63	146.95	4.26
46	0.643	136.33	108.10	136.20	83.80	137.09	51.86	135.18	43.07	130.22	10.24	124.37	4.90	140.02	3.37	147.02	4.77
47	0.658	138.33	108.23	138.08	82.90	137.95	51.47	135.66	42.69	130.06	9.99	124.08	4.77	140.11	3.75	146.99	4.26
48	0.672	140.14	108.48	139.38	82.90	138.74	51.60	136.19	42.69	130.47	10.12	124.36	5.03	140.01	3.25	147.01	4.01
49	0.686	142.20	107.97	140.67	82.78	139.15	51.09	136.73	42.57	130.49	9.99	124.64	5.28	140.16	3.12	147.04	4.52
50	0.701	144.05	107.46	142.02	82.65	140.11	51.09	137.05	42.57	130.56	9.99	124.20	4.90	139.60	3.63	146.85	4.01
51	0.715	146.01	107.08	143.85	82.01	140.92	50.58	137.87	42.06	130.49	10.24	124.76	4.77	140.03	3.50	147.28	3.75
52	0.729	148.58	107.46	146.04	82.40	142.22	50.84	138.66	42.44	131.27	10.24	125.17	5.28	140.18	3.63	147.18	4.14
53	0.744	149.92	106.57	147.51	81.63	143.05	50.58	139.23	41.93	130.96	10.12	125.24	5.03	140.25	3.37	147.25	4.01
54	0.758	152.34	106.32	148.90	81.50	143.69	50.71	140.12	42.57	131.47	10.63	125.49	5.66	140.50	3.37	147.50	3.88
55	0.772	153.94	105.81	150.50	81.12	144.27	50.46	140.45	42.95	131.03	10.63	124.80	6.17	140.07	2.99	146.94	3.75
56	0.786	156.15	105.30	152.46	81.00	145.59	50.71	141.52	42.31	131.34	10.63	125.61	5.92	140.50	3.37	147.24	3.88
57	0.801	158.65	104.92	154.57	81.00	147.07	50.58	142.61	42.18	131.80	10.75	125.94	7.06	140.70	3.50	147.45	3.75
58	0.815	160.50	104.66	156.30	80.61	148.41	50.71	143.83	43.20	132.12	12.15	125.89	7.70	140.65	4.01	147.39	4.14
59	0.829	162.61	104.41	157.90	80.10	149.75	50.33	144.66	42.31	131.94	12.03	125.96	7.83	140.34	3.75	146.95	3.75
60	0.844	164.58	104.03	159.88	80.23	151.22	50.33	146.26	42.82	132.26	12.66	126.03	8.46	140.41	3.75	147.02	3.88
61	0.858	166.66	103.77	162.08	80.23	153.30	50.33	147.96	42.69	132.69	12.92	126.58	9.61	140.20	4.26	146.69	4.26
62	0.872	168.90	104.03	164.19	79.60	155.41	50.07	150.06	42.57	133.39	13.93	126.90	10.50	140.01	3.88	146.76	4.01
63	0.887	171.00	103.90	166.42	79.85	157.01	50.07	152.30	42.69	133.97	14.83	127.48	12.28	140.21	3.88	146.83	3.75
64	0.901	173.53	103.14	169.07	78.96	160.29	49.06	154.95	42.06	135.48	14.95	128.73	13.04	140.95	4.14	147.06	3.50

(*continued*)

TABLE A.1 (*Continued*)

Frame	Time	Base Rib	Cage	Right	Hip	Right	Knee	Right	Fibula	Right	Ankle	Right	Heel	Right	Metat.	Right	Toe
	s	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*	*X*	*Y*
65	0.915	175.81	103.52	171.74	79.47	163.22	49.06	157.62	42.31	136.75	16.48	129.62	14.83	141.08	4.65	147.18	3.75
66	0.929	179.23	103.14	175.16	78.83	166.89	48.55	161.03	41.04	139.14	17.37	132.15	16.10	141.94	5.28	148.31	2.99
67	0.944	181.49	103.14	177.80	78.58	170.04	48.16	164.69	41.42	140.90	18.64	134.02	18.52	142.42	5.79	148.40	2.86
68	0.958	183.65	103.77	180.60	79.09	174.11	48.04	168.25	41.55	143.69	19.79	136.95	21.57	143.69	7.57	149.04	3.63
69	0.972	184.86	103.39	182.44	77.94	176.97	47.02	171.50	40.15	145.54	20.42	139.43	23.35	144.14	7.95	148.59	3.50
70	0.987	187.71	104.28	185.67	78.83	181.73	47.53	176.00	40.15	149.53	21.82	143.81	25.13	146.86	9.74	150.17	4.26
71	1.001	188.82	104.16	187.29	78.20	184.36	46.89	179.40	39.51	152.68	22.21	146.95	25.90	149.11	10.75	151.91	4.14
72	1.015	191.13	104.92	189.48	78.83	188.33	47.27	184.00	39.89	156.39	23.73	150.54	27.17	153.34	12.03	156.52	5.66
73	1.030	192.42	105.55	190.89	79.09	191.78	47.66	187.71	40.53	159.84	24.24	153.73	27.68	157.17	12.28	160.73	5.92
74	1.044	194.40	106.19	193.00	79.60	195.29	48.16	191.47	41.04	163.99	24.37	157.50	27.93	161.82	12.66	165.38	6.43
75	1.058	196.18	105.94	195.16	79.72	199.10	48.93	195.79	41.42	168.44	24.50	161.95	27.04	167.29	12.15	170.98	6.68
76	1.072	197.69	106.06	196.42	80.10	202.66	49.31	199.47	41.55	172.50	23.61	165.88	25.77	171.99	10.88	176.57	6.17
77	1.087	199.57	106.70	198.68	81.00	206.19	50.33	203.39	42.44	177.30	22.84	170.56	25.01	177.81	10.12	182.65	6.17
78	1.101	201.17	107.21	200.53	81.63	209.57	51.22	206.90	43.20	182.08	21.95	175.08	23.10	183.74	9.61	189.21	4.90
79	1.115	202.99	107.59	202.48	82.14	212.66	52.11	210.63	43.84	187.09	21.06	179.83	21.57	189.76	8.97	195.61	4.77
80	1.130	204.67	107.59	204.80	82.65	215.74	52.62	213.83	44.47	192.20	19.66	184.95	19.28	196.02	7.19	201.75	4.39
81	1.144	206.27	107.46	206.78	82.52	218.49	53.13	216.96	44.47	197.36	17.88	190.11	16.73	202.45	6.81	208.94	3.75
82	1.158	208.30	107.72	209.07	83.03	221.28	53.89	219.88	45.24	202.96	16.86	195.58	14.57	209.19	5.41	215.94	4.26
83	1.173	210.18	107.72	211.07	83.03	223.80	54.02	222.91	45.24	208.15	15.59	201.27	12.53	215.14	4.90	222.65	3.50
84	1.187	211.95	107.46	212.97	82.78	226.20	54.40	225.70	45.49	213.86	13.68	207.37	10.12	221.88	4.39	229.13	3.88
85	1.201	213.83	107.21	215.10	82.90	228.59	54.65	228.21	45.49	219.43	12.79	213.45	8.59	228.72	3.75	235.59	4.90
86	1.215	215.63	107.08	217.03	82.65	230.52	54.65	230.52	45.11	224.92	11.39	219.32	6.68	234.84	4.26	241.72	5.28
87	1.230	217.58	106.70	219.11	82.65	232.85	54.65	232.60	45.24	230.94	11.39	225.85	5.66	241.38	4.52	247.74	6.30
88	1.244	219.36	106.45	220.89	82.27	234.63	53.64	234.76	44.47	236.28	11.01	232.09	4.52	247.86	4.52	254.10	7.32
89	1.258	221.06	105.94	222.71	81.63	236.07	53.13	236.84	44.35	241.80	10.37	237.35	4.01	253.00	6.05	259.36	9.48
90	1.273	223.53	105.55	225.18	81.12	238.29	52.75	239.18	43.46	247.07	10.63	243.89	3.63	259.41	7.06	265.01	11.26
91	1.287	225.31	104.92	227.60	81.00	239.81	51.73	241.47	43.33	252.03	11.01	249.23	3.63	264.24	8.84	269.33	12.92
92	1.301	227.33	104.66	229.75	80.10	241.33	50.84	243.36	42.82	255.83	11.01	253.54	3.63	268.31	9.86	273.40	14.32
93	1.316	229.28	104.41	231.57	79.98	243.03	50.46	245.32	42.82	259.31	11.26	257.53	4.14	271.78	10.75	276.49	15.84
94	1.330	231.49	104.03	234.16	79.60	245.23	50.33	247.40	42.82	262.03	12.15	260.50	4.01	274.75	11.26	278.95	16.48

95	1.344	233.80	103.52	236.47	79.21	247.67	49.82	249.57	42.31	263.95	10.88	262.55	3.63	276.42	10.88	281.39	15.72
96	1.358	235.62	103.01	238.80	78.83	250.00	49.82	251.91	41.68	265.27	10.24	263.49	3.12	277.74	9.99	282.70	14.70
97	1.373	238.04	103.01	241.60	79.21	252.80	50.33	254.32	42.06	266.41	9.86	264.12	2.99	278.63	8.59	283.72	13.43
98	1.387	240.34	103.14	244.29	79.21	255.23	50.20	256.50	41.55	267.06	9.23	264.14	3.37	279.15	7.44	284.50	11.39
99	1.401	242.88	103.01	246.83	79.09	257.39	50.07	258.92	41.42	268.72	9.48	264.52	3.63	280.42	6.05	286.15	9.10
100	1.416	245.09	102.63	249.16	79.34	259.60	50.07	260.74	41.68	269.78	9.23	264.69	3.63	281.10	4.90	287.08	7.95
101	1.430	247.53	103.26	251.60	79.72	261.90	50.96	263.30	42.57	270.18	9.48	265.21	4.26	281.25	4.39	287.86	6.68
102	1.444	249.91	102.88	253.73	80.36	264.42	50.96	265.95	42.44	270.78	9.10	265.18	3.63	281.47	3.50	288.09	5.54
103	1.459	251.81	103.39	255.63	81.00	266.70	51.60	267.59	42.82	271.15	9.23	265.30	4.39	281.84	3.63	288.20	5.15
104	1.473	253.96	104.16	257.65	81.38	268.47	51.47	269.23	42.69	271.40	9.23	265.16	3.88	281.96	3.25	288.45	5.03
105	1.487	256.07	104.28	259.51	81.25	269.56	51.35	270.19	42.31	271.34	8.59	265.36	3.50	281.52	2.74	288.14	4.39
106	1.501	257.61	105.05	260.92	81.63	270.34	51.60	270.85	42.44	270.85	8.84	264.99	3.88	281.15	2.35	287.77	4.52

TABLE A.2a Filtered Marker Kinematics – Rib Cage and Greater Trochanter (Hip)

			Base Rib Cage						Right Hip					
	Frame	Time	X	VX	AX	Y	VY	AY	X	VX	AX	Y	VY	AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	0.4695	1.43	0.3	1.0435	0.22	1.0	0.4474	1.64	−2.2	0.7870	0.01	3.3
	2	0.014	0.4900	1.41	−1.6	1.0467	0.23	0.5	0.4706	1.59	−4.6	0.7875	0.07	4.3
	3	0.029	0.5100	1.38	−2.9	1.0501	0.24	0.4	0.4928	1.51	−5.6	0.7889	0.14	4.8
	4	0.043	0.5294	1.33	−3.4	1.0535	0.24	0.6	0.5138	1.43	−5.3	0.7914	0.21	4.7
	5	0.057	0.5481	1.28	−3.3	1.0570	0.25	0.6	0.5336	1.36	−3.9	0.7948	0.27	3.9
	6	0.072	0.5661	1.24	−2.5	1.0607	0.26	0.2	0.5526	1.31	−2.0	0.7991	0.32	2.6
	7	0.086	0.5836	1.21	−1.5	1.0644	0.26	−0.7	0.5712	1.30	−0.2	0.8039	0.34	1.2
	8	0.100	0.6007	1.20	−0.5	1.0680	0.24	−1.8	0.5898	1.31	1.0	0.8089	0.35	−0.2
	9	0.114	0.6178	1.20	0.2	1.0713	0.21	−2.8	0.6086	1.33	1.7	0.8139	0.34	−1.5
	10	0.129	0.6349	1.20	0.5	1.0739	0.16	−3.5	0.6278	1.36	1.9	0.8185	0.31	−2.8
	11	0.143	0.6521	1.21	0.5	1.0759	0.11	−3.7	0.6474	1.38	1.9	0.8226	0.26	−3.8
	12	0.157	0.6695	1.22	0.5	1.0770	0.05	−3.4	0.6673	1.41	1.6	0.8259	0.20	−4.4
	13	0.172	0.6869	1.22	0.5	1.0774	0.01	−2.7	0.6876	1.43	1.1	0.8283	0.13	−4.6
	14	0.186	0.7044	1.23	0.6	1.0773	−0.02	−2.1	0.7082	1.44	0.5	0.8298	0.07	−4.7
	15	0.200	0.7221	1.24	1.0	1.0767	−0.05	−1.9	0.7288	1.44	0.2	0.8302	0.00	−4.7
	16	0.215	0.7400	1.26	1.6	1.0758	−0.08	−2.1	0.7495	1.45	0.3	0.8298	−0.07	−4.6
	17	0.229	0.7581	1.29	2.2	1.0745	−0.11	−2.3	0.7702	1.45	0.7	0.8283	−0.13	−4.4
	18	0.243	0.7768	1.32	2.8	1.0727	−0.14	−2.5	0.7910	1.47	1.1	0.8260	−0.19	−4.0
	19	0.257	0.7960	1.37	3.0	1.0704	−0.18	−2.8	0.8121	1.48	1.4	0.8228	−0.25	−3.1
	20	0.272	0.8158	1.41	2.9	1.0675	−0.22	−2.8	0.8335	1.51	1.7	0.8189	−0.28	−1.7
	21	0.286	0.8363	1.45	2.5	1.0640	−0.26	−2.4	0.8552	1.53	1.8	0.8147	−0.30	0.1
	22	0.300	0.8573	1.48	2.0	1.0600	−0.29	−1.6	0.8773	1.56	1.9	0.8105	−0.28	1.6
	23	0.315	0.8787	1.51	1.4	1.0557	−0.31	−0.7	0.8997	1.59	2.1	0.8067	−0.25	2.3
	24	0.329	0.9003	1.52	1.1	1.0512	−0.31	0.3	0.9226	1.62	2.2	0.8033	−0.21	2.6
	25	0.343	0.9222	1.54	1.3	1.0468	−0.30	1.7	0.9460	1.65	2.1	0.8005	−0.18	2.8
	26	0.357	0.9443	1.56	1.6	1.0427	−0.26	3.2	0.9698	1.68	1.6	0.7983	−0.13	3.1
	27	0.372	0.9668	1.58	1.8	1.0392	−0.21	4.4	0.9940	1.69	0.8	0.7967	−0.09	3.7
HCR	28	0.386	0.9896	1.61	1.5	1.0367	−0.14	5.0	1.0183	1.70	−0.4	0.7959	−0.03	4.4
	29	0.400	1.0128	1.63	0.9	1.0353	−0.06	4.9	1.0425	1.68	−1.7	0.7959	0.04	4.9

30	0.415	1.0362	1.64	0.0	1.0349	0.00	4.3	1.0664	1.65	−3.0	0.7970	0.11	5.0
31	0.429	1.0596	1.63	−0.9	1.0353	0.06	3.7	1.0897	1.60	−4.0	0.7991	0.18	4.5
32	0.443	1.0828	1.61	−1.8	1.0366	0.11	3.2	1.1121	1.53	−4.7	0.8023	0.24	3.1
33	0.458	1.1056	1.58	−2.5	1.0384	0.15	2.9	1.1336	1.46	−4.7	0.8060	0.27	1.1
34	0.472	1.1279	1.54	−2.8	1.0408	0.19	2.7	1.1540	1.40	−3.8	0.8100	0.27	−0.6
35	0.486	1.1496	1.50	−2.6	1.0438	0.23	2.3	1.1736	1.36	−2.3	0.8139	0.25	−1.5
36	0.500	1.1707	1.46	−2.2	1.0473	0.26	1.8	1.1928	1.34	−1.0	0.8173	0.23	−1.4
37	0.515	1.1915	1.43	−2.0	1.0512	0.28	1.0	1.2118	1.33	−0.8	0.8205	0.21	−1.3
38	0.529	1.2118	1.41	−2.1	1.0552	0.29	0.3	1.2307	1.31	−1.4	0.8234	0.19	−1.6
39	0.543	1.2317	1.38	−2.1	1.0594	0.29	−0.2	1.2494	1.29	−2.4	0.8260	0.17	−2.0
40	0.558	1.2511	1.35	−1.8	1.0634	0.28	−0.7	1.2675	1.25	−3.2	0.8282	0.14	−2.0
41	0.572	1.2702	1.32	−1.3	1.0673	0.27	−1.1	1.2850	1.20	−3.6	0.8300	0.11	−1.7
42	0.586	1.2890	1.31	−0.7	1.0710	0.25	−1.6	1.3018	1.14	−3.3	0.8314	0.09	−1.6
43	0.601	1.3076	1.30	−0.2	1.0744	0.22	−2.4	1.3178	1.10	−2.6	0.8325	0.07	−1.9
44	0.615	1.3263	1.30	0.3	1.0773	0.18	−3.2	1.3332	1.07	−1.7	0.8333	0.03	−2.7
45	0.629	1.3450	1.31	0.7	1.0796	0.13	−4.0	1.3484	1.05	−0.9	0.8335	−0.01	−3.3
46	0.643	1.3638	1.33	1.0	1.0810	0.07	−4.6	1.3633	1.04	−0.3	0.8330	−0.06	−3.3
47	0.658	1.3829	1.34	1.1	1.0815	0.00	−5.0	1.3782	1.04	0.3	0.8318	−0.11	−2.6
48	0.672	1.4021	1.36	1.1	1.0809	−0.08	−4.8	1.3931	1.05	1.1	0.8300	−0.14	−1.8
49	0.686	1.4217	1.37	1.1	1.0793	−0.14	−4.0	1.4083	1.07	1.9	0.8279	−0.16	−1.2
50	0.701	1.4414	1.39	0.9	1.0769	−0.19	−3.0	1.4239	1.11	2.2	0.8255	−0.17	−0.8
51	0.715	1.4614	1.40	0.6	1.0739	−0.23	−2.3	1.4400	1.14	1.8	0.8230	−0.18	−0.5
52	0.729	1.4814	1.40	0.4	1.0704	−0.26	−1.9	1.4564	1.16	1.2	0.8204	−0.18	−0.2
53	0.744	1.5015	1.41	0.5	1.0666	−0.28	−1.3	1.4731	1.17	1.1	0.8178	−0.18	0.2
54	0.758	1.5217	1.42	0.8	1.0624	−0.29	−0.5	1.4900	1.19	1.4	0.8151	−0.18	0.5
55	0.772	1.5421	1.43	0.9	1.0582	−0.29	0.5	1.5071	1.21	1.8	0.8127	−0.17	0.6
56	0.786	1.5626	1.44	0.7	1.0541	−0.28	1.3	1.5247	1.24	1.9	0.8103	−0.16	0.5
57	0.801	1.5833	1.45	0.5	1.0503	−0.25	1.9	1.5426	1.27	2.3	0.8081	−0.15	0.4
58	0.815	1.6041	1.46	0.7	1.0468	−0.22	2.1	1.5610	1.31	3.1	0.8059	−0.15	0.4
59	0.829	1.6250	1.47	1.5	1.0439	−0.19	2.1	1.5800	1.36	4.5	0.8038	−0.14	0.3
60	0.844	1.6461	1.50	2.6	1.0413	−0.16	1.9	1.5999	1.44	6.0	0.8018	−0.14	0.0
61	0.858	1.6678	1.54	3.6	1.0392	−0.14	1.5	1.6210	1.53	7.2	0.7998	−0.14	−0.3
62	0.872	1.6903	1.60	4.0	1.0373	−0.12	1.2	1.6436	1.64	7.7	0.7977	−0.15	−0.3
63	0.887	1.7136	1.66	3.6	1.0357	−0.10	1.5	1.6679	1.75	7.3	0.7955	−0.15	−0.1

(*continued*)

TABLE A.2a (*Continued*)

	Frame	Time	Base Rib Cage X	VX	AX	Y	VY	AY	Right Hip X	VX	AX	Y	VY	AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	64	0.901	1.7377	1.70	1.8	1.0343	−0.08	2.3	1.6937	1.85	5.5	0.7933	−0.15	0.2
	65	0.915	1.7623	1.71	−1.0	1.0334	−0.04	3.2	1.7208	1.91	2.5	0.7911	−0.15	0.7
	66	0.929	1.7866	1.67	−4.0	1.0332	0.01	4.0	1.7483	1.92	−1.1	0.7891	−0.13	1.3
	67	0.944	1.8101	1.60	−6.0	1.0338	0.07	4.3	1.7756	1.88	−4.2	0.7873	−0.11	2.0
	68	0.958	1.8323	1.50	−6.4	1.0353	0.14	4.2	1.8020	1.80	−6.4	0.7859	−0.08	2.9
	69	0.972	1.8531	1.41	−5.8	1.0377	0.19	3.8	1.8270	1.69	−7.5	0.7851	−0.03	3.8
TOR	70	0.987	1.8727	1.34	−4.7	1.0409	0.24	3.0	1.8505	1.58	−7.7	0.7851	0.03	4.5
	71	1.001	1.8914	1.28	−3.5	1.0447	0.28	1.8	1.8723	1.48	−6.8	0.7861	0.10	4.8
	72	1.015	1.9093	1.24	−2.4	1.0489	0.30	0.4	1.8927	1.39	−5.0	0.7880	0.17	4.6
	73	1.030	1.9268	1.21	−1.4	1.0532	0.29	−0.8	1.9120	1.33	−3.0	0.7909	0.23	4.1
	74	1.044	1.9440	1.20	−0.7	1.0572	0.27	−1.3	1.9308	1.30	−1.1	0.7947	0.29	3.5
	75	1.058	1.9610	1.19	−0.3	1.0610	0.26	−1.1	1.9493	1.30	0.5	0.7991	0.33	2.8
	76	1.072	1.9780	1.19	0.1	1.0645	0.24	−1.1	1.9680	1.32	1.7	0.8042	0.37	1.5
	77	1.087	1.9951	1.19	0.4	1.0679	0.22	−1.6	1.9870	1.35	2.4	0.8096	0.38	−0.4
	78	1.101	2.0122	1.20	0.7	1.0710	0.19	−2.6	2.0066	1.39	2.5	0.8149	0.36	−2.3
	79	1.115	2.0294	1.21	1.0	1.0735	0.15	−3.4	2.0267	1.42	2.0	0.8198	0.31	−3.9
	80	1.130	2.0469	1.23	1.3	1.0753	0.10	−3.7	2.0473	1.45	1.1	0.8238	0.24	−4.8
	81	1.144	2.0647	1.25	1.3	1.0763	0.04	−3.8	2.0681	1.45	0.1	0.8268	0.17	−5.0
	82	1.158	2.0827	1.27	1.0	1.0765	−0.01	−3.7	2.0889	1.45	−0.8	0.8287	0.10	−4.9
	83	1.173	2.1009	1.28	0.7	1.0760	−0.06	−3.6	2.1095	1.43	−1.1	0.8297	0.03	−4.7
	84	1.187	2.1193	1.29	0.5	1.0748	−0.11	−3.4	2.1298	1.42	−1.1	0.8296	−0.03	−4.4
	85	1.201	2.1378	1.30	0.6	1.0728	−0.16	−3.0	2.1500	1.40	−0.7	0.8287	−0.09	−4.2
	86	1.215	2.1563	1.30	0.8	1.0702	−0.20	−2.7	2.1699	1.40	0.1	0.8270	−0.15	−4.1
	87	1.230	2.1751	1.32	1.1	1.0671	−0.24	−2.3	2.1899	1.40	1.0	0.8244	−0.21	−3.8
	88	1.244	2.1940	1.34	1.5	1.0635	−0.27	−1.8	2.2100	1.43	2.0	0.8210	−0.26	−3.1
	89	1.258	2.2133	1.36	1.6	1.0595	−0.29	−1.1	2.2306	1.46	2.6	0.8169	−0.30	−2.0
	90	1.273	2.2329	1.38	1.6	1.0552	−0.30	−0.3	2.2518	1.50	2.6	0.8124	−0.32	−0.7
	91	1.287	2.2529	1.41	1.7	1.0510	−0.30	0.4	2.2735	1.54	2.6	0.8078	−0.32	0.6
	92	1.301	2.2731	1.43	1.9	1.0468	−0.29	0.8	2.2957	1.57	2.8	0.8033	−0.30	1.8
	93	1.316	2.2938	1.46	2.2	1.0428	−0.27	1.3	2.3185	1.62	3.3	0.7992	−0.27	3.0

	94	1.330	2.3149	1.49	2.4	1.0390	−0.25	1.9	2.3419	1.67	3.5	0.7957	−0.22	4.0
	95	1.344	2.3365	1.53	2.6	1.0356	−0.22	2.8	2.3661	1.72	3.1	0.7930	−0.15	4.8
	96	1.358	2.3587	1.57	2.6	1.0328	−0.17	3.5	2.3910	1.75	2.0	0.7913	−0.08	5.3
HCR	97	1.373	2.3813	1.60	2.2	1.0307	−0.12	3.7	2.4163	1.77	0.2	0.7908	0.00	5.3
	98	1.387	2.4045	1.63	1.3	1.0294	−0.06	3.9	2.4417	1.76	−1.7	0.7913	0.07	5.2
	99	1.401	2.4279	1.64	0.1	1.0289	−0.01	4.1	2.4667	1.72	−3.5	0.7929	0.15	4.8
	100	1.416	2.4514	1.63	−1.1	1.0292	0.05	4.5	2.4909	1.66	−4.7	0.7955	0.21	4.0
	101	1.430	2.4746	1.61	−2.3	1.0304	0.12	4.7	2.5142	1.59	−5.2	0.7989	0.26	2.4
	102	1.444	2.4974	1.57	−3.2	1.0327	0.19	4.4	2.5364	1.51	−5.3	0.8029	0.28	0.6
	103	1.459	2.5194	1.52	−4.0	1.0358	0.25	3.5	2.5575	1.44	−5.2	0.8069	0.28	−1.0
	104	1.473	2.5407	1.45	−4.9	1.0398	0.29	2.1	2.5775	1.36	−5.2	0.8108	0.25	−1.8
	105	1.487	2.5610	1.37	−6.4	1.0441	0.31	0.4	2.5965	1.29	−6.0	0.8142	0.22	−2.0
	106	1.501	2.5800	1.27	−8.6	1.0485	0.30	−1.1	2.6143	1.19	−7.9	0.8171	0.19	−2.1

TABLE A.2b Filtered Marker Kinematics – Femoral Lateral Epicondyle (Knee) and Head of Fibula

			Right Knee						Right Fibula					
	Frame	Time	X	VX	AX	Y	VY	AY	X	VX	AX	Y	VY	AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	0.4075	2.66	6.0	0.4754	−0.13	5.9	0.3570	2.84	8.4	0.4052	−0.16	5.1
	2	0.014	0.4461	2.71	1.6	0.4741	−0.03	8.3	0.3985	2.93	3.7	0.4036	−0.07	7.1
	3	0.029	0.4850	2.71	−1.5	0.4746	0.10	9.5	0.4407	2.95	0.0	0.4034	0.05	8.2
	4	0.043	0.5235	2.67	−3.5	0.4771	0.24	9.4	0.4829	2.93	−2.8	0.4049	0.17	8.2
	5	0.057	0.5613	2.61	−4.8	0.4815	0.37	8.2	0.5244	2.87	−4.8	0.4082	0.28	7.1
	6	0.072	0.5980	2.53	−5.8	0.4877	0.48	6.1	0.5649	2.79	−5.9	0.4130	0.37	5.4
	7	0.086	0.6336	2.44	−6.5	0.4952	0.55	3.6	0.6041	2.70	−6.7	0.4189	0.44	3.4
	8	0.100	0.6678	2.34	−7.0	0.5034	0.58	1.2	0.6420	2.60	−7.2	0.4255	0.47	1.4
	9	0.114	0.7006	2.24	−7.4	0.5118	0.58	−0.7	0.6784	2.49	−7.7	0.4324	0.48	−0.7
	10	0.129	0.7319	2.13	−7.8	0.5200	0.56	−2.4	0.7133	2.38	−8.1	0.4391	0.45	−2.5
	11	0.143	0.7615	2.02	−8.0	0.5278	0.51	−4.1	0.7464	2.26	−8.3	0.4453	0.40	−4.1
	12	0.157	0.7896	1.90	−7.9	0.5347	0.44	−5.7	0.7779	2.14	−8.4	0.4506	0.34	−5.1
	13	0.172	0.8159	1.79	−7.6	0.5404	0.35	−7.0	0.8076	2.02	−8.3	0.4549	0.26	−5.7
	14	0.186	0.8407	1.68	−7.2	0.5447	0.24	−7.8	0.8357	1.90	−8.1	0.4580	0.17	−6.0
	15	0.200	0.8641	1.58	−6.7	0.5474	0.13	−8.3	0.8620	1.79	−7.9	0.4598	0.08	−6.1
	16	0.215	0.8860	1.49	−6.2	0.5483	0.01	−8.3	0.8868	1.68	−7.4	0.4604	0.00	−5.9
	17	0.229	0.9067	1.41	−5.6	0.5476	−0.11	−8.0	0.9100	1.57	−6.5	0.4597	−0.09	−5.6
	18	0.243	0.9263	1.33	−4.7	0.5451	−0.22	−7.1	0.9318	1.49	−5.1	0.4579	−0.16	−5.0
	19	0.257	0.9448	1.27	−3.2	0.5412	−0.31	−5.6	0.9526	1.43	−3.3	0.4551	−0.23	−3.9
	20	0.272	0.9627	1.24	−1.2	0.5361	−0.38	−3.6	0.9727	1.40	−1.4	0.4514	−0.27	−2.4
	21	0.286	0.9803	1.24	0.9	0.5303	−0.42	−0.9	0.9925	1.39	0.2	0.4472	−0.30	−0.6
	22	0.300	0.9981	1.27	2.8	0.5242	−0.41	1.8	1.0124	1.40	1.2	0.4430	−0.29	0.6
	23	0.315	1.0165	1.32	4.2	0.5186	−0.36	3.8	1.0326	1.42	1.7	0.4389	−0.28	1.1
	24	0.329	1.0358	1.39	5.0	0.5138	−0.30	4.7	1.0531	1.45	1.8	0.4350	−0.26	1.0
	25	0.343	1.0561	1.46	4.8	0.5100	−0.23	4.7	1.0741	1.48	1.6	0.4314	−0.25	1.0
	26	0.357	1.0776	1.52	3.7	0.5072	−0.17	4.4	1.0953	1.50	1.0	0.4279	−0.23	1.5
	27	0.372	1.0997	1.57	1.8	0.5053	−0.10	4.2	1.1169	1.51	0.2	0.4247	−0.21	2.6
HCR	28	0.386	1.1223	1.58	−0.2	0.5042	−0.05	4.1	1.1384	1.50	−0.8	0.4220	−0.16	4.0
	29	0.400	1.1448	1.56	−1.6	0.5039	0.01	4.2	1.1598	1.48	−1.5	0.4202	−0.09	4.9
	30	0.415	1.1669	1.53	−2.1	0.5046	0.07	4.1	1.1808	1.46	−1.9	0.4194	−0.02	4.8

31	0.429	1.1885	1.50	−2.3	0.5061	0.13	3.5	1.2015	1.43	−2.6	0.4196	0.05	3.7
32	0.443	1.2097	1.46	−3.2	0.5083	0.17	2.0	1.2216	1.38	−4.0	0.4207	0.09	1.9
33	0.458	1.2304	1.41	−5.3	0.5110	0.19	−0.1	1.2410	1.31	−6.2	0.4221	0.10	0.1
34	0.472	1.2500	1.31	−7.7	0.5138	0.17	−2.1	1.2592	1.21	−8.5	0.4235	0.09	−1.1
35	0.486	1.2679	1.19	−9.5	0.5160	0.13	−3.3	1.2755	1.07	−9.9	0.4247	0.07	−1.7
36	0.500	1.2839	1.04	−9.8	0.5175	0.08	−3.5	1.2898	0.92	−10.4	0.4255	0.04	−1.6
37	0.515	1.2977	0.90	−9.3	0.5182	0.03	−2.9	1.3019	0.77	−10.4	0.4260	0.02	−1.0
38	0.529	1.3097	0.78	−8.2	0.5183	−0.01	−2.0	1.3119	0.63	−9.8	0.4262	0.02	−0.1
39	0.543	1.3200	0.67	−7.0	0.5180	−0.03	−0.9	1.3198	0.49	−8.3	0.4265	0.02	0.6
40	0.558	1.3288	0.58	−5.4	0.5175	−0.03	0.3	1.3260	0.39	−6.0	0.4269	0.03	0.8
41	0.572	1.3365	0.51	−3.4	0.5171	−0.02	1.3	1.3309	0.32	−3.3	0.4274	0.04	0.4
42	0.586	1.3435	0.48	−1.5	0.5170	0.00	1.3	1.3352	0.29	−1.2	0.4281	0.05	−0.3
43	0.601	1.3502	0.47	−0.1	0.5172	0.02	0.2	1.3393	0.29	0.0	0.4287	0.04	−1.1
44	0.615	1.3570	0.48	0.7	0.5175	0.01	−1.4	1.3435	0.29	0.5	0.4291	0.01	−1.8
45	0.629	1.3639	0.49	1.0	0.5175	−0.02	−2.5	1.3477	0.30	0.8	0.4291	−0.02	−2.0
46	0.643	1.3710	0.51	1.0	0.5169	−0.06	−2.6	1.3521	0.32	1.1	0.4287	−0.04	−1.8
47	0.658	1.3784	0.52	1.0	0.5157	−0.10	−2.0	1.3567	0.33	1.4	0.4278	−0.07	−1.1
48	0.672	1.3859	0.54	1.2	0.5141	−0.12	−1.0	1.3617	0.35	1.6	0.4268	−0.08	−0.3
49	0.686	1.3937	0.55	1.4	0.5123	−0.13	0.1	1.3669	0.38	1.8	0.4256	−0.08	0.5
50	0.701	1.4018	0.58	1.4	0.5105	−0.12	1.1	1.3725	0.41	1.9	0.4246	−0.06	1.3
51	0.715	1.4102	0.59	1.1	0.5089	−0.10	1.7	1.3785	0.43	1.8	0.4238	−0.04	1.9
52	0.729	1.4188	0.61	1.1	0.5077	−0.07	1.8	1.3849	0.46	1.8	0.4235	−0.01	2.1
53	0.744	1.4275	0.62	1.8	0.5069	−0.04	1.6	1.3917	0.49	2.2	0.4236	0.02	1.6
54	0.758	1.4366	0.66	3.2	0.5065	−0.02	1.1	1.3988	0.52	3.0	0.4241	0.04	0.8
55	0.772	1.4463	0.72	4.7	0.5062	−0.01	0.5	1.4066	0.57	4.1	0.4247	0.04	0.2
56	0.786	1.4571	0.79	5.7	0.5061	−0.01	−0.3	1.4151	0.64	5.4	0.4254	0.04	−0.1
57	0.801	1.4690	0.88	6.5	0.5059	−0.02	−1.0	1.4248	0.73	6.8	0.4260	0.04	−0.4
58	0.815	1.4822	0.98	7.5	0.5055	−0.04	−1.5	1.4359	0.83	8.5	0.4265	0.03	−0.9
59	0.829	1.4969	1.09	8.9	0.5048	−0.06	−2.0	1.4486	0.97	10.4	0.4269	0.02	−1.5
60	0.844	1.5135	1.23	10.6	0.5037	−0.10	−2.5	1.4635	1.13	12.1	0.4270	−0.01	−2.1
61	0.858	1.5322	1.40	12.3	0.5021	−0.14	−3.1	1.4809	1.31	13.4	0.4267	−0.04	−2.7
62	0.872	1.5534	1.58	13.8	0.4999	−0.18	−3.2	1.5011	1.51	14.2	0.4257	−0.09	−3.2
63	0.887	1.5775	1.79	14.5	0.4969	−0.23	−2.8	1.5242	1.72	14.4	0.4242	−0.14	−3.3
64	0.901	1.6046	2.00	13.8	0.4933	−0.26	−1.9	1.5503	1.93	14.1	0.4219	−0.18	−2.9

(*continued*)

TABLE A.2b (*Continued*)

	Frame	Time	Right Knee X	Right Knee VX	Right Knee AX	Right Knee Y	Right Knee VY	Right Knee AY	Right Fibula X	Right Fibula VX	Right Fibula AX	Right Fibula Y	Right Fibula VY	Right Fibula AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	65	0.915	1.6346	2.19	11.7	0.4894	−0.28	−0.8	1.5793	2.12	13.0	0.4189	−0.22	−2.2
	66	0.929	1.6671	2.33	9.0	0.4853	−0.28	0.5	1.6110	2.30	11.4	0.4156	−0.24	−1.2
	67	0.944	1.7014	2.44	6.1	0.4812	−0.27	2.0	1.6450	2.45	9.4	0.4119	−0.26	−0.1
	68	0.958	1.7369	2.51	3.5	0.4776	−0.23	3.8	1.6810	2.57	7.6	0.4083	−0.25	1.8
	69	0.972	1.7732	2.54	1.6	0.4747	−0.16	5.6	1.7185	2.66	5.9	0.4049	−0.20	4.3
TOR	70	0.987	1.8097	2.55	0.3	0.4730	−0.07	7.1	1.7572	2.74	4.1	0.4024	−0.12	6.8
	71	1.001	1.8462	2.55	−0.3	0.4728	0.04	8.1	1.7967	2.78	2.3	0.4014	−0.01	8.2
	72	1.015	1.8826	2.54	−0.5	0.4743	0.16	8.2	1.8368	2.80	0.6	0.4021	0.11	8.0
	73	1.030	1.9190	2.54	−0.8	0.4775	0.28	7.6	1.8769	2.80	−0.9	0.4045	0.22	6.7
	74	1.044	1.9552	2.52	−1.5	0.4823	0.38	6.4	1.9168	2.78	−2.2	0.4083	0.30	5.1
	75	1.058	1.9911	2.49	−2.8	0.4884	0.46	4.8	1.9563	2.74	−3.6	0.4132	0.36	3.7
	76	1.072	2.0265	2.44	−4.3	0.4955	0.52	3.0	1.9951	2.68	−4.9	0.4188	0.41	2.3
	77	1.087	2.0610	2.37	−5.7	0.5032	0.55	1.0	2.0329	2.60	−6.1	0.4249	0.43	0.7
	78	1.101	2.0943	2.28	−6.8	0.5112	0.55	−1.1	2.0694	2.50	−7.0	0.4311	0.43	−1.2
	79	1.115	2.1262	2.18	−7.4	0.5189	0.52	−2.8	2.1044	2.40	−7.7	0.4371	0.40	−2.9
	80	1.130	2.1565	2.07	−7.7	0.5260	0.47	−4.1	2.1379	2.28	−8.0	0.4424	0.34	−4.2
	81	1.144	2.1853	1.96	−7.7	0.5323	0.40	−5.0	2.1697	2.17	−8.0	0.4469	0.28	−5.0
	82	1.158	2.2124	1.85	−7.6	0.5375	0.32	−6.0	2.1999	2.05	−8.1	0.4504	0.20	−5.7
	83	1.173	2.2381	1.74	−7.4	0.5416	0.23	−6.9	2.2284	1.94	−8.0	0.4527	0.12	−6.1
	84	1.187	2.2622	1.63	−7.2	0.5442	0.13	−7.9	2.2553	1.82	−7.7	0.4537	0.03	−6.2
	85	1.201	2.2848	1.53	−6.9	0.5452	0.01	−8.7	2.2805	1.72	−6.9	0.4534	−0.06	−5.9
	86	1.215	2.3060	1.44	−6.3	0.5443	−0.12	−8.9	2.3043	1.63	−5.5	0.4519	−0.14	−5.1
	87	1.230	2.3259	1.35	−5.4	0.5416	−0.25	−8.1	2.3270	1.56	−3.9	0.4493	−0.21	−4.0
	88	1.244	2.3446	1.28	−4.0	0.5372	−0.35	−6.2	2.3489	1.51	−2.5	0.4459	−0.26	−2.4
	89	1.258	2.3625	1.24	−2.0	0.5315	−0.43	−3.6	2.3703	1.49	−1.5	0.4420	−0.28	−0.8
	90	1.273	2.3800	1.22	0.3	0.5250	−0.46	−0.7	2.3914	1.47	−0.9	0.4380	−0.28	0.7
	91	1.287	2.3975	1.25	2.9	0.5184	−0.45	2.2	2.4124	1.46	−0.3	0.4341	−0.26	1.6
	92	1.301	2.4156	1.31	5.4	0.5122	−0.39	4.7	2.4333	1.46	0.7	0.4305	−0.23	1.7
	93	1.316	2.4349	1.40	6.9	0.5071	−0.31	6.2	2.4542	1.48	1.7	0.4274	−0.21	1.6
	94	1.330	2.4556	1.51	7.1	0.5033	−0.22	6.7	2.4756	1.51	2.5	0.4246	−0.19	1.7

	95	1.344	2.4779	1.60	5.8	0.5009	−0.12	6.4	2.4975	1.55	2.6	0.4220	−0.16	2.3
	96	1.358	2.5014	1.67	3.7	0.4998	−0.03	5.4	2.5200	1.59	2.1	0.4200	−0.12	3.3
HCR	97	1.373	2.5257	1.71	1.3	0.4999	0.03	4.3	2.5429	1.61	1.2	0.4186	−0.07	4.1
	98	1.387	2.5502	1.71	−0.7	0.5008	0.09	3.4	2.5661	1.62	−0.1	0.4180	0.00	4.4
	99	1.401	2.5746	1.69	−2.4	0.5025	0.13	2.7	2.5893	1.61	−1.9	0.4185	0.06	3.9
	100	1.416	2.5984	1.64	−4.3	0.5047	0.17	1.6	2.6121	1.57	−4.4	0.4197	0.11	2.2
	101	1.430	2.6214	1.56	−6.8	0.5072	0.18	0.0	2.6341	1.48	−7.5	0.4215	0.12	0.0
	102	1.444	2.6431	1.45	−9.5	0.5098	0.17	−1.5	2.6545	1.35	−10.6	0.4232	0.11	−1.7
	103	1.459	2.6628	1.29	−11.7	0.5120	0.13	−2.5	2.6727	1.18	−12.7	0.4246	0.07	−2.5
	104	1.473	2.6800	1.11	−12.7	0.5136	0.10	−2.6	2.6883	0.99	−13.3	0.4253	0.04	−2.2
	105	1.487	2.6945	0.93	−12.3	0.5147	0.06	−2.2	2.7010	0.80	−12.6	0.4256	0.01	−1.6
	106	1.501	2.7065	0.76	−11.4	0.5153	0.03	−1.7	2.7111	0.63	−11.4	0.4256	−0.01	−1.0

TABLE A.2c Filtered Marker Kinematics – Lateral Malleolus (Ankle) and Heel

			Right Ankle						Right Heel					
	Frame	Time	X	VX	AX	Y	VY	AY	X	VX	AX	Y	VY	AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	0.0939	2.24	19.1	0.2143	0.80	−6.3	0.0300	2.39	17.2	0.2360	1.32	−15.9
	2	0.014	0.1279	2.50	16.4	0.2251	0.68	−10.0	0.0659	2.59	12.0	0.2531	1.03	−22.6
	3	0.029	0.1653	2.71	13.5	0.2337	0.51	−13.2	0.1042	2.73	8.0	0.2655	0.67	−26.3
	4	0.043	0.2054	2.88	10.8	0.2396	0.30	−15.2	0.1440	2.82	5.7	0.2723	0.28	−27.2
	5	0.057	0.2477	3.02	8.7	0.2423	0.07	−16.0	0.1849	2.89	4.8	0.2734	−0.11	−26.1
	6	0.072	0.2917	3.13	7.4	0.2417	−0.16	−15.5	0.2267	2.96	5.4	0.2692	−0.47	−23.7
	7	0.086	0.3372	3.23	6.7	0.2379	−0.37	−14.1	0.2696	3.05	7.0	0.2600	−0.79	−20.4
	8	0.100	0.3841	3.32	6.4	0.2311	−0.56	−12.1	0.3138	3.16	9.0	0.2466	−1.05	−16.4
	9	0.114	0.4322	3.41	6.5	0.2219	−0.72	−9.7	0.3599	3.30	10.5	0.2299	−1.26	−11.9
	10	0.129	0.4817	3.51	6.7	0.2106	−0.84	−7.1	0.4083	3.46	11.3	0.2107	−1.39	−7.3
	11	0.143	0.5326	3.60	6.8	0.1979	−0.92	−4.0	0.4590	3.63	11.2	0.1900	−1.47	−3.0
	12	0.157	0.5848	3.70	6.4	0.1844	−0.95	−0.5	0.5120	3.78	10.5	0.1687	−1.48	1.0
	13	0.172	0.6384	3.79	5.4	0.1707	−0.93	3.0	0.5671	3.93	9.3	0.1477	−1.44	4.5
	14	0.186	0.6931	3.85	3.7	0.1577	−0.86	6.2	0.6242	4.05	7.5	0.1276	−1.35	7.3
	15	0.200	0.7486	3.89	1.3	0.1460	−0.76	8.6	0.6829	4.14	5.2	0.1090	−1.23	9.7
	16	0.215	0.8045	3.89	−1.6	0.1360	−0.62	10.1	0.7427	4.20	2.1	0.0925	−1.07	11.6
	17	0.229	0.8600	3.85	−5.1	0.1282	−0.47	10.7	0.8029	4.20	−1.7	0.0783	−0.90	12.9
	18	0.243	0.9145	3.75	−8.8	0.1226	−0.31	10.3	0.8628	4.15	−6.2	0.0668	−0.70	13.3
	19	0.257	0.9672	3.60	−12.5	0.1192	−0.17	8.9	0.9215	4.02	−11.4	0.0582	−0.51	12.6
	20	0.272	1.0173	3.39	−16.2	0.1177	−0.06	6.7	0.9779	3.82	−17.1	0.0521	−0.34	10.8
	21	0.286	1.0641	3.13	−20.0	0.1175	0.02	3.9	1.0309	3.53	−23.2	0.0483	−0.21	8.1
	22	0.300	1.1068	2.82	−23.6	0.1182	0.05	0.7	1.0790	3.16	−28.9	0.0462	−0.11	4.6
	23	0.315	1.1447	2.46	−26.3	0.1190	0.04	−2.5	1.1212	2.71	−33.1	0.0451	−0.07	1.1
	24	0.329	1.1771	2.07	−27.4	0.1193	−0.02	−5.1	1.1565	2.21	−35.2	0.0441	−0.08	−1.1
	25	0.343	1.2038	1.67	−26.3	0.1185	−0.11	−6.5	1.1844	1.70	−34.7	0.0428	−0.11	−1.2
	26	0.357	1.2249	1.31	−23.1	0.1162	−0.21	−6.1	1.2051	1.22	−31.6	0.0411	−0.12	0.4
	27	0.372	1.2413	1.01	−18.3	0.1126	−0.28	−3.7	1.2192	0.80	−26.1	0.0395	−0.09	2.4
HCR	28	0.386	1.2539	0.79	−13.3	0.1081	−0.31	−0.2	1.2280	0.47	−19.3	0.0384	−0.05	3.5
	29	0.400	1.2639	0.63	−9.2	0.1037	−0.29	3.0	1.2327	0.25	−12.5	0.0381	0.01	3.3
	30	0.415	1.2720	0.53	−6.7	0.0999	−0.23	4.6	1.2350	0.11	−6.8	0.0385	0.05	2.1

31	0.429	1.2789	0.44	−5.5	0.0972	−0.16	4.7	1.2360	0.05	−2.7	0.0394	0.07	0.9
32	0.443	1.2846	0.37	−5.1	0.0955	−0.09	3.7	1.2365	0.04	−0.4	0.0404	0.07	0.1
33	0.458	1.2894	0.30	−4.8	0.0945	−0.05	2.3	1.2370	0.04	0.5	0.0414	0.07	−0.3
34	0.472	1.2931	0.23	−4.3	0.0940	−0.03	1.2	1.2377	0.05	0.8	0.0424	0.06	−0.4
35	0.486	1.2960	0.18	−3.3	0.0937	−0.02	0.6	1.2384	0.06	0.9	0.0432	0.06	−0.3
36	0.500	1.2982	0.14	−2.4	0.0935	−0.01	0.5	1.2394	0.08	0.9	0.0440	0.05	−0.5
37	0.515	1.2999	0.11	−1.9	0.0934	0.00	0.5	1.2406	0.09	0.2	0.0448	0.05	−0.8
38	0.529	1.3012	0.08	−2.1	0.0935	0.01	0.4	1.2419	0.08	−1.1	0.0453	0.03	−1.0
39	0.543	1.3022	0.05	−2.2	0.0936	0.01	0.3	1.2429	0.06	−2.2	0.0457	0.02	−0.9
40	0.558	1.3026	0.02	−2.0	0.0938	0.01	0.5	1.2435	0.02	−2.5	0.0458	0.00	−0.6
41	0.572	1.3026	−0.01	−1.3	0.0940	0.02	1.0	1.2435	−0.01	−1.8	0.0458	0.00	0.0
42	0.586	1.3023	−0.02	−0.5	0.0945	0.04	1.3	1.2431	−0.03	−0.7	0.0458	0.00	0.6
43	0.601	1.3020	−0.02	0.3	0.0952	0.06	1.2	1.2425	−0.04	0.3	0.0459	0.02	1.1
44	0.615	1.3017	−0.01	0.9	0.0962	0.08	0.6	1.2421	−0.02	1.1	0.0463	0.04	1.0
45	0.629	1.3016	0.00	1.5	0.0974	0.08	−0.2	1.2418	0.00	1.6	0.0469	0.05	0.5
46	0.643	1.3018	0.03	1.7	0.0985	0.07	−0.9	1.2420	0.02	1.8	0.0477	0.05	−0.1
47	0.658	1.3024	0.05	1.6	0.0994	0.05	−1.1	1.2425	0.05	1.7	0.0484	0.04	−0.5
48	0.672	1.3034	0.07	1.3	0.1000	0.04	−0.9	1.2433	0.07	1.6	0.0489	0.03	−0.5
49	0.686	1.3046	0.09	0.9	0.1005	0.03	−0.3	1.2445	0.09	1.3	0.0494	0.03	0.1
50	0.701	1.3059	0.10	0.5	0.1009	0.03	0.3	1.2460	0.11	0.9	0.0498	0.04	1.2
51	0.715	1.3074	0.11	0.1	0.1014	0.04	0.8	1.2477	0.12	0.2	0.0504	0.06	2.3
52	0.729	1.3090	0.10	−0.4	0.1020	0.05	1.4	1.2494	0.12	−0.6	0.0516	0.10	3.1
53	0.744	1.3104	0.10	−0.5	0.1029	0.08	1.9	1.2510	0.10	−0.9	0.0534	0.15	3.4
54	0.758	1.3117	0.09	−0.2	0.1042	0.11	2.6	1.2523	0.09	−0.6	0.0559	0.20	3.5
55	0.772	1.3129	0.09	0.3	0.1061	0.15	3.2	1.2536	0.08	−0.3	0.0591	0.25	3.5
56	0.786	1.3142	0.10	0.8	0.1086	0.20	3.6	1.2547	0.08	−0.1	0.0631	0.30	3.5
57	0.801	1.3157	0.11	1.4	0.1118	0.25	3.6	1.2559	0.08	0.3	0.0677	0.35	3.8
58	0.815	1.3174	0.14	2.7	0.1158	0.30	3.2	1.2571	0.09	1.5	0.0731	0.41	4.6
59	0.829	1.3196	0.19	4.7	0.1205	0.35	3.2	1.2585	0.12	3.5	0.0794	0.48	5.9
60	0.844	1.3227	0.27	7.2	0.1257	0.39	3.5	1.2606	0.19	6.1	0.0869	0.58	7.4
61	0.858	1.3274	0.39	9.8	0.1317	0.45	4.0	1.2640	0.30	9.1	0.0960	0.69	8.5
62	0.872	1.3340	0.55	12.2	0.1385	0.51	4.4	1.2692	0.45	12.3	0.1068	0.82	9.3
63	0.887	1.3432	0.74	14.0	0.1463	0.57	4.5	1.2769	0.65	15.4	0.1195	0.96	9.9
64	0.901	1.3552	0.95	15.0	0.1549	0.64	4.4	1.2878	0.89	17.8	0.1343	1.11	10.0

(continued)

TABLE A.2c (*Continued*)

	Frame	Time	Right Ankle X	Right Ankle VX	Right Ankle AX	Right Ankle Y	Right Ankle VY	Right Ankle AY	Right Heel X	Right Heel VX	Right Heel AX	Right Heel Y	Right Heel VY	Right Heel AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	65	0.915	1.3704	1.17	15.3	0.1645	0.70	3.6	1.3024	1.16	19.1	0.1511	1.25	9.1
	66	0.929	1.3887	1.39	15.2	0.1749	0.74	2.2	1.3210	1.44	19.2	0.1700	1.37	6.1
	67	0.944	1.4101	1.61	15.1	0.1857	0.76	0.6	1.3436	1.71	18.0	0.1902	1.42	0.8
	68	0.958	1.4346	1.82	14.9	0.1967	0.76	−1.1	1.3699	1.95	15.8	0.2106	1.39	−5.7
	69	0.972	1.4622	2.03	14.4	0.2074	0.73	−3.0	1.3995	2.16	12.9	0.2299	1.26	−11.7
TOR	70	0.987	1.4928	2.23	13.4	0.2175	0.67	−5.3	1.4317	2.32	9.9	0.2467	1.05	−16.4
	71	1.001	1.5261	2.41	12.2	0.2266	0.58	−8.2	1.4659	2.44	7.8	0.2600	0.79	−19.6
	72	1.015	1.5619	2.58	11.3	0.2340	0.44	−11.3	1.5016	2.55	7.3	0.2693	0.49	−21.8
	73	1.030	1.5999	2.74	10.7	0.2391	0.25	−13.9	1.5387	2.65	7.8	0.2741	0.17	−22.9
	74	1.044	1.6401	2.89	10.2	0.2413	0.04	−15.2	1.5774	2.77	8.5	0.2741	−0.16	−22.6
	75	1.058	1.6824	3.03	9.7	0.2403	−0.18	−15.1	1.6178	2.89	9.0	0.2694	−0.48	−21.1
	76	1.072	1.7267	3.16	9.3	0.2361	−0.39	−13.8	1.6601	3.03	9.4	0.2604	−0.76	−18.8
	77	1.087	1.7729	3.29	8.8	0.2291	−0.58	−11.9	1.7044	3.16	9.8	0.2475	−1.02	−16.2
	78	1.101	1.8209	3.42	8.2	0.2197	−0.73	−9.7	1.7506	3.31	10.3	0.2314	−1.23	−13.3
	79	1.115	1.8706	3.53	7.5	0.2082	−0.85	−7.3	1.7990	3.46	10.8	0.2124	−1.40	−9.9
	80	1.130	1.9218	3.63	6.9	0.1953	−0.94	−4.5	1.8495	3.62	11.2	0.1914	−1.51	−5.8
	81	1.144	1.9745	3.72	6.2	0.1814	−0.98	−1.5	1.9024	3.78	11.3	0.1692	−1.56	−1.4
	82	1.158	2.0284	3.81	5.5	0.1672	−0.98	1.7	1.9576	3.94	10.8	0.1467	−1.55	3.0
	83	1.173	2.0834	3.88	4.6	0.1533	−0.93	5.0	2.0150	4.09	9.5	0.1249	−1.48	6.9
	84	1.187	2.1394	3.94	3.3	0.1405	−0.84	8.1	2.0745	4.21	7.4	0.1044	−1.35	10.2
	85	1.201	2.1961	3.98	1.3	0.1293	−0.70	10.6	2.1354	4.30	4.5	0.0862	−1.19	12.8
	86	1.215	2.2531	3.98	−1.7	0.1204	−0.54	11.8	2.1974	4.34	0.6	0.0705	−0.99	14.5
	87	1.230	2.3098	3.93	−5.8	0.1140	−0.36	11.6	2.2595	4.32	−4.2	0.0579	−0.77	15.0
	88	1.244	2.3654	3.81	−10.9	0.1100	−0.21	10.2	2.3208	4.22	−10.1	0.0485	−0.56	14.3
	89	1.258	2.4188	3.61	−16.6	0.1081	−0.07	7.9	2.3801	4.03	−16.9	0.0420	−0.36	12.4
	90	1.273	2.4688	3.33	−22.3	0.1079	0.02	4.8	2.4360	3.73	−24.2	0.0381	−0.20	9.5
	91	1.287	2.5142	2.98	−26.8	0.1087	0.07	1.1	2.4869	3.34	−30.9	0.0362	−0.09	6.0
	92	1.301	2.5539	2.57	−29.4	0.1098	0.05	−2.6	2.5314	2.85	−35.8	0.0356	−0.03	2.6
	93	1.316	2.5876	2.14	−29.4	0.1103	−0.01	−5.3	2.5684	2.31	−38.0	0.0354	−0.01	0.2
	94	1.330	2.6151	1.73	−27.1	0.1095	−0.10	−6.1	2.5975	1.76	−37.1	0.0352	−0.02	−0.5

	95	1.344	2.6369	1.36	−22.7	0.1074	−0.18	−4.5	2.6188	1.25	−33.1	0.0347	−0.03	0.3
	96	1.358	2.6541	1.08	−17.4	0.1043	−0.23	−1.6	2.6333	0.82	−26.6	0.0343	−0.01	1.7
HCR	97	1.373	2.6677	0.87	−12.5	0.1009	−0.23	1.2	2.6422	0.49	−19.1	0.0343	0.02	2.4
	98	1.387	2.6789	0.72	−9.3	0.0978	−0.19	2.9	2.6473	0.27	−12.1	0.0349	0.05	2.0
	99	1.401	2.6882	0.60	−8.0	0.0954	−0.15	3.1	2.6500	0.14	−6.9	0.0359	0.08	0.8
	100	1.416	2.6960	0.49	−7.8	0.0936	−0.10	2.5	2.6514	0.07	−3.7	0.0371	0.08	−0.6
	101	1.430	2.7023	0.38	−7.8	0.0924	−0.08	1.5	2.6521	0.04	−2.2	0.0381	0.06	−1.7
	102	1.444	2.7068	0.27	−7.7	0.0915	−0.06	0.7	2.6524	0.01	−1.5	0.0388	0.03	−2.3
	103	1.459	2.7098	0.16	−7.0	0.0907	−0.05	0.4	2.6525	0.00	−1.1	0.0390	0.00	−2.2
	104	1.473	2.7114	0.07	−5.5	0.0899	−0.05	0.5	2.6523	−0.02	−0.8	0.0387	−0.03	−1.5
	105	1.487	2.7117	0.00	−3.4	0.0893	−0.04	0.7	2.6519	−0.03	−0.4	0.0381	−0.05	−0.6
	106	1.501	2.7114	−0.03	−1.2	0.0888	−0.03	0.8	2.6515	−0.03	−0.1	0.0373	−0.05	0.2

TABLE A.2d Filtered Marker Kinematics – Fifth Metatarsal and Toe

			Right Metatarsal						Right Toe					
	Frame	Time	X	VX	AX	Y	VY	AY	X	VX	AX	Y	VY	AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	0.0845	1.63	33.7	0.0926	0.83	−3.5	0.1283	1.34	38.1	0.0464	0.35	5.6
	2	0.014	0.1114	2.12	33.5	0.1042	0.74	−9.6	0.1515	1.92	40.8	0.0521	0.40	1.3
	3	0.029	0.1452	2.59	30.8	0.1137	0.56	−14.4	0.1833	2.51	39.0	0.0580	0.38	−4.1
	4	0.043	0.1854	3.00	26.7	0.1201	0.33	−17.1	0.2232	3.04	34.1	0.0630	0.29	−8.4
	5	0.057	0.2311	3.35	22.4	0.1230	0.07	−17.4	0.2702	3.48	28.1	0.0662	0.14	−10.3
	6	0.072	0.2813	3.64	18.4	0.1222	−0.17	−15.9	0.3229	3.84	22.3	0.0671	−0.01	−9.8
	7	0.086	0.3353	3.88	15.0	0.1180	−0.38	−13.0	0.3801	4.12	17.1	0.0659	−0.14	−7.7
	8	0.100	0.3923	4.07	12.0	0.1112	−0.55	−9.2	0.4407	4.33	12.7	0.0631	−0.23	−4.6
	9	0.114	0.4517	4.22	9.4	0.1024	−0.65	−4.8	0.5039	4.48	8.9	0.0594	−0.27	−1.2
	10	0.129	0.5130	4.34	7.1	0.0927	−0.68	−0.5	0.5689	4.59	5.6	0.0554	−0.26	2.1
	11	0.143	0.5758	4.43	5.0	0.0828	−0.66	3.2	0.6351	4.64	2.8	0.0518	−0.21	5.2
	12	0.157	0.6396	4.48	3.0	0.0737	−0.59	6.5	0.7018	4.67	0.3	0.0493	−0.12	7.8
	13	0.172	0.7040	4.51	0.9	0.0659	−0.48	9.2	0.7685	4.65	−2.1	0.0485	0.01	10.1
	14	0.186	0.7686	4.51	−1.3	0.0601	−0.33	11.1	0.8349	4.61	−4.5	0.0497	0.17	11.7
	15	0.200	0.8330	4.47	−3.6	0.0565	−0.16	12.3	0.9003	4.52	−7.1	0.0534	0.35	12.6
	16	0.215	0.8966	4.41	−5.8	0.0555	0.02	12.7	0.9642	4.40	−9.9	0.0596	0.53	12.8
	17	0.229	0.9590	4.31	−8.5	0.0572	0.20	12.5	1.0262	4.24	−13.2	0.0686	0.71	11.9
	18	0.243	1.0197	4.16	−12.1	0.0614	0.38	11.2	1.0855	4.03	−16.8	0.0801	0.87	9.5
	19	0.257	1.0780	3.96	−16.6	0.0680	0.53	8.7	1.1414	3.76	−20.2	0.0936	0.99	5.7
	20	0.272	1.1330	3.69	−21.4	0.0765	0.63	4.5	1.1930	3.45	−23.1	0.1083	1.04	0.6
	21	0.286	1.1835	3.35	−25.7	0.0859	0.66	−0.9	1.2400	3.10	−25.2	0.1232	1.00	−5.5
	22	0.300	1.2288	2.95	−28.9	0.0952	0.60	−7.2	1.2817	2.73	−26.4	0.1370	0.88	−12.1
	23	0.315	1.2680	2.52	−30.8	0.1031	0.45	−13.2	1.3180	2.35	−26.8	0.1483	0.66	−18.5
	24	0.329	1.3009	2.07	−31.0	0.1081	0.22	−17.6	1.3488	1.96	−26.1	0.1558	0.35	−23.5
	25	0.343	1.3274	1.63	−29.5	0.1094	−0.05	−19.2	1.3741	1.60	−23.9	0.1583	−0.01	−25.5
	26	0.357	1.3477	1.23	−25.9	0.1066	−0.33	−17.7	1.3945	1.28	−20.3	0.1554	−0.38	−24.1
	27	0.372	1.3626	0.89	−21.0	0.1001	−0.56	−13.5	1.4107	1.02	−15.8	0.1474	−0.70	−19.4
HCR	28	0.386	1.3732	0.63	−15.6	0.0907	−0.71	−7.6	1.4237	0.83	−11.9	0.1353	−0.94	−12.6
	29	0.400	1.3807	0.45	−10.9	0.0797	−0.78	−1.4	1.4343	0.68	−9.2	0.1206	−1.06	−4.9
	30	0.415	1.3860	0.32	−7.3	0.0685	−0.75	4.0	1.4432	0.56	−7.7	0.1050	−1.08	2.4

31	0.429	1.3899	0.24	−5.1	0.0582	−0.66	7.9	1.4504	0.46	−6.9	0.0898	−0.99	8.2
32	0.443	1.3928	0.18	−3.7	0.0496	−0.53	9.9	1.4563	0.37	−6.2	0.0765	−0.84	11.9
33	0.458	1.3949	0.13	−2.8	0.0431	−0.38	10.2	1.4609	0.28	−5.6	0.0658	−0.65	13.2
34	0.472	1.3965	0.10	−1.9	0.0387	−0.24	9.0	1.4644	0.21	−4.8	0.0578	−0.46	12.3
35	0.486	1.3977	0.08	−1.0	0.0363	−0.12	6.8	1.4668	0.15	−3.7	0.0525	−0.30	10.3
36	0.500	1.3987	0.07	−0.4	0.0353	−0.04	4.2	1.4685	0.10	−2.5	0.0492	−0.17	7.7
37	0.515	1.3997	0.07	−0.5	0.0351	0.00	2.0	1.4697	0.07	−1.8	0.0476	−0.08	5.0
38	0.529	1.4006	0.06	−1.2	0.0353	0.01	0.5	1.4706	0.05	−1.7	0.0469	−0.03	2.4
39	0.543	1.4013	0.03	−1.9	0.0355	0.01	−0.1	1.4712	0.02	−1.7	0.0467	−0.01	0.5
40	0.558	1.4016	0.00	−1.8	0.0357	0.01	−0.2	1.4713	0.00	−1.4	0.0466	−0.01	−0.5
41	0.572	1.4014	−0.02	−1.2	0.0359	0.01	−0.1	1.4712	−0.02	−0.9	0.0463	−0.02	−0.7
42	0.586	1.4010	−0.03	−0.3	0.0360	0.01	−0.2	1.4709	−0.02	−0.4	0.0459	−0.03	−0.5
43	0.601	1.4005	−0.03	0.3	0.0361	0.00	−0.5	1.4705	−0.03	0.1	0.0453	−0.04	−0.1
44	0.615	1.4001	−0.02	0.6	0.0361	−0.01	−0.9	1.4701	−0.02	0.5	0.0448	−0.04	0.0
45	0.629	1.3998	−0.01	0.6	0.0359	−0.02	−0.8	1.4699	−0.01	0.7	0.0443	−0.04	−0.1
46	0.643	1.3997	0.00	0.5	0.0355	−0.03	−0.3	1.4698	0.00	0.7	0.0437	−0.04	−0.3
47	0.658	1.3997	0.00	0.3	0.0351	−0.03	0.2	1.4698	0.01	0.7	0.0431	−0.05	−0.3
48	0.672	1.3997	0.00	0.5	0.0346	−0.02	0.6	1.4700	0.02	0.7	0.0424	−0.05	−0.2
49	0.686	1.3998	0.01	0.9	0.0344	−0.01	0.6	1.4703	0.03	0.6	0.0417	−0.05	0.0
50	0.701	1.4001	0.03	1.2	0.0343	−0.01	0.2	1.4708	0.04	0.4	0.0410	−0.05	0.3
51	0.715	1.4006	0.05	1.0	0.0342	−0.01	−0.2	1.4714	0.04	0.0	0.0403	−0.04	0.5
52	0.729	1.4014	0.06	0.5	0.0340	−0.01	−0.1	1.4719	0.03	−0.5	0.0397	−0.03	0.6
53	0.744	1.4023	0.06	−0.1	0.0338	−0.01	0.6	1.4724	0.02	−0.8	0.0393	−0.03	0.6
54	0.758	1.4031	0.05	−0.6	0.0337	0.00	1.3	1.4726	0.01	−0.9	0.0390	−0.02	0.8
55	0.772	1.4038	0.04	−1.1	0.0339	0.03	1.6	1.4727	0.00	−1.0	0.0388	0.00	0.8
56	0.786	1.4043	0.02	−1.6	0.0345	0.05	1.1	1.4725	−0.02	−1.3	0.0389	0.01	0.7
57	0.801	1.4045	0.00	−1.9	0.0353	0.06	0.3	1.4721	−0.04	−1.4	0.0390	0.01	0.2
58	0.815	1.4042	−0.03	−1.5	0.0361	0.06	−0.4	1.4714	−0.06	−1.0	0.0393	0.01	−0.3
59	0.829	1.4035	−0.05	−0.4	0.0369	0.05	−0.5	1.4705	−0.07	−0.1	0.0394	0.00	−1.0
60	0.844	1.4028	−0.04	1.2	0.0374	0.04	0.2	1.4695	−0.06	1.0	0.0394	−0.02	−1.7
61	0.858	1.4023	−0.01	3.0	0.0380	0.05	1.8	1.4687	−0.04	2.1	0.0390	−0.04	−2.1
62	0.872	1.4024	0.04	4.6	0.0389	0.09	4.0	1.4684	0.00	2.8	0.0381	−0.08	−1.8
63	0.887	1.4035	0.12	5.8	0.0406	0.17	6.4	1.4687	0.04	3.2	0.0368	−0.10	−0.8
64	0.901	1.4058	0.21	7.2	0.0437	0.27	8.3	1.4696	0.09	3.5	0.0353	−0.10	1.0

(*continued*)

TABLE A.2d (*Continued*)

	Frame	Time	Right Metatarsal						Right Toe					
			X	VX	AX	Y	VY	AY	X	VX	AX	Y	VY	AY
		s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	65	0.915	1.4094	0.32	9.2	0.0485	0.40	9.3	1.4712	0.14	4.6	0.0340	−0.07	3.0
	66	0.929	1.4149	0.47	12.4	0.0552	0.54	9.0	1.4737	0.22	7.3	0.0333	−0.01	5.0
	67	0.944	1.4229	0.68	16.9	0.0639	0.66	7.3	1.4775	0.35	12.7	0.0337	0.07	6.3
	68	0.958	1.4343	0.95	22.1	0.0741	0.75	4.1	1.4837	0.58	20.7	0.0354	0.17	6.4
	69	0.972	1.4502	1.31	26.8	0.0852	0.78	−0.2	1.4941	0.94	29.5	0.0385	0.26	5.4
TOR	70	0.987	1.4717	1.72	29.7	0.0963	0.74	−5.4	1.5107	1.43	36.3	0.0428	0.32	3.4
	71	1.001	1.4994	2.16	30.2	0.1063	0.62	−10.6	1.5349	1.98	39.0	0.0478	0.35	0.3
	72	1.015	1.5335	2.59	28.5	0.1142	0.44	−14.7	1.5673	2.54	37.5	0.0530	0.33	−3.4
	73	1.030	1.5734	2.97	25.3	0.1188	0.20	−16.9	1.6075	3.05	33.2	0.0574	0.26	−7.0
	74	1.044	1.6185	3.31	21.7	0.1200	−0.05	−16.6	1.6546	3.49	27.9	0.0603	0.13	−9.3
	75	1.058	1.6681	3.59	18.2	0.1175	−0.27	−14.3	1.7074	3.85	22.5	0.0612	−0.01	−9.9
	76	1.072	1.7213	3.83	15.2	0.1121	−0.45	−10.8	1.7647	4.13	17.5	0.0601	−0.15	−8.5
	77	1.087	1.7776	4.03	12.7	0.1046	−0.58	−7.3	1.8256	4.35	13.2	0.0570	−0.25	−5.7
	78	1.101	1.8365	4.19	10.4	0.0955	−0.66	−4.3	1.8892	4.51	9.7	0.0528	−0.31	−2.1
	79	1.115	1.8975	4.33	8.3	0.0856	−0.70	−1.4	1.9547	4.63	6.7	0.0482	−0.31	1.5
	80	1.130	1.9603	4.43	6.4	0.0754	−0.70	1.7	2.0215	4.70	4.0	0.0439	−0.27	4.7
	81	1.144	2.0243	4.51	4.7	0.0655	−0.66	4.9	2.0892	4.74	1.0	0.0406	−0.18	7.5
	82	1.158	2.0892	4.57	3.1	0.0566	−0.56	8.1	2.1571	4.73	−1.9	0.0388	−0.05	9.9
	83	1.173	2.1548	4.60	1.5	0.0493	−0.43	10.9	2.2246	4.69	−4.5	0.0391	0.11	12.0
	84	1.187	2.2208	4.61	−0.6	0.0444	−0.25	13.2	2.2912	4.61	−6.5	0.0418	0.29	13.4
	85	1.201	2.2866	4.58	−3.4	0.0421	−0.05	14.7	2.3564	4.50	−8.4	0.0474	0.49	14.0
	86	1.215	2.3518	4.51	−7.0	0.0430	0.17	15.1	2.4199	4.37	−10.7	0.0558	0.69	13.7
	87	1.230	2.4155	4.38	−11.3	0.0469	0.38	14.2	2.4813	4.20	−13.9	0.0671	0.88	12.1
	88	1.244	2.4771	4.19	−16.3	0.0539	0.57	11.6	2.5399	3.97	−18.0	0.0811	1.04	8.4
	89	1.258	2.5353	3.91	−21.8	0.0634	0.71	6.8	2.5948	3.68	−22.4	0.0968	1.12	2.6
	90	1.273	2.5890	3.56	−27.2	0.0743	0.77	0.1	2.6451	3.33	−26.3	0.1132	1.11	−4.7
	91	1.287	2.6371	3.14	−31.4	0.0853	0.72	−7.3	2.6900	2.93	−28.8	0.1286	0.99	−12.6
	92	1.301	2.6787	2.66	−33.6	0.0948	0.56	−14.1	2.7289	2.51	−29.5	0.1415	0.75	−20.0
	93	1.316	2.7133	2.18	−33.4	0.1013	0.31	−18.8	2.7617	2.09	−28.2	0.1501	0.42	−25.4
	94	1.330	2.7410	1.71	−30.8	0.1038	0.02	−20.6	2.7886	1.70	−25.4	0.1534	0.02	−27.6

	95	1.344	2.7622	1.30	−26.1	0.1019	−0.28	−19.0	2.8103	1.36	−21.4	0.1508	−0.37	−25.8
	96	1.358	2.7781	0.96	−20.4	0.0959	−0.52	−14.5	2.8275	1.09	−16.8	0.1427	−0.72	−20.6
HCR	97	1.373	2.7898	0.71	−15.0	0.0869	−0.69	−8.3	2.8413	0.88	−12.6	0.1304	−0.96	−12.9
	98	1.387	2.7985	0.53	−11.0	0.0762	−0.76	−1.9	2.8527	0.73	−10.0	0.1153	−1.08	−4.2
	99	1.401	2.8050	0.40	−8.5	0.0651	−0.75	3.5	2.8621	0.60	−9.0	0.0994	−1.08	3.6
	100	1.416	2.8099	0.29	−7.2	0.0548	−0.66	7.1	2.8697	0.47	−9.0	0.0843	−0.98	9.3
	101	1.430	2.8134	0.20	−6.2	0.0461	−0.54	8.6	2.8755	0.34	−8.9	0.0714	−0.81	12.4
	102	1.444	2.8155	0.11	−5.4	0.0393	−0.42	8.5	2.8794	0.21	−8.0	0.0610	−0.63	12.9
	103	1.459	2.8166	0.04	−4.4	0.0342	−0.30	7.4	2.8816	0.11	−6.3	0.0535	−0.44	11.6
	104	1.473	2.8167	−0.01	−2.9	0.0306	−0.21	6.1	2.8825	0.03	−4.1	0.0483	−0.29	9.2
	105	1.487	2.8162	−0.04	−1.1	0.0283	−0.13	4.9	2.8825	−0.01	−1.8	0.0451	−0.18	6.5
	106	1.501	2.8155	−0.04	0.3	0.0270	−0.07	3.5	2.8822	−0.02	−0.1	0.0431	−0.11	4.1

TABLE A.3a Linear and Angular Kinematics – Foot

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	85.6	−5.01	118.27	0.089	1.937	26.39	0.153	0.814	−4.92
	2	0.014	82.2	−3.11	138.72	0.120	2.310	24.92	0.165	0.707	−9.81
	3	0.029	80.5	−1.04	142.01	0.155	2.650	22.11	0.174	0.533	−13.77
	4	0.043	80.5	0.96	132.71	0.195	2.942	18.72	0.180	0.313	−16.13
	5	0.057	82.1	2.75	115.36	0.239	3.185	15.54	0.183	0.072	−16.70
	6	0.072	85.0	4.25	93.06	0.287	3.386	12.91	0.182	−0.164	−15.72
	7	0.086	89.1	5.41	68.58	0.336	3.555	10.83	0.178	−0.378	−13.58
	8	0.100	93.9	6.22	44.92	0.388	3.696	9.20	0.171	−0.553	−10.64
	9	0.114	99.2	6.70	25.05	0.442	3.818	7.94	0.162	−0.682	−7.27
	10	0.129	104.9	6.93	10.88	0.497	3.923	6.90	0.152	−0.761	−3.80
	11	0.143	110.6	7.01	2.33	0.554	4.015	5.88	0.140	−0.791	−0.37
	12	0.157	116.3	7.00	−2.26	0.612	4.092	4.69	0.129	−0.771	2.99
	13	0.172	122.1	6.94	−4.67	0.671	4.149	3.15	0.118	−0.705	6.11
	14	0.186	127.7	6.87	−5.67	0.731	4.182	1.18	0.109	−0.597	8.65
	15	0.200	133.3	6.78	−5.01	0.791	4.183	−1.12	0.101	−0.457	10.42
	16	0.215	138.8	6.72	−3.24	0.851	4.150	−3.73	0.096	−0.299	11.39
	17	0.229	144.3	6.69	−3.16	0.909	4.076	−6.80	0.093	−0.132	11.57
	18	0.243	149.8	6.63	−8.11	0.967	3.955	−10.45	0.092	0.032	10.78
	19	0.257	155.2	6.46	−19.17	1.023	3.778	−14.54	0.094	0.177	8.80
	20	0.272	160.4	6.08	−34.96	1.075	3.539	−18.79	0.097	0.284	5.61
	21	0.286	165.2	5.46	−53.64	1.124	3.240	−22.83	0.102	0.337	1.47
	22	0.300	169.3	4.55	−73.48	1.168	2.886	−26.24	0.107	0.326	−3.24
	23	0.315	172.6	3.36	−91.19	1.206	2.490	−28.53	0.111	0.245	−7.83
	24	0.329	174.8	1.94	−102.29	1.239	2.070	−29.23	0.114	0.102	−11.35
	25	0.343	175.8	0.43	−104.08	1.266	1.654	−27.90	0.114	−0.080	−12.86
	26	0.357	175.5	−1.03	−97.50	1.286	1.272	−24.51	0.111	−0.266	−11.90
	27	0.372	174.1	−2.36	−85.55	1.302	0.953	−19.64	0.106	−0.420	−8.61
HCR	28	0.386	171.7	−3.48	−68.80	1.314	0.710	−14.43	0.099	−0.512	−3.92
	29	0.400	168.4	−4.32	−43.96	1.322	0.540	−10.02	0.092	−0.532	0.78
	30	0.415	164.6	−4.74	−10.50	1.329	0.424	−7.03	0.084	−0.490	4.32
	31	0.429	160.6	−4.63	24.51	1.334	0.339	−5.31	0.078	−0.409	6.26

32	0.443	157.0	−4.04	52.05	1.339	0.272	−4.39	0.073	−0.311	6.77
33	0.458	154.0	−3.14	67.24	1.342	0.214	−3.77	0.069	−0.215	6.26
34	0.472	151.9	−2.11	68.18	1.345	0.164	−3.06	0.066	−0.132	5.12
35	0.486	150.6	−1.19	55.74	1.347	0.126	−2.15	0.065	−0.069	3.71
36	0.500	149.9	−0.52	35.86	1.348	0.103	−1.38	0.064	−0.026	2.36
37	0.515	149.7	−0.16	17.08	1.350	0.087	−1.21	0.064	−0.001	1.25
38	0.529	149.7	−0.03	4.56	1.351	0.068	−1.64	0.064	0.010	0.46
39	0.543	149.7	−0.03	−1.79	1.352	0.040	−2.06	0.065	0.012	0.10
40	0.558	149.6	−0.08	−4.87	1.352	0.009	−1.91	0.065	0.013	0.16
41	0.572	149.5	−0.17	−7.39	1.352	−0.015	−1.21	0.065	0.017	0.43
42	0.586	149.3	−0.29	−10.28	1.352	−0.026	−0.40	0.065	0.025	0.54
43	0.601	149.0	−0.46	−12.53	1.351	−0.026	0.26	0.066	0.033	0.31
44	0.615	148.6	−0.65	−12.14	1.351	−0.018	0.76	0.066	0.034	−0.11
45	0.629	148.0	−0.81	−8.07	1.351	−0.004	1.07	0.067	0.029	−0.47
46	0.643	147.2	−0.88	−1.88	1.351	0.012	1.10	0.067	0.021	−0.60
47	0.658	146.5	−0.86	3.55	1.351	0.027	0.96	0.067	0.012	−0.45
48	0.672	145.8	−0.78	6.31	1.352	0.039	0.86	0.067	0.008	−0.13
49	0.686	145.2	−0.68	5.50	1.352	0.052	0.88	0.067	0.009	0.13
50	0.701	144.7	−0.62	1.74	1.353	0.065	0.85	0.068	0.012	0.22
51	0.715	144.2	−0.63	−2.68	1.354	0.076	0.55	0.068	0.015	0.32
52	0.729	143.7	−0.70	−5.69	1.355	0.080	0.05	0.068	0.021	0.66
53	0.744	143.1	−0.80	−7.38	1.356	0.077	−0.32	0.068	0.034	1.25
54	0.758	142.4	−0.91	−10.45	1.357	0.071	−0.42	0.069	0.056	1.91
55	0.772	141.6	−1.10	−17.81	1.358	0.065	−0.41	0.070	0.088	2.38
56	0.786	140.6	−1.42	−28.50	1.359	0.059	−0.42	0.072	0.125	2.38
57	0.801	139.2	−1.91	−38.25	1.360	0.053	−0.24	0.074	0.156	1.90
58	0.815	137.4	−2.52	−45.01	1.361	0.053	0.57	0.076	0.179	1.39
59	0.829	135.1	−3.20	−49.94	1.362	0.069	2.14	0.079	0.196	1.33
60	0.844	132.2	−3.94	−52.91	1.363	0.114	4.22	0.082	0.217	1.88
61	0.858	128.6	−4.71	−52.39	1.365	0.190	6.40	0.085	0.250	2.88
62	0.872	124.5	−5.44	−49.61	1.368	0.297	8.37	0.089	0.300	4.16
63	0.887	119.7	−6.13	−47.15	1.373	0.429	9.90	0.093	0.369	5.46
64	0.901	114.4	−6.79	−43.01	1.380	0.580	11.08	0.099	0.456	6.34
65	0.915	108.6	−7.36	−31.69	1.390	0.746	12.24	0.106	0.550	6.44

(continued)

TABLE A.3a (*Continued*)

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	66	0.929	102.4	−7.70	−10.02	1.402	0.930	13.82	0.115	0.640	5.61
	67	0.944	96.0	−7.65	21.54	1.417	1.141	16.00	0.125	0.711	3.93
	68	0.958	89.8	−7.08	60.42	1.434	1.388	18.51	0.135	0.752	1.50
	69	0.972	84.4	−5.92	100.83	1.456	1.671	20.60	0.146	0.754	−1.62
TOR	70	0.987	80.1	−4.20	133.15	1.482	1.977	21.55	0.157	0.706	−5.36
	71	1.001	77.5	−2.11	148.09	1.513	2.287	21.20	0.166	0.600	−9.39
	72	1.015	76.7	0.04	142.55	1.548	2.583	19.88	0.174	0.437	−13.01
	73	1.030	77.6	1.97	121.35	1.587	2.856	18.00	0.179	0.229	−15.37
	74	1.044	79.9	3.51	93.05	1.629	3.098	15.92	0.181	−0.002	−15.93
	75	1.058	83.3	4.63	65.70	1.675	3.311	13.95	0.179	−0.227	−14.68
	76	1.072	87.5	5.39	44.39	1.724	3.497	12.25	0.174	−0.422	−12.28
	77	1.087	92.1	5.90	29.92	1.775	3.661	10.74	0.167	−0.578	−9.59
	78	1.101	97.1	6.24	20.34	1.829	3.804	9.31	0.158	−0.697	−7.02
	79	1.115	102.4	6.48	13.82	1.884	3.927	7.94	0.147	−0.779	−4.37
	80	1.130	107.8	6.64	9.60	1.941	4.031	6.65	0.135	−0.822	−1.42
	81	1.144	113.3	6.75	7.77	1.999	4.118	5.44	0.123	−0.820	1.72
	82	1.158	118.8	6.86	7.86	2.059	4.187	4.29	0.112	−0.772	4.87
	83	1.173	124.5	6.98	7.28	2.119	4.240	3.03	0.101	−0.680	7.92
	84	1.187	130.3	7.07	3.78	2.180	4.274	1.33	0.092	−0.546	10.66
	85	1.201	136.1	7.09	−0.95	2.241	4.278	−1.08	0.086	−0.375	12.64
	86	1.215	141.9	7.04	−3.31	2.302	4.243	−4.36	0.082	−0.184	13.44
	87	1.230	147.6	6.99	−3.75	2.363	4.154	−8.57	0.080	0.009	12.89
	88	1.244	153.3	6.93	−8.63	2.421	3.998	−13.62	0.082	0.184	10.90
	89	1.258	159.0	6.74	−23.94	2.477	3.764	−19.21	0.086	0.321	7.37
	90	1.273	164.4	6.25	−48.50	2.529	3.448	−24.71	0.091	0.395	2.47
	91	1.287	169.2	5.36	−75.19	2.576	3.057	−29.09	0.097	0.391	−3.11
	92	1.301	173.2	4.10	−96.87	2.616	2.616	−31.47	0.102	0.306	−8.34
	93	1.316	175.9	2.59	−110.55	2.650	2.157	−31.41	0.106	0.153	−12.08
	94	1.330	177.4	0.94	−117.61	2.678	1.718	−28.90	0.107	−0.039	−13.34
	95	1.344	177.5	−0.78	−118.51	2.700	1.331	−24.40	0.105	−0.229	−11.78
	96	1.358	176.1	−2.45	−109.12	2.716	1.020	−18.90	0.100	−0.376	−8.07

HCR	97	1.373	173.5	−3.90	−84.77	2.729	0.790	−13.77	0.094	−0.460	−3.57
	98	1.387	169.7	−4.88	−46.31	2.739	0.626	−10.14	0.087	−0.478	0.46
	99	1.401	165.5	−5.22	−1.86	2.747	0.500	−8.25	0.080	−0.446	3.31
	100	1.416	161.2	−4.93	36.59	2.753	0.390	−7.47	0.074	−0.384	4.77
	101	1.430	157.4	−4.18	59.94	2.758	0.286	−7.03	0.069	−0.310	5.07
	102	1.444	154.3	−3.22	66.68	2.761	0.189	−6.52	0.065	−0.239	4.60
	103	1.459	152.1	−2.27	61.71	2.763	0.100	−5.67	0.062	−0.178	3.88
	104	1.473	150.6	−1.45	51.41	2.764	0.027	−4.22	0.060	−0.128	3.28
	105	1.487	149.7	−0.80	39.26	2.764	−0.021	−2.28	0.059	−0.084	2.79
	106	1.501	149.3	−0.33	25.63	2.763	−0.038	−0.45	0.058	−0.048	2.13

TABLE A.3b Linear and Angular Kinematics – Leg

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	39.8	−2.41	40.67	0.272	2.479	11.66	0.362	0.268	0.58
	2	0.014	38.0	−1.70	56.27	0.308	2.618	7.99	0.366	0.277	0.37
	3	0.029	37.0	−0.80	66.77	0.347	2.708	4.97	0.370	0.279	−0.29
	4	0.043	36.7	0.21	71.22	0.386	2.760	2.66	0.374	0.268	−1.25
	5	0.057	37.3	1.24	70.05	0.425	2.784	1.03	0.378	0.243	−2.27
	6	0.072	38.8	2.21	64.35	0.465	2.789	−0.07	0.381	0.203	−3.27
	7	0.086	41.0	3.08	55.75	0.505	2.782	−0.78	0.384	0.150	−4.09
	8	0.100	43.8	3.81	46.45	0.545	2.767	−1.20	0.385	0.086	−4.55
	9	0.114	47.2	4.41	38.02	0.584	2.748	−1.41	0.386	0.020	−4.61
	10	0.129	51.0	4.90	30.50	0.624	2.727	−1.52	0.386	−0.045	−4.42
	11	0.143	55.2	5.28	23.32	0.662	2.704	−1.61	0.385	−0.107	−4.07
	12	0.157	59.7	5.56	16.47	0.701	2.681	−1.73	0.383	−0.162	−3.47
	13	0.172	64.3	5.75	10.44	0.739	2.655	−1.99	0.380	−0.206	−2.63
	14	0.186	69.1	5.86	5.57	0.777	2.624	−2.49	0.377	−0.237	−1.74
	15	0.200	74.0	5.91	1.65	0.814	2.584	−3.23	0.374	−0.256	−0.98
	16	0.215	78.8	5.91	−2.00	0.851	2.531	−4.22	0.370	−0.265	−0.37
	17	0.229	83.6	5.85	−6.16	0.886	2.463	−5.38	0.366	−0.266	0.11
	18	0.243	88.4	5.73	−11.89	0.921	2.378	−6.44	0.362	−0.262	0.45
	19	0.257	93.0	5.51	−20.20	0.954	2.279	−7.20	0.358	−0.253	0.66
	20	0.272	97.4	5.16	−31.54	0.986	2.171	−7.72	0.355	−0.243	0.87
	21	0.286	101.5	4.61	−45.55	1.017	2.058	−8.18	0.352	−0.228	1.17
	22	0.300	105.0	3.85	−60.60	1.045	1.938	−8.64	0.348	−0.209	1.35
	23	0.315	107.8	2.88	−73.46	1.072	1.811	−8.98	0.346	−0.190	1.08
	24	0.329	109.7	1.75	−80.46	1.097	1.681	−9.03	0.343	−0.178	0.43
	25	0.343	110.7	0.58	−79.07	1.120	1.553	−8.67	0.340	−0.178	−0.16
	26	0.357	110.7	−0.51	−68.79	1.141	1.433	−7.92	0.338	−0.183	−0.14
	27	0.372	109.8	−1.39	−51.73	1.161	1.326	−6.93	0.335	−0.182	0.76
HCR	28	0.386	108.4	−1.99	−32.87	1.179	1.235	−5.87	0.333	−0.161	2.26
	29	0.400	106.6	−2.33	−18.01	1.196	1.158	−4.90	0.331	−0.117	3.67
	30	0.415	104.6	−2.50	−9.85	1.212	1.095	−4.10	0.329	−0.056	4.33
	31	0.429	102.5	−2.61	−6.21	1.228	1.041	−3.70	0.329	−0.007	3.99

32	0.443	100.3	−2.68	−2.78	1.242	0.989	−4.01	0.330	0.058	2.75
33	0.458	98.1	−2.69	2.90	1.256	0.926	−5.05	0.331	0.086	0.98
34	0.472	95.9	−2.60	10.00	1.269	0.844	−6.23	0.332	0.086	−0.67
35	0.486	93.8	−2.41	15.87	1.280	0.748	−6.81	0.333	0.066	−1.61
36	0.500	91.9	−2.15	18.47	1.290	0.650	−6.62	0.334	0.040	−1.74
37	0.515	90.3	−1.88	17.56	1.299	0.559	−6.09	0.334	0.016	−1.43
38	0.529	88.9	−1.64	14.51	1.306	0.475	−5.57	0.334	−0.001	−0.98
39	0.543	87.6	−1.46	10.97	1.312	0.399	−4.93	0.334	−0.012	−0.39
40	0.558	86.5	−1.33	7.87	1.317	0.334	−3.89	0.334	−0.012	0.40
41	0.572	85.4	−1.24	5.13	1.322	0.288	−2.49	0.334	0.000	1.14
42	0.586	84.4	−1.18	2.56	1.326	0.263	−1.08	0.334	0.020	1.31
43	0.601	83.5	−1.16	0.58	1.329	0.257	0.06	0.334	0.037	0.64
44	0.615	82.5	−1.17	−0.22	1.333	0.265	0.82	0.335	0.038	−0.53
45	0.629	81.6	−1.17	0.10	1.337	0.281	1.24	0.336	0.022	−1.50
46	0.643	80.6	−1.16	0.69	1.341	0.300	1.34	0.336	−0.005	−1.86
47	0.658	79.7	−1.15	0.65	1.345	0.319	1.27	0.335	−0.031	−1.62
48	0.672	78.7	−1.14	−0.22	1.350	0.336	1.21	0.335	−0.051	−0.95
49	0.686	77.8	−1.16	−1.19	1.355	0.353	1.17	0.334	−0.059	−0.09
50	0.701	76.8	−1.18	−1.64	1.360	0.370	1.00	0.333	−0.053	0.72
51	0.715	75.9	−1.20	−1.93	1.366	0.382	0.66	0.332	−0.038	1.30
52	0.729	74.9	−1.23	−2.99	1.371	0.389	0.45	0.332	−0.016	1.62
53	0.744	73.8	−1.29	−5.30	1.377	0.395	0.80	0.332	0.009	1.74
54	0.758	72.7	−1.39	−8.47	1.383	0.411	1.72	0.332	0.034	1.74
55	0.772	71.6	−1.53	−11.61	1.389	0.444	2.77	0.333	0.058	1.66
56	0.786	70.2	−1.72	−13.88	1.395	0.491	3.58	0.334	0.081	1.43
57	0.801	68.7	−1.93	−14.88	1.403	0.547	4.29	0.335	0.099	1.00
58	0.815	67.1	−2.14	−14.71	1.411	0.613	5.39	0.337	0.110	0.53
59	0.829	65.2	−2.35	−14.08	1.420	0.701	7.07	0.338	0.114	0.23
60	0.844	63.2	−2.55	−13.69	1.431	0.816	9.13	0.340	0.116	0.09
61	0.858	61.1	−2.74	−13.68	1.443	0.962	11.25	0.342	0.117	0.00
62	0.872	58.7	−2.94	−13.46	1.458	1.137	13.11	0.343	0.116	0.06
63	0.887	56.2	−3.13	−11.96	1.476	1.337	14.27	0.345	0.118	0.37
64	0.901	53.6	−3.28	−8.16	1.497	1.545	14.31	0.347	0.127	0.82
65	0.915	50.9	−3.36	−1.88	1.520	1.746	13.29	0.349	0.142	1.13

(*continued*)

TABLE A.3b (*Continued*)

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	66	0.929	48.1	−3.33	6.33	1.547	1.926	11.68	0.351	0.159	1.26
	67	0.944	45.4	−3.18	15.67	1.575	2.080	9.99	0.353	0.178	1.39
	68	0.958	42.9	−2.88	25.25	1.606	2.211	8.47	0.356	0.199	1.66
	69	0.972	40.7	−2.46	34.20	1.639	2.322	7.11	0.359	0.225	1.88
TOR	70	0.987	38.9	−1.91	42.20	1.672	2.415	5.93	0.362	0.253	1.75
	71	1.001	37.6	−1.25	49.54	1.708	2.492	5.10	0.366	0.275	1.03
	72	1.015	36.8	−0.49	56.03	1.744	2.560	4.60	0.370	0.282	−0.23
	73	1.030	36.8	0.35	60.38	1.781	2.624	4.17	0.374	0.269	−1.71
	74	1.044	37.4	1.24	61.40	1.819	2.680	3.53	0.378	0.234	−2.97
	75	1.058	38.8	2.11	58.96	1.857	2.725	2.64	0.381	0.184	−3.79
	76	1.072	40.9	2.92	53.72	1.897	2.755	1.59	0.383	0.125	−4.25
	77	1.087	43.6	3.65	46.69	1.936	2.770	0.57	0.385	0.062	−4.57
	78	1.101	46.8	4.26	38.96	1.976	2.771	−0.31	0.385	−0.006	−4.80
	79	1.115	50.6	4.76	31.29	2.016	2.761	−0.96	0.384	−0.075	−4.73
	80	1.130	54.6	5.15	24.07	2.055	2.744	−1.39	0.383	−0.141	−4.26
	81	1.144	59.0	5.45	17.67	2.094	2.722	−1.68	0.380	−0.197	−3.51
	82	1.158	63.6	5.66	12.52	2.133	2.696	−1.93	0.377	−0.241	−2.66
	83	1.173	68.3	5.81	8.73	2.171	2.667	−2.22	0.373	−0.273	−1.78
	84	1.187	73.1	5.91	6.21	2.209	2.632	−2.66	0.369	−0.292	−0.96
	85	1.201	78.0	5.98	4.43	2.246	2.590	−3.35	0.365	−0.301	−0.35
	86	1.215	82.9	6.04	1.90	2.283	2.536	−4.35	0.361	−0.302	0.05
	87	1.230	87.9	6.04	−3.69	2.319	2.466	−5.61	0.356	−0.299	0.43
	88	1.244	92.8	5.93	−14.21	2.354	2.376	−6.98	0.352	−0.290	0.92
	89	1.258	97.6	5.63	−29.76	2.387	2.266	−8.35	0.348	−0.273	1.38
	90	1.273	102.0	5.08	−48.61	2.418	2.137	−9.47	0.344	−0.250	1.67
	91	1.287	105.9	4.24	−67.64	2.448	1.996	−9.96	0.341	−0.225	1.72
	92	1.301	109.0	3.14	−82.66	2.476	1.853	−9.67	0.338	−0.201	1.54
	93	1.316	111.0	1.88	−89.51	2.501	1.719	−8.82	0.335	−0.181	1.24
	94	1.330	112.0	0.58	−85.71	2.525	1.600	−7.71	0.333	−0.166	1.20
	95	1.344	112.0	−0.57	−71.95	2.547	1.499	−6.53	0.331	−0.147	1.66
	96	1.358	111.1	−1.47	−52.52	2.568	1.414	−5.46	0.329	−0.118	2.38

HCR	97	1.373	109.6	−2.08	−33.63	2.587	1.343	−4.69	0.327	−0.079	2.96
	98	1.387	107.7	−2.43	−20.13	2.606	1.279	−4.45	0.326	−0.033	3.19
	99	1.401	105.6	−2.65	−12.52	2.624	1.215	−4.84	0.326	0.012	2.88
	100	1.416	103.4	−2.79	−7.44	2.641	1.141	−5.82	0.327	0.049	1.96
	101	1.430	101.0	−2.86	−1.54	2.656	1.049	−7.22	0.328	0.068	0.66
	102	1.444	98.7	−2.84	5.68	2.671	0.934	−8.71	0.329	0.068	−0.54
	103	1.459	96.4	−2.70	12.64	2.683	0.800	−9.67	0.330	0.053	−1.23
	104	1.473	94.2	−2.48	18.05	2.694	0.658	−9.59	0.330	0.033	−1.29
	105	1.487	92.3	−2.19	21.86	2.702	0.526	−8.48	0.330	0.016	−0.96
	106	1.501	90.7	−1.85	24.55	2.709	0.415	−6.99	0.331	0.005	−0.61

TABLE A.3c Linear and Angular Kinematics – Thigh

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	82.7	3.29	24.42	0.430	2.081	1.36	0.652	−0.052	4.40
	2	0.014	85.5	3.59	18.34	0.460	2.073	−1.89	0.652	0.026	6.06
	3	0.029	88.6	3.81	12.48	0.489	2.027	−3.83	0.653	0.122	6.86
	4	0.043	91.8	3.95	5.97	0.518	1.963	−4.53	0.655	0.222	6.72
	5	0.057	95.1	3.98	−1.93	0.546	1.898	−4.32	0.659	0.314	5.74
	6	0.072	98.3	3.89	−10.64	0.572	1.840	−3.64	0.664	0.386	4.11
	7	0.086	101.4	3.68	−18.57	0.598	1.793	−2.92	0.670	0.432	2.20
	8	0.100	104.3	3.36	−24.25	0.624	1.756	−2.45	0.677	0.449	0.42
	9	0.114	106.9	2.99	−27.38	0.648	1.724	−2.28	0.683	0.444	−1.15
	10	0.129	109.2	2.58	−28.82	0.673	1.691	−2.31	0.689	0.417	−2.60
	11	0.143	111.2	2.16	−29.42	0.697	1.658	−2.41	0.695	0.369	−3.94
	12	0.157	112.8	1.74	−29.22	0.720	1.622	−2.53	0.700	0.304	−4.97
	13	0.172	114.0	1.33	−27.99	0.743	1.585	−2.69	0.704	0.227	−5.63
	14	0.186	114.9	0.94	−26.18	0.766	1.545	−2.83	0.706	0.143	−6.02
	15	0.200	115.6	0.58	−24.68	0.787	1.504	−2.79	0.708	0.055	−6.22
	16	0.215	115.9	0.23	−23.85	0.809	1.466	−2.51	0.708	−0.035	−6.23
	17	0.229	115.9	−0.11	−23.01	0.829	1.433	−2.04	0.707	−0.123	−5.97
	18	0.243	115.7	−0.43	−20.96	0.850	1.407	−1.40	0.704	−0.206	−5.34
	19	0.257	115.2	−0.70	−16.94	0.870	1.393	−0.55	0.701	−0.276	−4.22
	20	0.272	114.6	−0.91	−11.04	0.889	1.392	0.42	0.696	−0.326	−2.51
	21	0.286	113.7	−1.02	−4.04	0.909	1.405	1.38	0.692	−0.348	−0.34
	22	0.300	112.9	−1.03	2.83	0.930	1.431	2.28	0.687	−0.336	1.68
	23	0.315	112.1	−0.94	8.12	0.950	1.470	3.03	0.682	−0.300	2.98
	24	0.329	111.3	−0.79	10.77	0.972	1.518	3.42	0.678	−0.251	3.50
	25	0.343	110.8	−0.63	10.49	0.994	1.568	3.26	0.675	−0.200	3.62
	26	0.357	110.3	−0.49	7.65	1.016	1.611	2.48	0.672	−0.148	3.69
	27	0.372	110.0	−0.41	3.48	1.040	1.639	1.20	0.671	−0.094	3.92
HCR	28	0.386	109.6	−0.39	0.13	1.063	1.645	−0.31	0.670	−0.036	4.29
	29	0.400	109.3	−0.41	−0.49	1.087	1.630	−1.67	0.669	0.029	4.62
	30	0.415	109.0	−0.41	1.69	1.110	1.597	−2.61	0.670	0.096	4.64
	31	0.429	108.6	−0.36	4.27	1.133	1.555	−3.28	0.672	0.161	4.04

32	0.443	108.4	−0.29	3.58	1.154	1.503	−4.05	0.675	0.212	2.63
33	0.458	108.2	−0.26	−2.78	1.175	1.439	−4.94	0.678	0.236	0.62
34	0.472	107.9	−0.37	−13.38	1.196	1.362	−5.49	0.682	0.230	−1.26
35	0.486	107.6	−0.64	−23.43	1.214	1.282	−5.38	0.685	0.200	−2.26
36	0.500	106.9	−1.04	−28.36	1.232	1.208	−4.85	0.687	0.165	−2.32
37	0.515	105.9	−1.45	−26.92	1.249	1.143	−4.43	0.690	0.134	−2.00
38	0.529	104.5	−1.81	−20.92	1.265	1.082	−4.34	0.691	0.108	−1.77
39	0.543	102.9	−2.05	−12.96	1.280	1.019	−4.36	0.693	0.083	−1.52
40	0.558	101.2	−2.18	−4.84	1.294	0.957	−4.13	0.694	0.064	−1.01
41	0.572	99.3	−2.19	2.16	1.307	0.901	−3.50	0.695	0.054	−0.44
42	0.586	97.6	−2.11	6.84	1.320	0.857	−2.56	0.695	0.052	−0.31
43	0.601	95.9	−1.99	8.52	1.332	0.828	−1.53	0.696	0.045	−0.97
44	0.615	94.3	−1.87	7.81	1.344	0.813	−0.65	0.697	0.024	−2.12
45	0.629	92.8	−1.77	6.10	1.355	0.809	−0.07	0.697	−0.015	−2.99
46	0.643	91.4	−1.70	4.22	1.367	0.811	0.26	0.696	−0.062	−3.02
47	0.658	90.0	−1.65	2.22	1.378	0.817	0.61	0.695	−0.102	−2.35
48	0.672	88.7	−1.63	0.01	1.390	0.828	1.15	0.693	−0.129	−1.46
49	0.686	87.4	−1.65	−2.10	1.402	0.850	1.70	0.691	−0.143	−0.63
50	0.701	86.0	−1.69	−3.38	1.414	0.877	1.85	0.689	−0.147	0.01
51	0.715	84.6	−1.75	−3.24	1.427	0.902	1.51	0.687	−0.143	0.45
52	0.729	83.1	−1.79	−1.71	1.440	0.920	1.16	0.685	−0.134	0.67
53	0.744	81.7	−1.79	0.98	1.453	0.936	1.38	0.683	−0.124	0.79
54	0.758	80.2	−1.76	4.63	1.467	0.960	2.18	0.681	−0.112	0.78
55	0.772	78.8	−1.66	8.63	1.481	0.998	3.02	0.680	−0.101	0.58
56	0.786	77.5	−1.51	11.95	1.495	1.046	3.58	0.679	−0.095	0.20
57	0.801	76.3	−1.32	13.92	1.511	1.100	4.10	0.677	−0.096	−0.19
58	0.815	75.3	−1.11	14.82	1.527	1.163	5.02	0.676	−0.101	−0.46
59	0.829	74.5	−0.90	15.41	1.544	1.244	6.42	0.674	−0.109	−0.73
60	0.844	73.8	−0.67	16.49	1.562	1.347	7.98	0.673	−0.121	−1.12
61	0.858	73.4	−0.43	18.57	1.583	1.472	9.40	0.671	−0.141	−1.49
62	0.872	73.1	−0.14	21.54	1.605	1.616	10.36	0.669	−0.164	−1.58
63	0.887	73.1	0.19	24.59	1.629	1.768	10.40	0.666	−0.186	−1.28
64	0.901	73.4	0.56	27.13	1.655	1.913	9.09	0.663	−0.201	−0.68
65	0.915	74.1	0.97	29.35	1.683	2.028	6.50	0.660	−0.205	0.06

(*continued*)

TABLE A.3c (*Continued*)

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	66	0.929	75.0	1.40	31.19	1.713	2.099	3.27	0.658	−0.199	0.94
	67	0.944	76.4	1.86	31.71	1.743	2.122	0.24	0.655	−0.179	2.00
	68	0.958	78.1	2.31	30.36	1.774	2.106	−2.10	0.652	−0.142	3.27
	69	0.972	80.2	2.73	27.73	1.804	2.062	−3.60	0.651	0.085	4.59
TOR	70	0.987	82.5	3.10	24.46	1.833	2.003	−4.23	0.650	−0.010	5.65
	71	1.001	85.2	3.43	20.30	1.861	1.941	−3.96	0.650	0.076	6.21
	72	1.015	88.2	3.68	14.67	1.888	1.889	−3.05	0.652	0.167	6.16
	73	1.030	91.3	3.85	7.52	1.915	1.854	−2.02	0.655	0.253	5.61
	74	1.044	94.5	3.90	−0.87	1.941	1.832	−1.28	0.659	0.328	4.76
	75	1.058	97.7	3.82	−10.14	1.967	1.817	−0.93	0.665	0.389	3.66
	76	1.072	100.7	3.61	−19.00	1.993	1.805	−0.88	0.671	0.432	2.15
	77	1.087	103.6	3.28	−25.45	2.019	1.792	−1.09	0.677	0.451	0.23
	78	1.101	106.1	2.88	−28.34	2.045	1.774	−1.51	0.683	0.439	−1.77
	79	1.115	108.3	2.47	−27.92	2.070	1.749	−2.07	0.689	0.400	−3.40
	80	1.130	110.1	2.08	−25.47	2.095	1.715	−2.71	0.695	0.342	−4.45
	81	1.144	111.7	1.74	−22.64	2.119	1.671	−3.30	0.699	0.273	−5.04
	82	1.158	113.0	1.43	−20.83	2.142	1.620	−3.72	0.703	0.198	−5.39
	83	1.173	114.0	1.14	−20.75	2.165	1.565	−3.87	0.705	0.119	−5.64
	84	1.187	114.9	0.84	−22.11	2.187	1.510	−3.74	0.706	0.036	−5.90
	85	1.201	115.4	0.51	−23.89	2.208	1.458	−3.37	0.706	−0.050	−6.16
	86	1.215	115.7	0.16	−24.95	2.229	1.413	−2.72	0.705	−0.140	−6.20
	87	1.230	115.7	−0.20	−24.36	2.249	1.380	−1.76	0.702	−0.227	−5.66
	88	1.244	115.4	−0.54	−21.36	2.268	1.363	−0.58	0.698	−0.302	−4.43
	89	1.258	114.8	−0.81	−15.42	2.288	1.364	0.58	0.693	−0.354	−2.69
	90	1.273	114.0	−0.98	−6.72	2.307	1.380	1.61	0.688	−0.379	−0.70
	91	1.287	113.2	−1.01	3.05	2.327	1.410	2.71	0.682	−0.374	1.30
	92	1.301	112.4	−0.89	10.92	2.348	1.457	3.92	0.677	−0.342	3.07
	93	1.316	111.7	−0.69	14.55	2.369	1.522	4.86	0.673	−0.287	4.38
	94	1.330	111.3	−0.48	13.64	2.391	1.596	5.04	0.669	−0.216	5.19
	95	1.344	110.9	−0.30	9.66	2.415	1.666	4.28	0.667	−0.138	5.50
	96	1.358	110.8	−0.20	5.05	2.439	1.718	2.72	0.665	−0.059	5.33

HCR	97	1.373	110.6	−0.16	2.00	2.464	1.744	0.71	0.665	0.014	4.88
	98	1.387	110.5	−0.14	1.10	2.489	1.739	−1.30	0.666	0.080	4.42
	99	1.401	110.4	−0.13	0.73	2.513	1.706	−3.03	0.667	0.141	3.89
	100	1.416	110.3	−0.12	−1.58	2.537	1.652	−4.52	0.670	0.192	2.92
	101	1.430	110.2	−0.17	−7.15	2.561	1.577	−5.90	0.673	0.224	1.39
	102	1.444	110.0	−0.33	−14.74	2.583	1.483	−7.13	0.676	0.231	−0.34
	103	1.459	109.6	−0.59	−21.06	2.603	1.373	−8.01	0.679	0.215	−1.65
	104	1.473	109.0	−0.93	−22.96	2.622	1.254	−8.45	0.682	0.184	−2.18
	105	1.487	108.1	−1.25	−19.15	2.639	1.132	−8.72	0.684	0.152	−2.11
	106	1.501	107.0	−1.48	−10.16	2.654	1.005	−9.40	0.686	0.124	−1.89

TABLE A.3d Linear and Angular Kinematics – ½ HAT

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
TOR	1	0.000	85.1	0.89	−11.61	0.473	1.377	1.23	1.080	0.035	2.76
	2	0.014	85.7	0.71	−13.17	0.492	1.379	−0.70	1.081	0.084	3.89
	3	0.029	86.2	0.52	−11.89	0.512	1.357	−2.11	1.083	0.146	4.49
	4	0.043	86.6	0.37	−8.26	0.531	1.319	−2.87	1.085	0.212	4.46
	5	0.057	86.8	0.28	−3.23	0.550	1.275	−2.98	1.089	0.274	3.80
	6	0.072	87.0	0.27	1.64	0.568	1.234	−2.50	1.093	0.321	2.60
	7	0.086	87.3	0.33	4.85	0.585	1.203	−1.64	1.098	0.348	1.19
	8	0.100	87.6	0.41	6.05	0.602	1.187	−0.74	1.103	0.355	−0.18
	9	0.114	88.0	0.50	6.17	0.619	1.182	−0.15	1.108	0.343	−1.50
	10	0.129	88.4	0.59	6.19	0.636	1.183	0.06	1.113	0.312	−2.81
	11	0.143	88.9	0.68	6.10	0.653	1.184	0.06	1.117	0.262	−3.89
	12	0.157	89.5	0.76	5.21	0.670	1.184	0.06	1.120	0.201	−4.54
	13	0.172	90.2	0.83	3.01	0.687	1.186	0.21	1.123	0.133	−4.81
	14	0.186	90.9	0.85	−0.12	0.704	1.190	0.56	1.124	0.063	−4.87
	15	0.200	91.6	0.82	−3.16	0.721	1.201	1.15	1.124	−0.007	−4.83
	16	0.215	92.2	0.76	−5.47	0.738	1.223	1.93	1.124	−0.075	−4.72
	17	0.229	92.8	0.67	−6.91	0.756	1.257	2.71	1.122	−0.142	−4.48
	18	0.243	93.3	0.56	−7.30	0.774	1.300	3.24	1.120	−0.203	−3.98
	19	0.257	93.7	0.46	−6.57	0.793	1.349	3.38	1.116	−0.255	−3.07
	20	0.272	94.1	0.37	−4.89	0.813	1.397	3.11	1.112	−0.291	−1.63
	21	0.286	94.3	0.32	−2.38	0.833	1.438	2.48	1.108	−0.302	0.10
	22	0.300	94.6	0.31	0.85	0.854	1.468	1.66	1.104	−0.288	1.53
	23	0.315	94.8	0.34	3.98	0.875	1.486	0.95	1.100	−0.259	2.21
	24	0.329	95.1	0.42	5.39	0.896	1.495	0.64	1.096	−0.225	2.40
	25	0.343	95.5	0.50	3.95	0.918	1.504	0.91	1.093	−0.190	2.60
	26	0.357	96.0	0.53	0.20	0.939	1.521	1.52	1.091	−0.151	3.07
	27	0.372	96.4	0.50	−4.14	0.961	1.548	1.98	1.089	−0.102	3.79
HCR	28	0.386	96.8	0.41	−7.80	0.984	1.578	1.91	1.088	−0.042	4.62
	29	0.400	97.1	0.28	−10.40	1.006	1.602	1.32	1.088	0.030	5.28
	30	0.415	97.2	0.12	−11.97	1.029	1.616	0.50	1.089	0.109	5.48
	31	0.429	97.3	−0.06	−12.71	1.053	1.617	−0.32	1.091	0.186	4.93

32	0.443	97.1	−0.25	−12.51	1.076	1.606	−1.04	1.094	0.250	3.50
33	0.458	96.9	−0.42	−10.48	1.098	1.587	−1.63	1.098	0.287	1.45
34	0.472	96.4	−0.55	−6.07	1.121	1.560	−1.99	1.102	0.291	−0.50
35	0.486	96.0	−0.59	−0.55	1.143	1.530	−2.10	1.106	0.272	−1.54
36	0.500	95.5	−0.56	3.64	1.165	1.500	−2.10	1.110	0.247	−1.63
37	0.515	95.0	−0.49	4.75	1.186	1.470	−2.14	1.114	0.226	−1.49
38	0.529	94.7	−0.43	2.67	1.207	1.439	−2.14	1.117	0.205	−1.69
39	0.543	94.3	−0.41	−1.41	1.227	1.409	−1.93	1.119	0.178	−1.99
40	0.558	94.0	−0.47	−5.74	1.247	1.384	−1.49	1.122	0.148	−1.98
41	0.572	93.6	−0.58	−8.82	1.267	1.366	−0.95	1.124	0.121	−1.69
42	0.586	93.0	−0.72	−9.90	1.286	1.356	−0.43	1.125	0.099	−1.57
43	0.601	92.4	−0.86	−9.06	1.305	1.354	0.04	1.126	0.076	−1.99
44	0.615	91.6	−0.98	−7.22	1.325	1.357	0.42	1.127	0.042	−2.88
45	0.629	90.8	−1.07	−5.50	1.344	1.366	0.71	1.128	−0.006	−3.64
46	0.643	89.9	−1.14	−4.17	1.364	1.378	0.90	1.127	−0.062	−3.69
47	0.658	88.9	−1.19	−2.51	1.384	1.392	1.02	1.126	−0.112	−3.04
48	0.672	87.9	−1.21	0.15	1.404	1.407	1.07	1.124	−0.149	−2.20
49	0.686	87.0	−1.18	3.16	1.424	1.422	1.00	1.122	−0.175	−1.53
50	0.701	86.0	−1.12	4.79	1.444	1.435	0.80	1.119	−0.192	−1.05
51	0.715	85.1	−1.05	4.15	1.465	1.445	0.57	1.116	−0.205	−0.71
52	0.729	84.3	−1.00	2.30	1.486	1.452	0.52	1.113	−0.213	−0.43
53	0.744	83.5	−0.98	1.15	1.506	1.460	0.71	1.110	−0.217	−0.07
54	0.758	82.7	−0.97	1.51	1.527	1.472	0.94	1.107	−0.215	0.31
55	0.772	81.9	−0.94	2.94	1.549	1.487	0.88	1.104	−0.208	0.52
56	0.786	81.2	−0.88	5.03	1.570	1.497	0.44	1.101	−0.200	0.55
57	0.801	80.5	−0.79	7.77	1.591	1.499	−0.02	1.098	−0.192	0.60
58	0.815	79.9	−0.66	10.87	1.613	1.497	−0.02	1.096	−0.183	0.79
59	0.829	79.4	−0.48	13.59	1.634	1.499	0.59	1.093	−0.169	0.91
60	0.844	79.1	−0.27	15.25	1.656	1.513	1.59	1.091	−0.156	0.79
61	0.858	78.9	−0.05	15.83	1.677	1.544	2.58	1.089	−0.147	0.58
62	0.872	79.0	0.18	15.92	1.700	1.587	3.10	1.086	−0.140	0.55
63	0.887	79.2	0.41	15.99	1.723	1.633	2.64	1.085	−0.131	0.71

(*continued*)

TABLE A.3d (***Continued***)

	Frame	Time	Theta	Omega	Alpha	Cofm-X	Vel-X	Acc-X	Cofm-Y	Vel-Y	Acc-Y
		s	Deg	R/s	R/s/s	m	m/s	m/s/s	m	m/s	m/s/s
	64	0.901	79.7	0.64	15.90	1.747	1.663	0.88	1.083	−0.119	0.95
	65	0.915	80.3	0.86	15.03	1.770	1.658	−1.90	1.081	−0.104	1.21
	66	0.929	81.1	1.07	12.34	1.794	1.608	−4.71	1.080	−0.085	1.50
	67	0.944	82.0	1.22	6.89	1.816	1.523	−6.30	1.079	−0.061	1.88
	68	0.958	83.1	1.27	−0.81	1.838	1.428	−6.22	1.078	−0.031	2.43
	69	0.972	84.1	1.19	−8.51	1.857	1.345	−5.08	1.078	0.008	3.16
TOR	70	0.987	85.0	1.02	−13.55	1.876	1.283	−3.73	1.078	0.059	3.86
	71	1.001	85.8	0.81	−14.47	1.894	1.239	−2.53	1.079	0.119	4.26
	72	1.015	86.3	0.61	−11.76	1.911	1.210	−1.56	1.082	0.181	4.25
	73	1.030	86.8	0.47	−7.13	1.929	1.194	−0.86	1.085	0.240	3.91
	74	1.044	87.1	0.40	−2.15	1.946	1.185	−0.45	1.089	0.293	3.44
	75	1.058	87.4	0.41	2.49	1.962	1.181	−0.25	1.093	0.339	2.73
	76	1.072	87.8	0.48	6.25	1.979	1.178	−0.13	1.098	0.371	1.47
	77	1.087	88.2	0.59	8.26	1.996	1.177	0.01	1.104	0.381	−0.38
	78	1.101	88.8	0.71	7.82	2.013	1.178	0.23	1.109	0.360	−2.42
	79	1.115	89.4	0.81	4.89	2.030	1.184	0.59	1.114	0.312	−4.04
	80	1.130	90.1	0.85	0.35	2.047	1.195	1.00	1.118	0.245	−4.95
	81	1.144	90.8	0.82	−4.02	2.064	1.212	1.25	1.121	0.170	−5.20
	82	1.158	91.4	0.74	−6.62	2.082	1.231	1.20	1.123	0.096	−5.05
	83	1.173	92.0	0.63	−7.20	2.099	1.247	0.97	1.124	0.026	−4.71
	84	1.187	92.5	0.53	−6.49	2.117	1.259	0.81	1.124	−0.039	−4.37
	85	1.201	92.9	0.45	−5.14	2.135	1.270	0.83	1.123	−0.099	−4.19
	86	1.215	93.2	0.38	−3.23	2.154	1.283	1.00	1.121	−0.159	−4.11
	87	1.230	93.5	0.35	−0.73	2.172	1.299	1.26	1.118	−0.217	−3.86
	88	1.244	93.8	0.36	1.86	2.191	1.319	1.47	1.115	−0.269	−3.18
	89	1.258	94.1	0.41	3.58	2.210	1.341	1.52	1.110	−0.308	−2.09
	90	1.273	94.4	0.47	3.89	2.229	1.362	1.48	1.106	−0.329	−0.83
	91	1.287	94.9	0.52	3.56	2.249	1.383	1.53	1.101	−0.332	0.43

	92	1.301	95.3	0.57	3.94	2.269	1.406	1.67	1.096	−0.317	1.62
	93	1.316	95.8	0.63	5.06	2.289	1.431	1.80	1.092	−0.285	2.69
	94	1.330	96.3	0.71	5.42	2.309	1.457	1.92	1.088	−0.240	3.68
	95	1.344	96.9	0.79	3.53	2.330	1.486	2.11	1.085	−0.180	4.53
	96	1.358	97.6	0.81	−0.84	2.352	1.517	2.27	1.083	−0.110	5.12
HCR	97	1.373	98.3	0.76	−6.35	2.374	1.551	2.12	1.082	−0.034	5.43
	98	1.387	98.9	0.63	−11.08	2.396	1.578	1.49	1.082	0.045	5.56
	99	1.401	99.3	0.45	−13.91	2.419	1.593	0.57	1.083	0.125	5.41
	100	1.416	99.6	0.23	−14.78	2.442	1.594	−0.38	1.086	0.200	4.65
	101	1.430	99.7	0.02	−13.99	2.465	1.582	−1.19	1.089	0.258	3.12
	102	1.444	99.6	−0.17	−11.75	2.487	1.560	−1.90	1.093	0.289	1.13
	103	1.459	99.4	−0.31	−8.21	2.509	1.528	−2.78	1.097	0.291	−0.64
	104	1.473	99.1	−0.40	−3.79	2.531	1.481	−4.11	1.101	0.271	−1.69
	105	1.487	98.8	−0.42	0.37	2.552	1.410	−6.06	1.105	0.242	−2.07
	106	1.501	98.4	−0.39	2.87	2.571	1.308	−8.68	1.108	0.212	−2.24

TABLE A.4 Relative Joint Angular Kinematics – Ankle, Knee, and Hip

			Ankle			Knee			Hip		
	Frame	Time	Theta	Omega	Alpha	Theta	Omega	Alpha	Theta	Omega	Alpha
		s	Deg	R/s	R/s/s	Deg	R/s	R/s/s	Deg	R/s	R/s/s
TOR	1	0.000	−15.2	−2.29	94.89	46.7	6.74	−21.91	−2.4	2.39	36.03
	2	0.014	−16.4	−0.82	98.72	52.1	6.23	−46.59	−0.2	2.89	31.50
	3	0.029	−16.5	0.54	84.63	56.9	5.41	−65.29	2.3	3.30	24.37
	4	0.043	−15.6	1.60	63.13	61.0	4.37	−77.66	5.2	3.58	14.23
	5	0.057	−13.9	2.34	41.31	64.1	3.19	−84.77	8.2	3.70	1.30
	6	0.072	−11.7	2.78	20.10	66.2	1.94	−87.62	11.3	3.62	−12.28
	7	0.086	−9.3	2.92	−0.74	67.3	0.68	−86.54	14.2	3.35	−23.42
	8	0.100	−6.9	2.76	−18.70	67.3	−0.53	−81.99	16.8	2.95	−30.30
	9	0.114	−4.8	2.38	−30.23	66.4	−1.66	−75.17	19.0	2.48	−33.55
	10	0.129	−3.0	1.90	−34.13	64.6	−2.68	−67.64	20.8	1.99	−35.01
	11	0.143	−1.7	1.41	−32.12	62.0	−3.60	−60.31	22.2	1.48	−35.53
	12	0.157	−0.7	0.98	−26.84	58.7	−4.41	−53.09	23.3	0.98	−34.42
	13	0.172	−0.1	0.64	−20.23	54.8	−5.11	−45.78	23.8	0.50	−30.99
	14	0.186	0.3	0.40	−13.59	50.3	−5.72	−38.95	24.1	0.09	−26.06
	15	0.200	0.6	0.25	−8.03	45.4	−6.23	−32.91	24.0	−0.25	−21.52
	16	0.215	0.7	0.17	−3.87	40.1	−6.66	−26.57	23.7	−0.53	−18.38
	17	0.229	0.9	0.14	−0.56	34.5	−6.99	−18.05	23.1	−0.77	−16.10
	18	0.243	1.0	0.15	2.33	28.7	−7.17	−5.87	22.4	−0.99	−13.65
	19	0.257	1.1	0.21	4.60	22.8	−7.16	10.65	21.5	−1.16	−10.37
	20	0.272	1.3	0.29	5.47	16.9	−6.87	31.09	20.5	−1.28	−6.16
	21	0.286	1.6	0.36	3.83	11.5	−6.27	53.59	19.4	−1.34	−1.66
	22	0.300	1.9	0.40	−0.84	6.7	−5.34	74.76	18.3	−1.33	1.98
	23	0.315	2.2	0.34	−8.36	2.7	−4.13	90.62	17.2	−1.28	4.14
	24	0.329	2.5	0.16	−19.22	−0.1	−2.75	97.96	16.2	−1.21	5.38
	25	0.343	2.5	−0.21	−33.28	−1.8	−1.33	95.28	15.2	−1.13	6.53
	26	0.357	2.1	−0.79	−46.39	−2.3	−0.02	82.64	14.3	−1.03	7.45
	27	0.372	1.2	−1.54	−50.65	−1.8	1.04	62.40	13.6	−0.92	7.62
HCR	28	0.386	−0.4	−2.24	−38.30	−0.6	1.76	40.55	12.8	−0.81	7.93
	29	0.400	−2.5	−2.63	−7.99	1.1	2.20	24.37	12.2	−0.69	9.91
	30	0.415	−4.7	−2.47	30.12	3.0	2.46	16.62	11.7	−0.52	13.66

31	0.429	−6.5	−1.77	59.60	5.1	2.67	12.93	11.4	−0.30	16.99
32	0.443	−7.6	−0.77	70.70	7.4	2.83	5.97	11.2	−0.04	16.09
33	0.458	−7.8	0.25	64.54	9.8	2.84	−8.17	11.3	0.16	7.69
34	0.472	−7.2	1.08	46.87	12.1	2.60	−27.06	11.5	0.18	−7.32
35	0.486	−6.0	1.59	24.33	14.0	2.07	−44.04	11.6	−0.05	−22.88
36	0.500	−4.6	1.77	4.14	15.4	1.34	−53.13	11.4	−0.47	−32.00
37	0.515	−3.1	1.71	−8.73	16.2	0.55	−52.57	10.8	−0.96	−31.67
38	0.529	−1.8	1.52	−13.85	16.3	−0.17	−44.30	9.8	−1.38	−23.59
39	0.543	−0.6	1.31	−13.32	15.9	−0.72	−31.51	8.6	−1.64	−11.54
40	0.558	0.4	1.14	−9.71	15.2	−1.07	−17.12	7.2	−1.71	0.90
41	0.572	1.2	1.04	−6.42	14.2	−1.21	−3.98	5.8	−1.61	10.98
42	0.586	2.1	0.96	−6.79	13.2	−1.18	5.07	4.5	−1.40	16.74
43	0.601	2.8	0.84	−10.34	12.3	−1.06	8.50	3.5	−1.13	17.58
44	0.615	3.4	0.66	−12.36	11.4	−0.94	7.49	2.7	−0.89	15.03
45	0.629	3.9	0.49	−9.52	10.7	−0.85	4.99	2.0	−0.70	11.60
46	0.643	4.2	0.39	−3.26	10.1	−0.80	3.03	1.5	−0.56	8.39
47	0.658	4.6	0.40	3.33	9.4	−0.76	1.90	1.1	−0.46	4.74
48	0.672	4.9	0.49	7.14	8.8	−0.74	1.14	0.8	−0.43	−0.13
49	0.686	5.3	0.60	4.92	8.2	−0.73	0.45	0.4	−0.47	−5.26
50	0.701	5.9	0.63	−3.15	7.6	−0.73	0.18	0.0	−0.58	−8.17
51	0.715	6.4	0.51	−11.50	7.0	−0.72	1.21	−0.5	−0.70	−7.39
52	0.729	6.7	0.30	−14.34	6.4	−0.69	4.06	−1.2	−0.79	−4.00
53	0.744	6.9	0.10	−10.68	5.9	−0.61	8.77	−1.8	−0.81	−0.17
54	0.758	6.9	−0.01	−4.17	5.4	−0.44	15.06	−2.5	−0.79	3.12
55	0.772	6.9	−0.02	0.00	5.2	−0.18	22.07	−3.1	−0.73	5.70
56	0.786	6.8	−0.01	−1.14	5.1	0.19	28.41	−3.7	−0.63	6.92
57	0.801	6.8	−0.05	−7.42	5.5	0.63	32.98	−4.2	−0.53	6.16
58	0.815	6.8	−0.22	−17.46	6.2	1.13	35.54	−4.6	−0.45	3.95
59	0.829	6.5	−0.55	−28.75	7.3	1.65	36.64	−4.9	−0.41	1.81
60	0.844	5.9	−1.04	−37.44	8.9	2.18	37.19	−5.2	−0.40	1.24
61	0.858	4.8	−1.62	−40.93	10.9	2.71	37.90	−5.6	−0.38	2.74
62	0.872	3.2	−2.21	−40.94	13.3	3.26	39.00	−5.9	−0.32	5.62
63	0.887	1.2	−2.79	−41.67	16.2	3.83	40.04	−6.1	−0.22	8.61

(continued)

TABLE A.4 (*Continued*)

	Frame	Time	Ankle Theta	Ankle Omega	Ankle Alpha	Knee Theta	Knee Omega	Knee Alpha	Hip Theta	Hip Omega	Hip Alpha
		s	Deg	R/s	R/s/s	Deg	R/s	R/s/s	Deg	R/s	R/s/s
	64	0.901	−1.4	−3.40	−43.21	19.6	4.41	39.95	−6.2	−0.08	11.23
	65	0.915	−4.4	−4.03	−38.18	23.5	4.97	37.59	−6.2	0.10	14.31
	66	0.929	−8.0	−4.50	−17.72	27.8	5.48	32.25	−6.0	0.33	18.85
	67	0.944	−11.8	−4.54	18.41	32.4	5.90	23.44	−5.7	0.64	24.81
	68	0.958	−15.4	−3.97	59.42	37.4	6.15	10.87	−5.0	1.04	31.17
	69	0.972	−18.3	−2.84	91.15	42.5	6.21	−4.80	−4.0	1.53	36.24
TOR	70	0.987	−20.1	−1.36	104.21	47.6	6.02	−21.92	−2.5	2.08	38.01
	71	1.001	−20.5	0.14	96.45	52.4	5.58	−38.75	−0.5	2.62	34.78
	72	1.015	−19.8	1.40	72.74	56.7	4.91	−53.72	1.8	3.07	26.42
	73	1.030	−18.2	2.22	42.50	60.4	4.04	−65.55	4.5	3.38	14.66
	74	1.044	−16.2	2.61	15.01	63.4	3.03	−74.32	7.3	3.49	1.29
	75	1.058	−14.0	2.65	−3.93	65.4	1.92	−80.71	10.2	3.41	−12.63
	76	1.072	−11.8	2.50	−13.24	66.5	0.73	−83.96	12.9	3.13	−25.25
	77	1.087	−9.9	2.28	−16.29	66.6	−0.48	−82.70	15.3	2.69	−33.71
	78	1.101	−8.1	2.03	−17.51	65.7	−1.64	−76.99	17.3	2.17	−36.16
	79	1.115	−6.5	1.77	−18.73	63.9	−2.69	−68.11	18.9	1.66	−32.81
	80	1.130	−5.2	1.50	−19.65	61.3	−3.59	−57.74	20.1	1.23	−25.82
	81	1.144	−4.1	1.21	−19.37	58.0	−4.34	−48.02	20.9	0.92	−18.62
	82	1.158	−3.2	0.94	−17.21	54.2	−4.96	−40.91	21.6	0.70	−14.21
	83	1.173	−2.5	0.72	−13.87	49.9	−5.51	−37.09	22.1	0.51	−13.55
	84	1.187	−2.0	0.55	−10.79	45.2	−6.02	−35.35	22.4	0.31	−15.62
	85	1.201	−1.6	0.41	−7.88	40.0	−6.52	−32.90	22.6	0.06	−18.75
	86	1.215	−1.3	0.32	−3.38	34.5	−6.96	−26.57	22.5	−0.23	−21.73
	87	1.230	−1.1	0.31	3.27	28.6	−7.28	−14.15	22.2	−0.56	−23.63
	88	1.244	−0.8	0.42	9.14	22.6	−7.37	4.76	21.6	−0.90	−23.22
	89	1.258	−0.4	0.58	10.06	16.5	−7.14	28.98	20.7	−1.22	−19.01
	90	1.273	0.1	0.70	4.80	10.9	−6.54	56.01	19.6	−1.45	−10.61

	91	1.287	0.7	0.71	−4.67	5.8	−5.54	81.46	18.3	−1.52	−0.51
	92	1.301	1.3	0.57	−16.27	1.8	−4.21	99.89	17.1	−1.46	6.98
	93	1.316	1.7	0.25	−29.67	−1.1	−2.68	107.07	15.9	−1.32	9.49
	94	1.330	1.7	−0.28	−44.20	−2.6	−1.15	101.58	14.9	−1.19	8.22
	95	1.344	1.2	−1.02	−54.99	−3.0	0.22	85.39	14.0	−1.09	6.14
	96	1.358	0.0	−1.85	−53.10	−2.3	1.30	63.88	13.1	−1.01	5.89
HCR	97	1.373	−1.8	−2.53	−31.93	−0.8	2.05	43.96	12.3	−0.92	8.35
	98	1.387	−4.1	−2.77	4.96	1.1	2.55	29.90	11.6	−0.78	12.18
	99	1.401	−6.4	−2.39	43.33	3.4	2.90	20.18	11.1	−0.57	14.65
	100	1.416	−8.0	−1.53	67.62	5.9	3.13	9.21	10.7	−0.36	13.20
	101	1.430	−8.9	−0.46	71.62	8.5	3.17	−6.47	10.5	−0.20	6.83
	102	1.444	−8.8	0.52	59.88	11.0	2.94	−24.49	10.4	−0.16	−2.99
	103	1.459	−8.0	1.25	41.13	13.3	2.47	−39.25	10.2	−0.28	−12.86
	104	1.473	−6.7	1.70	22.10	15.1	1.82	−46.90	9.9	−0.53	−19.17
	105	1.487	−5.2	1.89	5.23	16.3	1.13	−47.03	9.4	−0.83	−19.53
	106	1.501	−3.7	1.85	−10.10	16.9	0.48	−40.67	8.5	−1.09	−13.02

TABLE A.5a Reaction Forces and Moments of Force – Ankle and Knee

			Foot Segment						Leg Segment					
			Ground		Ankle		Ground	Ankle	Ankle		Knee		Ankle	Knee
	Frame	Time	RX	RY	RX	RY	Cof P X	Moment	RX	RY	RX	RY	Moment	Moment
		s	N	N	N	N	m	N · m	N	N	N	N	N · m	N · m
TOR	1	0.000	0.0	0.0	20.9	3.9	0.000	1.6	−20.9	−3.9	52.0	31.6	−1.6	7.0
	2	0.014	0.0	0.0	19.8	0.0	0.000	1.6	−19.8	0.0	41.1	27.1	−1.6	7.0
	3	0.029	0.0	0.0	17.6	−3.1	0.000	1.5	−17.6	3.1	30.8	22.2	−1.5	6.9
	4	0.043	0.0	0.0	14.9	−5.0	0.000	1.3	−14.9	5.0	22.0	17.8	−1.3	6.5
	5	0.057	0.0	0.0	12.3	−5.5	0.000	1.1	−12.3	5.5	15.1	14.6	−1.1	5.8
	6	0.072	0.0	0.0	10.3	−4.7	0.000	0.9	−10.3	4.7	10.1	12.7	−0.9	4.8
	7	0.086	0.0	0.0	8.6	−3.0	0.000	0.7	−8.6	3.0	6.5	12.2	−0.7	3.6
	8	0.100	0.0	0.0	7.3	−0.7	0.000	0.6	−7.3	0.7	4.1	13.4	−0.6	2.3
	9	0.114	0.0	0.0	6.3	2.0	0.000	0.5	−6.3	−2.0	2.5	15.9	−0.5	1.0
	10	0.129	0.0	0.0	5.5	4.8	0.000	0.4	−5.5	−4.8	1.4	19.1	−0.4	−0.1
	11	0.143	0.0	0.0	4.7	7.5	0.000	0.4	−4.7	−7.5	0.4	22.8	−0.4	−1.0
	12	0.157	0.0	0.0	3.7	10.2	0.000	0.5	−3.7	−10.2	−0.9	27.1	−0.5	−1.9
	13	0.172	0.0	0.0	2.5	12.6	0.000	0.5	−2.5	−12.6	−2.8	31.8	−0.5	−2.7
	14	0.186	0.0	0.0	0.9	14.7	0.000	0.6	−0.9	−14.7	−5.7	36.2	−0.6	−3.5
	15	0.200	0.0	0.0	−0.9	16.1	0.000	0.6	0.9	−16.1	−9.5	39.6	−0.6	−4.2
	16	0.215	0.0	0.0	−3.0	16.8	0.000	0.6	3.0	−16.8	−14.2	42.0	−0.6	−4.9
	17	0.229	0.0	0.0	−5.4	17.0	0.000	0.6	5.4	−17.0	−19.7	43.4	−0.6	−5.8
	18	0.243	0.0	0.0	−8.3	16.3	0.000	0.6	8.3	−16.3	−25.5	43.7	−0.6	−6.8
	19	0.257	0.0	0.0	−11.5	14.8	0.000	0.5	11.5	−14.8	−30.7	42.7	−0.5	−8.0
	20	0.272	0.0	0.0	−14.9	12.2	0.000	0.3	14.9	−12.2	−35.5	40.7	−0.3	−9.4
	21	0.286	0.0	0.0	−18.1	9.0	0.000	0.1	18.1	−9.0	−39.9	38.2	−0.1	−11.1
	22	0.300	0.0	0.0	−20.8	5.2	0.000	−0.1	20.8	−5.2	−43.9	34.9	0.1	−12.8
	23	0.315	0.0	0.0	−22.7	1.6	0.000	−0.3	22.7	−1.6	−46.6	30.6	0.3	−14.3
	24	0.329	0.0	0.0	−23.2	−1.2	0.000	−0.5	23.2	1.2	−47.3	26.1	0.5	−15.1
	25	0.343	0.0	0.0	−22.2	−2.4	0.000	−0.5	22.2	2.4	−45.3	23.3	0.5	−14.6
	26	0.357	0.0	0.0	−19.5	−1.7	0.000	−0.5	19.5	1.7	−40.6	24.1	0.5	−12.7
	27	0.372	0.0	0.0	−15.6	1.0	0.000	−0.3	15.6	−1.0	−34.1	29.1	0.3	−9.5

HCR	28	0.386	37.3	87.1	−48.7	−82.4	1.227	−1.7	48.7	82.4	−64.3	−50.2	1.7	−33.8
	29	0.400	−4.2	192.6	−3.8	−184.1	1.244	4.6	3.8	184.1	−16.8	−148.2	−4.6	−19.9
	30	0.415	−43.7	304.1	38.2	−292.9	1.261	8.3	−38.2	292.9	27.2	−255.2	−8.3	−7.6
	31	0.429	−74.0	404.2	69.8	−391.4	1.291	2.8	−69.8	391.4	59.9	−354.6	−2.8	−4.5
	32	0.443	−91.9	476.6	88.4	−463.4	1.290	6.9	−88.4	463.4	77.7	−429.9	−6.9	7.8
	33	0.458	−102.7	521.7	99.7	−508.9	1.301	4.6	−99.7	508.9	86.3	−480.2	−4.6	14.5
	34	0.472	−110.5	552.9	108.1	−541.1	1.304	5.0	−108.1	541.1	91.5	−516.7	−5.0	24.8
	35	0.486	−114.2	579.8	112.5	−569.0	1.311	2.6	−112.5	569.0	94.3	−547.2	−2.6	31.6
	36	0.500	−110.5	599.6	109.4	−589.9	1.317	−0.5	−109.4	589.9	91.7	−568.4	0.5	35.0
	37	0.515	−98.1	604.5	97.1	−595.7	1.316	0.2	−97.1	595.7	80.9	−573.4	−0.2	37.8
	38	0.529	−79.8	589.9	78.5	−581.7	1.321	−3.5	−78.5	581.7	63.7	−558.2	3.5	32.5
	39	0.543	−62.0	558.1	60.3	−550.2	1.322	−5.1	−60.3	550.2	47.2	−525.1	5.1	28.1
	40	0.558	−48.7	516.7	47.2	−508.7	1.325	−6.6	−47.2	508.7	36.8	−481.5	6.6	24.8
	41	0.572	−39.8	473.4	38.8	−465.3	1.333	−10.5	−38.8	465.3	32.2	−436.1	10.5	20.2
	42	0.586	−33.0	433.4	32.7	−425.2	1.335	−10.6	−32.7	425.2	29.8	−395.6	10.6	19.8
	43	0.601	−27.0	400.3	27.2	−392.2	1.345	−14.1	−27.2	392.2	27.4	−364.4	14.1	15.8
	44	0.615	−21.8	377.1	22.4	−369.4	1.358	−18.7	−22.4	369.4	24.6	−344.7	18.7	10.9
	45	0.629	−18.1	365.0	19.0	−357.6	1.369	−22.5	−19.0	357.6	22.3	−335.4	22.5	7.8
	46	0.643	−16.6	362.3	17.4	−355.0	1.377	−25.3	−17.4	355.0	21.0	−333.8	25.3	6.6
	47	0.658	−16.8	366.5	17.6	−359.1	1.384	−27.9	−17.6	359.1	21.0	−337.2	27.9	6.6
	48	0.672	−16.8	375.0	17.5	−367.3	1.390	−30.3	−17.5	367.3	20.7	−343.7	30.3	7.0
	49	0.686	−14.5	386.1	15.2	−378.2	1.396	−33.5	−15.2	378.2	18.3	−352.3	33.5	5.9
	50	0.701	−9.6	400.3	10.3	−392.4	1.406	−38.8	−10.3	392.4	13.0	−364.3	38.8	2.3
	51	0.715	−3.8	418.8	4.2	−410.8	1.417	−45.1	−4.2	410.8	6.0	−381.2	45.1	−2.3
	52	0.729	2.2	441.0	−2.1	−432.7	1.412	−45.4	2.1	432.7	−0.9	−402.2	45.4	−0.1
	53	0.744	8.7	465.8	−9.0	−457.0	1.424	−53.5	9.0	457.0	−6.8	−426.3	53.5	−5.0
	54	0.758	16.5	493.7	−16.8	−484.4	1.428	−58.8	16.8	484.4	−12.3	−453.7	58.8	−6.2
	55	0.772	26.8	523.9	−27.1	−514.2	1.432	−64.8	27.1	514.2	−19.7	−483.6	64.8	−7.9
	56	0.786	40.1	552.6	−40.4	−542.9	1.436	−71.2	40.4	542.9	−30.9	−513.0	71.2	−10.4
	57	0.801	54.8	576.8	−55.0	−567.5	1.440	−77.3	55.0	567.5	−43.5	−538.7	77.3	−12.5
	58	0.815	68.1	595.8	−67.6	−586.9	1.444	−82.7	67.6	586.9	−53.3	−559.3	82.7	−12.4
	59	0.829	79.6	608.6	−77.9	−599.7	1.447	−87.0	77.9	599.7	−59.1	−573.0	87.0	−10.0
	60	0.844	90.8	612.1	−87.4	−602.8	1.451	−89.7	87.4	602.8	−63.1	−576.4	89.7	−6.4

(*continued*)

TABLE A.5a (*Continued*)

			Foot Segment						Leg Segment					
			Ground		Ankle		Ground	Ankle	Ankle		Knee		Ankle	Knee
	Frame	Time	RX	RY	RX	RY	Cof P X	Moment	RX	RY	RX	RY	Moment	Moment
		s	N	N	N	N	m	N · m	N	N	N	N	N · m	N · m
	61	0.858	101.5	602.3	−96.4	−592.2	1.455	−89.8	96.4	592.2	−66.5	−566.1	89.8	−2.2
	62	0.872	110.4	576.1	−103.8	−565.0	1.459	−86.7	103.8	565.0	−68.9	−538.7	86.7	2.3
	63	0.887	115.6	530.3	−107.7	−518.2	1.463	−79.7	107.7	518.2	−69.7	−491.1	79.7	6.5
	64	0.901	114.5	463.0	−105.7	−450.2	1.467	−68.6	105.7	450.2	−67.5	−421.9	68.6	10.1
	65	0.915	105.2	377.3	−95.5	−364.4	1.470	−54.3	95.5	364.4	−60.1	−335.2	54.3	12.5
	66	0.929	88.2	282.1	−77.2	−269.9	1.474	−38.8	77.2	269.9	−46.1	−240.4	38.8	13.3
	67	0.944	65.7	190.1	−53.0	−179.1	1.478	−24.2	53.0	179.1	−26.4	−149.3	24.2	12.5
	68	0.958	41.4	110.8	−26.7	−101.8	1.482	−12.3	26.7	101.8	−4.1	−71.2	12.3	10.6
	69	0.972	18.4	44.4	−2.1	−37.9	1.486	−3.6	2.1	37.9	16.9	−6.8	3.6	6.8
TOR	70	0.987	0.0	0.0	17.1	3.5	0.000	1.4	−17.1	−3.5	32.9	34.3	−1.4	3.6
	71	1.001	0.0	0.0	16.8	0.3	0.000	1.4	−16.8	−0.3	30.4	29.2	−1.4	4.6
	72	1.015	0.0	0.0	15.8	−2.5	0.000	1.4	−15.8	2.5	28.0	23.0	−1.4	5.7
	73	1.030	0.0	0.0	14.3	−4.4	0.000	1.3	−14.3	4.4	25.4	17.2	−1.3	6.4
	74	1.044	0.0	0.0	12.6	−4.9	0.000	1.1	−12.6	4.9	22.1	13.4	−1.1	6.3
	75	1.058	0.0	0.0	11.1	−3.9	0.000	0.9	−11.1	3.9	18.1	12.2	−0.9	5.5
	76	1.072	0.0	0.0	9.7	−2.0	0.000	0.7	−9.7	2.0	14.0	12.8	−0.7	4.3
	77	1.087	0.0	0.0	8.5	0.2	0.000	0.6	−8.5	−0.2	10.0	14.1	−0.6	3.0
	78	1.101	0.0	0.0	7.4	2.2	0.000	0.5	−7.4	−2.2	6.6	15.6	−0.5	1.8
	79	1.115	0.0	0.0	6.3	4.3	0.000	0.5	−6.3	−4.3	3.7	17.8	−0.5	0.6
	80	1.130	0.0	0.0	5.3	6.7	0.000	0.5	−5.3	−6.7	1.6	21.5	−0.5	−0.5
	81	1.144	0.0	0.0	4.3	9.2	0.000	0.5	−4.3	−9.2	−0.2	25.9	−0.5	−1.5
	82	1.158	0.0	0.0	3.4	11.7	0.000	0.6	−3.4	−11.7	−1.7	30.7	−0.6	−2.2
	83	1.173	0.0	0.0	2.4	14.1	0.000	0.6	−2.4	−14.1	−3.5	35.5	−0.6	−2.7
	84	1.187	0.0	0.0	1.1	16.3	0.000	0.7	−1.1	−16.3	−6.0	39.8	−0.7	−3.1
	85	1.201	0.0	0.0	−0.9	17.8	0.000	0.8	0.9	−17.8	−9.8	43.0	−0.8	−3.6
	86	1.215	0.0	0.0	−3.5	18.5	0.000	0.8	3.5	−18.5	−15.1	44.7	−0.8	−4.3

	87	1.230	0.0	0.0	−6.8	18.0	0.000	0.7	6.8	−18.0	−21.7	45.3	−0.7	−5.6
	88	1.244	0.0	0.0	−10.8	16.4	0.000	0.6	10.8	−16.4	−29.4	45.0	−0.6	−7.4
	89	1.258	0.0	0.0	−15.3	13.6	0.000	0.4	15.3	−13.6	−37.5	43.5	−0.4	−9.7
	90	1.273	0.0	0.0	−19.6	9.7	0.000	0.1	19.6	−9.7	−44.8	40.3	−0.1	−12.3
	91	1.287	0.0	0.0	−23.1	5.3	0.000	−0.2	23.1	−5.3	−49.6	36.0	0.2	−14.5
	92	1.301	0.0	0.0	−25.0	1.2	0.000	−0.4	25.0	−1.2	−50.8	31.4	0.4	−15.9
	93	1.316	0.0	0.0	−24.9	−1.8	0.000	−0.5	24.9	1.8	−48.4	27.6	0.5	−15.9
	94	1.330	0.0	0.0	−22.9	−2.8	0.000	−0.6	22.9	2.8	−43.5	26.5	0.6	−14.5
	95	1.344	0.0	0.0	−19.4	−1.6	0.000	−0.5	19.4	1.6	−36.8	29.0	0.5	−11.7
	96	1.358	0.0	0.0	−15.0	1.4	0.000	−0.3	15.0	−1.4	−29.5	33.9	0.3	−8.2
HCR	97	1.373	0.0	0.0	−10.9	4.9	0.000	0.0	10.9	−4.9	−23.4	39.0	0.0	−4.9
	98	1.387	0.0	0.0	−8.1	8.2	0.000	0.3	8.1	−8.2	−19.9	42.8	−0.3	−2.8
	99	1.401	0.0	0.0	−6.5	10.4	0.000	0.5	6.5	−10.4	−19.4	44.2	−0.5	−2.0
	100	1.416	0.0	0.0	−5.9	11.6	0.000	0.6	5.9	−11.6	−21.4	42.9	−0.6	−2.4
	101	1.430	0.0	0.0	−5.6	11.8	0.000	0.7	5.6	−11.8	−24.8	39.7	−0.7	−3.2
	102	1.444	0.0	0.0	−5.2	11.4	0.000	0.7	5.2	−11.4	−28.4	36.1	−0.7	−4.1
	103	1.459	0.0	0.0	−4.5	10.9	0.000	0.6	4.5	−10.9	−30.3	33.7	−0.6	−4.6
	104	1.473	0.0	0.0	−3.4	10.4	0.000	0.6	3.4	−10.4	−28.9	33.1	−0.6	−4.2
	105	1.487	0.0	0.0	−1.8	10.0	0.000	0.6	1.8	−10.0	−24.4	33.6	−0.6	−3.2
	106	1.501	0.0	0.0	−0.4	9.5	0.000	0.6	0.4	−9.5	−19.0	34.0	−0.6	−2.1

TABLE A.5b Reaction Forces and Moments of Force – Hip

			Thigh Segment					
			Knee		Hip		Knee	Hip
	Frame	Time	RX	RY	RX	RY	Moment	Moment
		s	N	N	N	N	N · m	N · m
TOR	1	0.000	−52.0	−31.6	59.7	112.1	−7.0	22.9
	2	0.014	−41.1	−27.1	30.3	117.1	−7.0	17.8
	3	0.029	−30.8	−22.2	9.1	116.7	−6.9	13.8
	4	0.043	−22.0	−17.8	−3.8	111.5	−6.5	10.8
	5	0.057	−15.1	−14.6	−9.4	102.8	−5.8	8.5
	6	0.072	−10.1	−12.7	−10.6	91.7	−4.8	6.7
	7	0.086	−6.5	−12.2	−10.0	80.4	−3.6	5.0
	8	0.100	−4.1	−13.4	−9.8	71.3	−2.3	3.4
	9	0.114	−2.5	−15.9	−10.4	65.0	−1.0	2.0
	10	0.129	−1.4	−19.1	−11.7	60.0	0.1	0.9
	11	0.143	−0.4	−22.8	−13.3	56.1	1.0	0.0
	12	0.157	0.9	−27.1	−15.2	54.5	1.9	−0.8
	13	0.172	2.8	−31.8	−18.1	55.5	2.7	−1.5
	14	0.186	5.7	−36.2	−21.7	57.6	3.5	−2.5
	15	0.200	9.5	−39.6	−25.3	59.9	4.2	−3.6
	16	0.215	14.2	−42.0	−28.4	62.3	4.9	−5.0
	17	0.229	19.7	−43.4	−31.3	65.1	5.8	−6.7
	18	0.243	25.5	−43.7	−33.4	69.0	6.8	−8.7
	19	0.257	30.7	−42.7	−33.9	74.4	8.0	−10.5
	20	0.272	35.5	−40.7	−33.1	82.1	9.4	−12.2
	21	0.286	39.9	−38.2	−32.1	91.9	11.1	−14.0
	22	0.300	43.9	−34.9	−30.9	100.1	12.8	−16.0
	23	0.315	46.6	−30.6	−29.4	103.1	14.3	−17.9
	24	0.329	47.3	−26.1	−27.9	101.5	15.1	−19.1
	25	0.343	45.3	−23.3	−26.8	99.4	14.6	−18.7
	26	0.357	40.6	−24.1	−26.5	100.7	12.7	−16.1
	27	0.372	34.1	−29.1	−27.3	106.9	9.5	−11.7

HCR	28	0.386	64.3	50.2	−66.1	29.7	33.8	−54.4
	29	0.400	16.8	148.2	−26.3	−66.4	19.9	−37.6
	30	0.415	−27.2	255.2	12.4	−173.3	7.6	−23.5
	31	0.429	−59.9	354.6	41.3	−276.1	4.5	−20.8
	32	0.443	−77.7	429.9	54.7	−359.4	−7.8	−11.1
	33	0.458	−86.3	480.2	58.3	−421.0	−14.5	−7.8
	34	0.472	−91.5	516.7	60.4	−468.3	−24.8	−0.4
	35	0.486	−94.3	547.2	63.8	−504.4	−31.6	4.7
	36	0.500	−91.7	568.4	64.2	−526.0	−35.0	7.3
	37	0.515	−80.9	573.4	55.7	−529.1	−37.8	9.9
	38	0.529	−63.7	558.2	39.0	−512.7	−32.5	5.0
	39	0.543	−47.2	525.1	22.5	−478.1	−28.1	3.0
	40	0.558	−36.8	481.5	13.4	−431.6	−24.8	4.7
	41	0.572	−32.2	436.1	12.3	−383.0	−20.2	6.5
	42	0.586	−29.8	395.6	15.2	−341.7	−19.8	12.0
	43	0.601	−27.4	364.4	18.7	−314.3	−15.8	12.5
	44	0.615	−24.6	344.7	21.0	−301.1	−10.9	10.9
	45	0.629	−22.3	335.4	21.9	−296.7	−7.8	10.1
	46	0.643	−21.0	333.8	22.5	−295.3	−6.6	11.2
	47	0.658	−21.0	337.2	24.4	−295.0	−6.6	13.8
	48	0.672	−20.7	343.7	27.3	−296.3	−7.0	16.7
	49	0.686	−18.3	352.3	27.9	−300.2	−5.9	17.7
	50	0.701	−13.0	364.3	23.4	−308.6	−2.3	15.2
	51	0.715	−6.0	381.2	14.5	−323.0	2.3	11.2
	52	0.729	0.9	402.2	5.7	−342.8	0.1	14.6
	53	0.744	6.8	426.3	1.0	−366.2	5.0	12.2
	54	0.758	12.3	453.7	0.1	−393.6	6.2	14.7
	55	0.772	19.7	483.6	−2.5	−424.7	7.9	16.6
	56	0.786	30.9	513.0	−10.6	−456.2	10.4	16.5
	57	0.801	43.5	538.7	−20.3	−484.2	12.5	16.1
	58	0.815	53.3	559.3	−24.8	−506.3	12.4	18.3
	59	0.829	59.1	573.0	−22.7	−521.5	10.0	23.6
	60	0.844	63.1	576.4	−17.8	−527.1	6.4	29.5

(*continued*)

TABLE A.5b (*Continued*)

			Thigh Segment					
			Knee		Hip		Knee	Hip
	Frame	Time	RX	RY	RX	RY	Moment	Moment
		s	N	N	N	N	N · m	N · m
	61	0.858	66.5	566.1	−13.2	−519.0	2.2	34.4
	62	0.872	68.9	538.7	−10.1	−492.1	−2.3	37.3
	63	0.887	69.7	491.1	−10.7	−442.7	−6.5	37.2
	64	0.901	67.5	421.9	−16.0	−370.1	−10.1	33.6
	65	0.915	60.1	335.2	−23.2	−279.2	−12.5	27.5
	66	0.929	46.1	240.4	−27.5	−179.5	−13.3	20.7
	67	0.944	26.4	149.3	−25.0	−82.3	−12.5	15.2
	68	0.958	4.1	71.2	−16.0	2.9	−10.6	11.9
	69	0.972	−16.9	6.8	−3.5	74.9	−6.8	9.3
TOR	70	0.987	−32.9	−34.3	8.9	122.0	−3.6	9.0
	71	1.001	−30.4	−29.2	7.9	120.0	−4.6	10.4
	72	1.015	−28.0	−23.0	10.7	113.5	−5.7	12.3
	73	1.030	−25.4	−17.2	14.0	104.6	−6.4	13.6
	74	1.044	−22.1	−13.4	14.8	96.0	−6.3	13.4
	75	1.058	−18.1	−12.2	12.8	88.5	−5.5	11.8
	76	1.072	−14.0	−12.8	9.0	80.7	−4.3	9.4
	77	1.087	−10.0	−14.1	3.9	71.1	−3.0	6.8
	78	1.101	−6.6	−15.6	−2.0	61.1	−1.8	4.2
	79	1.115	−3.7	−17.8	−8.0	54.2	−0.6	2.0
	80	1.130	−1.6	−21.5	−13.8	51.8	0.5	0.4
	81	1.144	0.2	−25.9	−18.9	53.0	1.5	−0.7
	82	1.158	1.7	−30.7	−22.8	55.8	2.2	−1.4
	83	1.173	3.5	−35.5	−25.5	59.1	2.7	−1.7
	84	1.187	6.0	−39.8	−27.3	62.0	3.1	−2.1
	85	1.201	9.8	−43.0	−28.9	63.7	3.6	−3.0
	86	1.215	15.1	−44.7	−30.5	65.2	4.3	−4.5

	87	1.230	21.7	−45.3	−31.7	68.8	5.6	−6.7
	88	1.244	29.4	−45.0	−32.7	75.5	7.4	−9.4
	89	1.258	37.5	−43.5	−34.2	83.8	9.7	−12.8
	90	1.273	44.8	−40.3	−35.7	92.0	12.3	−16.4
	91	1.287	49.6	−36.0	−34.2	99.0	14.5	−19.0
	92	1.301	50.8	−31.4	−28.5	104.4	15.9	−19.7
	93	1.316	48.4	−27.6	−20.9	108.1	15.9	−18.6
	94	1.330	43.5	−26.5	−14.9	111.6	14.5	−15.7
	95	1.344	36.8	−29.0	−12.5	115.8	11.7	−11.5
	96	1.358	29.5	−33.9	−14.1	119.7	8.2	−6.7
HCR	97	1.373	23.4	−39.0	−19.4	122.3	4.9	−2.9
	98	1.387	19.9	−42.8	−27.3	123.5	2.8	−1.0
	99	1.401	19.4	−44.2	−36.7	121.9	2.0	−1.4
	100	1.416	21.4	−42.9	−47.1	115.1	2.4	−3.9
	101	1.430	24.8	−39.7	−58.3	103.2	3.2	−7.8
	102	1.444	28.4	−36.1	−68.8	89.8	4.1	−12.0
	103	1.459	30.3	−33.7	−75.7	80.0	4.6	−14.7
	104	1.473	28.9	−33.1	−76.9	76.4	4.2	−14.9
	105	1.487	24.4	−33.6	−73.9	77.2	3.2	−12.8
	106	1.501	19.0	−34.0	−72.3	78.9	2.1	−10.4

TABLE A.6 Segment Potential, Kinetic, and Total Energies – Foot, Leg, Thigh, and ½ HAT

		Foot Segment				Leg Segment				Thigh Segment				HAT Segment			
Frame	Time	PE	TKE	RKE	Total	PE	TKE	RKE	Total	PE	TKE	RKE	Total	PE	TKE	RKE	Total
	s	J	J	J	J	J	J	J	J	J	J	J	J	J	J	J	J
1	0.000	1.2	1.8	0.0	3.0	9.5	8.3	0.1	17.9	36.3	12.3	0.3	48.8	203.6	18.2	0.4	222.3
2	0.014	1.3	2.3	0.0	3.6	9.6	9.2	0.1	18.9	36.2	12.2	0.3	48.8	203.8	18.3	0.3	222.4
3	0.029	1.4	2.9	0.0	4.3	9.7	9.9	0.0	19.6	36.3	11.7	0.4	48.4	204.1	17.9	0.1	222.1
4	0.043	1.4	3.5	0.0	4.9	9.8	10.2	0.0	20.0	36.4	11.1	0.4	47.9	204.6	17.2	0.1	221.8
5	0.057	1.4	4.0	0.0	5.5	9.9	10.4	0.0	20.3	36.7	10.5	0.4	47.6	205.2	16.3	0.0	221.6
6	0.072	1.4	4.6	0.0	6.0	10.0	10.4	0.1	20.5	36.9	10.0	0.4	47.4	206.0	15.6	0.0	221.7
7	0.086	1.4	5.1	0.0	6.5	10.0	10.3	0.2	20.5	37.3	9.6	0.4	47.3	207.0	15.1	0.1	222.1
8	0.100	1.3	5.5	0.1	6.9	10.1	10.2	0.3	20.5	37.6	9.3	0.3	47.2	207.9	14.8	0.1	222.8
9	0.114	1.3	6.0	0.1	7.3	10.1	10.1	0.3	20.5	38.0	9.0	0.2	47.2	208.9	14.6	0.1	223.6
10	0.129	1.2	6.3	0.1	7.6	10.1	9.9	0.4	20.4	38.3	8.6	0.2	47.1	209.8	14.4	0.2	224.3
11	0.143	1.1	6.6	0.1	7.8	10.1	9.8	0.5	20.3	38.7	8.2	0.1	47.0	210.6	14.1	0.2	224.9
12	0.157	1.0	6.9	0.1	8.0	10.0	9.6	0.5	20.2	38.9	7.7	0.1	46.7	211.2	13.9	0.3	225.4
13	0.172	0.9	7.0	0.1	8.0	9.9	9.4	0.6	20.0	39.1	7.3	0.0	46.5	211.6	13.7	0.4	225.7
14	0.186	0.8	7.1	0.1	8.0	9.9	9.2	0.6	19.7	39.3	6.8	0.0	46.1	211.9	13.7	0.4	225.9
15	0.200	0.8	7.0	0.1	7.9	9.8	9.0	0.6	19.4	39.4	6.4	0.0	45.8	212.0	13.9	0.4	226.2
16	0.215	0.7	6.9	0.1	7.7	9.7	8.6	0.6	18.9	39.4	6.1	0.0	45.5	211.9	14.4	0.3	226.6
17	0.229	0.7	6.6	0.1	7.4	9.6	8.2	0.6	18.3	39.3	5.9	0.0	45.2	211.6	15.4	0.2	227.2
18	0.243	0.7	6.2	0.1	7.0	9.5	7.6	0.6	17.7	39.2	5.7	0.0	44.9	211.1	16.7	0.2	227.9
19	0.257	0.7	5.7	0.1	6.5	9.4	7.0	0.5	16.9	39.0	5.7	0.0	44.7	210.5	18.1	0.1	228.7
20	0.272	0.8	5.0	0.1	5.8	9.3	6.4	0.5	16.1	38.7	5.8	0.0	44.6	209.7	19.6	0.1	229.4
21	0.286	0.8	4.2	0.0	5.0	9.2	5.7	0.4	15.3	38.5	5.9	0.0	44.4	208.9	20.8	0.1	229.7
22	0.300	0.8	3.3	0.0	4.2	9.1	5.1	0.3	14.4	38.2	6.1	0.0	44.3	208.1	21.5	0.0	229.7
23	0.315	0.9	2.5	0.0	3.4	9.0	4.4	0.1	13.6	37.9	6.4	0.0	44.3	207.4	21.9	0.1	229.3
24	0.329	0.9	1.7	0.0	2.6	9.0	3.8	0.1	12.8	37.7	6.7	0.0	44.4	206.7	22.0	0.1	228.8
25	0.343	0.9	1.1	0.0	2.0	8.9	3.3	0.0	12.2	37.5	7.1	0.0	44.6	206.1	22.1	0.1	228.4
26	0.357	0.9	0.7	0.0	1.5	8.8	2.8	0.0	11.6	37.4	7.4	0.0	44.8	205.7	22.5	0.1	228.3
27	0.372	0.8	0.4	0.0	1.3	8.8	2.4	0.0	11.2	37.3	7.6	0.0	44.9	205.3	23.1	0.1	228.6
28	0.386	0.8	0.3	0.0	1.1	8.7	2.1	0.1	10.8	37.2	7.7	0.0	44.9	205.1	23.9	0.1	229.2
29	0.400	0.7	0.2	0.0	1.0	8.6	1.8	0.1	10.5	37.2	7.5	0.0	44.8	205.1	24.7	0.0	229.8
30	0.415	0.7	0.2	0.0	0.9	8.6	1.6	0.1	10.3	37.3	7.3	0.0	44.5	205.3	25.2	0.0	230.5

31	0.429	0.6	0.1	0.0	0.7	8.6	1.4	0.1	10.2	37.4	6.9	0.0	44.3	205.7	25.5	0.0	231.1
32	0.443	0.6	0.1	0.0	0.7	8.6	1.3	0.1	10.0	37.5	6.5	0.0	44.1	206.3	25.4	0.0	231.7
33	0.458	0.5	0.0	0.0	0.6	8.6	1.2	0.1	9.9	37.7	6.0	0.0	43.8	207.0	25.0	0.1	232.1
34	0.472	0.5	0.0	0.0	0.5	8.7	1.0	0.1	9.8	37.9	5.4	0.0	43.3	207.8	24.2	0.2	232.2
35	0.486	0.5	0.0	0.0	0.5	8.7	0.8	0.1	9.6	38.1	4.8	0.0	42.9	208.6	23.2	0.2	232.0
36	0.500	0.5	0.0	0.0	0.5	8.7	0.6	0.1	9.4	38.2	4.2	0.0	42.5	209.3	22.2	0.2	231.7
37	0.515	0.5	0.0	0.0	0.5	8.7	0.4	0.1	9.2	38.4	3.8	0.1	42.2	209.9	21.3	0.1	231.3
38	0.529	0.5	0.0	0.0	0.5	8.7	0.3	0.0	9.1	38.4	3.3	0.1	41.9	210.5	20.3	0.1	230.9
39	0.543	0.5	0.0	0.0	0.5	8.7	0.2	0.0	9.0	38.5	3.0	0.1	41.6	211.0	19.4	0.1	230.5
40	0.558	0.5	0.0	0.0	0.5	8.7	0.1	0.0	8.9	38.6	2.6	0.1	41.3	211.5	18.6	0.1	230.2
41	0.572	0.5	0.0	0.0	0.5	8.7	0.1	0.0	8.9	38.6	2.3	0.1	41.1	211.8	18.1	0.2	230.1
42	0.586	0.5	0.0	0.0	0.5	8.7	0.1	0.0	8.8	38.7	2.1	0.1	40.9	212.1	17.8	0.3	230.2
43	0.601	0.5	0.0	0.0	0.5	8.7	0.1	0.0	8.9	38.7	1.9	0.1	40.8	212.4	17.7	0.4	230.4
44	0.615	0.5	0.0	0.0	0.5	8.8	0.1	0.0	8.9	38.7	1.9	0.1	40.7	212.6	17.7	0.5	230.8
45	0.629	0.5	0.0	0.0	0.5	8.8	0.1	0.0	8.9	38.7	1.9	0.1	40.7	212.6	17.9	0.6	231.1
46	0.643	0.5	0.0	0.0	0.5	8.8	0.1	0.0	8.9	38.7	1.9	0.1	40.7	212.5	18.3	0.7	231.5
47	0.658	0.5	0.0	0.0	0.5	8.8	0.1	0.0	8.9	38.6	1.9	0.1	40.6	212.3	18.7	0.7	231.7
48	0.672	0.5	0.0	0.0	0.5	8.8	0.2	0.0	8.9	38.6	2.0	0.1	40.6	211.9	19.2	0.8	231.9
49	0.686	0.5	0.0	0.0	0.5	8.7	0.2	0.0	8.9	38.4	2.1	0.1	40.6	211.5	19.7	0.7	231.9
50	0.701	0.5	0.0	0.0	0.5	8.7	0.2	0.0	8.9	38.3	2.2	0.1	40.6	211.0	20.2	0.6	231.8
51	0.715	0.5	0.0	0.0	0.5	8.7	0.2	0.0	8.9	38.2	2.4	0.1	40.7	210.4	20.5	0.6	231.5
52	0.729	0.5	0.0	0.0	0.5	8.7	0.2	0.0	8.9	38.1	2.5	0.1	40.6	209.9	20.7	0.5	231.1
53	0.744	0.5	0.0	0.0	0.5	8.7	0.2	0.0	8.9	38.0	2.5	0.1	40.6	209.3	20.9	0.5	230.7
54	0.758	0.5	0.0	0.0	0.5	8.7	0.2	0.0	8.9	37.9	2.6	0.1	40.6	208.7	21.3	0.5	230.5
55	0.772	0.5	0.0	0.0	0.6	8.7	0.3	0.0	9.0	37.8	2.9	0.1	40.7	208.1	21.7	0.5	230.2
56	0.786	0.6	0.0	0.0	0.6	8.7	0.3	0.1	9.1	37.7	3.1	0.1	40.9	207.6	21.9	0.4	229.9
57	0.801	0.6	0.0	0.0	0.6	8.8	0.4	0.1	9.2	37.7	3.5	0.0	41.2	207.1	22.0	0.3	229.3
58	0.815	0.6	0.0	0.0	0.6	8.8	0.5	0.1	9.4	37.6	3.9	0.0	41.5	206.5	21.9	0.2	228.6
59	0.829	0.6	0.0	0.0	0.6	8.8	0.7	0.1	9.6	37.5	4.4	0.0	41.9	206.1	21.9	0.1	228.1
60	0.844	0.6	0.0	0.0	0.7	8.9	0.9	0.1	9.9	37.4	5.2	0.0	42.6	205.6	22.3	0.0	227.9
61	0.858	0.7	0.0	0.0	0.7	8.9	1.3	0.1	10.3	37.3	6.2	0.0	43.5	205.2	23.1	0.0	228.3
62	0.872	0.7	0.1	0.0	0.8	9.0	1.7	0.2	10.9	37.2	7.5	0.0	44.7	204.8	24.4	0.0	229.3

(*continued*)

TABLE A.6 (*Continued*)

Frame	Time	Foot Segment				Leg Segment				Thigh Segment				HAT Segment			
		PE	TKE	RKE	Total	PE	TKE	RKE	Total	PE	TKE	RKE	Total	PE	TKE	RKE	Total
	s	J	J	J	J	J	J	J	J	J	J	J	J	J	J	J	J
63	0.887	0.7	0.1	0.1	0.9	9.0	2.4	0.2	11.6	37.1	9.0	0.0	46.0	204.5	25.8	0.1	230.4
64	0.901	0.8	0.2	0.1	1.1	9.1	3.2	0.2	12.5	36.9	10.5	0.0	47.4	204.1	26.7	0.2	231.1
65	0.915	0.8	0.3	0.1	1.2	9.1	4.1	0.2	13.4	36.7	11.8	0.0	48.5	203.8	26.5	0.4	230.7
66	0.929	0.9	0.5	0.1	1.5	9.2	5.0	0.2	14.3	36.6	12.6	0.1	49.2	203.6	24.9	0.6	229.1
67	0.944	1.0	0.7	0.1	1.8	9.2	5.8	0.2	15.2	36.4	12.9	0.1	49.4	203.4	22.3	0.8	226.5
68	0.958	1.1	1.0	0.1	2.1	9.3	6.6	0.1	16.0	36.3	12.6	0.1	49.1	203.2	19.6	0.8	223.7
69	0.972	1.1	1.3	0.0	2.5	9.4	7.3	0.1	16.7	36.2	12.1	0.2	48.5	203.2	17.4	0.7	221.3
70	0.987	1.2	1.7	0.0	3.0	9.5	7.9	0.1	17.4	36.1	11.4	0.3	47.8	203.3	15.8	0.5	219.7
71	1.001	1.3	2.2	0.0	3.5	9.6	8.4	0.0	18.0	36.2	10.7	0.3	47.2	203.5	14.9	0.3	218.7
72	1.015	1.4	2.7	0.0	4.1	9.7	8.8	0.0	18.5	36.3	10.2	0.4	46.8	203.9	14.4	0.2	218.5
73	1.030	1.4	3.3	0.0	4.7	9.8	9.3	0.0	19.1	36.4	9.9	0.4	46.8	204.5	14.3	0.1	218.9
74	1.044	1.4	3.8	0.0	5.2	9.9	9.6	0.0	19.5	36.7	9.8	0.4	46.9	205.2	14.3	0.1	219.6
75	1.058	1.4	4.4	0.0	5.8	10.0	9.9	0.1	20.0	37.0	9.8	0.4	47.1	206.1	14.5	0.1	220.7
76	1.072	1.4	4.9	0.0	6.3	10.0	10.1	0.2	20.3	37.3	9.8	0.3	47.4	207.0	14.7	0.1	221.8
77	1.087	1.3	5.5	0.0	6.8	10.1	10.2	0.2	20.5	37.6	9.7	0.3	47.6	208.1	14.7	0.2	223.0
78	1.101	1.2	5.9	0.1	7.2	10.1	10.2	0.3	20.6	38.0	9.5	0.2	47.7	209.1	14.6	0.3	224.0
79	1.115	1.1	6.4	0.1	7.6	10.0	10.2	0.4	20.6	38.3	9.1	0.2	47.6	210.0	14.4	0.3	224.8
80	1.130	1.1	6.7	0.1	7.8	10.0	10.1	0.5	20.5	38.6	8.7	0.1	47.4	210.8	14.3	0.4	225.5
81	1.144	1.0	7.0	0.1	8.0	9.9	9.9	0.5	20.4	38.9	8.1	0.1	47.1	211.3	14.4	0.4	226.1
82	1.158	0.9	7.2	0.1	8.1	9.9	9.8	0.6	20.2	39.1	7.6	0.1	46.7	211.7	14.7	0.3	226.6
83	1.173	0.8	7.3	0.1	8.2	9.8	9.6	0.6	19.9	39.2	7.0	0.0	46.2	211.9	14.9	0.2	227.0
84	1.187	0.7	7.4	0.1	8.2	9.7	9.3	0.6	19.6	39.3	6.5	0.0	45.8	211.8	15.2	0.1	227.2
85	1.201	0.7	7.3	0.1	8.1	9.5	9.1	0.6	19.2	39.3	6.0	0.0	45.3	211.6	15.6	0.1	227.4
86	1.215	0.6	7.2	0.1	7.9	9.4	8.7	0.6	18.8	39.2	5.7	0.0	44.9	211.3	16.1	0.1	227.4
87	1.230	0.6	6.8	0.1	7.5	9.3	8.2	0.6	18.2	39.0	5.5	0.0	44.6	210.8	16.7	0.1	227.5
88	1.244	0.6	6.4	0.1	7.1	9.2	7.6	0.6	17.5	38.8	5.5	0.0	44.4	210.1	17.4	0.1	227.6
89	1.258	0.7	5.7	0.1	6.4	9.1	6.9	0.6	16.6	38.6	5.6	0.0	44.2	209.3	18.2	0.1	227.6
90	1.273	0.7	4.8	0.1	5.5	9.0	6.2	0.5	15.6	38.3	5.8	0.0	44.1	208.5	18.9	0.1	227.5

91	1.287	0.8	3.8	0.0	4.6	8.9	5.4	0.3	14.6	38.0	6.0	0.0	44.0	207.6	19.4	0.1	227.2
92	1.301	0.8	2.8	0.0	3.6	8.8	4.6	0.2	13.6	37.7	6.4	0.0	44.0	206.7	20.0	0.2	226.8
93	1.316	0.8	1.9	0.0	2.7	8.8	4.0	0.1	12.8	37.4	6.8	0.0	44.2	205.9	20.5	0.2	226.5
94	1.330	0.8	1.2	0.0	2.0	8.7	3.5	0.0	12.2	37.2	7.4	0.0	44.6	205.1	21.0	0.3	226.4
95	1.344	0.8	0.7	0.0	1.5	8.6	3.0	0.0	11.7	37.1	7.9	0.0	45.0	204.6	21.5	0.3	226.4
96	1.358	0.8	0.5	0.0	1.3	8.6	2.7	0.0	11.3	37.0	8.4	0.0	45.4	204.2	22.3	0.3	226.8
97	1.373	0.7	0.3	0.0	1.1	8.6	2.4	0.1	11.0	37.0	8.6	0.0	45.6	204.0	23.1	0.3	227.4
98	1.387	0.7	0.2	0.0	1.0	8.5	2.2	0.1	10.8	37.0	8.6	0.0	45.6	204.0	24.0	0.2	228.2
99	1.401	0.6	0.2	0.0	0.8	8.5	2.0	0.1	10.6	37.1	8.3	0.0	45.4	204.2	24.6	0.1	228.9
100	1.416	0.6	0.1	0.0	0.7	8.5	1.7	0.1	10.4	37.2	7.8	0.0	45.1	204.7	24.8	0.0	229.5
101	1.430	0.5	0.1	0.0	0.6	8.6	1.5	0.1	10.2	37.4	7.2	0.0	44.6	205.3	24.7	0.0	230.0
102	1.444	0.5	0.0	0.0	0.6	8.6	1.2	0.1	9.9	37.6	6.4	0.0	44.0	206.1	24.2	0.0	230.3
103	1.459	0.5	0.0	0.0	0.5	8.6	0.9	0.1	9.6	37.8	5.5	0.0	43.3	206.9	23.3	0.1	230.2
104	1.473	0.5	0.0	0.0	0.5	8.6	0.6	0.1	9.3	37.9	4.6	0.0	42.5	207.6	21.8	0.1	229.5
105	1.487	0.5	0.0	0.0	0.5	8.6	0.4	0.1	9.1	38.1	3.7	0.0	41.8	208.3	19.7	0.1	228.1
106	1.501	0.5	0.0	0.0	0.5	8.6	0.2	0.1	8.9	38.2	2.9	0.1	41.1	208.9	16.9	0.1	225.9

TABLE A.7 Power Generation/Absorption and Transfer – Ankle, Knee, and Hip

		Muscle Power Gen(+)/ABS(−)			Rate of Transfer Across Joints and Muscle						Segment Angular Velocity			
					Leg to Foot		Thigh to Leg		Pelvis to Thigh					
Frame	Time	Ankle	Knee	Hip	Joint	Muscle	Joint	Muscle	Joint	Muscle	Foot	Leg	Thigh	HAT
	s	W	W	W	W	W	W	W	W	W	R/s	R/s	R/s	R/s
2	0.014	−2.8	−37.3	51.4	49.4	−2.7	110.5	0.0	56.1	12.6	−3.46	−1.70	3.59	0.71
3	0.029	−0.4	−32.0	45.6	46.0	−1.2	85.6	0.0	29.6	7.2	−1.06	−0.80	3.81	0.52
4	0.043	1.3	−24.4	38.7	41.3	0.3	62.9	1.4	17.6	4.0	1.18	0.21	3.95	0.37
5	0.057	2.1	−15.9	31.6	36.8	1.4	44.7	7.2	14.8	2.4	3.14	1.24	3.98	0.28
6	0.072	2.3	−8.0	24.3	32.8	2.0	31.6	10.6	15.2	1.8	4.74	2.21	3.89	0.27
7	0.086	2.0	−2.1	16.8	28.9	2.2	22.6	11.0	14.6	1.6	5.91	3.08	3.68	0.33
8	0.100	1.6	1.0	10.0	24.6	2.1	17.4	7.6	12.2	1.4	6.66	3.81	3.36	0.41
9	0.114	1.2	1.4	5.1	20.1	2.1	14.9	3.0	8.2	1.0	7.03	4.41	2.99	0.50
10	0.129	1.0	−0.2	1.9	15.2	2.1	13.7	−0.2	2.6	0.6	7.16	4.90	2.58	0.59
11	0.143	0.8	−3.3	0.0	10.0	2.3	12.5	−2.3	−3.8	0.0	7.17	5.28	2.16	0.68
12	0.157	0.7	−7.3	−0.8	4.1	2.7	10.3	−3.3	−10.6	−0.6	7.13	5.56	1.74	0.76
13	0.172	0.7	−12.1	−0.8	−2.3	3.1	6.1	−3.6	−18.4	−1.3	7.08	5.75	1.33	0.83
14	0.186	0.7	−17.3	−0.2	−9.1	3.4	−0.8	−3.3	−27.4	−2.1	7.06	5.86	0.94	0.85
15	0.200	0.7	−22.4	0.9	−15.6	3.7	−10.0	−2.4	−36.6	−2.1	7.06	5.91	0.58	0.82
16	0.215	0.7	−27.9	2.6	−22.0	3.8	−20.9	−1.1	−45.3	−1.2	7.06	5.91	0.23	0.76
17	0.229	0.7	−34.4	5.2	−28.7	3.7	−32.6	0.0	−54.1	0.0	7.02	5.85	−0.11	0.67
18	0.243	0.7	−42.0	8.5	−36.2	3.3	−43.6	0.0	−62.3	0.0	6.90	5.73	−0.43	0.56
19	0.257	0.5	−49.9	12.2	−44.1	2.6	−52.6	0.0	−68.6	0.0	6.66	5.51	−0.70	0.46
20	0.272	0.3	−57.2	15.7	−51.3	1.6	−59.6	0.0	−73.1	0.0	6.25	5.16	−0.91	0.37
21	0.286	0.1	−62.3	18.7	−56.6	0.4	−65.4	0.0	−76.3	0.0	5.61	4.61	−1.02	0.32
22	0.300	−0.1	−62.4	21.3	−58.4	−0.5	−69.8	0.0	−76.3	0.0	4.71	3.85	−1.03	0.31
23	0.315	−0.2	−54.6	23.0	−55.6	−1.0	−72.5	0.0	−72.4	0.0	3.53	2.88	−0.94	0.34
24	0.329	−0.2	−38.4	23.2	−47.9	−0.9	−73.4	0.0	−66.8	0.0	2.11	1.75	−0.79	0.42
25	0.343	0.0	−17.7	21.1	−36.8	−0.3	−71.5	0.0	−61.7	0.0	0.49	0.58	−0.63	0.50
26	0.357	0.4	0.2	16.6	−25.2	0.2	−65.9	6.3	−58.0	0.0	−1.27	−0.51	−0.49	0.53
27	0.372	0.4	9.3	10.8	−16.1	0.4	−56.4	3.9	−55.4	0.0	−2.99	−1.39	−0.41	0.50

HCR	28	0.386	4.0	53.9	44.0	−12.7	3.3	−99.0	13.3	−113.1	0.0	−4.40	−1.99	−0.39	0.41
	29	0.400	−13.3	38.2	25.9	50.6	−10.7	−28.3	8.1	−47.0	0.0	−5.24	−2.33	−0.41	0.28
	30	0.415	−23.5	16.0	12.3	86.6	−20.7	22.6	3.1	0.8	0.0	−5.34	−2.50	−0.41	0.12
	31	0.429	−6.2	10.2	6.2	91.8	−7.4	43.2	1.6	15.1	1.3	−4.80	−2.61	−0.36	−0.06
	32	0.443	−8.3	−18.6	0.4	76.0	−18.5	39.0	−2.2	−2.7	2.7	−3.88	−2.68	−0.29	−0.25
	33	0.458	−0.7	−35.3	−1.3	55.5	−12.4	30.3	−3.8	−29.2	2.0	−2.85	−2.69	−0.26	−0.42
	34	0.472	3.5	−55.3	−0.1	39.6	−9.3	31.1	−9.1	−43.5	0.1	−1.88	−2.60	−0.37	−0.55
	35	0.486	3.3	−55.8	−0.2	29.2	−2.9	40.9	−20.3	−41.8	−2.8	−1.12	−2.41	−0.64	−0.59
	36	0.500	−0.8	−38.8	−3.5	20.5	0.3	51.5	−36.2	−36.0	−4.1	−0.60	−2.15	−1.04	−0.56
	37	0.515	0.3	−16.0	−9.5	11.5	−0.1	55.7	−54.9	−38.9	−4.9	−0.29	−1.88	−1.45	−0.49
	38	0.529	−5.4	5.3	−6.9	3.2	0.4	52.9	−53.3	−48.4	−2.1	−0.12	−1.64	−1.81	−0.43
	39	0.543	−7.3	16.6	−5.0	−2.5	0.1	46.3	−41.1	−51.6	−1.3	−0.02	−1.46	−2.05	−0.41
	40	0.558	−9.0	21.0	−8.0	−6.7	0.0	37.1	−33.0	−42.9	−2.2	0.03	−1.33	−2.18	−0.47
	41	0.572	−13.6	19.3	−10.4	−11.9	0.0	25.0	−25.0	−27.6	−3.7	0.06	−1.24	−2.19	−0.58
	42	0.586	−12.8	18.4	−16.7	−18.5	0.0	12.9	−23.4	−12.7	−8.6	0.03	−1.18	−2.11	−0.72
	43	0.601	−15.2	13.1	−14.2	−24.7	1.2	6.0	−18.3	−0.1	−10.8	−0.09	−1.16	−1.99	−0.86
	44	0.615	−16.8	7.7	−9.7	−28.2	5.0	8.3	−12.8	12.1	−10.7	−0.27	−1.17	−1.87	−0.98
	45	0.629	−16.6	4.7	−7.1	−28.3	9.7	18.3	−9.1	26.1	−10.8	−0.43	−1.17	−1.77	−1.07
	46	0.643	−16.5	3.5	−6.3	−24.7	12.9	31.5	−7.7	41.5	−12.7	−0.51	−1.16	−1.70	−1.14
	47	0.658	−18.4	3.3	−6.4	−18.7	13.7	43.7	−7.6	56.4	−16.4	−0.49	−1.15	−1.65	−1.19
	48	0.672	−22.5	3.4	−7.1	−13.0	12.3	52.2	−8.0	69.0	−20.2	−0.41	−1.14	−1.63	−1.21
	49	0.686	−28.1	2.9	−8.3	−10.0	10.7	54.6	−6.9	76.9	−20.9	−0.32	−1.16	−1.65	−1.18
	50	0.701	−32.6	1.2	−8.7	−10.9	13.1	50.2	−2.7	78.3	−16.9	−0.34	−1.18	−1.69	−1.12
	51	0.715	−31.2	−1.2	−7.8	−15.4	23.1	40.2	2.7	74.2	−11.7	−0.51	−1.20	−1.75	−1.05
	52	0.729	−20.0	0.0	−11.5	−23.7	36.0	27.5	0.1	69.5	−14.6	−0.79	−1.23	−1.79	−1.00
	53	0.744	−10.8	−2.5	−10.0	−36.4	58.2	14.7	6.4	68.6	−12.0	−1.09	−1.29	−1.79	−0.98
	54	0.758	−3.8	−2.3	−11.7	−54.6	77.7	3.0	8.6	70.4	−14.2	−1.32	−1.39	−1.76	−0.97
	55	0.772	−1.8	−1.0	−12.1	−80.0	97.4	−8.1	12.1	68.7	−15.6	−1.50	−1.53	−1.66	−0.94
	56	0.786	−0.8	2.2	−10.4	−113.2	121.4	−19.1	15.7	59.9	−14.6	−1.71	−1.72	−1.51	−0.88
	57	0.801	6.1	7.6	−8.5	−150.5	149.1	−27.8	16.4	48.6	−12.8	−2.01	−1.93	−1.32	−0.79
	58	0.815	26.5	12.8	−8.3	−187.0	177.3	−31.0	13.8	42.8	−12.1	−2.46	−2.14	−1.11	−0.66
	59	0.829	65.3	14.5	−9.8	−222.4	204.4	−28.5	9.0	44.0	−11.4	−3.10	−2.35	−0.90	−0.48
	60	0.844	120.8	12.0	−11.8	−260.9	228.2	−22.8	4.3	49.0	−8.0	−3.89	−2.55	−0.67	−0.27
	61	0.858	181.5	5.1	−13.0	−303.1	246.0	−15.9	0.9	54.8	−1.6	−4.76	−2.74	−0.43	−0.05

(*continued*)

TABLE A.7 (*Continued*)

	Frame	Time	Muscle Power Gen(+)/ABS(−)			Rate of Transfer Across Joints and Muscle						Segment Angular Velocity			
						Leg to Foot		Thigh to Leg		Pelvis to Thigh					
			Ankle	Knee	Hip	Joint	Muscle	Joint	Muscle	Joint	Muscle	Foot	Leg	Thigh	HAT
		s	W	W	W	W	W	W	W	W	W	R/s	R/s	R/s	R/s
	62	0.872	232.2	−6.3	−12.0	−344.4	254.6	−10.7	−0.3	57.2	0.0	−5.62	−2.94	−0.14	0.18
	63	0.887	263.5	−21.7	−8.1	−376.4	249.1	−12.8	0.0	49.2	7.1	−6.43	−3.13	0.19	0.41
	64	0.901	272.4	−38.9	−2.6	−387.8	225.0	−24.0	0.0	27.1	18.9	−7.25	−3.28	0.56	0.64
	65	0.915	254.1	−54.0	2.8	−366.0	182.6	−36.6	0.0	−3.4	23.8	−8.03	−3.36	0.97	0.86
	66	0.929	203.4	−62.7	6.9	−307.3	129.2	−39.1	0.0	−28.9	22.2	−8.58	−3.33	1.40	1.07
	67	0.944	130.7	−62.9	9.8	−221.6	77.0	−24.4	0.0	−37.9	18.5	−8.57	−3.18	1.86	1.22
	68	0.958	60.7	−54.9	12.4	−125.8	35.5	5.8	0.0	−29.0	15.0	−7.81	−2.88	2.31	1.27
	69	0.972	14.0	−35.3	14.2	−31.8	8.9	44.1	0.0	−8.1	11.0	−6.31	−2.46	2.73	1.19
TOR	70	0.987	−3.3	−18.2	18.7	40.6	−2.6	81.8	0.0	18.1	9.2	−4.28	−1.91	3.10	1.02

APPENDIX B

UNITS AND DEFINITIONS RELATED TO BIOMECHANICAL AND ELECTROMYOGRAPHICAL MEASUREMENTS

All units used are Système International d'Unités (SI). The system is based on seven well-defined base units and two supplementary units. Only one measurement unit is needed for any physical quantity, whether the quantity is a base unit or a derived unit (which is the product and/or quotient of two or more of the base units).

TABLE B.1 Base SI Units

Physical Quantity	Symbol	Name of SI Unit	Symbol of Unit
Length	l	Meter	m
Mass	m	Kilogram	kg
Time	t	Second	s
Electric current	I	Ampere	A
Temperature	T	Kelvin	K
Amount of substance	n	Mole	mol
Luminous intensity	I	Candela	cd
Plane angle	$\boldsymbol{\theta}$, $\boldsymbol{\varphi}$, etc.	Radian	rad
Solid angle	$\boldsymbol{\Omega}$	Steradian	sr

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TABLE B.2 Derived SI Units

Physical Quantity	Symbol	Name of SI Unit	Definition
Velocity	v	m/s	Time rate of change of position
Acceleration	a	m/s^2	Time rate of change of velocity
Acceleration	g	m/s^2	Acceleration of a freely falling body in a vacuum because of gravity. At sea level $g = 9.80665\ m/s^2$
Angular velocity	$\boldsymbol{\omega}$	rad/s	Time rate of change of orientation of a line segment in a plane
Angular acceleration	$\boldsymbol{\alpha}$	rad/s^2	Time rate of change of angular velocity
Angular displacement	$\boldsymbol{\theta}$	Radian (rad)	Change in orientation of a line segment, which is given by the plane angle between initial and final orientations
Period	T	Second (s)	Time to complete one cycle of a periodic event or, more generally, time duration of any event or phase of an event
Frequency	f	Hertz (Hz)	Number of repetitions of a periodic event that occurs in a given time interval. 1 Hz equals one repetition or cycle per second (1 Hz = 1/s)
Density	$\boldsymbol{\rho}$	kg/m^3	Mass per unit volume of an object or substance
Specific gravity	d		Ratio of the density of a substance to the density of water at 4 °C
Force	F	Newton (N)	Effect of one body on another that causes the bodies to accelerate relative to an inertial reference frame. 1 N is that force that, when applied to 1 kg of mass, causes it to accelerate at 1 m/s^2 in the direction of the force application relative to the inertial reference frame ($1\ N = 1\ kg \cdot m/s^2$)
Weight	G	N	Force exerted on a mass because of gravitational attraction; equal to the product of the mass of the body and the acceleration from gravity ($G = m \cdot g$)
Mass moment of inertia	I	$kg \cdot m^2$	Measure of a body's resistance to accelerated angular motion about a given axis; equal to the sum of the products of the masses of its differential elements and the squares of their individual distances from that axis
Linear momentum	p	$kg \cdot m/s$	Vector quantity possessed by a moving rigid body, quantified by the product of its mass and the velocity of its mass center
Angular momentum	L	$kg \cdot m^2/s$	Vector of the linear momentum of a rigid body about a point; the product of the linear momentum and the perpendicular distance of the linear momentum from the point. For planar movement, the angular momentum is the product of the moment of inertia in the plane about its centroid and the angular velocity in the plane
Moment of force	M	$N \cdot m$	Turning effect of a force about a point; the product of the force and the perpendicular distance from its line of action to that point
Pressure, normal stress, shear stress	p	Pascal (Pa)	Intensity of a force applied to, or distributed over, a surface; the force per unit area. 1 Pa is the pressure resulting from 1 N applied uniformly and in a directional perpendicular over an area of 1 m^2 ($1\ Pa = 1\ N/m^2$)
Linear strain	$\boldsymbol{\varepsilon}$		Deformation resulting from stress measured by the percentage change in length of a line (linear strain) or the change in strain)
Shear strain	$\boldsymbol{\gamma}$		

TABLE B.2 ***(Continued)***

Physical Quantity	Symbol	Name of SI Unit	Definition
Young's modulus Shear modulus	E G	Pa	Ratio of stress to strain over the initial linear portion of a stress–strain curve
Work	W	Joule (J)	Energy change over a period of time as a result of a force acting through a displacement in the direction of the force. 1 J is the work done when a force of 1 N is displaced a distance of 1 m in the direction of the force ($1\ J = 1\ N \cdot m$). Work is also the time integral of power ($1\ J = 1\ W \cdot s$)
Mechanical energy	E	J	Capacity of a rigid body to do work; quantified by the sum of its potential and kinetic energies
Potential energy	V	J	Energy of a mass or spring associated with its position or configuration relative to a spatial reference Gravitational potential energy of a mass m raised a distance h above the reference level equals mgh, where g is the acceleration from gravity. Elastic potential energy of a linear spring with stiffness k stretched or compressed a distance e equals $k \cdot e^2/2$
Kinetic energy	T	J	Energy of a mass associated with its translational and rotational velocities. The translational kinetic energy T of a mass m with a velocity v is $1/2\ mv^2$. The rotational kinetic energy T of a body rotating in a plane with a moment of inertia I rotating with a angular velocity $\boldsymbol{\omega}$ is $1/2\ I\boldsymbol{\omega}^2$
Power	P	Watt (W)	Rate at which work is done or energy is expended. The power generated by a force is the dot product of the force and the velocity at the point of application of the force ($P = F \cdot V$). The power generated by a moment is the dot product of the moment and the angular velocity of the rigid body ($P = M \cdot \boldsymbol{\omega}$)
Coefficient of friction	$\boldsymbol{\mu}$	—	For two objects in contact over a surface, the ratio of the contact force parallel to the surface to the contact force perpendicular to the surface
Coefficient of viscosity	$\boldsymbol{\eta}$	$N \cdot s/m^2$	Resistance of a substance to change in form; calculated by the ratio of shear stress to its rate of deformation
Electrical charge	q	Coulomb (C)	Quantity of a negative or positive charge on any mass. The charge on an electron or proton is 1.602×10^{-19} C, or 1 C, has the charge of 6.242×10^{18} electrons or protons (1 A [ampere] = 1 C/s)
Voltage, electrical potential	E	Volt (V)	Potential for an electrical charge to do work (1 V = 1 J/C)
Electrical resistance	R	Ohm (**Ω**)	Property of a conducting element that opposes the flow of electrical charge in response to an applied voltage (1 **Ω** = 1 V/A)
Electrical capacitance	C	Farad (F)	Property of an electrical element that quantifies its ability to store electrical charge. A capacitance of 1 F means that 1 C of charge is stored with a voltage change of 1 V (1 F = 1 C/V)

NOTES TO TABLES B.1 AND B.2

1. Prefixes are used to designate multiples or submultiples of units.

Prefix	Multiplier	Symbol	Examples
Mega	10^6	M	Megahertz (MHz)
Kilo	10^3	k	Kilowatt (kW)
Centi	10^{-2}	c	Centimeter (cm)
Milli	10^{-3}	m	Millisecond (ms)
Micro	10^{-6}	μ	Microvolt (μV)

2. (a) When a compound unit is formed by multiplication of two or more units, the symbol for the compound unit can be indicated as follows:

$$\mathrm{N{\cdot}m \text{ or } N\ m, \text{ but not } Nm}$$

(b) When a compound unit is formed by dividing one unit by another, the symbol for the compound unit can be indicated as follows:

$$\mathrm{kg/m^3} \text{ or as a produce of kg and } \mathrm{m^{-3}}, \mathrm{kg{\cdot}m^{-3}}$$

3. Because the symbol for second is s, not sec, the pluralization of symbols should not be done; for example, kgs may be mistaken for kg · s, or cms may be mistaken for cm · s.

INDEX

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